

MetaTrader 5 Trading Platform / Platform Setup / Reports / Credit Facility

Credit Facility Report

Credit Facility Report is a report on credit operations ("Credit" balance operation) for the specified period.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report will be created;
- **Period** starting and ending date of the period, for which credit funds accrual/withdrawal will be requested.

The following data on credit funds accrual/withdrawal is provided:

- **Deal** credit funds accrual/withdrawal deal ticket;
- **Login** number of the account, at which the deal has been performed;
- **Name** account holder name;

- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;

- **Time** transaction time;
- **Comment** comments on the transaction;
- **Profit** transaction size. A positive value means that the funds have been credited, while a negative one means that the funds have been withdrawn;
- **Currency** transaction currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily

Daily Report

Daily Report is a report on the financial state of the requested accounts as of the end of a day.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;
- **Period** starting and ending date of the period. The report will be generated for each day in the selected period.

Daily reports generation must be enabled on the server for the group containing the necessary account to generate this type of reports.

The report contains the following data:

- **Time** date and time, the data is generated for;
- **Login** account number;
- **Name** account holder name;

- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;

- **Prev Balance** balance at the end of the previous day;
- **Deposit** total funds deposited/withdrawn from the account per day;
- **Closed P/L** total closed profit/loss at all deals per day.
- **Balance** balance at the end of the day;
- **Credit** credit funds deposited/withdrawn from the account per day;
- **Floating P/L** floating profit/loss of all open positions at the end of the day;
- **Equity** equity at the end of the day;
- **Margin** money required to cover open positions as of the end of the day;
- **Free Margin** amount of free margin volume as of the end of the day;
- **Currency** deposit currency.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Reports](#) / Daily Detailed

Daily Detailed Report

Daily Detailed Report is a detailed report on the financial state of the one selected account as of the end of a day.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** number of the account, for which the report must be generated;
- **Period** a day, for which the report must be generated. Since the daily data, based on which the report is created, is only generated at the end of the trading day, a current day report cannot be requested.

- Daily data generation should be enabled for the client or group for which the request is created. Otherwise, the report on them will not be available.
- If the report for the selected day is not available for some reason (for example, daily data generation was disabled on the server), the system will try to show the next available report.

The report is divided into several blocks.

Header

The following data is specified here:

- **Brokerage company name;**
- **Account number;**
- **Account holder name;**
- **Deposit currency;**
- **Report generation date.**

Orders

This block contains the table displaying the orders set during the specified day. The table displays all data fields available by orders in the "[Orders](#)" section.

Deals

The deals performed during the selected day are displayed here. The table displays all data fields available by deals in the "[Deals](#)" section. Additional parameters are displayed at the bottom of the orders block:

- **Closed P/L** total closed profit/loss at all deals per day.

- **Deposit/Withdrawal** total funds deposited/withdrawn from the account per day;
- **Credit Facility** credit funds deposited/withdrawn from the account per day;
- **Round Commission** commission by orders and positions accumulated during a day/month. Depending on the settings (specified for the group in the administrator terminal), preliminary commission calculation is performed during a day/month and the appropriate funds are blocked in the account and displayed here. Final commission calculation is performed at the end of a day/month and the appropriate sum is withdrawn from the account by the balance operation.
In case commission is charged immediately during a deal, its value is shown in the "Commission" field.
- **Additional Operations** financial result of other transactions performed in the account. These are additional charges, corrections, bonuses, agent commissions and annual interests;
- **Total** total sum for the above operations.

Positions

This block shows the positions remaining open in the account at the moment of the report generation (the end of the selected day). The table displays all data fields available by positions in the "[Positions](#)" section. Additional parameter is displayed at the bottom of the positions block:

- **Floating P/L** floating profit/loss of all open positions at the end of the day.

Working Orders

Active orders (pending and unfilled market orders) are displayed in this block.

A/C Summary

This block displays the aggregate account figures:

- **Closed P/L** total closed profit/loss at all deals per day.
- **Deposit/Withdrawal** total funds deposited/withdrawn from the account per day;
- **Credit Facility** credit funds deposited/withdrawn from the account per day;
- **Round Commission** commission by orders and positions accumulated during a day/month. Depending on the settings (specified for the group in the administrator terminal), preliminary commission calculation is performed during a day/month and the appropriate funds are blocked in the account and displayed here. Final commission calculation is performed at the end of a day/month and the appropriate sum is withdrawn from the account by the balance operation.
- **Additional Operations** financial result of other transactions performed in the account. These are additional charges, corrections, bonuses, agent commissions and annual interests;
- **Total** total sum for the above operations;
- **Previous Ledger Balance** balance as of the end of the previous day;
- **Previous Equity** equity as of the end of the previous day;

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- **Balance** balance at the end of the day;
- **Equity** equity at the end of the day;
- **Floating P/L** floating profit/loss of all open positions at the end of the day;
- **Margin Requirements** money required to cover open positions as of the end of the day;
- **Free Margin** amount of free margin volume as of the end of the day.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Server Logs

Daily Server Logs Report

Daily Server Report is a statistical report on the platform operation for a specified day.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Period** a day, for which the report must be generated.

The report contains several diagrams and tables:

- **Server Messages** total number of entries in the trading server for the day, as well as the number of critical messages for the day;
- **Critical Errors** list of critical errors for the day is displayed in the table. Description and the number of cases are shown for each error.
- **Top Flooders** this diagram shows the users, from whom the server log contains most messages.
- **Manager Connections** list of manager connections to the server for the day is displayed in this table. Manager login and IP address, from which the connection has been performed, are shown.
- **Client Builds** this diagram shows distribution of connections to the server by builds and terminal types: terminal PC version, mobile terminals for iPhone and Android.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Dealing

Daily Dealing Report

Daily Dealing Report is a daily report on dealing activity. The report contains three tables.

Setup in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Period** a day, for which the report must be generated.

Setup in MetaTrader 5 Administrator

Additional parameters are set in the report configuration of MetaTrader 5 Administrator:

- **Currency** currencies used for displaying dealer's profit in the report;
- **Groups** groups, for which the report is generated. "*" mask and "!" negation symbol can be used to specify the groups. *,!demo*,!contest* value is assigned to this parameter by default. It means that a report is prepared for all groups except those that begin with "demo" and "contest" words.
- **Dangerous Tick Deviation** value of a deviation (in pips in either direction) of a price thrown in by a dealer from the current price, above which the throw is deemed to be dangerous. Dangerous Ticks Thrown in by Dealers table is built based on that parameter.
- **Dangerous Profit** value of the profit (definite currency is specified in Currency field) obtained by a dealer/client after the prices have been thrown in by the dealer, above which dealer's actions are deemed to be dangerous. Dangerous Actions of Dealers table is built based on that parameter.
- **Max Dangerous Actions Shown** maximum number of entries displayed in Dangerous Actions of Dealers table.

Managers

This table shows the list of managers who have connected the server during a day.

- **Login** manager's account number;
- **Name** manager's name;
- **Connections** number of connections to the server per day;
- **Requests** number of requests received by a dealer per day;
- **Requested** amount of new prices sent by a dealer in response to a request (requote);
- **Rejected** number of requests rejected by a manager per day;
- **Dealer profit, currency** profit earned by a manager in a specified currency;
- **Dealer profit, pips** profit earned by a manager in pips.

Dangerous Actions of Dealers

The following table contains the list of suspicious deals confirmed by dealers:

- **Manager** dealer's account number;
- **Name** dealer's name;
- **Dealer profit, currency** profit earned by a dealer in a specified currency;
- **Dealer profit, pips** profit earned by a dealer in pips;
- **Details** server log entry describing dealer's actions.

Dangerous Ticks Thrown in by Dealers

The following table contains the list of suspicious quotes thrown in by dealers:

- **Manager** dealer's account number;
- **Name** dealer's name;
- **Symbol** symbol, for which a quote has been thrown in;
- **Bid deviation, Pips** deviation of the thrown in quote Bid price from the market Bid price that was actual during the throw;
- **Ask deviation, Pips** deviation of the thrown in quote Ask price from the market Ask price that was actual during the throw.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Trades

Daily Trade Report

Daily Trade Report is a report on trading operations for a specified day.

Setup in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report will be created;
- **Period** a day, for which the report must be generated.

Setup in MetaTrader 5 Administrator

Additional parameters are set in the report configuration of MetaTrader 5 Administrator:

- **Currency** currency used for displaying various parameters in the report (Balance, Profit, etc.).
- **Top Profit Deals** the number of the most profitable deals to be displayed in the relevant report table. The maximum value is 10,000.
- **Top Loss Deals** the number of deals with the largest losses to be displayed in the relevant report table. The maximum value is 10,000.
- **Top Profit Positions** the number of the most profitable positions to be displayed in the relevant report table. The maximum value is 10,000.
- **Top Loss Positions** the number of positions with the largest losses to be displayed in the relevant report table. The maximum value is 10,000.
- **Top Profit Accounts** the number of accounts with the highest profits to be displayed in the relevant report table. The maximum value is 10,000.

Diagrams

The report contains several diagrams:

- **Profit and Loss of Clients** distribution of closed profit, closed loss and net profit by accounts.
- **Number of Client Trades** distribution of the number of clients' profitable and loss-making deals.
- **Total Profit/Loss of Current Client Positions** total profit/loss of the current clients' positions.

Tables

The report contains two tables:

Daily trades

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This report shows 10 most profitable and loss-making deals per day and 10 accounts having the greatest profit for the day.

The following data on the deals is shown:

- **Deal** deal ticket;
- **Login** number of the account, at which the deal has been performed;
- **Name** account holder name;
- **Symbol** the symbol, for which the deal is executed;
- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;
- **Type** deal type (buy/sell);
- **Volume** deal volume in lots;
- **Price** price, at which the deal was executed;
- **Swap** swap size;
- **Profit, currency** profit/loss in a specified currency.

The following data on the most profitable accounts is displayed:

- **Login** account number;
- **Name** account holder name
- **Group** group the account belongs to;
- **Leverage** leverage;
- **Placed Orders** number of active pending orders in the account at the moment of report request;
- **Orders** number of orders in the account history for the selected day;
- **Deals** number of performed deals;
- **Balance, currency** account balance at the moment of the report request;
- **Floating P/L, currency** floating profit/loss of open positions in the account at the moment of the report request;
- **Closed P/L, currency** closed profit/loss per day.

Open positions

This table shows 10 most profitable and 10 most loss-making positions as of the end of the day. The following data is displayed for each item:

- **Login** number of an account having an open position;
- **Name** account holder name;
- **Symbol** symbol, by which the position has been opened;
- **Group** group the account belongs to;
- **Type** position direction (buy/sell);
- **Volume** position volume in lots;
- **Open Price** weighted average position open price;
- **S/L** Stop Loss level;
- **T/P** Take Profit level;
- **Market Price** financial instrument price at the moment of a request.
- **Swap** swap size;

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- **Points** profit/loss in points at the moment of a request;
- **Profit, currency** profit/loss in a specified currency at the moment of a request.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Orders

Daily Orders Report

Daily Orders Report features information about market and pending orders which remained open (active) at the end of the selected day.

Configuration in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting a report:

- **Groups** the report will be generated on accounts listed in these groups.
- **Period** a day, for which the report must be generated.

Daily reports must be enabled on the server for the group containing the necessary accounts, to generate this type of reports.

Data in the report

The report is presented as a table displaying the following order data:

- **Order** ticket number (unique number) of a trade operation.
- **ID** order ID in the external system.
- **Position** ticket of the position opened, modified or closed due to this order.
For "close by" orders, two tickets are displayed: closed position ticket and opposite position ticket.
- **Login** account number.
- **Name** account holder's name.

- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Setup time** order setup time. The record has a format of YYYY.MM.DD HH:MM.MSC (year.month.day hour:minute.millisecond).
- **Type** order type: "Buy", "Sell", "Buy Limit", "Sell Limit", "Buy Stop", "Sell Stop", "Buy Stop Limit", "Sell Stop Limit" or "Close By".
- **Symbol** the financial instrument for which the order is placed.
- **Volume** volume requested in the order/executed volume, lots.
- **Price** price specified in the order at which the trade operation should be executed.
- **S/L** Stop Loss level.
- **T/P** Take Profit level.
- **Done time** order execution time. The record has a format of YYYY.MM.DD HH:MM.MSC (year.month.day hour:minute.millisecond).
- **Current price** the current price of a financial instrument for which the order was placed.
- **Reason** reason for placing the order.

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- **State** order state at the end of the day for which the report is generated.
- **Comment** a comment to the order.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Positions

Daily Positions

Daily Positions Report features information about trading positions which remained open at the end of the selected day.

Configuration in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting a report:

- **Groups** the report will be generated on accounts listed in these groups.
- **Period** a day, for which the report must be generated.

Daily reports must be enabled on the server for the group containing the necessary accounts, to generate this type of reports.

Data in the report

The report is presented as a table displaying the following position data:

- **Position** position ticket (unique number).
- **ID** position identifier in the external system.
- **Login** the account of the number on which the position was opened.
- **Name** account holder's name.

- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Creation time** position opening time. The record has a format of YYYY.MM.DD HH:MM.MSC (year.month.day hour:minute.millisecond).
- **Action** position type, Buy or Sell.
- **Symbol** financial instrument for which the position was opened.
- **Volume** position volume in lots.
- **Price** weight-average position opening price: $(\text{price of deal 1} * \text{volume of deal 1} + \dots + \text{price of deal N} * \text{volume of deal N}) / (\text{volume of deal 1} + \dots + \text{volume of deal N})$. The rounding accuracy of the average weighted price is equal to the number of decimal places in the symbol price plus three additional digits.
- **S/L** Stop Loss level.
- **T/P** Take Profit level.
- **Commission** the amount of position withheld on the position. It is calculated as the sum of commissions on deals by which the position was formed.
- **Fee** total position fees. It is calculated as the sum of fees on deals by which the position was formed.
- **Swap** calculated swaps.
- **Profit** position profit at the end of the selected day.
- **Currency** the currency in which the profit is indicated.

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- **Reason** reason for opening a position.
- **Comment** a comment to position.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Daily Expert Advisors

Daily Expert Advisors Report

Daily Expert Report is a report on trading operations performed by clients' Expert Advisors. The report is generated for seven days.

Configuration in MetaTrader 5 Manager

The following parameters should be specified before requesting a report in the manager terminal:

- **Groups** groups with the accounts used for generating a report;
- **Period** reports are generated for the last seven days up to the specified day.

Configuration in MetaTrader 5 Administrator

Additional parameters are specified in the report configuration of MetaTrader 5 Administrator:

- **Group** a group or several groups (separated by commas) having the accounts used for reports generation. "*" mask and "!" negation symbol can be used when specifying the groups. For example, if *,!demo*,!contest* are specified, the groups with the names beginning from **demo** and **contest** symbols will not be considered in generated reports.
- **Currency** currency used when displaying various parameters in the report (Balance, Profit, etc.).

Diagrams

The report contains several diagrams:

- **Deals of Clients** total number of deals performed by clients, as well as the number of deals performed by Expert Advisors.
- **Orders of Clients** total number of orders placed by clients, as well as the number of orders placed by Expert Advisors.
- **Profit and Loss of Clients** total profit (Profit)/loss (Loss) gained by clients, as well as profit/loss obtained by Expert Advisors.

Tables

The report contains the following tables:

Users with the highest number of deals performed by Expert Advisors

The report contains 10 users having the highest number of deals performed by their Expert Advisors for a specified day. The table displays only profitable accounts (for a specified day).

The following data is displayed in the table:

- **Login** account number;
- **Name** name of the account holder;
- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;
- **Balance, currency** account balance at the time of the report request;
- **Floating P/L, currency** floating profit/loss on the account at the time of the report request;
- **Experts profit, currency** profit made by an Expert Advisor for a specified day in a specified currency;
- **Total profit, currency** total profit/loss for a specified day in a specified currency;
- **Experts deals** number of deals performed by Expert Advisors for a specified day;
- **Total deals** total number of deals performed by a user for a specified day.

Users with the highest number of orders placed by Expert Advisors

The report contains 10 users having the highest number of orders placed by their Expert Advisors for a specified day. The table displays only profitable accounts (for a specified day).

The following data is displayed in the table:

- **Login** account number;
- **Name** name of the account holder;
- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;
- **Balance, currency** account balance at the time of the report request;
- **Floating P/L, currency** floating profit/loss on the account at the time of the report request;
- **Experts profit, currency** profit made by an Expert Advisor for a specified day in a specified currency;
- **Total profit, currency** total profit/loss for a specified day in a specified currency;
- **Experts orders** number of orders placed by Expert Advisors for a specified day;
- **Total orders** total number of orders placed by a user for a specified day.

Users with the highest profit earned by Expert Advisors

The report contains 10 users having the highest profit as a result of their Expert Advisors' operation for a specified day. The table displays only profitable accounts (for a specified day).

The following data is displayed in the table:

- **Login** account number;
- **Name** name of the account holder;
- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;
- **Balance, currency** account balance at the time of the report request;
- **Floating P/L, currency** floating profit/loss on the account at the time of the report request;
- **Experts deals** number of deals performed by Expert Advisors for a specified day;
- **Experts orders** number of orders placed by Expert Advisors for a specified day;
- **Experts profit, currency** profit made by an Expert Advisor for a specified day in a specified currency;
- **Total profit, currency** total profit/loss for a specified day in a specified currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deals History

Deals History Report

Deals History Report is a summary report on the deals over the selected period.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

The following data is displayed for each deal:

- **Deal** ticket number (a unique identifier) of a deal.
- **ID** ID of a deal in an external trading system.
- **Order** ticket of the order, as a result of which the deal has been performed.
- **Login** number of the account, at which the deal has been performed.
- **Name** account holder name.

- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Time** time of the deal. The record is represented as YYYY.MM.DD HH:MM (year.month.day hour:minute).
- **Type** type of a trade operation: "Buy" a buy deal, "Sell" a sell deal.
- **Entry** direction of the deal: in (market entry), out (market exit), in/out (position reversal).
- **Symbol** a financial instrument of the deal.
- **Volume** volume of an executed deal in lots.
- **Price** price, the deal was executed at.
- **Reason** the reason for executing the deal.
- **Commission** commission charged for the deal execution.
- **Fee** fee charged for the deal execution.
- **Swap** swap size.
- **Profit** financial result of the deal. For entry deals, zero profit is shown.
- **Dealer** number of the dealer's account who processed this deal. "0" specified in this field means that the deal was processed without a dealer.
- **Currency** account deposit currency.
- **Comment** a comment on a deal.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deals Profit

Deals Profit

A detailed deal report for the selected period. It consists of several diagrams: by deal volume with distribution by profitable and losing trades, by profit, loss, total number and other metrics.

Deals Profit

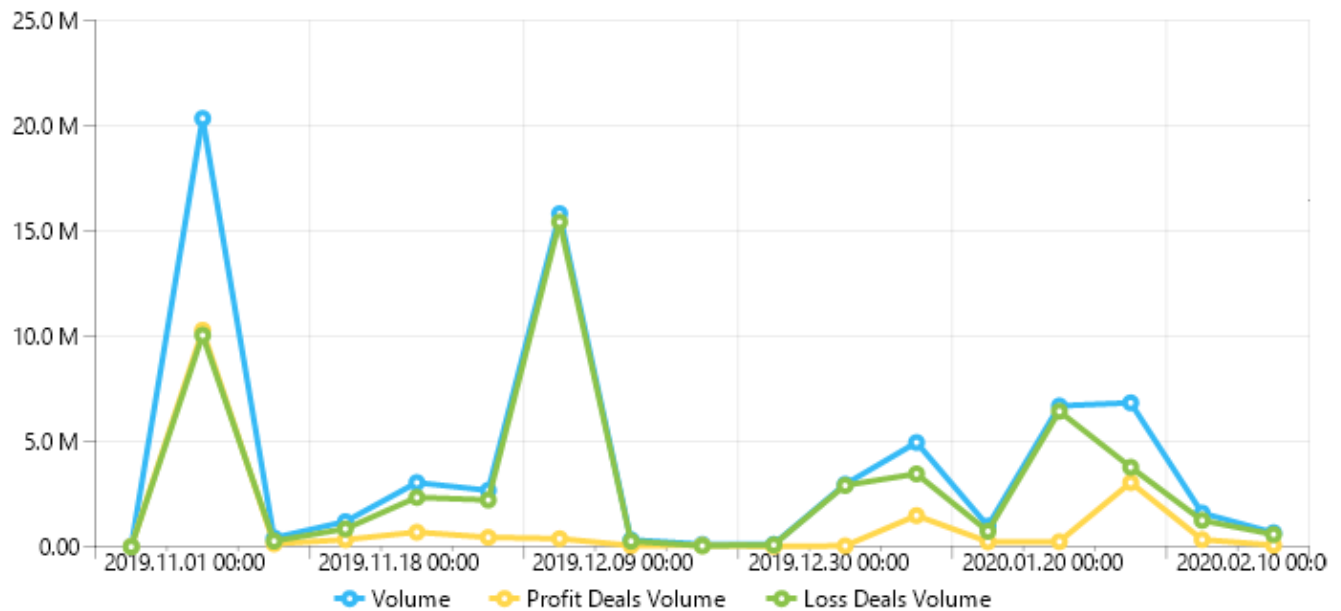
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2019.11.01 - 2020.02.19

All Lead Campaign

All Lead Source

Volume, USD



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.
- **Lead Campaign** the name of an advertising campaign by which the client, who has executed the deal, was attracted.
- **Lead Source** the website from which the client, who has executed the deal, came.

Charts in the report

The report includes multiple charts:

- **Volume** distribution of the total volume of deals, the volume of profitable deals and the volume of losing deals.
- **Profit Amount** distribution of deals profits and losses.
- **Profit Deals Count** distribution of the number of profitable and losing deals.
- **Profit Deals Volume Average** distribution of the average volume of profitable and losing deals.
- **Profit-Loss** the difference between the total profit and total loss from deals.
- **Profit-Loss Accumulated** the accumulated difference between the total profit and total loss from deals.

The volume of deals is indicated in money. Volumes, profit and loss values are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

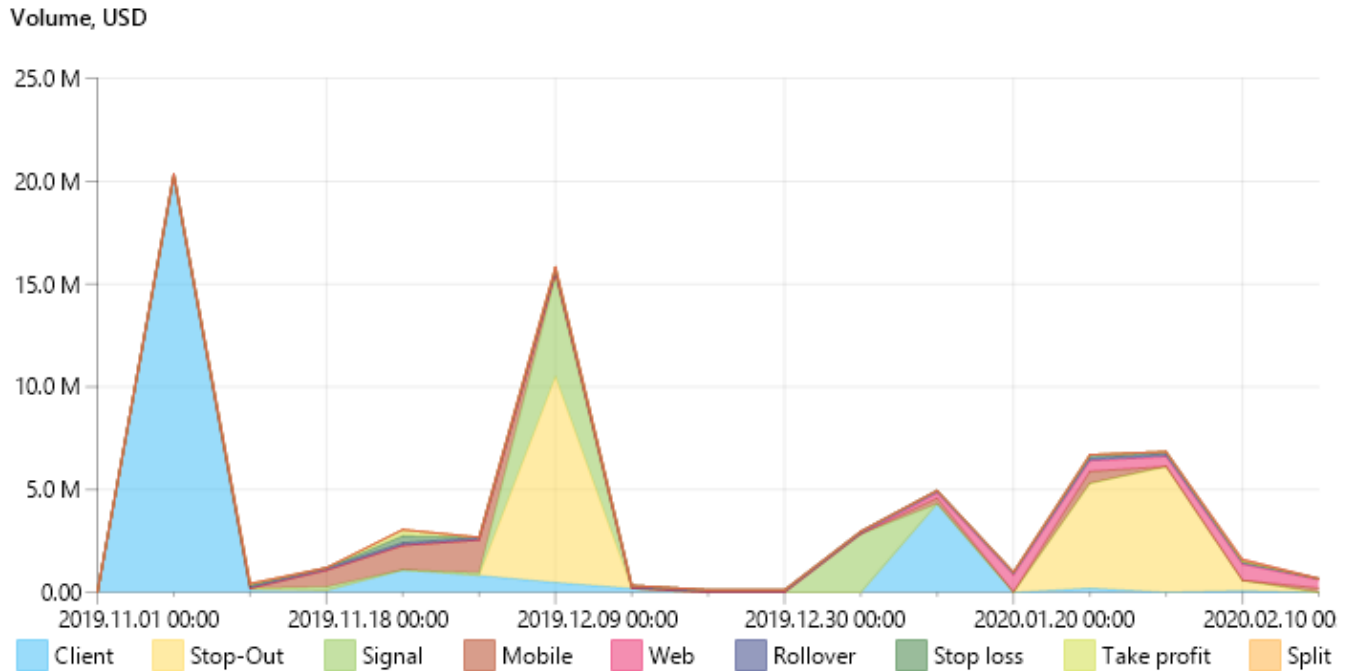
Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deals Initiators

Deals Initiators

Deal Initiators (reasons) report. The report demonstrates how traders prefer to perform operations: manually from the desktop or mobile terminal, using trading robots or copying signals. In addition, it shows which methods generate the greatest profit and loss.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.
- **Lead Campaign** the name of an advertising campaign by which the client, who has executed the deal, was attracted.
- **Lead Source** the website from which the client, who has executed the deal, came.

Charts in the report

The report includes multiple charts grouped into categories:

- **Volume** distribution of the total volume of deals execution reasons and by time.
- **Profit Deals Volume** distribution of the volume of profitable deals.
- **Loss Deals Volume** distribution of the volume of losing deals.
- **Profit Amount** distribution of deal profits.
- **Loss Amount** distribution of deal losses.
- **Profit Deals Count** distribution of the number of profitable deals.
- **Loss Deals Count** distribution of the number of losing deals.
- **Profit Deals Volume Average** distribution of the average volume of profitable deals.
- **Loss Deals Volume Average** distribution of the average volume of losing deals.
- **Profit-Loss** the difference between the total profit and total loss from deals.
- **Profit-Loss Accumulated** the accumulated difference between the total profit and total loss from deals.

The volume of deals is indicated in money. Volumes, profit and loss values are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

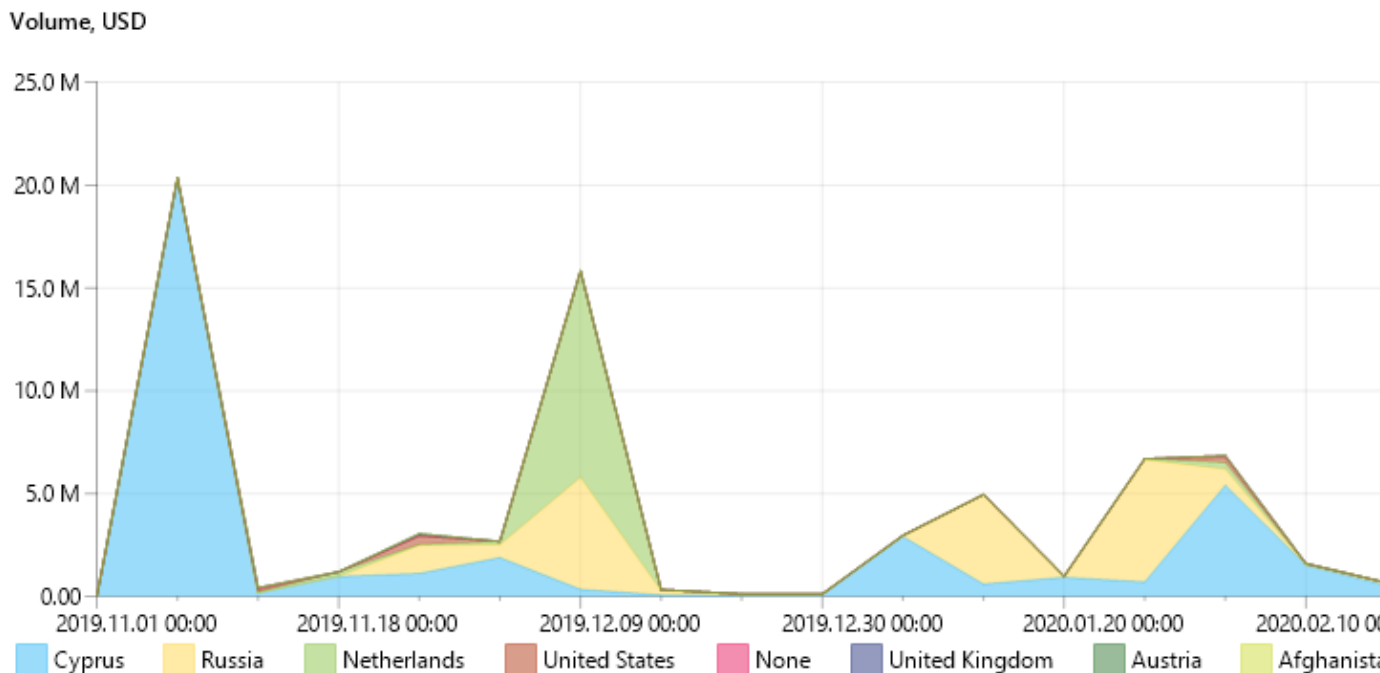
Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deals
Geography

Deals Geography

The report shows trading activity distribution by country. The information is provided with a breakdown by different aspects: by number, by volume and by deal results.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.
- **Lead Campaign** the name of an advertising campaign by which the client, who has executed the deal, was attracted.
- **Lead Source** the website from which the client, who has executed the deal, came.

Charts in the report

The report includes multiple charts grouped into categories:

- **Volume** distribution of the total volume of deals by country and time.
- **Profit Deals Volume** distribution of the volume of profitable deals.
- **Loss Deals Volume** distribution of the volume of losing deals.

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- **Profit Amount** distribution of deal profits.
- **Loss Amount** distribution of deal losses.
- **Profit Deals Count** distribution of the number of profitable deals.
- **Loss Deals Count** distribution of the number of losing deals.
- **Profit Deals Volume Average** distribution of the average volume of profitable deals.
- **Loss Deals Volume Average** distribution of the average volume of losing deals.
- **Profit-Loss** the difference between the total profit and total loss from deals.
- **Profit-Loss Accumulated** the accumulated difference between the total profit and total loss from deals.

The volume of deals is indicated in money. Volumes, profit and loss values are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deals Weekly

Deals Weekly

Use the report to assess your traders' activity. It displays information on executed deals grouped by week. Data can be filtered by groups.

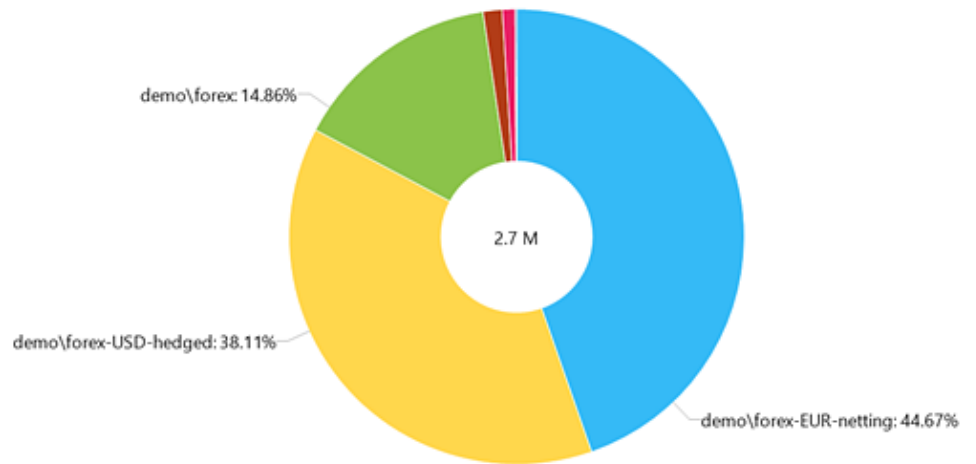
Deals Weekly

Groups filter *

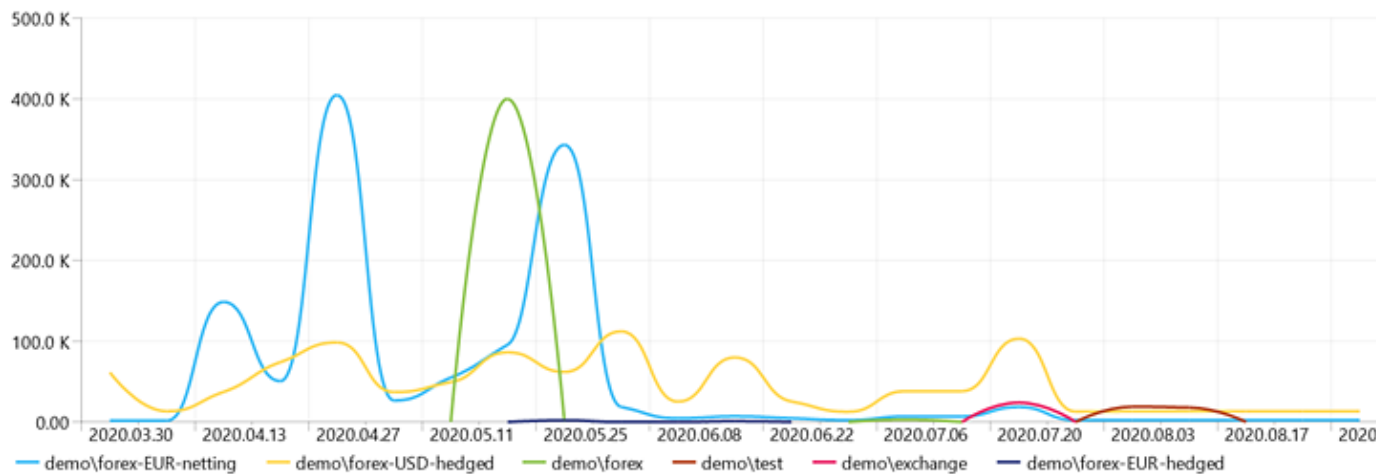
2020.04.01 - 2020.09.09

Volume, USD

- demo\forex-EUR-netting
- demo\forex-USD-hedged
- demo\forex
- demo\test
- demo\exchange
- demo\forex-EUR-hedged



Volume, USD



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes multiple charts grouped into categories:

- **Volume** distribution of deals volume: a total volume diagram and distribution by weeks.
- **Profit Deals Volume** distribution of the volume of profitable deals.
- **Loss Deals Volume** distribution of the volume of losing deals.
- **Profit Amount** distribution of deal profits.
- **Loss Amount** distribution of deal losses.
- **Profit Deals Count** distribution of the number of profitable deals.
- **Loss Deals Count** distribution of the number of losing deals.
- **Profit Deals Volume Average** distribution of the average volume of profitable deals.
- **Loss Deals Volume Average** distribution of the average volume of losing deals.
- **Profit-Loss** the difference between the total profit and total loss from deals.
- **Profit-Loss Accumulated** the accumulated difference between the total profit and total loss from deals.

The volume of deals is indicated in money. Volumes, profit and loss values are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Deposit and Withdrawal

Deposit and Withdrawal Report

Deposit and Withdrawal Report is a report on deposit and withdrawal operations over the selected period.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;
- **Period** starting and ending date of the period, for which the report will be generated.

The following data is displayed for each balance operation:

- **Deal** ticket of an executed balance operation deal;
- **Login** number of the account, at which the deal has been performed;
- **Name** account holder name;

- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;

- **Time** transaction time;
- **Comment** comments on the transaction;
- **Amount** transaction amount. A positive value means that the funds have been credited, while a negative one means that the funds have been withdrawn;
- **Currency** account deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Equity

Equity Report

Equity Report is a report on the financial state of the requested accounts as of the end of a day. The accounts are grouped according to their "Comment" field value. This report allows to generate reports on clients' accounts grouping them by custom criteria.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;
- **Period** starting and ending date of the period, for which the report will be generated. The report is based on daily reports. Therefore, it cannot be generated for the current day, as daily reports are generated only at the end of a day.

Daily reports generation must be enabled on the server for the group containing the necessary account to generate this type of reports.

The report contains the following data:

- **Bank** bank name. Accounts are grouped by a value of this field. The value is taken from the "Comment" field of the client record.
- **Accounts** number of accounts in a group;
- **Prev Balance** sum of accounts balances as of the end of each previous day included in the selected period. It is calculated as Prev Balance for day 1 + Prev Balance for day 2, etc.;
- **Closed P/L** total closed profit/loss in the accounts for the selected period;
- **Deposit** summary total of balance transactions in the accounts for the selected period;
- **Balance** sum of accounts balances as of the end of each day included in the specified period. It is calculated as Balance for day 1 + Balance for day 2, etc.;
- **Floating P/L** total floating profit/loss as of the end of each day included in the specified period. It is calculated as Floating P/L for day 1 + Floating P/L for day 2, etc.;
- **Credit** amount of credit transactions in the accounts for the specified period;
- **Equity** total amount of funds in the accounts as of the end of each day included in the specified period. It is calculated as Equity for day 1 + Equity for day 2, etc.;
- **Margin** amount of funds necessary to cover open positions in the accounts as of the end of each day included in the specified period. It is calculated as Margin for day 1 + Margin for day 2, etc.;
- **Free Margin** amount of free margin in the accounts as of the end of each day included in the specified period. It is calculated as Free Margin for day 1 + Free

MetaTrader 5 trading platform

Margin for day 2, etc.;

- **Currency** currency for parameters display.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Reports](#) / Margin Calls

Margin Calls Report

Margin Call Report is a report on the state of margin call or stop out accounts.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;

The report contains the following data concerning the accounts:

- **Login** account number;
- **Name** account holder name;

- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;

- **Leverage** leverage;
- **Balance** current balance;
- **Credit** current amount of credit funds;
- **Floating P/L** floating profit/loss of all open positions in the account;
- **Equity** current amount of funds;
- **Margin** current amount of funds necessary to cover open positions;
- **Free Margin** current amount of free margin;
- **Margin Limits** a margin level, at which the account enters the state of Margin Call/Stop Out;
- **Margin Level** margin level. It corresponds to the account equity or equity percentage to the margin volume ($\text{Equity} / \text{Margin} * 100$) depending on the group settings on the server;
- **Add. Margin** amount of funds necessary to be deposited in the account to be out of Margin Call state;
- **Currency** account deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Orders History

Orders History Report

Orders History Report is a summary report on the orders over the selected period. This report includes all pending orders (filled, canceled, expired, etc.) except for currently active orders (placed).

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;
- **Period** starting and ending date of the period, for which the report will be generated.

The report contains the following data on the orders:

- **Order** ticket number (unique number) of a trade operation;
- **Login** account number;
- **Name** account holder name;

- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;

- **Setup time** order setup time. The record is represented as YYYY.MM.DD HH:MM (year.month.day hour:minute);
- **Type** type of a trade operation: "Buy" a long position, "Sell" a short position or names of pending orders "Sell Stop", "Sell Limit", "Buy Stop", "Buy Limit", "Buy Stop Limit" or "Sell Stop Limit".
- **Symbol** a financial instrument of the order;
- **Volume** volume requested in the order/executed volume, lots;
- **Price** price specified in the order at which the trade operation should be executed;
- **Done time** order execution time;
- **Comment** comments on the order.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Positions History

Positions History Report

Positions History Report is a summary report on positions for the selected period.

The platform does not explicitly store the history of positions. The history is presented based on deals executed for the selected period. The platform collects data on deals related to a position (position opening, additional volume, partial and full closure), and then combines the data into one record.

The report only shows the positions that were closed during the selected period. If the position is still open or its close time does not fall within the selected interval, it will not be displayed in history.

Setup

The following parameters must be specified in the manager terminal before requesting the report:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

The report displays the following information on positions:

- **Position** position ticket (unique number). Usually, the position ticket matches the ticket of the order, as a result of which the position was opened.
- **Login** account number on which the position was opened.
- **Name** account holder's name.

- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Open Time** position opening time. The time is determined based on the first deal.
- **Type** position type, buy or sell.
- **Volume** position volume. For partially closed positions, the report shows the closed volume and the original volume.
- **Symbol** the symbol for which the position was opened.
- **Open Price** the weighted average position opening time: $(\text{price of deal 1} * \text{volume of deal 1} + \dots + \text{price of deal N} * \text{volume of deal N}) / (\text{volume of deal 1} + \dots + \text{volume of deal N})$.
- **S / L** position Stop Loss. The value is filled in accordance with the Stop Loss value of the deals by which the position was opened and closed.
- **T / P** position Take Profit. The value is filled in accordance with the Take Profit value of the deals by which the position was opened and closed.

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- **Close Time** position closing time. The time is set according to the last deal.
- **Close Price** position closing price.
- **Reason** reason for opening a position.
- **Commission** total commission for all deals related to the position.
- **Fee** total fee for all deals related to the position.
- **Swap** total swap for all deals related to the position.
- **Profit** total profit from all deals related to the position.
- **Currency** account deposit currency.
- **Comment** comments on the transaction.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Segregated

Segregated Report

Segregated Report is a report on the change of financial state of the requested accounts for the specified period. The report is generated on the basis of daily reports. Thus each rate is calculated as the difference between its value in the daily report for the last date and the value in the daily report for the first date specified when requesting the report. Summarized data of the accounts (for example, balance, equity and floating p/l) is taken from the daily reports generated for the last date within the time range the report is requested for. If there is no report generated for that date, zero values will be displayed.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas;
- **Period** starting and ending date of the period, for which the report will be generated.

Daily reports generation must be enabled on the server for the group containing the necessary account to generate this type of reports.

The report contains the following data:

- **Login** account number.
- **Name** account holder name.

- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Deposit** total funds deposited/withdrawn from the account per day.
- **Credit** final amount of credit funds submitted for the selected period (the difference between provided and withdrawn funds).
- **Commission** commission in the account for the specified period. Both blocked commission (in case they are accumulated within a day/a month and then withdrawn in a single balance deal), and commission charged immediately during a transaction are considered here.
- **Swap** total amount of swaps accumulated during the selected period. The amount is calculated based on swap values in deals.
- **Profit** net profit (profit - loss) obtained for the selected period.
- **Interest** annual interest accrued for the selected period.
- **Balance** balance at the moment of generation of the daily report for the last date within the time range the report is requested for.

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- **Floating P/L** floating profit/loss in the account at the moment of generation of the daily report for the last date within the time range the report is requested for.
- **Equity** amount of funds in the account at the moment of generation of the daily report for the last date within the time range the report is requested for.
- **Currency** account deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Summary

Summary Report

Summary Report is a report on a summary motion of funds and funds cycles in the requested accounts over the selected period.

Setup

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

The report contains the following data:

- **Login** account number.
- **Name** account holder name.
- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.
- **Deposit** amount of deposited funds for the specified period.
- **Withdrawal** amount of withdrawn funds for the specified period.
- **In/Out** difference between Deposit and Withdrawal parameters.
- **Credit** total amount of credit funds submitted to the account for the selected period (difference between provided and withdrawn funds).
- **Additional** total amount of additional transactions in the account for the selected period. These are additional charges, corrections, bonuses, agent commissions and annual interests.
- **Volume** total amount of all the above mentioned transactions in the account.
- **Commission** commissions debited from the account for the specified period.
- **Fee** amount of fees debited from the account for the specified period.
- **Swap** swaps withdrawn from the account for the specified period. The amount is calculated based on swap values in deals.
- **Profit** net profit obtained for the selected period.
- **Cur. Balance** account balance at the moment of the report request.
- **Cur. Equity** account equity at the moment of the report request.
- **Currency** account deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Gateways
Turnover

Gateways Turnover Report

Gateways Turnover Report is a report on volumes of the deals handled by gateways.

Setup in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Period** starting and ending date of the period, for which the report will be generated.

Setup in MetaTrader 5 Administrator

- **Currency** currency used for displaying deals volumes processed by a gateway.

The following data on each gateway is displayed in the report:

- **Symbol** symbol, for which turnover is displayed;
- **Deals** number of deals processed by a gateway for this symbol;
- **Lots** deals volume (in lots) processed by a gateway for this symbol;
- **Amount** volume of the deals (in base currency) processed by a gateway for this symbol. Calculation is performed according to the equations:
 - for "Forex" calculation type = Volume (in lots) * Contract Size
 - for "Futures", "Exchange Futures", "FORTS Futures" calculation types = [Price * Volume (in lots) * Tick value] / Tick size
 - for "CFD", "CFD Index", "CFD Leverage" and "Exchange Stocks" calculation types = Price * Volume (in lots) * Contract Size;
- **Currency** symbol base currency;
- **Amount Rate** conversion rate of a base currency to the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator);
- **Amount, currency** deals volume in the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator).

Volume in the parameters display currency (Amount, currency) is calculated considering exchange rates as of the moment of a report generation and does not reflect an accurate volume at the time of actual deals.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Gateways Profit

Gateways Profit Report

Gateways Profit Report is a report on volumes of the deals handled by gateways, as well as the profit earned by a broker when handling these deals.

Setup in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Period** starting and ending date of the period, for which the report will be generated.

Setup in MetaTrader 5 Administrator

- **Currency** currency used for displaying deals volumes and profit.

The following data on each gateway is displayed in the report:

- **Symbol** symbol, for which turnover and profit are displayed;
- **Deals** number of deals processed by a gateway for this symbol;
- **Lots** deals volume (in lots) processed by a gateway for this symbol;
- **Amount** volume of the deals (in base currency) processed by a gateway for this symbol. Calculation is performed according to the equations:
 - for "Forex" calculation type = Volume (in lots) * Contract Size
 - for "Futures", "Exchange Futures", "FORTS Futures" calculation types = [Price * Volume (in lots) * Tick value] / Tick size
 - for "CFD", "CFD Index", "CFD Leverage" and "Exchange Stocks" calculation types = Price * Volume (in lots) * Contract Size;
- **Currency** symbol base currency;
- **Amount Rate** conversion rate of a base currency to the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator);
- **Amount, currency** deals volume in the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator);
- **Profit, pips** broker's profit in pips. Calculated by the following equations:
 - for BUY deal = Deal price - gateway price
 - for SELL deal = Gateway price - deal price
- **Profit** broker's profit in symbol profit currency. Calculated by the following equations:

symbols with "Forex", "CFD", "CFD Index", "CFD Leverage" or "Exchange Stocks" calculation types:

- for BUY deals = [Deal price * Volume (in lots) * Contract Size] - [Gateway price * Volume (in lots) * Contract Size]
- for SELL deals = [Deal price * Volume (in lots) * Contract Size] - [Gateway price * Volume (in lots) * Contract Size]

symbols with "Futures", "Exchange Futures" and "FORTS Futures" calculation types:

- for BUY deals ("Futures") = [Deal price - Gateway price] * Volume (in lots) * Tick value
- for SELL deals ("Futures") = [Gateway price - Deal price] * Volume (in lots) * Tick value

- **Currency** symbol profit currency;
- **Profit Rate** conversion rate of a profit currency to the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator);
- **Profit, currency** broker's profit in the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator).

- Volume in the parameters display currency (Amount, currency) is calculated considering exchange rates as of the moment of a report generation and does not reflect an accurate volume at the time of actual deals.
- Profit in the parameters display currency (Profit, currency) is calculated considering exchange rates as of the moment of a report generation and does not reflect accurate profit at the time of actual deals. Besides, actual conversion rates from an exchange side may differ from trading server ones. Therefore, an exact profit earned by a broker is displayed only in Profit, pips and Profit columns.

Total parameters for all gateways are displayed at the bottom of the report.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Gateways
White Label

Gateways White Label Report

Gateways White Label Report is a report on volumes of the deals performed on a trading server by external brokers via MetaTrader 5 Gateway (standard or White Label version).

The algorithm of working with MetaTrader 5 Gateway can be presented the following way: an account for an external brokerage company is opened on a trading server. The company can use this account for performing trade operations. The external brokerage company configures MetaTrader 5 Gateways for connecting to the trading server.

This report allows to view the volume of deals performed by external brokers on a trading server.

Setup in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Period** starting and ending date of the period, for which the report will be generated.

Setup in MetaTrader 5 Administrator

- **Currency** currency used for displaying deals volumes and profit.

The report is generated on each account, transactions on which are handled via MetaTrader 5 Gateway. The following account data is displayed:

- **Symbol** symbol, for which trading operations data is displayed;
- **Deals** number of deals for this symbol;
- **Lots** deals volume (in lots) for this symbol;
- **Amount** volume of the deals (in base currency) for this symbol. Calculated by the following equations:
 - for "Forex" calculation type = Volume (in lots) * Contract Size
 - for "Futures", "Exchange Futures", "FORTS Futures" calculation types = [Price * Volume (in lots) * Tick value] / Tick size
 - for "CFD", "CFD Index", "CFD Leverage", "Exchange Stocks" calculation types = Price * Volume (in lots) * Contract Size
- **Currency** symbol base currency;
- **Amount Rate** conversion rate of a base currency to the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator);
- **Amount, currency** deals volume in the currency used for parameters display (specified in report settings in MetaTrader 5 Administrator).

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Volume in the parameters display currency (Amount, currency) is calculated considering exchange rates as of the moment of a report generation and does not reflect an accurate volume at the time of actual deals.

Total data for all accounts using MetaTrader 5 Gateway for performing transactions is displayed at the bottom of the report.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Execution Types

Execution Types Report

Execution Type Report a report on the number, volume and types of executed trades.

Configuration in MetaTrader 5 Manager

The following parameter should be configured in the manager terminal before requesting the report:

- **Period** reports are generated for seven days preceding the specified day.

Configuration in MetaTrader 5 Administrator

An additional parameter is configured in the report configuration in MetaTrader 5 Administrator:

- **Currency** the currency used when displaying various parameters in the report (Balance, Profit, etc.).

Report Data

The report is divided into four units: information on accounts, a diagram on the number of trades, a diagram on the volume of trades, a summary of trades.

Account Information

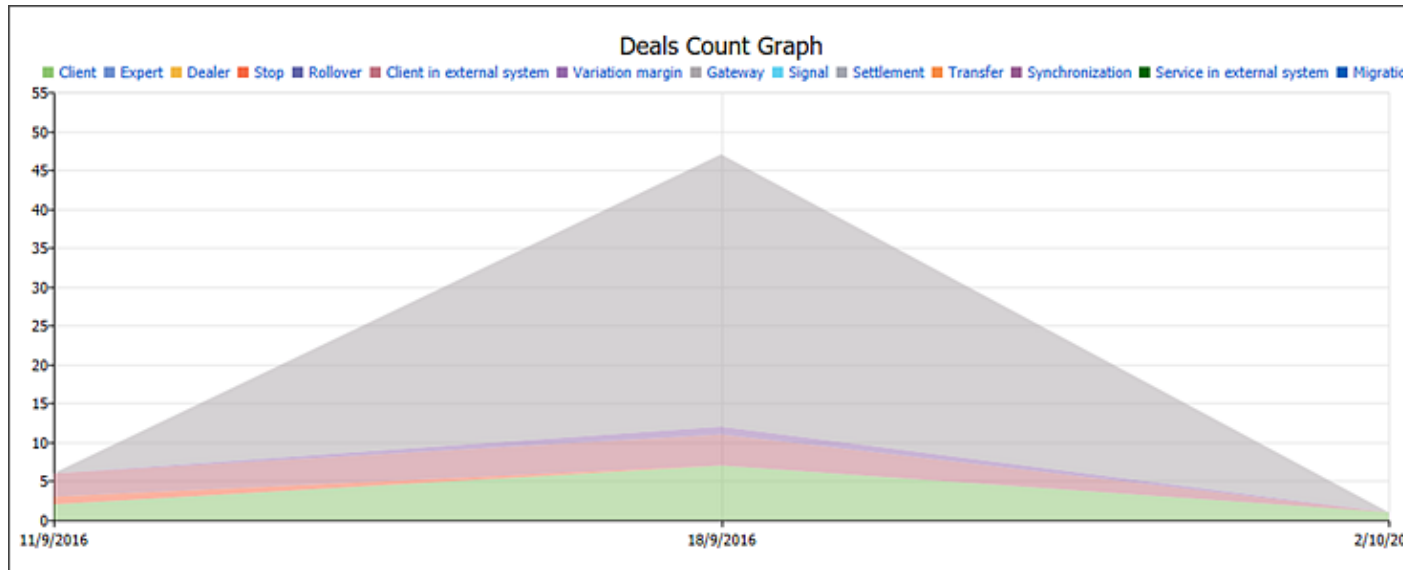
This unit provides the following information for each client group:

- Group name
- The number of clients in the group
- The percentage of active accounts
- The total balance of all accounts
- The total floating profit/loss
- The total equity amount
- The group deposit currency

Deals Count Graph

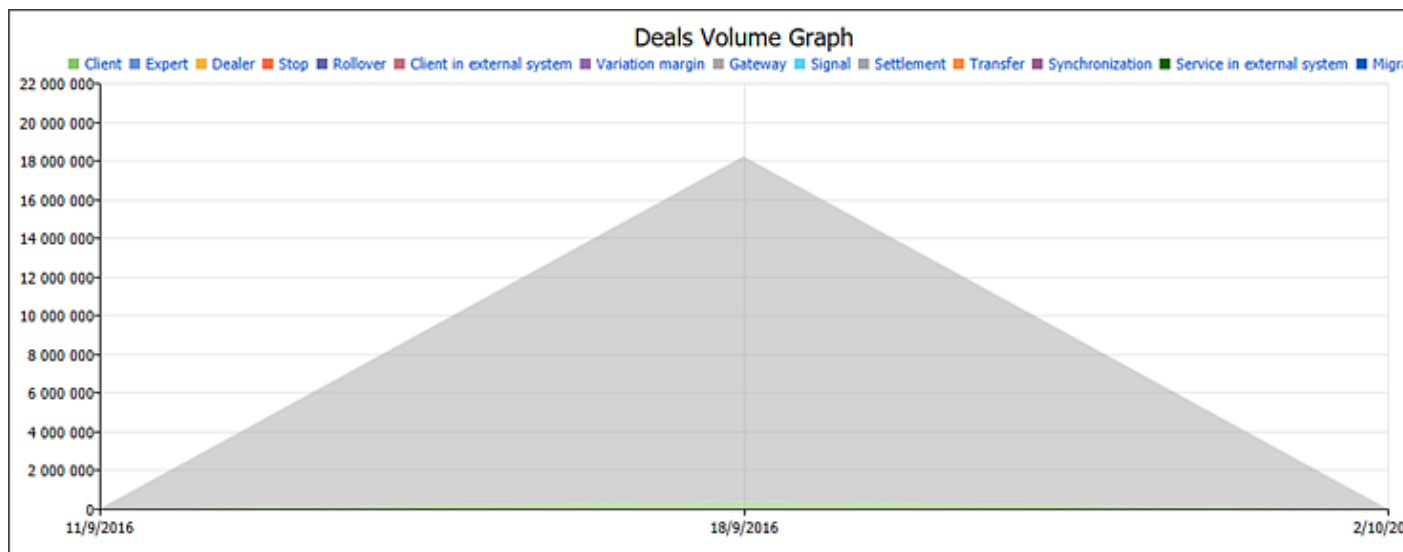
The chart shows the number of executed trades of each type separately: performed by clients, by dealers, executed for rollover and variation margin charges, etc.

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Deals Volume Graph

The chart shows the volume of executed trades of each type separately: performed by clients, by dealers, executed for rollover and variation margin charges, etc.



A Summary Report on Trades

This unit features information on the number and volume of each type of trades for separate dates.

- **Client** the deal was performed by a client manually through the client terminal;
- **Expert** the deal was performed by a client using an Expert Advisor;
- **Dealer** the deal was performed by a dealer through the manager terminal;
- **Stop** the deal was performed when the client reached the Stop out level;
- **Rollover** the deal was performed to calculate swaps when reopening a position;
- **External** the deal was performed by a client from an external trading system;
- **Variation margin** the deal was performed to charge variation margin;
- **Gateway** the deal was performed by the MetaTrader 5 Gateway connected to the platform.
- **Signal** the deal was performed as a result of copying a trading signal as per a subscription in the client terminal.
- **Settlement** the deal was executed as a result of operations associated with a futures/option delivery date coming into effect. It is currently not used.
- **Transfer** the deal was performed as a result of relocating a position at a calculated price to a new symbol with the same underlying asset. It is currently not used.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Reports](#) / Trade Accounts

Trade Accounts Report

Trade Accounts Report a report on the current trading state of client accounts.

Configuration in MetaTrader 5 Manager

The following parameters must be set in the manager terminal before requesting the report:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.

Report Data

The report is divided into four units that provide information on modified open orders, history orders, deals and positions:

- **Login** client account number.
- **Name** account holder's name.
- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.
- **Balance** the current account balance.
- **Credit** credit funds on the account.
- **Equity** current account equity.
- **Margin** margin required to maintain open positions on the account.
- **Free Margin** the amount of free margin on the account.
- **Margin Level** the margin level on the account. It corresponds to the account equity or equity percentage to the margin volume ($\text{Equity} / \text{Margin} * 100$) depending on the group settings on the server.
- **Blocked Comm.** commission by orders and positions accumulated during a day/month. Depending on the [settings](#), preliminary commission calculation is performed during a day/month and the appropriate amount of money is blocked in the account and displayed here. The final commission calculation is performed at the end of a day/month and the appropriate sum is withdrawn from the account in a balance operation (a separate deal of the 'Daily/Monthly commission' type), and the blocked amount is unblocked.
- **Blocked Profit** if a client group is configured to only include [floating loss](#) into free margin calculation, the amount of profit received by a client during a trading day will be recorded in a separate field 'Blocked Profit'. At the end of the trading day, the accumulated profits is released (reset) and is reflected on the user's balance (included into free margin calculation).
- **Profit** profit on the account.
- **Swap** the value of swaps on current client positions.
- **Floating P/L** amount of current floating profit or loss.

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- **Currency** client's deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Trade Transactions

Trade Transaction Report

Trade Transaction Report provides detailed data on trading operations at MetaTrader 5 servers. This report is useful not only for a broker but also when providing data to regulatory bodies.

The report is implemented as TradesTransactionReports.dll file (TradesTransactionReports64.dll for 64-bit platform version) combining the report and the server plugin. The plugin collects data on report transactions and saves them in Reports\Trades.Transaction.Reports directory in trading server's folder, while the report provides collected data in the appropriate format. The plugin should be enabled in the settings of the appropriate trading server. If the plugin is not enabled, report data will not be updated and the report will provide data only for the plugin's operation period.

TradesTransactionReports.dll file is included into the standard delivery and located in Reports directory of MetaTrader 5 trading server.

Plugin Installation

The appropriate plugin should be installed to collect the data used in the report. To install the plugin, enter Plugins section in MetaTrader 5 Administrator. After the server to install the plugin has been selected, click "+ Add".

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Plugin: Trade Transaction Report

Common Information

 The server plugin is intended for enrichment of the server functionality and represents an external program with remote configuration management.

☒ Enable ☒ Configurable by managers

☐ Enable plugin profiling

Name: Trade Transaction Report

Module: Trades.Transaction.Reports64.dll Trade Main, 1

Parameters:

Parameter	Value
 Groups	*
 Base Directory	Trade Transaction Report
 Symbols	*
 Daily Report Path	confirms

Add Edit Delete

OK Cancel Help

The following settings are available:

- **Groups** group filter. Groups are separated with a comma. To specify groups, you may also use the mask "*" and the negation sign "!". For example, the value "demoforex,democfd" means that the plugin will collect the transaction details for users in demoforex and democfd groups. The value "*" is for all groups. The default value is "*,!demo*,!contest*" which means all groups except those whose names start with demo and contest.
- **Base Directory** name of the directory where the database of transactions is saved for the report. The data is saved for each day. This directory is automatically created in Reports\Trade.Transaction.Report. In most cases, this parameter should save a default value.
- **Symbol** symbol filter. Groups are separated with a comma. To specify groups, you may also use the mask "*" and the negation sign "!". For example, the value "EURUSD,GBPUSD" means that the plugin will collect the transaction details for EURUSD and GBPUSD symbols. The value "*" is for all symbols. *,!CFD* value is for all symbols except those in the CFD group.

It is not recommended to use the same Base Directory parameters for different plugins. In such case, correct operation of those plugins is not guaranteed.

Report Installation

To install the plugin, enter Reports section in MetaTrader 5 Administrator. After the server to install the report has been selected, click "+ Add".

Report: Trade Transactions

Common Information

Reports are server plugins providing built-in or custom reports. Please specify report name, select report plugin, and specify additional parameters if needed.

☒ Enable

Name: Trade Transactions

Type: Trade Transactions Trade Main, 1

Parameter	Value
ab Base Directory	Trade Transaction Report
ab Show group owner	Yes
ab Show daily transactions	Yes
ab Show reason	No
ab Show retcode	No
ab Show rejected	No
ab Daily Report Path	Value
ab Daily Report Time	Value

Add Edit Delete

OK Cancel Help

The following specific settings are available for the report:

- **Base Directory** the directory where transaction databases are searched for. This parameter should be similar to the same-name parameter of the corresponding plugin. If the plugin with a similar Base Directory parameter does not exist, the report will be empty. Several reports may refer to the similar transaction databases.
 - **Show group owner** the flag of Group Owner column display in the report. On default, the column is displayed (Yes). When set to No, Group Owner is not displayed in the report.
 - **Show daily transactions** whether to show the state of orders and positions as of the end of the trading day (the "daily order" and "daily deal" fields) in the report.
 - **Show reason** whether to show reasons for the execution of trading operations (the "reason" field) in the report.
 - **Show retcode** whether to show codes of operation execution results (the "retcode" field) in the report.
 - **Show rejected** showing rejected operation in the report.
 - **Daily Report Path, Daily Report Time** once a day, the report plugin unloads data on all transactions for the past day to a CSV file.
-
- In the "Daily Report Path" parameter, set the name of the folder in the trade server installation directory, in which files with transactions will be saved. The "confirms" folder is used by default. In this case, files are saved at the following path: [trade server directory]\confirms\\YYYY.MM.DD.trans\\"Base Directory".csv, where "Base Directory" is the value of plugin's "Base Directory" parameter.
 - Specify the time, when the file with transactions should be generated, in the "Daily Report Time" parameter. The time is not set by default, so the file is generated at the end of the trading day (after completion of all day closing procedures).

The standard parameters are described in the <u>"Reports"</u> section.
--

Using the Report

MetaTrader 5 Manager is used for requesting the report. The requested data can be filtered by group, logins, period and symbol mask. The report is generated in real time and the data can be updated by new transactions at each new request.

Find out more about working with the reports in the appropriate section of MetaTrader 5 Manager documentation.

MetaTrader 5 trading platform

1863 - MetaQuotes - John Smith

File Edit View Tools Window Help

Automation

Навигатор

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Trade Transactions ☆ 2023.02.01 - 2023.02.17

* Auto refresh

Login	Deal	Type	Entry	Sy
3856277	5028330918	buy	in	EU
3856277	5028330919	sell	in	EU
3856277	5028330920	buy	in	AL

* Delayed
CFD
Forex
Collateral
Currenex
Hotspot
Integral
Metals
Preliminary
LMAX
ECN
DGCX

For Help, press F1 17.2 / 0.1 MB

MetaTrader 5 trading platform

The request is resulted in the table which can be saved in HTML, CSV or Open XML (MS Office Excel 2007) formats:

Login ▲	Deal	Action	Type	Entry	Symbol	Lots	Time	Pri
3856277	5028330918	market	buy	in	EURUSD	1	2023.02.17 14:11:29	1.062
3856277	5028330919	market	sell	in	EURCHF	1	2023.02.17 14:11:31	0.989
3856277	5028330920	market	buy	in	AUDUSD	1	2023.02.17 14:11:34	0.683
3856277	5028330921	market	sell	in	EURCAD	1	2023.02.17 14:39:36	1.435
3856277	5028330922	market	buy	in	AUDCHF	1	2023.02.17 14:39:39	0.635
3856277	5028330923	market	buy	in	CHFJPY	1	2023.02.17 14:39:42	144.8
3856277	5028330924	market	sell	in	EURUSD	1	2023.02.17 14:39:48	1.063
3881139	0	pending expiration	sell limit		EURUSD	0.05	2023.02.17 14:22:00	1.100
3881154	0	pending expiration	buy stop		EURUSD	1.1	2023.02.17 14:22:00	1.500
3881157	0	pending expiration	sell stop		EURUSD	0.05	2023.02.17 14:22:00	0.500
3881158	0	pending expiration	buy limit		EURUSD	0.05	2023.02.17 14:20:00	0.500
3881162	0	pending expiration	buy stop limit		EURUSD	0.05	2023.02.17 14:22:00	1.500
3881165	0	pending expiration	buy limit		EURUSD	0.05	2023.02.17 14:32:00	0.500
3881165	0	pending expiration	sell limit		EURUSD	0.05	2023.02.17 14:32:00	1.100
3881170	0	pending expiration	buy stop limit		EURUSD	0.05	2023.02.17 14:35:00	1.500
3881170	0	pending expiration	sell stop limit		EURUSD	0.05	2023.02.17 14:35:00	0.900
3881172	0	pending expiration	sell stop limit		EURUSD	0.05	2023.02.17 14:27:00	0.900

The report includes the following information:

- **Login** user account's number;
- **Group** group the account belongs to;
- **Country** client's country of residence;
- **Account comment** comment to the trading account;
- **Leverage** user account's leverage;
- **Dealer** login of the dealer that placed the order. Indicated only if the request is placed by a dealer;
- **IP** user's IP address;
- **Currency** account deposit's currency;
- **Order** order number (ticket);
- **Order ID** order identifier in a external trading system. The identifier is changed in case the order is updated in the external trading system by canceling or placing a new order;
- **Deal** deal number (ticket);
- **Deal ID** deal identifier in an external trading system;
- **Position** the ticket of a deal;
- **Position ID** the ID of a position in an external trading system;
- **Action** action type (see below);
- **Type** trading operation type (see below);
- **Entry** direction of the operation: in (market entry), out (market exit), in/out (position reversal);
- **Symbol** trading symbol;
- **Position By** the ticket of the opposite position. It is shown if the source position (Position) was closed by an opposite one;

- **Lots** volume in lots;
- **Amount** volume;
- **Time** transaction time;
- **Price** transaction price;
- **Bid** bid market price at the moment of transaction;
- **Ask** ask market price at the moment of transaction;
- **Position Price** position price for position closing transactions;
- **SL** Stop Loss price;
- **TL** Take Profit price;
- **Margin Rate** margin rate at the moment of transaction;
- **Margin Amount** margin volume at the moment of transaction without considering the leverage;
- **Commission** deal commission;
- **Swap** swap size;
- **Profit** profit/loss size;
- **Profit Currency** profit currency;
- **Profit Rate** conversion rate of the profit currency to deposit currency;
- **Raw Profit** the size of profit/loss obtained as a result of the operation. Indicated in the profit currency of the operation symbol;
- **Amount Closed** closed volume in amounts;
- **Gateway Price** price in external trade system;
- **Reason** reason of the operation execution;
- **Retcode** result of execution;
- **Group Owner** name of the company the client is serviced by. It corresponds to Company field in client group configuration.

"Action" field reflects the transaction type. It may contain the following values:

- **market** market operation (entering, exiting, reversal);
- **position modify** position's Stop Loss and Take Profit modification;
- **pending** placing a pending order;
- **pending modify** modifying a pending order;
- **pending cancel** canceling a pending order;
- **pending expiration** pending order expiration;
- **pending activation** pending order activation;
- **stop loss** closing position by Stop Loss;
- **take profit** closing position by Take Profit;
- **deposit** balance operation;
- **order stopout** canceling order by Stop Out;
- **position stopout** closing position by Stop Out;
- **stoplimit** Stop Limit order activation;
- **daily order** the state of the order as of the end of the trading day;
- **daily position** the state of the position as of the end of the trading day;
- **close by** closing of a position by an opposite one;
- **rollover** transferring of a position on the next trading day;
- **market by dealer** a market operation performed by a dealer;
- **position modify by dealer** modification of the Stop Loss or Take Profit levels of a position performed by a dealer;
- **pending modify by dealer** modification of a pending order by a dealer;
- **pending cancel by dealer** canceling of a pending order by a dealer;
- **pending activation by dealer** activation of a pending order by a dealer;

- **dealer close by** the operation of closing of a position by an opposite one performed by a dealer;
- **variation margin** variation margin calculation.

"Type" field displays order or transaction type:

- **buy** order for buying at the current market price;
- **sell** order for selling at the current market price;
- **buy limit** buy limit order;
- **sell limit** sell limit order;
- **buy stop** buy stop order;
- **sell stop** sell stop order;
- **buy stop limit** buy stop limit order;
- **sell stop limit** sell stop limit order;
- **deposit** deposit or withdrawal operation;
- **credit** credit operation;
- **charge** charging commissions;
- **correction** correction operations;
- **bonus** adding bonuses;
- **commission** commission size;
- **commission daily** commission per day;
- **commission monthly** commission per month;
- **agent daily** charging a daily agent commission;
- **agent monthly** charging a monthly agent commission;
- **interest rate** accrual of annual interest;
- **buy canceled** a canceled buy deal;
- **sell canceled** a canceled sell deal;
- **dividend** payment of taxable dividend;
- **dividend franked** payment of nontaxable dividend;
- **tax** charging a tax;
- **agent** accrual of instant agent commission.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Trade Modifications

Trade Modification Report

Trade Modification Report report on manually modified orders, trades and positions. It allows you to easily monitor the work of the platform administrators and managers.

Setup in MetaTrader 5 Manager

The following parameters should be set in the manager terminal before requesting the report:

- **Groups** groups containing the accounts, on which the report should be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated. The period is only taken into account for the history of orders and deals. All open and modified positions and orders are included in the report, regardless of their opening date.

Report Data

The report is divided into four blocks containing data on modified open orders, history orders, deals and positions:

- **Ticket** trade operation number (ticket).
- **Login** client account number.
- **Open Time** trade operation creation time.
- **Type** trade operation type.
- **Lots** operation volume in lots.
- **Symbol** financial instrument of the deal.
- **Open Price** open price.
- **Market Price** current market price.
- **Swap** swap.
- **Profit** profit target in the client deposit currency.
- **Modified** operation modification attributes:
 - ◆ **Administrator** changed by an administrator.
 - ◆ **Manager** open price changed by a manager.
 - ◆ **Restore** restored from the archive.
 - ◆ **Position** modification attributes are inherited by the deal from the position closed by that deal.
 - ◆ **Admin API** changed via Manager API administrator interface.
 - ◆ **Manager API** changed via Manager API manager interface.
 - ◆ **Server API** changed via Server API.
 - ◆ **Gateway API** changed via Gateway API.

Apart from the data on modified operations, reports contain excerpts from the related entries in the trade server journal.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Trade Performance Summary

Trade Performance Summary

The Trade Performance Summary Report is a consolidated report on the trading performance of each account for the selected period.

- The trading metrics provide a comprehensive analysis of the trading strategy. They assess overall trading activity and intensity, financial results, including profits and losses, and the risk-reward ratio.
- Trade performance and long/short performance metrics help identify the strengths and weaknesses of the strategy.
- The analysis of extreme values and profit/loss sequences helps identify potential risks and the strategy's resilience to market changes.

Configuration in MetaTrader 5 Manager

The following parameters should be configured in the Manager terminal before requesting the report:

- **Groups** groups for which account reports should be generated. You can specify one or several accounts separated by commas.
- **Period** the start and end dates of the period for which the report will be generated.

Information in the report

The report contains the following information:

- **Login** account number.
- **Name** account holder's name.
- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Terminal** type of the terminal from which the last trading operation was performed.
- **Registration** registration date.
- **Last access** last connection date.
- **Lifetime days** the number of days from the date of registration to the date of last connection.
- **Total deals** the total number of deals.
- **Total trades** the total number of trades (deals that recorded a financial result).
- **Total net profit** total profit/loss for the entire trading period. Considers all profitable and losing trades.
- **Gross profit** the total value of all profitable trades in monetary units.
- **Gross loss** the total value of all losing trades in monetary units.
- **Profit factor** the ratio of the total gross profit to the total gross loss: Gross profit / Gross loss. One means that the gross profit is equal to the gross loss.

- **Expected payoff** the anticipated return from a deal. Calculated as Total net profit / Total trades and reflects the average profit/loss per trade.
- **Payoff ratio** the ratio of the average profit per trade to the average loss per trade: Average profit trade / Average loss trade.
- **Tharp expectancy** expected profitability of the trading system per trade: Expected payoff / -Average loss trade. Reflects the average return per dollar at risk.
- **Profit trades** the number of profitable trades.
- **Profit (%)** the percentage of profitable trades relative to the total number of trades.
- **Loss trades** the number of losing trades.
- **Loss (%)** the percentage of losing trades relative to the total number of trades.
- **Long trades** the number of Buy trades
- **Long trades profit** profit from Buy trades
- **Long trades loss** loss from Buy trades.
- **Long (%)** the percentage of profitable Buy trades.
- **Short trades** the number of Sell trades.
- **Short trades profit** profit from Sell trades.
- **Short trades loss** loss form Sell trades.
- **Short (%)** the percent of Sell trades.
- **Best trade** the trade with the highest profit.
- **Worst trade** the trade with the highest loss.
- **Average profit trade** average profit per profitable trade: Gross profit / Profit trades.
- **Average loss trade** average loss per losing trade: Gross loss / Loss trades.
- **Max Consecutive Wins** the maximum number of consecutive profitable trades.
- **Max Consecutive Losses** the maximum number of consecutive losing trades.
- **Max Consecutive Profit** the maximum profit in a series of consecutive trades.
- **Max Consecutive Loss** the maximum loss in a series of consecutive trades.
- **Account Comment** comment to the accounts.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Trade Group Statistics

Trade Group Statistics

The comprehensive report provides a wide range of statistical information by groups, such as the number of accounts, commission amounts, the share of trades by terminal types and sources, and more.

Configuration in MetaTrader 5 Manager

The following parameters should be configured in the Manager terminal before requesting the report:

- **Groups** groups for which account reports should be generated. You can specify one or several accounts separated by commas.
- **Period** the start and end dates of the period for which the report will be generated.

Information in the report

The information in the report is divided into blocks by group types: real, demo and contest.

- **Group** the group the account belongs to.
- **Accounts** the number of accounts in the group.
- **Deals** the number of deals executed on accounts within the groups during the selected period.
- **Commissions** the total commissions charged for deals during the selected period.
- **Mean commission** average commissions per deal.
- **Currency** group's deposit currency.
- **Trade: Manual** the share of trades executed manually.
- **Trade: Algo** the share of trades executed using Expert Advisors.
- **Trade: Signals** the share of trades copied through the Signals service.
- **Trade: Other** the share of trades executed by other methods.
- **Terminal: Desktop** the share of trades executed through the desktop trading terminal.
- **Terminal: Mobile** the share of trades executed through the mobile trading terminal.
- **Terminal: Web** the share of trades executed through the web-version of the trading terminal.
- **Terminal: Other** the share of trades executed through other client applications.

MetaTrader 5 Trading Platform / Platform Setup / Reports / StopOut
Compensations

StopOuts Compensation Report

StopOuts Compensation Report reflects operations relating to the compensation of a negative balance after the Stop Out state. All trades with the "so compensation" type are included into this report.

The report features sorting the requested data. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

The following data on each compensation deal is available in the report:

- **Deal** the ticket of the deal.
- **Login** account number.
- **Name** account holder's name.

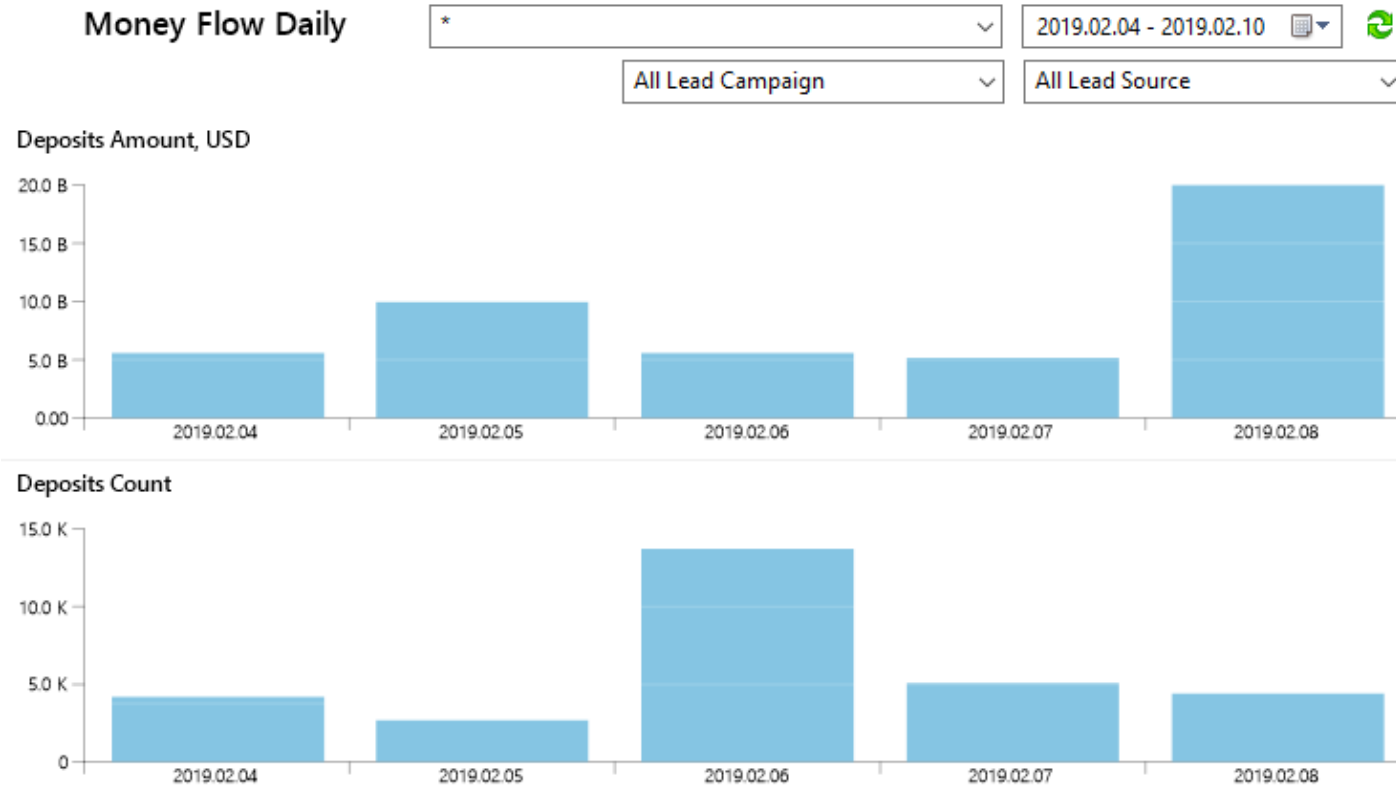
- **Group** group the account belongs to.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.

- **Group** group the account belongs to.
- **Time** time of the compensation operation.
- **Amount** the compensation amount.
- **Currency** the currency, in which the compensation was performed.
Corresponds to the account deposit currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Money Flow Daily

Money Flow Daily

The report presents the distribution of deposits and withdrawals by days, as well as the difference between them. Data can be filtered by Lead Source and Lead Campaign, and thus you can evaluate the real effectiveness of advertising campaigns and marketing channel.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated. For periods less than two months, data is displayed by days; for periods between 2 and 4 months, data is displayed by weeks (aggregated); data is presented by months for periods over 4 months.
- **Lead Campaign** the name of an advertising campaign by which the client was attracted.
- **Lead Source** the website from which the client came.

Charts in the report

The report includes multiple charts:

- **Deposits Amount / Withdrawals Amount** distribution of deposit and withdrawal amounts over time.
- **Deposits Count / Withdrawals Count** distribution of the number of deposit and withdrawal operations over time.
- **Deposits Medium / Withdrawals Medium** the average deposit and withdrawal amount over time.
- **Balance-Withdrawal** difference between deposit and withdrawal amounts over time.
- **Balance-Withdrawal Accumulated** accumulated difference between deposit and withdrawal amounts over time.

Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

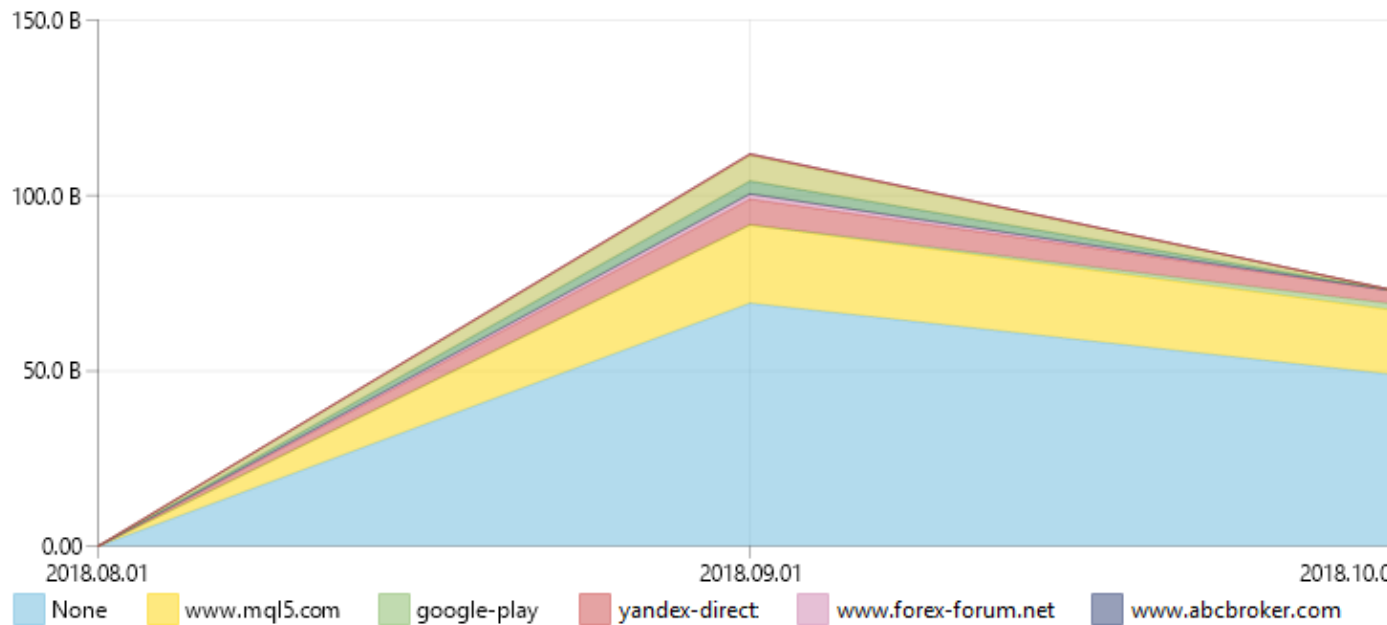
Money Flow Daily by Lead Source / Lead Campaign / Country

These reports present the distribution of deposits and withdrawals not only by time, but also by attraction channels, marketing campaigns and countries. Select a group and time period and find out which advertisement drives the most of client deposits and which advertising sites provide the highest CTR.

Money Flow Daily by Lead Source

2018.08.01 - 2018.10.31

Deposits Amount, USD



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated. For periods less than two months, data is displayed by days; for periods between 2 and 4 months, data is displayed by weeks (aggregated); data is presented by months for periods over 4 months.

Charts in the report

The report includes multiple charts:

- **Deposits Amount / Withdrawals Amount** distribution of deposit and withdrawal amounts over time, grouped by Lead Campaign, Lead Source or country.
- **Deposits Count / Withdrawals Count** distribution of the number of deposit and withdrawal operations over time, grouped by Lead Campaign, Lead Source or country.
- **Deposits Medium / Withdrawals Medium** average deposit and withdrawal amount over time, grouped by Lead Campaign, Lead Source or country.
- **Balance-Withdrawal** difference between deposit and withdrawal amounts over time, grouped by Lead Campaign, Lead Source or country.

MetaTrader 5 trading platform

- **Balance-Withdrawal Accumulated** accumulated difference between deposit and withdrawal amounts over time, grouped by Lead Campaign, Lead Source or country.

- Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.
- Lead Campaign and Lead Source data are available in appropriate fields of trading accounts.

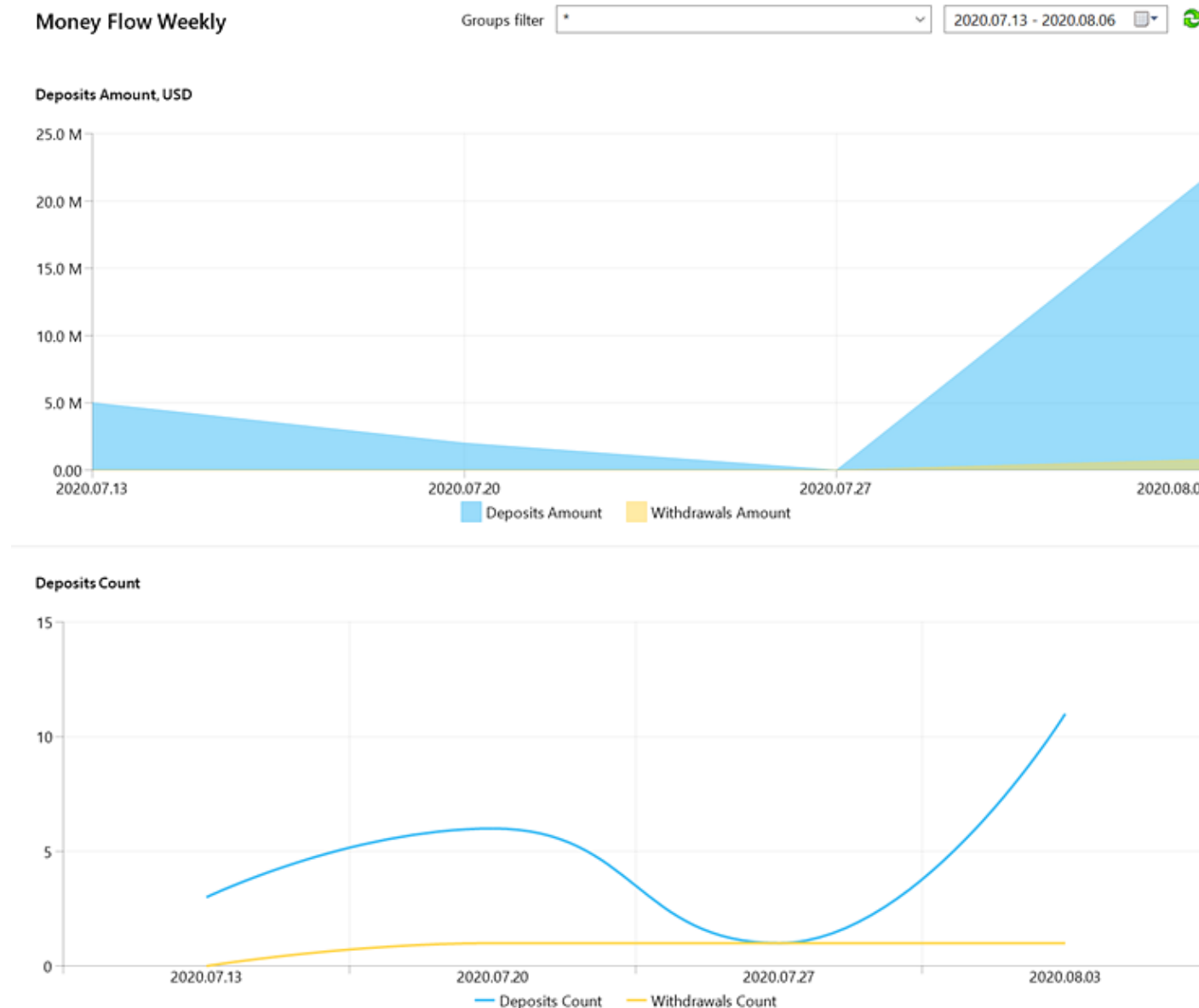
Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Money Flow Weekly

Money Flow Weekly

The report enables the evaluation of financial operations on customer accounts. It reflects the distribution of deposits and withdrawals by week and the difference between the two operation types. Data can be filtered by groups.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes multiple charts:

- **Deposits Amount** distribution of deposit and withdrawal amounts by time.
- **Deposits Count** distribution of the number of deposits and withdrawals by time.
- **Deposits Average** the average deposit and withdrawal amount by time.
- **Deposits-Withdrawals** the difference between deposit and withdrawal amounts by time.
- **Deposits-Withdrawals Accumulated** the accumulated difference between deposit and withdrawal amounts by time.

Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

Setup in MetaTrader 5 Administrator

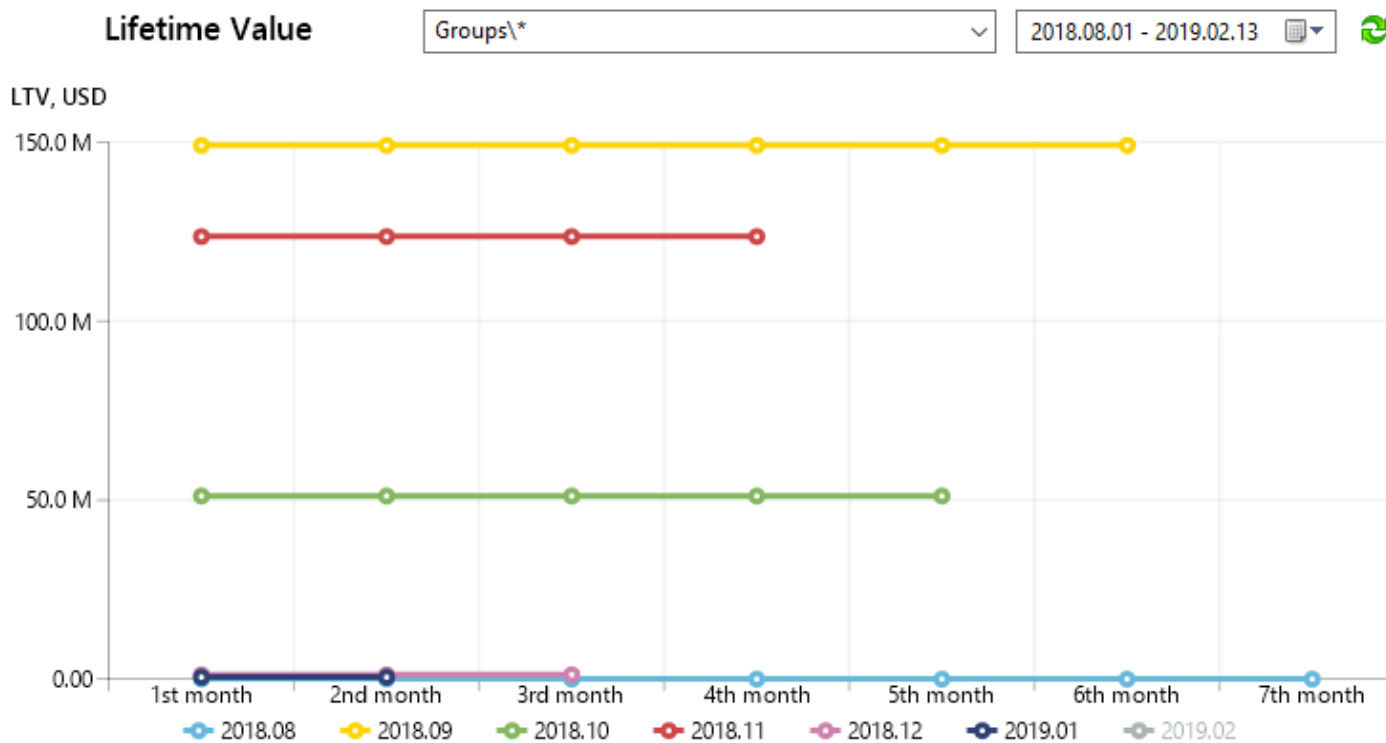
The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Lifetime Value

Lifetime Value

The report presents the generalized LTV picture for accounts from a selected group, which were registered in a certain period.

The report enables the analysis of changes in the client's lifetime value over time, after account registration. Also, the report provides data on the amount and number of deposit operations. Select the distribution type for information display: by Lead Source, by Lead Campaign and by countries. Thus you can evaluate the real effectiveness of each advertising campaign and attraction channel.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes multiple charts:

- **LTV by Lifetime and Lead Source / Lead Campaign / Country** the life-time value of accounts is presented with the distribution by Lead Source, Lead Campaign and Country (in three different charts). The lifetime value is calculated as (amount of deposits)/(number of deposits). For example, a period of 2018.08.01-2019.02.01 is selected for the report. The report retrieves data on accounts created during this period (taking into account specified group filtration parameters). The system calculates LTV for each month of account existence with the data accumulation: LTV for the first month, the first two months, the first three months, etc.
- **Deposits Amount by Lifetime and Lead Source / Lead Campaign / Country** deposit values over the entire lifetime, distributed by Lead Source, Lead Campaign and Country (in three different charts). Information is presented separately for each month of the account existence.
- **Deposits Count by Lifetime and Lead Source / Lead Campaign / Country** number of deposit operations over the entire lifetime, distributed by Lead Source, Lead Campaign and Country (in three different charts). Information is presented separately for each month of the account existence.
- **Inactive Clients Count by Lead Source / Lead Campaign / Country** the number of clients who did not make any deposit over the selected period, distributed by Lead Source, Lead Campaign and Country (in three different charts).

- Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.
- Lead Campaign and Lead Source data are available in appropriate fields of trading accounts.

Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

Lifetime Value Daily

This report is similar to the previous one, except that the LTVs and data on deposits are distributed in accordance with client registration dates (account opening dates). Additional filters are available in the report: by Lead Campaign and by Lead Source.

Lifetime Value Detailed by Client

This report features FTD (first-time deposit) and LTV (lifetime value) metrics concerning your clients rather than separate accounts. The report collects data on all accounts belonging to a client and generates summary information: the date and amount of the first deposit, as well as the amount of deposits on all accounts over a selected period of time.

Lifetime Value Detailed by Client							
				Groups*	2018.08.01 - 2019.02.13		
				All Lead Campaign	All Lead Source		
Client Id	Name	Registration Time	First Deposit Time	Trading Accounts	Deposits	First Depos...	Lifetime Value
1080	Richardson Timothy	2018.08.11 05:39:47	2018.08.11 05:39:47	1	1	10 000.00	10 000.00
1081	James Steven	2018.09.28 18:33:29	2018.09.28 18:33:29	8	8	1 000 000.00	22 395 900.00
1082	Jackson Adam	2018.09.30 16:05:57	2018.09.30 16:05:57	1	1	100 000.00	100 000.00
1084	Johnson Justin	2018.09.30 16:32:41	2018.09.30 16:32:41	2	2	3 000.00	6 000.00
1085	Watson Jordan	2018.09.30 16:36:05	2018.09.30 16:36:05	1	1	5 000 000.00	5 000 000.00
1086	Wood Jose	2018.09.30 16:39:41	2018.09.30 16:39:41	1	1	1 000 000.00	1 000 000.00
1087	Evans Alejandro	2018.09.30 16:51:59	2018.09.30 16:51:59	1	1	1 000 000.00	1 000 000.00
1088	Wilson Jeremiah	2018.09.30 17:02:31	2018.09.30 17:02:31	1	1	5 000.00	5 000.00
1089	Jones Nicholas	2018.09.30 17:03:02	2018.09.30 17:03:02	1	1	1 000 000.00	1 000 000.00
1090	Carter Jordan	2018.09.30 17:03:07	2018.09.30 17:03:07	2	2	1 000 000.00	1 010 000.00
1091	James Jonathan	2018.09.30 17:07:07	2018.09.30 17:07:07	4	4	10 000.00	1 016 000.00
1092	Lee Adrian	2018.09.30 17:25:57	2018.09.30 17:25:57	5	5	3 000.00	1 012 000.00
1093	Wilson Mason	2018.09.30 17:38:08	2018.09.30 17:38:08	2	2	5 000.00	10 000.00
1094	Patterson Landon	2018.09.30 17:45:32	2018.09.30 17:45:32	1	1	1 000 000.00	1 000 000.00
1095	Cooper Aiden	2018.09.30 18:07:45	2018.09.30 18:07:45	2	2	1 000 000.00	2 000 000.00
1096	Smith Jayden	2018.09.30 18:17:22	2018.09.30 18:17:22	1	1	1 000 000.00	1 000 000.00
1097	Martin Ashton	2018.09.30 18:18:52	2018.09.30 18:18:52	1	1	3 000.00	3 000.00

Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.
- **Lead Campaign** the name of an advertising campaign by which the client was attracted.
- **Lead Source** the website from which the client came.

Data in the report

The report presented as table displaying the following data:

- **Client Id** client identifier.
- **Name** client's name.
- **Country** client's country of residence.
- **Registration Time** client registration date (when the first account owned by the client was opened).
- **First Deposit Time** the date when the first deposit was made.
- **Trading Accounts** the number of the client's trading accounts.
- **Deposits** the number of deposit operations made by the client on all his or her accounts.
- **First Deposit** the amount of the first deposit.
- **Lifetime Value** lifetime value information on all accounts owned by the client. LTV is calculated as (amount of deposits + amount of withdrawals)/(number of deposits + number of withdrawals).

Double click on a table row to open a window with the detailed client information.

Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

Lifetime Value Detailed by Trading Account

This report is similar to the previous one, except that the data is provided by separate trading accounts, not clients.

Lifetime Value Detailed by Lead Source / Lead Campaign / Country

These reports present summary tables of FTD and LTV, the number and total amount of deposits and withdrawals. The data can be presented in three modes: by attraction channels, campaigns and countries.

Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Data in the report

The report presented as table displaying the following data:

- **Lead Campaign** the report presents data concerning this lead campaign.
- **Lead Source** the report presents data concerning this lead source.
- **Country** data concerning this country is presented in the report.
- **Login** the account data for which is presented in the report.
- **LTV** lifetime value. LTV is calculated as $(\text{amount of deposits} + \text{amount of withdrawals}) / (\text{number of deposits} + \text{number of withdrawals})$.
- **Deposits** the number of deposit operations.
- **Deposit Amount** the total amount of all deposits.
- **Deposit Medium** the average deposit amount. The value is calculated as $(\text{amount of deposits}) / (\text{number of deposits})$.
- **Withdrawals** the number of withdrawal operations.
- **Withdrawals Amount** the total amount of all withdrawal operations.
- **Withdrawals Medium** the average amount of withdrawals. The value is calculated as $(\text{amount of withdrawals}) / (\text{number of withdrawals})$.

Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.

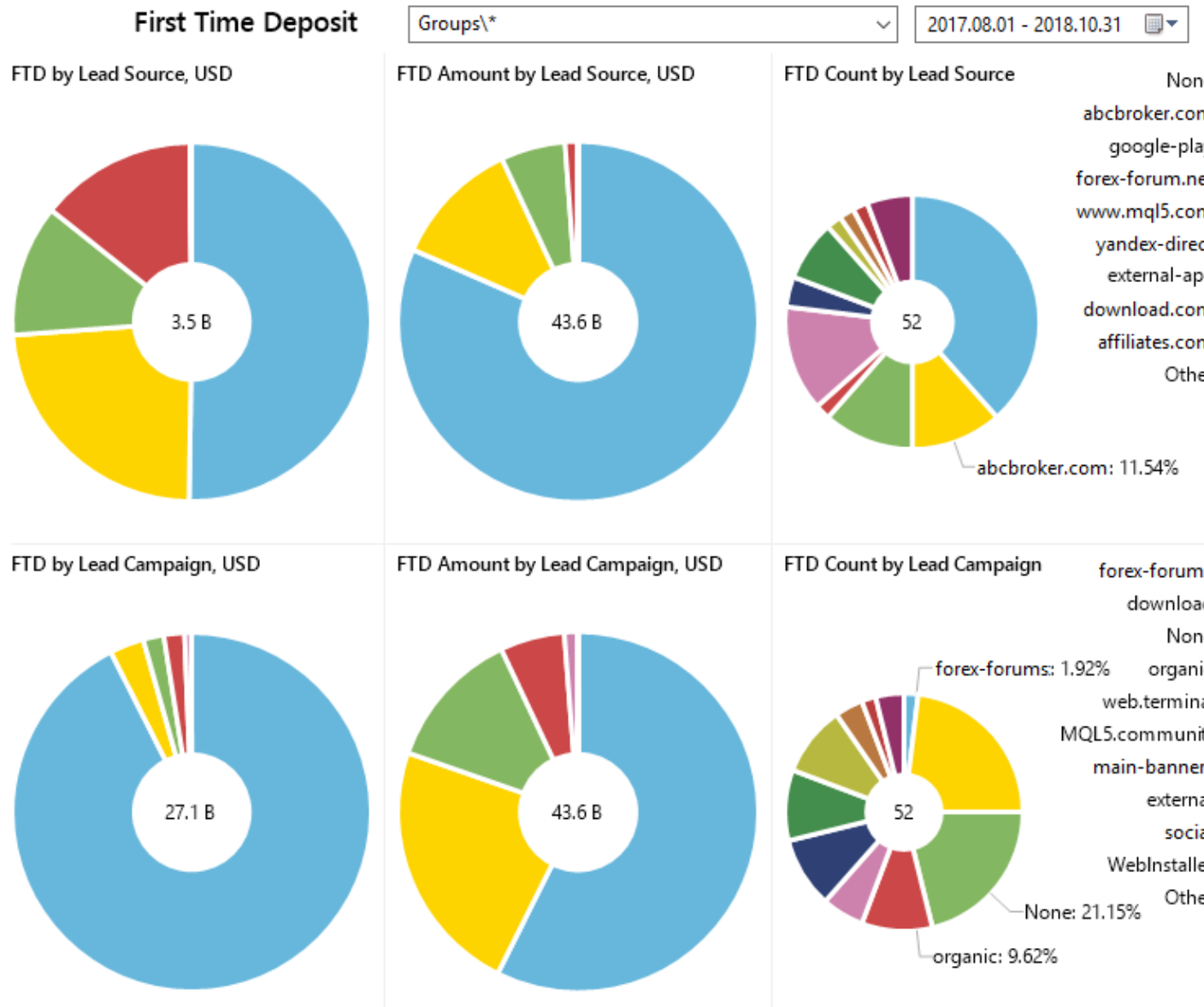
Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / First Time Deposit

First Time Deposit

The report is available as a set of diagrams presenting information about first-time deposits made by clients.



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the report will be generated for accounts from the specified groups. You can specify one or several groups separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes three types of charts:

- **FTD** the average value of clients' first deposits.
- **FTD Amount** the total amount of clients' first-time deposits over the selected period.
- **FTD Count** the number of clients' first-time deposits over the selected period.

The first-time deposit is registered per client. A client may have multiple trading accounts, so the report will reflect only one deposit (the first one) on one of these accounts.

In addition to the diagram type, you can select from various distribution options based on different parameters:

- **Lead Campaign** the name of an advertising campaign by which the client was attracted.
- **Lead Source** the website from which the client came.
- **Country** the client's country of residence.
- **Value** first-time deposit (different ranges).

- Balance operation values reflected in charts are automatically converted to the same currency in accordance with current quotes. The report currency is configured on the trade server side.
- Lead Campaign and Lead Source data are available in appropriate fields of trading accounts.

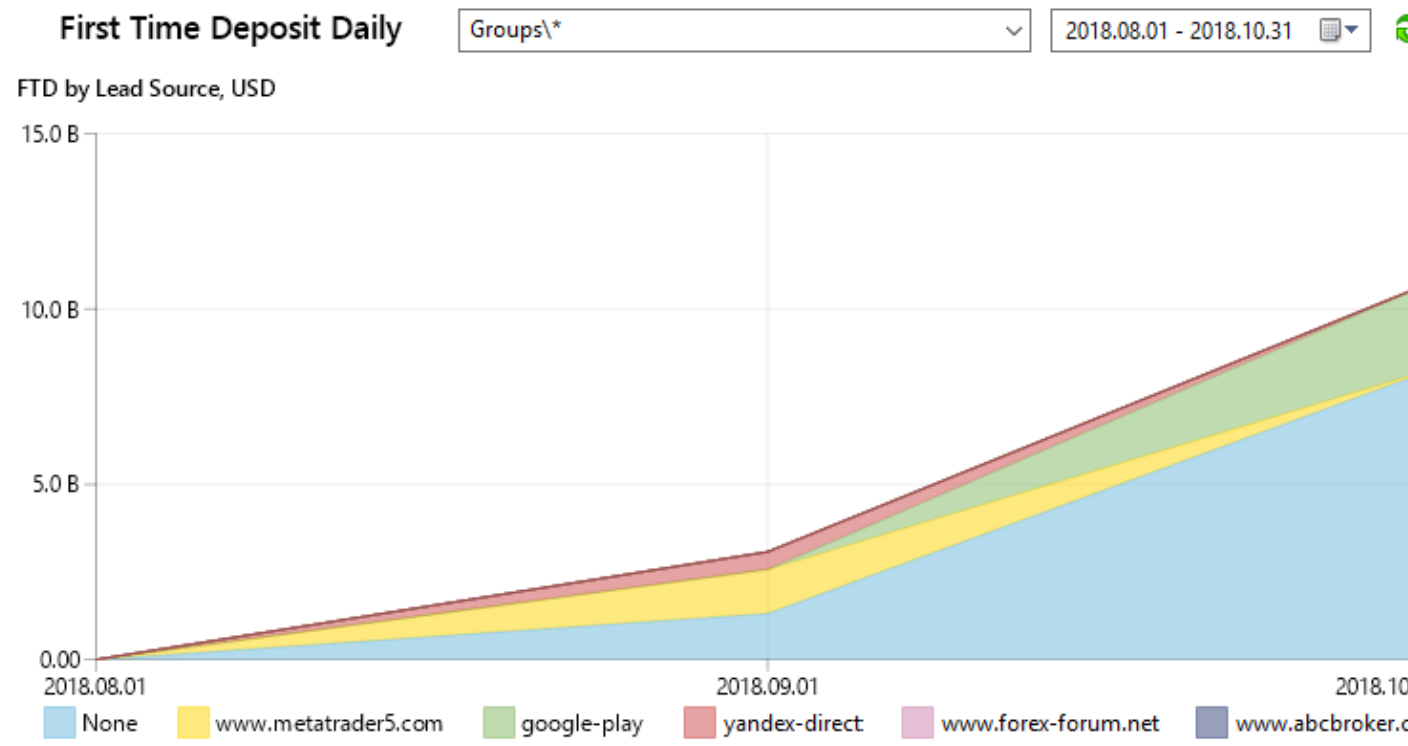
Setup in MetaTrader 5 Administrator

The additional 'Currency' parameter can be set in the report configuration in MetaTrader 5 Administrator. Appropriate values in reports will be displayed in this currency. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

First Time Deposit Detailed

MetaTrader 5 trading platform

This report features the distribution of clients' first-time deposits based on the client property and based on the time.



The report features the same metrics and parameters while additionally enabling the analysis of first-time deposit distribution over time.

First Time Deposit Amount

This report features the distribution of clients' first-time deposits according to different related parameters:

- **Lead Campaign** the name of an advertising campaign by which the client was attracted.
- **Lead Source** the website from which the client came.
- **Country** the client's country of residence.

In addition to groups and dates, the report provides filtering by campaign and lead source.

First Time Deposit Amount

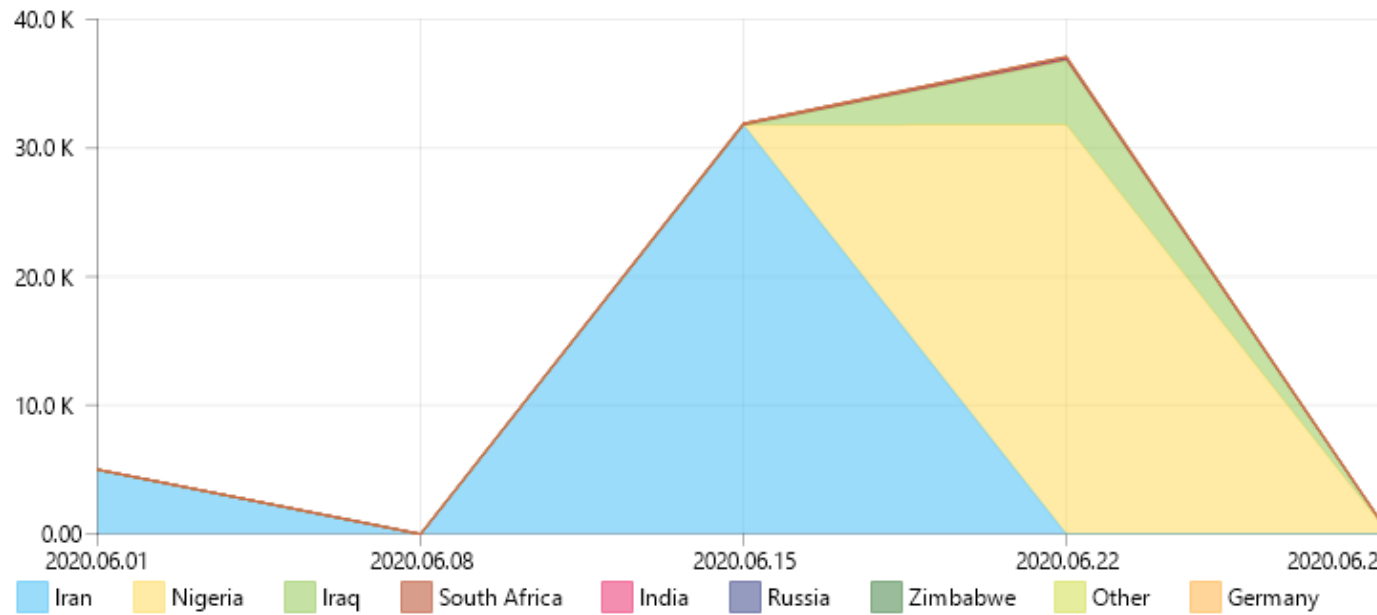
2020.06.01 - 2020.07.01

Bce Lead Campaign

Bce Lead Source

*

FTD Amount by Country, USD



MetaTrader 5 Trading Platform / Platform Setup / Reports / Retention of Trading Accounts

Retention of Trading Accounts

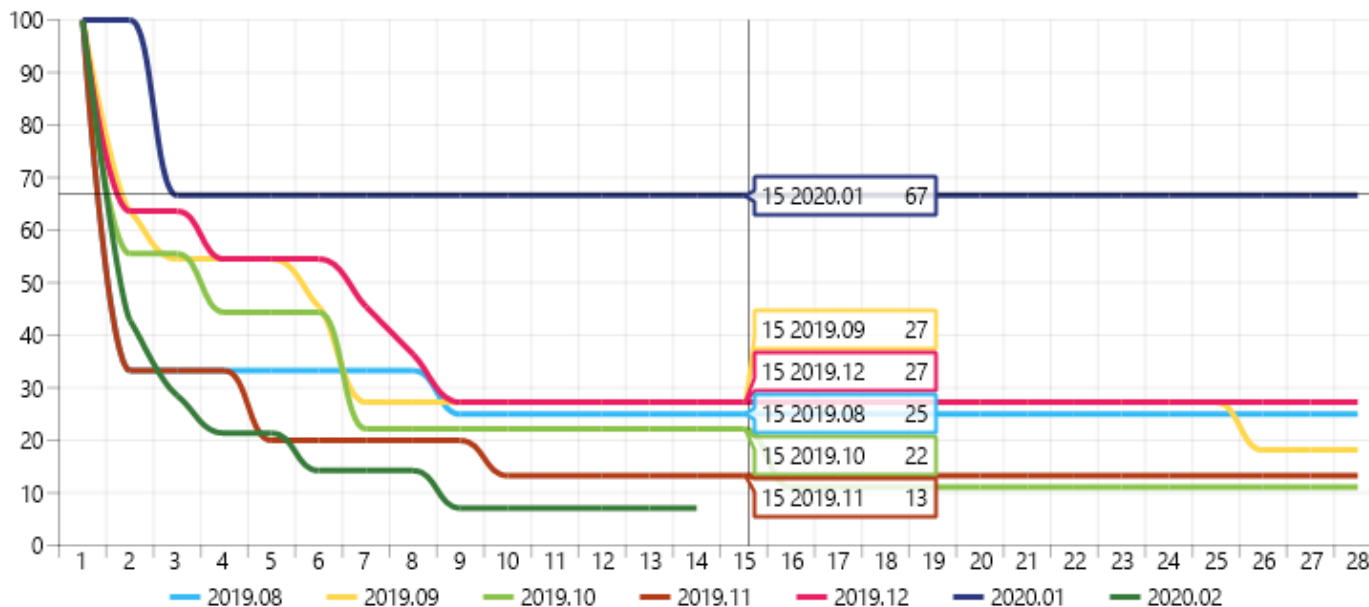
The report visualizes the efficiency of your client retention efforts: activity dynamics during the first month after registration, i.e. the percentage of registered accounts, which connected to the server on the first day, on the second day and so on until the 28th day. The information is displayed in four modes: by registration month, by Lead Source, by Lead Campaign and by country.

Retention of Trading Accounts

Groups*

2019.08.01 - 2020.02.19

Retention by Registration Date, %



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes multiple charts:

- **Retention by Registration Date** the percentage of accounts with the specified registration date, which connected to the server on the 1st day after registration, on the 2nd day and so on until the 28th day.
- **Retention by Lead Campaign** the percentage of accounts with the specified campaign, which connected to the server on the 1st day, on the 2nd day and so on until the 28th day.
- **Retention by Lead Source** the percentage of accounts with the specified lead source, which connected to the server on the 1st day, on the 2nd day and so on until the 28th day.
- **Retention by Country** the percentage of accounts from the specified country, which connected to the server on the 1st day, on the 2nd day and so on until the 28th day.

Lead Campaign and Lead Source data are available in appropriate fields of trading accounts.

Setup in MetaTrader 5 Administrator

The following parameters can be specified in the report configuration in MetaTrader 5 Administrator:

- **Top Count** the number of data series which will be shown in the report: registration dates, campaigns, sources and countries. The default is 10. Data series with the highest values are selected for the report, while all other series are summed up and are displayed as the "Other" variable.
- **Cache Rebuild Period, sec** the maximum time by which data in the report can lag. Time is specified in seconds. The default is 30. If the latest data in the server cache is within the specified interval from the report request time, the data cache will not be refreshed. It means that the data in the report will not be refreshed in the case of frequent repeated requests. This enables fast generation of reports while avoiding extra server load.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Retention of Clients

Retention of Clients

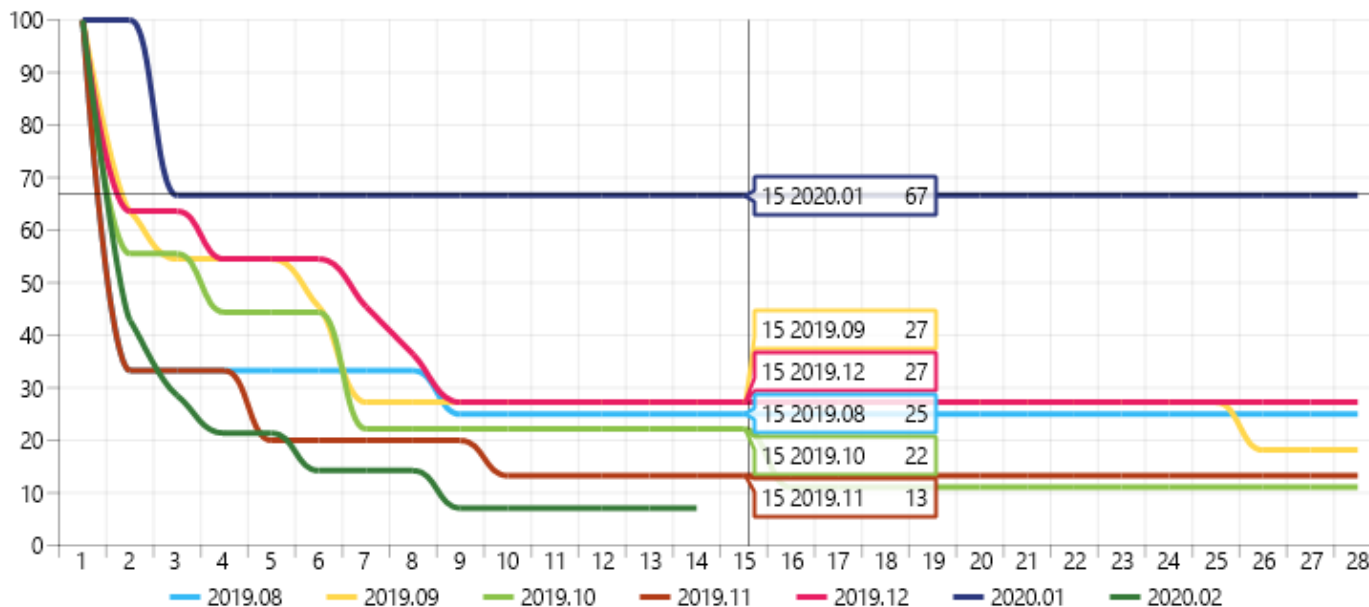
The report visualizes the efficiency of your client retention efforts: trading account activity dynamics during the first month after registration, i.e. the percentage of registered accounts, which connected to the server on the first day, on the second day and so on until the 28th day. The information is displayed in four modes: by registration month, by Lead Source, by Lead Campaign and by country.

Retention of Clients

Groups*

2019.08.01 - 2020.02.19

Retention by Registration Date, %



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated.

Charts in the report

The report includes multiple charts:

- **Retention by Registration Date** the percentage of clients who connected to the server on the 1st day, on the 2nd day and so on until the 28th day.
- **Retention by Lead Campaign** the percentage of clients with a specified campaign, who connected to the server on the 1st day, on the 2nd day and so on until the 28th day.
- **Retention by Lead Source** the percentage of clients with the specified lead source, who connected to the server on the 1st day, on the 2nd day and so on until the 28th day.
- **Retention by Country** the percentage of clients from the specified country, who connected to the server on the 1st day, on the 2nd day and so on until the 28th day.

The report is generated based on clients. A client can have multiple trading accounts, and thus only one (the first) connection from any of the accounts will be reflected in the report.

Lead Campaign and Lead Source data are available in appropriate fields of trading accounts.

Setup in MetaTrader 5 Administrator

The following parameters can be specified in the report configuration in MetaTrader 5 Administrator:

- **Top Count** the number of data series which will be shown in the report: registration dates, campaigns, sources and countries. The default is 10. Data series with the highest values are selected for the report, while all other series are summed up and are displayed as the "Other" variable.
- **Cache Rebuild Period, sec** the maximum time by which data in the report can lag. Time is specified in seconds. The default is 30. If the latest data in the server cache is within the specified interval from the report request time, the data cache will not be refreshed. It means that the data in the report will not be refreshed in the case of frequent repeated requests. This enables fast generation of reports while avoiding extra server load.

MetaTrader 5 Trading Platform / Platform Setup / Reports / NFA

NFA Reports

U.S. regulator National Futures Association (NFA) has specific requirements concerning financial control and reporting for U.S.-registered brokerages. To simplify the compliance with the regulator's requirements, the NFA Reports is included in the platform standard delivery. It automatically generates a number of reports to be submitted to NFA.

The reports are generated according to the regulator requirements described in Forex Data Layout Overview and Implementation Reference Version 1.8.9. The applied format consists of text ASCII files using field separators similar to CSV. The format is used for easy parsing and analysis of reports on the NFA side.

The report itself is implemented as the NFA.Reports64.dll file combining the report and the server plugin. The plugin collects the necessary data and saves it in the trade server directory, while the report provides collected data in the appropriate format. The plugin should be enabled in the settings of the appropriate trading server. If the plugin is not enabled, report data will not be updated and the report will provide data only for the plugin's operation period.

The NFA.Reports64.dll file is included into the standard delivery and located in Reports directory of MetaTrader 5 trading server.

Available reports

In accordance with the NFA Rule 2-48, the plugin generates the following report types:

- Orderbook Changes a list of order transactions (Buy/Sell market orders and pending orders).
- Trade list of transactions.
- Customer Info list of active clients.
- Adjustments list of transaction modifications (modification of an open price/volume of an open position or modification of an open price/close price/volume/profit of a closed position).
- Ad Hoc Tick Data list of ticks for a day.

The following types of reports are also generated by the plugin, but they do not contain any data since there is no corresponding functionality in the MetaTrader 5 platform. These reports are generated because they should still be submitted to the regulator even if there is no data.

- Money Manager Info a list of NFA-accredited account managers.
- Customer/Manager Concordance a list of customers to account manager concordance.
- Market Events a list of significant market events (e.g. a failure of a trade server).

MetaTrader 5 trading platform

All generated reports are placed in the directory of the server the plugin is installed for: [trade server installation directory]\NFA\Export\[trading day date]. A trading day means the period from 17:00 (05:00 PM) of the the previous day to 17:00 (05:00 PM) of the current day in the EST time zone. For example:

- An event that occurred on 13.12.2010 at 16:50 applies to the trading day of 13.12.2010.
- An event that occurred on 13.12.2010 at 17:50 applies to the trading day of 14.12.2010.

In addition, all the time values given in the statement are converted to the EST time zone.

The detailed description of all reports is provided in the "Reported data" section below.

- The platform time zone does not affect report generation since the NFA Export plugin converts any time of the trade server to the EST time zone and works only with this time.
- The trading platform does not support market events. Therefore, the empty Market Events report is generated.
- The trading platform does not support account managers. Therefore, the plugin generates empty Money Manager and Customer/Manager Concordance reports. If your company uses implementation of the account managers from third parties, you should contact them to develop the appropriate reports.

Exporting stop loss and take profit data

Among other things, the Orderbook Changes and Trade reports should contain data on stop loss and take profit orders. However, in MetaTrader 5, they are represented as position fields/properties rather than separate orders. Therefore, when generating the reports, the plugin exports these fields as Stop and Limit orders on its own. An operation ID (ORDER_ID) is generated for them in a special way:

- Position take profit is exported as a Limit order with the ID [Position ticket]_TP_[counter]
- Position stop loss is exported as a Stop order with the ID [Position ticket]_SL_[counter]

Here, the ticket is a position ID on the platform side, while the counter is a serial number of a stop loss or a take profit set for a specific position. Example:

1. A stop loss is set to 1000 for a position with the ticket 12345. In the report, the operation is displayed as ORDER ADD 12345_SL_1.
2. The stop loss is then removed. This operation is displayed as ORDER CANCELLED 12345_SL_1.
3. A new stop loss for the position is set to 500. This operation is displayed as ORDER ADD 12345_SL_2.

MetaTrader 5 trading platform

4. The stop loss is changed to 700. This change is displayed as ORDER MODIFY 12345_SL_2 in the report.

Thus, the counter value is changed only if a stop loss/take profit is removed and a new one is placed, which means an old order is removed, while a new one is placed. When changing the levels, it is assumed that the modification is applied to the same order.

Plugin installation

The appropriate plugin should be installed to collect the data used in the report. To install the plugin, enter Plugins section in MetaTrader 5 Administrator. After the server with the installed plugin has been selected, click + Add.

Plugin: NFA Reports

Common Information

The server plugin is intended for enrichment of the server functionality and represents an external program with remote configuration management.

☒ Enable ☐ Configurable by managers

☐ Enable plugin profiling

Name: NFA Reports

Module: NFA.Reports64.dll Trade Main, 1

Parameters:

Parameter	Value
Symbols	*
Groups	*
Forex Dealer Member ID	BRK
Server ID	Main
Base Directory	NFA.Reports
SFTP Address	
SFTP Port	1
SFTP Login	
SFTP Password	
SFTP Private Key Path	
Report Time	17:00:00
Send Time	18:00:00

Add Edit Delete

OK Cancel Help

The following settings are available:

- **Symbol** sort by symbols. Symbols are separated with a comma. To specify symbols, you may also use the mask "*" and the negation sign "!". For example, the value "EURUSD,GBPUSD" means that the plugin will collect the transaction details for EURUSD and GBPUSD symbols. The value "*" is for all symbols. *,!CFD* value is for all symbols except those in the CFD group.
- **Groups** sort by groups. Groups are separated with a comma. To specify groups, you may also use the mask "*" and the negation sign "!". For example, the value "demoforex,democfd" means that the plugin will collect the transaction details for users in "demoforex" and "democfd" groups. The value "*" is for all groups. The default value is "*,!demo*,!contest*" which means all groups except those whose names start with "demo" and "contest".
- **Forex Dealer Member ID** a seven-digit identifier of a broker given by NFA (FDM's NFA ID).
- **Server ID** trade server name specified in the report.
- **Base Directory** name of the directory where the database of transactions is saved for the report. The data is saved by days. This directory is automatically created in Reports\NFA.Reports. In most cases, this parameter should save a default value.
- **SFTP Address** address for connecting to Secured FTP server the reports are to be uploaded to.
- **SFTP Port** port for connecting to Secured FTP server.
- **SFTP Login** login for connecting to Secured FTP server.
- **SFTP Password** password for connecting to Secured FTP server.
- **SFTP Private Key Path** path for the private key for authorization on Secured FTP server.
- **Report Time** report generation time (EST).
- **Send Time** time of sending reports to the SFTP server (EST).

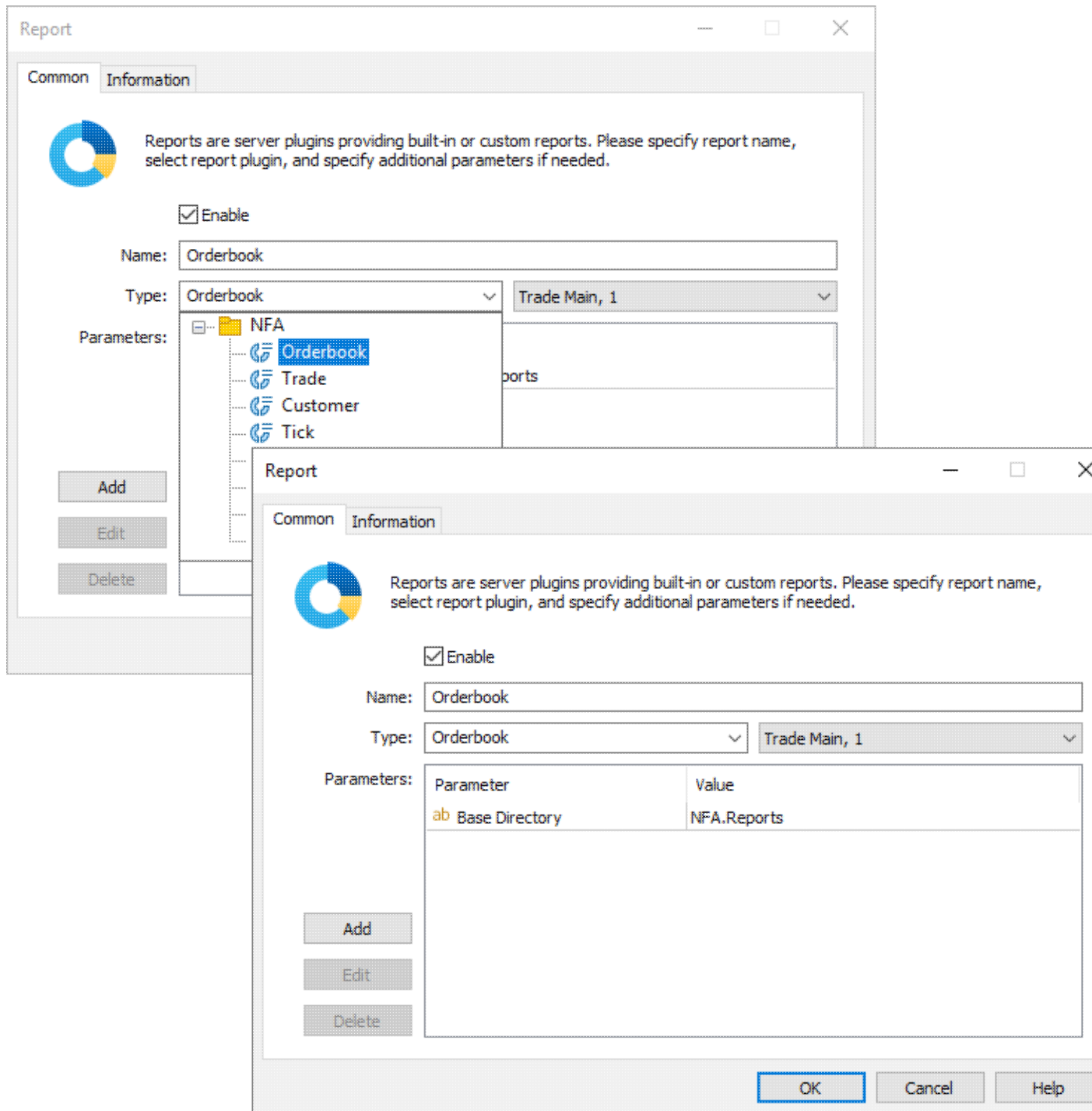
It is not recommended to use the same Base Directory parameters for different plugins. In this case, correct operation of those plugins is not guaranteed.

Report installation

Reports are generated and sent to NFA by the server plugin. The report configuration stage is needed only to be able to request and view them via the Manager terminal.

To install the report, enter Reports section in MetaTrader 5 Administrator. After the server to install the report has been selected, click + Add.

MetaTrader 5 trading platform



Each of the NFA Reports is configured separately. If you want to view the Addhoc Tick Data Requests report via the Manager terminal, create a separate configuration for it. The same should be done if you need the Orders report. There is no point in creating configurations for the Money Managers, Customer/Manager Concordance and Market Events reports since they always remain empty and are necessary only when reporting to NFA, which is performed by the plugin.

After adding a configuration, select the necessary report in the Type field and set the only additional parameter Base Directory for it. This is a path to the directory where the search for report data is performed. It should correspond to the path specified in the Base Directory parameter in the plugin settings. If not a single plugin saves data at the specified path, the report remains empty. Multiple reports may take data from the same directory.

The standard parameters are described in the Reports section.

Viewing the reports

To view the report request, use MetaTrader 5 Manager. Simply select the necessary report in the Server Reports section to retrieve data from the server. The report data can be sorted by group, logins, period and symbol mask.

Find out more about working with the reports in the appropriate section of MetaTrader 5 Manager documentation.

MetaTrader 5 trading platform

1026605 - TradeServer - John Smith

File Edit View Tools Window Help

Automation

Filter

Navigator

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 - Tick
 - Adjustments
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 - Dealing
 - Groups (177)
 - Plugins (5)
 - Support Center
 - App Store

Orderbook

2019.08.12 - 2019.08.16

* Groups*

ID	ORDER_ID	FDM_ID	PRODUCT_CODE	ORDER_TYPE	BID_PRICE	ASK_PRICE	DURATION
942	12065623	BRK	EURUSD	LIMIT	1.121110	1.121150	FOK
943	12065623	BRK	EURUSD	LIMIT	1.121110	1.121150	FOK
944	12065624	BRK	EURUSD	LIMIT	1.121110	1.121150	FOK
945	12065624	BRK	EURUSD	LIMIT	1.121110	1.121150	FOK
946	12065625	BRK	EURUSD	LIMIT	1.121020	1.121050	FOK
947	12065625	BRK	EURUSD	LIMIT	1.121020	1.121050	FOK
948	12065626	BRK	EURUSD	LIMIT	1.120990	1.121020	FOK
949	12065626	BRK	EURUSD	LIMIT	1.120990	1.121020	FOK
950	12065627	BRK	EURUSD	LIMIT	1.120800	1.120850	FOK
951	12065627	BRK	EURUSD	LIMIT	1.120800	1.120850	FOK
952	12065630	BRK	AUDCHF	LIMIT	0.659190	0.659990	FOK
953	12065630	BRK	AUDCHF	LIMIT	0.659190	0.659990	FOK
954	12065631	BRK	AUDCHF	LIMIT	0.659200	0.660000	FOK
955	12065631	BRK	AUDCHF	LIMIT	0.659200	0.660000	FOK
956	12065632	BRK	CHFJPY	LIMIT	108.396000	108.456000	FOK
957	12065632	BRK	CHFJPY	LIMIT	108.396000	108.456000	FOK
958	12065633	BRK	AUDCHF	LIMIT	0.659090	0.659890	FOK
959	12065633	BRK	AUDCHF	LIMIT	0.659090	0.659890	FOK
960	12065636	BRK	EURUSD	LIMIT	1.120730	1.120790	FOK

For Help, press F1

1452 / 23 KB

No data is generated when requesting reports. In fact, the Manager terminal simply reads data from the server files generated by the plugin. If no data is generated by the plugin, the reports remain empty.

Sending the reports to NFA

To submit the reports to NFA, the appropriate files should be uploaded to the regulator's special SFTP server, except for Adhoc Tick Data Requests which is requested by NFA only if necessary. NFA Reports plugin is able to send reports to the server automatically. To achieve this, specify the sending time and the SFTP server connection data in its settings.

To get SFTP access (address, login and password), contact NFA in accordance with Security Request Procedure described in Forex Data Layout Overview and Implementation Reference.

Apart from a login and a password for accessing SFTP, you also need to create a bunch of RSA keys (private/public) and send the public key to NFA. To do this, follow the stages described in Forex SFTP Connectivity Implementation Reference:

1. On the <http://www.putty.org/> website, download the puttygen.exe (PuTTY SSH key generation utility) application.
2. Launch puttygen.exe and select SSH-2 RSA key type.
3. Click Generate and follow the application instructions.
4. Click "Save public key" and specify the name of the public key file.
5. Click "Save private key" and specify the name of the private key file.
6. Send the public key to NFA for its subsequent installation on the SFTP server.

- NFA requires a public key file name to contain Forex Dealer Member ID.
- The private key is an additional means of authentication on the SFTP server. Keep it in a safe place and do not submit it to anyone.
- When saving the private key, do not specify "passphrase". Otherwise, the plugin will not be able to automatically send reports to the SFTP server due to incompatibility of pagent.exe and Windows services.

After NFA installs the public key on its SFTP server, you will be able to send reports via the plugin automatically. For configuration:

1. Specify data for connecting to Secured FTP server in the plugin settings (address, login, password and path to the private key).
2. Set report sending time in the plugin settings.

Report data

The contents of all reports are described in details in Forex Data Layout Overview and Implementation Reference. NFA Reports is fully compliant with the requirements. The general description is provided below for more convenience.

The time in all reports is indicated according to the EST time zone.

Orderbook Changes

- ID unique operation ID within a single report (for one day).
- BATCH_DATE report date.
- TRANSACTION_DATETIME operation date and time.
- ORDER_ID order ticket/ID on MetaTrader 5 side.
- FDM_ID broker ID in NFA.
- MKT_SEG_ID market segment ID. The field remains blank since there is no such entity in MetaTrader 5.
- PRODUCT_CODE financial instrument name.
- PRODUCT_CAT product category. C call option, P put option, F Forex symbol, R forward contract.
- CONTRACT_YEAR forward/option contract execution year.
- CONTRACT_MONTH forward/option contract execution month.
- CONTRACT_DAY forward/option contract execution day.
- STRIKE strike option price.
- OPTION_TYPE option type: standard or exotic.
- TRANSACTION_TYPE transaction type: entry, modification, execution, etc.
- ORDER_TYPE order type.
- VERB operation direction: buy or sell.
- BID_PRICE Bid price at the time of operation.
- ASK_PRICE Ask price at the time of operation.
- QUANTITY operation volume in the number of contracts. The value cannot be fractional. If the calculated volume has a fractional part, it will be rounded to the nearest whole number.
- REMAINING_QTY volume of an unexecuted part of an order (filled only for partially executed orders). The value cannot be fractional. If the calculated volume has a fractional part, it will be rounded to the nearest whole number.
- PRICE Limit price for limit orders and option price.
- STOP_PRICE stop price in orders.
- STOP_PRODUCT_CODE dependent product code. The field remains blank since there is no such entity in MetaTrader 5.
- TRAIL_AMT trailing size. The field is always set to zero since a trailing stop works only on the client terminal side.
- LIMIT_OFFSET distance to a limit price in trailing stop limit orders. The field is always set to zero since there is no such entity in MetaTrader 5.
- DURATION order type by expiration type: GTC, up to a given date, time, etc.
- EXPIRY_DATE order expiration date.
- ORDER_ORIGIN order source. SYSTEM created by the platform, DLR created by a dealer, CUST created by a client.
- MANAGER_ID capital manager ID. The field remains blank since there is no such entity in MetaTrader 5.
- CUSTACCT_ID client account number.
- SERVER_ID unique server ID (set in the [plugin settings](#)).
- CUST_GROUP client category. The field remains blank since there is no such entity in MetaTrader 5.
- LINKED_ORDER_ID linked order ID (for example, for OCO orders). The field remains blank since there is no such entity in MetaTrader 5.
- LINK_REASON reason an additional order is linked to a main one. The field remains blank since there is no such entity in MetaTrader 5.

MetaTrader 5 trading platform

- OPEN_CLOSE currently not used, always remains blank.

Trade

- ID unique operation ID within a single report (for one day).
- BATCH_DATE report date.
- TRADE_ID trade ticket/ID on MetaTrader 5 side.
- ORDER_ID order ticket/ID on MetaTrader 5 side.
- FDM_ID broker ID in NFA.
- MKT_SEG_ID market segment ID. The field remains blank since there is no such entity in MetaTrader 5.
- TIME_TRADEMATCH trade execution time.
- PRODUCT_CAT product category. C call option, P put option, F Forex symbol, R forward contract.
- PRODUCT_CODE financial instrument name.
- CONTRACT_YEAR forward/option contract execution year.
- CONTRACT_MONTH forward/option contract execution month.
- CONTRACT_DAY forward/option contract execution day.
- STRIKE strike option price.
- VERB operation direction: buy or sell.
- BID_PRICE Bid price at the time of operation.
- ASK_PRICE Ask price at the time of operation.
- QUANTITY operation volume in the number of contracts. The value cannot be fractional. If the calculated volume has a fractional part, it will be rounded to the nearest whole number.
- REMAINING_QTY volume of an unexecuted part of a trade (filled only for partially executed trades). The value cannot be fractional. If the calculated volume has a fractional part, it will be rounded to the nearest whole number.
- FILL_PRICE trade price.
- CONTRA_FILL_PRICE initial trade price at a liquidity provider side (if trades are sent to external systems).
- SERVER_ID unique server ID (set in the [plugin settings](#)).
- IMPLIED_VOLATILITY implied option volatility.
- IB_REBATE broker markup for a trade.
- COMMISSION commission for a trade.

Customer Info

- BATCH_DATE report date.
- CUSTACCT_ID client account number.
- FDM_ID broker ID in NFA.
- CUST_NAME customer's name.
- CUST_BIRTH_DATE customer's date of birth. The value is copied from the [client record](#) to which the trading account is linked.
- CUST_TYPE customer type. ECP counterparty, RETAIL retail client, OMNIBUS white label.
- CUST_GROUP client category. The field remains blank since there is no such entity in MetaTrader 5.
- COUNTRY_TYPE customer type by country. FOR foreign, DOM local.
- NET_LIQUIDATING_VALUE account liquidation value.
- ACCT_OPEN_DATE account opening date.

Adjustments

- BATCH_DATE report date.
- FDM_ID broker ID in NFA.
- TRADE_ID trade ticket/ID on MetaTrader 5 side.
- ADJUSTMENT_DATETIME adjustment time.
- ADJUSTMENT_TYPE adjustment type. PRICE price change, QTY volume change, CASH monetary outcome change.
- PRICE new price after an adjustment.
- QUANTITY new operation volume in the number of contracts.
- CASH_AMOUNT amount charged/debited from an account.
- NOTE adjustment reason.

Ad Hoc Tick Data

- FDM_ID broker ID in NFA.
- BATCH_DATE report date.
- PRODUCT_CODE financial instrument name.
- QUOTE_DATETIME quote date and time.
- BID_PRICE Bid price.
- ASK_PRICE Ask price.

Money Manager Info, Customer/Manager Concordance, Market Events

These reports always remain empty.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Reports](#) / EMIR

EMIR Report

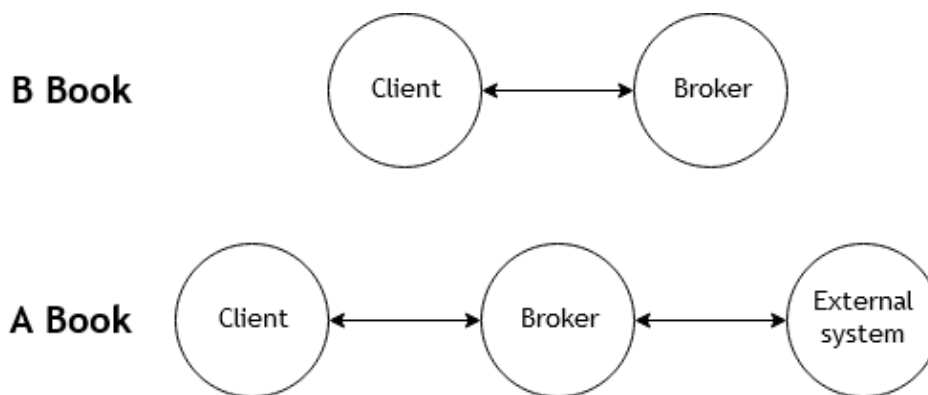
EMIR (European Market Infrastructure Regulation) is a regulation by European Parliament and Council of the European Union, which sets unified requirements for over-the-counter derivatives trading, aims at reducing related risks and trade reporting. EMIR came into force on August 16, 2012. For more details please visit the [ESMA website](#).

One of the EMIR requirements concerns reporting derivative trades to licensed trade repositories. The standard "EMIR Reports" report is available in the platform package to assist brokers in meeting this requirement. It enables the automatic generation of reports in the appropriate format and sending them to the repository's FTP server.

Among all the repositories to which the reports can be sent, the largest and the most popular one is [CME Group](#). MetaTrader 5 EMIR Reports are designed specifically for operations with this repository. The report specification is available at the [CME Group website](#).

A Book and B Book trades in the report

The main difference in the trades presentation in the report concerns the number of rows corresponding to each of them. B Book deals which are processed directly in the MetaTrader 5 platform and not sent to external trading systems, are reflected in a single report table row. A Book deals are shown in two rows: one of them reflects the internal part of the deal between the trader and the broker (intermediary party) and the second row shows the external part between the broker and the external system (exchange). In other words, B Book trades have only the internal part, while an A Book trade consists of an internal and an external part.



In addition, some of the report fields are filled in different ways, depending on the deal part:

Field	The B Book trade or the internal part of the A Book trade	External part of the A Book trade
NonReportingPartyIDType	Value "LEI", if the broker's client who has performed the trade is <u>marked</u> as "Corporate" or "Fund". Value "CLC" for all other client types.	Value "LEI".
NonReportingPartyLEI	If "LEI" is specified in the NonReportingPartyIDType field, the value of "LEI number for EMIR reports" from the <u>client record</u> will be used. If "CLC" is specified in "NonReportingPartyIDType", the value is determined as follows (in order of priority): • External client ID from <u>trading account settings</u>. • Internal client ID, which is equal to the <u>client record</u> ID in MetaTrader 5. • Trading account ID, i.e. <u>account</u> login/number.	LEI of the liquidity provider in accordance with the <u>gateway settings</u> .
NonReportingPartyCountry	The value is determined as follows: • For legal entities from the "Registration country" field of the <u>client record</u>. • For individuals from the "Country" field of the <u>client record</u>.	The "EMIR Report Country" parameter value from the <u>gateway settings</u> .
UTI	Consists of three parts: [E02][LEI][Transaction ID]:	Consists of three parts: [E02][LEI][Transaction

Field	The B Book trade or the internal part of the A Book trade	External part of the A Book trade
	• E02 the first part always consists of these three characters. • LEI broker's LEI from the "Broker LEI" parameter in the <u>plugin settings</u>. • Transaction ID <u>deal ticket</u> in MetaTrader 5.	ID]: • E02 the first part always consists of these three characters. • LEI liquidity provider's LEI from the <u>gateway settings</u>. • Transaction ID <u>deal ID in the external system</u>.
ExecutionVenue	Value "XXXX".	The "EMIR Report MIC" parameter value from the <u>gateway settings</u> .

The detailed description of the report contents is provided in the "Reported data" section below.

Plugin Installation and Features

The report is implemented as the EMIR.Reports64.dll file combining two report modules and the server plugin:

- The plugin collects data required for report sending via FTP, and saves this data in the trade server directory.
- Reports enable the convenient view of information in the created files and the manual generation of relevant reports.

The EMIR.Reports64.dll file is included into the standard platform package and is located under the Reports directory of the MetaTrader 5 trading server.

The plugin generates the reports daily, based on previous-day data. Report creation time is set in the plugin settings. Along with generation, the plugin can automatically send reports to the regulator's SFTP server, and thus you do not have to submit them manually. Report sending time and address are also specified in the settings.

The generated files are saved in the directory of the server for which the plugin is installed: [trading server installation directory]\Reports\EMIR.Reports\CME\Outbox. Once a file is sent to the SFTP server, it is moved from the "Outbox" folder to "Sent", located under the same directory.

To install the report, navigate to the Plugins section in MetaTrader 5 Administrator. After the server with the installed plugin has been selected, click +Add.

MetaTrader 5 trading platform

Plugin: EMIR Reports

Common

Information



The server plugin is intended for enrichment of the server functionality and represents an external program with remote configuration management.

☒ Enable
 ☐ Configurable by managers

☐ Enable plugin profiling

Name:

Module:

Parameters:

Parameter	Value
 Report Time	12:36:00
 Symbols	*
 Groups	*
 Send Time	15:00:00
 CME firm's SFTP account ID	<account ID>
 SFTP Host	address
 SFTP Port	port
 SFTP Login	login
 SFTP Password	password
 SFTP Key Path	C:\KeyPath
 Broker LEI	LEI
 1.6 CorporateSector	F

Add

Edit

Delete

OK

Cancel

Help

The following parameters are available here:

- **Report Time** report generation time in the format of HH:MM:SS. This is the MetaTrader 5 trading platform site.
- **Symbol** sort by symbols. A comma separated list of symbols. The mask "*" and the negation sign "!" can be used here. For example, the value "EURUSD,GBPUSD" means that the plugin will collect information related to deals for EURUSD and GBPUSD symbols. The value "*" is for all symbols. *,!CFD* value is for all symbols except those in the CFD group.
- **Groups** sort by groups. A comma separated list of groups. The mask "*" and the negation sign "!" can be used here. For example, the value "demoforex,democfd" means that the plugin will collect the transaction details for users in demoforex and democfd groups. The value "*" is for all groups. The default value is "*,!demo*,!contest*" which means all groups except those whose names start with demo and contest.
- **Send Time** time when the report will be send to the regulator's FTP server, HH:MM:SS. This is the MetaTrader 5 trading platform site.
- **CME firm's SFTP account ID** broker's ID in the Secured FTP server of the repository, to which reports will be uploaded. If the reports are to be sent manually, this field can be left blank.
- **SFTP Host** address for connecting to Secured FTP server. If the reports are to be sent manually, this field can be left blank.
- **SFTP Port** port for connecting to Secured FTP server. If the reports are to be sent manually, this field can be left blank.
- **SFTP Login** login for connecting to Secured FTP server. If the reports are to be sent manually, this field can be left blank.
- **SFTP Password** password for connecting to Secured FTP server. If the reports are to be sent manually, this field can be left blank.
- **SFTP Private Key Path** path to the private key for authorization on the Secured FTP server. If the reports are to be sent manually, this field can be left blank.
- **Broker LEI** broker's Legal Entity Identifier (LEI), consists of a 20-digit alphanumeric code. The value is used in various fields of the report.
- **1.6 CorporateSector** the sector in which the company operates. For example, F an investment company, C a credit company, etc. The value is directly substituted in the appropriate report field. The default value is F.
- **1.7 FinancialOrNonFinancialCounterParty** type of the reporting company. For example, F financial counterparty, N non-financial counterparty, etc. The value is directly substituted in the appropriate report field. The default value is F.
- **1.9 ReportingEntityLEI** Legal Entity Identifier (LEI) of the reporting entity. The value is directly substituted in the appropriate report field.
- **2.1 ContractType CD** symbol path mask, according to which it will be assigned to type "CD" (Financial contracts for difference). It is used to fill the "2.1 ContractType" field in the report. For example, if *\cfd* is specified in the parameter, then the "CD" value will be set in the "2.1 ContractType" field for all symbols, having \cfd\ in their path. For example, you can use this mask if contracts for difference in your platform are created in the CFD\, CME\CFD\, Real\CFD\Main\ and similar sections.
- **2.1 ContractType FU** the same for the "FU" type (Futures).
- **2.1 ContractType FW** the same for the "FW" type (Forwards).

- **2.1 ContractType OP** the same for the "OP" type (Option).
- **2.1 ContractType SB** the same for the "SB" type (Spreadbet).
- **2.1 ContractType SW** the same for the "SW" type (Swap).
- **2.1 ContractType ST** the same for the "ST" type (Swaption).
- **2.1 ContractType OT** the same for the "OT" (Other).
- **2.2 AssetClass CO** symbol path mask, according to which it will be assigned to "CO" assets class (Commodity and emission allowances). It is used to fill the "2.2 AssetClass" field in the report. For example, if `*\commodity\*` is specified in the parameter, then the "CO" value will be set in the "2.2 AssetClass" field for all symbols, having `\commodity\` in their path. For example, you can use this mask if contracts for difference in your platform are created in the `Commodity\`, `CME\Commodity\`, `Real\Commodity\Main\` and similar sections.
- **2.2 AssetClass CR** the same for the "CR" class (Credit).
- **2.2 AssetClass CU** the same for the "CU" class (Currency).
- **2.2 AssetClass EQ** the same for the "EQ" class (Equity).
- **2.2 AssetClass IR** the same for the "IR" class (Interest Rate).
- **2.38 Intragroup** determines whether both counterparts of the deal belong to the same company. One of the two values : Y belong to one company, N do not belong. The value is directly substituted in the appropriate report field. The default value is N.
- **Mail Login List** the list of manager logins, to which report generation errors will be emailed.
- **Stop On Error** report generation mode in case data is not enough. If "Yes" (default), the report will not be generated if at least one of the fields has no data. If "No", the report will be generated even if there is missing data. The report fields having no data will be left empty. Additionally, the list of relevant errors will be added to the "Comment" field of each unfilled row.

- Data from the plugin parameters is extracted at the report generation time and is not stored in deal details. If you change plugin parameters after the execution of the relevant deals but prior to report generation, then the new parameter values as of the report generation time will be used for the report.
- For the detailed description of all report fields please visit the [ESMA](#) site.

Handling Report Generation Errors

The report generation is based on data from different settings and databases of the platform, including trading accounts, client records and gateway settings. If any of the data is missing, the generation report will be interrupted by default, i.e. the report file will not be generated and will not be sent to the SFTP server. This prevents the sending of incorrect data to the trading repository.

There is the possibility to disable report generation interruption in case of missing data. To do this, set "No" in Stop On Error. In this case, the report generation will not be interrupted, but some of the report fields will be left blank. For example, if only client LEI is not available for a transaction, the transaction row in the report will have one field empty, while all other fields will be filled. In this case you can add the missing data manually and send the report to the repository.

Errors

All missing data errors are recorded to the trade server journal. To request relevant logs, use the "EMIR Reporter" and "EMIR CME Manual" keywords. Error reports are additionally sent via the internal email system.

Data on encountered issues is also reflected in the reports. A brief description of errors which occurred when filling the fields is displayed in the Comment fields of appropriate report rows.

For EMIR CME Manual report, summary information is additionally shown in the last row. This includes the total number of rows in the report and the number of rows with missing data.

Notifications

After each report generation attempt, the plugin sends a message with the attempt results to managers, via the internal email system. The list of logins to whom the emails will be sent is specified in the Mail Login List parameter of the plugin.

The email subject is "EMIR report automated generation status". It contains the following details:

- If the report has been generated without errors, the email will contain the following message: "successfully generated X rows for the report by specification "CME"", where X is the number of records in the created report, excluding the table header.
- If the report has been created, but one or more fields could not be filled, the email will contain the following: "generated X rows for the report by specification "CME", Y rows have warnings (see trade server log by keyword EMIR for details)". X the number of report records excluding the table header, Y the number of records with warnings. The warnings are displayed in the "Comment" column of appropriate rows in the table.
- If the report could not be created, the email will state: "failed to generate report (see trade server log by keyword EMIR for details)". In this case, you should request the trade server journal using the "EMIR" keyword.

After each attempt to upload a report to the FTP server, the plugin also sends notifications to managers from the "Mail Login List" parameter. A separate email is generated for each sent file.

The email subject is "EMIR report uploading status". It contains the name of the file and its sending status:

- When uploaded successfully, the message is "successfully uploaded".
- If error occurs, one of the following statuses will be delivered depending on the error:
 - "internal error (see trade server log by keyword EMIR for details)" the error could not be identified.
 - "failed to initialize SFTP connection (see trade server log by keyword EMIR for details)" repository server connection error.

- "failed to upload (see trade server log by keyword EMIR for details)" file uploading error.

Configure gateways for reporting A-book transactions

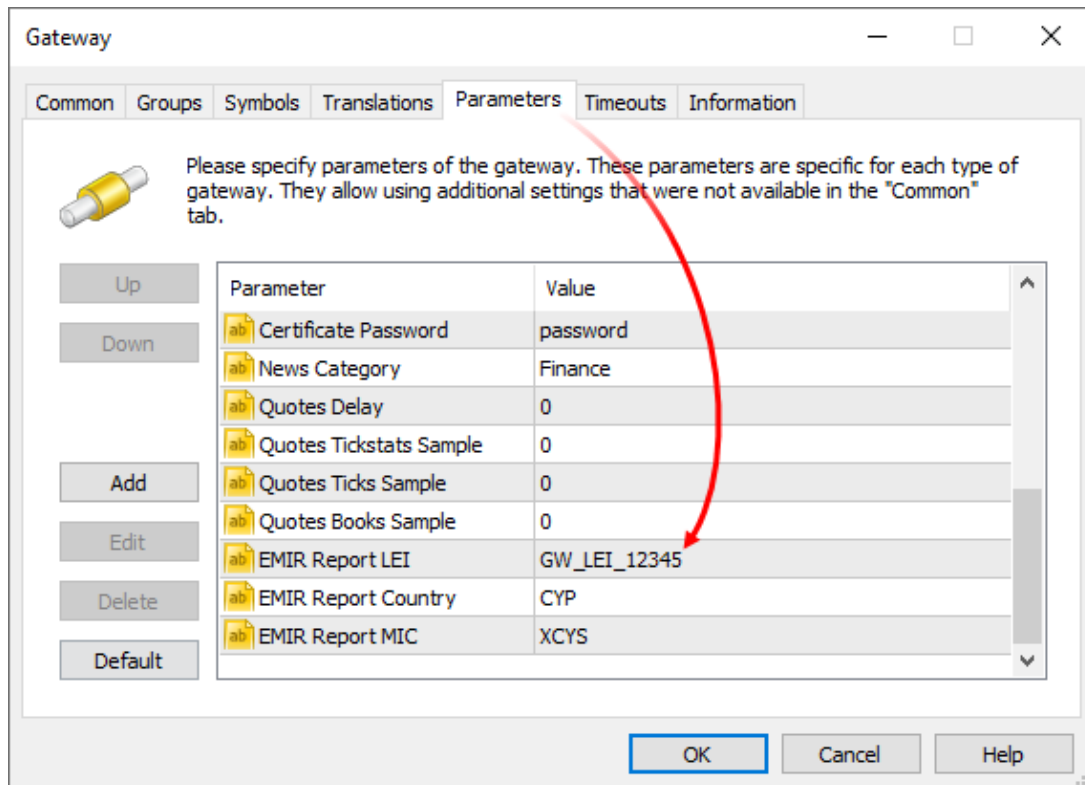
For operations sent to external trading systems through gateways (operations with an external counterparty), additional information is indicated in the EMIR report:

- ReportingPartyLEI LEI of the reporting company. It is always filled with the "ReportingEntityLEI" parameter value from the plugin settings.
- BeneficiaryLEI LEI of the company which acts as a beneficiary in the operation. It is always filled with the "ReportingEntityLEI" parameter value from the plugin settings.
- NonReportingPartyLEI and NonReportingPartyCountry LEI and country code of the counterparty.
- ExecutionVenue code of the execution venue (exchange) to which the gateway is connected.

Since the broker can work with different external systems, the values for these two parameters should be specified in the settings of the gateway via which the deals are forwarded.

Open the gateway settings and add the following parameters:

- EMIR Report LEI LEI of the liquidity provider to which the gateway is connected. Used to fill the NonReportingPartyLEI report field.
- EMIR Report Country code of liquidity provider's registration country. It is used in accordance with the ISO 3166 standard. Used to fill the NonReportingPartyCountry report field.
- EMIR Report MIC the code of the trading exchange to which the gateway is connected. It is indicated in accordance with the ISO 10383 standard. Used to fill the ExecutionVenue report field. If the parameter is not specified, the "XXXX" value will be inserted into the report field.



These parameters can be added to any gateway. They will only be used for the report and do not affect the gateway operation.

Data from the gateway parameters is extracted at the report generation time and are not stored in deal details. If you change gateway parameters after the execution of the relevant deals but prior to report generation, then the new parameter values as of the report generation time will be used for the report.

Report Installation

EMIR Report includes two report modules: EMIR CME Automated and EMIR CME Manual.

EMIR CME Automated

This report allows viewing data from the file generated by the plugin. These include files from two directories

- [trade server installation directory]\Reports\EMIR.Reports\CME\Outbox the generated files which has not yet been sent
- [trade server installation directory]\Reports\EMIR.Reports\CME\Sent the files which were successfully sent to the repository

- To start collecting information, the plugin must be enabled in the appropriate trade server settings. If the plugin is not enabled, report data will not be updated and the report will provide data only for the plugin operation period.

- When the EMIR CME Automated report is used in the Manager terminal, no requests are sent to the server database. The report only presents data from the files generated by the plugin.

EMIR CME Manual

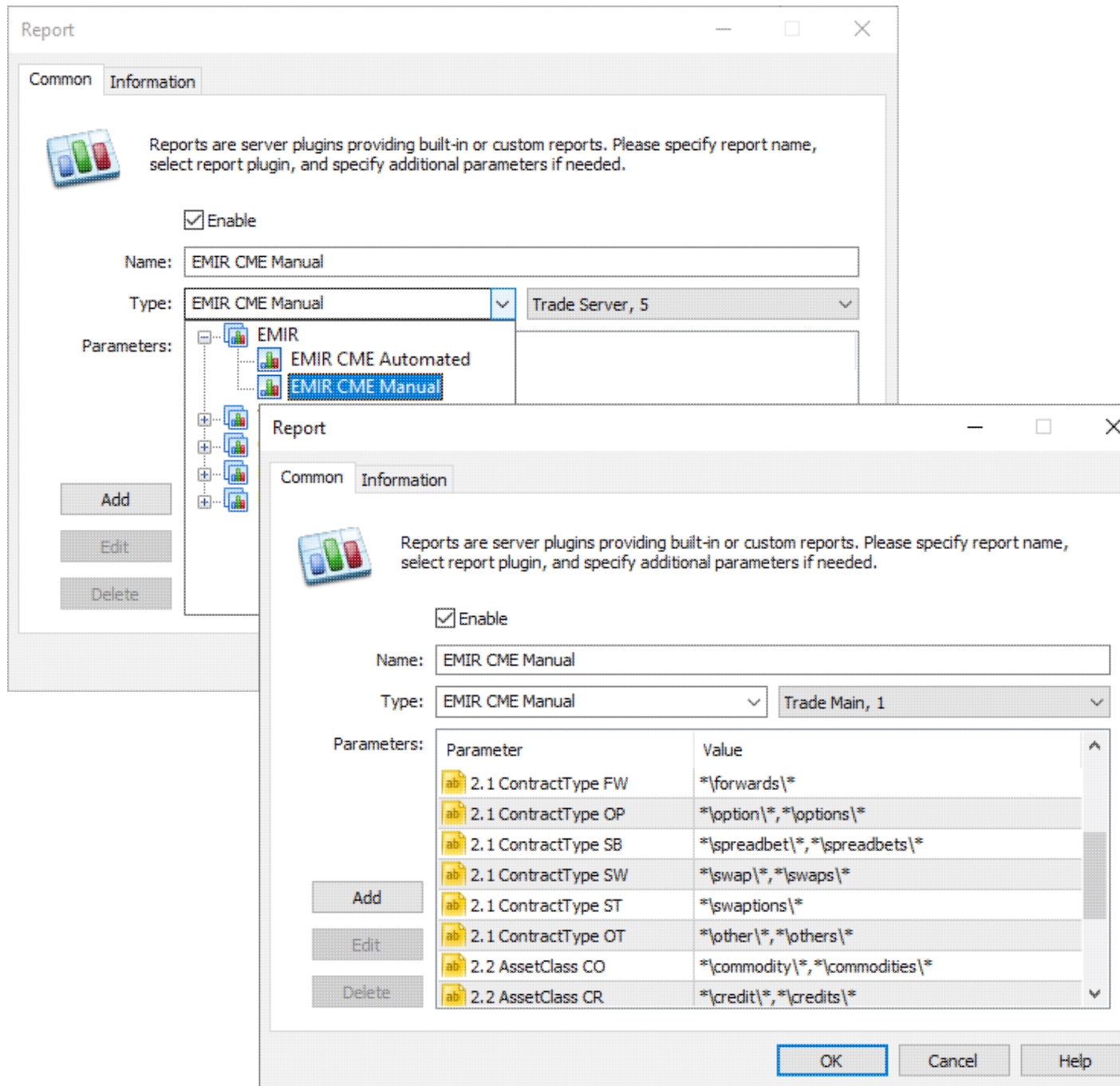
This module enables manual report generation for any time frame. This feature can be useful if the automated report was not generated for any of the previous days, for some reason. Set the required time interval, request report generation from the Manager terminal, export the received data to file and upload it to the repository manually.

Unlike the Automated report, this mode does not access the generated files and does not depend on whether the server plugin is installed or not. The data is requested directly from the server database.

Installation

To install the reports, open the Reports section of MetaTrader 5 Administrator. After selecting a server for report installation, **+Add**.

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Once the configuration is added, select the required report in the Type field. No additional parameters are used for EMIR CME Automatic.

The set of parameters for the EMIR CME Manual almost coincides with plugin parameters. These values are used to fill the report.

- **Broker LEI** broker's Legal Entity Identifier (LEI), consists of a 20-digit alphanumeric code. The value is used in various fields of the report.
- **1.6 CorporateSector** the sector in which the company operates. For example, F an investment company, C a credit company, etc. The value is directly substituted in the appropriate report field. The default value is F.
- **1.7 FinancialOrNonFinancialCounterParty** type of the reporting company. For example, F financial counterparty, N non-financial counterparty, etc. The value is directly substituted in the appropriate report field. The default value is F.
- **1.9 ReportingEntityLEI** Legal Entity Identifier (LEI) of the reporting entity. The value is directly substituted in the appropriate report field.
- **2.1 ContractType CD** symbol path mask, according to which it will be assigned to type "CD" (Financial contracts for difference). It is used to fill the "2.1 ContractType" field in the report. For example, if *\cfd* is specified in the parameter, then the "CD" value will be set in the "2.1 ContractType" field for all symbols, having \cfd\ in their path. For example, you can use this mask if contracts for difference in your platform are created in the CFD\, CME\CFD\, Real\CFD\Main\ and similar sections.
- **2.1 ContractType FU** the same for the "FU" type (Futures).
- **2.1 ContractType FW** the same for the "FW" type (Forwards).
- **2.1 ContractType OP** the same for the "OP" type (Option).
- **2.1 ContractType SB** the same for the "SB" type (Spreadbet).
- **2.1 ContractType SW** the same for the "SW" type (Swap).
- **2.1 ContractType ST** the same for the "ST" type (Swaption).
- **2.1 ContractType OT** the same for the "OT" (Other).
- **2.2 AssetClass CO** symbol path mask, according to which it will be assigned to "CO" assets class (Commodity and emission allowances). It is used to fill the "2.2 AssetClass" field in the report. For example, if *\commodity* is specified in the parameter, then the "CO" value will be set in the "2.2 AssetClass" field for all symbols, having \commodity\ in their path. For example, you can use this mask if contracts for difference in your platform are created in the Commodity\, CME\Commodity\, Real\Commodity\Main\ and similar sections.
- **2.2 AssetClass CR** the same for the "CR" class (Credit).
- **2.2 AssetClass CU** the same for the "CU" class (Currency).
- **2.2 AssetClass EQ** the same for the "EQ" class (Equity).
- **2.2 AssetClass IR** the same for the "IR" class (Interest Rate).
- **2.38 Intragroup** determines whether both counterparts of the deal belong to the same company. One of the two values : Y belong to one company, N do not belong. The value is directly substituted in the appropriate report field. The default value is N.

The Symbols and Groups parameters are not used here, since filtering by groups and symbols can be set when requesting a report from the Manager terminal.

Viewing the Reports

MetaTrader 5 trading platform

To view the report request, use MetaTrader 5 Manager. Simply select the necessary report in the Server Reports section to retrieve data from the server.

For report operation details, please visit the appropriate section of MetaTrader 5 Manager documentation.

The screenshot displays the MetaTrader 5 Manager application window. The title bar reads "1863 - MetaQuotes - John Smith". The menu bar includes "File", "Edit", "View", "Tools", "Window", and "Help". Below the menu bar is a toolbar with icons for "New Account", "Connect", "Automation", and others. The left sidebar, titled "Navigator", shows a tree view of the application's structure. The "Server Reports" folder is expanded, and the "EMIR CME" report is selected. The main panel displays the "EMIR CME" report for the date range "2019.09.23 - 2019.09.25". The report is a table with the following columns: "Action", "Level", "ReportingTimestamp", "NonReportingPartyIDType", "NonReportingPartyLEI", and "Cleared". The table contains 15 rows of data, all with "Action" set to "N" and "Level" set to "T". The "ReportingTimestamp" for all rows is "2019-09-25T14:23:29Z". The "NonReportingPartyIDType" is "CLC" for all rows. The "NonReportingPartyLEI" values are: 5621, 5872, 6590, 6969, 5446, 7094, 5709, 6797, 6832, 6611, 6711, 5235, 7152, 6018, and 5957. The "Cleared" column is empty for all rows.

Action	Level	ReportingTimestamp	NonReportingPartyIDType	NonReportingPartyLEI	Cleared
N	T	2019-09-25T14:23:29Z	CLC	5621	N
N	T	2019-09-25T14:23:29Z	CLC	5872	N
N	T	2019-09-25T14:23:29Z	CLC	6590	N
N	T	2019-09-25T14:23:29Z	CLC	6969	N
N	T	2019-09-25T14:23:29Z	CLC	5446	N
N	T	2019-09-25T14:23:29Z	CLC	7094	N
N	T	2019-09-25T14:23:29Z	CLC	5709	N
N	T	2019-09-25T14:23:29Z	CLC	6797	N
N	T	2019-09-25T14:23:29Z	CLC	6832	N
N	T	2019-09-25T14:23:29Z	CLC	6611	N
N	T	2019-09-25T14:23:29Z	CLC	6711	N
N	T	2019-09-25T14:23:29Z	CLC	5235	N
N	T	2019-09-25T14:23:29Z	CLC	7152	N
N	T	2019-09-25T14:23:29Z	CLC	6018	N
N	T	2019-09-25T14:23:29Z	CLC	5957	N

For Help, press F1

The plugin access to server data is not restricted. To generate files, the plugin can use any available information on groups and symbols. In contrast, the EMIR CME Manual report has restricted access to data in accordance with the manager account permissions and settings of the group to which this account belongs.

Sending Reports to Repository

To send the reports to the repository, the corresponding files need to be uploaded to a special SFTP server. The EMIR Reports plugin is able to send reports to the server automatically. For this, you should specify report sending time and SFTP server connection details, in the plugin settings.

Contact an appropriate repository to get SFTP access (address, login and password). For example, server access specifications for the CME Group are available at <https://www.cmegroup.com/trading/global-repository-services/files/ftp-connectivity.pdf>.

The plugin create file reports to be sent, in the [trade server installation directory]\Reports\EMIR.Reports\CME\Outbox. If sending to repository is not activated in plugin settings by the time of report generation (the "SFTP Host" field is empty), the files will be simply saved in this directory. Once you set the required settings, all the saved files will be uploaded to the repository at the next report sending time.

If sending to repository fails for any reason, its file is saved under [trade server installation directory]\Reports\EMIR.Reports\CME\Outbox. The next day the plugin will retry to send the file at the scheduled time.

Report Data

Some fields in the report are always empty. Data for this fields may not be available in MetaTrader 5 or may be optional. Only the fields marked in the specification as "FX only" or "All" are implemented in the report. Fields "Commodities only", "Credit only" and "Rates only" are not available, because they are not applicable in MetaTrader 5.

For the detailed description of the reports please visit the [ESMA](#) site. The general description is provided below for more convenience.

Field	Description	Value
Comment	Description of errors that occurred during the formation of this row (or empty if there are no errors)	A list of errors or an empty value if there are no errors
Action	The action type performed by the transaction	N new operation

Field	Description	Value
Level	Indicates the object which the action refers to.	T – trade, P – position
ReportingTimestamp	Report generation date and time	Report generation date and time
ReportingPartyLEI	LEI of the reporting company	The value of the "Broker LEI" parameter from the <u>plugin settings</u> ; if the parameter is not filled, an error will be generated and the field will be blank
NonReportingPartyIDType	Code for the external counterparty ID type	For the rows describing the external transaction part the value "LEI" is always used. For the rows describing the transaction on the MetaTrader 5 side: • The value "LEI" if the broker is the client who performed the transaction is <u>marked</u> as "Corporate" or "Fund" • Value "CLC" for all other client types
NonReportingPartyLEI	LEI of the second party of the transaction	For the rows describing the external deal part, liquidity provider's LEI is specified in accordance with the <u>gateways settings</u>. For the rows describing the transaction on the MetaTrader 5 side the value depends in the NonReportingPartyIDType field: If "LEI" is specified in the NonReportingPartyIDType field, the value of "LEI number for EMIR report" from the <u>client record</u> will be used

Field	Description	Value
		If "CLC" is specified in "NonReportingPartyIDType", the value is determined as follows (in order of priority): • External client ID from <u>trade account settings</u>. • Internal client ID, which is equal to the <u>client record</u> ID in MetaTrader 5. • Trading account ID, i.e. <u>account login/number</u>. If LEI is specified in the NonReportingPartyIDType field, but NonReportingPartyLEI field value cannot be determined, and error will be generated and the field will be left blank. If gateway configuration via which transaction was executed is not found for an external deal (for example the configuration was deleted), an error will be generated and the field will be left blank.
NonReportingPartyCountry	Country of the second party of the transaction	For the rows describing the external part of the transaction, the "EMIR Report Country" parameter value from the <u>gateway settings</u> is specified. For the rows describing the transaction on the MetaTrader 5 side, the value is determined as follows: • For legal entities from the "Registration country" field of the <u>client record</u>. • For individuals from the "Country" field of the <u>client record</u>.

Field	Description	Value
		If the appropriate gateway or client record parameter is not filled, an error will be generated and the field will be left blank. If gateway configuration via which the transaction was executed is not found for an external deal (for example the configuration was deleted), an error will be generated and the field will be left blank.
CorporateSector	The sector in which the company operates	The "1.6 CorporateSector" parameter value from the <u>plugin settings</u> . If the parameter is not filled, an error will be generated and the field will be left blank
FinancialOrNonFinancialCounterParty	Type of reporting company	The "1.7 FinancialOrNonFinancialCounterParty" parameter value from <u>plugin settings</u> . If the parameter is not filled, an error will be generated and the field will be left blank
ReportingBrokerLEI	Broker's LEI	Not filled
ReportingAssetManagerLEI	Asset manager's LEI	Not filled
ReportingEntityLEI	LEI of the reporting broker, if the report is submitted via a third party	The "1.9 ReportingEntityLEI" parameter value from the <u>plugin settings</u> . If the parameter is not filled, an error will be generated and the field will be left blank
ClearingMemberLEI	Clearing member's LEI	Not filled
BeneficiaryIDType	Code for specifying the identifier of the beneficiary: LEI or CLC (client code)	"LEI" is always used
BeneficiaryLEI	Transaction beneficiary's	The value of the "Broker LEI" parameter from the <u>plugin settings</u>

MetaTrader 5 trading platform

Field	Description	Value
	LEI	the parameter is not filled, an error will be generated and the field will be blank
TradingCapacity	How the transaction was performed: personally (P) or via an agent (A)	The value "A" is always used
CountryPartySide	Indicates who the counterparty is: buyer (B) or seller (S)	The "B" value for buy deals and "S" for sell deals
CommercialOrTreasuryLink	Determines whether the object is associated with the financial activity of the counterparty	Not filled
ClearingThreshold	Determines whether the counterparty has exceeded the clearing threshold	Not filled
MTM	The last price of the transaction asset at the trading session close time, multiplied by deal size	The last price as at the report generation time, which is equal to the "Report Time" from the <u>plugin settings</u> . If the last price is unavailable for any reason, the average price between ask and bid is used
MTMCurrency	Currency in which the value of the "MTM" field is indicated	The quote currency of the transaction symbol
ValuationDateTime	The date for which the value of the "MTM" field is indicated	Report generation date, which is equal to the "Report Time" from the <u>plugin settings</u>

Field	Description	Value
ValuationMethod	"MTM" valuation method: Mark-to-market (M), Mark-to-model (O), CCP's valuation (C)	The value "M" is always used
Collateralizaion	Determines whether a collateral agreement is concluded between the parties	Not filled
CollateralPortfolio	Determines if the collateral is a portfolio	Not filled
CollateralPortfolioCode	Portfolio code	Not filled
ProductType	Financial product type	Filled in accordance with the deal <u>symbol type</u> : • Forex, Forex No Leverage • FXForward • Futures, Exchange Futures, Exchange FORTS Futures • CFD, CFD Index, CFD Leverage • CFD • Exchange Options, Exchange Margin Options • Option
ExchangeTraded	The exchange in which the product is traded	Not filled
ContractType	Contract type	Filled in accordance with the plugin <u>ContractType* settings</u> . If no mask matches a symbol, or more than one mask match, an error will be generated and the field will be left blank
AssetClass	Asset class	Filled in accordance with the plugin <u>AssetClass* settings</u> . If no mask matches a symbol, or more than one mask match, an error will be generated

Field	Description	Value
		and the field will be left blank
ProductClassificationType	Product classification type: CFI (C) or UPI (U)	The value "C" is always used
ProductClassification	Product classification in accordance with the <u>ISO 10692</u> standard (for CFI)	Filled using the "CFI" field from the relevant <u>financial instrument settings</u>
ProductIdentificationType	Product identification type	Not filled
ProductIdentification	Product ID	Not filled
UnderlyingType	Underlying asset type	Not filled
Underlying	Underlying asset identifier	Not filled
NotionalCurrency1	Base currency	Base currency of the transaction symbol
NotionalCurrency2	Quote currency	Quote currency of the transaction symbol
SettlementCurrency	Settlement currency	Quote currency of the transaction symbol
UTI	Unique Transaction Identifier	Consists of three parts: [E02][LEI][Transaction ID]: • E02 the first part always consists of these three characters • LEI for the deals forwarded to an external system it is the liquidity provider's LEI from the <u>gateway settings</u>. For B-book deals broker's LEI from the "Broker LEI" parameter in <u>gateway settings</u> • Deal ID for A-book it is the <u>ID In the external system</u>, for B-book it is <u>deal ticket</u> in MetaTrader 5

MetaTrader 5 trading platform

Field	Description	Value
		Deals related to one position have same UTI. The above formula uses ID of the deal which opened the position. Positions have the same UTI as the deals by which the positions were opened.
ReportTrackingNumber	Report group identifier	Not filled
ComplexTradeComponentID	Internal identifier for report grouping	Not filled
ExecutionVenue	Execution venue code	For the rows describing the external part of the transaction, it is filled in accordance with the "EMIR Report MIC" parameter from <u>gateway settings</u> . If not specified, the "XXXX" value is used. For the rows describing the transaction on the MetaTrader 5 side "XXXX" is always used
Compression	Determines whether the operation is the result of portfolio compression	The value "N" is always used (it is
PriceOrRate	Derivative price minus fees and accumulate profit	Transaction price
PriceNotation	Units in which the price is indicated	The value "U" (units) is always used
PriceCurrency	Currency in which the price is indicated	Quote currency of the transaction
NotionalAmount	Amount of product for which the price is specified	Volume of a deal

MetaTrader 5 trading platform

Field	Description	Value
PriceMultiplier	Contract size	Transaction contract size
Quantity	The volume of operations in lots	The volume of operations in lots
UpfrontFee	Upfront payment amount	Not filled
SettlementMethod	Settlement method: monetary (C), physical (P), optional (O), U (cash)	The value "U" is always used
ExecutionDateTime	Contract execution date	Transaction date and time
StartDate	Contract effective date	Transaction date and time
MaturityDate	Contract maturity date	Not filled
TerminationDate	Contract termination date	Not filled
SettlementDate	Settlement date	Not filled
MasterAgreementType	Link to the master agreement	Not filled
MasterAgreementVersion	Year of the master agreement	Not filled
ConfirmDateTime	Contract confirmation date	Transaction date and time
ConfirmType	Contract confirmation type: non-electronic (Y), not-conformed (N), electronic (E)	The value "E" is always used

MetaTrader 5 trading platform

Field	Description	Value
ClearingObligation	Indicates the need of clearing for the operation	The value "N" is always used (no n
Cleared	Indicates if clearing was performed	The value "N" is always used (was performed)
ClearingDateTime	Clearing date	Not filled
ClearingVenueLEI	Clearing organization LEI	Not filled
Intragroup	Determines whether both counterparts of the deal belong to the same company	"2.38 Intragroup" parameter value <u>plugin settings</u>
FinalSettleCcy	Final settlement currency	Quote currency of the transaction
ExchangeRate	Price as on the date of the transaction	Transaction price
ForwardExchangeRate	Forward price	Transaction price
ExchangeRateBasis	Exchanged currencies	The symbol, for which the deal is executed
OptionType	Type of option	Not filled
OptionExerciseType	Option exercise type	Not filled
OptionStrikePrice	The strike price of the option	Not filled
StrikePriceNotation	Units in which the option strike price is indicated	Not filled
UnderlyingMaturityDate	Option expiration date	Not filled

MetaTrader 5 Trading Platform / Platform Setup / Reports / Fund Overview

Fund Overview

If you are using the MetaTrader 5 platform for the operation of an investment fund, this report will show the key fund performance characteristics, including Assets Under Management and Net Asset Value, as well as information about fund investors and managers.

ENTLOW - Entropy Player Low Risk

2018.01.05 - 2019.06.05



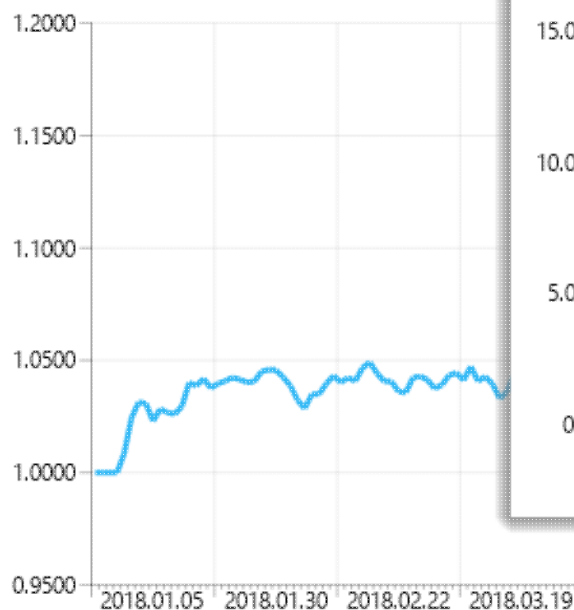
Fund Value, USD

Shareholders

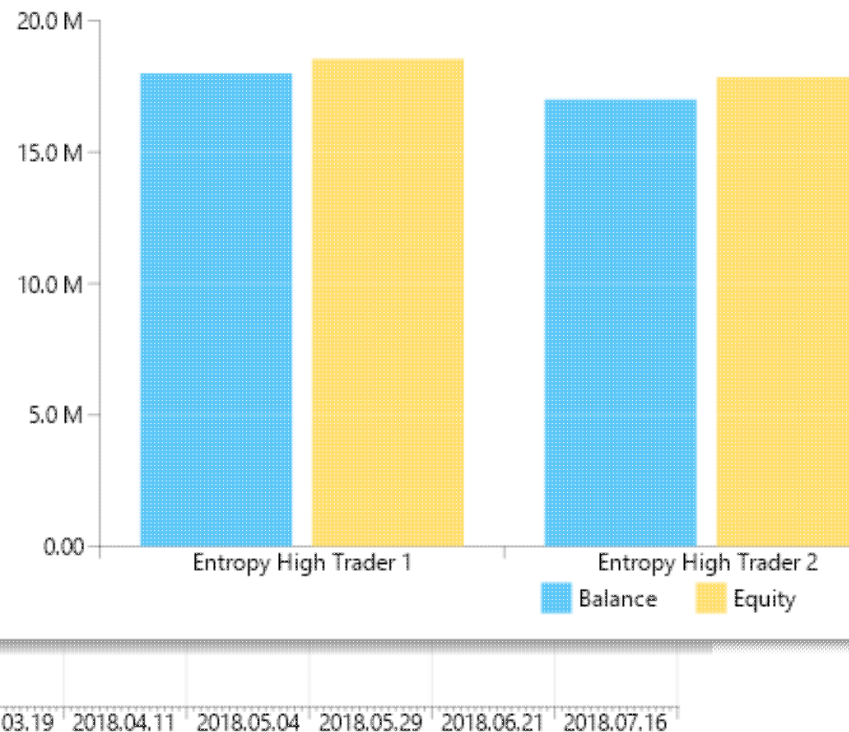
44.9 M

69

Fund Value



Trading Accounts Value, USD



Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Period** starting and ending date of the period, for which the report will be generated.

Report Data

The report provides the following information:

- **Fund Value** Asset Under Management (AUM).
- **Shareholders** the number of the fund shareholders.
- **Fund value (diagram)** Net Asset Value (NAV) dynamics.
- **Trading Accounts Value** current balance and equity on the accounts of fund managers.
- **Investors Volume** the share of each shareholder. The size of the corresponding position (in lots) for the fund symbol is displayed on each shareholder's account.

Setup in MetaTrader 5 Administrator

The following parameters can be specified in the report configuration in MetaTrader 5 Administrator:

- **Fund name** the name of the fund for which the report is generated. The name can be obtained from the appropriate section of MetaTrader 5 Administrator settings.
- **Currency** the currency in which the amounts are displayed in the report. Conversion to the specified currency is performed in accordance with the current quotes in the platform. For example, EURUSD quotes are used to convert an amount in EUR to USD.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Risk Appetite

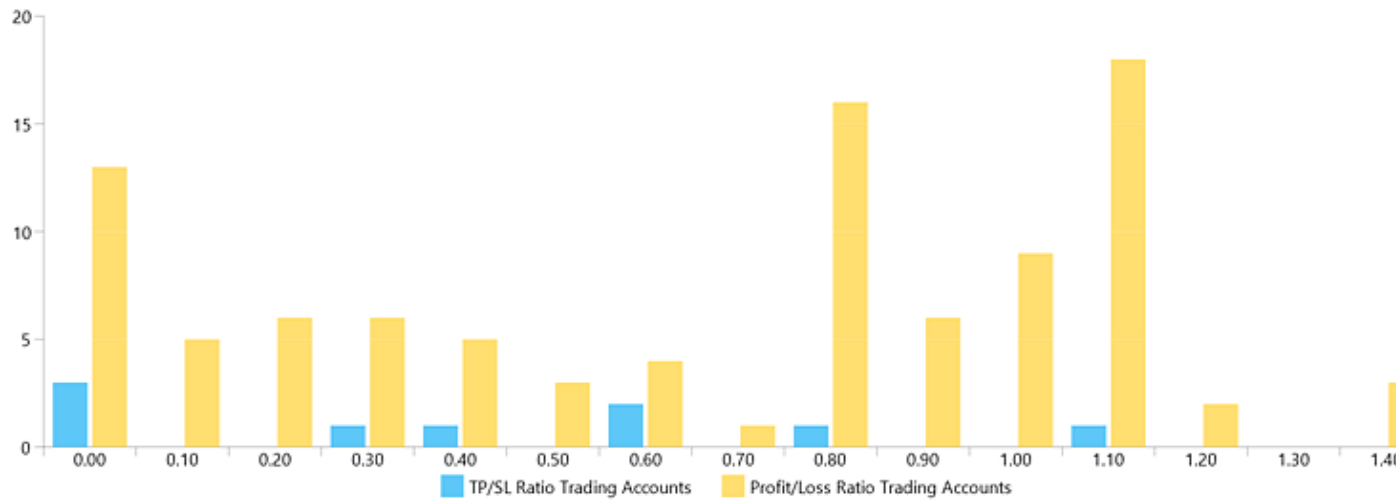
Risk Appetite

The "Risk Appetite" report reflects the risk level accepted by traders.

Risk Appetite

All Lead Source ▾ All Lead Campaign ▾ * ▾ 2020.06.01 - 2020.11.25 📅

TP/SL and Profit/Loss Ratio Distribution



Top-100 Trading Accounts by Profit/Loss Ratio

Login	Name	Registration Time	Stop Loss, %	Take Profit, %	TP/SL Ratio	Loss, %	Profit, %	Profit/Loss R
1818186	Tina Steen	2020.09.28 13:37:14				0.00	0.64	25
1818678	Prince Asante	2020.11.17 12:43:52		0.05		0.00	0.04	4
1817514	Waliet Albertyn	2020.07.09 15:53:09	0.14	2.80	0.05	0.21	0.81	3
1817661	Keith Jacobs	2020.07.26 05:00:00	0.18			0.04	0.13	3
1817515	Tina Steen	2020.07.09 17:24:04		0.00		0.08	1.06	1
1817992	John Smith	2020.09.01 13:05:57	0.39	0.16	2.50	0.28	0.59	2
1817514	Ashish Agarwal	2020.07.09 15:53:09	0.14	2.80	0.05	0.21	0.81	3
1817661	Dotun Atoyebi	2020.07.26 05:00:00	0.18			0.04	0.13	3

Report Data

The report consists of a chart showing the ratio of total profit to total loss and the ratio of Take Profit to Stop Loss, along with a table with detailed variables for each account:

- **Stop Loss, %** the average percentile between the position open price and the Stop Loss level.
- **Take Profit, %** the average percentile between the position open price and the Take Profit level.
- **Loss, %** the average percentile between the position open price and the losing position close price.
- **Profit, %** the average percentile between the position open price and the winning position close price.
- **TP/SL Ratio** the ratio between the earlier mentioned 'Take Profit, %' and 'Stop Loss, %' values.
- **Profit/Loss Ratio** the ratio between 'Profit, %' and 'Loss, %'.

The y-axis shows the number of accounts. Thus, it reflects how many accounts have certain values of profit-to-loss and Take Profit-to-Stop Loss ratio.

The table contains 100 accounts with the highest profit-to-loss ratio.

Filters

Data requested for the report can be filtered. Specify the following parameters before requesting a report in the manager terminal:

- **Lead Source** the website from which the client came.
- **Lead Campaign** the name of an advertising campaign by which the client was attracted.
- **Groups** the groups containing the accounts, on which the report must be created. You can specify one or several accounts separating them by commas.
- **Period** starting and ending date of the period, for which the report will be generated. The filter sets the time frame for which deals are analyzed.

Setup in MetaTrader 5 Administrator

The following parameters can be specified in the report configuration in MetaTrader 5 Administrator:

- **Top Count** the number of data series which will be shown in the report: profit-to-risk ratio and Take Profit-to-Stop Loss ratio. The default is 10. Data series with the highest values are selected for the report, while all other series are summed up and are displayed as the "Other" variable.
- **Cache Rebuild Period, sec** the maximum time by which data in the report can lag. This is done in seconds. 30 on default. If the latest data in the server cache is within the specified interval from the report request time, the data cache will not be refreshed. It means that the data in the report will not be refreshed in the case of frequent repeated requests. This enables fast generation of reports while avoiding extra server load.

- **Distribution Percentile, %** parameter indicating the data display percentile. The parameter cuts off the specified percentage of low chart values. The value is set to 80 by default. This means that only 20% of all values are shown in the report, which include the largest profit-to-loss and Take Profit-to-Stop Loss ratios.
- **Minimum Deals** parameter for the minimum number of deals required for an account to be included in the report. Only out deals are counted. The default value is 10. The ratio calculation is considered irrelevant for less deals and the appropriate value is excluded from the report.

MetaTrader 5 Trading Platform / Platform Setup / Reports / Fast Profit Deals

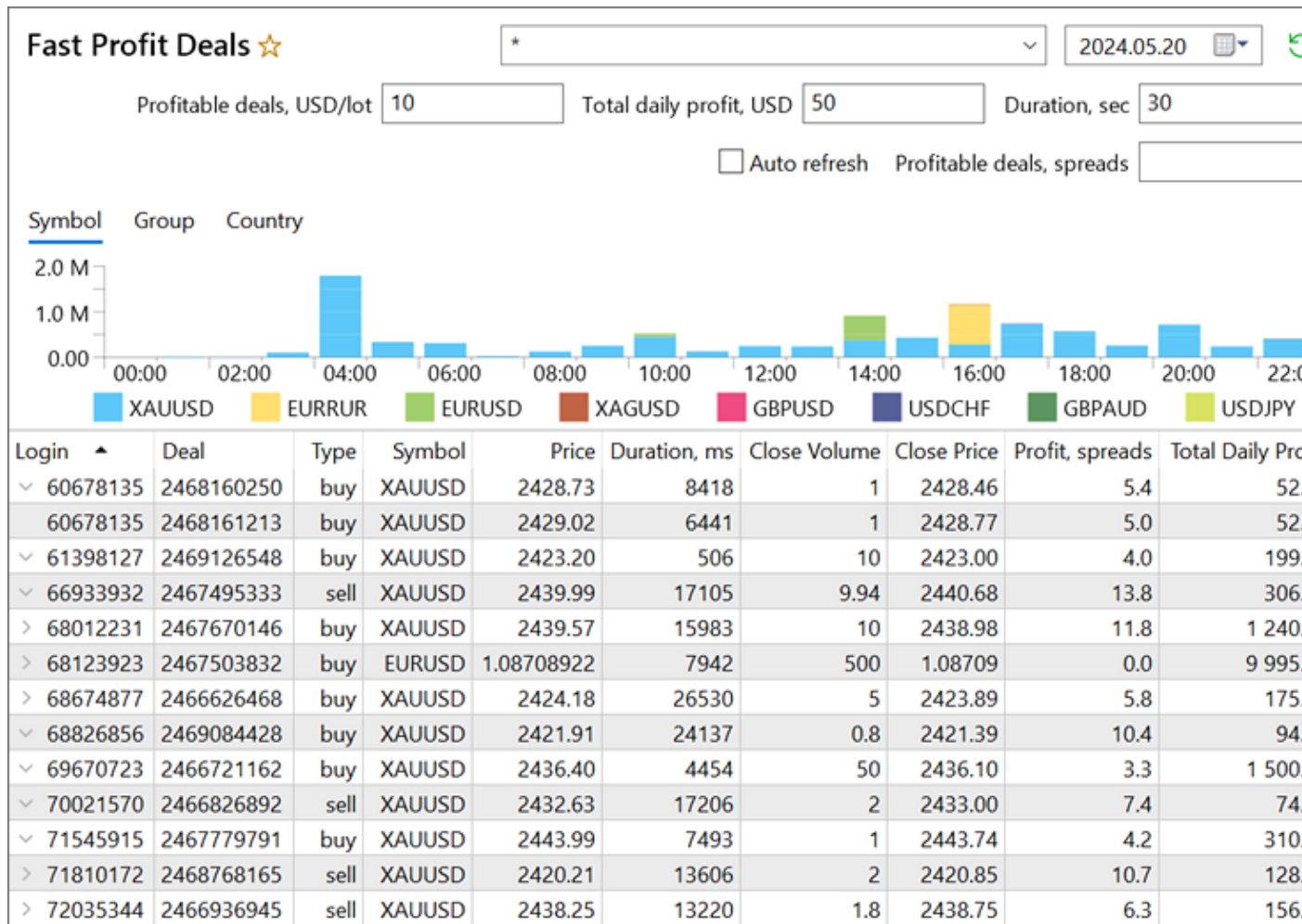
Fast Profit Deals

This report is designed to detect traders exploiting arbitrage opportunities through quote delays. Due to various technical limitations, such as delays on the data provider side and network constraints, there may be slight variations in the speed at which quotes are received on brokerage servers. Quotes from one server may react to important news slightly faster than those from another. Traders may attempt to exploit this difference by finding a faster source, observing price movement, and quickly opening a trade in a favorable direction. This effect can be particularly significant during important economic news releases, when prices may move substantially in a short timeframe.

Typically, traders in such scenarios follow a certain pattern: upon the news release, they execute a short-term trade lasting no more than a minute, aiming to capture profit within a few spreads.

The 'Fast Profit Deals' report will assist in detecting such behavior. It shows all deals with a profit higher than the specified profit value and a duration shorter than the specified duration value. Specify the desired parameters in the report settings:

MetaTrader 5 trading platform



Duration is calculated in seconds between consecutive entry and exit deals based on the nearest previous entry preceding the closing deal within the same position. Profit is defined as the average amount in USD per position lot. This setup also allows determining those traders who accumulate large volumes/sums through small trades.

Additionally, information in the report can be filtered by the following parameters:

- **Profitable trades, spreads** profit from the deals in the number of spreads of the corresponding instrument.
- **Total daily profit** total profit on the trader's account for the selected day.
- **Groups** account groups for which the report is generated. You can specify one or several accounts separating them by commas.
- **Period** report day. A pair of deals is only included in the report only if both the entry and exit deals fall on the specified day.

Profit Graph

The graph displays the profit for all deals included in the report, broken down by hour. It will allow you to trace the connection of suspicious deals with any events. Data can be analyzed from three perspectives: by symbol, group and country.

To filter the data in the table, select the desired time interval on the chart.

List of Deals

Deals in the table are grouped by account. If multiple deals from the same account are included in the report, they will be shown together as a submenu that can be collapsed and expanded using arrows in the "Login" field.

The following information is available for the deals:

- **Login** account number on which the deal was performed.
- **Name** account holder's name.
- **Group** group to which the account belongs.
- **Country** client's country of residence.
- **Account comment** comment to the trading account.
- **Deal** deal ticket number (a unique identifier).
- **ID** ID of a deal in an external trading system.
- **Order** ticket of the order, as a result of which the deal has been performed.
- **Position** position ticket (unique number) to which the deal belongs.
- **Time** time of the position entry deal. Time is specified in the form of YYYY.MM.DD HH:MM (year.month.day hour:minute).
- **Type** trading operation type: Buy or Sell.
- **Symbol** financial instrument for which the deal was performed.
- **Volume** total volume of the initial position in lots.
- **Price** price at which the entry deal was executed.
- **S/L** Stop Loss level.
- **T/P** Take Profit level.

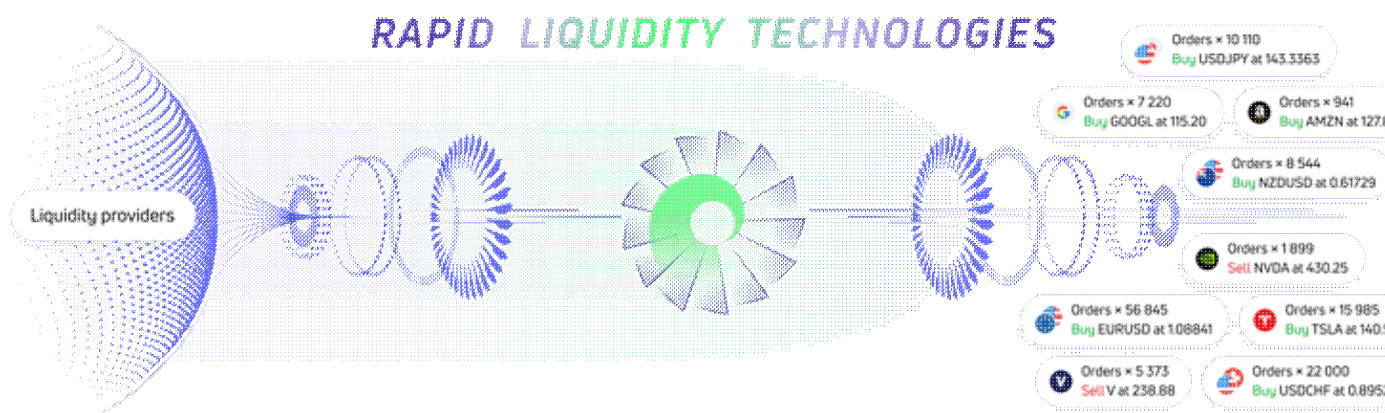
MetaTrader 5 trading platform

- **Close Time** time of the position exit deal. Time is specified in the form of YYYY.MM.DD HH:MM (year.month.day hour:minute).
- **Close Volume** exit deal volume in lots.
- **Close Price** price at which the exit deal was executed.
- **Duration** deal duration in milliseconds. Calculated as the difference between the Close Time and Time parameters.
- **Reason** reason for executing the deal.
- **Commission** commission charged for the deal execution.
- **Fee** fee charged for the deal execution.
- **Swap** swap amount.
- **Profit** financial result of the deal in the account deposit currency.
- **Currency** account deposit currency.
- **Profit, USD** financial result of the deal in USD. The amount is converted at the exchange rate of the deposit currency to USD at the time of the deal.
- **Profit, spreads** financial result of the deal in the number of spreads of the corresponding instrument. The value is calculated using the following formula: (Closing deal price - Position opening price) \ (Market Ask from the deal - Market Bid from the deal).
- **Total Daily Profit, USD** total profit on the trader's account for the selected day.
- **Comment** deal comment.

MetaTrader 5 Trading Platform / Platform Setup / Ultency

Ultency Matching Engine: Built-in liquidity aggregation and order execution

Ultency Matching Engine is an ultra-low latency solution for liquidity aggregation, trade order matching, and risk management. This technological platform enables reliable connection to multiple providers simultaneously, consolidating their prices and execution to deliver the best trading conditions for traders. The system is highly customizable to meet the unique requirements of each broker, offering a comprehensive suite of risk management and reporting tools.



Ultency for Brokers

Ultency offers unique advantages for brokers:

- Access multiple liquidity providers through an intuitive gallery with detailed descriptions of offerings and connect in a few clicks.
- Choose the best prices from various sources to improve order execution for traders.
- Lower spreads by accessing multiple liquidity providers. This makes trading more cost-efficient for clients and helps attract new traders.
- Save on infrastructure costs by avoiding individual connections to each provider. Ultency reduces development and maintenance costs, simplifying scalability.
- Access various order types and a wide range of risk management tools. Select optimal execution strategies based on market volatility and client needs.
- Efficiently process orders during news releases and fast market movements. The platform tracks prices from multiple providers, minimizing delays or price gaps.

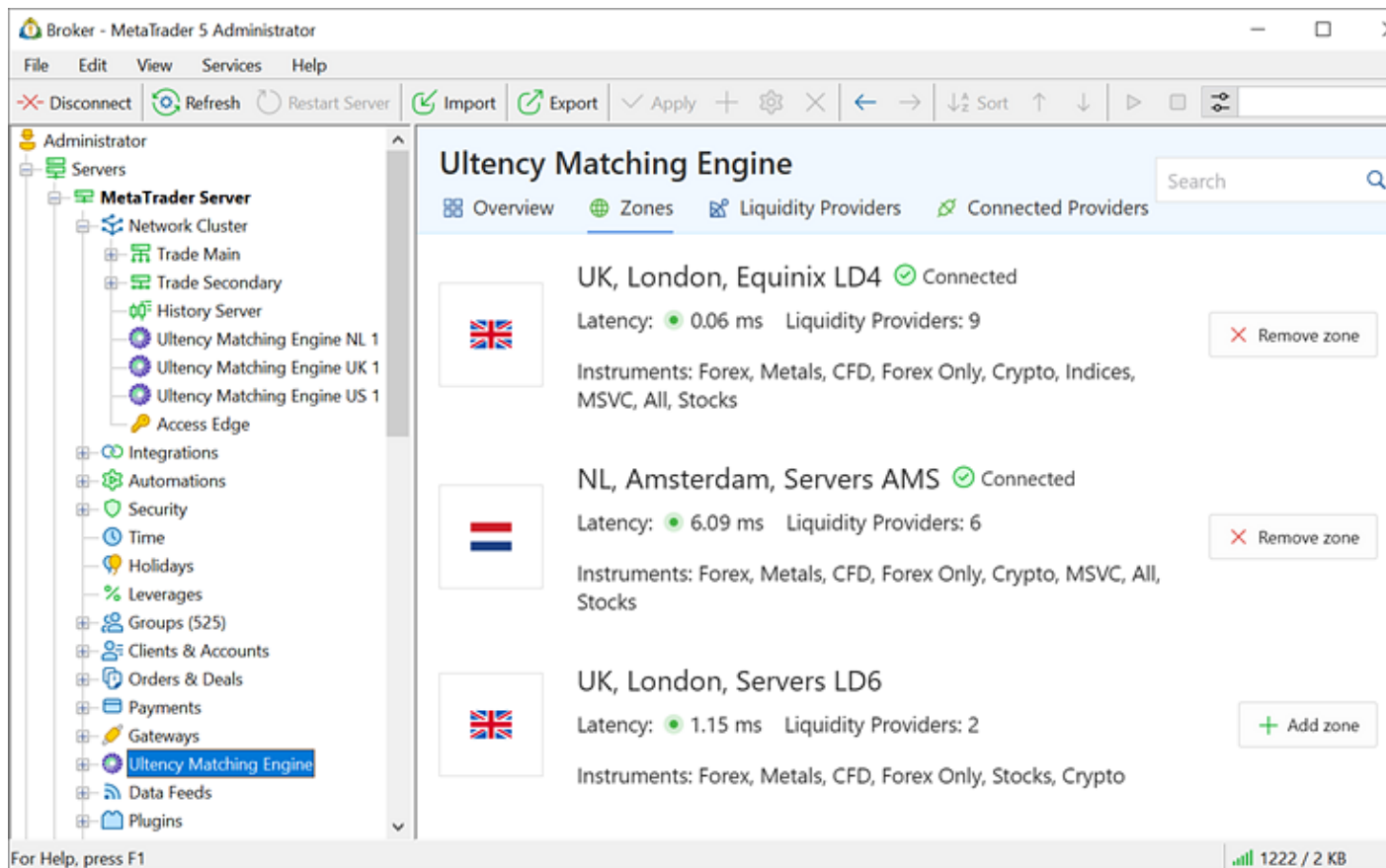
Ultency Features

Ultency is a native solution for the MetaTrader 5 platform::

MetaTrader 5 trading platform

- The engine uses highly efficient internal data transfer protocols between cluster components, significantly speeding up order processing and enhancing system resilience.
- The main instrument and trading session parameters are configured once in MetaTrader 5, eliminating the need to monitor their compliance in multiple systems. All order execution, client and configuration change logs are stored in one place, accessible only to the broker.

Each broker gets a dedicated platform that processes only their orders. You can choose from multiple hosting locations depending on which liquidity providers you want to connect to and the markets you offer. For example, to offer access to various exchanges, you can use a server in London for Forex providers and another one in New York for NYSE providers.

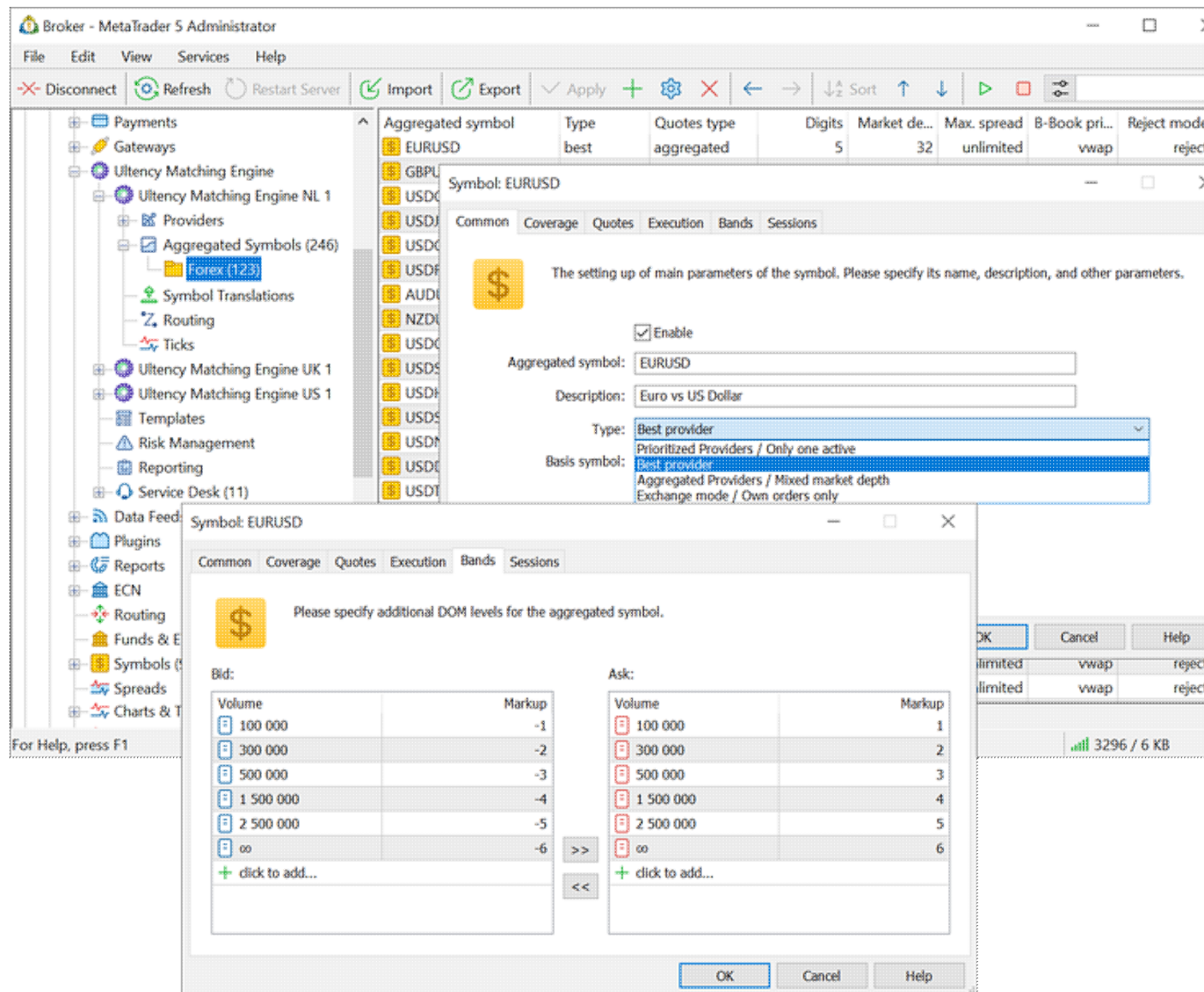


Highly Configurable

Ultency allows detailed configuration of price aggregation and order execution parameters:

- For each instrument, you can specify individual liquidity providers to aggregate prices for clients, configure custom order book rules, spread adjustments, and execution parameters.
- For each client/group, you can assign liquidity providers to execute their orders, as well as individual markups. You can manage order routing between A-Book, B-Book, and C-Book executions (partial hedge, overhedge, reverse hedge).

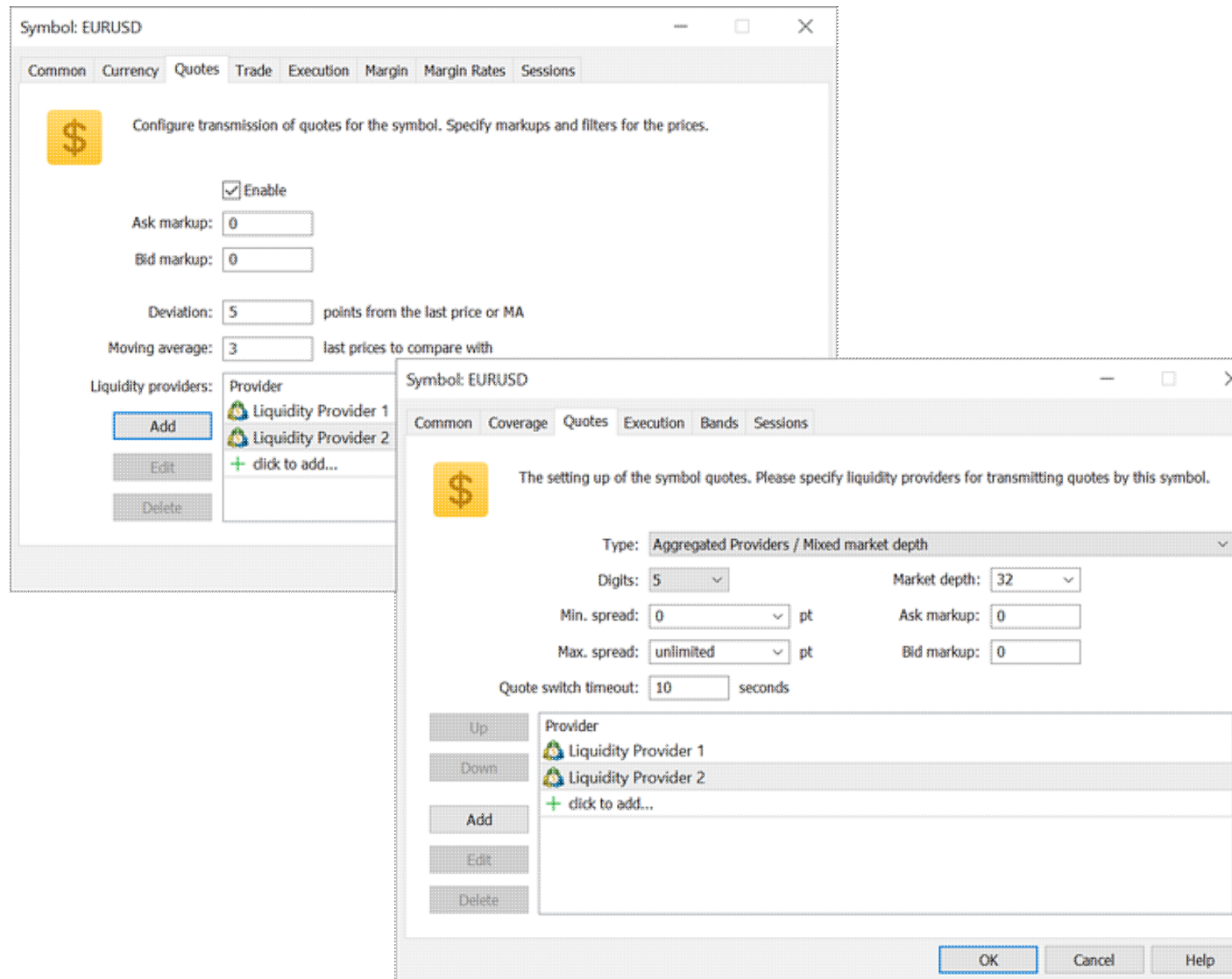
With the flexible template system, you can save markups, price feed configurations, routing rules, and order execution settings. Create templates for different conditions and apply them automatically based on time, market events, or risk management rules.



Protection and Risk Management

Intelligent filtering protects against incorrect prices. Each price entering the system is validated and compared with prices from other providers. Based on the validation results, the system decides whether to accept or reject the quote.

The system supports automatic risk management rules, including coverage based on position volume, instrument, currency, turnover, profit, and more. Rules can be set for individual clients, client groups, or the broker's entire portfolio.



Monitoring and Reporting

Ultency supports special coverage accounts for monitoring key metrics, including funds, profit/loss, total positions, and more. The data can be analyzed by liquidity providers, clients, client groups, and the entire portfolio, and used to configure automatic risk management rules.

A dedicated dashboard allows dealers to monitor performance parameters in real-time, including the volume of orders routed to providers, risk management rules, and the broker's portfolio.

The built-in reports enable the analysis of client trading results, A-Book execution efficiency, earnings from markups, commissions, and swaps. Reports can be segmented by instruments and client groups, allowing you to timely identify toxic clients and take appropriate measures.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Connection

Connecting Ultency

Ultency is a ready-to-use solution. You don't need to rent your own servers or install additional software. By connecting to Ultency, you receive a dedicated platform, tailored specifically for you, which will aggregate prices from your selected liquidity providers and process only your orders.

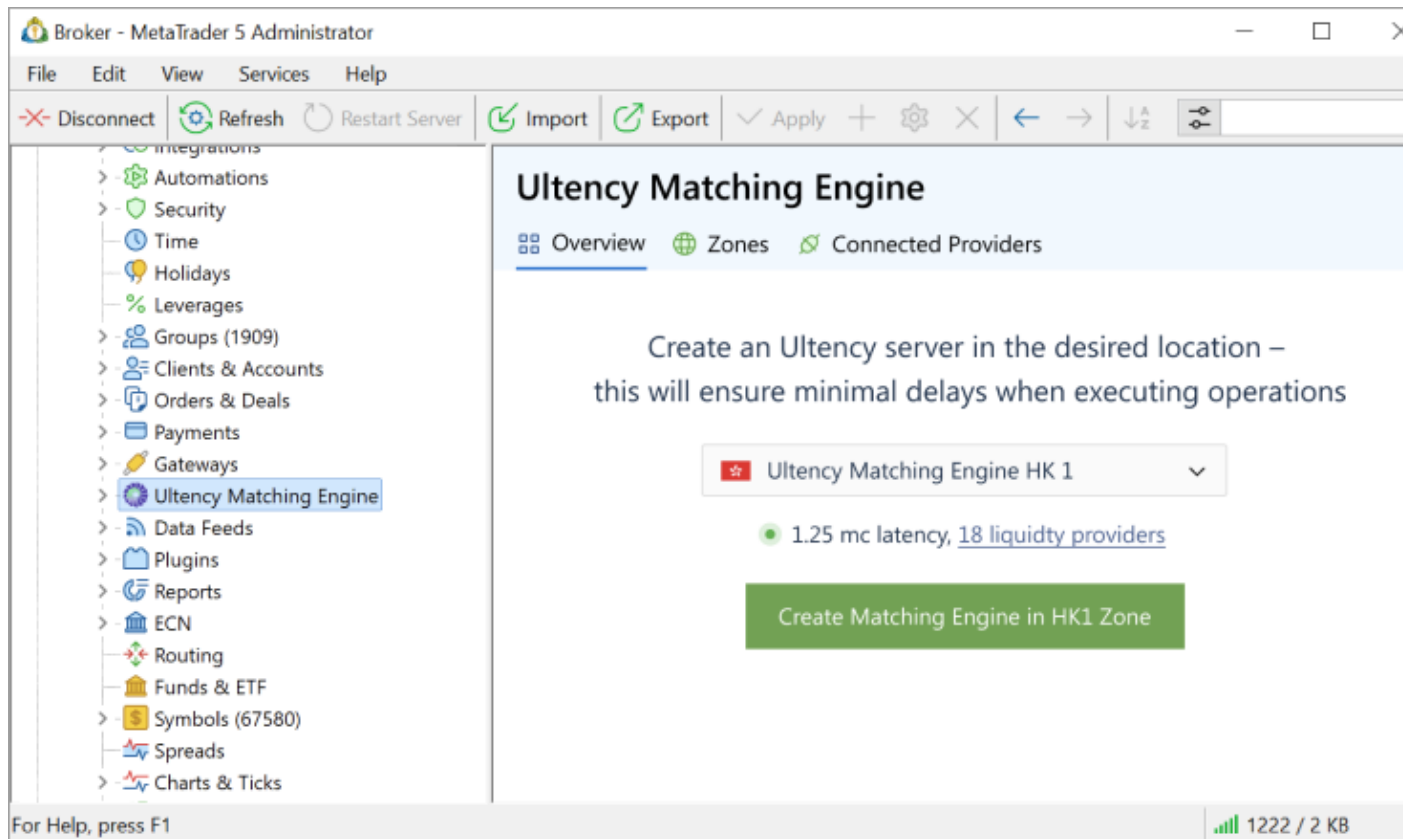
Steps to connect Ultency:

1. Obtain an Ultency server by selecting an appropriate geo-zone.
2. Contact liquidity providers via the built-in communication system and sign a service agreement.
3. Connect providers to your zone using the provided credentials.

Zones

Zones are geographic locations where your Ultency servers can be hosted. Each zone has its own list of available liquidity providers to whom you can connect. Select a zone based on the liquidity providers you intend to use and the markets in which you wish to offer trading. You can view all available zones in the "Overview" section. By default it suggests a zone with the fastest connection to your platform. Check if the zone has suitable liquidity providers and click "Create Matching Engine". The component will be added to your cluster. The server rental cost will be included in your company's invoice.

MetaTrader 5 trading platform



Go to the "Zones" section to manage your zones. The list displays both available and connected zones, with key information such as: network latency between the Ultency server and your platform, the number of available liquidity providers, and available categories of financial instruments.

MetaTrader 5 trading platform

The screenshot displays the 'Broker - MetaTrader 5 Administrator' window. The left sidebar shows a tree view with 'Servers' expanded, containing 'MetaTrader Server Dev' and several 'Ulteny Matching Engine' instances. The main panel is titled 'Ulteny Matching Engine' and shows three configured zones. Each zone includes a flag icon, a name, a 'Connected' status, latency, the number of liquidity providers, and a list of supported instruments. Action buttons like 'Remove zone' and 'Add zone' are present for each entry.

Zone Name	Status	Latency	Liquidity Providers	Instruments	Action
UK, London, Equinix LD4	Connected	5.04 ms	9	Forex, Metals, CFD, Forex Only, Crypto, Indices, MSVC, All, Stocks	Remove zone
NL, Amsterdam, Servers AMS	Connected	5.06 ms	6	Forex, Metals, CFD, Forex Only, Crypto, MSVC, All, Stocks	Remove zone
UK, London, Servers LD6		6.28 ms	2	Forex, Metals, CFD, Forex Only, Stocks, Crypto	Add zone

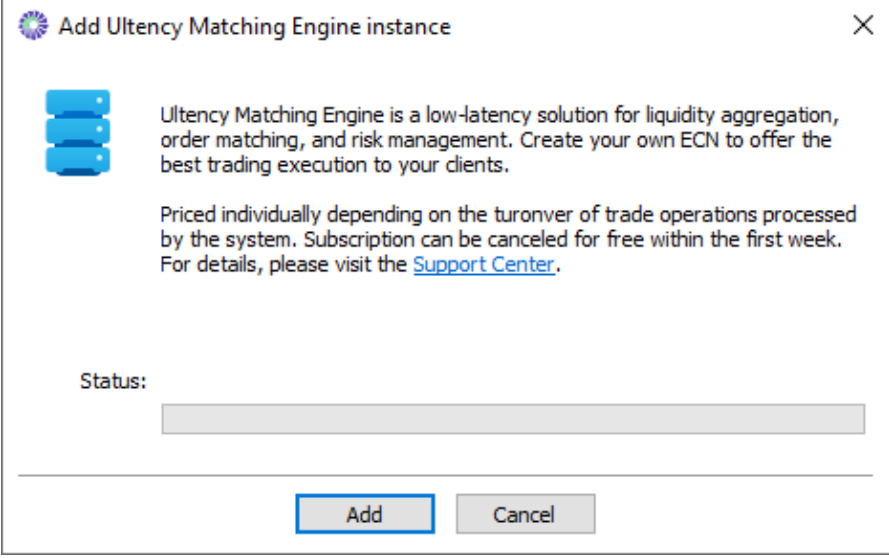
For Help, press F1

43.0 / 0.1 MB

MetaTrader 5 trading platform

If you want to provide access to multiple markets simultaneously, you can connect to several zones. For example, you may use an Ultency server in London for Forex providers and another in New York for NYSE providers.

Click "Add zone" and review the terms. Next, add the zone:



Add Ultency Matching Engine instance

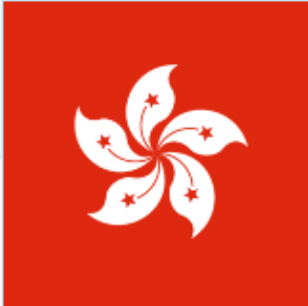
Ultency Matching Engine is a low-latency solution for liquidity aggregation, order matching, and risk management. Create your own ECN to offer the best trading execution to your clients.

Priced individually depending on the turnover of trade operations processed by the system. Subscription can be canceled for free within the first week. For details, please visit the [Support Center](#).

Status:

Liquidity Providers

Once you have obtained an Ultency server in your selected zone, you need to contact a liquidity provider, i.e. the company that will supply price feeds and process your orders. A separate service agreement must be signed with each provider. To view detailed information about available liquidity providers and their terms, go to the relevant section or click on any zone to see its providers.



Ultency Matching Engine HK1

Latency: ● 0.18 ms Connected Providers: 2 Available: 7

Available liquidity providers:

 Liquidity Provider 1 ● 68 ms latency www.provider1.com	 Liquidity Provider ● 12 ms latency www.provide33.com
 Liquidity Provider 2 ● 10 ms latency www.provider2.com	 Liquidity Provider ● 61.88 ms latency www.provider4.com

MetaTrader 5 trading platform

On each provider's page, you will find:

- General description.
- List of zones where the provider is available.
- Service details: available instruments, turnover volume, commission structure (per trade, periodic fees), and service costs.

Utlency offers Service Desk, a built-in communication system, to facilitate direct contact with providers. Select the desired service and click "Request". This will create a private dialogue with the provider where you can discuss all necessary details.


**Liquidity
Provider**

Utlency Liquidity Provider

www.metaquotes.net Cyprus

support@metaquotes.net +35712345678

Utlency Liquidity Provider 0.18 ms ultme.utlency.com 0.26 ms Your Server

Contact the provider Connect with credentials

Volume based offer ^

Request

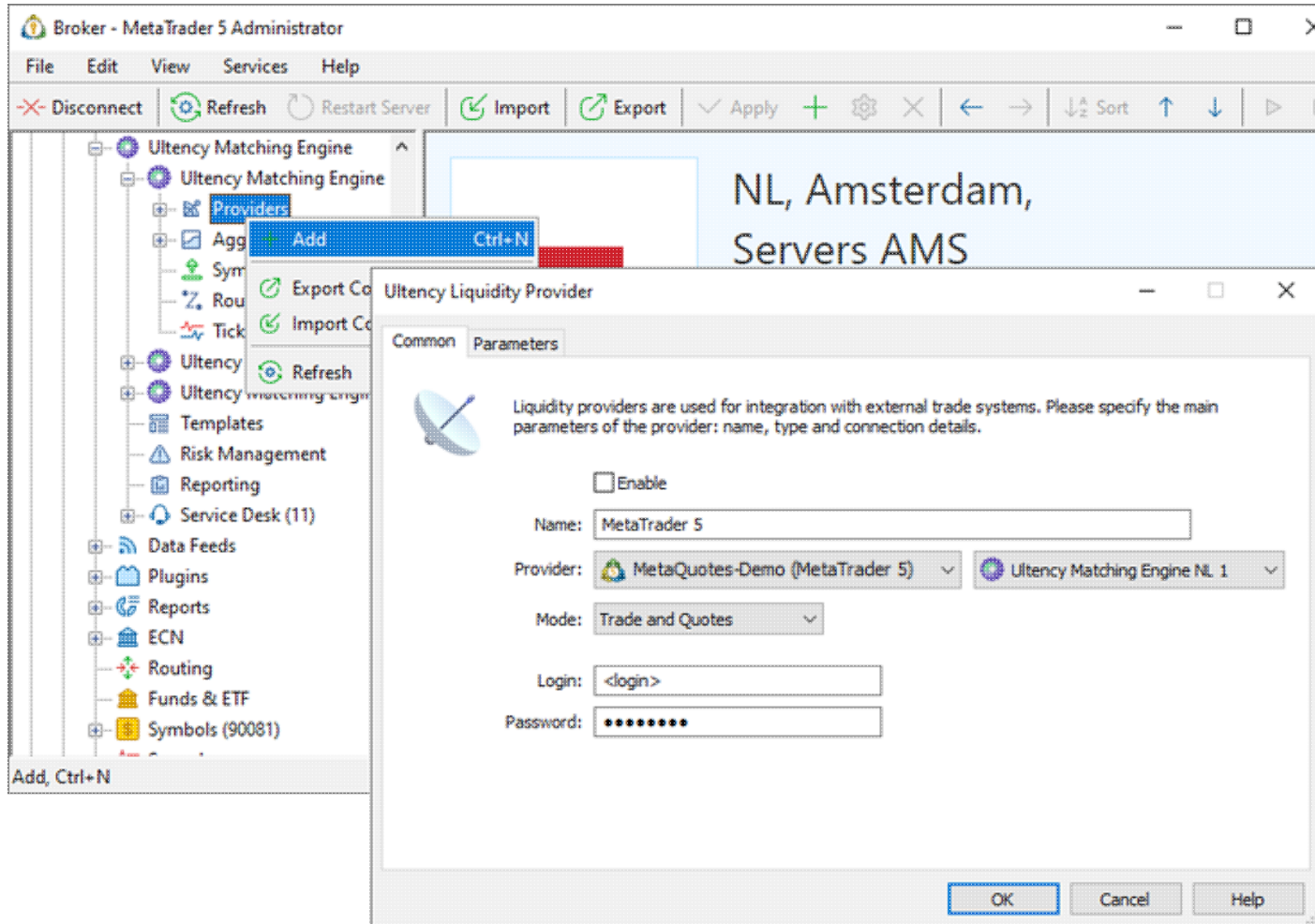
Min. deposit	Account currencies	Requires licenses	Available instruments
50000	USD, EUR	All	Forex, Crypto

Instruments	Volume	Commission	Type	Min. monthly fee
Forex	<5 bln	5 USD/mln	per monthly volume	1000 USD
	5 - 10 bln	4 USD/mln	per monthly volume	
	10 - 20 bln	3 USD/mln	per monthly volume	
	≥20 bln	2 USD/mln	per monthly volume	
Crypto	<1 bln	7 USD/mln	per monthly volume	500 USD
	1 - 5 bln	6 USD/mln	per monthly volume	
	5 - 10 bln	5 USD/mln	per monthly volume	
	≥10 bln	5 USD/mln	per monthly volume	

MetaTrader 5 trading platform

Connecting a Liquidity Provider

Once a service agreement is signed, the liquidity provider will supply you with connection details, including server address, login, and password. Create a provider configuration:

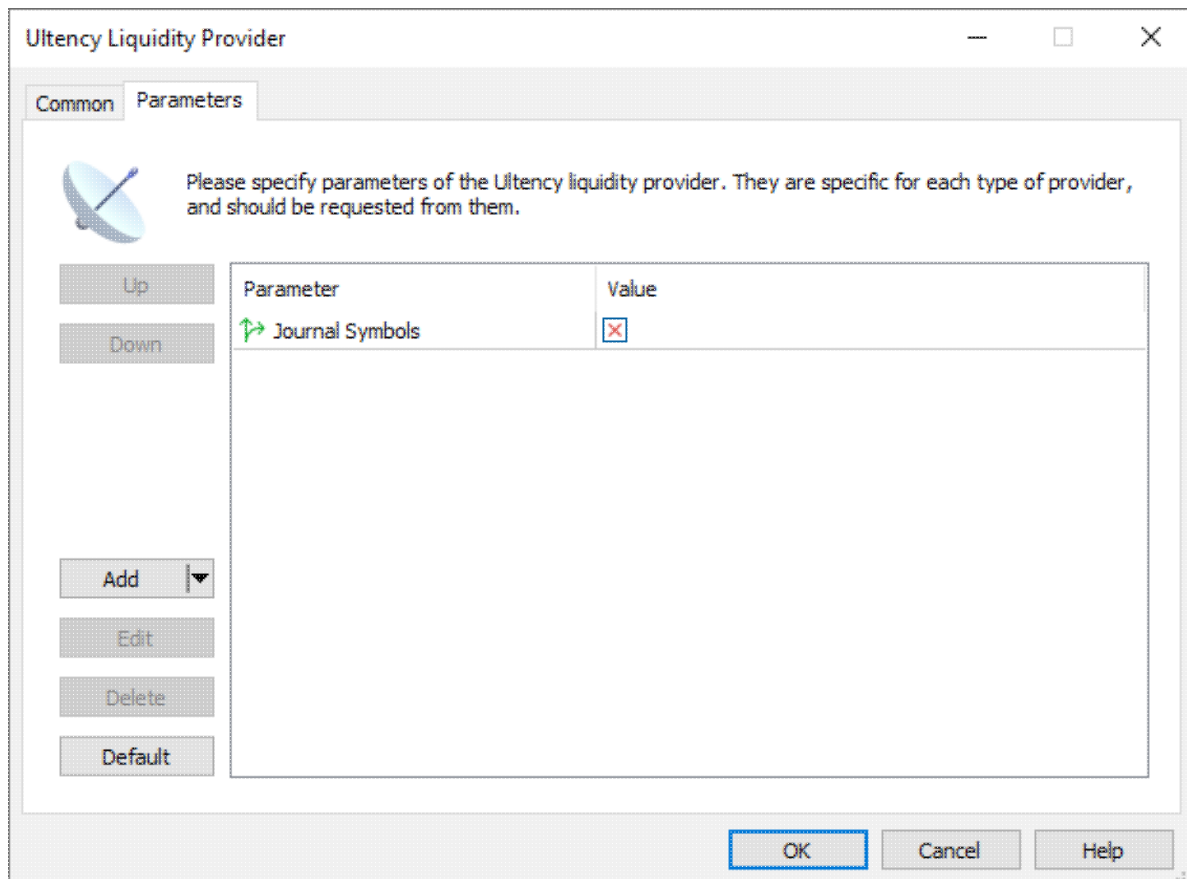


Fill in the parameters:

- Name the name of the configuration.
- Provider the provider you have a contract with and the Ultency server where it will be connected.
- Mode the provider's operating mode: quotes only, order processing only, both quotes and orders.
- Server provider's server address.
- Login login credentials for the provider.
- Password password for the provider.
- Coverage group and account when adding a liquidity provider, the system automatically creates a special group and a trading account within that group. All client orders routed to this provider will be duplicated to this account for risk management and monitoring purposes.

Under the "Parameters" tab, set additional options:

- Journal Symbols controls whether symbol changes from the provider are logged in the trading server journal. Adding a new provider may import tens of thousands of symbols, which could clutter the journal during the initial setup. It is recommended to disable "Journal Symbols" during initial configuration. You can enable it later for monitoring and debugging purposes.



Ultency Liquidity Provider

Common Parameters

Please specify parameters of the Ultency liquidity provider. They are specific for each type of provider, and should be requested from them.

Parameter	Value
Journal Symbols	X

Up Down

Add Edit Delete Default

OK Cancel Help

MetaTrader 5 trading platform

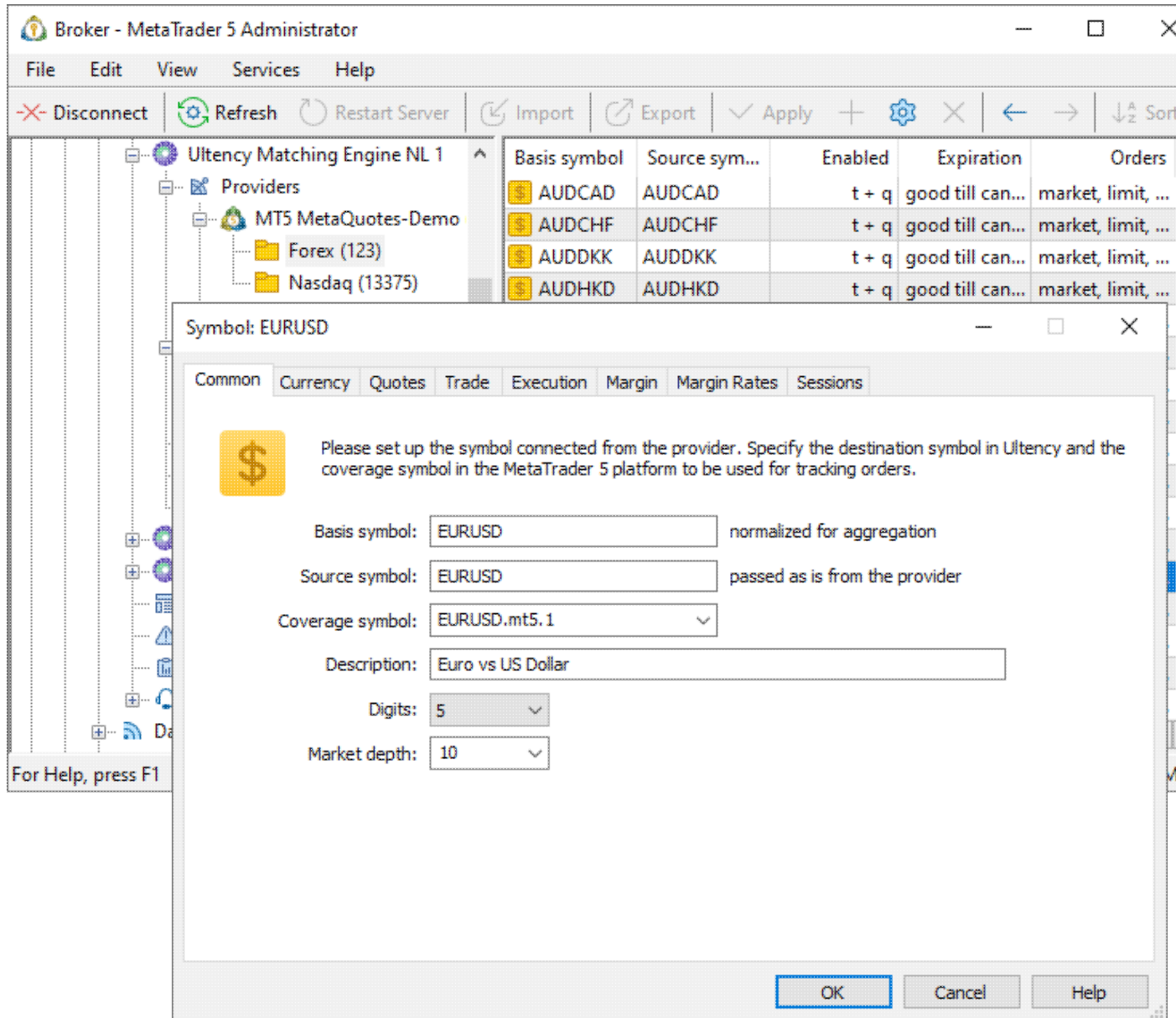
Once the provider is successfully connected, the provider's available financial instruments will be imported into Ultency. You can proceed with instrument configuration for subsequent aggregation.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Provider Symbols

Provider Symbols

Ultency imports financial instrument settings from liquidity providers and subsequently uses them for price data aggregation and order execution routed to the provider.

Common



The following settings are available in the "Common" section:

- **Basis symbol** the name of the instrument associated with the symbol received from a provider. It is used to link the provider's instrument to an aggregated symbol.

Each aggregated symbol must have a designated basis symbol, which defines the instrument for which data is collected, as well as the list of providers whose data should be aggregated. Ultency scans all selected providers and aggregates data from instruments with the specified basis symbol. The basis symbol is also used for routing the requests.

A single provider cannot have multiple symbols with the same basis symbol.

To bulk modify or remove a basis symbol, select the desired records, click "Edit" in the menu, and then specify or delete the postfix in the corresponding field. The postfix must begin with a period (.). For example, if you select records with the basis symbols EURUSD, USDJPY, and GBPUSD and add a postfix .x (EURUSD.x) in the bulk editing window, all three base symbols will be changed to EURUSD.x, USDJPY.x, and GBPUSD.x, respectively.

- **Source symbol** the name of the instrument on the provider side. This field is populated automatically when the provider is connected.
- **Coverage symbol** a symbol used to duplicate all trades executed through this provider.
- **Description** a brief description of the symbol.
- **Digits** the number of decimal places in the symbol prices. This is used for indication only on the provider's side. Prices are rounded only when transmitted to MetaTrader 5, while Ultency works with prices according to the original number of digits from the liquidity provider.
- **Market depth** the number of price levels (on each side) to be transmitted for this symbol. Other levels will be excluded from the aggregated book. If Market depth is disabled on Ultency or provider side, ticks for this symbol will be processed with the maximum execution volume specified in the settings and participate in order matching and B-Book book formation by levels.

A liquidity provider must not have multiple instruments with the same basis symbol. For example, a regular EURUSD symbol and a similar symbol with different settings and some suffix in the name (EURUSD.x, etc.). This may cause errors or undefined behavior in Ultency. If duplicate symbols are available for the account used for connecting to the provider, contact the provider to ensure only one set is available.

Currency

Parameters in this section are informational and should not be modified. They reflect how the liquidity provider handles transactions and which settings they use.

- **Base currency** the base currency of the instrument.
- **Profit currency** the currency in which profits from trades on this symbol will be calculated.

- **Margin currency** the currency used to calculate margin requirements for the symbol.

Symbol: EURUSD

Common Currency Quotes Trade Execution Margin Margin Rates Sessions

 Configure base, profit and margin currencies of the symbol.

Base currency: EUR

Base currency digits: 2

Profit currency: USD

Profit currency digits: 2

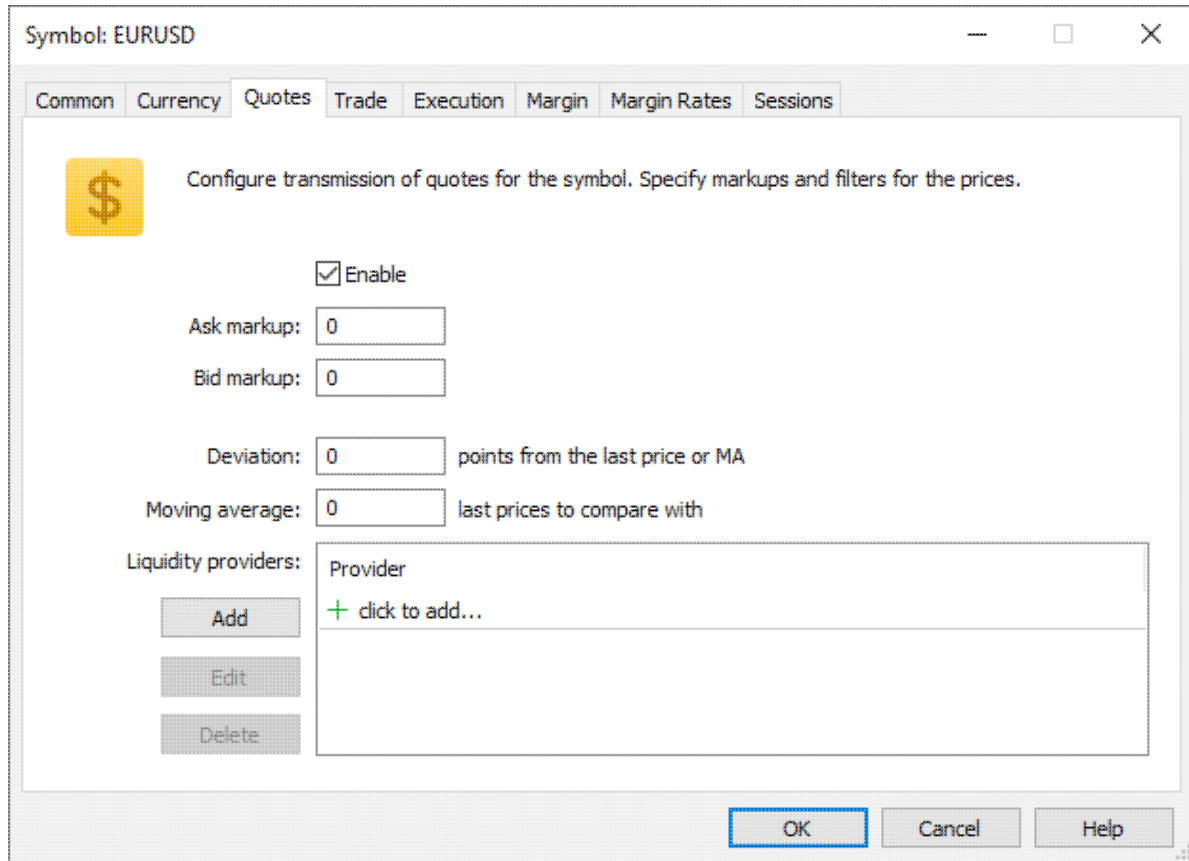
Margin currency: EUR

Margin currency digits: 2

OK Cancel Help

Quotes

This section defines settings for receiving quotes from the provider:



The screenshot shows the 'Quotes' tab in the MetaTrader 5 settings window for the EURUSD symbol. The window has a title bar 'Symbol: EURUSD' and standard window controls. Below the tabs, there is a yellow square icon with a dollar sign. The main text reads: 'Configure transmission of quotes for the symbol. Specify markups and filters for the prices.'

The settings include:

- ☒ Enable
- Ask markup: 0
- Bid markup: 0
- Deviation: 0 points from the last price or MA
- Moving average: 0 last prices to compare with
- Liquidity providers: A list box with the header 'Provider' and a single entry '+ click to add...'. To the left of the list box are three buttons: 'Add', 'Edit', and 'Delete'.

At the bottom right, there are three buttons: 'OK', 'Cancel', and 'Help'.

Specify the following parameters:

- **Enable** enables reception of quotes and depth of market for the symbol.
- **Bid markup** the number of points to adjust the Bid price received from the provider before passing it to the aggregated symbol. Positive values increase, negative values decrease the price. The value is in points of the source symbol's price.
- **Ask markup** the number of points to adjust the Ask price received from the provider before passing it to the aggregated symbol. Positive values increase, negative values decrease the price. The value is in points of the source symbol's price.

Price Filtering

Filters are used to control the correctness of quotes from the liquidity provider:

- **Deviation** maximum allowed difference (in points) between the received price and the previous or average price (if "Moving average" is non-zero). Prices exceeding this deviation are discarded.
- **Moving average** number of previous prices used to calculate the average. If the current price deviates from this average by more than the "Deviation" value, it is discarded. A zero value means the current price is compared to the last price.
- **Liquidity providers** a list of liquidity providers whose prices are used to cross-check quotes from the current provider. When receiving a quote, the system will compare it with the latest price for the same basis symbol from the specified providers. If the price differs by more than the "Deviation" value, it is automatically discarded.

Trading

Parameters in this section are informational and should not be modified. They reflect how the liquidity provider handles transactions and which settings they use.

If necessary, you can only narrow the settings within the limits allowed by the provider. For example, you can disable one of the supported execution modes (but not enable the unsupported one) or increase the minimum volume.

The screenshot shows the 'Symbol Properties' window for the EURUSD symbol, specifically the 'Trade' tab. The window has a title bar 'Symbol: EURUSD' and standard window controls. Below the title bar are tabs: 'Common', 'Currency', 'Quotes', 'Trade' (selected), 'Execution', 'Margin', 'Margin Rates', and 'Sessions'. The 'Trade' tab contains a yellow dollar sign icon and the text 'Configure trade parameters of the symbol: calculation type, allowed order types and volume limits.' The parameters are as follows:

- Contract size: 100000
- Calculation: Forex (dropdown)
- Filling: Fill or Kill (dropdown)
- Expiration: All (dropdown)
- Orders: All (dropdown)
- Minimum volume: 0.01 (dropdown)
- Volume step: 0.01 (dropdown)
- Maximum volume: 500 (dropdown)
- Volume limit: (empty dropdown)

At the bottom right are three buttons: 'OK' (highlighted with a blue border), 'Cancel', and 'Help'.

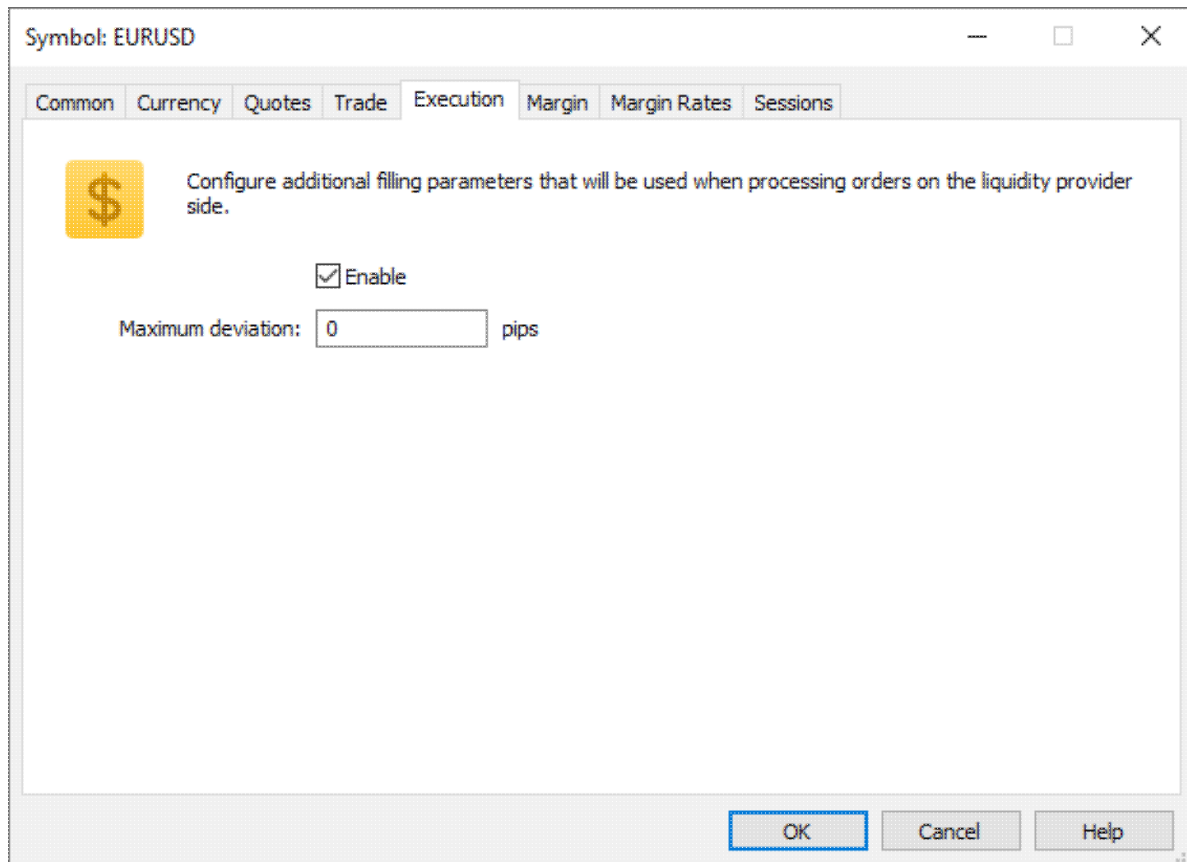
The following settings are available:

- **Contract size** – the amount of the base asset per lot/contract (for currency pairs it is actually the value of one lot in the base currency of the instrument).
- **Calculation** – symbol margin requirements and profit calculation type. This parameter also affects the calculation of swaps.
- **Filling** – additional order filling rules which can be specified in Ultency. To allow a particular fill type, check the relevant box:
 - ◆ **Fill or Kill** – order can only be executed in the specified volume.
 - ◆ **Immediate or Cancel** – order can be executed in the maximum volume available in the market within the requested volume. If the request cannot be filled completely, an order with the available volume will be executed, and the remaining volume will be canceled.
 - ◆ **Passive** – order is only placed in the market depth. If the order can be filled immediately when placed, this order is canceled.
 - ◆ **Return** – this mode is not available in the list. It is always allowed for market orders (Buy and Sell) in the "Exchange execution" mode and for limit and stop-limit orders in the "Market Execution" and "Exchange Execution" modes.
- **Expiration** – order expiration conditions. To allow an expiration type, check the relevant box:
 - ◆ **Good Till Canceled** – the order will remain in the queue until it is canceled manually.
 - ◆ **Day** – the order will be valid only during the current trading day. When processed on the MetaTrader 5 platform side, day orders are removed at

the end of the calendar day (24 hours) taking into account trading server time and symbol trading sessions. For example, if trading ends at 21:00, untriggered day orders will be canceled at the same time. When orders are processed in external systems via gateways, the expiration rules can be different and depend on the specific system.

- ◆ Specified time the order will be valid until the specified date.
- ◆ Specified day the order will be valid until 00:00 of the specified day. If outside a trading session, it will expire at the next available session.
- Orders allowed order types: Market, Limit, Stop, Stop Limit, Stop Loss, Take Profit, Close By. Enabling Close By orders does not affect netting accounts, as Close By orders can only be used on hedging accounts.
- Minimum volume minimum order size for the symbol. The parameter is ignored when closing positions.
- Maximum volume maximum order size for the symbol. Applied when placing orders and when closing positions by Stop Out.
- Volume step volume change step.
- Volume limit maximum total volume of an open position and pending orders in one direction (buy or sell) per one symbol. For example, a limit of 5 lots allows you to have an open 5-lot position and place a 5-lot Sell Limit order. But in this case, placing a pending Buy Limit order is not allowed, since the total volume in one direction would exceed the limit. Also, if there is a 5-lot Buy position, Ulteny cannot place a Sell Limit order with a volume exceeding 5 lots: if the pending order triggered after the closing of the initial position, the system would have a short position exceeding the specified limit.

Execution



The section defines additional execution parameters for orders routed to the liquidity provider.

Maximum deviation

This parameter is only used when operations are sent as Instant Execution and Request Execution orders.

In case of an instant execution, an acceptable deviation is specified in the request sent to the external provider.

When a requote from an external provider is received, Ultency checks if the new price is within the specified deviation. If so, Ultency sends a new request to the provider with the prices from the requote. If the provider accepts the prices, the request is executed. In case of a repeated requote, the deviation will be checked according to the first request (i.e. the price of the original order).

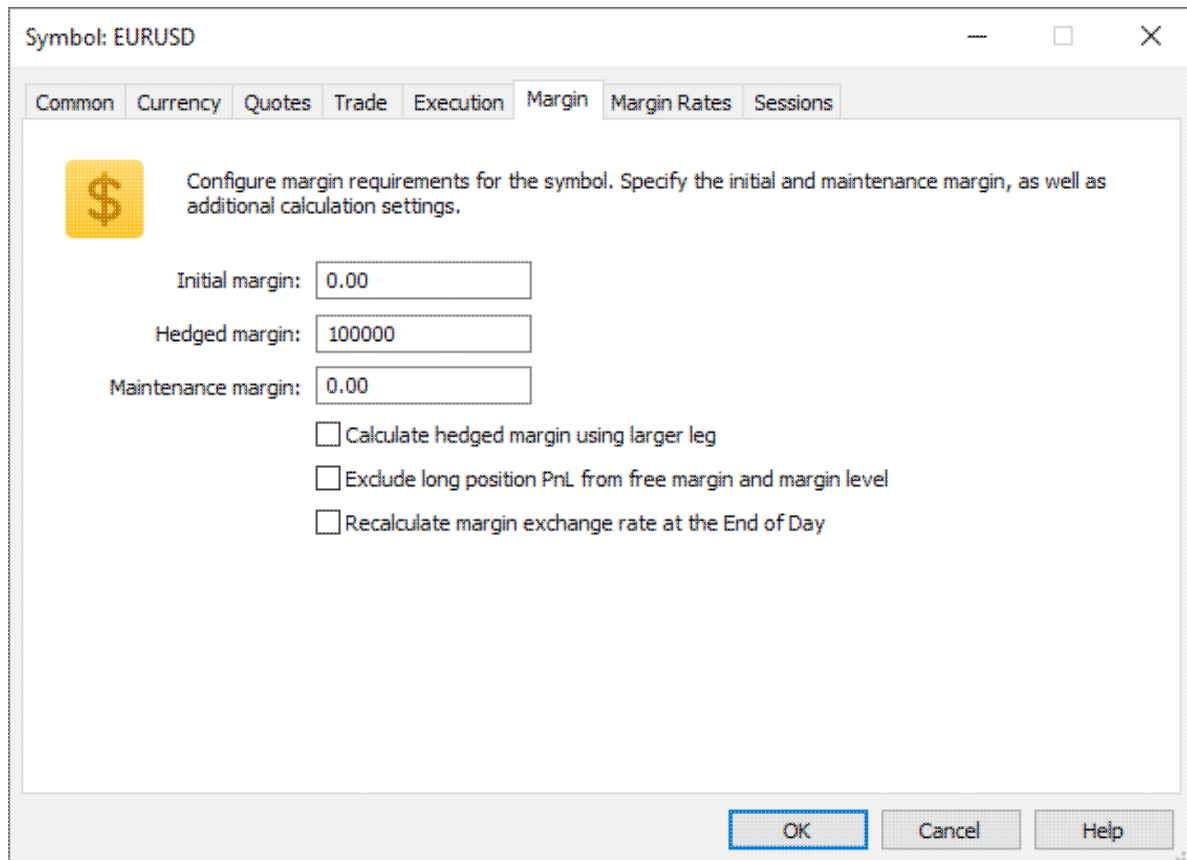
If deviation is exceeded, execution is stopped, and the order is rejected.

Margin

Parameters in this section are informational and should not be modified. They reflect how the liquidity provider handles transactions and which settings they use. These settings are also used to calculate the coverage account margin to align conditions with the liquidity provider's terms.

MetaTrader 5 trading platform

When trading on the MetaTrader 5 side, your clients will use the same margin settings in your platform. Align them with the providers' settings to avoid increased risks.



The screenshot shows the 'Margin' tab of the MetaTrader 5 settings window for the EURUSD symbol. The window has a title bar 'Symbol: EURUSD' and standard window controls. Below the title bar is a tabbed interface with 'Common', 'Currency', 'Quotes', 'Trade', 'Execution', 'Margin', 'Margin Rates', and 'Sessions'. The 'Margin' tab is active, displaying a yellow dollar sign icon and the instruction: 'Configure margin requirements for the symbol. Specify the initial and maintenance margin, as well as additional calculation settings.' There are three input fields: 'Initial margin:' with the value '0.00', 'Hedged margin:' with the value '100000', and 'Maintenance margin:' with the value '0.00'. Below these fields are three unchecked checkboxes: 'Calculate hedged margin using larger leg', 'Exclude long position PnL from free margin and margin level', and 'Recalculate margin exchange rate at the End of Day'. At the bottom right are three buttons: 'OK' (highlighted with a blue border), 'Cancel', and 'Help'.

Symbol: EURUSD

Common Currency Quotes Trade Execution **Margin** Margin Rates Sessions

 Configure margin requirements for the symbol. Specify the initial and maintenance margin, as well as additional calculation settings.

Initial margin:

Hedged margin:

Maintenance margin:

☐ Calculate hedged margin using larger leg

☐ Exclude long position PnL from free margin and margin level

☐ Recalculate margin exchange rate at the End of Day

OK Cancel Help

Margin Rates

Parameters in this section are informational and should not be modified. They reflect how the liquidity provider handles transactions and which settings they use. These settings are also used to calculate the coverage account margin to align conditions with the liquidity provider's terms.

When trading on the MetaTrader 5 side, your clients will use the same margin settings in your platform. Align them with the providers' settings to avoid increased risks.

Symbol: EURUSD

Common Currency Quotes Trade Execution Margin **Margin Rates** Sessions

 Configure margin multiplies for each type of trade operations.

Liquidity margin rate:

Currency margin rate:

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.00000000	0.00000000	0.00000000	0.00000000
 Sell	1.00000000	0.00000000	0.00000000	0.00000000
Maintenance margin				
 Buy	0.00000000	0.00000000	0.00000000	0.00000000
 Sell	0.00000000	0.00000000	0.00000000	0.00000000

OK Cancel Help

Sessions

This section contains the provider's default quoting and trading sessions. You may narrow (but not expand) these sessions to restrict quoting and order routing.

Symbol: EURUSD

Common Currency Quotes Trade Execution Margin Margin Rates Sessions

 Configure trade and quotation sessions of the symbol by days. During quotation session it is possible to view the price charts, but trading is prohibited.

Day	Quotes	Trade
 Sunday	23:00-24:00	23:00-24:00
 Monday	00:00-24:00	00:00-24:00
 Tuesday	00:00-24:00	00:00-24:00
 Wednesday	00:00-24:00	00:00-24:00
 Thursday	00:00-24:00	00:00-24:00
 Friday	00:00-23:00	00:00-23:00
 Saturday		

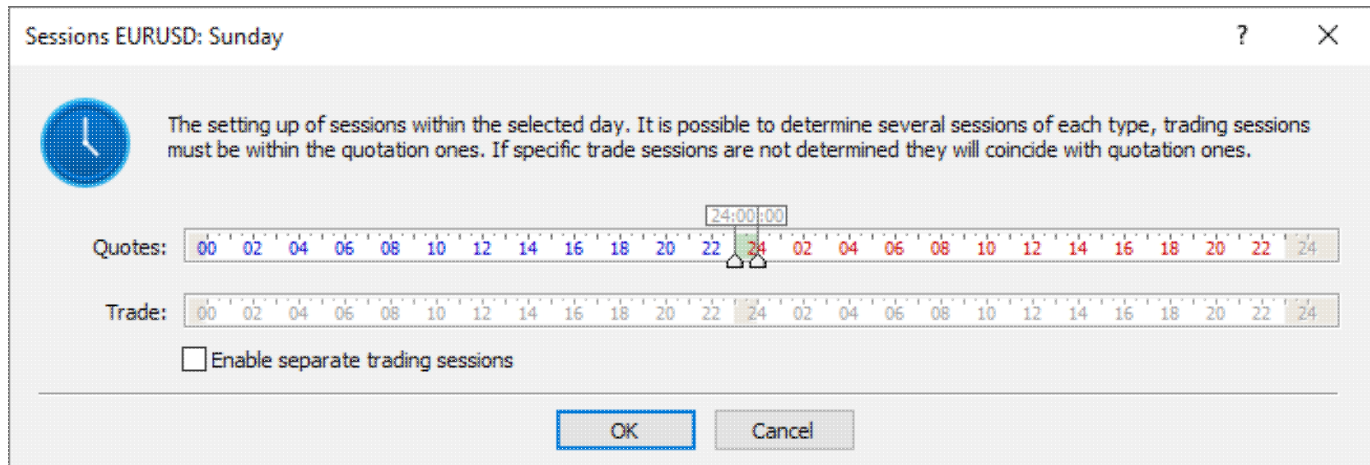
Edit

From:  To: 

OK Cancel Help

MetaTrader 5 trading platform

Click on a day to adjust its session settings:



To configure the time of quote and trading sessions separately, enable the corresponding option. Next, set up the sessions. The sliders that control session beginning and end times can be moved with the mouse and with the arrow keys. You can hold down the Shift key to slow down the slider speed. This enables precision up to minutes when configuring sessions.

If the symbol should only be available for a specific time, set From and To parameters. Outside this range, trading will be disabled for this symbol. Clients will receive a "Trading disabled" message upon attempting to place orders.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Aggregated Symbols

Aggregated Symbols

Ultency receives price data from multiple liquidity providers and consolidates it into a single stream. As a result, your clients get better execution prices. This makes trading more attractive and cost-efficient for clients and helps you to attract new traders. In addition to external provider data, Ultency allows you to aggregate your own clients' orders into the order book, enabling the broker to operate as a fully functional ECN.

To collect price data, you need to create aggregated symbols. To avoid creating them manually, simply select the symbols imported from a provider and choose "Copy to Aggregated Symbols":

The screenshot shows the 'Broker - MetaTrader 5 Administrator' window. On the left, a tree view shows the 'Ultency Matching Engine' structure, including 'Providers' (Forex (123), Nasdaq (13375)), 'Aggregated Symbols (246)', and other components. The 'Forex (123)' provider is selected. On the right, a table lists symbols and their settings. A context menu is open over the 'AUDCHF' row, with 'Copy to Aggregated Symbols' highlighted. The status bar at the bottom indicates 'Copy the selected Ultency symbols to the aggregated symbols'.

Basis ...	Source sym...	Enabled	Expiration	Orders
AUDCAD	AUDCAD	t + q	good till can...	market, limit, ...
AUDCHF	AUDCHF	t + q	good till can...	market, limit, ...
AUDDKK	AUDDKK	t + q	good till can...	market, limit, ...
AUDHKD	AUDHKD	t + q	good till can...	market, limit, ...
AUDHUF	AUDHUF	t + q	good till can...	market, limit, ...
AUDJPY	AUDJPY	t + q	good till can...	market, limit, ...
AUDNO	AUDNO	t + q	good till can...	market, limit, ...
AUDNZD	AUDNZD	t + q	good till can...	market, limit, ...
AUDPLN	AUDPLN	t + q	good till can...	market, limit, ...
AUDSEK	AUDSEK	t + q	good till can...	market, limit, ...
AUDSGD	AUDSGD	t + q	good till can...	market, limit, ...
AUDTHE	AUDTHE	t + q	good till can...	market, limit, ...
AUDUSD	AUDUSD	t + q	good till can...	market, limit, ...
AUDZAR	AUDZAR	t + q	good till can...	market, limit, ...
CADCHF	CADCHF	t + q	good till can...	market, limit, ...
CADDKK	CADDKK	t + q	good till can...	market, limit, ...
CADJPY	CADJPY	t + q	good till can...	market, limit, ...

Copy the selected Ultency symbols to the aggregated symbols

MetaTrader 5 trading platform

As a result, the selected symbols will be automatically added to the Aggregated Symbols section, within the same subgroup and with the same settings, including name, description, basis symbol, and trading sessions. Price feeds from the selected provider will also be enabled for these symbols by default.

The screenshot displays the 'Broker - MetaTrader 5 Administrator' window. The left sidebar shows a tree view of the configuration structure, with 'Forex (123)' selected under 'Aggregated Symbols (246)'. The main panel on the right shows a table of aggregated symbols.

Aggregated symbol	Quotes t...	Digits	Market...	B-Boo...	Reject ...
EURUSD	aggregated	5	32	vwap	reject
GBPUSD	aggregated	5	32	vwap	reject
USDCHF	aggregated	5	32	vwap	reject
USDJPY	aggregated	3	32	vwap	reject
USDCNH	aggregated	5	32	vwap	reject
USDRUB	aggregated	3	32	vwap	reject
AUDUSD	aggregated	5	32	vwap	reject
NZDUSD	aggregated	5	32	vwap	reject
USDCAD	aggregated	5	32	vwap	reject
USDSEK	aggregated	5	32	vwap	reject
USDHKD	aggregated	5	32	vwap	reject
USDSGD	aggregated	5	32	vwap	reject
USDNOK	aggregated	5	32	vwap	reject
USDDKK	aggregated	5	32	vwap	reject
USDTRY	aggregated	5	32	vwap	reject
USDZAR	aggregated	5	32	vwap	reject
USDCZK	aggregated	5	32	vwap	reject

At the bottom of the table, there are two tabs: 'Aggregated Symbols Forex (123)' (selected) and 'All Aggregated Symbols (246)'.

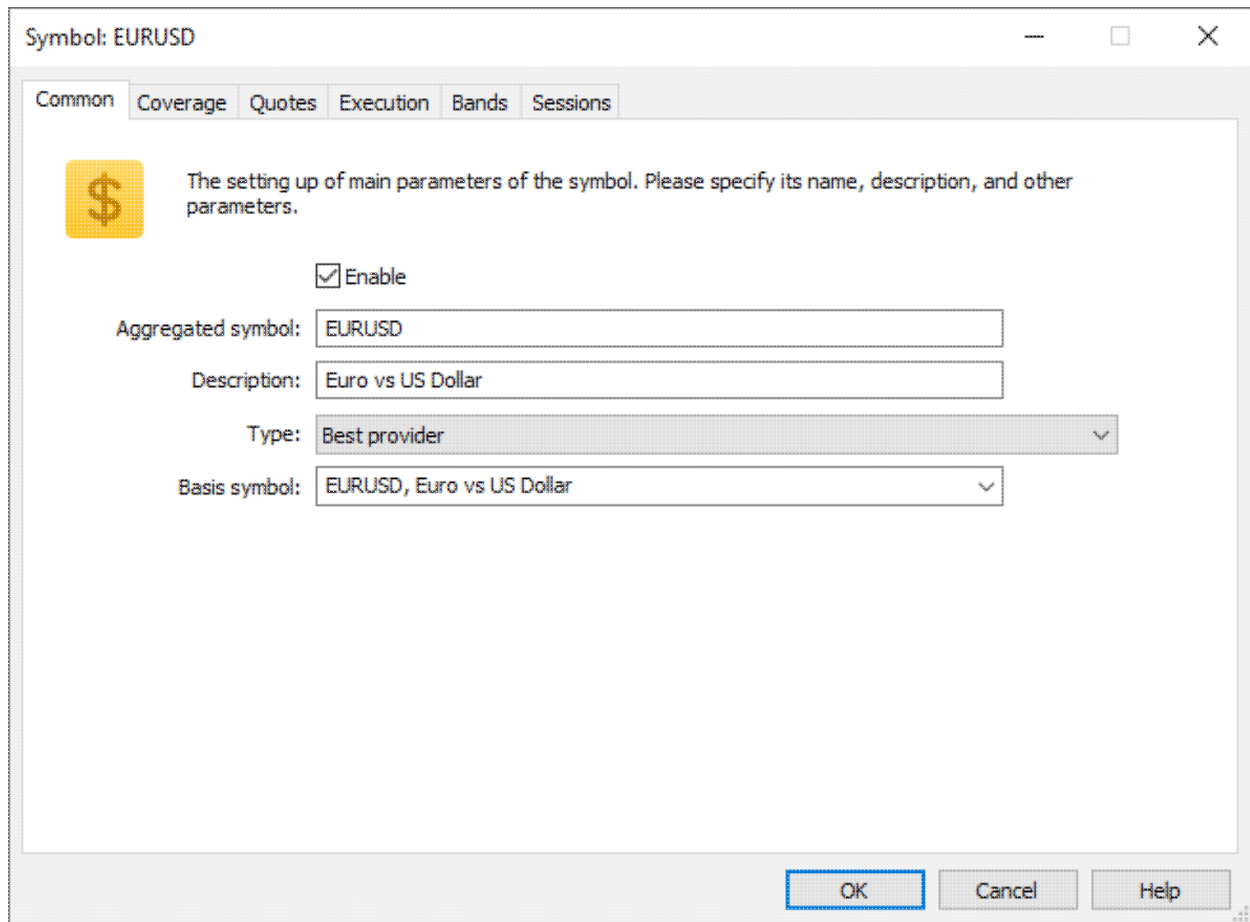
For Help, press F1

44.0 / 0.0 M

MetaTrader 5 trading platform

Once this is done, proceed to configure the remaining parameters.

Common



The screenshot shows the 'Symbol: EURUSD' dialog box with the 'Common' tab selected. The dialog has a title bar with standard window controls. Below the title bar is a tabbed interface with 'Common', 'Coverage', 'Quotes', 'Execution', 'Bands', and 'Sessions'. The 'Common' tab contains a yellow square icon with a dollar sign. To the right of the icon is the text: 'The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.' Below this text is a checkbox labeled 'Enable' which is checked. Further down are four input fields: 'Aggregated symbol:' with the value 'EURUSD', 'Description:' with the value 'Euro vs US Dollar', 'Type:' with a dropdown menu showing 'Best provider', and 'Basis symbol:' with a dropdown menu showing 'EURUSD, Euro vs US Dollar'. At the bottom right of the dialog are three buttons: 'OK', 'Cancel', and 'Help'.

Symbol: EURUSD

Common Coverage Quotes Execution Bands Sessions

 The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.

☒ Enable

Aggregated symbol: EURUSD

Description: Euro vs US Dollar

Type: Best provider

Basis symbol: EURUSD, Euro vs US Dollar

OK Cancel Help

The following settings are available in the "Common" section:

- Aggregated symbol the name of the aggregated symbol.
- Description a brief description of the symbol.
- Type order execution type for the symbol:
 - ◆ Prioritized providers / Only one active only one provider is used, the first available from the list in the Execution section. If the first provider is unavailable, disabled, or trading is not possible (for example, if there is no active trading session at the moment), the system will automatically attempt to route the order through the next available provider.
 - ◆ Best provider orders are executed through the provider (from the list in the "Execution" section) that offers the best available price for the requested volume.
 - ◆ Aggregated providers / Mixed market depth orders are executed through providers from the list in the "Execution" section. Orders can be executed through different providers, including partially. FOK orders are always executed through only one provider, which currently offers the best available price for the requested volume (as in the "Best provider" mode) and which supports the FOK fill mode.
 - ◆ Exchange mode / Own orders only orders are not routed to external providers. The order book is formed exclusively from your clients' orders. The mode is currently under development.
- Basis symbol the name of the instrument associated with the symbol received from a provider. Each aggregated symbol must have a designated basis symbol, which defines the instrument for which data is collected, as well as the list of providers whose data should be aggregated. Ultency scans all selected providers and aggregates data from instruments that share the specified basis symbol. The basis symbol is also used for routing requests.

A single provider cannot have multiple symbols with the same basis symbol.

Coverage

MetaTrader 5 trading platform

Symbol: EURUSD

Common Coverage Quotes Execution Bands Sessions

 The setting up of the symbol coverage. Please specify coverage symbol, groups and accounts.

☒ Coverage on client symbol

Coverage symbol: EURUSD.ME.24

A-Book coverage group: coverage\ultency\ME.24\A_book

A-Book coverage account: 4598687

B-Book coverage group: coverage\ultency\ME.24\B_book

B-Book coverage account: 4598688

OK Cancel Help

To ensure precise control over the execution of trades for an aggregated symbol, all operations related to that symbol are duplicated to separate coverage accounts. One account collects all transactions routed to external providers (A-Book), while another collects all operations executed within the platform (B-Book).

If you do not specify coverage groups/accounts when adding the configuration for an aggregated symbol, Ultency will automatically copy them from the configurations of other symbols. If the groups/accounts are not yet specified anywhere, Ultency will automatically create new groups and accounts and add them to the aggregated symbol's configuration. The groups are created with the netting accounting system, allowing you to view the net positions for all clients. These groups and accounts will be immediately added to the aggregated symbol's settings. You can control system operations by requesting reports for the relevant groups and accounts through the Manager terminal.

- Coverage symbol if this option is enabled, coverage deals for the aggregated symbol will be executed on the original MetaTrader 5 symbols from which client requests are routed to this aggregated symbol. If you wish to cover trades on a different symbol, disable this option and specify the symbol in the following field.
- Coverage symbol the name of the symbol where trades will be duplicated. It is used only when the "Coverage on client symbol" is disabled.
- A-Book coverage group the group where the account for duplicating A-Book operations is located.
- A-Book coverage account the account where A-Book operations are duplicated.
- B-Book coverage group the group where the account for duplicating B-Book operations is located.
- B-Book coverage account the account where B-Book operations are duplicated.

Ensure that the settings are correctly aligned: the specified coverage account must belong to the specified group. If you move the account or replace it with an account from a different group, be sure to update the group in the aggregated symbol settings accordingly.

Quotes

MetaTrader 5 trading platform

Symbol: EURUSD

Common Coverage **Quotes** Execution Bands Sessions

 The setting up of the symbol quotes. Please specify liquidity providers for transmitting quotes by this symbol.

Type: Aggregated Providers / Mixed market depth

Digits: 5 Market depth: 32

Min. spread: 0 pt Ask markup: 0

Max. spread: unlimited pt Bid markup: 0

Quote switch timeout: 10 seconds

Up
Down
Add
Edit
Delete

Provider

 MT5 MetaQuotes-Demo

+ click to add...

OK Cancel Help

This section defines the settings for quote aggregation:

- **Type** type of quote aggregation:
 - ◆ **Prioritized providers / Only one active** only one provider is used, the first available one from the [list](#). If the first provider remains unavailable for the duration specified in the "Quote switch timeout" parameter, the system will switch to the next provider in the list.
 - ◆ **Aggregated providers / Mixed market depth** the market depth is formed from the prices of providers in the [list](#).
 - ◆ **Aggregated providers with own orders / Mixed market depth** the market depth is formed from the prices of providers in the [list](#) and client requests inside the platform. The mode is currently under development.
- **Digits** the number of decimal places to which prices received from sources will be rounded.
- **Market depth** the number of orders/levels displayed in the order book. Depth is set for one direction. For example, if set to 16, the book will display up to 16 buy orders and the same number of sell orders.
- **Min. spread** the minimum spread that will be maintained in the instrument's price book, even when orders come in at better prices. More detailed information is provided [below](#).
- **Max. spread** the maximum spread that will be maintained in the instrument's price book, even when orders come in at worse prices. More detailed information is provided [below](#).
- **Ask markup** the number of points by which all ask prices (the ask side of the order book) will be shifted. A positive or negative value can be specified. The markup is applied before the minimum and maximum spread settings.
- **Bid markup** the number of points by which all bid prices (the bid side of the order book) will be shifted. A positive or negative value can be specified. The markup is applied before the minimum and maximum spread settings.
- **Quote switch timeout** used only for aggregation in "Prioritized providers / Only one active" mode. If quotes do not arrive from the prioritized provider within the specified number of seconds, the stream switches to the next provider in the [list](#). When quotes from the first provider resume, the stream automatically switches back. Upon the end of a quoting session for the prioritized provider, the switch to the next provider occurs immediately, without the timeout.

Providers

This section defines the list of providers whose quotes can be aggregated for the given instrument. The order of providers in the list determines their priority. The first provider is considered the highest priority.

Minimum and maximum spread

When aggregating order books from multiple sources and combining orders within the cluster, situations may arise where both sides of a trade are willing to execute at worse prices than actually available. For example, one provider may offer to buy an asset at a price of 100, while another may offer to sell the same asset at 90. If the actual spread for the symbol becomes narrower than the minimum allowed, Ultency will adjust the displayed prices in the order book to maintain the required spread.

Actual order prices remain unchanged, and trades are still executed at provider prices.

The adjustment to maintain the minimum spread is done using the following algorithm:

- The current spread is calculated as the difference between the best Bid and Ask. The calculation applies all markups specified for the aggregated symbol.
- If the current spread is smaller than the defined minimum, the system calculates the spread delta. It is the amount by which the spread needs to be widened.
- New Bid and Ask prices are computed. $\text{Bid New} = \text{Bid} - \text{floor}(\text{spread_delta}/2)$ points. $\text{Ask New} = \text{Ask} + \text{ceil}(\text{spread_delta}/2)$ points. Here 'floor' and 'ceil' are standard functions for getting the nearest integer value from below and above.
- For all buy orders with prices higher than Bid New, the visible price is adjusted to Bid New.
- For all sell orders with prices lower than Ask New, the visible price is adjusted to Ask New.

The Maximum Spread parameter works in a similar way. If the actual spread exceeds the specified maximum, all orders are shifted to reduce the spread.

Execution

Symbol: EURUSD

Common Coverage Quotes Execution Bands Sessions

 The setting up of the symbol execution. Please specify liquidity providers for executing orders by this symbol.

B-Book price mode: VWAP Reject mode: Reject

Maximum slippages: Profitable: off pips Losing: off pips

Limit and Take Profit orders activation: Mode: Market Timeout: 0 milliseconds

☐ Merge execution deals

Up Down Add Edit Delete

Provider

 MT5 MetaQuotes-Demo

+ click to add...

OK Cancel Help

This section defines order execution settings for the aggregated symbol:

B-Book price mode

Ultency allows partial coverage of orders. In the routing settings, you can specify what portion of each order should be executed via external providers (A-Book) and what portion should be executed internally within the cluster (B-Book). The "B-Book price mode" parameter defines how the non-covered (B-Book) portion of the order is executed:

- VWAP the uncovered volume is executed at prices from the aggregated order book.
- A-Book the volume is executed at the same price as the covered volume with the external provider. This mode provides additional protection from off-market prices, assuming the liquidity provider (especially a large one) uses robust filtering mechanisms to avoid incorrect executions.

Reject mode

This setting determines how an order will be handled if rejected by the current prioritized provider:

- Reject the order is rejected.
- B-Book the order is executed internally within the platform without routing to external providers.

This setting is applied in all execution modes except Exchange mode / Own orders only.

Slippages

This section defines how much the execution price in Ultency can deviate from the client's requested price.

- Profitable the number of points by which the price may deviate from the requested one, in a profitable direction for the client.
- Losing the number of points by which the price may deviate from the requested one, in an unprofitable direction for the client.

The function has a different behavior depending on the execution mode:

- A-Book after an actual deal is executed with the external provider, Ultency compares the execution price with the initially requested price (considering translation settings). If the executed price is within the allowed slippage range, the deal is executed on the platform side at the client's original requested price. If the slippage exceeds the limit, Ultency will execute the order on the platform side at the actual execution price applied on the provider side.

Suppose both profitable and losing slippage values are set to 5 points. Ultency sends a Buy Limit order at 1.14053 to the external provider, but the provider fills it at 1.14051 (2 points better). This slippage is acceptable. Ultency will fill

the client's order at the original price, 1.14053.

If the provider's execution price exceeds the slippage limit, Ultency will fill the order at that price, i.e. 1.14051.

- **B-Book** the price requested by the client is compared to the prices in the aggregated order book, considering the order fill type. For example, if a FOK order cannot be fully filled at acceptable prices, it will be completely rejected. An IOC order will be partially filled within available volumes at acceptable prices.

Merge execution deals

Depending on the settings, deals may be split across different sources, partially in B-Book, partially in A-Book with multiple providers. Ultency records the entire execution chain and can display it to the trader: the trading history on the trader's side will display all deals that executed the order. To hide execution details from traders, enable "Merge execution deals". In this case, all trades used to fill the order on the Ultency side will be presented in the platform as a single combined operation, showing total volume and weighted average price. Thus, the trader will also see a single trader in the account history.

Merging deals affects slippage parameters. If merging is disabled, slippage is checked for each individual deal that executes the order. If enabled, slippage is checked only for one combined deal (at the weighted average price).

Limit and Take Profit orders activation

Configure how limit orders, not previously routed to Ultency, and Take Profit activations are handled:

- **Mode** type of order to be sent to Ultency upon activation of Take Profit levels and limit orders not previously routed to Ultency. Options: "Limit" and "Market".
- **Timeout** the lifetime of an order routed to Ultency (in milliseconds). If the order is not executed within this time, this order is canceled. In case of the re-activation of a limit order or the Take Profit level, a new order with the specified timeout will be sent to Ultency.

By default, Ultency displays limit orders with a lifetime of 5 seconds.

These settings do not apply to Stop Loss activations, which are always sent as market orders.

Processing limit orders outside Ultency

You can configure limit orders to remain off Ultency until triggered. Configure routing rules to prevent such orders from being sent for matching before their activation.

Add a rule to process placing of limit and stop limit orders, as well as the stop limit order activation without forwarding to Ultency. Please note that routing rules are processed top-downward. For example, if you have a general rule according to which

MetaTrader 5 trading platform

all requests are sent to Ultency, then a new rule for processing limit orders should be located above the general one in the list.

Routing

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action:

Delay in milliseconds

▼

Where request is:

Pending order, Order activation

▼

Where order is:

Buy Limit, Sell Limit, Buy Stop, Sell Stop, Buy Stop Limit, Sell Stop Limit

▼

Where conditions are:

Type	Condition	Value
ab Request\Symbols	Equal (=)	Ultency*

Add

Edit

Delete

OK

Cancel

Help

Providers

This section defines a list of providers through which orders for this instrument can be executed. The order of providers in the list determines their priority. The first provider is considered the highest priority.

Levels

Use these settings to fine-tune aggregation and display of order book levels: combine aggregated levels from different sources by volume for a cleaner, more user-friendly order book. This can be used when processing orders in B-Book mode, when you act as a market maker and there's no need to show all the initial levels from providers.

Symbol: EURUSD

Common Coverage Quotes Execution **Bands** Sessions

 Please specify additional DOM levels for the aggregated symbol.

Bid:

Volume	Markup
☐ 100 000	-1
☐ 300 000	-2
☐ 500 000	-3
☐ 1 500 000	-4
☐ 2 500 000	-5
☐ ∞	-6
+ click to add...	

Ask:

Volume	Markup
☐ 100 000	1
☐ 300 000	2
☐ 500 000	3
☐ 1 500 000	4
☐ 2 500 000	5
☐ ∞	6
+ click to add...	

>> <<

OK Cancel Help

MetaTrader 5 trading platform

Set volume per level in the number of contracts. The levels are set separately for each side (Bid/Ask). Example for Bid side:

- First level = 100 000, markup = -1. Ultency will collect closest orders aggregated to 100,000 contracts into one level. The level price will be set to the volume-weighted average price minus 1 point. The volume is 100,000.
- Second level = 300,000, markup = -2. Ultency will collect closest orders aggregated to 300,000 contracts into one level. The level price will be set to the volume-weighted average price minus 2 points. The volume is 300,000. Please note that the volume is always counted from the top of the order book (the price closest to the market), and not from the previous level.
- The same applies for the other levels.
- Level ∞ . This is a technical level. Its price is calculated as a weighted average of all orders used to form the previous level, adjusted by its own markup. It is not visible in the market depth but is used for order execution – all orders that do not fall into the other levels are executed at this price.

Important: If there isn't enough actual volume to fill a level, the full configured volume is still displayed in the market depth.

Let's consider order merging using two symbols as an example: EURUSD.raw, which is the original aggregated market depth with no level settings specified, and EURUSD with level settings used.

MetaTrader 5 trading platform

Symbol: EURUSD.raw

Common

Coverage

Quotes

Execution

Bands

Sessions

\$

Please specify additional DOM levels for the aggregated symbol.

Bid:

Volume

+ click to add

Symbol: EURUSD

Common

Coverage

Quotes

Execution

Bands

Sessions

\$

Please specify additional DOM levels for the aggregated symbol.

Bid:

Volume	Markup
10 000	
50 000	
200 000	1
1 000 000	1
5 000 000	2
20 000 000	3
50 000 000	4
∞	5
+ click to add...	

Ask:

Volume	Markup
10 000	
50 000	
200 000	-1
1 000 000	-1
5 000 000	-2
20 000 000	-3
50 000 000	-4
∞	-5
+ click to add...	

>>

<<

OK

Cancel

Help

MetaTrader 5 trading platform

The orders of the original market depth will be displayed as follows:

MetaTrader 5 trading platform

EURUSD.raw, Euro vs US Dollar		
Volume	Price	Trade
200.00	1.08330	
	1.08329	
	1.08328	
	1.08327	
	1.08326	
	1.08325	
	1.08324	
	1.08323	
100.00	1.08322	
	1.08321	
19 600 040...	1.08320	
	1.08319	
40.00	1.08318	
10.00	1.08317	
	1.08316	
10.00	1.08315	
1 490 000.00	1.08314	
1 140 000.00	1.08313	13000*1
510 000.00	1.08312	50000*1
50 000.00	1.08311	137000*
13 000.00	1.08310	/ 200000
13 000.00	1.08310	
50 000.00	1.08309	13000*1
	1.08308	50000*1
510 000.00	1.08307	510000*1
1 140 000.00	1.08306	1140000*
1 490 010.00	1.08305	1490010*
	1.08304	50*1.083
50.00	1.08303	40*1.083
	1.08302	1796900*
40.00	1.08301	/ 500000
19 600 000...	1.08300	
	1.08299	
100.00	1.08298	
	1.08297	
	1.08296	
	1.08295	
	1.08294	
	1.08293	
	1.08292	
	1.08291	
200.00	1.08290	

[illegible]

Sessions

Define quote and trade sessions for the aggregated symbol. These sessions can be independent of provider-side sessions, but make sure providers collectively cover the time to avoid gaps in quotes and execution.

Symbol: EURUSD

Common Coverage Quotes Execution Bands Sessions

 The setting up of trade and quotation sessions of the symbol by days. During the quotation session it is possible to view the price dynamics but trading is prohibited.

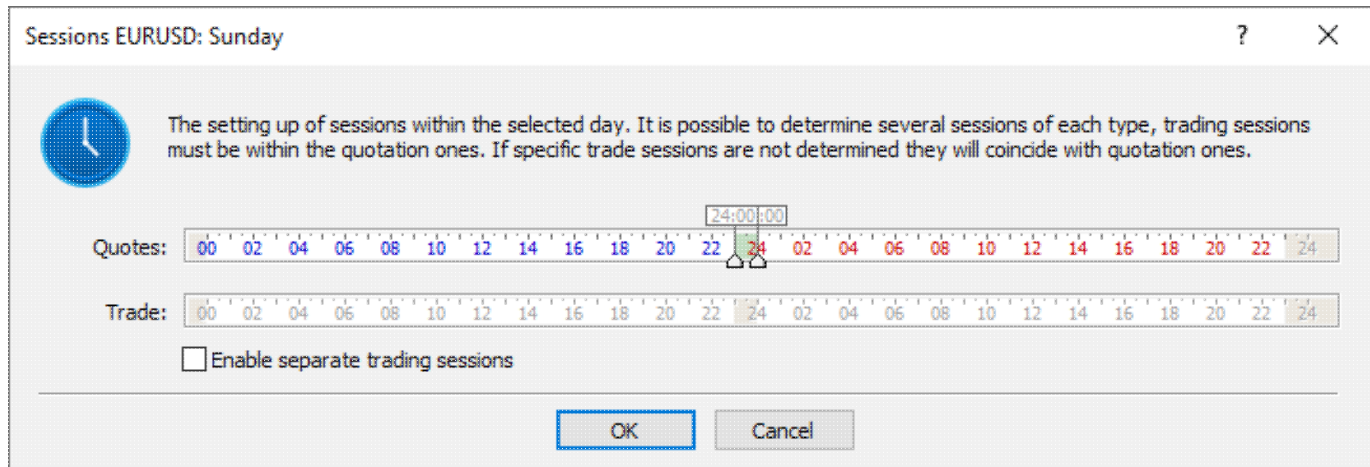
Day	Quotes	Trade
 Sunday	23:00-24:00	23:00-24:00
 Monday	00:00-24:00	00:00-24:00
 Tuesday	00:00-24:00	00:00-24:00
 Wednesday	00:00-24:00	00:00-24:00
 Thursday	00:00-24:00	00:00-24:00
 Friday	00:00-23:00	00:00-23:00
 Saturday		

Edit

OK Cancel Help

MetaTrader 5 trading platform

Click on a day to adjust its session settings:



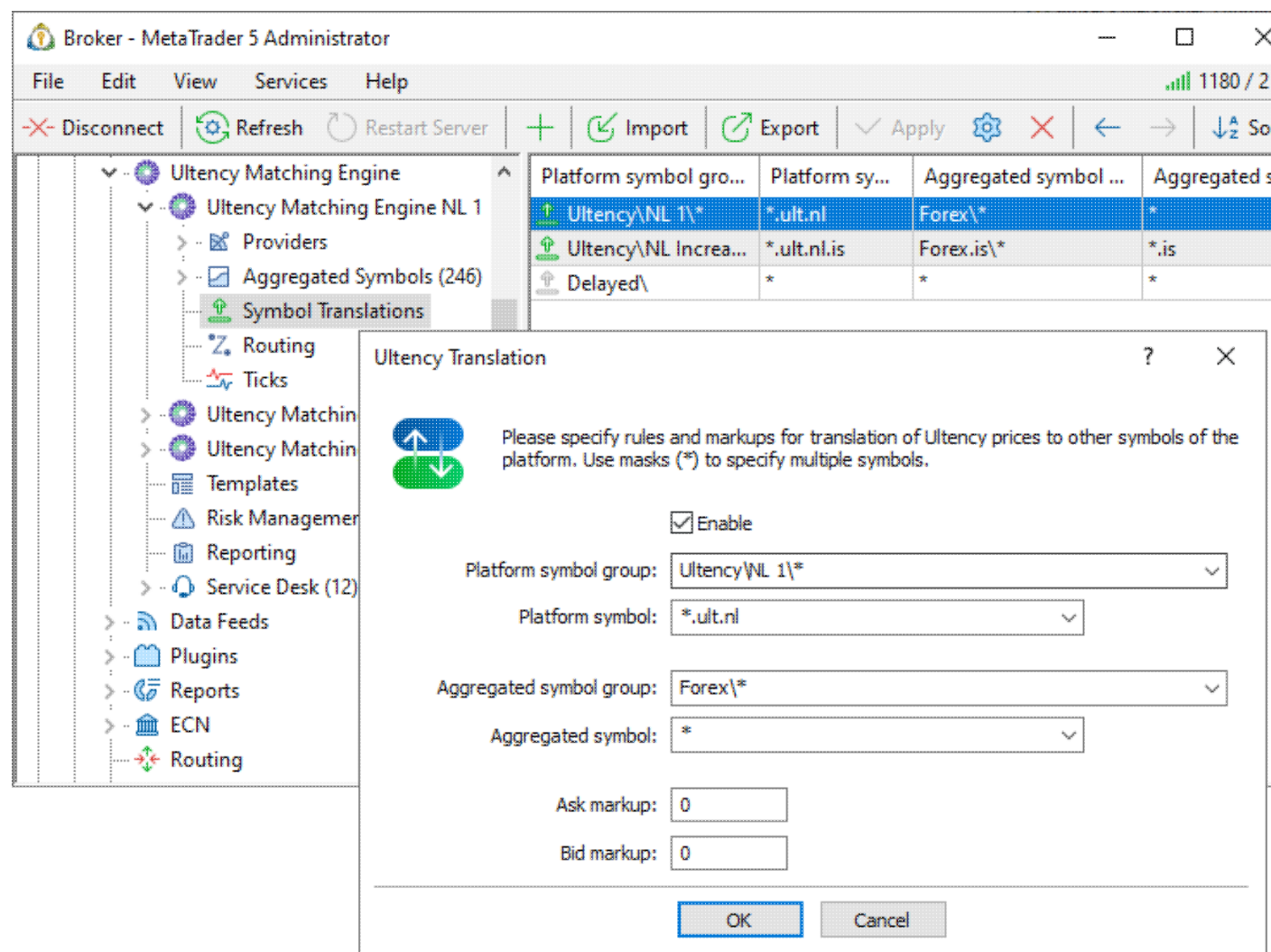
To configure the time of quote and trading sessions separately, enable the corresponding option. Next, set up the sessions. The sliders that control session beginning and end times can be moved with the mouse and with the arrow keys. You can hold down the Shift key to slow down the slider speed. This enables precision up to minutes when configuring sessions.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Translations of Symbols and Quotes

Translations of Symbols and Quotes

Use the "Translation" section to map aggregated symbols in Ultency to trading symbols on the MetaTrader 5 platform.

Your traders do not have direct access to Ultency symbols. They can only trade using the platform's trading symbols, which are configured in the respective section of the platform. To ensure quotes flow from Ultency to the correct symbols in your platform, set up symbol translation rules under.



MetaTrader 5 trading platform

By creating multiple translation rules, you can retransmit Ultency quotes into several sets of symbols simultaneously. For example, with different markups, trading conditions, or delivery modes (real-time or delayed).

Translation rule parameters:

- Platform symbol the symbol that will receive data from Ultency. This symbol must be pre-created in the platform.
- Aggregated symbol the aggregated symbol in Ultency from which data will be transmitted.
- Bid and Ask markups the number of points to adjust the corresponding prices. Positive values increase, negative values decrease the price. The value is in points of the source symbol's price. If multiple symbols fall under one group rule but have different precisions (number of decimal places), the markup will be applied according to each source symbol's precision. For example, if a symbol has 4 decimal places, a markup of "1" will adjust the price by 0.0001; if a symbol has 5 decimal places, a markup of "1" will adjust the price by 0.00001.

You can use the wildcard mask "*" for bulk retranslations and conversions of source and destination symbol names. In this case, all pairs of source-target symbols that have matching patterns represented by the mask will be retransmitted. The "*" mask is also used to link symbols based on patterns. For example:

- If aggregated symbols are specified as "*" and platform symbols as "*.ult.nl", all aggregated symbols will be retransmitted to platform symbols with the same names and a ".ult.nl" suffix. Example: if the aggregated symbol is EURUSD, the corresponding platform symbol will be EURUSD.ult.nl.
- If aggregated symbols are specified as "*.is" and platform symbols as "*.ult.nl", all aggregated symbols ending with names ending in ".is" will be retransmitted to platform symbols with the same names, replacing ".is" with ".ult.nl.is". Example: if the aggregated symbol is EURUSD.is, the corresponding platform symbol will be EURUSD.ult.nl.is. If the aggregated symbol is EURUSD, it will not be retransmitted to the platform.

The parameters "Platform symbol group" and "Aggregated symbol group" serve as additional filters. For example, if the platform symbol is specified as "*.ult.nl", this means "all symbols ending with .ult.nl", regardless of their folder location. For example:

- Ultency\NL 1 symbols with standard conditions.
- Ultency\NL Increased spread symbols with increased spreads.

If you want to restrict translation only to "*.ult.nl" symbols in a specific folder, you can set a path in "Platform symbol group", such as "Ultency\NL 1*".

- Each aggregated symbol must have at least one translation rule configured. Otherwise, traders will not receive quotes from Ultency.
- You can translate quotes from a single Ultency symbol to multiple MetaTrader 5 symbols. However, each platform symbol can only have one

MetaTrader 5 trading platform

Ultency aggregated symbol. If multiple translation rules are set for the same platform symbol, only the first rule will be applied. If you have multiple Ultency servers, and they contain translation rules for the same MetaTrader 5 symbol, only the first rule from the first server will be used.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Routing

Routing

To ensure client trading requests are forwarded to Ultency for execution, define routing rules. In the Common settings, specify which requests should be routed to Ultency. For example, all requests for symbols matching Ultency*. Select "Send to Ultency" for the action and specify the Ultency server that will handle execution.

Routing: Ultency Execution

Common Dealers

 Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action:

Where request is:

Where order is:

Where conditions are:

Type	Condition	Value
ab Request\Symbols	Equal (=)	Ultency*

Add Edit Delete

OK Cancel Help

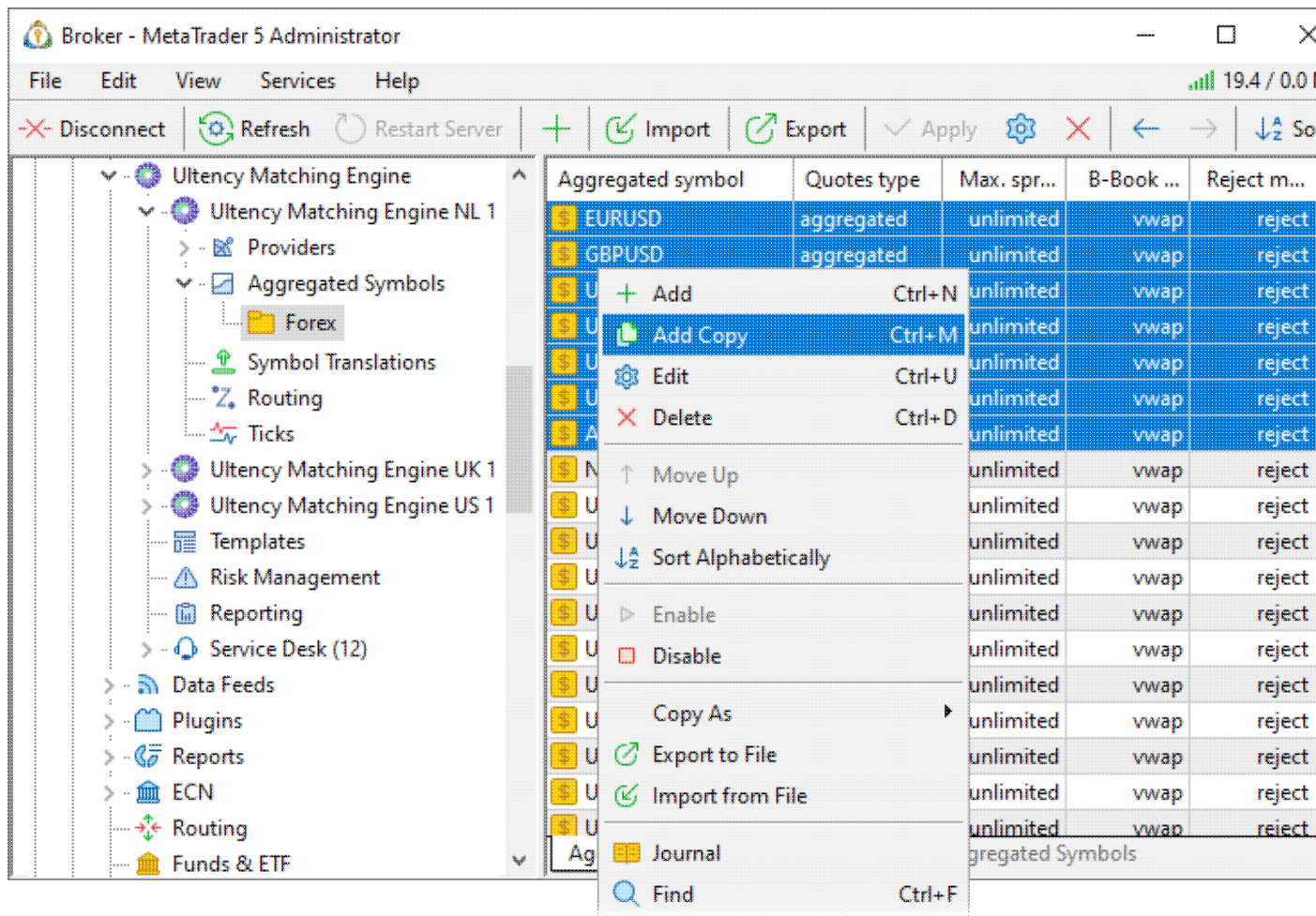
MetaTrader 5 trading platform

Requests forwarded to Ultency in accordance with the routing rules will be processed according to Ultency's internal rules.

Internal Routing in Ultency

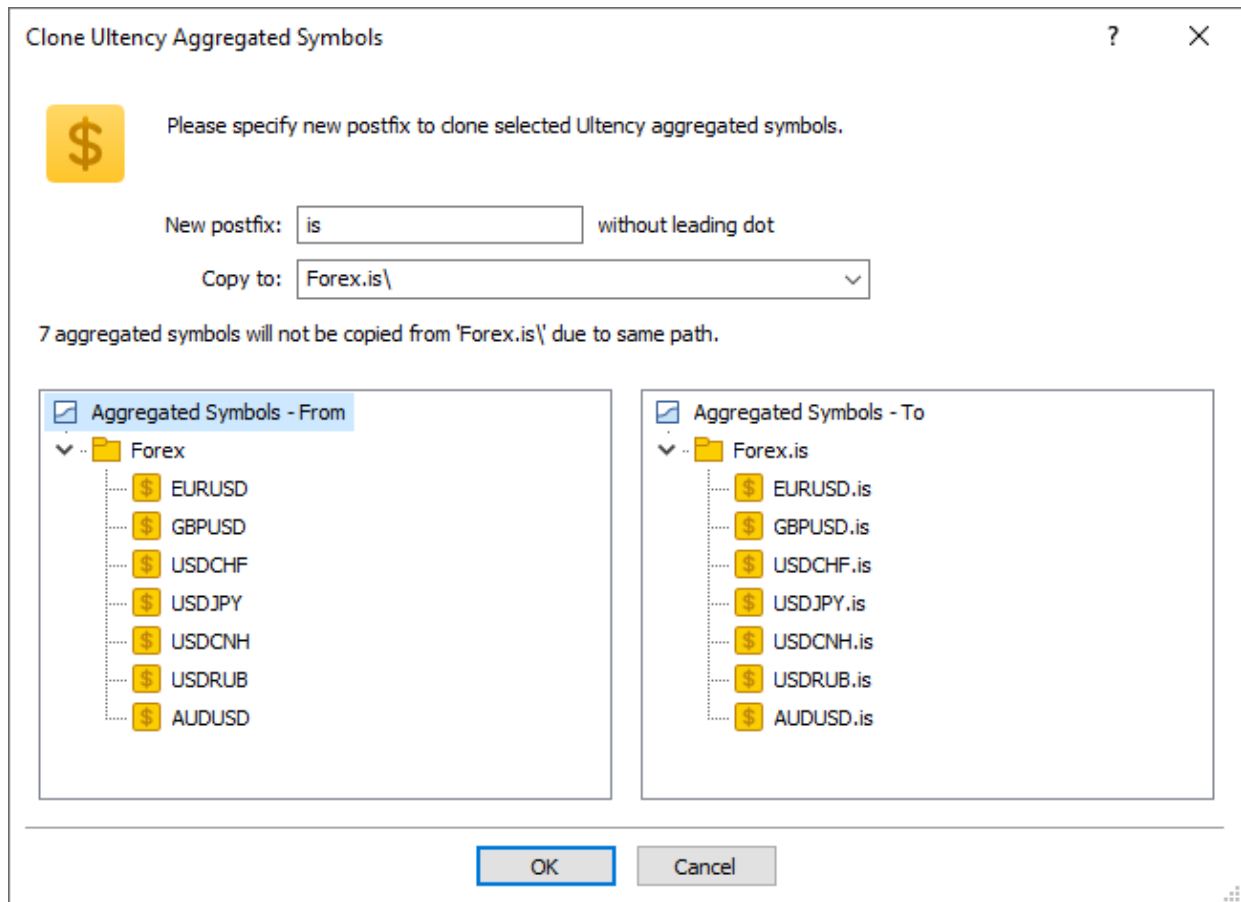
To define how trading requests are processed inside Ultency, use the "Routing" section in the settings of the respective Ultency server. Depending on various conditions (symbol, volume, trader group, etc.), you can manage trade execution flexibly, such as routing orders to different providers or rejecting them.

To implement different execution conditions for the same symbols, create separate sets of aggregated instruments. Select the required symbols from the list and choose "Add Copy" from the context menu.



MetaTrader 5 trading platform

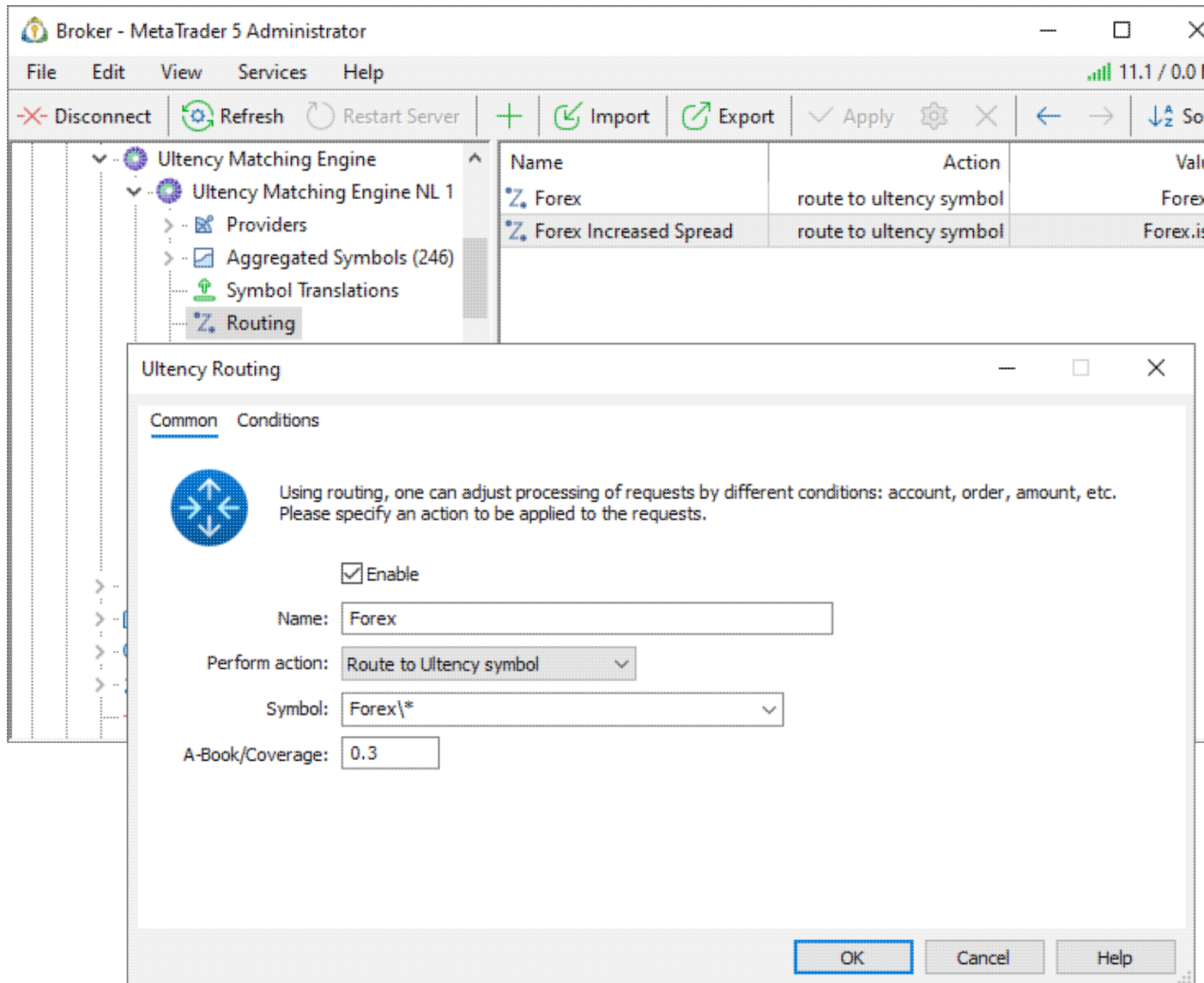
Next, define a postfix that will be added to all symbol names in the new set. You should also specify a subfolder name where these symbol copies will be placed.



MetaTrader 5 trading platform

After creating copies, adjust their settings as needed.

Once you have prepared the necessary sets of aggregated symbols, proceed to configure routing rules:



Create a rule and specify the following parameters:

- Name name of the routing rule.
- Perform action action to be taken when a request matches the rule conditions.
 - ◆ Route to Ultency symbol the request will be executed based on the settings of the specified aggregated symbol.
 - ◆ Reject the request will be rejected.
- Symbol aggregated symbol according to which the request will be executed. For convenience, symbols can be specified as subgroups. For example, Forex*. Based on the symbol in the incoming request, Ultency will automatically select a matching aggregated symbol from this subgroup. See symbol selection logic below.
- A-Book/Coverage portion of the order to be sent to an external provider (A-Book share). 0 indicates that the entire order will be executed internally (B-Book), 1 the entire order will be routed to an external liquidity provider as per execution settings.

Rule Conditions

To define which requests should be processed according to the rule, specify a set of conditions:

Ultency Routing

Common Conditions

Please specify detailed conditions for the request processing rule. For the rule to be applied, all conditions must be met.

Type	Condition	Value
Request\Symbol	Equal (=)	Ultency\NL 1*
Schedule\Date and time	Equal (=)	2024.10

Conditions tree:

- Conditions
 - Schedule
 - Date and time
 - Date
 - Time
 - Accounts
 - Request
 - Position
 - Order

Buttons: Add, Edit, Delete, Help

A detailed description of available conditions is provided in a [separate section](#).

Rule Execution Order

Rules are executed from top downwards. If an incoming request matches the conditions of the first rule, it will be processed according to that rule. If it doesn't match, the system checks the next rule, and so on.

To reorder rules, use the "↑ Move Up" and "↓ Move Down" options in the context menu or the corresponding commands in the [Edit](#) menu and [Standard Toolbar](#).

How Aggregated Symbols Are Selected

Routing rules enable different handling of operations on the same instrument, depending on the conditions. For example, EURUSD orders from clients in the real/regular group may be routed to one liquidity provider, while forwarding orders from real/vip clients to another. To implement such differentiation, create separate sets of aggregated symbols as mentioned earlier. Then, define a specific routing rule for each client group, indicating the corresponding subgroup of aggregated symbols as the symbol.

Example: Assume that Ultency has two sets of aggregated instruments:

- Forex\Liquidity Provider 1*. The subgroup contains EURUSD.lp1 which has EURUSD set as its [basis symbol](#).
- Forex\Liquidity Provider 2*. This subgroup contains EURUSD.lp2 which also has EURUSD as the basis symbol.

Each symbol is configured to [execute](#) through its respective liquidity provider.

Through [symbol translation settings](#), both subgroups are mapped to a single symbol set on the trading platform side:

- Forex\Liquidity Provider 1* -> Ultency\Forex*
- Forex\Liquidity Provider 2* -> Ultency\Forex*

As a result, the trading platform features a single symbol, Ultency\Forex\EURUSD, available for client trading. A routing rule is then set up as follows:

MetaTrader 5 trading platform

Ultency Routing

Common Conditions

Using routing, one can adjust processing of requests by different conditions: account, order, amount, etc.
Please specify an action to be applied to the requests.

☒ Enable

Name:

Perform action:

Symbol:

Coverage: **Ultency Routing**

Common Conditions

Please specify detailed conditions for the request processing rule. For the rule to be applied, all conditions must be met.

Type	Condition	Value
Accounts\Group	Equal (=)	real\vip
Request\Symbol	Equal (=)	Ultency\Forex*

Add Edit Delete

OK Cancel Help

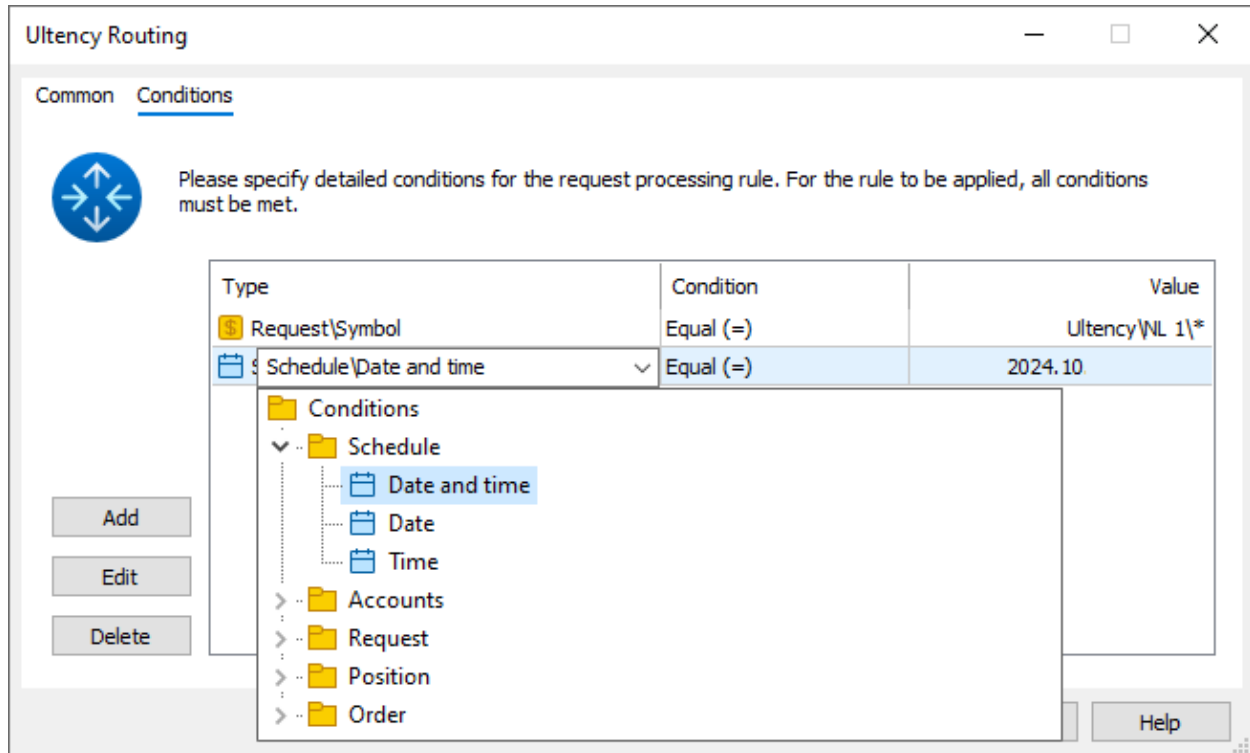
How the selection works:

1. When a trading request for EURUSD is received, the system checks the routing rules from top to bottom and identifies the first rule that matches the request conditions. Let's say the rule specified above applies.
2. The system checks the group of aggregated symbols specified in the rule. In this case, it is Forex\Liquidity Provider 1*.
3. Ultency checks the list of instruments in the Forex\Liquidity Provider 1* subgroup and selects the one whose basis symbol matches the symbol in the request EURUSD. In this case, it is EURUSD.lp1. This symbol will be used to execute the client's order.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Routing /
Conditions

Routing Conditions

The routing rule settings specify a set of conditions defining which trade requests will be processed according to this rule.



Comparison Types

Standard comparison types are available for all conditions:

- Equal (=)
- Not equal (!=)
- Greater than (>)
- Greater than or equal (>=)
- Less than (<)
- Less than or equal (<=)

For string conditions, such as country name, symbol, or email, the Match Mask condition is also available. Here you can specify strings with the "*" mask. For example:

- Condition "Symbol", value "Forex\USD*" all symbols from the Forex group, where the underlying currency is USD.
- Condition "Email", value "*@mailbox.com" applies to all email addresses on the mailbox.com domain.

For login conditions, you can specify multiple values and value ranges. For example: Account\Login = 100,102,107,120-130.

Time

These parameters set date and time conditions. A rule will be triggered if a request arrives on the specified date or within the specified time range.

Accounts

These settings specify conditions based on account parameters:

- Login the trading request was submitted from the specified account.
- Group, Country, City, Language, Phone, Email, Color, Status, Company the trading request was submitted from an account that belongs to the specified group, country, city, etc.
- Comment the comment added by the client to the trading request (order).
- Registration the condition is set relative to the account creation date in the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for accounts registered on the specified date, earlier or later than this date.
- Last visit the condition is set relative to the last connection of the account to the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for account connected on the specified date, earlier or later than this date.
- Days since registration the condition is set relative to the number of days that have passed since the account was created. The number of days is calculated based on the number of times the clock crosses midnight (00:00) since the event. The principle of operation and purpose of this condition are similar to the "Registration" condition.

MetaTrader 5 trading platform

- Days since last login the condition is set relative to the number of days that have elapsed since the last connection to the account. The number of days is calculated based on the number of times the clock crosses midnight (00:00) since the event. The principle of operation and purpose of this condition are similar to the "Last visit" condition.
- Days since last trade activity condition is set relative to the number of days that have elapsed since the last operation on the account. This considers any trading operations, commission charges, etc. Balance operations are not taken into account. In addition, the system checks if the account has any open positions or active pending orders. When using this condition, please note that newly registered accounts may also fall under this condition. For example, a trader may have created an account 1-2 days ago and has not yet performed any trading operations. If you set a condition based on the absence of trading activity in the last 180 days, this trader will also be included in the automation action. To avoid such situations, use the trading activity condition in conjunction with other conditions, such as "Registration" or "Days since registration". The calculated number of days is rounded down. For example, if 5 days and 23 hours have passed since the event, this will be counted as 5 days.
- Online condition is set relative to the status of account connection to the trade server. Possible values are "true" and "false", i.e. the account is either connected (any connection type: client terminal, mobile terminal and so on) or not.
- Leverage the condition is set relative to the account leverage. For example, the regulator's requirements have changed, and you need to disable trading with a leverage greater than 1:20 on accounts from the US. Set two conditions: "Country = USA", Leverage > 1:20. Assign the action "Decline" to this rule.
- Balance the condition is set relative to the current account balance, the amount is indicated in the deposit currency.
- Credit is similar to the "Balance" condition. In this case, the amount of credit funds on the account is checked.
- Total positions the condition is set by the number of open positions which currently exist on the account. This action allows you to control extremely active accounts. For example, set "Total positions > 100", "Group = real*". Assign the action "Decline" to this rule.
- Total orders similar to the "Total positions" condition, it checks the number of active pending orders on the account.
- Floating profit, Equity, Margin, Free margin, Margin level these conditions are similar to the "Balance" condition. Appropriate trading account states are checked here.
- Deposit currency the request arrived from an account with the specified deposit currency.
- Lead Source, Lead Campaign lead source and campaign parameters specified in the account.
- Enabled the condition is specified relative to the "Enable this account" setting.
- Trading enabled, Algo trading by Expert Advisors enabled the conditions are set relative to the corresponding trading settings.
- Agent account the condition is set relative to the agent account specified in the trading account settings.
- Own funds percentage when trading, clients are able to use their own funds they deposited on their accounts, as well as the credit and bonuses provided by a broker. Both types of funds increase their Equity parameter. The "Own funds

percentage" condition allows configuring the automation task depending on the own funds share on a client's account. 100% means the account has client's funds only with no credit and bonuses. The value of 0% indicates that only credit and bonus funds are used for trading.

The value is calculated as $(1 - (\text{Credit} / \text{Equity})) * 100$.

The fractional part is rounded down during calculation: 1.5% becomes 1%, 0.9% becomes 0%. The parameter value cannot be greater than 100% or less than 0%. If an action is required after the client has spent all their funds, it is more efficient to use the "Own funds volume" condition.

- **Own funds volume** the parameter works similarly to the previous condition, although it applies an absolute own funds value rather than a relative one. Specified in the deposit currency. The calculation equation: $\text{Equity} - \text{Credit}$.

Request

These settings define conditions based on the parameters of the trading request received from the client.

- **Symbol** a symbol or a group of symbols for which the rule will apply. You can use masks containing "*" and "!" to define symbols.
- **Volume** the trade volume requested in the order (in lots). This parameter allows configuring rules based on the requested volume, for example, automatically processing requests with a volume less than 1 lot.
- **Comment** allows comparing the request comment with a specified value. When condition "=" is specified, an exact match is checked. Condition ">" or ">=" searches for the specified substring in a comment line. Condition "<" or "<=" searches for the comment substring in the specified line.
- **Reason** the reason for the creation of the request: whether the order was submitted by a client, Expert Advisor, dealer, etc.
- **Price** the price in the request. In Instant and Request execution modes, this is the price specified by the trader in the order. In Market and Exchange modes, this is the current market price of the instrument. Since prices vary significantly between instruments, it is recommended to use this condition in combination with the "Symbol" condition. It can also be used to set the general price threshold. For example, you can decline all requests with a cost of less than one dollar.
- **Value** the value of the requested operation in the symbol's base currency. Value calculation depends on the symbol's margin/profit calculation mode.

Position

These settings define conditions based on the parameters of the position affected by the received trading request.

- **Ticket** unique position ticket.
- **Volume** the current position volume for the symbol specified in the request.
- **Type** position type: buy or sell.

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- Open price weight-average price of the position opening: $(\text{price of deal 1} * \text{volume of deal 1} + \dots + \text{price of deal N} * \text{volume of deal N}) / (\text{volume of deal 1} + \dots + \text{volume of deal N})$.
- Current price the price of the financial symbol for which the position has been opened, at the trigger activation time.
- Current profit the current floating profit from the position for which the request was received. Specified in the client's deposit currency.
- Reason reason for position opening.
- Creation time position opening time.
- Days since creation time passed since the position was opened.
- Update time time when the position was last modified (when its volume was changed).
- Days since modification the number of days passed since the last position modification (a change in its volume).
- Expert ID identifier (magic number) of an Expert Advisor by which a position was opened in the client terminal.
- Comment a text comment to the position.

Order

These settings define conditions based on order parameters.

- Ticket the unique order ticket.
- ID order identifier in the external system.
- Setup time time of order placing by a client.
- Days since setup the number of days that have elapsed since the client placed the order. The calculated number of days is rounded down. For example, if the order was placed 3 days and 22 hours ago, this will be counted as 3 days.
- Expiration time order expiration date, if it was set by the client.
- Type order type: "Buy", "Sell", "Buy Limit", "Sell Limit", "Buy Stop", "Sell Stop", "Buy Stop Limit", "Sell Stop Limit" or "Close By".
- Order price price specified by trader for execution of the order.
- Trigger price this field is used for the "Buy Stop Limit" and "Sell Stop Limit" orders. It sets the price level at which the orders trigger and the relevant limit orders are placed.
- Current price the price of the financial symbol for which the order has been opened, at the request arrival time.
- Stop Loss the Stop Loss level.
- Take Profit the Take Profit level.
- Initial volume volume requested in the order.
- Remained volume if the order is not filled in the volume requested by trader this field will display the remainder volume.
- State current order state: Started, Placed, Partially Filled, Rejected, Filled, etc.
- Expert ID identifier (magic number) of an Expert Advisor which placed the order in the client terminal.
- Position the ticket of the position that will be modified or closed as a result of the execution of this order.

MetaTrader 5 trading platform

- **Comment** a text comment to the order.
- **Contract size** the contract size of the symbol, for which an order was placed.
- **Currency** the deposit currency of the client who has placed the order.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Ticks

Ticks

This section allows you to view the tick history for aggregated symbols all quotes received from liquidity providers that affected the best Bid and Ask prices.

The screenshot shows the MetaTrader 5 Administrator window. The left sidebar contains a tree view with the following structure:

- Payments
- Gateways
- Ultency Matching Engine
 - Ultency Matching Engine 1
 - Providers
 - Aggregated Symbols (93)
 - Symbol Translations
 - Routing
 - Trades
 - Quotes & Books
 - Ticks** (selected)
 - Execution Books
 - Liquidity Offers
 - Templates
 - Risk Management
 - Reporting
 - Service Desk (12)
 - Data Feeds
 - Plugins

The main panel displays a table of tick data. The table has three columns: Time, Bid, and Ask. The data is filtered for the date 2025.03.12. The table shows a series of ticks with their corresponding Bid and Ask prices.

Time	Bid	Ask
2025.03.12 00:00:36.463145	1.09138000	1.09156000
2025.03.12 00:00:38.703127	1.09137000	1.09154000
2025.03.12 00:00:41.664125	1.09136000	1.09154000
2025.03.12 00:00:42.503128	1.09134000	1.09151000
2025.03.12 00:00:48.518130	1.09135000	1.09152000
2025.03.12 00:00:54.583137	1.09134000	1.09152000
2025.03.12 00:00:54.583137	1.09134000	1.09152000
2025.03.12 00:00:54.916132	1.09134000	1.09151000
2025.03.12 00:00:55.616156	1.09133000	1.09151000
2025.03.12 00:00:59.377131	1.09133000	1.09150000
2025.03.12 00:01:09.164152	1.09137000	1.09154000
2025.03.12 00:01:11.300135	1.09138000	1.09155000
2025.03.12 00:01:11.651129	1.09137000	1.09154000
2025.03.12 00:01:15.212127	1.09138000	1.09155000
2025.03.12 00:01:15.566132	1.09137000	1.09154000
2025.03.12 00:01:18.247137	1.09138000	1.09156000

At the bottom of the table, there is a search bar and a filter section. The filter section includes a dropdown menu set to 'Today', a date range from '2025.03.12 00:00' to '2025.03.12 23:59', and a 'Request' button.

Each tick may contain the following data:

- **Date** date and time when the tick was received.
- **Bid** the Bid price.
- **Ask** the Ask price.
- **Last** price of the last executed deal.
- **Volume** volume of the last performed transaction.
- **Direction** direction of the deal as a result of which the tick was created: Buy or Sell.

Requesting Tick Data

To request ticks for a symbol, follow these steps:

- **Symbol Selection**

In the first field, specify one of financial symbols available in the system. You can either enter the symbol manually or select it from the list by clicking the drop-down arrow.

- **Period Selection**

Specify the time period for which you want to request ticks. You can select one of the predefined periods by clicking on  (today, last 3 days, last week, last month, last 3 months, last 6 months or the entire history) or define custom period.

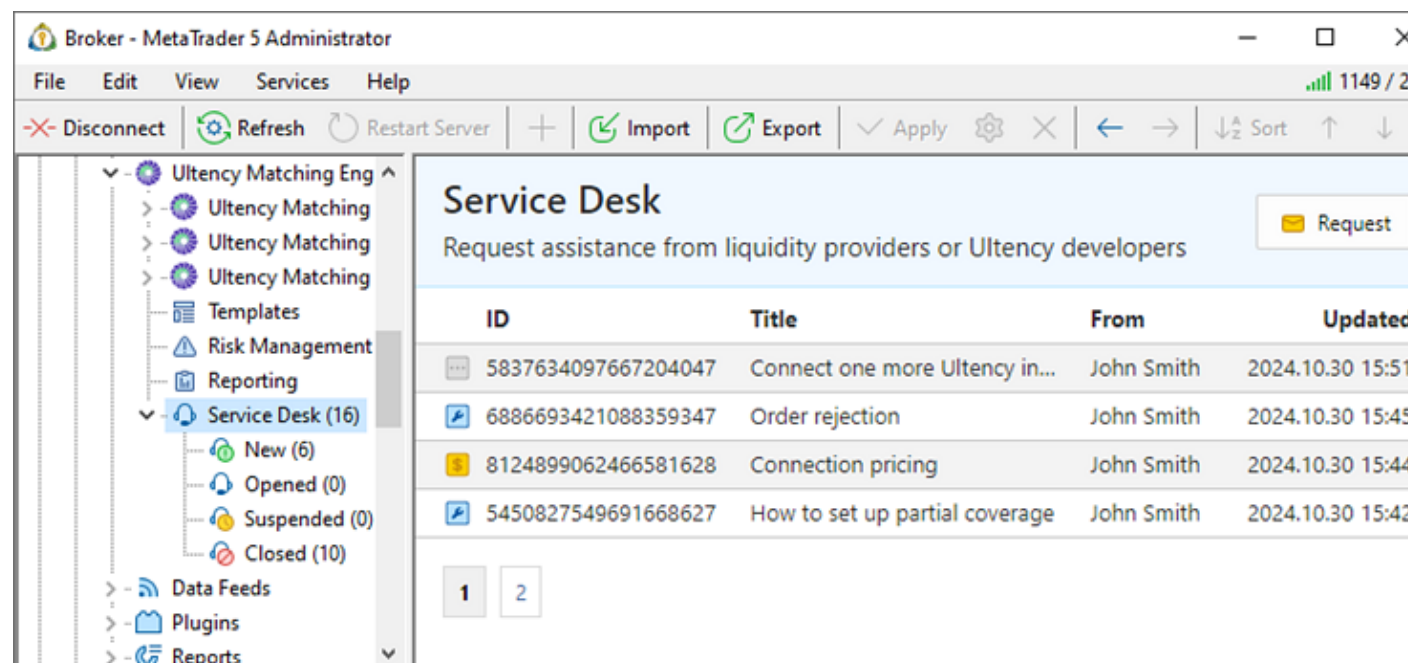
- **Request Execution**

To retrieve the tick data, click the "Request" button or use the corresponding context menu command .

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Service Desk

Service Desk

Service Desk is a built-in support and communication system within Ultency, designed to simplify interactions with liquidity providers. Whenever you encounter any questions or issues, there's no need to search for contact information. Simply reach out to the provider directly through Service Desk. You can also use it to contact Ultency developers for any technical assistance.



All requests to providers are organized into categories for convenience:

- **New** requests that have not yet received a response.
- **Open** ongoing conversations and unresolved requests.
- **Suspended** requests that have been postponed for various reasons.
- **Closed** requests that have been resolved and where correspondence is completed.

Creating a Request

To contact a liquidity provider, click "Request". Select the provider to whom the request will be sent, choose a category, and briefly describe the issue in the subject field. Then, provide a detailed description of the issue. If necessary, attach additional files such as logs, screenshots, etc. The more detailed your description, the faster you will receive a solution.

Create a request

Provide as many details as possible to help us solve your problem quickly

From:	John Smith	To:	Ultency Support Team 
	Technical 		Ultency Support Team
Title:	Financial		Just2Trade
	Technical		MetaQuotes-Demo
	Other		FXCM Prime
			PrimeXM

Hello,

I need to execute only 50% of order volume using liquidity providers and the rest with the platform. How can I achieve that?

 Attach file

MetaTrader 5 trading platform

If the issue requires urgent attention, click ⓘ to the right of the subject field. The request will be marked as priority and displayed with a corresponding label in the list.

Correspondence

To view a provider's response or add more details, open the request.

Order rejection

From: John Smith To: Lime Trading (CY) Ltd |  Technical | 2024.10.30 15:45 | #688669342108 Close

My orders get rejected all the time

Add comment

</>

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Please find attached the log file

1. 📁 20241030.zip / 1.16 kb ×

📎 Attach file 🗑️ Clear attachments list Add Cancel

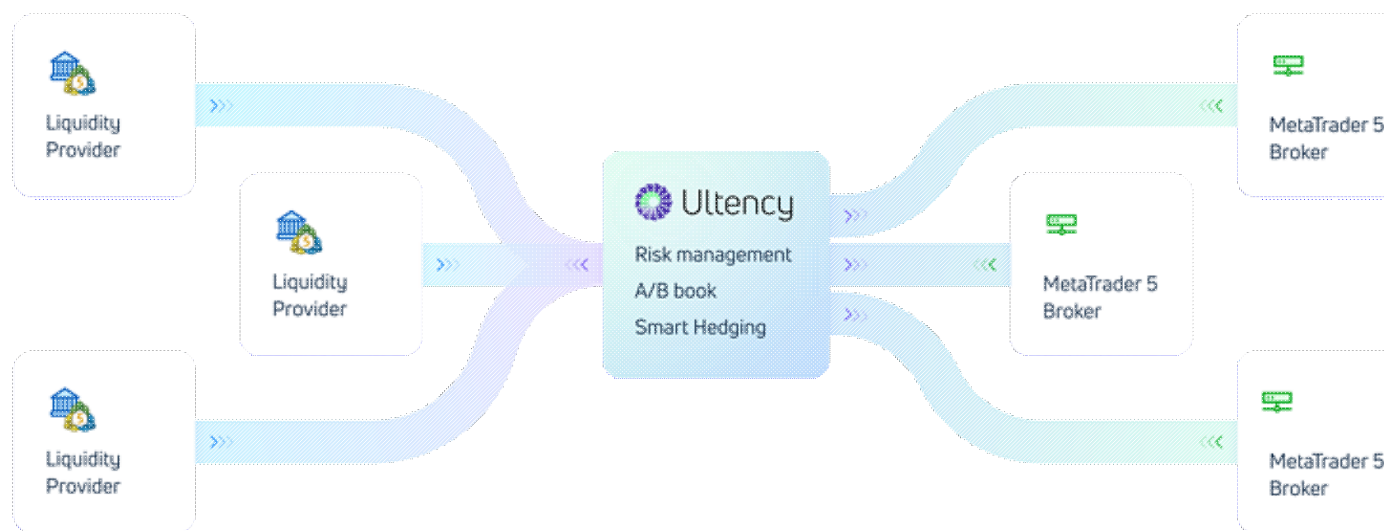
MetaTrader 5 trading platform

Once the issue is resolved, click "Close" in the top-right corner. The request will then be moved to the Closed category.

MetaTrader 5 Trading Platform / Platform Setup / Ultency / Become a Provider

Become a Liquidity Provider in Ultency

Become a liquidity provider and generate income from executing trades on behalf of other brokers. You will earn direct commissions on executed trades while simultaneously increasing the volume of trading operations.



Access a Vast Pool of Potential Clients

Ultency provides access to an extensive audience of potential clients. Your offering will be showcased within the Administrator terminals of all MetaTrader 5 brokers. Additionally, Ultency ensures seamless interaction with clients through an integrated communication system. Negotiations and technical issue resolutions can be conducted directly within MetaTrader 5 Administrator.

Become a Provider in Just 5 Minutes

Registering with Ultency takes only a few minutes. Complete a short company profile and outline your terms, including available instruments, trading volumes, fees, and more. As a decentralized system, Ultency enables you to select your regions of operation, ensuring minimal execution latency for connected brokers.

How to Get Started

- Register as a liquidity provider and await approval
- Create offers detailing your terms
- Upon receiving a request and finalizing terms with a broker, open a standard trading account in MetaTrader 5 for that broker
- The broker will connect via Ultency using the designated account to process client trades through it
- Monitor account activity using all available MetaTrader 5 tools: view current positions and trading history, create reports

Registering as a Liquidity Provider

Navigate to the "Ultency\Liquidity Offers" section and complete a brief registration form with your company details:

Liquidity Offers

Overview
 Brokers

Become a liquidity provider and start earning

Become a liquidity provider and earn commission from executing deals. Your offer will be displayed in the administrator terminals of all MetaTrader 5 brokers. Fill out a simple form with your company information, and describe your offer: available instruments, volumes, commissions, etc. Since Ultency is a distributed system, you can choose regions where you offer is available, to ensure minimal latency for brokers.

Provider name	Liquidity Provider		
Logotype	Choose file	200x200 px, JPG or PNG	logo.png ✕
Regulations	CySec		
Countries	EU		
Address	1 Anexartisias str, 3036		
Description	As a trusted brokerage company, we are excited to offer our services as a liquidity provider on this leading platform. We ensure deep market depth, tight spreads, and fast execution, giving traders access to high-quality, reliable liquidity across a variety of asset classes. With advanced technology and robust risk management strategies, we guarantee seamless and efficient trading, supporting both retail and institutional		

Create a request

To register, provide the following:

- Company name
- Logo 200*200
- List of regulatory licenses: CySec, FCA, NFA, etc.
- Countries where you provide services
- Company address and contact information: phone, website, email
- Links to legal documents: business terms, service agreements, order execution policy
- A brief company and service description for display in the marketplace

After completing the form, click "Create a request" to send your application for review. If further details are needed, our team will contact you.

Liquidity Offers

 Overview

 Brokers



Liquidity
Provider

Liquidity Provider

 www.broker.com  Cyprus  support@broker.com  +35799999999

 Edit

As a trusted brokerage company, we are excited to offer our services as a liquidity provider on this leading platform. We ensure deep market depth, tight spreads, and fast execution, giving traders access to high-quality, reliable liquidity across a variety of asset classes. With advanced technology and robust risk management strategies, we guarantee seamless and efficient trading, supporting both retail and institutional clients. Partner with us for unparalleled liquidity and a competitive edge in the global financial markets.



Your application for registration is under review

We will notify you about the status soon. If you have any questions, please contact [Service Desk](#), [Create offers](#) describing your prices and conditions right now without waiting approval. This will allow you to start as quickly as possible.

 Create Offer

MetaTrader 5 trading platform

While your application is under review, you may edit your company details at any time.

You can also start creating service offers immediately, without waiting for the verification to complete. This will allow you to get started faster.

Creating Offers

An offer is a description of your services: prices, conditions, available tools, etc. This information will be displayed in the [list of liquidity providers](#) enabling brokers to evaluate and choose their preferred providers.

Click "Create Offer":

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Provide the following information:

- Specify the zones where your offer will be available. Zones are geographic locations where other brokers' Ulteny servers may be located. From these locations, brokers will connect to you to receive liquidity. Select zones depending on where your platform is located.
- Service name.
- The minimum deposit required for a broker's account to process their traders' operations.
- The minimum monthly fee a base fee charged to brokers regardless of liquidity usage. Additional pricing configurations are provided below.
- Minimum turnover (volume of processed deals) in millions of currency units.
- The available currencies in which brokers can maintain accounts.
- Required regulatory compliance CySec, FCA, NFA, etc.
- Supported trading instruments.
- Link to additional service information.
- Brief description of the service.

In the block below, define our liquidity provision rates. You may charge commissions based on trade volume or per operation.

Instruments	USD	Volume, mln	Commission, USD/mln	Type	Min, USD	Max, USD
Forex		0 – 5	5	per monthly volume ▾	1	10
		5 – 10	4	per monthly volume ▾	1	10
		10 – 20	3	per monthly volume ▾	1	10
		20 – 1000	2	per monthly volume ▾	1	10
 Add section						
Save						

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Add a section to describe fees for a specific group of instruments, such as Forex, Crypto, etc. Multiple sections can be added.

Define commission levels within each section based on volume or turnover. To add a new level, hover over an existing row and select "Add Row".

- **Instruments** description of instruments to which commissions apply.
- **Volume** volume range for the applicable commission rates. Select the units to display the values from the dropdown menu: lots, millions, or trillions (if the volume is specified in nominal value).
- **Commission** commission amount. Select units to display the values from the dropdown menu: currency, currency per million units of volume, percentage of the deal value/turnover, or points.
- **Type** commission type: per trade volume, daily turnover, or monthly turnover.
- **Min/Max** minimum and maximum commission thresholds. The units in which the value is specified depend on how the commission is calculated.
- **Min. monthly fee** a baseline fee charged to brokers regardless of liquidity usage.

The appropriate commission settings must be configured for the group that will include the accounts provided to brokers for liquidity access.

Configuring Broker Account Groups

Brokers will connect to your liquidity services via Ultency using standard trading accounts. To connect a broker, simply create an account for them and provide the login credentials.

Create a dedicated account group with the following settings:

- Access to required trading instruments
- Commission structure in accordance with your offer
- Netting accounting system. Broker accounts will consolidate multiple client trades into aggregate positions supported by the netting system.

MetaTrader 5 Trading Platform / Platform Setup / ECN

Electronic Communication Network

The trading platform allows you to implement your own Electronic Communication Network (ECN). The ECN allows you to:

- Provide liquidity and best execution prices to your clients.
- Profit from the matching of orders and spread widening when forwarding operations to an external system.
- Become a full-fledged liquidity provider for other companies.
- Implement your own trading facility.
- Avoid using expensive third-party solutions.

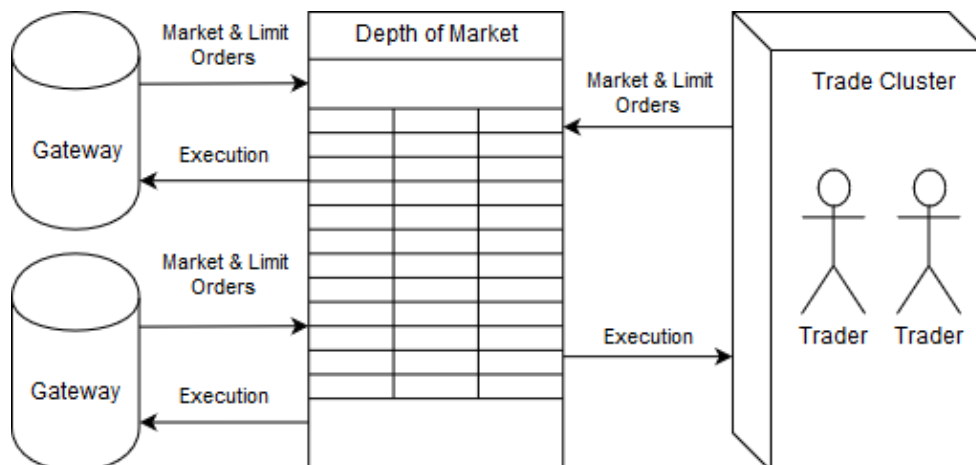
Market Depth formation algorithm

The ECN allows combining data from an unlimited number of external sources and this form a single quotes stream and the Market Depth feature for symbols.

The system receives Market Depth data from gateways and data feeds, aggregates them and uploads to ordinary symbols. The entire operation is performed within the framework of a standard scheme of price data delivery to the platform, so no additional integration is required for the ECN.

Price data aggregation is flexibly configurable. The broker selects data sources and data to be collected. For example, you may choose to use Buy requests from one ECN gateway and Sell requests from a different one. Data can be additionally filtered by request volume, time, etc.

In addition to data aggregation from external sources, the ECN can collect orders placed by clients within a trading cluster. Thus, the broker's clients can see their own market and limit orders in the Market Depth. Separate settings are provided in the ECN for the aggregation of internal orders. The broker determines orders from specific groups or even individual clients which can be added to the Market Depth.

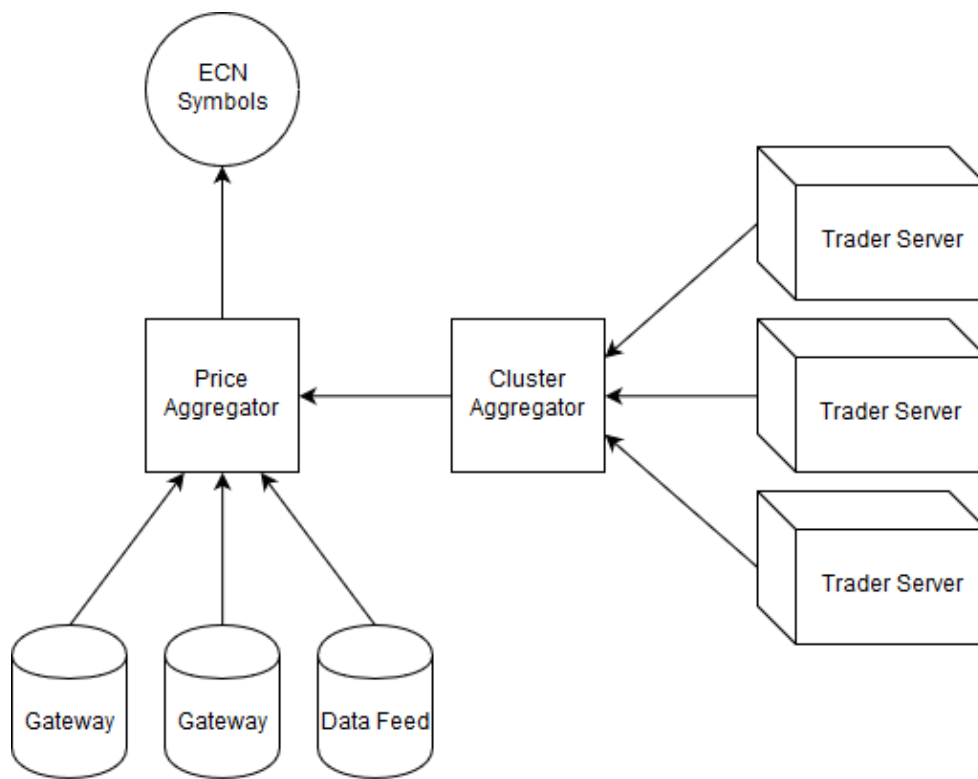


MetaTrader 5 trading platform

Price data in the ECN are aggregated using the special Price Aggregator engine. It directly connects to gateways and data feeds, and collects Market Depth changes from them in accordance with translation settings.

Another engine Cluster Aggregator collects price data from traders within the cluster. During server start, the engine collects the list of all open orders on all trade servers within the cluster and then maintains this data up to date in accordance with all trade transactions performed on servers. Cluster Aggregator notifies Price Aggregator about changes in order status. Based on such notifications, Price Aggregator maintains the Market Depth feature in an up-to-date state.

All received data is recorded in ECN symbols.



Enabling of ECN and operation statistics

The ECN can be enabled or disabled under the Status section. The same section contains general system operating statistics:

- **Total requests** the total number of requests received by the ECN for processing. These include not only market operation requests, but also requests to modify or delete orders.
- **Total active orders** the number of orders currently waiting to be matched (which are currently being processed).
- **Total deals in ECN** the total number of deals, which have been created as a result of order matching within the cluster.
- **Total deals in Gateways** the total number of deals, which have been created as a result of order matching with external liquidity providers.
- **Price ticks** the number of ticks which have been generated by the ECN. In fact, these are the ticks which are created during deal execution (they reflect the Last price changes).
- **Price books** the number of order book changes which have been generated by the ECN as a result of aggregation.

The screenshot shows the MetaTrader 5 Administrator window. The left sidebar contains a tree view with categories like Servers, Integrations, Security, Automations, Time, Holidays, Groups (674), Clients & Accounts, Orders & Deals, Gateways, Data Feeds, Plugins, Reports, ECN, Symbols (17), Translations, Routing, Funds & ETF, and Symbols (84217). The main panel is titled 'Electronic Communication Network' and shows the version as 5.00 build 2470 (29 May 2020), last boot: 2020.05.30 05:39. A green 'Disable' button is in the top right. Below the title, there are statistics for Total requests (45306), Total active orders (8), Total deals in ECN (14453), Total deals in Gateways (13654), Price ticks (12279253), and Price books (6243539). A text block explains that the trading platform allows implementing your own ECN, followed by a list of benefits: Provide liquidity and best execution prices to your clients, Profit from order matching and spread widening for operations forwarded to an external system, Become a full-fledged liquidity provider for other companies, Implement your own trading facility, and Stop using expensive third-party solutions. A link 'How to build your own ECN in a few minutes' is provided. At the bottom, there is a status bar with tabs for Status, Book Journal, Matching, Matching History, Ticks, and Trade. The status bar also shows 'For Help, press F1' and '1208 / 4 Kb'.

Statistic	Value
Total requests	45306
Total active orders	8
Total deals in ECN	14453
Total deals in Gateways	13654
Price ticks	12279253
Price books	6243539

MetaTrader 5 trading platform

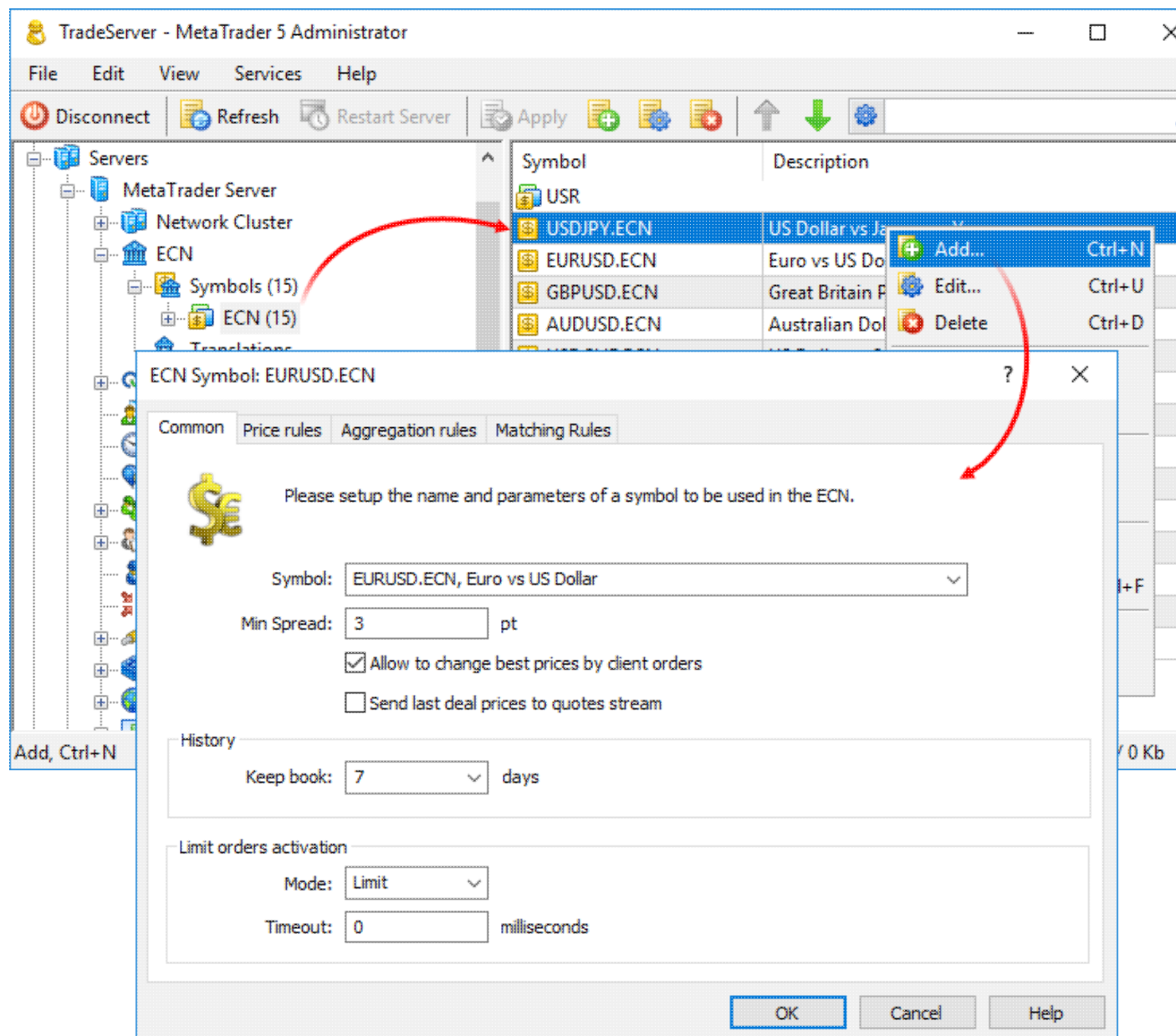
- In case of server restart, the existing ECN operation statistics is reset and collection of statistics starts from scratch.
- When disconnected, the ECN does not collect any price data and it does not process orders, which are forwarded to the ECN in accordance with routing rules.

MetaTrader 5 Trading Platform / Platform Setup / ECN / Forming Market Depth

How the Market Depth is Formed

The ECN receives price data from different sources and aggregates them into a single Market Depth. Such data sources include gateways, datafeeds and the trade servers of the cluster. In other words, the ECN collects data not only from external systems, but also from its own clients.

Traditional symbols are used for collecting data. Create a separate subgroup of symbols and add these symbols to the ECN section. No special configuration is required for symbols used in the ECN. Set parameters similar to normal trading instruments.



Common Settings

In common settings, you should specify an earlier created ECN symbol and the following additional parameters:

- **Minimum spread** the minimum spread to be supported in the symbol's Market Depth even when requests at better prices are received. See more details [below](#).
- **Allow to change best prices by client orders** the Market Depth of an ECN instrument can be formed of liquidity providers (gateways) and of orders placed by clients within the trading cluster. The best prices in the Market Depth must always be represented by liquid orders, i.e. by orders which can really be executed. If you have configured the ECN so that client orders are not considered to be valid and do not participate in matching, disable the possibility to change best prices for such orders.
- **Send last deal prices to quotes stream** if this option is disabled, then the prices of ECN symbol deals performed by traders within the cluster will not be sent to the ECN symbol quote stream.

History keeping settings

Specify how long the ECN should store the Market Depth formation and orders matching history.

The Market Depth history is stored as separate Journals, in which each price change is recorded. These changes can be viewed in the separate [Market Depth Journal](#) section. Since there can be a lot of changes (millions of entries per day), Journals can take up a large amount of disk space. Therefore, it is not recommended to set the parameter to "unlimited".

Access to the matching history is currency under development.

Minimum spread

When receiving the Market Depth from different sources and aggregating orders within the cluster, a situation may occur in which both parties are ready to execute the deal at worse conditions than it is possible. For example, a trader on one server is ready to buy an asset at a price of 100, and another trader on a different server is ready to sell the same asset at 90. When matching such requests, the ECN organizing broker will receive profit in the amount of the spread. In this case, source prices must not be sent to the Market Depth. For that purpose the ECN provides the so called "visible price" and the "Minimum spread" parameter. If the actual symbol spread is less than the minimum allowed value, the ECN will modify request prices displayed in the Market Depth. Actual request prices will not be changed, so trades will be performed using actual prices.

Price conversion to meet the minimum spread condition is performed according to the following algorithm:

- The current Market Depth spread is calculated as the difference between the best Bid and Ask.

MetaTrader 5 trading platform

- If the current spread is less than the specified minimum value, a difference between them (delta) is calculated. This is the amount by which the current spread should be increased.
- New Bid and Ask prices are calculated. $\text{Bid New} = \text{Bid} - \text{floor}(\text{spread_delta}/2)$ points. $\text{Ask New} = \text{Ask} + \text{ceil}(\text{spread_delta}/2)$ points. Here floor and ceil are standard functions for obtaining the nearest higher and lower integer value.
- The visible price is set to Bid New for all Buy orders with the price greater than Bid New.
- The visible price is set to Ask New for all Sell orders with the price lower than Ask New.

Example:

1. The Market Depth of the EURUSD ECN symbol is formed according to the following rules: Sell items are used from the LMAX source (the EURUSDLMAX symbol), Buy items are provided by Currenex (the EURUSDCNX symbol).
2. The minimum spread is set to 10 points.
3. The actual Bid price of EURUSDCNX is 1.17723, while the Ask price of EURUSDLMAX is 1.17725.
4. When aggregating these prices to the Market Depth, the final spread will be equal to 2 points, i.e. less than the minimum allowable value.
5. The spread must be increased by 8 points (10 - 2).
6. $\text{Bid New} = 1.17723 - \text{floor}(8/2) \text{ points} = 1.17719$.
7. $\text{Ask New} = 1.17725 + \text{ceil}(8/2) \text{ points} = 1.17729$.
8. The total volume of Buy orders having the price greater than Bid New is transferred to the Bid New level.
9. The total volume of Sell orders having the price less than Ask New is transferred to the Ask New level.

MetaTrader 5 trading platform

[illegible]

EURUSDCNX, Currenex EUR/USD

Volume	Price	Trade
10.00	1.17737	
30.00	1.17735	
5.00	1.17730	
40.00	1.17729	
20.00	1.17728	
30.00	1.17727	
10.00	1.17726	
20.00	1.17723	
20.00	1.17722	
40.00	1.17721	
20.00	1.17720	
10.00	1.17719	
5.00	1.17715	
30.00	1.17711	

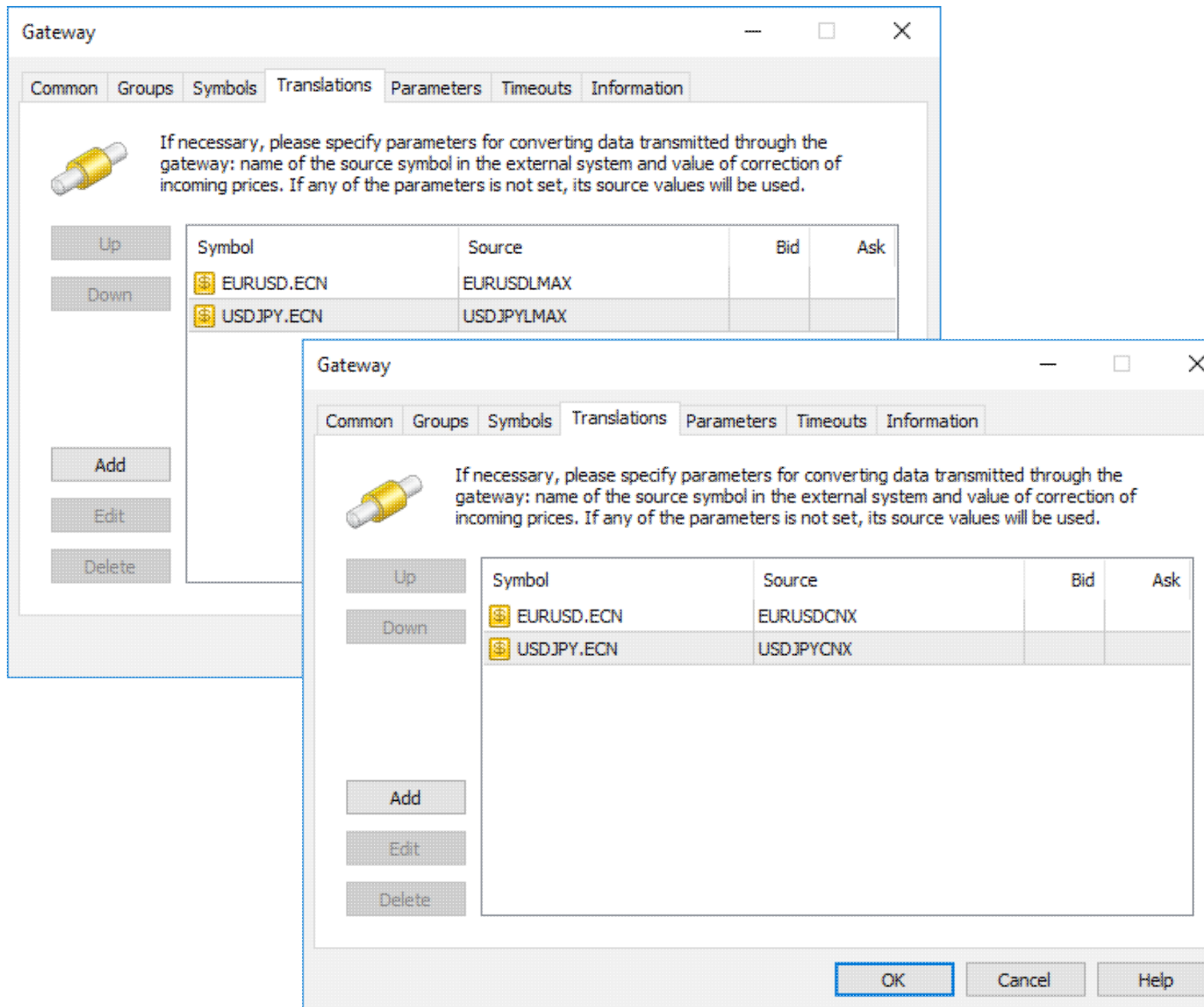
sl 0 5.00 tp 0

Sell Close Buy

EURUSD LMAX, LMAX EUR/USD			
Volume	Price	Trade	
10.00	1.18968	▼	
0.01	1.18875	▼	
0.01	1.18824	▼	
0.01	1.18776	▼	
308.15	1.18610	▼	
0.04	1.17899	▼	
0.65	1.17896	▼	
30.50	1.17729	▼	
21.25	1.17728	▼	
25.00	1.17727	▼	
30.00	1.17726	▼	
9.95	1.17725	▼	
2.00	1.17723	▼	
3.00	1.17722	▼	
20.01	1.17721	▼	
20.50	1.17720	▼	
23.75	1.17719	▼	
0.01	1.17692	▼	
0.50	1.17200	▼	
0.01	1.17174	▼	
0.10	1.17000	▼	
0.50	1.16948	▼	
0.15	1.15878	▼	

Configuration of gateways and datafeeds

You need to configure gateways and datafeeds to enable passing of quotes and Market Depth changes to ECN symbols. Each ECN symbol must be appropriately linked to the appropriate symbol on the external system side via gateway/datafeed translation settings:

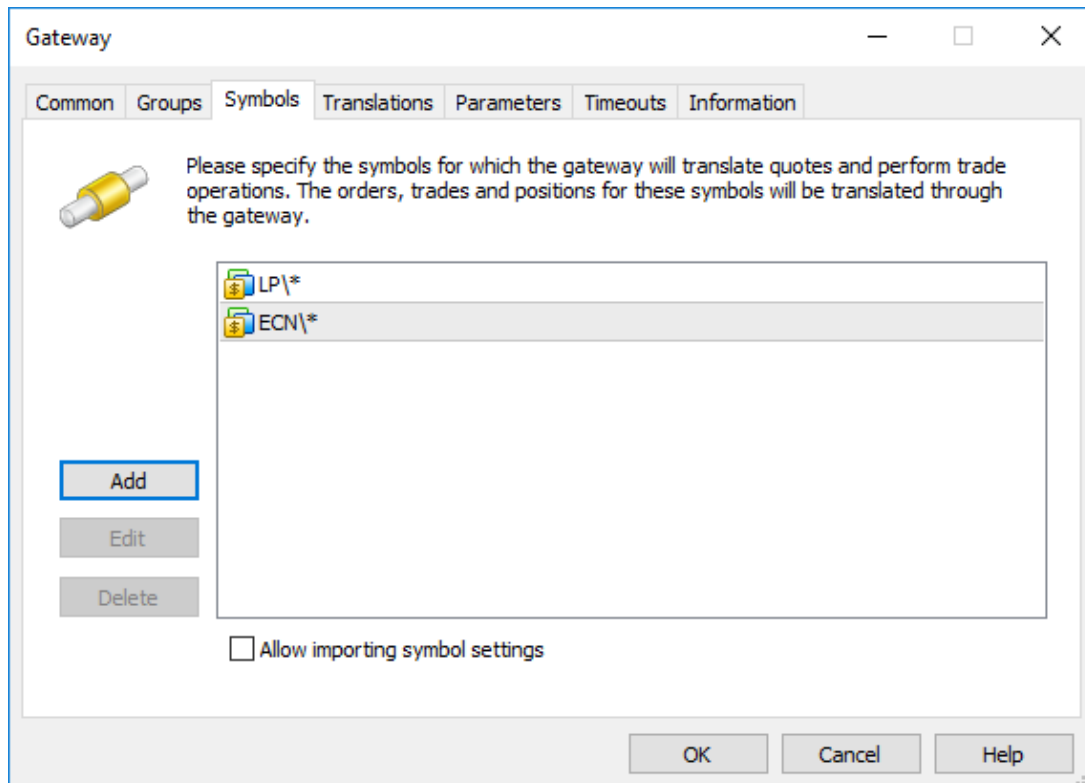


MetaTrader 5 trading platform

With such settings, all quotes and Market Depth changes for the specified symbol will be passed from the gateway to the ECN. Detailed filtering can be configured in ECN symbol parameters.

If the symbol name in an external system matches the ECN symbol in MetaTrader 5, translation settings are not required.

Also, the gateway must have access to the ECN symbol group. Specify a group of symbols using the mask:

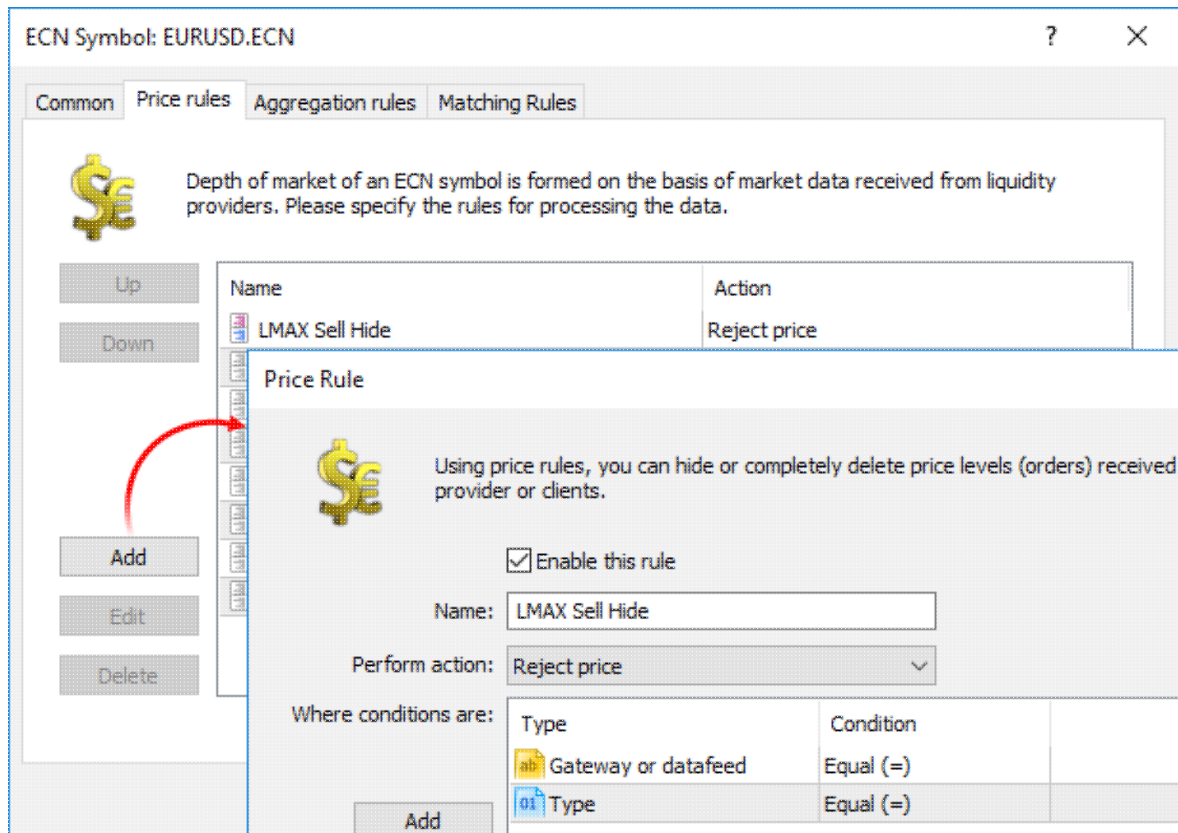


Configuring data aggregation from gateways and datafeeds

Receiving of Market Depth changes from external sources can be configured in detail for each ECN symbol. This can be done in symbol settings, in the "Price rules" tab.

- If the rules are not set, all data from gateways/datafeeds with appropriate translation settings will be added to the Market Depth. By default, all levels from gateways are considered to be liquid, those from datafeeds are considered to be illiquid. Liquid levels can be used for matching orders and forming best prices in the Market Depth.
- Gateway prices are not considered liquid unless this gateway is connected as a dealer or it has passed the symbol settings to the external system. The last condition is required for the proper processing of prices and trading operations when symbol settings on the platform side and on the external system side differ (different number of decimal places, contract size, etc.).
- If a gateway or a data feed does not support the order book feature and only translates the best bid/ask prices, its prices will be displayed in the aggregated Market Depth with the volume of 1 lot. If such a price is considered liquid, the ECN will consider the volume of this liquidity to be unlimited. Orders will be forwarded to these sources in any requested volume. The log will contain a clear indication that the request was matched with a tick element of the aggregated Market Depth. Example: "matched 10 at 1.09496, #3245040 buy 10 EURUSD.ECN at market vs sell limit (tick) 1 at 1.09496 on 'LMAX Gateway', by rule 'LMAX'".

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Click "Add" and set the rule name. In the "Perform action" field, select the method to process the price:

- **Reject price** the level will be totally rejected. It will not be displayed in the Market Depth and will not be saved in the ECN symbol history.
- **Use the price as liquidity, without displaying it** the level will be shown in the Market Depth, but will not be accepted by the platform. The level will be saved in the database, so it can be used for matching orders.
- **Display price only, without using it as liquidity** the level can appear in the Market Depth but it cannot be used for matching of orders. Such levels are considered to be illiquid and they cannot change the best prices in the order book. The application of this rule to a level does not guarantee that the level will be visible to traders in the Market Depth. This means that the level will not be hidden. For more details please visit the "[Market Depth availability to clients](#)" section.

Price rules cannot turn illiquid prices into liquid or make invisible prices be visible. For example, datafeed prices cannot be marked as liquid, as well as prices hidden due to minimum spread settings cannot be displayed.

Set the conditions that the price must meet in order to be processed according to this rule.

- **Date and time** processing prices received on the specific day and time.
- **Time** processing prices received at a specific time.
- **Day of week** processing prices received on a particular day of the week.
- **Type** processing prices depending on the direction of the offer: Buy or Sell.
- **Volume** processing prices depending on offer volume. For example, you may choose to hide large-volume price levels from the Market Depth.
- **Price** processing specific levels.
- **Gateway or datafeed** processing prices received from the selected gateway or datafeed.
- **Client login** processing request from the specific client trading within the cluster.

All specified rules are checked for each price passed to an ECN symbol.

Configuring data aggregation from clients within the cluster

For each ECN symbol you can set detailed rules, according to which the orders placed by clients within the cluster but not processed by the ECN module directly, will be added to the aggregated Market Depth. Open the "Aggregation rule" tab in ECN symbol settings.

- The settings do not affect the orders, which were forwarded to the ECN module according to the corresponding routing rules. Such orders are unconditionally displayed in the ECN symbol's Market Depth.

MetaTrader 5 trading platform

- These settings only affect the display of client orders in the Market Depth. The orders are considered to be illiquid, so they are not used for matching orders and cannot change the best prices in the Market Depth.
- If the rules are not set, orders from clients within the cluster will not be added to the Market Depth.

ECN Symbol: EURUSD.ECN

Common Price rules Aggregation rules Matching Rules

 Depth of market of an ECN symbol can also include orders from your MetaTrader 5 cluster. Please specify the rules to be used to select such orders.

Up
Down

Name	Symbol	Bid	Ask
 Orders from EURUSD.ECN	EURUSD.ECN	0	0
 Order			

Add
Edit
Delete

Aggregation Rule

 Using aggregation rules, you can define, which orders from the MetaTrader 5 cluster should be used in building the depth of market of an ECN symbol.

☒ Enable this rule

Name:

Bid markup: Ask markup:

Where symbol is:

Where order is:

Where conditions are:

Type	Condition	Value
 Client group	Equal (=)	real*

Add
Edit
Delete

OK Cancel

Click "Add" and set the rule name.

Configuring transformations

You can define markups for client levels, which enable conversion of client orders when displayed in the Market Depth. Specify the desired number of points in the "Bid" and "Ask" fields. A positive value is used to increase the price, a negative one reduces it. For example, if the value of "-3" is specified in the "Bid" field and the client places a Buy Limit order at the price of 1.12286, then the request will appear as 1.12283 in the Market Depth.

Please note that price markups in the aggregation rule have a higher priority than similar settings in translation rules. For example, EURUSD.ECN data are forwarded to EURUSD.USR with the markups of -2/+2. Clients trade EURUSD.USR at converted prices. The following aggregation rule is set: client requests for the EURUSD.USR symbol are aggregated in the EURUSD.ECN symbol. If zero markups are specified in this rule, during aggregation the ECN will automatically convert client EURUSD.USR requests to source EURUSD.ECN prices. If non-zero markups are specified, the ECN will use them. An example with specific prices:

- Source prices in EURUSD.ECN: 1.15651 / 1.15659
- When passed to EURUSD.USR, the prices are converted according to the markup rule -2/+2: 1.15649 / 1.15661
- The client sees EURUSD.USR prices and places a Buy Limit orders 20 points below the market, at 1.15629.
- If zero markups are set in aggregation rules, back conversion will be applied to the client's order: EURUSD.USR -> EURUSD.ECN. Two points will be added to the price: $1.15629 + 0.00002 = 1.15631$.
- If markups are specified in the aggregation rule, for example -5/+5, the client's order will be passed to EURUSD.ECN at a 5 point less price: $1.15629 - 0.00005 = 1.15624$.

Conditions

In the "Where symbol is" field, specify the symbol, which orders will be added to the Market Depth. Order types should be specified in the "Where order is": Buy, Sell. Only limit orders are displayed in the Market Depth. So, the Buy option enables the display of Buy Limit orders, while Sell enables Sell Limit orders.

Set the conditions which the orders must meet in order to be processed according to this rule.

- **Date and time** processing orders received on the specific day and time.
- **Time** processing orders received at a specific time.
- **Day of week** processing orders received on a particular day of the week.
- **Volume** processing orders depending on their volume. For example, you may choose to not display large-volume orders from the Market Depth.
- **Client group** processing orders placed by clients from the specific client group. For example, you may set the rule to only accept orders placed by clients from real groups.
- **Client login** processing orders from the specific client.

In the example shown in the image above, all EURUSD.ECN Market and Limit orders placed by clients from real groups will be added to the EURUSD.ECN Market Depth. Thus, clients trading the ECN symbol will be able to see their own orders in the Market Depth.

Availability of Market Depth (order book) for clients

Two order book types (states) are possible in the ECN:

- **Full book** is the internal state of the Market Depth in ECN. Prices are passed to it in accordance with the aggregation settings from gateways and the trading cluster.
- **Aggregated book** is the Market Depth which is actually shown to clients. Compared to the internal order book, in this state some of the prices can be hidden due to minimum spread settings, order book depth limitation and price rules, as well as depending on price liquidity.

Both order book states can be viewed in the journal. Use the context menu to switch between them.

What is price liquidity and how it is determined

The order book can contain two types of orders: liquid and illiquid. According to this property, these orders can be visible to traders (shown in the aggregated order book) or hidden from them.

- **Liquid orders:** these orders can be used to fill client orders (used in matching).
- **Illiquid orders:** these orders cannot be used in matching.

Order liquidity is determined by its source and ECN settings:

- **Orders from gateways:** an order is liquid if the following conditions are met:
 - The gateway must pass to ECN symbol settings in the external system (all standard liquidity gateways support this function).
 - The gateway must be connected to the ECN as a dealer using routing rules.
 - The gateway must be specified as a liquidity provider in matching rules.
- **Orders from datafeeds:** these orders are never considered as liquid.
- **Orders from clients within the cluster:** these orders are always liquid.

How price liquidity affects its availability to traders

Only liquid orders can form best prices in the order book for traders (Best Bid/Best Ask). If the price is illiquid and it is currently the best one among all prices, then it will not be shown in the Market Depth for traders. In other words, illiquid prices are shown in traders' order book only if they are not the best.

Liquid prices by default are shown to traders in the order book. In contrast to illiquid prices, such prices are displayed without a check of whether they are currently the best prices. Liquid prices can be forcibly hidden from the traders' order book using price rules (Perform action = Use price as liquidity, without showing).

If the Market Depth does not contain any liquid price (for example, you have not specified any gateway in matching rules), the traders will see an empty order book.

Aggregation rules for symbols having different price accuracy on the platform and external system side

If the accuracy of security prices on the external system side differs from that in the trading platform, ECN converts these prices when sending to the aggregated Market Depth.

If the accuracy on the platform side is higher, decimal places are added to the prices provided by the external system. Consider the following example. The price accuracy on the platform side is 5 decimal places, while the external system provides prices with 4 decimal places. The price of 1.1329 received from the external system will be displayed as 1.13290 in the aggregated Market Depth feature.

If the accuracy on the platform side is lower, the prices provided by the external system will be rounded according to the following rules:

- Buy orders are rounded down to the symbol accuracy on the platform side.
- Sell orders are rounded up to the symbol accuracy on the platform side.

Example: EURUSD.ECN symbol with 4 decimal places exists in the platform. It features aggregated EURUSD prices from the LMAX gateway which provides the prices with 5 decimal places. The Market Depth feature provided to clients will be formed as follows:

MetaTrader 5 trading platform

EURUSD.ECN, LMAX EUR/USD				EURUSD.LMAX, LMAX EUR/USD			
Volume	Price	Trade		Volume	Price	Trade	
0.10	1.1328	▼	▲	0.20	1.13131	▼	▲
10.10	1.1326	▼	▲	15.00	1.13103	▼	▲
0.20	1.1322	▼	▲	0.50	1.13091	▼	▲
0.47	1.1321	▼	▲	0.10	1.13018	▼	▲
0.20	1.1314	▼	▲	0.31	1.13015	▼	▲
15.00	1.1311	▼	▲	0.01	1.13010	▼	▲
0.50	1.1310	▼	▲	30.50	1.12932	▼	▲
0.41	1.1302	▼	▲	30.00	1.12931	▼	▲
0.01	1.1301	▼	▲	33.00	1.12930	▼	▲
60.50	1.1294	▼	▲	30.01	1.12929	▼	▲
75.51	1.1293	▼	▲	12.50	1.12928	▼	▲
150.01	1.1292	▼	▲	46.51	1.12925	▼	▲
30.00	1.1284	▼	▲	18.00	1.12924	▼	▲
30.00	1.1278	▼	▲	20.00	1.12923	▼	▲
30.00	1.1277	▼	▲	20.00	1.12922	▼	▲
30.00	1.1276	▼	▲	45.50	1.12921	▼	▲
60.00	1.1274	▼	▲	30.00	1.12849	▼	▲
0.10	1.1273	▼	▲	30.00	1.12782	▼	▲
30.00	1.1272	▼	▲	30.00	1.12771	▼	▲
30.00	1.1271	▼	▲	30.00	1.12769	▼	▲
30.00	1.1270	▼	▲	30.00	1.12748	▼	▲
30.00	1.1268	▼	▲	30.00	1.12742	▼	▲
sl 0	1.00	tp 0		sl 0	1.00	tp 0	
Sell	Close	Buy		Sell	Close	Buy	

MetaTrader 5 trading platform

Prices will be converted to the accuracy of 4 decimal points, then levels with the same prices will be combined.

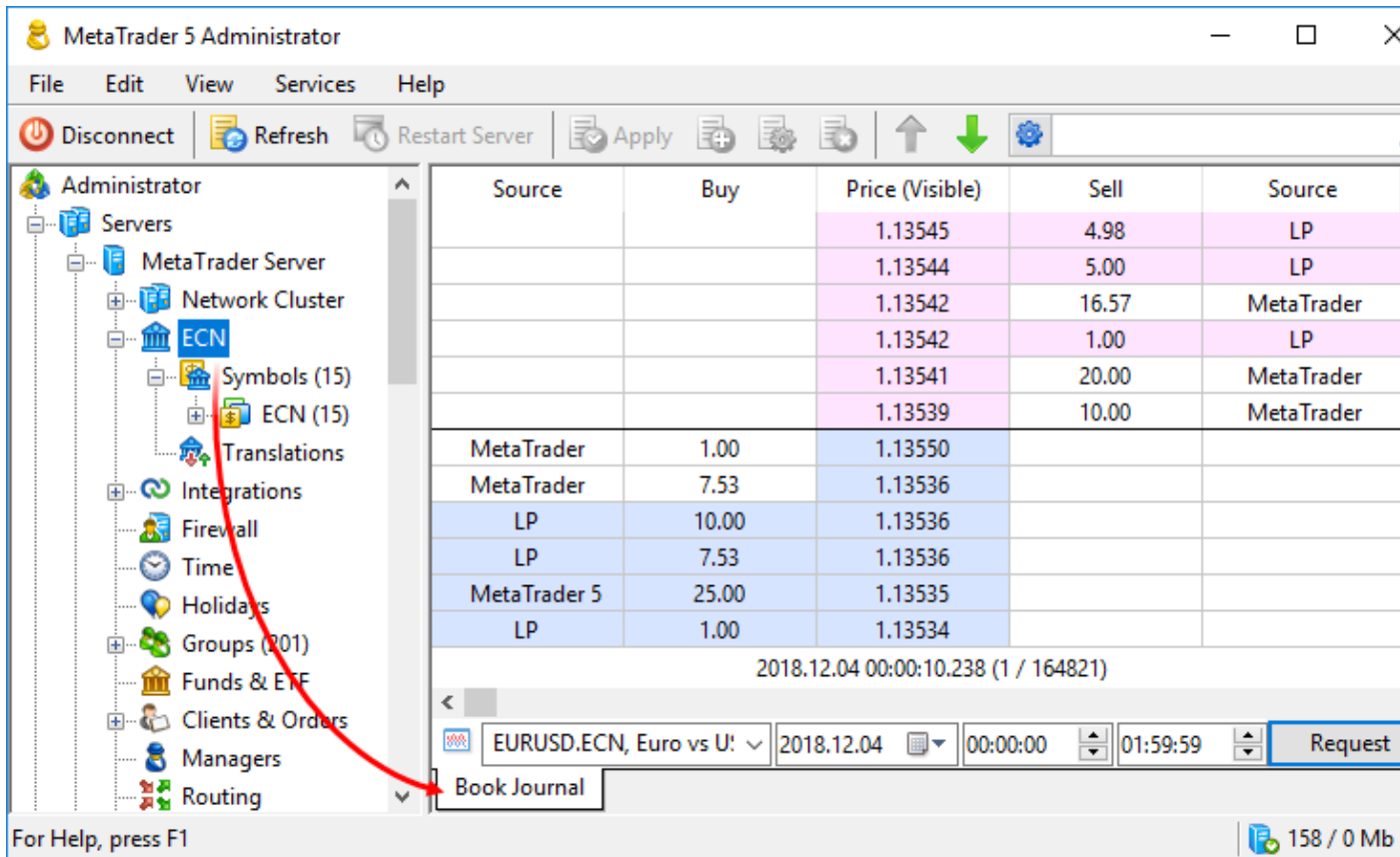
Original data feed prices can be viewed in the Market Depth journal. The original price and the price displayed to clients (in brackets) are shown in the Price column:

Flags	Source	Buy	Price (Visible)	Sell	Source	Flags
			1.13205 (1.1321)	0.47	LMAX Gateway	Liquid, Visible
			1.13131 (1.1314)	0.20	LMAX Gateway	Liquid, Visible
			1.13103 (1.1311)	15.00	LMAX Gateway	Liquid, Visible
			1.13091 (1.1310)	0.50	LMAX Gateway	Liquid, Visible
			1.13018 (1.1302)	0.10	LMAX Gateway	Liquid, Visible
			1.13015 (1.1302)	0.31	LMAX Gateway	Liquid, Visible
			1.13010 (1.1301)	0.01	LMAX Gateway	Liquid, Visible
			1.12925 (1.1293)	23.00	LMAX Gateway	Liquid, Visible
			1.12924 (1.1293)	30.20	LMAX Gateway	Liquid, Visible
			1.12923 (1.1293)	30.00	LMAX Gateway	Liquid, Visible
			1.12922 (1.1293)	30.01	LMAX Gateway	Liquid, Visible
			1.12921 (1.1293)	17.49	LMAX Gateway	Liquid, Visible
Liquid, Visible	LMAX Gateway	10.01	1.12919 (1.1291)			
Liquid, Visible	LMAX Gateway	17.50	1.12918 (1.1291)			
Liquid, Visible	LMAX Gateway	33.00	1.12917 (1.1291)			
Liquid, Visible	LMAX Gateway	50.50	1.12916 (1.1291)			
Liquid, Visible	LMAX Gateway	25.00	1.12915 (1.1291)			
Liquid, Visible	LMAX Gateway	0.67	1.12867 (1.1286)			
Liquid, Visible	LMAX Gateway	30.00	1.12849 (1.1284)			
Liquid, Visible	LMAX Gateway	30.00	1.12782 (1.1278)			
Liquid, Visible	LMAX Gateway	30.00	1.12771 (1.1277)			
Liquid, Visible	LMAX Gateway	30.00	1.12769 (1.1276)			
Liquid, Visible	LMAX Gateway	30.00	1.12748 (1.1274)			
Liquid, Visible	LMAX Gateway	30.00	1.12742 (1.1274)			

MetaTrader 5 Trading Platform / Platform Setup / ECN / Market Depth Journal

Market Depth Journal

The trading platform stores the Marked Depth formation history, showing what levels from what providers were accepted, what levels were rejected or merged. Since the history volume is large, the special log visualizer tool is provided for history analysis.



The screenshot shows the MetaTrader 5 Administrator interface. On the left, a tree view under 'Administrator' includes 'Servers', 'MetaTrader Server', 'Network Cluster', 'ECN', 'Symbols (15)', 'ECN (15)', 'Translations', 'Integrations', 'Firewall', 'Time', 'Holidays', 'Groups (201)', 'Funds & ETF', 'Clients & Orders', 'Managers', and 'Routing'. A red arrow points from the 'ECN' folder to the 'Book Journal' button at the bottom. The main area displays a table of market depth data.

Source	Buy	Price (Visible)	Sell	Source
		1.13545	4.98	LP
		1.13544	5.00	LP
		1.13542	16.57	MetaTrader
		1.13542	1.00	LP
		1.13541	20.00	MetaTrader
		1.13539	10.00	MetaTrader
MetaTrader	1.00	1.13550		
MetaTrader	7.53	1.13536		
LP	10.00	1.13536		
LP	7.53	1.13536		
MetaTrader 5	25.00	1.13535		
LP	1.00	1.13534		

Below the table, the date and time are shown: 2018.12.04 00:00:10.238 (1 / 164821). At the bottom, there is a 'Book Journal' button and a 'Request' button. The status bar at the bottom indicates 'For Help, press F1' and '158 / 0 Mb'.

Select the symbol, day and time, for which you want to view the history, and click "Request". After that, you will see the Market Depth status as of the initial request time. To view Market Depth changes over time, use the slider below.

Due to the large amount of information, you can request history for only one symbol and for one day at a time.

The time of the presented Market Depth state is shown below, as well as the ordinal number of the Market Depth change and the total number of changes for the selected time period.

The following Market Depth information is provided in the Journal:

- **Source** liquidity provider from which the Buy/Sell level was received. The source is the gateway configuration name. When you hover over it, a tooltip appears featuring the provider's basic information, such as the used module, the default name, the gateway ID, version of the gateway and used Gateway API.
- **Buy/Sell** the lot volume of buy/sell orders at the specified price.
- **Price** the price of Buy and Sell orders. In the "Full book" mode, the column may contain an additional price shown in brackets. This price appears if the original price was modified, for example due to minimum spread settings or due to different number of decimal places used in the trading platform and by the liquidity provider. This is the price, which was visible to clients.
- **Flags** additional data on the price level:
 - **Liquid** the level is liquid, a trading operation (order matching) can be performed at this level.
 - **Visible** the level is visible in the internal ECN order book. This flag is set only based on price rules and does not mean that the level will be visible to traders. It means that the level can be visible to traders under certain conditions, such as compliance with the minimum spread and market depth limitation requirements. For details, please visit the Availability of Market Depth (order book) for clients section.
 - **Tick** the level is formed at the best prices. Such a flag is provided in case the data source, for whatever reason, starts broadcasting only the best prices, without the Market Depth information.

Use the context menu to manage the Market Depth appearance. You can change the Market Depth type, choose volume mode (lots and units), as well as show and hide columns and grid. The following commands are also available in the context menu:

- **Invisible levels** show price levels, which are hidden from clients in accordance with aggregation settings.
- **Full book** show all levels, which were passed to the ECN symbol from gateways and the trading cluster in accordance with the aggregation settings.
- **Aggregated book** show the Market Depth, which is actually displayed to clients. Minimum spread settings are applied to this Market Depth state, while invisible levels are hidden from it.

MetaTrader 5 Trading Platform / Platform Setup / ECN / Order Matching

Order Matching

The ECN feature allows matching of client orders placed from within the trading cluster. For example, if a client wants to buy a symbol at a certain price, the ECN will match an opposite order with an offer to sell the symbol at the specified price, and perform a trade. A counterparty in such a trade is either another client within the cluster or an external liquidity provider.

The ECN maintains a list of trading orders waiting to be matched. The list of orders is formed in accordance with individual ECN symbol settings. Orders can be selected based on the direction, client group, time, volume and other parameters. Once an opposite order is found for an order from the list, the ECN will match these orders.

Matching is only allowed for operations with ECN symbol.
--

Trade request routing in ECN

To forward client requests to ECN for matching, set the appropriate routing rule. In general settings, specify which requests should be sent to ECN. For example, you may specify here all requests for ECN* symbols. The action to be applied to requests should be set to "Process to dealers". Next, on the "Dealers" tab select "ECN".

MetaTrader 5 trading platform

Routing: To ECN

Common

Dealers

Up

Down

Add

Edit

Delete

Please specify dealers who will process requests that meet the rule conditions.

Login

ECN

Name

ECN Dealer

Routing: To ECN

Common

Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

To ECN

Perform action:

Process to dealers

☐ skip this rule if no dealers online

Where request is:

All

Where order is:

All

Where conditions are:

Type	Condition	Value
Symbols	Equal (=)	ECN*

Add

Edit

Delete

OK

Cancel

Help

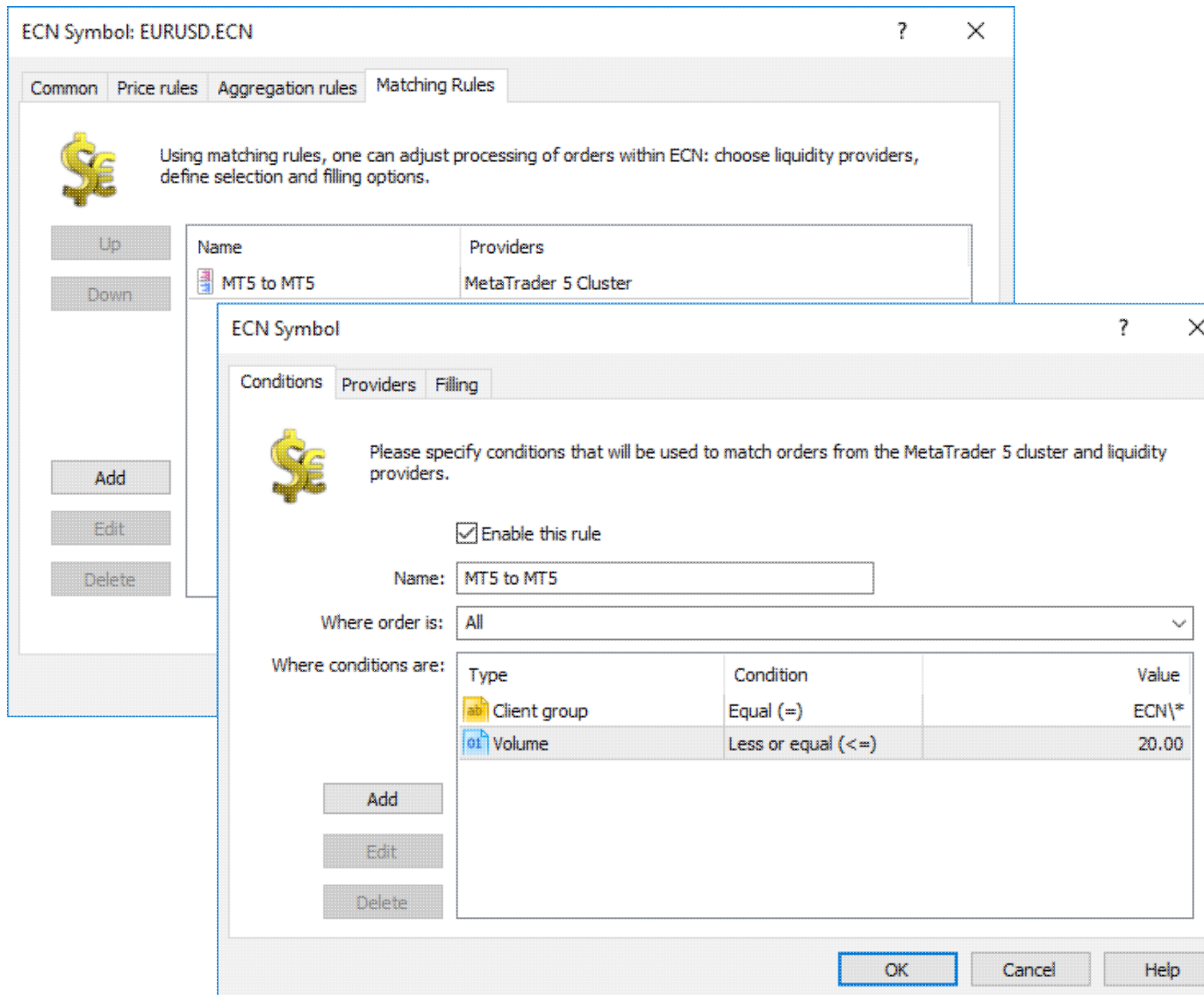
1820

MetaTrader 5 trading platform

Requests forwarded to ECN in accordance with the routing rules, will be processed according to matching rules.

Matching rules

Matching settings are specified separately for each ECN symbol.



Conditions

Add a new rule and set the conditions for selecting operations to be matched.

Select the operation type in the "Where order is" field: Buy, Sell. If you select both options, the ECN will match market Buy and Sell orders.

Set the conditions that the order must meet in order to be added to the matching list:

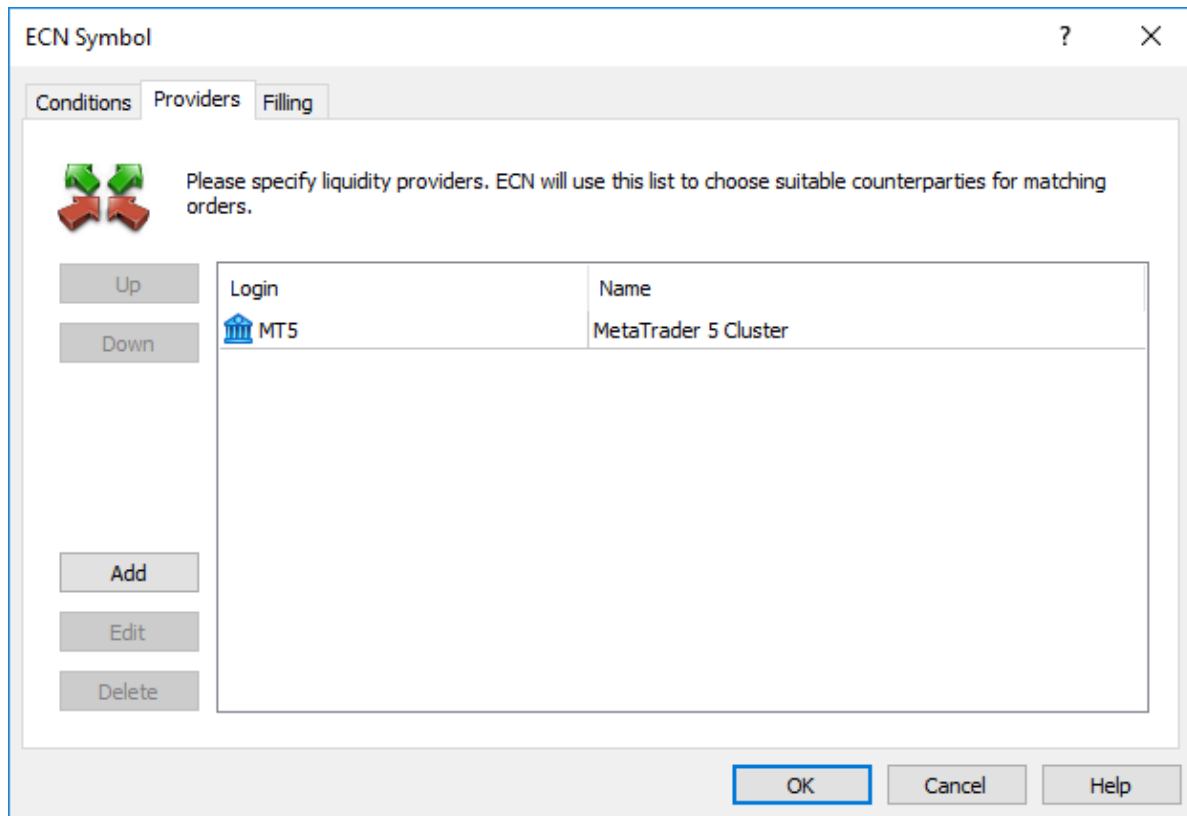
- **Date and time** processing orders received on the specific day and time.
- **Time** processing orders received at a specific time.
- **Day of week** processing orders received on a particular day of the week.
- **Client login** processing orders from the specific client.
- **Client group** processing orders placed by clients from the specific client group.
- **Volume** processing orders depending on their volume.

In the above example, Buy and Sell orders with the volume of 20 lots or less, placed by clients from ECN* groups will be added to the matching list.

Providers

In the "Providers" tab, select providers who can serve as counterparties in client trades. You can select here clients from within the cluster, as well as external liquidity providers. Matching of orders with external providers is performed via gateways, so you can select gateway configurations existing in the trading platform for external providers.

To match client orders within the cluster, select the 'MT5' option.



Within one rule, all providers have the same priority, while their position in the list does not matter. The procedure for choosing a provider during matching is described in the [relevant section](#).

Filling

In the "Filling" section, set additional rules to forward orders to external liquidity providers. Please view the ["Order Filling"](#) section for more details concerning these settings.

Activation of limit orders and Take Profit levels

In common settings of the ECN symbol, you should specify the execution mode for limit orders which have not been [previously sent to the ECN](#), as well as the Take Profit activation handling mode.

Set two parameters:

- **Mode** the type of the order which will be sent to ECN upon the activation of Take Profit orders, as well as limit orders which have not been previously sent to ECN. The mode can be "Limit" or "Market".
- **Timeout** the lifetime of an order sent to the ECN, in milliseconds. If the order is not executed within this time, this order should be canceled. In case of the re-activation of a limit order or the Take Profit level, a new order with the specified timeout will be sent to the ECN.

Limit orders sent to the ECN have the default timeout of 5 seconds.

The settings in this section do not affect the activation of Stop Losses. They are always sent as market orders.

Processing limit orders outside ECN

You can configure the ECN operation so as not to forward limit orders from within the cluster to be matched in the ECN before the activation of such orders. This can be done through the configuration of appropriate routing rules.

Add a rule to process placing of limit and stop limit orders, as well as the stop limit order activation without forwarding to a dealer (ECN). Note that the rules are processed in the top-down order. For example, if you have a general rule according to which all requests are sent to ECN, then a new rule for processing limit orders should be located above the general one in the list.

Routing

Common **Dealers**

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action:

Where request is:

Where order is:

Where conditions are:

Type	Condition	Value
Symbols	Equal (=)	ECN*

Request matching algorithm

The requests which can be matched via ECN are determined for each ECN symbol in accordance with the matching rules. The matching process is launched whenever the Market Depth is changed, i.e. when prices from a gateway are received, when a new order is placed, when symbol settings are changed, etc.

Before the matching is launched, the best liquid Bid and Ask prices are received from a non-aggregated Market Depth. If the prices are not received, it means that the Market Depth is empty, and the matching process is completed.

The non-aggregated Market Depth contains all levels (including hidden ones), which were passed to the ECN symbol from gateways and the trading cluster in accordance with the aggregation settings. Minimum spread settings are not applied to this Market Depth state.

Defining the request to be executed first

The system receives 4 element queues from the current non-aggregated Market Depth:

- A queue of Market Buy orders
- A queue of Market Sell orders
- A queue of Limit Buy orders

- A queue of Limit Sell orders

Orders in these queues are sorted in the price worsening order: Buy orders are sorted by price in descending order, Sell orders are sorted in ascending order. Limit orders within the same price level, as well as all market orders are sorted by age: those submitted earlier are located first.

ECN tries to execute market orders first. To do this, the system receives one oldest deal from each of the market order queues. If both orders are received, the oldest of them will be executed.

If there are no market orders, the system takes one oldest order with the best price from each limit order queue. Suppose, this is the order #1 Buy Limit with the price A and order #2 Sell Limit with the price B.

- Distance of order #1 Buy Limit from the market price is calculated, i.e. at how many points above the market the client is ready to buy the financial instrument: $\Delta A = |A - \text{Ask}|$.
- Distance of order #2 Sell Limit from the market price is calculated, i.e. at how many points below the market the client is ready to Sell the financial instrument: $\Delta B = |B - \text{Bid}|$.
- If $\Delta A > \Delta B$, order #1 will be executed first.
- If $\Delta B > \Delta A$, order #2 will be executed first.
- If $\Delta A = \Delta B$, the order placed earlier will be executed first.

If neither a market nor a limit order could be received, the current matching cycle ends. A new cycle will be launched after the next change in the Market Depth state.

Determining the second party of the deal

The second side of a deals can be determined using the following two algorithms: time-based and Round-Robin. The version to be used should be specified in matching rules. In the first case, the side is selected according to the age of orders: preference is given to the orders which were placed in the Market Depth earlier. Within the Round-Robin algorithm, the second part for matched orders is determined evenly among all available liquidity providers. This algorithm ignores the opposite order creation time. The system matches bids and asks in a circular order, among all available providers, without time priority.

The system checks the matching rules for the specified symbol and finds the first one suitable for the order, which has been selected for execution. If there is no such a rule, ECN will select the next oldest order from the queues in accordance with the earlier described algorithm.

- If the 'Fill or Kill' (FOK) or 'Immediate or Cancel' (IOC) policy is used for the order for which no rule was found, such an order will be canceled at the next matching iteration, and "No liquidity" comment will be added. The above policies require immediate execution. If such an execution is not possible, the appropriate order must be canceled.
- If the order has the 'Return' policy, such an order will stay in the Market Depth until the matching rule appears or until expiration (if such an expiration is

specified in the order), or until it is used as the second party for another order's deal.

The system receives a list of liquidity providers from the selected matching rule. The following actions are performed for each provider:

Time	Round-robin
A provider with the highest priority is determined – the one that offers the best execution price is selected. Arrays of opposite orders are checked in the order of age, the oldest order is taken first. The price is additionally checked for limit orders – an order with the best price is always used for them. The following is checked to make sure that the order can be used: • The system checks if the prices of the two orders match. The price is not checked for market orders. • The provider's order volume is additionally checked during the execution of a FOK order. It must be greater than or equal to the requested volume, so that the order can be completely filled in accordance with the policy. This check is not performed for 'Immediate or Cancel' and 'Return' orders. • If the provider's order has the FOK policy, its volume must be less than or equal to the current order, which is being executed. • If any of the checks fails, the next provider's order is	The system checks arrays of opposite orders and selects orders with the best price among all providers. If one and the same best price in the Market Depth is represented by orders from several providers, the system selects all of them. The selected orders are sorted by the volumes of deals that were previously executed under the same contract through the relevant providers: • When matching orders, the ECN keeps an internal record of deal volumes executed via each of the provider. The volumes are accounted separately for each financial asset, for the entire period of the instrument circulation. Buy and sell deal volumes are accounted separately. • Orders of the providers, which have smaller volumes of previously matched deals the provider has, are more preferable for current matching. • For example, to match a Buy order, the system selects three sell orders with the best price of 1.0001, from three different providers. Earlier, 50 buy lots were matched to the first provider, 20 lots were matched to the second one, and 30 lots were matched to the third provider. Since the second provider has the smallest volume of matched orders, this provider's order will be selected as the second side for matching. This algorithm provides even distribution of matched orders between all providers.

Time	Round-robin
selected. If all checks have passed successfully, the selected order will be used for executing the source order. The current unfilled volume of the executed order will be reduced by the provider's order volume. The system will further go through the provider's orders till his or her last order, or until the matched order is fully executed. If after all the provider's orders the matched order is still not filled: • For IOC and Return orders, the system switches to the next provider to fill the required volume. • For FOK orders, the system resets the list of selected suitable orders and switches to the next provider. The policy requires immediate execution in the full volume, that is why all opposite orders belong to the same provider. After going through all providers, the system checks if a proper list of orders has been collected for the execution of the source order: partial filling is enough for IOC and Return orders, while the full	The following is checked to make sure that the order can be used: • The system checks if the prices of the two orders match. The price is not checked for market orders. • The provider's order volume is additionally checked during the execution of a FOK order. It must be greater than or equal to the requested volume, so that the order can be completely filled in accordance with the policy. This check is not performed for 'Immediate or Cancel' and 'Return' orders. • If the provider's order has the FOK policy, its volume must be less than or equal to the current order, which is being executed. • If any of the checks fails, the next provider's order is selected. • If all checks have passed successfully, the selected order will be used for executing the source order. The current unfilled volume of the executed order will be reduced by the provider's order volume. The system will further go through the providers till their last order, or until the matched order is fully executed. If after all the providers the matched order is still not filled: • For IOC and Return orders, the system switches to the next provider to fill the

Time	Round-robin
execution is required for FOK orders. If the list has been successfully compiled, it is sent to corresponding liquidity provider for execution.	required volume. If there are no more providers, the system switches to the next price level. Again, the ECN sorts available orders by previously matched volume through these providers. • For FOK orders, the system resets the list of selected suitable orders of the current level and switches to the next one. The policy requires immediate execution in full volume, that is why all opposite orders must belong to the same provider and have the same price. After going through all providers, the system checks if a proper list of orders has been collected for the execution of the source order: partial filling is enough for IOC and Return orders, while the full execution is required for FOK orders. If the list has been successfully compiled, it is sent to corresponding liquidity provider for execution.

Prices used for matching

When two orders are matched, the deal price can be equal to:

- The price of the opposite limit order if one of the orders is a market order.
- The price of the older order (the one placed earlier) if both orders are of the Limit type.

Execution of further orders

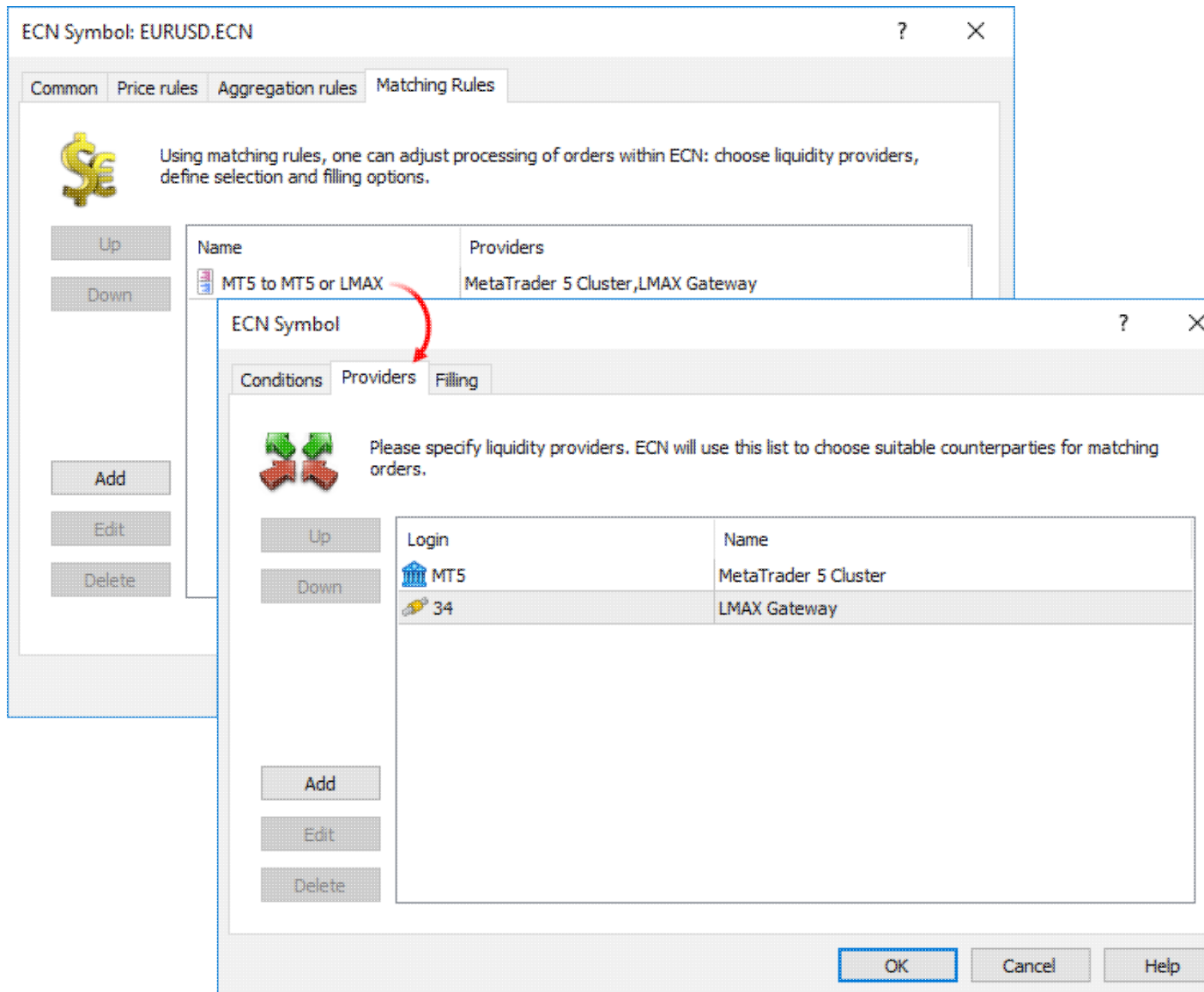
Selection and filling of further orders is performed in accordance with the above algorithm until an iteration happens, during which no pair of orders is received for matched execution.

After the matching cycle, the system once again goes through all orders in ECN. Among them the system selects all market and limit orders with the IOC and FOK fill policy, which is currently not used in matching. These orders are canceled with the "No liquidity" comment, since their fill policy requires instant execution.

Examples

One matching rule is specified. Two providers are specified in it: the MetaTrader 5 cluster itself and the LMAX gateway ECN orders can be matched with these providers.

MetaTrader 5 trading platform



Let us consider a few market situations.

Normal execution through the first provider					
Source	Buy	Price (Visible)	Sell	Source	Source
		1.14700	10.00	1003 : #105	
		1.14644 (1.14671)	20.00	LMAX Gateway	
		1.14643 (1.14671)	27.50	LMAX Gateway	
		1.14642 (1.14671)	15.01	LMAX Gateway	
		1.14641 (1.14671)	4.90	LMAX Gateway	
1001 : #106	10.00	1.14700 (1.14671)			1001 : #108
LMAX Gateway	25.50	1.14639			LMAX Gateway
LMAX Gateway	23.01	1.14638			LMAX Gateway
LMAX Gateway	5.00	1.14637			LMAX Gateway
LMAX Gateway	25.00	1.14636			LMAX Gateway
The best liquid price is bid=1.14700, ask=1.14641. Since bid>ask, the matching process is launched.					The best liquid price is bid=1.14700, ask=1.14641. Since bid>ask, the matching process is launched.
The following queues of ECN orders are formed from the Market Depth:					The following queues of ECN orders are formed from the Market Depth:
• Market Buy: empty • Market Sell: empty • Limit Buy: [#106 10.00 at 1.14700] • Limit Sell: [#105 10.00 at 1.14700]					• Market Buy: empty • Market Sell: empty • Limit Buy: [#106 10.00 at 1.14700] • Limit Sell: [#105 10.00 at 1.14700]
Since there are no market orders, the system uses the limit order queues and receives the oldest orders with the best price:					
• #105 sell limit 10.00 at 1.14700 • #106 buy limit 10.00 at 1.14700					
To choose between two orders, their distance from the market is calculated:					The only available order is #106.
• $\Delta A = \#105.\text{price} - \text{bid} = 1.14700 - 1.14700 = 0 \text{ points}$ • $\Delta B = \#106.\text{price} - \text{ask} = 1.14700 - 1.14641 = 59 \text{ points}$					
$\Delta B > \Delta A$, order #106 is selected for execution.					
The only matching rule 'MT5 to MT5 or LMAX' corresponds to order #106. The system starts checking providers in this rule.					The only matching rule 'MT5 to MT5 or LMAX' corresponds to order #106. The system starts checking providers in this rule.
The MT5 provider is processed. The system searches for the provider's orders in the opposite Market Depth side and finds the order #105.					The MT5 provider is processed. The system searches for the provider's orders in the opposite Market Depth side and finds the order #105.
					• sell limit 10.00 at 1.14700 • sell limit 20.00 at 1.14644 • sell limit 27.50 at 1.14643

Normal execution through the first provider					
					• sell limit
Matching is performed and the following entry is added to the log: matched 10.00 at 1.14700, #106 buy limit 10.00 EURUSD.ECN at 1.14700 vs #105 sell limit 10.00 at 1.14700 on '1003', by rule 'MT5 to MT5 or LMAX'					Matching is p matched 17.0 vs sell limit 2 matched 5.00 vs sell limit 5 matched 15.0 vs sell limit 1 matched 13.0 vs sell limit 5
Execution using several orders of different providers					
Source	Buy	Price (Visible)	Sell	Source	Source
		1.14700	10.00	1003 : #109	
		1.14632 (1.14665)	34.00	LMAX Gateway	
		1.14631 (1.14665)	37.50	LMAX Gateway	
		1.14630 (1.14665)	7.51	LMAX Gateway	
		1.14629 (1.14665)	2.50	LMAX Gateway	
1001 : #110	50.00	1.14700 (1.14665)			1001 : #1
LMAX Gateway	27.51	1.14626			LMAX Gate
LMAX Gateway	13.00	1.14625			LMAX Gate
LMAX Gateway	36.00	1.14624			LMAX Gate
LMAX Gateway	52.50	1.14623			LMAX Gate
The best liquid price: bid=1.14700, ask=1.14629. Since bid>ask, the matching process is launched.					The best process i
The following queues of ECN orders are formed from the Market Depth: • Market Buy: empty • Market Sell: empty • Limit Buy: [#110 50.00 at 1.14700] • Limit Sell: [#109 10.00 at 1.14700]					The follo • M • M • Li • Li
Since there are no market orders, the system uses the limit order queues and receives the oldest orders with the best price: • #109 sell limit 10.00 at 1.14700 • #110 buy limit 50.00 at 1.14700					Since the receives • #1 • #1
To choose between two orders, their distance from the market is calculated: • $\Delta A = \#109.\text{price} - \text{bid} = 1.14700 - 1.14700 = 0$ points • $\Delta B = \#110.\text{price} - \text{ask} = 1.14700 - 1.14629 = 71$ points $\Delta B > \Delta A$, order #110 is selected for execution.					To choos • ΔA • ΔB $\Delta B > \Delta A$,

Execution using several orders of different providers	
The only matching rule 'MT5 to MT5 or LMAX' corresponds to order #110. The system starts checking providers in this rule.	The only system s
The MT5 provider is processed. The system searches for the provider's orders in the opposite Market Depth side and finds the order #109.	The MT5 the oppo has the F the syste 14622'.
Matching is performed and the following entry is added to the log: matched 10.00 at 1.14700, #110 buy limit 50.00 EURUSD.ECN at 1.14700 vs #109 sell limit 10.00 at 1.14700 on '1003', by rule 'MT5 to MT5 or LMAX'	Matching matched vs sell li
Since the source order is not fully executed after matching, the system switches to the next provider. The following requests are found for executing the remaining 40 lots: • sell limit 2.50 at 14629 • sell limit 7.51 at 14630 • sell limit 37.50 at 14631	
Matching is performed and the following entries are added to the log: matched 2.50 at 1.14629, #110 buy limit 50.00 EURUSD.ECN at 1.14700 vs sell limit 2.50 at 1.14629 on 'LMAX Gateway', by rule 'MT5 to MT5 or LMAX' matched 7.51 at 1.14630, #110 buy limit 50.00 EURUSD.ECN at 1.14700 vs sell limit 7.51 at 1.14630 on 'LMAX Gateway', by rule 'MT5 to MT5 or LMAX' matched 29.99 at 1.14631, #110 buy limit 50.00 EURUSD.ECN at 1.14700 vs sell limit 37.50 at 1.14631 on 'LMAX Gateway', by rule 'MT5 to MT5 or LMAX'	

MetaTrader 5 Trading Platform / Platform Setup / ECN / Order Execution

Order Filling

As a result of matching, an opposite order of another client from within the MetaTrader 5 cluster or an order from an external system (liquidity provider) can be selected for a client order execution.

Order execution within the trade cluster

When two orders are matched within the trade cluster, two opposite deals are created to fill these orders. In this case:

- The deal volume is equal to the lowest volume of the two orders (taking into account the remaining unfilled order volume). For example, when matching orders Buy 2.00/1.00 and Sell 2.00/0.00, the volume of the deals will be equal to 1 lot.
- The price of the deals depends on the type of the matched orders:
 - If both are limit orders, the prices of both deals will be equal to the price of the order which was placed in the ECN earlier.
 - If only one of the orders is of the limit type, its price will be used for both deals.

Order execution at external liquidity providers

Once an order is matched with the price level of an external liquidity provider, it is forwarded to that provider for execution. For this, the ECN module creates a trade request to place a new order in accordance with symbol settings in the external system, which were received from the gateway, as well as filling parameters specified in the triggered matching rules.

Trading requests generated by ECN for liquidity providers are added to the request queue and are processed by the gateways similarly to regular requests from trade servers in the cluster. There are some specific features connected with the MetaTrader 5 Gateway API, which should be taken into account when developing gateways for the platform:

1. Instead of the login of the client who created the request, the ECN specifies its own dealer identifier, which is equal to the UINT64_MAX constant.
2. The ECN does not pass information related to the client, trading account, order and position when sending a trade request.
3. The gateway must not use the source ECN symbol configuration in the platform to check a request or make any decisions, since the ECN generates requests which are not linked to the platform and it takes into account the symbol configuration in the external system.

The ECN does not store information about MetaTrader 5-side client positions. Any operation which the ECN forwards to an external liquidity provider is a regular Buy

or Sell order. It does not matter which action the client performs: enters or exits the market. Therefore, we recommended using only netting accounts to connect gateways to external systems. In this case, all individual operations on the same symbol forwarded to an external system will be treated as one cumulative position. If you use a hedging account, new positions will be created for each order, while old positions will not be closed.

Execution settings

Set up execution parameters to be applied with liquidity providers using matching rules:

The screenshot shows the 'ECN Symbol' dialog box with the 'Filling' tab selected. The dialog contains the following settings:

- Matching method:** Time
- Filling method:** Limit Order
- Filling attempts:** 0
- Filling maximum time:** 60000 milliseconds
- Maximum deviation:** 0 pips
- Maximum slippages:**
 - Profitable:** off pips
 - Losing:** unlimited pips

At the bottom right, there are three buttons: OK, Cancel, and Help.

Matching method

The ECN supports two methods for selecting the second side of a deal in matching: time-based and Round-Robin. In the first case, the side is selected according to the age of orders: preference is given to the orders which were placed in the Market Depth earlier. Within the Round-Robin algorithm, the second part for matched orders is determined evenly among all available liquidity providers. This algorithm ignores the opposite order creation time. The system matches bids and asks in a circular order, among all available providers, without time priority.

For more details on matching please visit the [appropriate section](#).

Filling method

In the "Filling method" field, you should specify the type of requests to be forwarded to the external liquidity provider:

- **Limit orders** operations will be forwarded as limit orders with an expiration as per the "Filling maximum time" setting. The price will be equal to the one specified in the client order or the current market price (depends on the [execution mode](#) selected for the ECN symbol). This mode helps in avoiding large slippage.
- **Market orders** operations will be forwarded as orders with the Market Execution mode.
- **Source order type** the type of orders forwarded to an external system will be selected in accordance with the initial order type: market orders will be forwarded as market operations, and limit orders will be sent as limit orders.

Filling attempts

Here you specify the maximum number of attempts to fill an order in accordance with this rule.

Matching of trading orders can fail. For example, an external system may offer for the operation the price that exceeds [maximum deviation](#). In this case, the source order is returned to the Market Depth and its filling attempts counter is increased.

If the limit of attempts is exceeded, ECN will switch to processing the next rule, which is suitable for the given order.

- 0 means no limitations.
- If only one matching rule is set, the number of filling attempts is actually unlimited. Upon the limit excess, the ECN will always proceed to the same matching rule processing.

Filling maximum time

This is the maximum time, within which the external system (provider) is allowed to fill the matched trade request. If the order remains unfilled in the external system after the specified time, a command is sent to the external system to delete the order.

After order cancellation in the external system, the remaining volume is returned to the Marked Depth for further matching. The filling attempts counter of the source ECN order (the client request for which matching is performed) is increased by one in this case.

Deviation

This parameter is used only when operations are sent as Instant Execution and Request Execution orders.

In case of an instant execution, an acceptable deviation is specified in the request sent to the external system.

When a requote from an external system is received, ECN checks if the new price is within the specified deviation. If the deviation is not exceeded, the ECN sends to the provider a new request at the requoted prices. If the provider accepts the prices, the request is executed. In case of a repeated requote, the deviation will be checked according to the first request (i.e. the price of the original order).

If the deviation is exceeded, the execution is terminated. In this case, the order is returned to the Market Depth and an error is written to the log:

```
2018.12.05 15:40:57.714 ECN '2002': filling order #417689 on 'MetaTrader 5 Gateway ECN'
[instant sell 1.00 EURCAD.ECN at 1.50744]
```

The number of filling attempts is increased by one. Then, the ECN will try to resend the order.

During request execution, the ECN first requests prices from the external system. After receiving the price, the ECN checks if the price is within the allowed deviation. If the deviation is not exceeded, the ECN sends to the provider an order at the requoted prices.

If the deviation is exceeded, the execution is terminated. In this case, the order is returned to the Market Depth and an error is written to the log:

```
2018.12.05 15:49:08.370 ECN '2002': filling order #417690 on 'MetaTrader 5 Gateway ECN'
[prices for EURCAD.ECN 4.00]
```

The number of filling attempts is increased by one. Then, the ECN will try to resend the order.

Maximum slippages

Specify here the maximum deviation of the execution price provided by the external system from the requested price.

- **Profitable** the number of points by which the price may deviate from the requested one, in the profitable direction for the client.
- **Losing** the number of points by which the price may deviate from the requested one, in the unprofitable direction for the client.

After the actual deal execution in the external system, the ECN compares its price with the initially requested one (taking into account translation settings). If the price is within the allowed deviation, the deal will be executed on the platform side at the requested price. If deviation is exceeded, the ECN executes the deal at the price of actual execution on the external provider side.

Suppose, the values of 5 are set both for the profitable and losing slippage. The ECN passes to the external system the Buy Limit order at the price of 1.14053, and the system executes a deal at the price of 1.1405, i.e. at a 2-point better price. Such slippage is acceptable. On the platform side, the order will be executed at the initial price of 1.14053.

If the deal price exceeded the deviation, the ECN would execute the order in the platform at that specific price (1.14051).

Execution rules for symbols having different price accuracy on the platform and external system side

If the accuracy of security prices on the external system side differs from that in the trading platform, this difference will be taken into account when creating trading requests to be sent to gateways and when processing deals from external systems.

Lower accuracy in the platform

If the accuracy on the platform side is lower, decimal places are added to the prices sent to the external system. Consider the following example. The price accuracy on the platform side is 4 decimal places, while the external system provides prices with 5 decimal places. When sending an order with the price of 1.1329, the price will be converted to 1.13290.

The following rules apply for deal price rounding:

- Buy deal prices are rounded up to the symbol accuracy on the platform side.
- Sell deal prices are rounded down to the symbol accuracy on the platform side.

Thus, the broker can only have a positive balance discrepancy with the external system, in contrast to rounding performed by the common rules.

Higher accuracy in the platform

If the accuracy in the platform is higher, during order sending to an external system the prices will be rounded according to the following rules:

- Buy orders are rounded down to the symbol accuracy on the platform side.
- Sell orders are rounded up to the symbol accuracy on the platform side.

Thus clients' limit orders will always be filled on the provider's side and in MetaTrader 5 at a price which is not worse than that specified in the order. This is not possible when rounding according to common rules.

MetaTrader 5 trading platform

Consider the following example. The price accuracy on the platform side is 5 decimal places, while the external system provides prices with 4 decimal places. When executing a client Sell order at the price of 1.12919, an order with the price of 1.1291 (rounded down) will be placed in the external system. A similar Buy request would be placed as an order with the price of 1.1292 (rounded up).

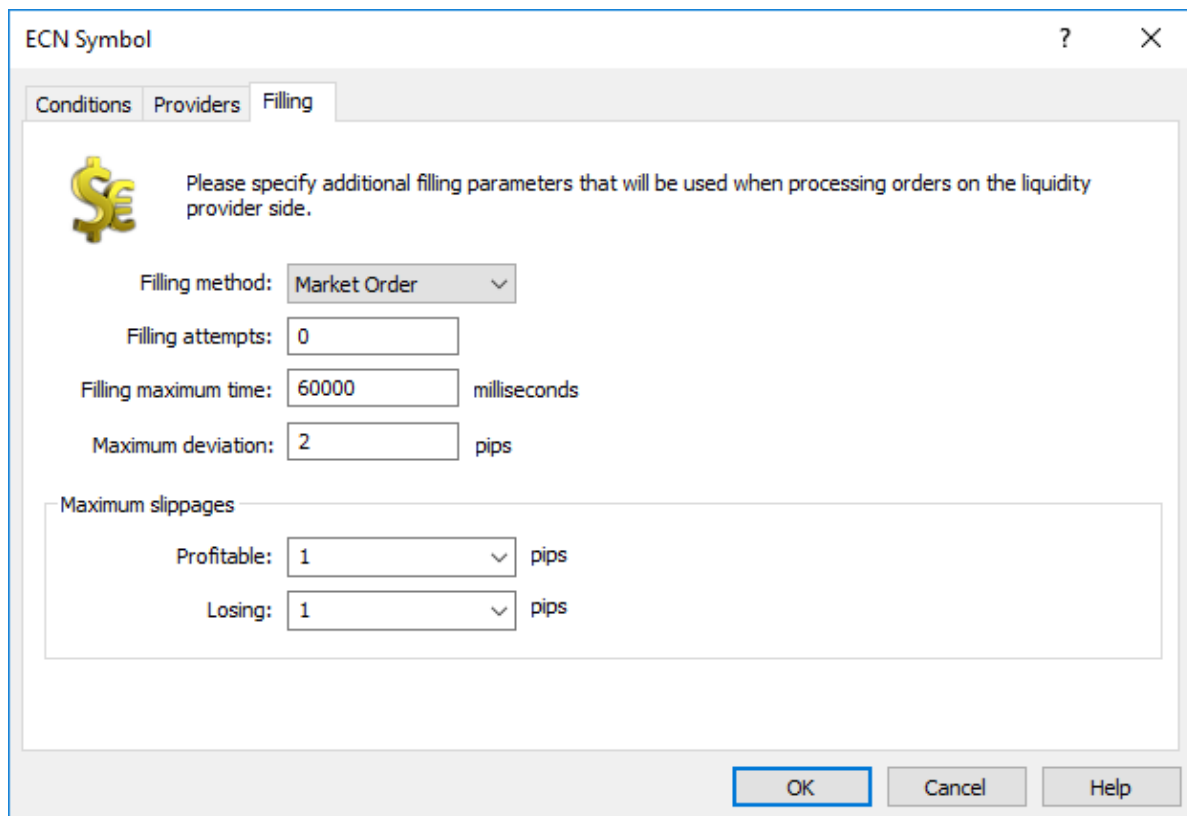
Check of the maximum deviation of the execution price in the "Instant" and "Request" modes as well as of the price slippage is performed based on the symbol accuracy set in the MetaTrader 5 trading platform (whenever applicable, the prices are rounded according to the above rules).

The acceptable price deviation in the request sent to the liquidity provider is specified with the provider's accuracy (taking into account the rounding of the request price and the allowed prices according to the above rules).

Examples: Lower accuracy in the platform

The accuracy of EURCAD.ECN in the MetaTrader 5 trading platform is equal to 4 decimal places. The liquidity provider has a 5-digit accuracy. The symbol execution mode is Instant.

Maximum deviation is 2 points, maximum profitable/losing slippage = 1/1.



The screenshot shows the 'ECN Symbol' dialog box with the 'Filling' tab selected. The dialog contains the following fields and controls:

- Conditions** | **Providers** | **Filling** (selected)
-  Please specify additional filling parameters that will be used when processing orders on the liquidity provider side.
- Filling method:
- Filling attempts:
- Filling maximum time: milliseconds
- Maximum deviation: pips
- Maximum slippages
 - Profitable: pips
 - Losing: pips
- Buttons: **OK**, **Cancel**, **Help**

MetaTrader 5 trading platform

1. The client places an order: #418239 buy limit 0.10 EURCAD.ECN at 1.4977, ECN matches it with 'MetaTrader 5 Gateway ECN':

2019.02.12 11:35:27.553	ECN	'2002': request received (#418239 buy limit 0.10 EURCAD.ECN at 1.4977)
2019.02.12 11:35:27.553	ECN	'2002': order #418239 buy limit 0.10 EURCAD.ECN at 1.4977
2019.02.12 11:35:27.556	ECN	'2002': request answered - Placed (#418239 buy limit 0.10 EURCAD.ECN at 1.4977)
2019.02.12 11:35:27.561	ECN	'2002': execution sent - request new order #418239
2019.02.12 11:35:27.561	ECN	'2002': execution sent - added order #418239, buy limit 0.10 EURCAD.ECN at 1.4977
2019.02.12 11:35:27.561	ECN	'2002': matched 0.10 at 1.49764, #418239 buy limit 0.10 EURCAD.ECN at 1.4977

The Instant request with the price of 1.49770 is formed for the gateway, which then returns a re-quote with the price of 1.49754:

2019.02.12 11:35:27.588	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754
2019.02.12 11:35:27.588	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754
2019.02.12 11:35:41.393	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754

ECN rounds the potential buy deal price of 1.49754 up to 4-digit accuracy 1.4976.

The deviation from initial price (1 point) is within the allowed deviation of 2 points and thus a new Instant request for the gateway is formed using new prices:

2019.02.12 11:35:41.393	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754
2019.02.12 11:35:41.395	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754
2019.02.12 11:35:41.396	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754
2019.02.12 11:35:43.191	ECN	'2002': filling order #418239 on 'MetaTrader 5 Gateway ECN' at 1.49754

The gateway confirms the buy request at a new price of 1.49754. The ECN rounds up the deal price in MetaTrader 5 to 4 decimal places: 1.4976. The resulting profitable slippage from the order price of 1.4977 (1 point) is within the allowed value of 1 point, so the order is filled at the initial price of 1.4977.

2019.02.12 11:35:43.191	ECN	'2002': order filled 0.10 at 1.4977 on 'MetaTrader 5 Gateway ECN'
2019.02.12 11:35:43.193	ECN	'2002': execution sent - filled order #418239, buy 0.10 EURCAD.ECN at 1.4977

2. The client places an order: #418242 instant sell 1.00 EURCAD.ECN at 1.4967, ECN matches it with 'MetaTrader 5 Gateway ECN':

2019.02.12 12:04:31.961	ECN	'2002': request received (#418242 instant sell 1.00 EURCAD.ECN at 1.4967)
2019.02.12 12:04:31.961	ECN	'2002': order #418242 sell limit 1.00 EURCAD.ECN at 1.4967
2019.02.12 12:04:31.964	ECN	'2002': request answered - Placed (#418242 instant sell 1.00 EURCAD.ECN at 1.4967)
2019.02.12 12:04:31.964	ECN	'2002': execution sent - request new order #418242
2019.02.12 12:04:31.965	ECN	'2002': execution sent - added order #418242, sell 1.00 EURCAD.ECN at 1.4967
2019.02.12 12:04:31.965	ECN	'2002': matched 1.00 at 1.49679, #418242 sell limit 1.00 EURCAD.ECN at 1.4967

The Instant request with the price of 1.49670 is formed for the gateway, which then returns a re-quote with the price of 1.49679:

2019.02.12 12:04:31.965	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway ECN' at 1.49679
2019.02.12 12:04:31.966	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway ECN' at 1.49679
2019.02.12 12:04:31.968	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway ECN' at 1.49679

ECN rounds the potential buy deal price of 1.49679 down, to 4-digit accuracy 1.4967. There is no deviation from the initial order price of 1.4967, so Instant request

MetaTrader 5 trading platform

with the new prices is formed for the gateway. This request is again requoted with the price of 1.49701:

2019.02.12 12:04:31.968	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway E
2019.02.12 12:04:31.969	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway E
2019.02.12 12:04:31.969	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway E
2019.02.12 12:04:42.768	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway E

ECN rounds the potential buy deal price of 1.49701 down, to 4-digit accuracy 1.4970. The deviation from the original order price of 1.4967 is 3 points, which is greater than the allowed value of 2 points. The order is returned to the Market Depth and is then totally canceled, since its fill policy is Fill or Kill:

2019.02.12 12:04:48.350	ECN	'2002': filling order #418242 on 'MetaTrader 5 Gateway E
2019.02.12 12:04:48.352	ECN	'2002': failed to fill 1.00 at 1.49679 on MetaTrader 5 Gate
2019.02.12 12:04:48.353	ECN	'2002': order #418242 sell limit 1.00 EURCAD.ECN at 1.4
2019.02.12 12:04:48.354	ECN	'2002': order #418242 sell limit 1.00 EURCAD.ECN at 1.4

Examples: Higher accuracy in the platform

The accuracy of EURCAD.ECN in the MetaTrader 5 trading platform is equal to 6 decimal places. The liquidity provider has a 5-digit accuracy. The symbol execution mode is Instant.

Maximum deviation is 9 points, maximum profitable/losing slippage = 5/5.

ECN Symbol

Conditions Providers Filling

Please specify additional filling parameters that will be used when processing orders on the liquidity provider side.

Filling method: Market Order

Filling attempts: 0

Filling maximum time: 60000 milliseconds

Maximum deviation: 9 pips

Maximum slippages

Profitable: 5 pips

Losing: 5 pips

OK Cancel Help

MetaTrader 5 trading platform

1. The client places an order: #418249 buy limit 0.50 EURCAD.ECN at 1.497566, ECN matches it with 'MetaTrader 5 Gateway ECN':

2019.02.12 12:42:29.148	ECN	'2002': request received (#418249 buy limit 0.50 EURCAD.ECN at 1.497566)
2019.02.12 12:42:29.148	ECN	'2002': order #418249 buy limit 0.50 EURCAD.ECN at 1.497566
2019.02.12 12:42:29.150	ECN	'2002': request answered - Placed (#418249 buy limit 0.50 EURCAD.ECN at 1.497566)
2019.02.12 12:42:29.151	ECN	'2002': execution sent - request new order #418249
2019.02.12 12:42:29.151	ECN	'2002': execution sent - added order #418249, buy limit 0.50 EURCAD.ECN at 1.497566
2019.02.12 12:42:51.141	ECN	'2002': matched 0.50 at 1.49756, #418249 buy limit 0.50 EURCAD.ECN

An Instant request is formed for the gateway with the rounded down price of 1.49756. The gateway confirms it at this price:

2019.02.12 12:42:51.142	ECN	'2002': filling order #418249 on 'MetaTrader 5 Gateway ECN' at 1.49756
2019.02.12 12:42:51.145	ECN	'2002': filling order #418249 on 'MetaTrader 5 Gateway ECN' at 1.49756
2019.02.12 12:43:12.878	ECN	'2002': filling order #418249 on 'MetaTrader 5 Gateway ECN' at 1.49756

The price converted to 6 decimal places 1.497560 provides a positive slippage of 6 points from the initial price of 1.497566. It is beyond the maximum profitable slippage of 5 points and the order is executed at the price of 1.497560:

2019.02.12 12:43:12.878	ECN	'2002': order filled 0.50 at 1.497560 on 'MetaTrader 5 Gateway ECN'
2019.02.12 12:43:12.879	ECN	'2002': execution sent - filled order #418249, buy 0.50 EURCAD.ECN at 1.497560

2. The client placed an order: #418259 instant sell 0.50 EURCAD.ECN at 1.496100, ECN matched it with 'MetaTrader 5 Gateway ECN':

2019.02.12 13:23:46.316	ECN	'2002': request received (#418259 instant sell 0.50 EURCAD.ECN at 1.496100)
2019.02.12 13:23:46.316	ECN	'2002': order #418259 sell limit 0.50 EURCAD.ECN at 1.496100
2019.02.12 13:23:46.326	ECN	'2002': request answered - Placed (#418259 instant sell 0.50 EURCAD.ECN at 1.496100)
2019.02.12 13:23:46.333	ECN	'2002': execution sent - request new order #418259
2019.02.12 13:23:46.334	ECN	'2002': execution sent - added order #418259, sell 0.50 EURCAD.ECN at 1.496100
2019.02.12 13:23:46.338	ECN	'2002': matched 0.50 at 1.49610, #418259 sell limit 0.50 EURCAD.ECN

An Instant request is formed for the gateway with the rounded up price of 1.496106. The gateway sends a re-quote of 1.49611:

2019.02.12 13:23:51.373	ECN	'2002': filling order #418259 on 'MetaTrader 5 Gateway ECN' at 1.49611
2019.02.12 13:23:51.373	ECN	'2002': filling order #418259 on 'MetaTrader 5 Gateway ECN' at 1.49611
2019.02.12 13:23:51.378	ECN	'2002': filling order #418259 on 'MetaTrader 5 Gateway ECN' at 1.49611

The potential deal price is converted to 6 decimal places 1.496110. Deviation from the original order price 1.496100 is 10 points, which is greater than the allowed value of 9 points. The order is returned to the Market Depth and is then totally canceled, since its fill policy is Fill or Kill:

2019.02.12 13:23:51.380	ECN	'2002': filling order #418259 on 'MetaTrader 5 Gateway ECN' at 1.49611
2019.02.12 13:23:51.384	ECN	'2002': failed to fill 0.50 at 1.49610 on MetaTrader 5 Gateway ECN
2019.02.12 13:23:51.394	ECN	'2002': order #418259 sell limit 0.50 EURCAD.ECN at 1.496100
2019.02.12 13:23:51.397	ECN	'2002': order #418259 sell limit 0.50 EURCAD.ECN at 1.496100

MetaTrader 5 Trading Platform / Platform Setup / ECN / Price Translations

Price Translations

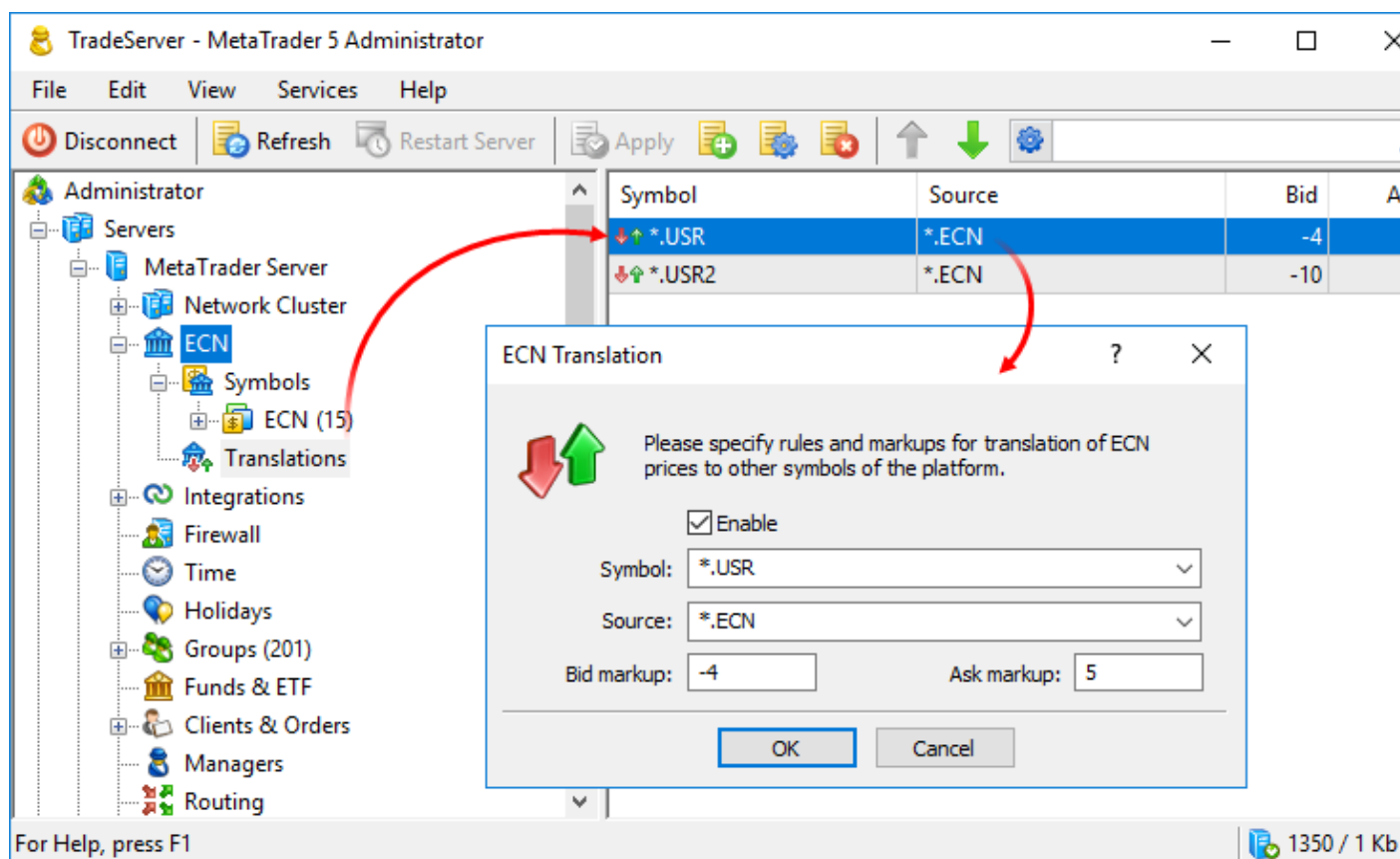
In the "Translations" section, you can configure translation of ECN symbol prices to other symbols. Markup settings can be automatically applied to such prices. This enables liquidity providing brokers to implement different ECN trading conditions for different client groups.

Similar to gateways, the ECN converts prices and trading requests.

For example, EURUSD.ECN data are forwarded to EURUSD.USR with the markups of -2/+2. Clients trade EURUSD.USR at converted prices. When forwarding orders from these clients back to external systems, the ECN will use original unchanged prices.

- The gateway passes original prices from the external system to the ECN symbol EURUSD.ECN: 1.15651 / 1.15659
- Translation of prices from EURUSD.ECN into EURUSD.USR with the conversion of -2/+2 is set up in ECN. Prices are converted to 1.15649 / 1.15661
- The client places the EURUSD.USR Buy order at a price of 1.15661
- The order is passed to the ECN for processing
- The ECN converts the price to the original one 1.15659 and passes it to the external system
- If the order is successfully executed in the external system, the ECN sends a response to the client, informing that the order was executed at the requested (converted) price of 1.15661

The broker earns the profit of 2 pips in this case.



The following settings are available:

- **Symbol** the symbols, to which data from the ECN symbol will be passed. Any non-ECN symbol or the source symbol can be specified here. If an ECN symbol has a translation setting, in which this symbol is specified as a target, a marked up price stream will be provided for such a symbol. If there are no such settings, the source price stream will be provided for the ECN symbol.
- **Source** the ECN symbol, which data will be passed.
- **Bid, Ask** the number of markup points to convert appropriate prices. A positive value increases the price, a negative one decrease it. The values are specified in the source symbol price points.

You may use the mask "*" in the source symbol names for mass translations and conversions. In this case, the translation affects all source-target pairs, in which the asterisk replaces the same set of characters. For example:

- If symbols EURUSD.ECN, EURJPY.ECN, EURJPY.USR and EURGBP.USR exist in the platform, the rule EUR*.ECN -> EUR*.USR will create the translation EURJPY.ECN -> EURJPY.USR.
- If symbols EURUSD.ECN, EURJPY.ECN, EURUSD.USR and EURJPY.USR exist in the platform, the rule *.ECN -> *.USR will create translations EURUSD.ECN -> EURUSD.USR and EURJPY.ECN -> EURJPY.USR.

If several symbols meeting one group rule have different number of decimal places in prices, different price markups will be applied to them in accordance with their

specific accuracy. For example, if a symbol has 4 decimal places, the markup of "1" will change prices by 0.0001; for symbols with 5 decimal places the markup will be equal to 0.00001.

- If several rules are configured for one target symbol, only the first rule in the list will be applied to it.
- Price markup settings are not applied for the ECN symbol, when translated to other symbols. For example, the rule is EURUSD.ECN -> EURUSD.ECN, Bid -1, Ask +1. If EURUSD.ECN is used as a source in other translation rules, original prices without the markup will be translated from it.
- Translation rules can be changed without server restart. Changes take effect immediately after saving. If the rules for a specific symbol have changed, the Market Depth of such a symbol will be reset in accordance with the new settings.
- Translation rules only apply to data from gateways and datafeeds. Trade orders for source symbols placed by clients within the cluster are not passed to target symbols. In order to show cluster orders in the target symbol, set it as an ECN symbol and configure aggregation rules for it.

Features of trading operations

If only one translation parameter or no parameters at all are set for the ECN, trades are conducted without any peculiarities since the ECN is able to match symbols on its side with the ones in the external trading system.

If more than one translation parameter is set for the ECN, the platform attempts to match the symbols correctly. For example, the two translation parameters are set for the ECN:

- EURUSD.ECN -> EURUSD.USR1
- EURUSD.ECN -> EURUSD.USR2

If an order is placed on the platform side, a symbol in the external system response is defined by the original order ticket. For example, a trader places an order #145269 Buy 1.00 EURUSD.USR.2. It is sent to the external system as #145269 Buy 1.00 EURUSD. When receiving a response, the ECN converts the symbol back to EURUSD.USR2 by the original order ticket.

However, there are some trading operations/events that are not preceded by placing an order on the trading platform side. For example, charging a variation margin in an external system. In that case, the ECN cannot clearly define which of the two symbols the event is related to: EURUSD.USR1 or EURUSD.USR2.

In this situation, the ECN attempts to select the correct symbol by its availability to certain accounts the event is related to. For example, if only EURUSD.USR2 is available for an account, the ECN performs an operation for it. Availability of symbols for clients is defined on the "Symbols" tab of the group settings.

If both symbols are available for the account, the operation is performed for the first one in the translation list. It is EURUSD.USR1 in our example.

Thus, if there are multiple translation settings, the following rule is used: trading operations are performed for the first symbol available for a client group. All subsequent translation settings are used only to pass price data.

The general price translation scheme for ECN trading

The placing and execution price of the orders sent to the ECN is affected not only by the above translation settings. The following platform settings also affect the prices:

- Symbol spread settings: both general settings and those specific of separate client groups.
- Translation settings specified in the gateways.
- Slippage settings specified in the matching rules.

Consider an example of all price translations:

1. The Client

The user creates a request based on the prices displayed in the terminal. For example, Buy Limit at 1.14059:

```
2019.01.04 11:26:57.369 Trades '2002': buy limit 1.00 EURUSD.ECN at 1.14059
2019.01.04 11:26:57.370 Trades '2002': accepted buy limit 1.00 EURUSD.ECN at 1.14059
```

2. The Trade server

The Trade Server receives the request and forwards it to the ECN, in accordance with the routing rule:

```
2019.01.04 12:15:40.530 192.168.0.1 '2002': buy limit 1.00 EURUSD.ECN at 1.14059 (1.14
2019.01.04 12:15:40.530 192.168.0.1 '2002': request transferred to dealers, rule 'ECN' (bu
```

3. The Trade server

Spread balance settings specified for the client group are applied to the request price. For example, the value of 2 bid / 2 ask is specified:

MetaTrader 5 trading platform

Symbol: *

Common Trade Execution Margin Margin Rates Swaps

Please set up main parameters of symbols for the group.

Symbol: *

☒ Enable market depth Market depth limit: unlimited

☐ Use default spread

Spread difference: 4 pt Difference balance: 2 bid / 2 ask

☒ Use default volumes

Minimum: 1.00 Step: 1.00 Maximum: 1000.00

☒ Use default limit

Limit: 0.00

OK Cancel Help

The price of 1.14059 is reduced by 2 points, to 1.14057.

4. The Trade server

Also, ECN price markup settings are applied to the request price. For example, Bid Markup = -4, Ask Markup = 4.

ECN Translation

Please specify rules and markups for translation of ECN prices to other symbols of the platform.

☒ Enable

Symbol: EURUSD.ECN

Source: EURUSD.ECN

Bid markup: -4 Ask markup: 4

OK Cancel

The price of 1.14057 is reduced by two more points, to **1.14053**. A request with this price is sent to the ECN.

```
2019.01.04 12:15:40.531 ECN '2002': request received (#417761 buy limit 1.00 EURUSD.E
```

In this case, the Trade server performs the back translation – it reduces the Ask price instead of increasing it, because the client is already using the adjusted prices. 1.14053 is the initial price for the ECN.

5. The ECN

The ECN places an order to be matched, at the price of **1.14053**, confirms it and sends trade executions to the server, for creating the order in the database (similar to gateways):

```
2019.01.04 12:15:40.532 ECN '2002': order #417761 buy limit 1.00 EURUSD.ECN at 1.140
2019.01.04 12:15:40.534 ECN '2002': request answered - Placed (#417761 buy limit 1.00 E
2019.01.04 12:15:40.536 ECN '2002': execution sent - request new order #417761
2019.01.04 12:15:40.538 ECN '2002': execution sent - added order #417761, buy limit 1.00
```

6. The Trade server

After receiving a request from the ECN, the trade server increases its price back to the client price of **1.14059**. To the price of 1.14053, the server adds 2 points of the "Spread balance" specified in symbol settings for the client group and 4 points from ECN markup settings.

Thus, in the trade database and on the client side the order has the price adjusted in accordance with the group and ECN settings (1.14059). It corresponds to the quotes delivered to the client. However, the ECN itself uses the translated price (1.14053).

```
2019.01.04 12:15:40.535 192.168.0.1 '2002': order placed for execution [#417761 buy limit
2019.01.04 12:15:40.536          '2002': order #417761 buy limit 1.00 EURUSD.ECN at 1.1
                                total time: 5.84 ms [route: 0.26 ms, ecn: 5.26 ms, ecn place: 0.0
2019.01.04 12:15:40.538          '2002': order #417761 buy limit 1.00 EURUSD.ECN at 1.1
                                total time: 7.67 ms [route: 0.26 ms, ecn: 7.15 ms, ecn place: 0.0
```

7. The ECN

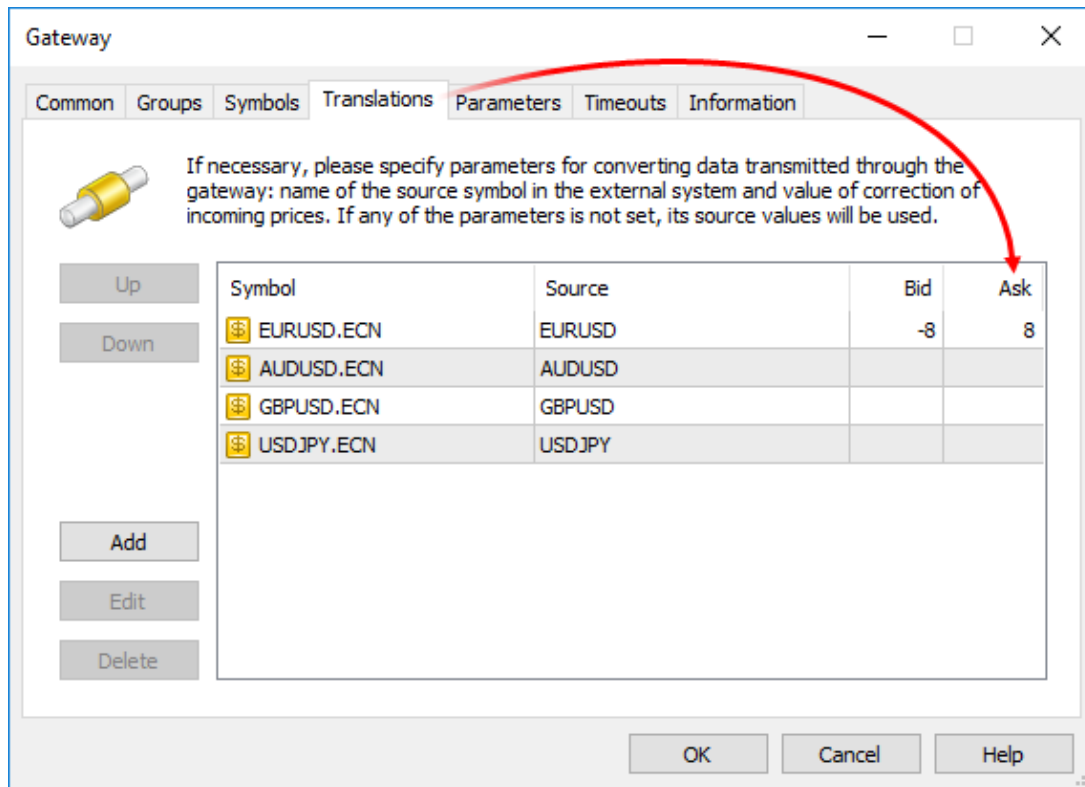
Using the converted price of **1.14053**, the ECN matches the client order with an opposite request received from the gateway.

```
2019.01.04 12:15:40.539 ECN '2002': matched 1.00 at 1.14044, #417761 buy limit 1.00 EU
2019.01.04 12:15:40.541 ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN
```

8. Gateway

The gateway receives the trade request from the ECN and adjusts it in accordance with the translation settings:

MetaTrader 5 trading platform



The order price 1.14053 is reduced by 8 points to 1.14045. A request at this price is sent to the external system.

2019.01.04 12:15:40.550 Gateway '141156': request #3090901 received (#1844674407200)

In this case, the gateway performs the back translation it reduces the Ask price instead of increasing it, because the ECN is already using the adjusted prices. 1.14045 is the initial price for the external system.

9. Gateway

After sending an order to the external system, the gateway generates an appropriate confirmation and a trade execution for the trading platform. The request price is again translated during this operation it is increased by 8 points, to 1.140453 (the external system price is translated into MetaTrader 5 prices).

2019.01.04 12:15:40.552 Gateway '18446744073709551615': request #3090901 answered

2019.01.04 12:15:40.567 Gateway execution sending complete - request new order #1844674407

2019.01.04 12:15:40.567 Gateway execution sending complete - added order #1844674407

10. The ECN

The ECN receives from the gateway the confirmation and trade executions having the price of 1.14053:

2019.01.04 12:15:40.552 ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN'

2019.01.04 12:15:40.553 ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN'

2019.01.04 12:15:40.567 ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN'

MetaTrader 5 trading platform

2019.01.04 12:15:40.568	[request new order #18446744072000000129], time: 0.07 ms ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN' [added order #18446744072000000129, buy limit 1.00 EURUSD.E
-------------------------	--

11. Gateway

Suppose, the gateway received a notification from the external system that the order was actually executed at the price of **1.14043** (2 points lower than the initially requested price of 1.14045). The following execution is formed in this case:

2019.01.04 12:15:41.833	Gateway execution sending complete - filled order #18446744072
-------------------------	--

12. The ECN

The ECN matches the order at the price of **1.14051**: the actual deal execution price (1.14043) is widened by 8 points in accordance with the gateway settings.

2019.01.04 12:15:41.841	ECN '2002': filling order #417761 on 'MetaTrader 5 Gateway ECN' [filled order #18446744072000000129, buy 1.00 EURUSD.ECN at 1.
-------------------------	---

- | |
|--|
| <ul style="list-style-type: none">• If several translation settings are specified for the same symbol in the gateway configuration and the gateway cannot determine which one to use for a particular request, the platform will try to determine it itself. For more information, see the "Price and symbol translation" section.• Some gateways may not support trading operation price conversion options. This feature is determined by the developer. In this case, the price conversion settings will only affect the displayed quotes. |
|--|

13. The ECN

When executing the order, the platform checks allowable slippage in ECN symbol settings. Suppose, the values of 5 are set both for the profitable and losing slippage:

MetaTrader 5 trading platform

ECN Symbol

Conditions Providers Filling

Please specify additional filling parameters that will be used when processing orders on the liquidity provider side.

Filling method: Limit Order

Filling attempts: 0

Filling maximum time: 60000 milliseconds

Maximum deviation: 0 pips

Maximum slippages

Profitable: 5 pips

Losing: 5 pips

OK Cancel Help

The deal price of 1.14051 is 2 points better than 1.14053, which is within the allowable profitable slippage. In this case the order is executed at its initial price in the ECN, i.e 1.14053 (point 5):

```
2019.01.04 12:15:41.834 ECN '2002': order filled 1.00 at 1.14053 on 'MetaTrader 5 Gate
2019.01.04 12:15:41.836 ECN '2002': execution sent - filled order #417761, buy 1.00 EU
```

If the deal price exceeded the deviation, the ECN would execute the order in the platform at that specific price (1.14051).

14. The Trade server

After request processing in the ECN, the price of 1.14053 is translated into the client price. The trade server increases it by 2 points in accordance with the "Spread balance" parameter of the client group and by 4 more points in accordance with the ECN translation settings (points 3 and 4). The resulting deal in the database will be formed at the price of 1.14059:

```
2019.01.04 12:15:41.836 '2002': deal performed [#129507 buy 1.00 EURUSD.ECN at 1.140
2019.01.04 12:15:41.836 '2002': order performed buy 1.00 at 1.14059 [#417761 buy limit 1
2019.01.04 12:15:41.836 '2002': order #417761 buy limit 1.00 EURUSD.ECN at 1.14059 fill
```

15. The Client

The client receives a deal at the price of 1.14059:

2019.01.04 12:15:40.539	Trades	'2002': order #417761 buy limit 1.00 / 1.00 EURUSD.EC
2019.01.04 12:15:41.836	Trades	'2002': deal #129507 buy 1.00 EURUSD.ECN at 1.1405

Thus, the price undergoes the following transformations during the entire order execution process:

- 1.14059 the order price on the client and trade server sides
- 1.14053 the order price in the ECN, which resulted from the order price conversion in accordance with the "Spread balance" specified for the trade group and the ECN markup
- 1.14045 the order price at the gateway, which resulted from the reduction of the order price in ECN by the translation value specified in the gateway settings
- 1.14043 the deal price at the gateway
- 1.14051 the deal price in the ECN, which resulted from the increase of the actual deal execution price in the external system by the translation value specified in the gateway settings
- 1.14053 the deal price at the gateway adjusted to the order price in accordance with the slippage settings
- 1.14059 the deal price on the trade server and client side, which resulted from deal price increase by the "Spread balance" specified for the trade group and the ECN markup

The client placed an order at the price of 1.14059 and received its execution at the requested price. The real order execution price in the external system was 1.14043. The broker's profit was 16 points.

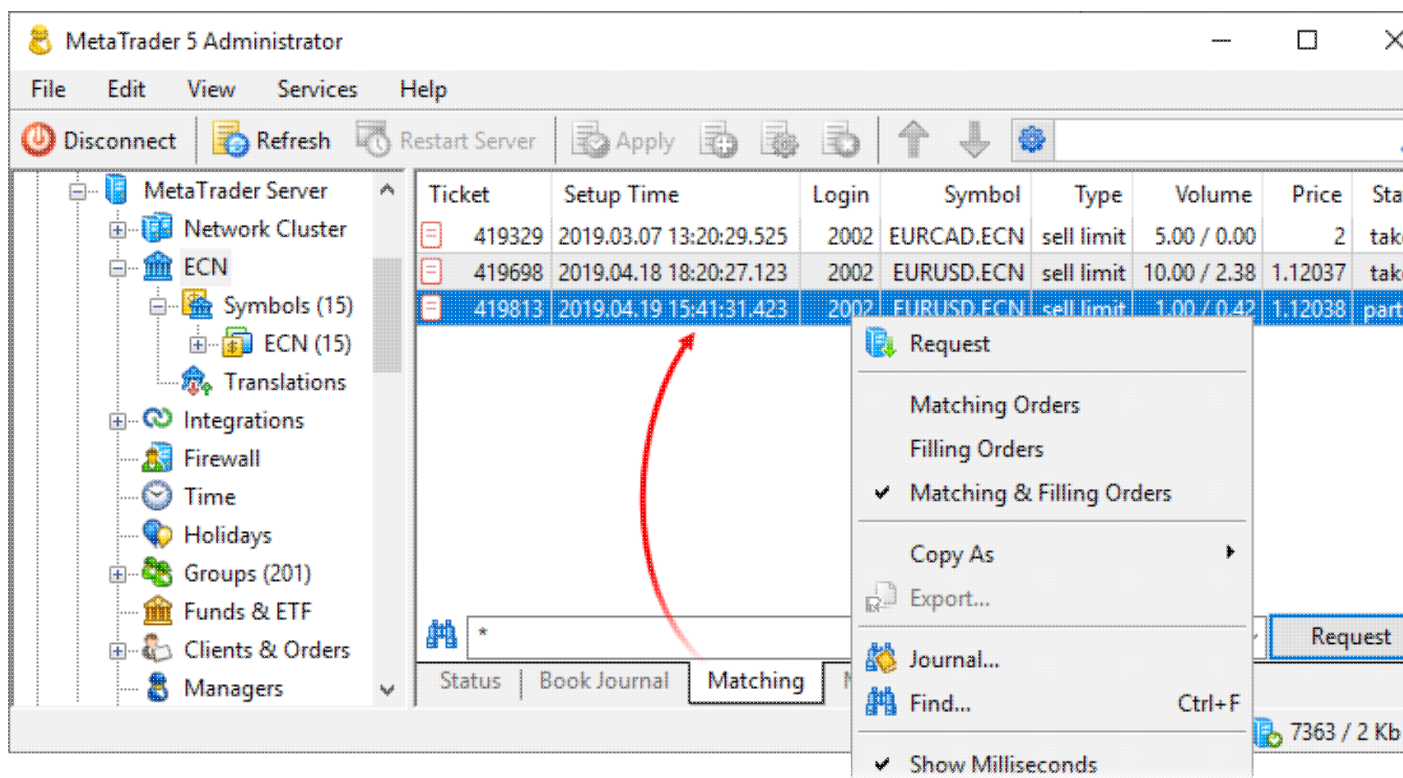
MetaTrader 5 Trading Platform / Platform Setup / ECN / Matching History

Current Matching State and Matching History

A separate tool is provided to control the order matching in ECN. You can query the current status of the pending requests queue, as well as the entire history of matching.

Current Matching State

Navigate to "ECN \ Matching", specify the groups of accounts, for which you want to view data, and click "Request".



Data can be presented in three modes. Use context menu to switch between them.

- **Matching Orders** source client requests, which were sent to ECN.
- **Filling Orders** service orders created within the ECN to execute client requests. They are formed on the basis of client orders and liquidity provider bids for execution at the respective gateway.
- **Matching Orders and Filling Orders** a tree view of the source and execution orders, which allows you to view how requests are processed in the ECN.

The required basic data is available for each order type:

Matching Orders

- **Setup Time** time of order setup in the ECN.
- **Login** client login.
- **Order** order ticket in the platform.
- **Server** the name of the trade server, on which the order was placed.
- **Group** the group to which the client account belongs.
- **Symbol** the name of the financial symbol for which the order is placed.
- **Type** type of an order placed to ECN market depth: buy, sell, buy limit or sell limit.
- **Order Volume** the volume of the order which was created in the ECN to execute the client order: the initial operation volume requested by the client in the order and the current filled volume in the ECN. The volume value is displayed in lots.
- **Client Volume** the volume of the order which was created by the client: source volume and currently executed volume. The volume value is displayed in lots.
- **Order Price** the price of the order which was created in the ECN to execute the client order.
- **Client Price** the price specified in the original client order.
- **Order Filling** order filling policy.
- **Order Expiration** order expiration mode.
- **State** current order state in the ECN.

If necessary, you can manually delete orders to be matched from the ECN. This operation might be necessary, for example, if the source trader order was canceled but the gateway did not sent the relevant transaction due to an error. To do this, execute the "Delete" command in the context menu. The command is only available for users with the "Manager" and "Edit orders, positions, and deals" permissions.

Filling Orders

- **Setup Time** time of order creation in the external system.
- **Login** client login in the MetaTrader 5 platform.
- **Order** order ticket in the MetaTrader 5 platform.
- **ID** order ID in the external system. The ID is filled by the gateway through which the order is sent for execution.
- **Server** the name of the trade server, on which the order was placed.
- **Symbol** the name of the financial symbol for which the filling order is created.

MetaTrader 5 trading platform

- **Type** the type of the order which was created for sending to the external system: buy, sell, buy limit or sell limit.
- **Volume** the volume of the order that was created in the external system system to execute the client order: requested and current filled volume. The volume value is displayed in lots.
- **Order Price** the price in the order on the external system side.
- **Client Price** the price specified in the original client order.
- **Deviation** allowable deviation for order execution. It is specified in [filling settings](#).
- **Filling** [fill policy](#) applicable to the order. It is determined by the original client order.
- **Expiration** [expiration mode](#) of the filling order.
- **State** the current state of the filling order.
- **Provider** the gateway which is used to forward the order to the external system.
- **Comment** additional information.

Matching History

ECN stores the full operation history and thus you can access complete information related to order execution, such as execution providers and volumes. To obtain this data, navigate to the "ECN\Matching History" section, set the filter by group and dates and execute the query.

MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Network Cluster

- ECN
 - Symbols (15)
 - ECN (15)
 - Translations
- Integrations
- Firewall
- Time
- Holidays
- Groups (201)
- Funds & ETF
- Clients & Orders
- Managers
- Routing's

Ticket	Execution Time	Login	Symbol	Volume	Price	State
419544	2019.04.10 13:19:45	2002	EURUSD.ECN	10.00 / 10.00	market	done
82	2019.04.10 13:19:45	2002	EURUSD.ECN			Request
31	2019.04.10 13:19:45	2002	EURUSD.ECN			
419697	2019.04.18 18:40:36	2002	EURUSD.ECN			Matching Orders
1	2019.04.18 18:40:34	2002	EURUSD.ECN			Filling Orders
32	2019.04.18 18:40:21	2002	EURUSD.ECN			Deals
33	2019.04.18 18:40:27	2002	EURUSD.ECN			Orders & Deals
34	2019.04.18 18:40:33	2002	EURUSD.ECN			
2	2019.04.18 18:40:35	2002	EURUSD.ECN			Copy As
3	2019.04.18 18:40:36	2002	EURUSD.ECN			Export...
35	2019.04.18 18:40:36	2002	EURUSD.ECN			Journal...
*						Find...

Status Book Journal Matching Matching History

For Help, press F1

Show Milliseconds

Similarly to the current matching state, orders can be presented in several modes:

- **Matching orders** traders' source orders.
- **Filling Orders** service orders created within ECN to execute client requests at gateways. The fields "Matching Login" and "Matching Order" in this mode are filled only for orders matched within the cluster (not sent to external providers)
- **Deals** the deals which were performed as a result of order execution at the gateways. These deals contain data on execution at the corresponding gateway and the client order execution parameters taking into account slippage settings.
- **Orders and deals** a tree view of all order processing stages: the initial client order, the order in ECN, the resulting deal.
One and the same client order can be executed in parts with different liquidity providers. It means that multiple filling orders can be created based on one order, as well multiple deals can be executed via different gateways.

Detailed information is available for each trading operation type:

Matching Orders

This presentation mode provides information about original client orders as well as filling orders, i.e. service orders created within ECN to execute client requests at gateways.

- **Setup Time** time of order setup in the ECN.
- **Execution time** order execution time in the ECN.
- **Login** client login.
- **Order** order ticket in the platform.
- **Server** the name of the trade server, on which the order was placed.
- **Order Symbol** the name of the financial symbol for which the order in the ECN was placed.
- **Client Symbol** the name of the financial symbol for which the client order was placed.
- **Order Type** the type of the order which was placed in the ECN: buy, sell, buy limit or sell limit.
- **Client Type** type of the order placed by the client.
- **Order Volume** the volume of the order which was created in the ECN to execute the client order: the initial operation volume requested by the client in the order and the current filled volume in the ECN. The volume value is displayed in lots.
- **Client Volume** the volume of the order which was created by the client: source volume and currently executed volume. The volume value is displayed in lots.
- **Order Price** the price of the order which was created in the ECN to execute the client order.
- **Client Price** the price specified in the original client order.
- **Order Filling** the fill policy of the order, which was created in the ECN.
- **Client Filling** the fill policy of the order, specified in the original client order.
- **Order Expiration** expiration mode of the order which was created in the ECN.

MetaTrader 5 trading platform

- **Client Expiration** expiration mode of the order, which is specified in the original client order.
- **State** current order state in the ECN.

Filling Orders

- **Setup Time** time of order creation in the external system.
- **Execution time** order execution time in the external system.
- **Login** client login.
- **Order** order ticket in the MetaTrader 5 platform.
- **Matching Login** the login of the client, who placed an opposite order which is being used to fill the current order. It is filled in only if the order is matched within the cluster, with the counter order of another client in MetaTrader 5.
- **Matching Order** the ticket of the matching order which is used to execute the current order. It is only filled if the order is matched within the cluster, with the counter order of another client in MetaTrader 5.
- **Gateway Order** executing order ticket, only used within the ECN.
- **ID** order ID in the external system. The ID is filled by the gateway through which the order is sent for execution.
- **Server** the name of the trade server, on which the order was placed.
- **Symbol** the name of the financial symbol for which the filling order is created.
- **Type** the type of the order which was created for sending to the external system: buy, sell, buy limit or sell limit.
- **Volume** the volume of filling orders: the original volume created by the client and the current filled volume. The volume value is displayed in lots.
- **Order Price** price in the filling order.
- **Trigger** the actual price at which the order was executed in the external system.
- **Deviation** allowed deviation during order execution. It is specified in filling settings.
- **Filling** fill policy applicable to the order. It is determined by the original client order.
- **Expiration** expiration mode of the filling order.
- **State** the current state of the filling order.
- **Provider** the gateway through which the order was forwarded to the external system.
- **Comment** additional information.

Deals

- **Time** deal execution time at the gateway.
- **Login** client login.
- **Order** the ticket of the source client order in the platform.
- **Gateway Order** the internal ticket of the execution order; it is only used in ECN for internal purposes.
- **Gateway Deal** the internal ticket of the deal; it is only used in ECN for internal purposes.
- **ID** ID of a deal in an external system. The ID is filled by the gateway through which the order is sent for execution.
- **Server** the name of the trade server, on which the source order was placed.
- **Symbol** the name of the financial symbol for which the deal was executed.

- **Type** deal type: buy or sell.
- **Volume** the executed deal volume in lots.
- **Deal Price** the price at which the deal was executed at the platform side, taking into account price translation settings.
- **Gateway Price** the price at which the deal was actually executed at the external system side.
- **Commission** commission charged by the external system for the deal. The value is filled by gateways.
- **Provider** the gateway via which the deal was executed.

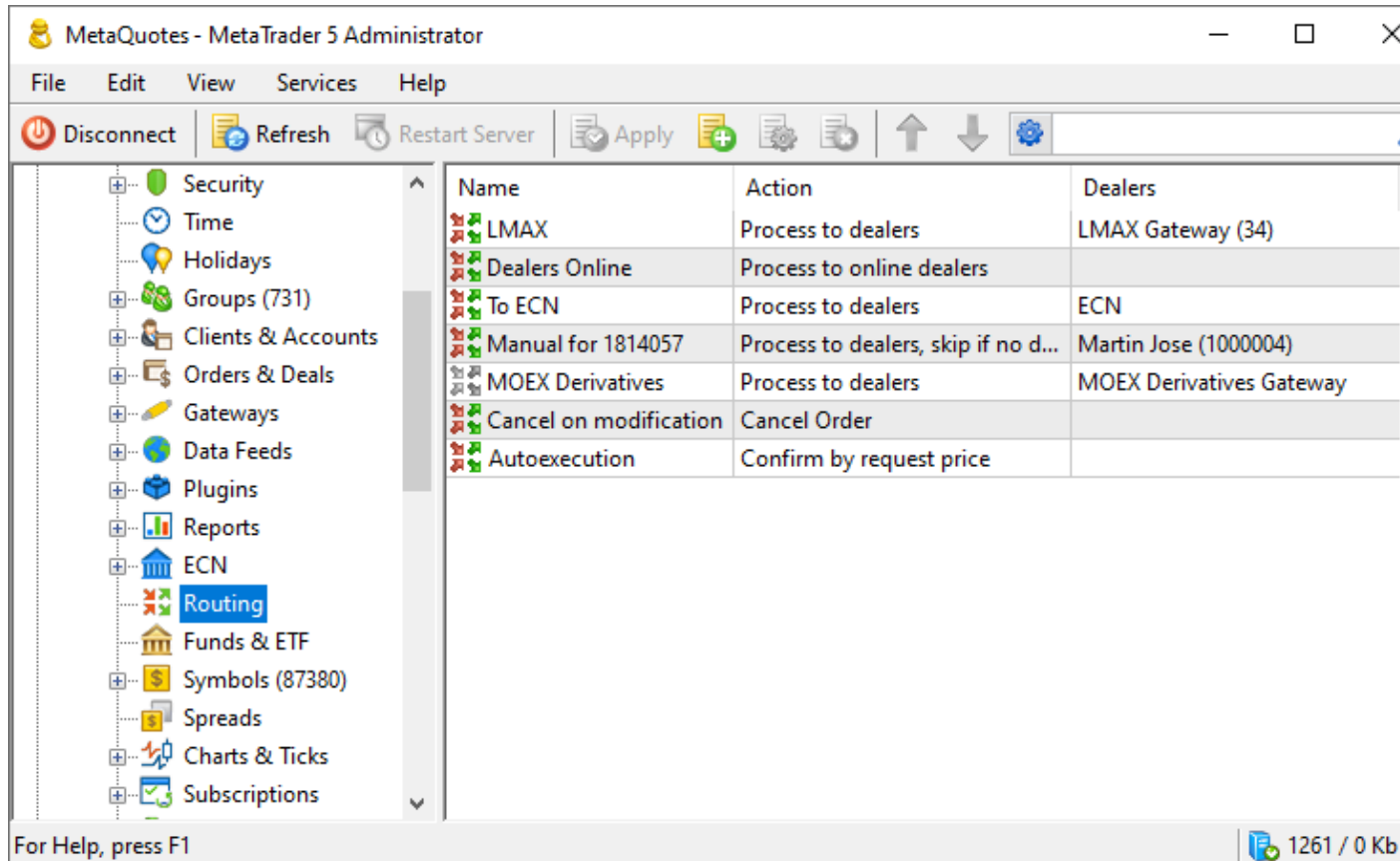
Orders and Deals

- **Ticker** order/deal ticket. Depending on the operation type, the ticket of the source client order, of the filling order and of the internal deal (used only in ECN) is shown.
- **Setup Time** time of order creation by the client and of filling order creation in the ECN.
- **Execution time** order execution time.
- **Login** client login.
- **ID** operation ID in the external system. The ID is filled by the gateway through which the order is sent for execution.
- **Symbol** name of a trading symbol.
- **Type** trading operation type.
- **Price** depending on the operation type, this shows the price of the source order, of the filling order or of the executed deal.
- **AW Price** average weighted order execution price by external liquidity providers. This value is used when an order was executed partially with different providers, otherwise it is equal to the Price field. It is calculated as (execution price at gateway 1 * volume of deal 1 + ... + execution price at gateway N * volume of deal N) / (volume of deal 1 + ... + volume of deal N).
- **State** order state.
- **Commission** commission charged by the external system for the deal. The value is filled by gateways.
- **Provider** the gateway via which the order was executed.
- **Comment** additional information.

MetaTrader 5 Trading Platform / Platform Setup / Routing

Routing

This section allows managing rules of processing clients' trade requests depending on different conditions.



The screenshot shows the 'MetaQuotes - MetaTrader 5 Administrator' window. The left sidebar contains a tree view with various settings categories. The 'Routing' category is selected and highlighted in blue. The main area displays a table of routing rules.

Name	Action	Dealers
LMAX	Process to dealers	LMAX Gateway (34)
Dealers Online	Process to online dealers	
To ECN	Process to dealers	ECN
Manual for 1814057	Process to dealers, skip if no d...	Martin Jose (1000004)
MOEX Derivatives	Process to dealers	MOEX Derivatives Gateway
Cancel on modification	Cancel Order	
Autoexecution	Confirm by request price	

At the bottom of the window, there is a status bar that reads 'For Help, press F1' and a file size indicator '1261 / 0 Kb'.

The table of rules contains the following details:

- **Name** names of routing rules;
- **Dealers** list of dealers who process request that meet the conditions.

Execution of Rules

Rules are executed in the direction from top to bottom. If an incoming request meets the conditions of the upper rule, it is processed according to this rule; otherwise the request is checked for meeting conditions of the second rule, and so on. A request is processed according to the rules until it is executed or passed to a dealer. For example, meeting the rule directing to delay the request execution for some time, the request will be delayed and then checked for correspondence to further rules.

Any change in the routing table leads to that all request currently processed according to these rules are moved to the beginning of the table and go through all rules again.

To change the position of rules in the list, use context menu commands: "↑ Move Up" and "↓ Move Down", the same commands of the Edit menu or buttons on the Standard toolbar.

If you need to add a rule or change an existing one, press "+ Add" or "⚙ Edit", respectively. They are available in the Edit menu, on the Standard toolbar and in the context menu. As soon as you press the button, the rule configuration window with two tabs will be opened.

Additional general information about working with configuration records is given in the "Working with Instructions" section.

Common

Routing: Delay for 2 seconds

Common | Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Delay for 2 seconds

Perform action: Delay in milliseconds 2000

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
ab Symbols	Equal (=)	Forex*

Add Edit Delete

OK Cancel Help

On the "Common" tab you set common parameters of the rule, as well as conditions of requests, under which they will be processed according to this rule:

- **Enable this rule** to make a rule effective, tick this field;
- **Name** name of the rule;
- **Perform action** select the action that should be executed with regard to the request that meets the rule conditions. For some actions an additional field appears to the right of this one (for example, number of seconds and ticks for a delay);
- **Where request is** select a rule condition by the type of request. In order to select or deselect all elements in the list, click the right mouse button;
- **Where order is** select a rule condition by the type of order. In order to select or deselect all elements in the list, click the right mouse button;
- **Where conditions are** select additional conditions to choose requests.

- The rule conditions are taken into account according to the principle of "AND". In other words, in order for a request to be processed according to the rule, it must meet all the conditions specified in this rule.
- The rule conditions must not contradict each other. For example, if you indicate the following two group conditions in one rule: \real* and demo*, none of the trading requests will correspond to this rule. A request cannot belong to a demo account and a real one at the same time.

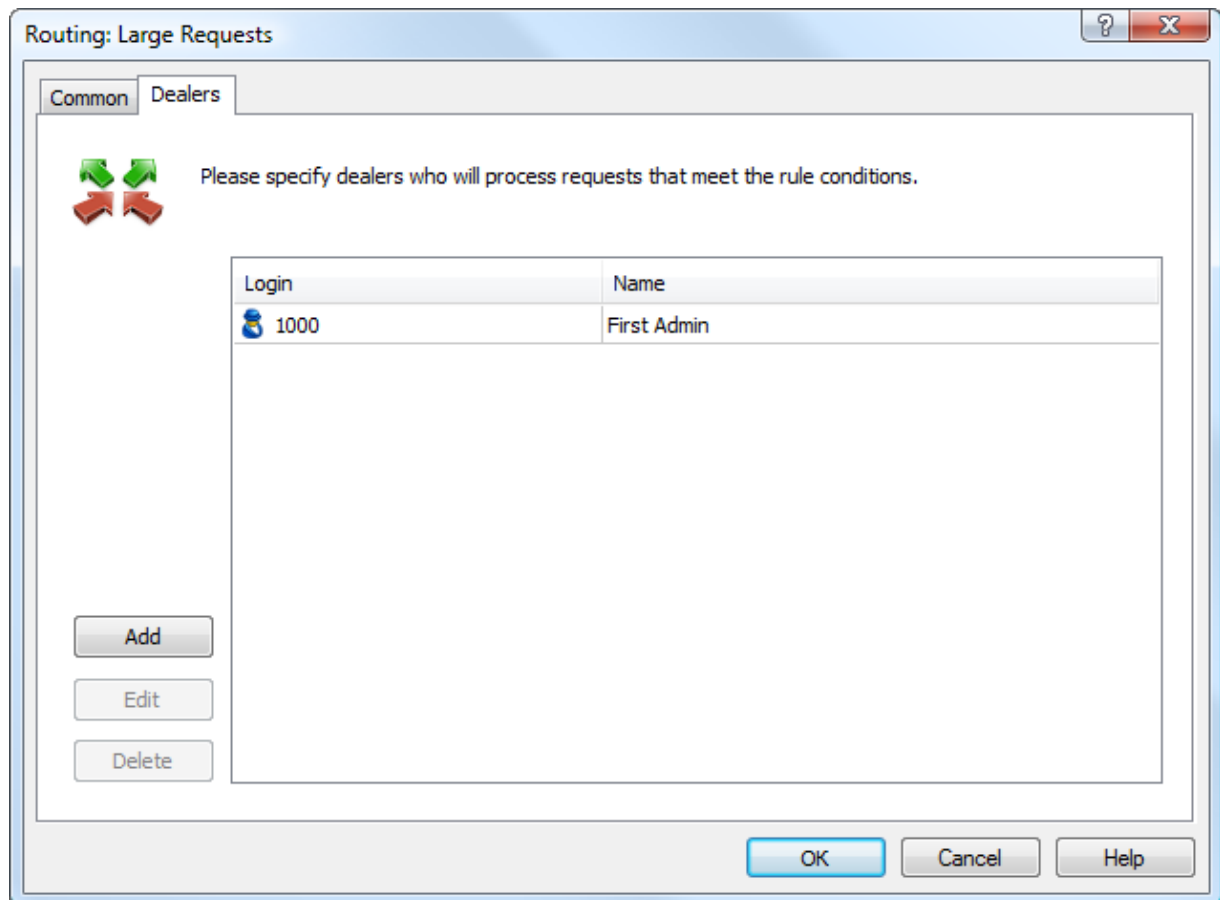
A separate block is included for adding additional conditions. If you press "Add" a new line will be created in the table. Indicate the following in it:

- **Type** parameter, according to which the condition will be checked;
- **Condition** condition of rule triggering (equal, not equal, etc.);
- **Value** value the condition parameter is compared to.

Be attentive when specifying conditions, For example, if you select the "Symbol" type, you can't set the condition "more than or equal to" ($\geq$), etc. Otherwise the condition is incorrect and is not processed.

To edit an additional condition, select it and press "Edit" or double click on one of its parameters. To delete a condition press the "Delete" button or the Delete key.

Dealers



There is an action performed by the request, at which it is sent to a dealer or a gateway to be handled - "Process to dealers". Requests processed by such a rule are enqueued to be processed by dealers specified in this tab.

- **Add** add a dealer. A new line appears in the table as soon as you press this button. In the "Login" field select one of managers' accounts from the list. Only accounts with permissions to modify orders are shown in this list. The "Name" field is filled out automatically;
- **Delete** delete a selected dealer. The same action can be performed by pressing the "Delete" key;
- **Edit** edit a selected point. The same action can be performed by a double click on the field the login is specified in.

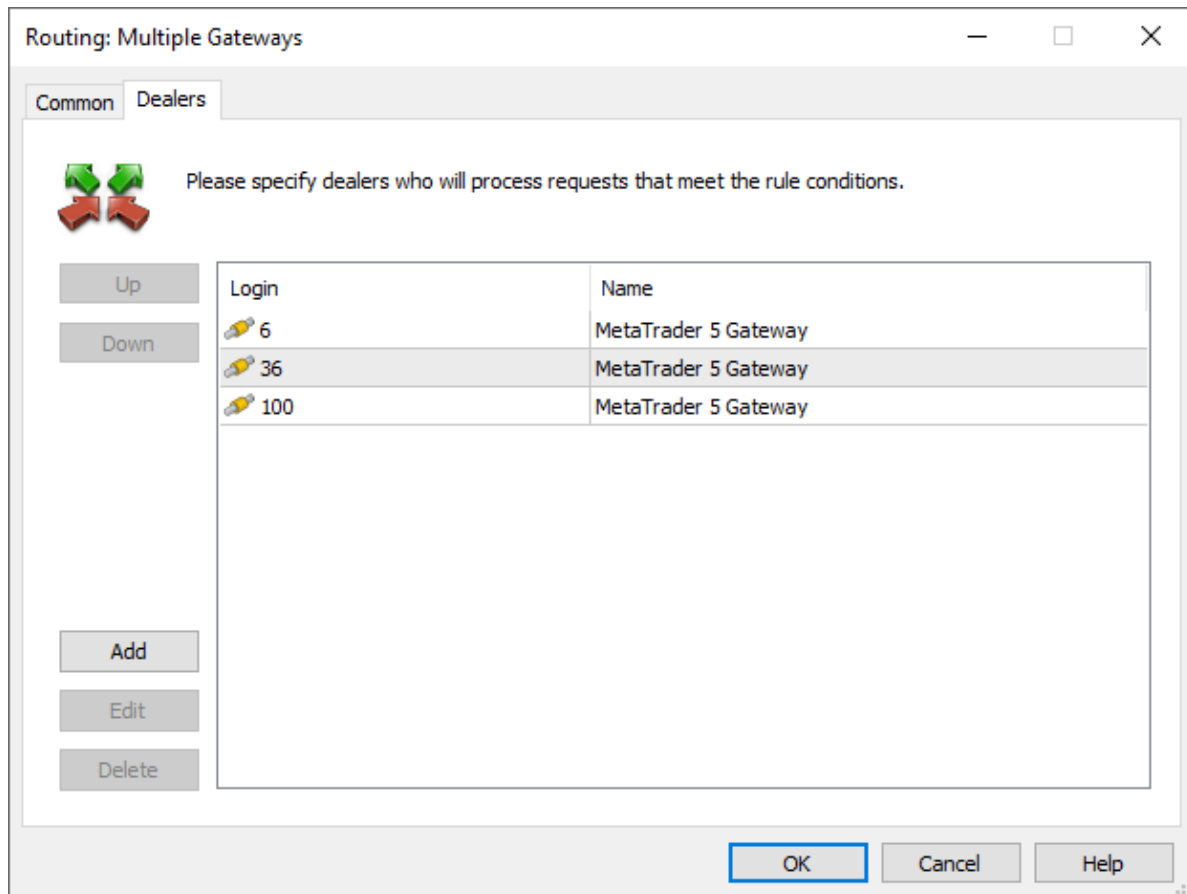
- Only a gateway or a manager account with the "Dealing" permission can be added to the list of dealers.
- The list of dealers provides a special "ECN" item. Use this option to forward trade requests to ECN matching.

To save the settings press "OK". If you press "Cancel", the changes will not be saved.

Sending requests to multiple dealers

You can configure routing rules to forward request to multiple dealers/gateways at a time. For example, you can be using several gateway instances to connect to the same external system. in this case, each of the gateways uses its own account in the trading system, to which only operations from certain MetaTrader 5 accounts are forwarded.

MetaTrader 5 trading platform



In such cases, the routing system attempts to automatically send the request to the appropriate gateway.

1. The server determines gateway/dealer priority: to which one the request should be sent in the first place. This is done based on the external accounts data of the account from which the request is received.

MetaTrader 5 trading platform

Account: 3019575, Carter Aiden, USD, 1:100, Netting

Overview Personal **Account** Profile Security

Group:

Color: ☐ None ☐ None

Leverage:

Bank account:

Agent account:

☒ Enable this account

☒ Enable trading

☒ Enable to change password

☒ Enable trailing stops

☒ Enable one-time password

☒ Enable trading by Expert Advisors

☒ Enable Web API connections

☒ Enable daily reports

☐ Change password at next login

☐ Sponsored MetaTrader Virtual Hosting

Trade accounts:

Gateway ID	Account
 6, MetaTrader 5 Gateway	1009336696

If the gateway from the routing rule is indicated in symbol settings, the request will be forwarded to this gateway.

If the gateway refuses the request for some reason or returns it, the request will be forwarded to other gateways. The gateway that captures the request first will process it.

2. If a certain dealer/gateway refuses to process the request (for example, if processing contradicts the gateway's internal logic), the request is deleted from the dealer's/gateway's list. After that the request can be picked up by another dealer/gateway from the rule list.

For example:

- The rule sets the action "Process to online dealers". Two gateways are indicated in the dealers list: gateway 1 and gateway 2.
- Both gateways are currently online. A request is assigned to these connections.
- Gateway 1 picks up the request and refuses it.
- The request is removed from the gateway 1 list and is forwarded to gateway 2.
- Gateway 2 captures the request and processes it.

If gateway 2 also refuses the request, it will be deleted from its list as well. Since there will be no gateways left to process the request, the server will reject the request and will add a corresponding log:

request rejected, due all assigned dealers returned request in queue (instant buy 4 EURUSD)

General example for the operation scheme:

- If the request is assigned to only one gateway (and to no one else) in the rule, the request goes to this gateway.
- If multiple gateways are assigned in the rule (and only gateways), the request goes to the one whose ID is found in the external accounts list. Gateways are checked in the same order as they are indicated in the rule.
- If there is no such a gateway, the request is sent to the first gateway in the rule list (for which the request has not been deleted and is available).
- If multiple gateways and dealers are assigned in the rule, gateways are checked first (in the same order used in group settings): gateways specified in the external accounts list of the account are searched. If such a gateway is found, the request is sent to that gateway.
- If the gateway refuses the request, it will be processed according to the rules specified above.
- If no priority gateway is found, the request will be passed to the gateway or dealer who first captures the request from the queue.

Context Menu

The context menu of the "Routing" tab allows executing the following commands:

-  **Add** add a new rule;
-  **Edit** edit a selected rule;

-  **Delete** delete a selected rule;
-  **Move Up** move a selected rule up relative to others;
-  **Move Down** move a selected rule down relative to others;
-  **Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
-  **Enable** enable the selected configuration.
-  **Disable** disable the selected configuration.
- **Automation triggers** create an automation task for the selected event or edit an existing one. The menu displays only the triggers and tasks associated with the current section.
- **Automation actions** add an automation action to an existing task or create a new task based on the action. The menu displays only the actions associated with the current section.
-  **Export to File** export routing settings to a file.
-  **Import from File** import routing settings to a file.
-  **Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration automatically specified in the query field. You will only need to press the request button.
-  **Find** open the search window;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators in the table with the rules.

MetaTrader 5 Trading Platform / Platform Setup / Routing / Actions and Conditions

Actions and Conditions

In the routing rules you set actions that are taken in accordance with trade requests, as well as conditions of selecting requests, meeting which these actions are taken.

Actions

An action performed to a trade request, is selected in the "Perform action" field of the "Common" tab:

- **Delay in milliseconds** delay request execution by the specified number of milliseconds. When selecting this action, an additional field for specifying the number of milliseconds will appear to the right of it. After this action is performed, the request execution continues in accordance with the created rules located below.
- **Delay in ticks** delay request execution by the specified number of ticks. When selecting this action, an additional field for specifying the number of ticks will appear to the right of it. After this action is performed, the request execution continues in accordance with the created rules located below. Maximum allowed delay is 60 ticks.
- **Clear TP** clear the Take Profit level set in the order.
- **Clear SL** clear the Stop Loss level set in the order.
- **Clear SLTP** clear the Stop Loss and Take Profit levels set in the order.
- **Process to dealers** enqueue the request to be processed by a dealer specified in the "Dealers" tab.
- **Process to online dealers** enqueue the request to be processed by dealers who are currently online. At that, the list of dealers from the "Dealers" is ignored. The request is passed to any online dealer that has a right to process it.
- **Reject** reject the request. An additional field is available for this action, allowing you to specify the reason for rejecting a request. This reason will be displayed in the trading dialog in client terminals when an operation is rejected in accordance with this rule. Providing a detailed description will help eliminate unnecessary questions from clients. The maximum message length is 31 characters.

MetaTrader 5 trading platform

Position: #5074015223 sell 0.5 EURUSD 1.07123

Symbol: EURUSD, Euro vs US Dollar

Type: Instant Execution

Volume: 0.50 50 000 EUR

Stop Loss: 0 Take Profit: 0

Comment:

Rejected
(Custom rejection reason)

OK Print

- **Requote** send the current market price responding to the request.
- **Confirm by request price** confirm the order execution at the requested price.
- **Confirm by market price** confirm the order execution at the current market price. The action is applied to both client orders and orders placed by dealers.
- **Cancel order** cancel a pending order during its activation or modification. For example, if a pending order has triggered, but the client has already reached the maximum position volume and a new position cannot be opened, the routing rule will remove this order. Otherwise, the order would have continued to trigger on each new tick. When removing an order by this rule, "deleted [by dealer]" is added to the order comment. An entry about the routing rule that canceled the order is also added to the server journal:

```
'1813415': modify order #100103968 buy limit 1.00 EURUSD at 1.06927 sl: 0.00000 tp: 0.00000
'1813415': request rejected, order #100103968 will be canceled, rule 'Cancel on modification'
'1813415': order canceled - by routing rule [#100103968 buy limit 1.00 EURUSD at 1.06927]
```

- An additional option of "skip this rules if no dealers online" is available for actions that pass requests to dealers. If this option is enabled, the rule will be ignored, if none of dealers is online.
- The "Cancel order" action can only be applied during pending order activation or modification (including modification buy a dealer). The rule does not affect other trade requests.

Conditions by the Request Type

This type of conditions is specified in the "Where request is" field. Here you can specify the types of client requests, which will be proceeded due to the given rule. Any number of elements can be selected in this field.

- **Price** request price (for request execution).
- **Request execution** confirmation of order execution at the dealer's price in the request execution mode (if the option of order confirmation is enabled).
- **Instant execution** placing of an order in the instant execution mode.
- **Market execution** placing of an order in the market execution mode.
- **Exchange execution** placing of an order in the exchange execution mode.
- **Pending order** placing of a pending order.
- **SL & TP modification** modification of Stop Loss and Take Profit of a position.
- **Order modification** modification of a pending order.
- **Order removal** removal of a pending order.
- **Close By** an operation of closing two oppositely directed positions at a single symbol.
- **Order activation** activation of a pending order.
- **Stop Limit activation** activation of Stop Limit.
- **SL activation** activation of a Stop Loss order.
- **TP activation** activation of a Take Profit order.
- **Stop-Out order** request to delete a pending order as a stop-out level is reached (is used, if margin requirements are set for pending orders).
- **Stop-Out position** request to forced closing of a position as the stop-out level is reached.
- **Order expiration** removing an order due to its expiration.
- **Execution by dealer** execution of a trade operation by a dealer.
- **Pending order by dealer** placing of a pending order by a dealer.
- **Position modification by dealer** modification of a position by a dealer.
- **Order modification by dealer** modification of an order by a dealer.
- **Order removing by dealer** deletion of an order by a dealer.
- **Pending order activation by dealer** forced activation of a pending order by a dealer.
- **Stop-Limit activation by dealer** forced activation of a Stop-Limit order by a dealer. As a result of this, the order becomes a Limit order.
- **Close By by dealer** an operation of closing two oppositely directed positions at a single symbol performed by a dealer.

Conditions by the Order Type

This type of conditions is specified in the "Where order is" field. Here you can specify the types of orders which will be proceeded due to the given rule. Unlimited number of elements can also be selected in this field.

- **Buy** a buy order;
- **Sell** a sell order;
- **Buy Limit** a limit order to buy;
- **Sell Limit** a limit order to sell;
- **Buy Stop** a stop order to buy;
- **Sell Stop** a stop order to sell;
- **Buy Stop Limit** Buy Stop Limit order;
- **Sell Stop Limit** Sell Stop Limit order.

For "SL and TP modification", "SL activation", "TP activation", "Stop-Out Position" and "Position modification by dealer" operations, the condition by order type sets the direction of positions associated with these operations. For example, to route an operation closing a Buy position by Take Profit, the rule should be specified as "For orders: Buy", despite the fact that a Sell operation is actually performed to close the position.

Additional Conditions

You can set additional conditions for a rule in the "Where conditions are" field. In additional conditions you can set a parameter to compare, the condition (equal, not equal, etc.) and the value to which the parameter is compared. The following parameters can be set:

Request

- **Date and time** using this parameter you can compare the date and time when a request was received with a date and time specified in the "Value" field;
- **Symbols** using this parameter you can specify a symbol or a group of symbols, requests by which will be processed according to the rule. Symbols can be specified using masks containing "*" and "!";
- **Request volume** deal volume requested in the order (in lots). This parameter allows setting up a rule depending on the request volume. For example, it can be the automatic processing of requests with volume less than 1 lot;
- **Deviation from market, points** this condition works for the instant and market execution of orders, and for triggering of pending orders and SL/TP orders. It considers the difference between the price of a client's request and the current market price. In the market execution mode, when a client doesn't specify a price in the order, the condition considers the difference between the market price at the time of placing the order and the current market price. For buy trades, the deviation is calculated as ("Current Ask Price" - "Price of Client's Request"), and for sell trades it is ("Price of Client's Request" - "Current Bid Price"). For example, if a client wants to make a buy deal at 1.2000, and the current Ask price is 1.2008, then the deviation is $1.2008 - 1.2000 = 8$ points. Thus, when setting up this condition, note the following if deviation is positive, then opening by the request price is for benefit of the client, if deviation is negative then for benefit of the broker. Another example: if the deviation is set to < -8 , then buy requests, in which the requested price is greater than the current price by more than 8 points, will fall under this condition;
- **Time** using this parameter you can compare the time when a request was received with a time specified in the "Value" field;
- **Day of week** this parameter allows to route requests depending on a day of the week;
- **Request comment** this parameter allows to compare a request comment with a specified one. If "=" condition is specified, exact match of a comment is checked. If ">" or ">=" conditions are set, a specified substring is searched in a comment string. If "<" or "<=" conditions are set, comment substring is searched in a specified string;
- **Placed by expert** this parameter allows to route requests placed by MQL5 programs.

- **Placed by signal** all trade operations copied by a client terminal according to a subscription to a trade signal are marked in a special way. This parameter allows routing trade requests created by trade signals. If it is equal to "true", the rule will trigger for all signal operations. If it is equal to false, it will trigger for all operations except for signal ones.
- **Dealer processed request** this parameter allows applying the routing rules depending on a dealer (or gateway) identifier specified in an order or position. The dealer identifier is specified in an order after it has been confirmed (processed) by the dealer/gateway. Due to it, this rule can be applied only when modifying/deleting an order, and not for newly created orders as they do not have a dealer identifier. For positions, the dealer identifier is specified according to the dealer identifier of the order, whose execution resulted in the position opening.
This parameter can be used when processing trade operations for a symbol through several gateways simultaneously. An order or position created through a specific gateway must be further processed through the same gateway.
- **Dealer placed request** the parameter allows routing requests by a login of a dealer who set a request on a client's behalf.
- **Deviation from market, spreads** this condition works when executing orders in the Instant or Market modes, as well as when pending orders and Stop Loss/Take Profit orders are triggered. It takes into account the difference between the price of a client's request and the current market price. During a market execution, when a client does not set a price in the order, the difference between the market price during the request and the current market price is taken into account.
The deviation is set in spreads. For floating-spread symbols, the current spread valid during the request check is used. For fixed-spread symbols, a spread value from the symbol settings is used.
For Buy trades the deviation is calculated as ("Current Ask price" - "Client's request price"), for Sell trades it is equal to ("Client's request price" - "Current Bid price"). For example, if a client wants to buy at 1.2000, and the current Ask is 1.2008, then the deviation is equal to $1.2008 - 1.2000 = 8$ points. The deviation value is divided by the current spread value and the result is compared with the value in the rule.
When setting the condition, keep in mind that if the deviation is positive, opening at the request price is performed in the client's favor, if the deviation is negative, opening is performed against the client. Another example: if we set < -1 , the condition corresponds to the buy requests where a request price exceeds the current price by more than 1 spread.
- **Request price** order price. In the Instant and Request execution modes, the price which the trader has specified in the order is checked; in the Market or Exchange execution modes, the current symbol price is checked. Since prices of different symbols differ considerably, this condition should be used together with the "Symbol" condition. It can also be used to set the general price threshold. For example, you can decline all requests with a cost of less than one dollar.
- **Value** the value of the requested operation in the symbol's base currency. The value calculation depends on the symbol's margin/profit calculation mode.

For symbols with the Forex calculation mode, the value is equal to the product of the contract size and the volume. For example, for EURUSD with the contract

size of 100,000, the value of 1 lot is equal to EUR 100,000.

For symbols with the CFD, CFD Leverage, CFD Index and Futures calculation modes, the contract size is not expressed in money, but, for example, in the number of shares. Therefore, the contract size is additionally multiplied by the symbol value to obtain the value in monetary terms. For Futures symbols, the final value is additionally multiplied by the ratio of the tick value to the tick size. For example, if a Futures symbol has USD as the base currency, the contract size is equal to 100, the cost is 33, and the tick value to tick size ratio is 1/0.1, then the value of one lot of the position will be equal to $100 \times 33 \times 10 = \text{USD } 33,000$. For a CFD symbol with the same parameters, one lot size will be $100 \times 33 = \text{USD } 3,300$.

Account

- **Login** client's account number. Using this parameter, you can create individual rules for specific accounts;
- **Group** group to which the client is included. This parameter allows setting up rules for specific groups of accounts. For specifying groups, the mask "*" can be used. For example, if you specify "demo*" it will indicate all groups, whose name (including path) starts with "demo" (for example, demo\forex, demoforex, demoforex\usd). If you specify "*forex", it will indicate all groups, whose name ends with "forex" (for example, demo\forex, real\forex). You can also specify several groups or group masks separate by comma.
- **Country** in this parameter you can specify a country of a client. All the clients who live in it will be processed according to the given rule.
- **City** use this parameter to apply the rules to the clients who live in the specified city;
- **ZIP code** use this parameter to apply the rule for clients with the specified zip code.
- **Color** use this parameter to apply the rules to the clients marked by the specified color;
- **Leverage** use this parameter to apply the rules to the clients with the specified leverage;
- **Comment** use this parameter to apply the rules to the clients with the specified (the "Comment" field in the client record);
- **Status** this parameter allows request routing based on the client status. For example, by combining it with the Country and Symbol parameters, you can prevent non-US clients from trading US stocks.
- **Margin** using this parameter you can set up the rule to be applied depending on the size of margin reserved at the moment (in deposit currency);
- **Margin level** this parameter allows applying the rule depending on the current margin level (in percents);
- **Free margin** this parameter allows applying the rule depending on the current size of free margin (in deposit currency);
- **Equity** use this parameter to apply the rules depending on the current size of client's equity (in deposit currency);
- **Balance** use this parameter to apply the rules depending on the current balance of the client (in deposit currency);

Position

- **Position volume** current position volume for the symbol of the request;
- **Position profit** current profit on the position of the symbol of the request.
- **Position value** position value on the symbol's base currency. The value calculation depends on the symbol's margin/profit calculation mode.

For symbols with the Forex calculation mode, the value is equal to the product of the contract size and the volume. For example, for EURUSD with the contract size of 100,000, the value of 1 lot is equal to EUR 100,000.

For symbols with the CFD, CFD Leverage, CFD Index and Futures calculation modes, the contract size is not expressed in money, but, for example, in the number of shares. Therefore, the contract size is additionally multiplied by the symbol value to obtain the value in monetary terms. For Futures symbols, the final value is additionally multiplied by the ratio of the tick value to the tick size. For example, if a Futures symbol has USD as the base currency, the contract size is equal to 100, the cost is 33, and the tick value to tick size ratio is 1/0.1, then the value of one lot of the position will be equal to $100 \times 33 \times 10 = \text{USD } 33,000$. For a CFD symbol with the same parameters, one lot size will be $100 \times 33 = \text{USD } 3,300$.

- **Age of position** using this parameter, you can specify time in seconds that has passed since the opening of position for the symbol, the request for which is being processed. This parameter allows tracking positions by the time of holding;
- **Last change of position** using this parameter, you can specify time in seconds that has passed since the last change of position for the symbol, request for which is being processed. A change of position includes increasing its volume and partial closing. This parameter allows avoiding evasion of the previous one ("Age of position") through manipulating one position by increasing and decreasing its volume;
- **Avg lifetime of position** this parameter allows tracking the average lifetime of a position for the symbol, the request for is being processed. The Average lifetime is calculated in seconds using the following formula: $\text{current time} ((\text{time of opening} + \text{time of changing})/2)$.
- **Position SL touched** condition is triggered when the market price touches the stop loss of a position (while the stop loss may not be activated yet). Possible values true or false.
- **Position TP touched** condition is triggered when the market price touches the take profit of a position (while the take profit may not be activated yet). Possible values true or false.
- **Positions total** this parameter allows tracking the total number of a client's open positions on all symbols. For example, you can set the platform to reject trade requests to open new positions, if the client has reached the specified limit.
- **Client positions total by symbol** the parameter allows tracking the number of positions on the symbol that is specified in the current trade request. For example, if a client has placed an order on EURUSD, this condition will check the current number of the client's open positions on EURUSD.

Order

- **Order SL touched** condition is triggered when the market price touches the stop loss of a pending order. Possible values true or false. In combination with the "Order activation" condition, it allows you to track the simultaneous breakthrough of a pending order (trigger) and its stop loss level. This may occur during a release of important news or after a weekend when a large price gap is formed.
- **Order TP touched** condition is triggered when the market price touches the stop loss of a pending order. Possible values true or false. In combination with the "Order activation" condition, it allows you to track the simultaneous breakthrough of a pending order (trigger) and its take profit level. This may occur during a release of important news or after a weekend when a large price gap is formed.
- **Orders total** this parameter allows tracking the total number of a client's orders on all symbols. All orders are taken into account, including pending and history orders. Each opening and closing of a position (including partial closure), as well as placing of a pending order increases this counter.
- **Client orders total by symbol** the parameter allows tracking the number of orders on the symbol that is specified in the current trade request. For example, if a client has placed an order on EURUSD, this condition will check the current number of the client's orders (both active and history) on EURUSD.
- **Order In** orders, as a result of which the trader enters the market, increases the current position or reverses it. When a new order is received, the platform calculates the effect of the order execution on the current account state: whether a new position will be opened or existing one will be closed or reversed. The check applies to market orders (the effect of the order execution) and for pending orders (the effect of the order triggering and execution at a price specified therein).
- On hedging accounts: any pending order is considered an entry order, since its execution can only lead to the opening of a new position.
- On hedging accounts: triggering of Stop Loss, Take Profit and Stop Out is considered an exit order, since this always leads to closing of existing positions.
- On netting accounts: order direction is determined relative to the current uncovered volume of the symbol, taking into account the existing open position and all pending orders for that symbol. For example, the account has a Buy 1.00 position, a Buy Limit 2.00 order and a Sell Limit 1.00 order. If both pending orders were triggered, the trader would have the resulting position Buy 2.00. This resulting position is taken into account when checking the "Order In" rule. For example, if the trader places a Buy Limit order with the volume of 1.00, the system considers it an entry order. A Sell Limit 1.00 order would be considered an exit order.
- On netting accounts: for the rule checks performed for pending orders, a position having a Stop Loss or Take Profit level, is considered to be fully covered, so the calculated final uncovered position volume is considered to be zero. For example, the account has a Buy 1.00 position with a specified Stop Loss, a Buy Limit 2.00 order and a Sell Limit 1.00 order. If both pending orders were triggered, the trader would have the resulting position Buy 1.00. This resulting position is taken into account when checking the "Order In" rule for pending orders.
- **Order Out** orders, as a result of which the trader exits the market, partially closes the current position or reverses it. The rule is similar to the "Order In"

condition.

Statistics

- **Daily deals** this parameter allows applying the rules depending on the number of deals executed for the client during the previous and current day (including holidays and weekend);
- **Daily deal period** frequency of trades during the day. Calculated on the basis of 8 last deals (average time in seconds between deals);
- **Daily profit** profit of the client whose request is being processed for the previous and current day (including holidays and weekend);

Symbol

- **Gap mode** this parameter allows processing trade requests in a special way under the market conditions that differ from normal ones. For example, after a gap, client requests can be rejected or requoted during a certain number of subsequent ticks. The gap mode is defined separately for each symbol according to its settings.
The parameter may take two values true or false (enabled/disabled). If the gap mode is active when checking a request according to the selected symbol routing rule, actions set in this rule are applied to it.
The gap mode is checked by an instrument's Bid and Ask prices. If a gap is detected at least on one of the prices, the rule is triggered.
- **Current spread** the parameter allows applying routing rules depending on the difference between the current Bid and Ask prices (as at the time of check) of the financial symbol from the trade request. For example, you can forward trade requests to a dealer if the spread increases a certain value.

For string type parameters (for example, City, Request comment, etc.), exact values match is checked when "=" condition is specified. If ">" or ">=" conditions are set, a specified substring is searched in the parameter. If "<" or "<=" conditions are set, parameter value substring is searched in a specified string;

MetaTrader 5 Trading Platform / Platform Setup / Routing / Example of Rules

Example of Rules

In this section we will consider a set of routing rules for processing most common situations that appear in the processing of trade requests. The set consists of eight rules:

Name	Dealers
Automate All Demo	
Allow Modification	
Large Requests	First Admin (1000)
Large Positions	First Admin (1000)
Medium Requests	John Smith (1111), Joe Black (1136)
Medium Positions	John Smith (1111), Joe Black (1136)
Requote Cheaters	
Automate Other Requests	

Automate All Demo

This rule allows to automatically process all requests that come from demo accounts without any changes.

Routing: Automate All Demo

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Automate All Demo

Perform action: Confirm by request price

Where request is: Price, Request Execution, Instant Execution, Market Execution, Pending order, SL & TP r

Where order is: All

Where conditions are:

Type	Condition	Value
Client group	Equal (=)	demo*

Add Edit Delete

OK Cancel Help

The following parameters should be set up on the "Common" tab of the rule:

- **Perform action: Confirm by request price**
Confirm requests at prices that are specified in orders.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: Client group = demo***
For all sub-groups of the "demo" group.

Allow Modification

This rule allows modifying positions and pending orders without any limitation. All operations of this type will be processed automatically without any changes.

Routing: Allow Modification

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Allow Modification

Perform action: Confirm by request price

Where request is: SL & TP modification, Order modification, Order removal, Order activation, Stop-Limit act

Where order is: All

Where conditions are:

Type	Condition	Value
------	-----------	-------

Add Edit Delete

OK Cancel Help

The following parameters are specified for this rule:

- **Perform action: Confirm by request prices**
Confirm requests at prices that are specified in orders.
- **Where request is: SL & TP modification, Order modification, Order removal**
For requests to change levels of Stop Loss and Take Profit of open positions, as well as to change and delete pending orders.
- **Where order is: All**
For all types of orders.
- **Where conditions are: No**
No additional conditions are required.

Large Requests

This rule allows routing large trade requests to be processed by dealers.

Routing: Large Requests

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Large Requests

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
Request volume	Greater or equal (>=)	10.00

Add Edit Delete

OK Cancel Help

The following parameters are specified in this rule:

- **Perform action: Process to dealers**
Pass the processing to dealers specified in the "Dealers" tab.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: Request volume ≥ 10**
For requests with the volume of 10 lots and more.

Large Positions

This rule helps to prevent omission of the previous rule that limits automatic execution of requests larger than 10 lots. Some clients may attempt to open large positions by sending several requests of smaller volume. This rule analyzes the size of a client's position of a symbol.

Routing: Large Positions

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Large Positions

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
01 Position volume	Greater or equal ($\geq$)	10.00

Add Edit Delete

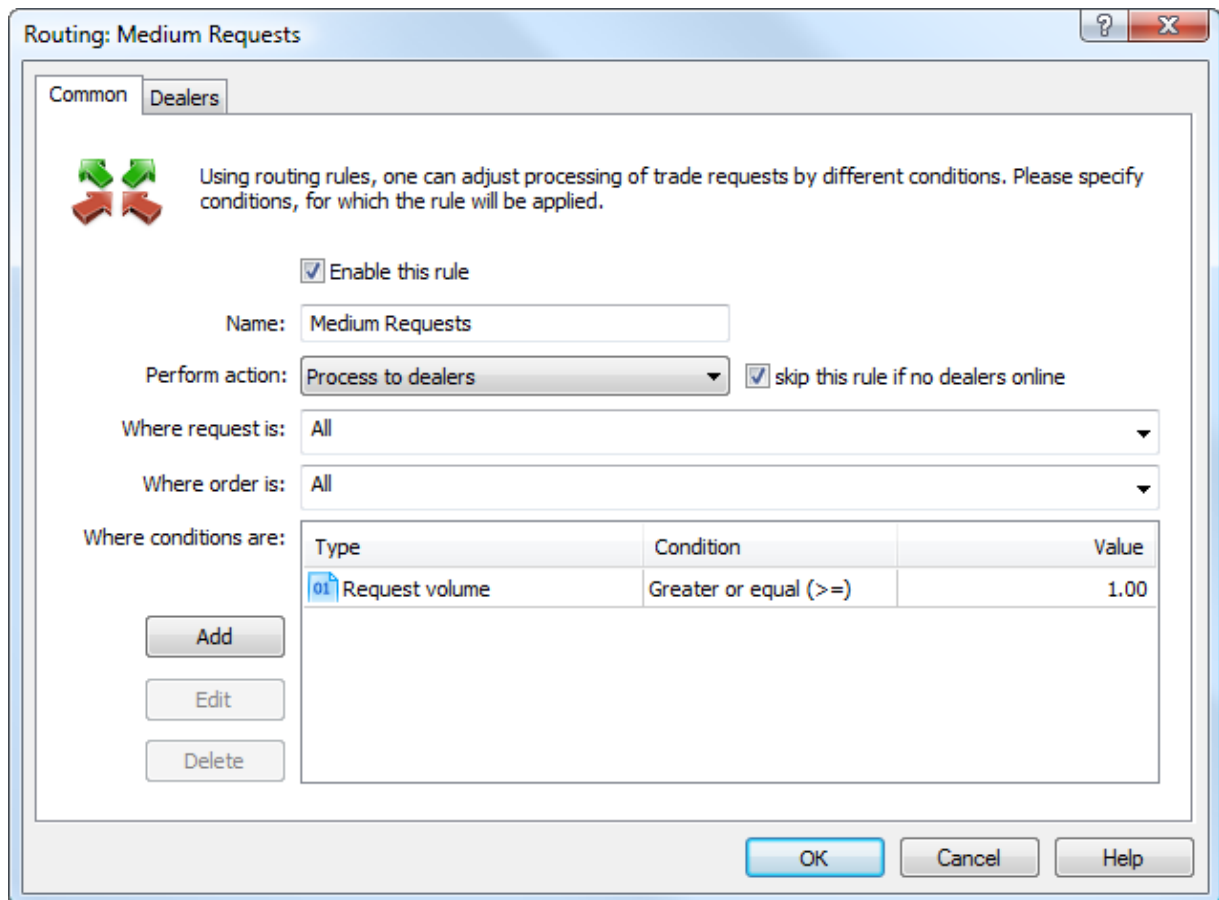
OK Cancel Help

The following parameters are specified in this rule:

- **Perform action: Process to dealers**
Pass the processing to dealers specified in the "Dealers" tab.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: Position volume >= 10**
For requests, which being fulfilled, the symbol position will be equal to 10 lots or even larger.

Medium Requests

This rule allows routing trade requests of average volumes to be processed by dealers. In this case this is a different group of dealers, other than that on the previous rule.



Routing: Medium Requests

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Medium Requests

Perform action: Process to dealers ☒ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
Request volume	Greater or equal (>=)	1.00

Add Edit Delete

OK Cancel Help

The following parameters are specified in this rule:

- **Perform action: Pass to dealers. Skip this rule if no dealers online.**
Pass the processing to dealers specified in the "Dealers" tab. The additional condition is set - the rule will be skipped if there are no dealers online.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: Request volume >= 1**
For requests with the volume equal to or larger than 1 lot.

Medium Positions

This rule helps to prevent omission of the previous rule that limits automatic execution of requests larger than 1 lot. Some clients may attempt to open large positions by sending several requests of smaller volume. This rule analyzes the size of a client's position on this symbol.

Routing: Medium Positions

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action: ☒ skip this rule if no dealers online

Where request is:

Where order is:

Where conditions are:

Type	Condition	Value
01 Position volume	Greater or equal (>=)	1.00

Add

Edit

Delete

OK

Cancel

Help

The following parameters are specified in this rule:

- **Perform action: Pass to dealers. Skip this rule if no dealers online.**
Pass the processing to dealers specified in the "Dealers" tab. The additional condition is set - the rule will be skipped if there are no dealers online.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: Position volume >= 1**
For requests, which being fulfilled, the symbol position will be equal to 1 lot or even larger.

Requote Cheaters

This routing rule allows to block trade requests that are at non-market prices. For the additional condition, this rule takes into account the number of non-market requests sent by the client for this day.

Routing: Requote Cheaters

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action:

Where request is:

Where order is:

Where conditions are:

Type	Condition	Value
01 Deviation from market	Greater or equal (>=)	10

Add

Edit

Delete

OK

Cancel

Help

The following parameters are specified in this rule:

- **Perform action: Requote**
Send current market prices replying to the request.
- **Where request is: Request Execution, Instant Execution, Market Execution, Exchange Execution**
This is the rule for the above execution types.
- **Where order is: Buy, Sell**
For Buy and Sell orders.
- **Where conditions are: Deviation from market ≥ 10**
For requests with the price that differs from the market price by 10 or more points. Clients who have sent 30 requests at non-market prices or more are taken into account.

Automate Other Requests

This rule allows to automatically process requests that have not been processed according to the above rules. If such a rule is not specified at the end of the list, such requests will not be processed by the server.

Routing: Automate Other Requests

Common Dealers

 Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Automate Other Requests

Perform action: Confirm by market price

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value

Add

Edit

Delete

OK Cancel Help

The following parameters are specified in this rule:

- **Perform action: Confirm by market price**
Confirm by prices specified in orders.
- **Where request is: All**
For all types of requests.
- **Where order is: All**
For all types of orders.
- **Where conditions are: No**
No additional conditions are required.

MetaTrader 5 Trading Platform / Platform Setup / Funds & ETF

Funds and ETFs

An investment fund is a way of investing money, in which investors do not have to trade in the market themselves. They entrust their money to an organization, which then manages the funds, for example by investing them in certain securities. The profit obtained from the funds turnover is further distributed among investors, while the fund receives a certain fee for the investment and management activities.

MetaTrader 5 enables the automation of fund and ETF operation:

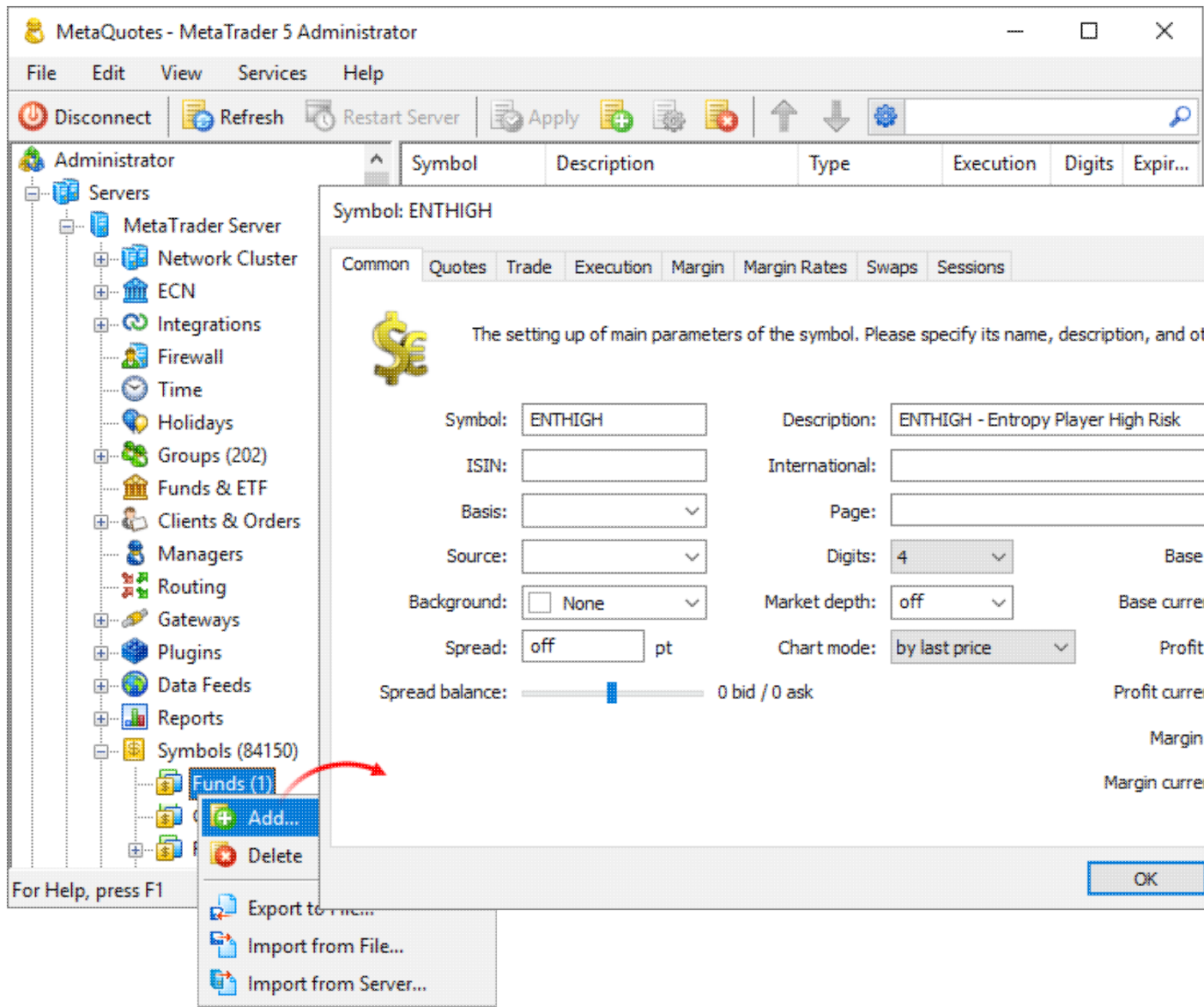
- Manage investors and keep detailed records of investments. Fund management is performed according to a traditional scheme adopted in the platform: you create investor accounts, add relevant investment amounts to their balances and add these accounts to a special fund investors section.
- Work with the fund managers. Similarly to investors, managers are added in the system in the form of regular trading accounts. Invested assets are credited to these accounts and are further used to perform trading operations.
- Automatically build AUM (Assets Under Management) charts. The money invested is credited to regular trading accounts, in which it is used by the fund managers for trading (management). The platform automatically builds the AUM charts based on the account equity data and taking into account applicable management and success fees. Investors can view this chart in the desktop, mobile and web terminals by connecting using their trading accounts, similar to regular traders.
- Automate the calculation of fees and payments to investors. You do not need to use third-party accounting systems. The platform provides flexible fund settings, which include the hurdle rate, fees, fee calculation frequency and payment mode among others.
- Provide real-time fund performance information to investors. Investors can view the fund charts and the current value of the shares by simply connecting to the trading account via the MetaTrader 5 desktop, mobile or web terminal.
- Receive fund performance reports.

Create symbols to display the fund yield

For each fund, you should create three financial instruments: to display NAV (Net Asset Value) chart, AUM (Assets Under Management) chart and fund performance. In the future, NAV displaying symbols will also be used for buying shares, i.e. it will be possible to invest by simply opening a regular trading position via the client terminal.

Add a separate funds section under the "Symbols" section and create financial instruments. They do not require any specific settings.

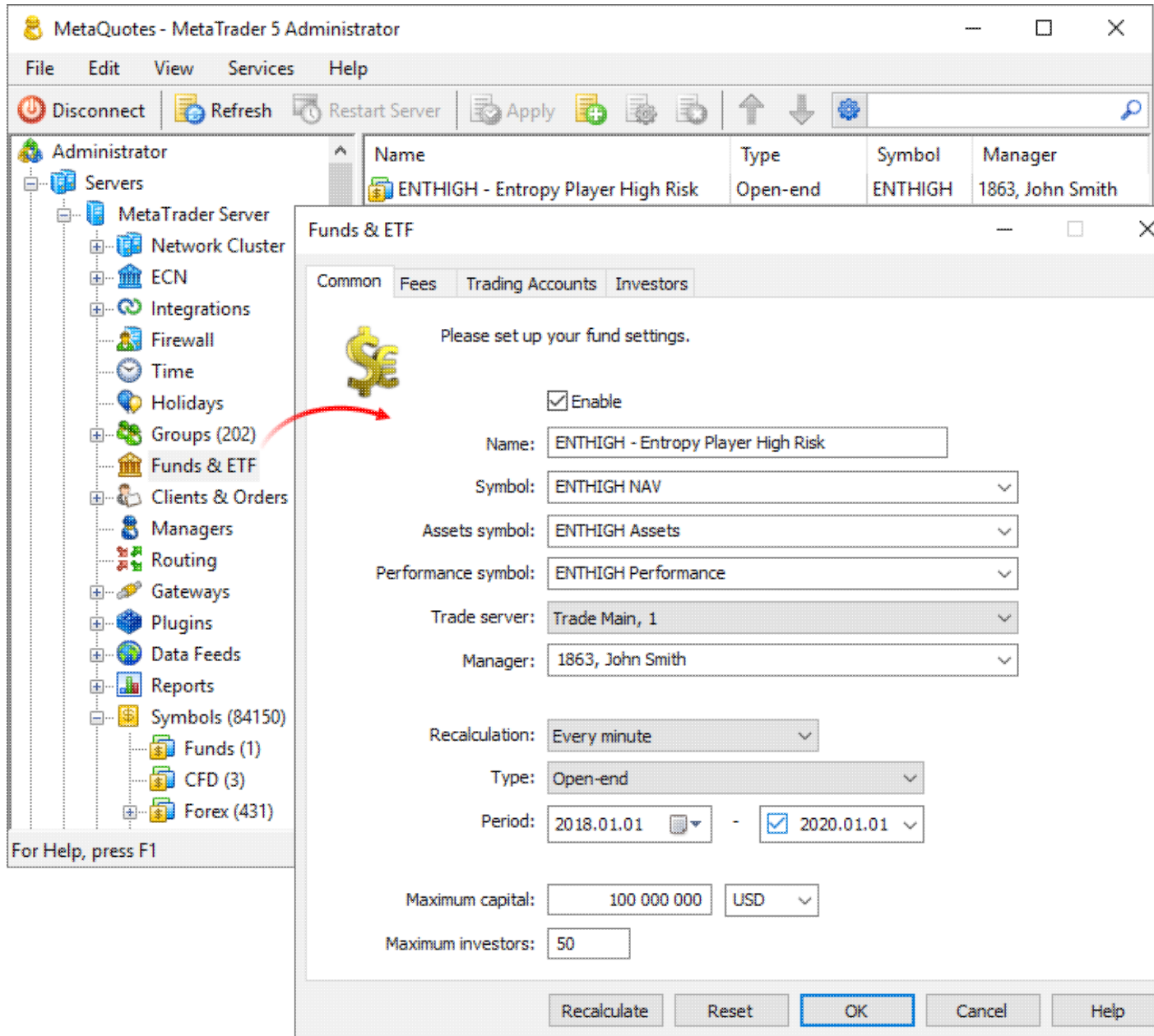
MetaTrader 5 trading platform



MetaTrader 5 trading platform

Create funds and configure general settings

Create a configuration for each fund under the "Funds and ETF" section:



Set up the following parameters under the Common section:

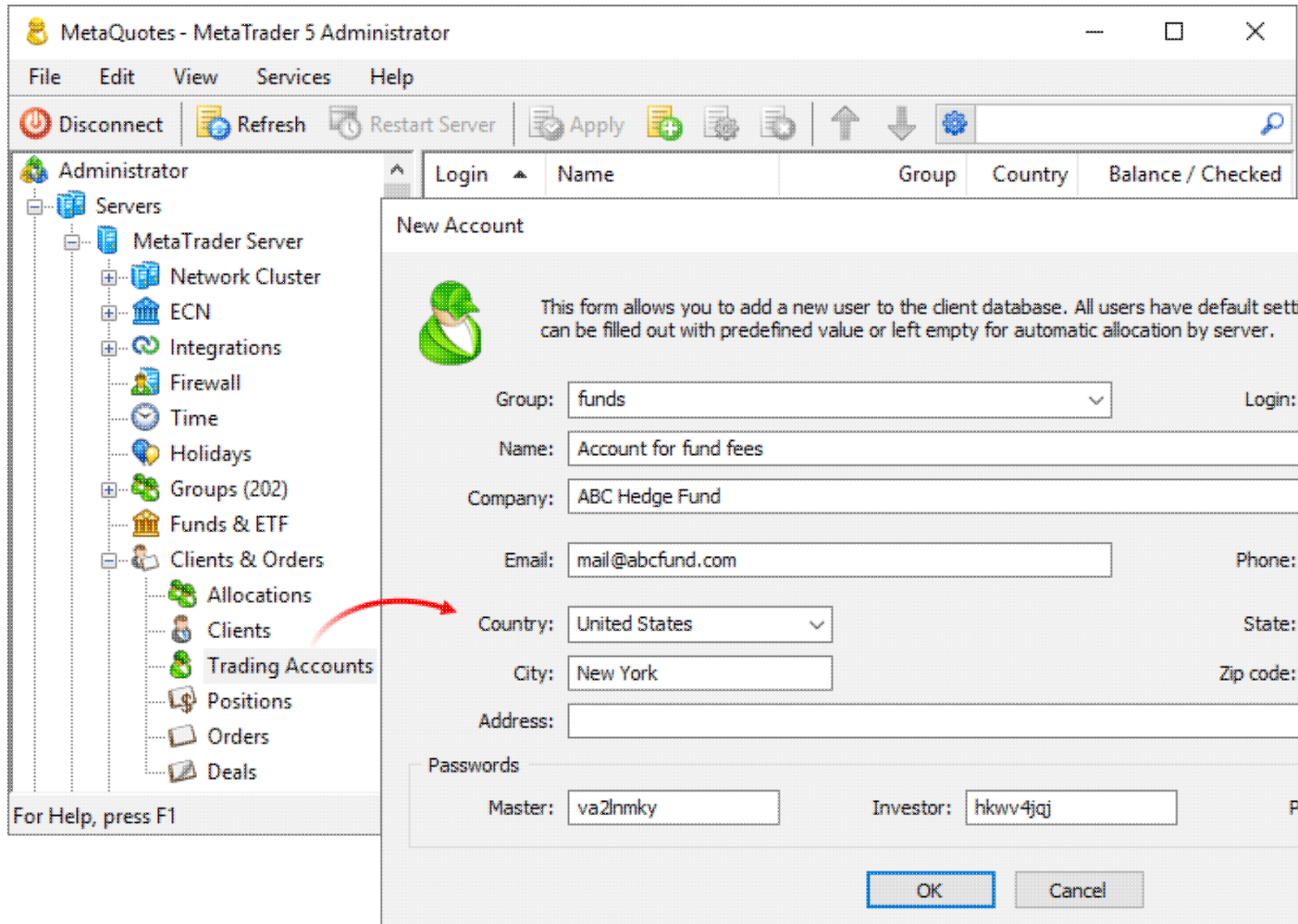
- **Name** the fund name.
- **Symbol** the name of the trading symbol, in which the NAV (Net Asset Value) will be reflected.
- **Assets symbol** the name of the trading symbol, in which a change in the AUM (Assets Under Managements) will be shown.
- **Performance symbol** the name of the trading symbol, which will reflect the fund performance chart.
- **Trade server** the trade server, on which the fund will be managed. Only accounts opened on the selected server can be used to created investors and fund managers. They cannot be created on different servers.
- **Manager** the account of the manager, who will administer the fund. Currently the field is not used. In the future, the field will be used to grant separate permissions to access the fund reports, as well as to change the lists of investors and managers.
- **Recalculation** fund charts calculation mode:
 - **Every minute** the fund chart is recalculated every minute, after which appropriate quotes are added to the symbols' price stream.
 - **Every hour** the fund chart is recalculated every hour, after which appropriate quotes are added to the symbols' price stream.
 - **Daily** the fund chart is recalculated at the end of each trading session, after which appropriate quotes are added to the symbols' price stream.
 - **Manually** the platform does not calculate fund performance variables and does not generate charts. This mode should be used when broadcasting fund data from external sources via gateways or data feeds.
- **Type** fund type:
 - **Open-end** the shares of such a fund can be purchased or sold at any day. The assets of the fund can be freely increased or decreased.
 - **Closed-end** unlike the previous type, the capital of such a fund cannot be changed. The fund issues a limited number of shares, which will not change in the future.
 - **Period** the fund lifetime. The platform performs any operations concerning the fund (such as calculations, charting, fee charging, etc.) only within the specified period.
 - **Maximum capital** the maximum allowable amount of investments within the fund. Investors and their shares are managed under a separate section of fund settings. When the investment amount reaches the specified value, the platform will prevent from further adding of new investors or from increase of shares of existing investors.
 - **Currency** the currency in which all the fund variables are calculated: prices, commissions, etc. For example, AUM (Assets under Management) is calculated based on the funds on trading accounts. If the trading account deposit currency is GBP, while the fund currency is USD, the funds will be converted at the current GBPUSD rate when calculating AUM.
 - **Maximum investors** the maximum number of investors who can purchase the fund shares (specified under a separate section of fund settings).

MetaTrader 5 trading platform

Configure payments and fees

The MetaTrader 5 trading platform enables the automation of all calculations concerning the fund: management fees and payments of returns.

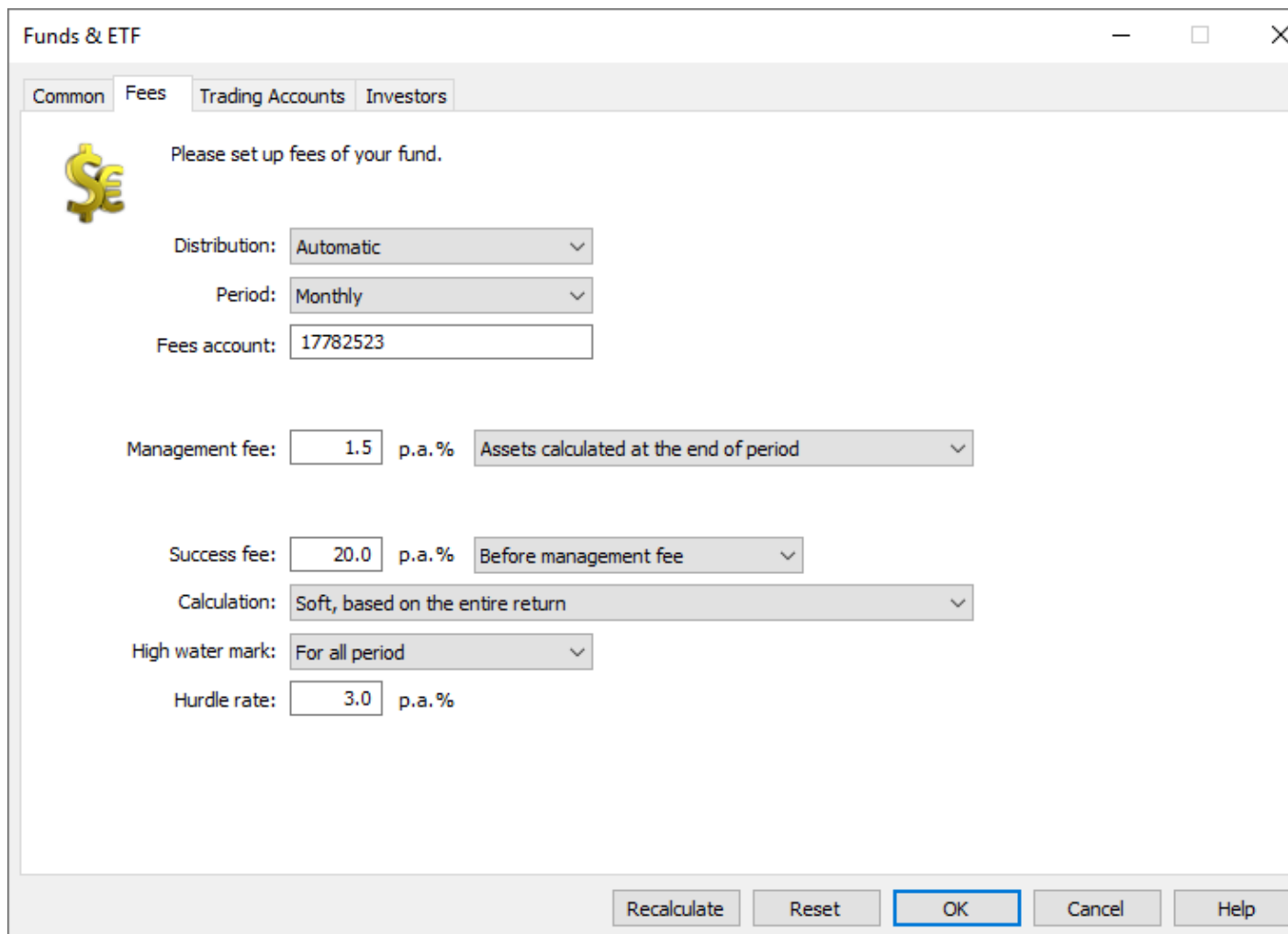
Before configuring payments, you should create a separate trading account for each fund: all relevant payments will be performed in this account ("Fees account"). It is recommended to create a separate group for such accounts, while this enables more convenient work with accounts (including granting of appropriate access permissions to managers).



Management fees

The company which manages the fund, charges investors a certain fee for the services. This is usually equal to a percentage of the Assets Under Management (AUM). This is referred to as the "Management fees". All relevant calculation parameters are specified under the fund's "Fees" section:

- **Distribution** fee distribution mode. If "Automatic", the fee will be calculated by the platform. The "Report" mode enables manual calculation using external data.
- **Period** fee calculation and distribution frequency: daily, monthly, quarterly or yearly.
- **Fees account** a special trading account, to which fund management fees will be credited.
- **Management fee** the fund management fee as a percentage of AUM. The calculation can also be based on assets values as of the beginning of the calculation period, end of period, or as an average value over a period. For example, if you select quarterly or period beginning calculation, the specified percentage will be calculated based on AUM as of the end of the previous quarter.



Funds & ETF

Common Fees Trading Accounts Investors

Please set up fees of your fund.



Distribution: Automatic

Period: Monthly

Fees account: 17782523

Management fee: 1.5 p.a. % Assets calculated at the end of period

Success fee: 20.0 p.a. % Before management fee

Calculation: Soft, based on the entire return

High water mark: For all period

Hurdle rate: 3.0 p.a. %

Recalculate Reset OK Cancel Help

MetaTrader 5 trading platform

Management Fee is calculated using the following formula:

$$\text{Management Fee} = (\text{Fund Equity} * \text{Days} * \text{Management Fee Percent}/100)/365$$

where:

- 'Fund Equity' is the fund value at the time of calculation
- 'Days' is the calculation period
- 'Management Fee Percent' is the management fee in percent

On a selected basis, the platform will calculate the management fees and will transfer the relevant amount to the special account, which is specified in the fund settings ("Fees account"). The accrual is performed as a balance operation of Commission type, with a comment "MF '[Fund Name]'". For example, "MF 'ENTHIGH'".

The fees account balance is taken into account when calculating fund values, including its AUM. Actually, each management fee transfer reduces the fund's AUM by an appropriate amount. Nothing is deducted from investor accounts.

Success fee

In the case of successful fund management (i.e. profitable trading), the company may charge an additional fee, the so-called "Success fee". If during the selected period the AUM value increases by the amount, which is equal to or exceeds the threshold value (Hurdle rate), additional commission will be deducted from the fees account. All related calculation parameters are specified under the "Fees" section.

- Distribution, Period, Fees accounts these parameters are common for management fee and access fee.
- Success fee fee as a percentage of the total returns for the current period or of the profit above the Hurdle Rate (defined in the "Calculation" field). The returns are determined separately for each share: actually it is performed by the position on the investor account. The success fee calculation mode is also specified here: before or after management fee calculation. The management fee affects the AUM value, which is taken into account during performance calculation. Therefore the success fee value is affected by its calculation time.
- Calculation determines the amount based on which success fee will be calculated: the total yield amount for the current period or the amount above the hurdle rate.
- High water mark the fund management success can be determined using different methods. Currently only the most popular method is supported in the platform: "High water mark". In accordance with this method, management in the calculation period is considered successful if the increase per share (positions on the investor account) exceeds the hurdle rate. You can select how the excess shall be determined: over the entire fund management period, the last year or the last quarter.
- Hurdle rate is the threshold rate of return. If exceeded during the selected period (specified in "High water mark"), the fund management is considered successful and a commission fee will be charged.

MetaTrader 5 trading platform

The calculated success fee is deducted from investor accounts as a balance operation of "Commission" type with a comment "SF '[Fund Name]'". For example, "SF 'ENTHIGH'". Further this amount is transferred to the "Fees account" as a Commission operation. A comment for this operation is specified as "SF '[Fund Name]'-'[Investor account from which the commission has been debited]'". For example, "SF 'ENTHIGH'-'1235966'".

For each position on investor accounts, the platform performs the following actions:

- Determines the Hurdle rate of the share (position): "Maximum share value (NAV) over the High water mark period" + "NAV Hurdle Rate". The platform calculates the maximum position value over the entire time, over the last year or over the last quarter, depending on the "High water mark" parameter. The position value is calculated as volume multiplied by NAV, its maximum value is searched in the price history of the appropriate symbol. Then "Hurdle rate" percentage of this amount is calculated and added to the total value.
- Then the platform checks if share profit exceeded the hurdle rate: "NAV at the end of the period" - "Hurdle rate". It means that the platform uses the position value as at the calculation time (before or after charging the management fee) and deducts the hurdle rate calculated at the previous stage, from the position value.
- The share value (NAV) is determined: "NAV at the end of the period" - "Maximum NAV over the High water mark period". The values are calculated as shown in previous points.
- If the share profit is negative (loss), zero, or it does not exceed the Hurdle Rate, the success fee is not charged.
- If the share profit exceeds the Hurdle Rate, the success fee is calculated in accordance with the Calculation field value:
- Soft: Profit per share * Success fee. The fee is calculated as percentage of share profit calculated above.
- Hard: Profit beyond the Hurdle Rate * Success fee.

- The "Fees account" balance is taken into account when calculating the AUM. Thus, charging of any management fees reduces the AUM value.
 - Use a separate trading account for each fund's "Fees account". Do not perform any other operations on this account.

Set up trading accounts for fund managers

Common trading accounts are used for managing invested funds. You distribute funds between these accounts and then the managers connect to these accounts and perform trading operations.

Create a separate trading account for each fund manager. It is recommended to create a separate group for such accounts, while this enables more convenient work with the accounts (for example, generation of relevant reports). Then transfer the investors' funds to these accounts, for example, by performing balance operations using the Manager terminal.

MetaTrader 5 trading platform

Once you have prepared the accounts, specify them in the fund settings, under the "Trading Accounts" section:

Funds & ETF

Common Fees Trading Accounts Investors

 Please set up trading accounts of your fund.

Login	Name	Balance	Equity
 17368650	Entropy High Trader 1	18 000.00 USD	18 546.00 USD
 17368661	Entropy High Trader 2	17 000.00 USD	17 860.00 USD
 17368677	Entropy High Trader 3	15 000.00 USD	15 768.00 USD
		50 000.00 USD	52 174.00 USD

Add

Edit

Delete

Recalculate Reset OK Cancel Help

MetaTrader 5 trading platform

The current balances and equity amounts will be instantly shown for each of the manager accounts.

The current fund AUM value is calculated based on the equity amounts available on the trading manager accounts.

Set the investors

The fund investors are also managed using trading accounts.

Create a separate trading account for each investor. It is recommended to create a separate group for such accounts. Make sure to use hedging position accounting for an investor group. Once the share purchaser invests the required funds, distribute them between the trading accounts of the trading managers. Next, to allocate a share, open on the investors' s account a long trading position for the fund symbol, while the position size should be equal to the relevant share. For example, if the share size is 1000, then the position size on the investor' s account should be 1000 lots (provided that the symbol contract size is 1).

Thus, the investor will be able to monitor their investments by connecting to the trading account via the client terminal. The investor will have access to three fund charts, as well as to their share state, which is displayed as a trading position. The trading position state will be updated in accordance with the current total fund value (the update frequency depends on the "Recalculation" parameter).

To open positions on investor accounts, use the Manager terminal:

MetaTrader 5 trading platform

Account: 1008, John Smith, EUR, 1:200, Hedge

Overview Exposure Personal Account Profile Subscriptions Balance Trade History Security

Symbol: ENTLOW

Type: Market Order

Volume: 10.00000000 10 ENTLOW

Fill Policy: Return

At Price: 1975.8158 Update ☐ Auto

Stop Loss: 0.0000 Take Profit: 0.0000

Comment:

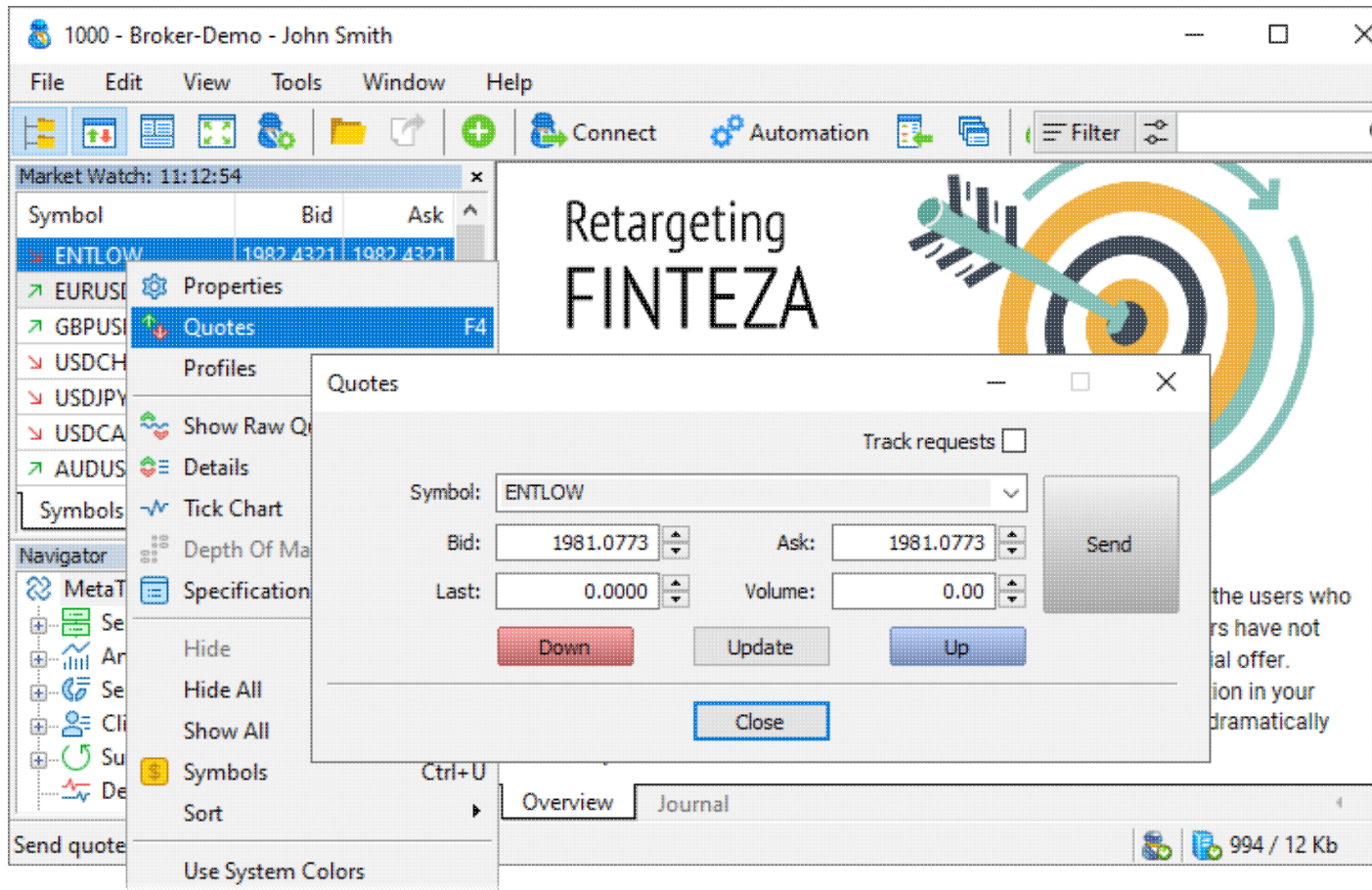
1981.8558 / 1981.8558

Sell at 1975.8158 Buy at 1975.8158

Client Properties... Update Cancel Help

MetaTrader 5 trading platform

If there are no prices for the fund yet, drop the first price into the stream, otherwise an attempt to open a position will cause the terminal to display the "no prices" error. Once all the shares have been allocated and the relevant funds have been added to the fund managers' accounts, the prices will be calculated automatically.



MetaTrader 5 trading platform

Once you have prepared the accounts, specify them in the fund settings, under the "Investors" section:

Funds & ETF

Common Fees Trading Accounts **Investors**

Please set up investors of your fund.

Login	Name	Shares
943907	Mitchell David	2.48
943908	Walker Jose	4.91
943909	James Jose	8.18
943910	Martin Dominic	3.68
		19.25

Add
Edit
Delete

943907, Mitchell David

Ticket	Volume	Price	Time	Value
14270859	2.48	1.06765	2019.10.23 16:30	20 285.35
HWM	2.48	1.06976	2019.10.07 23:19	20 325.45

OK Cancel

MetaTrader 5 trading platform

By clicking on the investor name you can view the current state of the investor's share (position). Also, maximum share value (High water mark) over the current calculation period is shown for each share.

- The volume of shares is calculated only for positions which are opened for the symbol corresponding to the current fund. No other positions opened on these accounts are taken into account in this section.
- Investor accounts should be located in hedging position accounting groups. Otherwise, calculations in the fund may be performed incorrectly.

Allocation of additional shares

If you need to allocate additional shares within an already running fund, please pay attention to the following:

- After allocating shares for new investors (opening positions on their accounts) and adding investors to the list, do not forget to distribute their invested funds between relevant managers' accounts. Otherwise, the NAV chart will be incorrect as the number of units has increased, while the management funds have not. Funds are distributed between managers at your discretion.
- To avoid incorrect NAV values before you distribute funds between manager accounts, you can temporarily disable the fund. This will pause the calculation of charts. After completing the setup, turn the fund back on.
- If incorrect prices were still added to the fund charts, you can delete such prices via the "1 Minute History Chars" and "Bid/Ask/Last Ticks" sections.

Assets Under Management and Net Asset Value; fund performance chart

The trading platform allows tracking the fund yield dynamics in real time, similarly to tracking of common symbol quotes.

As described above, ordinary trading instruments are created for the fund. The platform automatically records the changes in NAV (Net Asset Value), the AUM value (assets under management) value and profitability into the price history of these symbols.

These charts are calculated at intervals, specified in the "Recalculation" field. After each calculation, a quote with the appropriate fund value is added to the symbols' price history.

Assets Under Management (AUM)

The current fund value is used for calculating the fund management fee and the success fee, as well as for calculating all other charts. It is calculated according to the following formula:

(The total amount of equity on the fund's trading accounts - management fees)

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The platform sums the equity of all trading accounts of the fund and deducts the amount of charged management fees. The trading account deposit currency differing from the fund currency is converted at the current rate.

Net Asset Value (NAV)

The value is calculated according to the following formula:

AUM / Number of fund shares

AUM value calculated by the previous formula is divided by the current number of shares allocated by the fund to its investors.

Fund Performance

The value is calculated according to the following formula:

Previous-period Performance + $\log(\text{Current-period AUM} / \text{Previous-period AUM})$

Here:

- Previous-period Performance for the first value on the chart this formula element is ignored. For the first period, the previous period performance will be zero.
- Current-period AUM is calculated according to the AUM formula specified above.
- Previous-period AUM is also calculated according to the above AUM formula, but for the previous period. The amount of funds deposited during the calculation period is added to this value (i.e. the money distributed at the beginning of the calculation period between managers).

Period refers to the chart calculation periodicity, specified in the Recalculation field. If charts are recalculated once a minute, the previous period is equal to the previous minute.

Reset the Fund State

The fund state can be reset using two commands in the settings dialog:

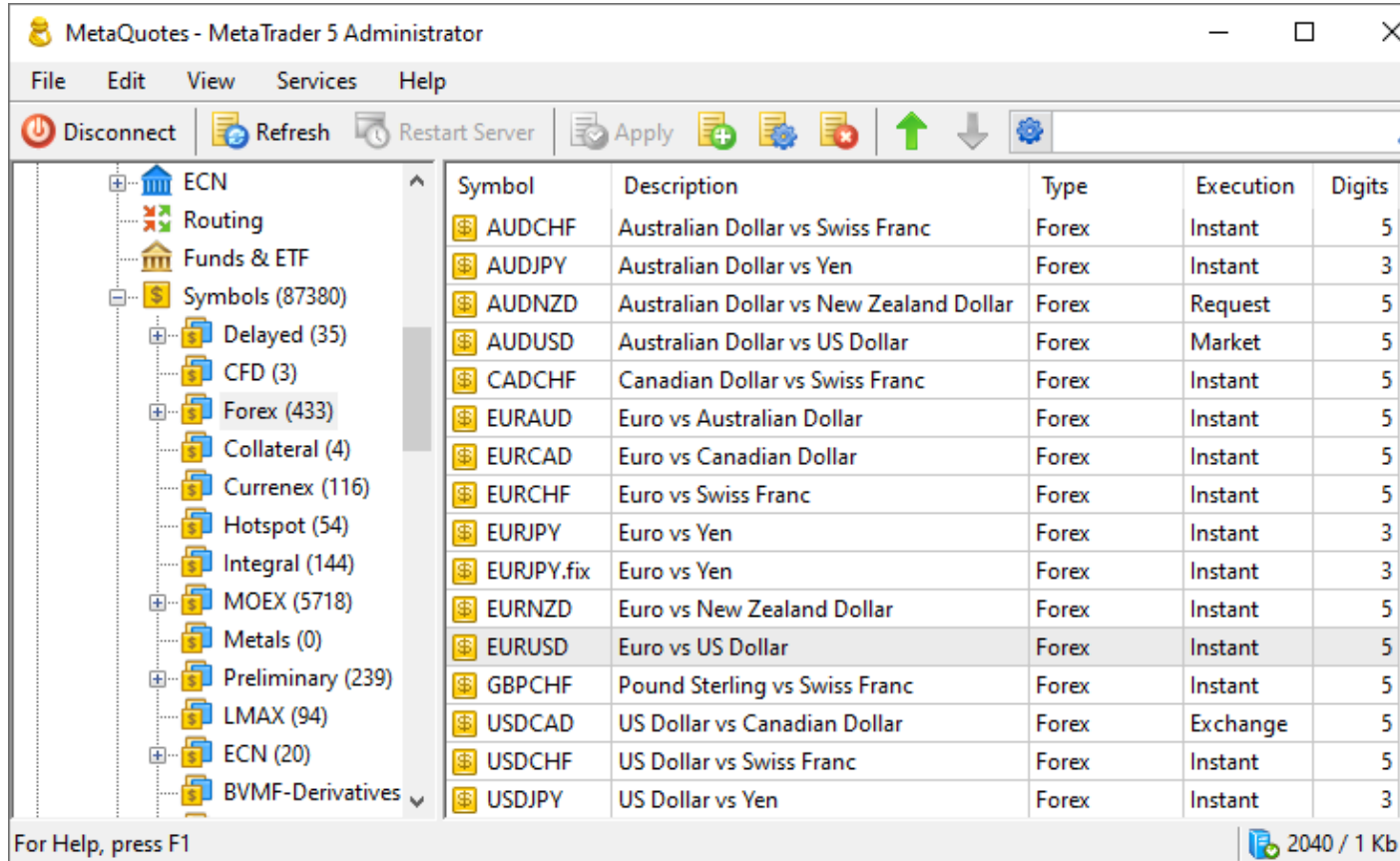
- Recalculation full recalculation of fund charts: NAV, AUM and performance. The platform reproduces the entire fund history based on manager and investor accounts: investment operations, share purchases and other operations. The graph history is calculated and recorded based on this data.
- Reset resetting accumulated statistics used for the calculation of fund parameters, including commissions. Use this option if you need to restart the fund and exclude previously accumulated values from further calculations. For example, this operation may be needed after testing.

Use these commands with caution. Deleted values cannot be recovered.
--

MetaTrader 5 Trading Platform / Platform Setup / Symbols

Symbols

This section is devoted to the management of symbols in the trading system. The possibility of structuring symbols in groups is available here.



The screenshot shows the 'MetaQuotes - MetaTrader 5 Administrator' window. The left sidebar displays a tree view of symbol groups: ECN, Routing, Funds & ETF, Symbols (87380), Delayed (35), CFD (3), Forex (433), Collateral (4), Currenex (116), Hotspot (54), Integral (144), MOEX (5718), Metals (0), Preliminary (239), LMAX (94), ECN (20), and BVMF-Derivatives. The main panel displays a table of symbols.

Symbol	Description	Type	Execution	Digits
AUDCHF	Australian Dollar vs Swiss Franc	Forex	Instant	5
AUDJPY	Australian Dollar vs Yen	Forex	Instant	3
AUDNZD	Australian Dollar vs New Zealand Dollar	Forex	Request	5
AUDUSD	Australian Dollar vs US Dollar	Forex	Market	5
CADCHF	Canadian Dollar vs Swiss Franc	Forex	Instant	5
EURAUD	Euro vs Australian Dollar	Forex	Instant	5
EURCAD	Euro vs Canadian Dollar	Forex	Instant	5
EURCHF	Euro vs Swiss Franc	Forex	Instant	5
EURJPY	Euro vs Yen	Forex	Instant	3
EURJPY.fix	Euro vs Yen	Forex	Instant	3
EURNZD	Euro vs New Zealand Dollar	Forex	Instant	5
EURUSD	Euro vs US Dollar	Forex	Instant	5
GBPCHE	Pound Sterling vs Swiss Franc	Forex	Instant	5
USDCAD	US Dollar vs Canadian Dollar	Forex	Exchange	5
USDCHF	US Dollar vs Swiss Franc	Forex	Instant	5
USDJPY	US Dollar vs Yen	Forex	Instant	3

For Help, press F1

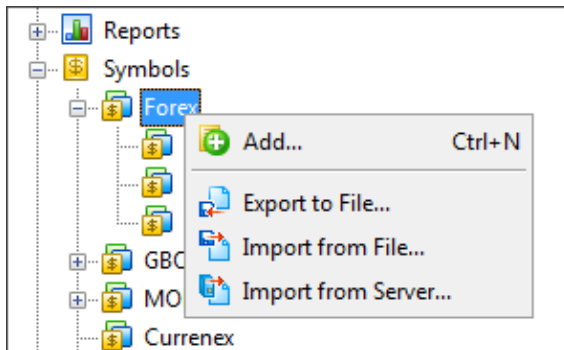
2040 / 1 Kb

The structure of symbols can be viewed in the left part of the administrator terminal. The right part reflects group or symbols included into a selected group of a higher order. Information about symbols is represented in four columns:

- **Symbol** symbol name;
- **Type** group of the highest order to which the symbol is included;
- **Execution** type of trade operation execution for this symbol;
- **Digits** number of decimal places in the symbol price.

Managing Groups of Symbols

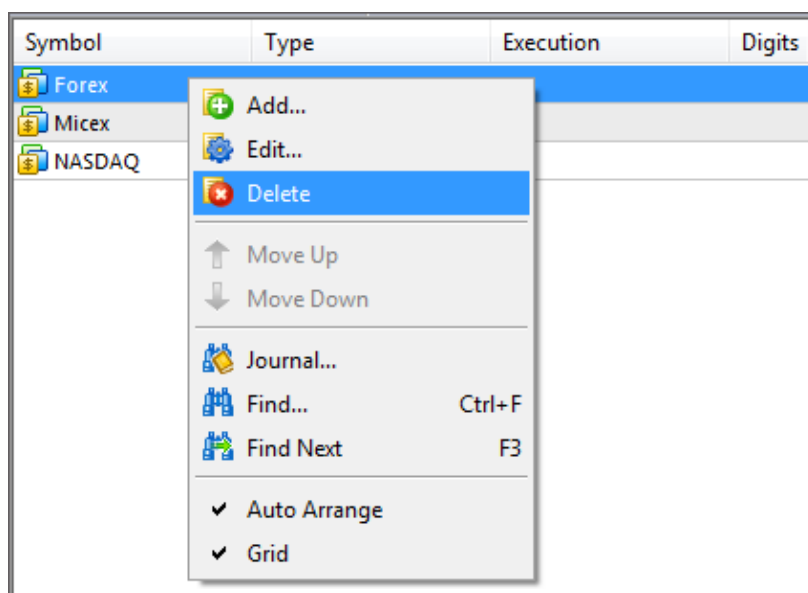
To create a new group, select a group of higher level, in which the new group should be created. To create a root group, select the Symbols section. After that call the context menu and execute the "+ Add" command specifying the name of the new group.



- After a group of symbols has been created, you will not be able to change its name.
- Symbol group name cannot contain any punctuation marks or special characters (allowed are ".", "_", "&" and "#"). It's not recommended to use the following characters in the symbol names: <, >, :, ", /, |, ?, *.

If you need to delete a group, select it in the right part of the "Symbols" section and press "X Delete" in the Edit menu, on the Standard toolbar or in the context menu:

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A group that contains at least one symbol cannot be deleted. In order to delete a group, first you should delete all symbols from it.

How to Manage Symbols

Symbols are managed in the right part of the administrator terminal.

- **Adding**

In order to add a symbol, enter the necessary group and execute the "+ Add" command in the Edit menu, on the toolbar or in the context menu. After that the window of symbol settings will be opened.

- **Editing**

In order to edit a symbol, select it and execute the "⚙ Edit" command of the Edit menu, on the Standard toolbar or in the context menu. After that the window of symbol settings will be opened. The symbol editing window can be opened by double clicking on the symbol in the list. You can also edit several symbols at the same time. To do this, select symbols using the mouse and "Ctrl" or "Shift", and then execute the "⚙ Edit" command.

- **Deleting**

In order to delete a symbol, select it and execute the "✕ Delete" command in the Edit menu, on the Standard toolbar or in the context menu. You can delete several symbols at the same time, selecting them using the mouse and "Ctrl" or "Shift".

- **Sorting**

Symbols and symbol groups can be sorted alphabetically. The feature of sorting is that it is performed on the server, not on the local MetaTrader 5 Administrator. To sort symbols execute the "↕ Sort Alphabetically" command in the context menu. Once it is done, the confirmation window will appear. In it you can choose whether the symbol groups should also be sorted or not.

- We strongly recommend to perform changes of financial symbols only on weekends or holidays, when markets are closed.
- Additional general information about working with configuration records is given in the ["Working with Instructions"](#) section.

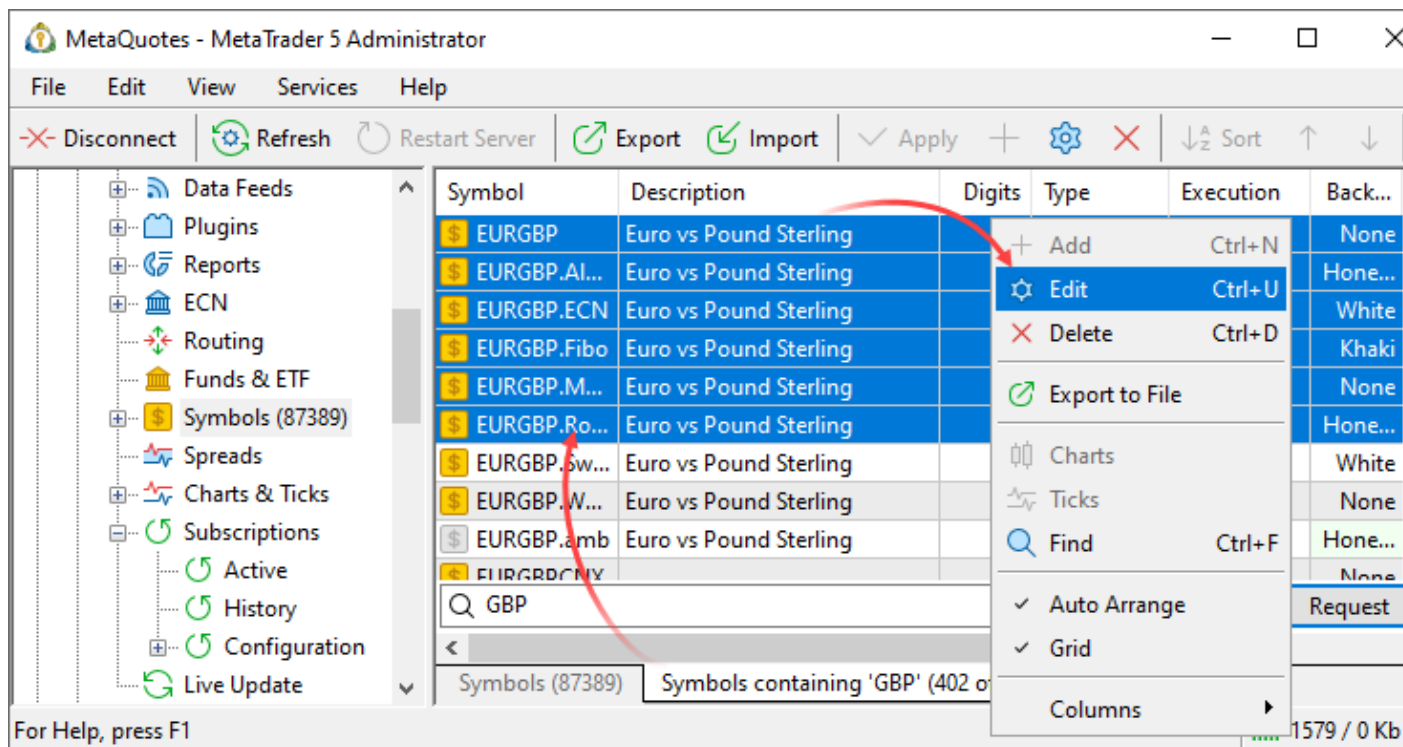
Symbol filtering and batch editing

You can [edit the settings of multiple symbols](#) at a time. Select the required symbols in the list and click "Edit" in the context menu.

To bulk edit symbols which are in different subcategories, use the filter tab at the bottom of the window. Specify a comma separated list of symbols or a mask, for example:

- EURUSD, GBPJPY two specified symbols
- EUR*, GBP* the symbols with the names beginning with EUR and GBP
- *USD, !GBPUSD all symbols with USD as the second currency except for GBPUSD

Select the desired symbols from the filtered list and edit:



To create copies of multiple symbols, select the desired symbols, click " Edit" in the menu, and then specify the postfix in the "Symbol" field. The postfix must begin with a period (.). For example, if you select the symbols EURUSD, USDJPY, GBPUSD and add the postfix .x (EURUSD.x) in the group editing window, you will get copies of the symbols: EURUSD.x, USDJPY.x and GBPUSD.x.

Context Menu

The context menu of the "Symbols: section contains the following commands:

-  **Add** add a new symbol.
-  **Edit** edit a selected symbol or open a selected group.
-  **Delete** delete a selected symbol or group.
-  **Move Up** move a selected symbol up relative to others. This sorting is saved on the server. To sort the list of symbols temporarily, click on the heading of any column. When temporary sorting is enabled, the symbol moving up/down buttons become inactive.
-  **Move Down** move a selected symbol down relative to others. This sorting is saved on the server. To sort the list of symbols temporarily, click on the heading of any column. When temporary sorting is enabled, the symbol moving up/down buttons become inactive.
-  **Sort alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
- **Automation triggers** create an automation task for the selected event or edit an existing one. The menu displays only the triggers and tasks associated with the current section.
- **Automation actions** add an automation action to an existing task or create a new task based on the action. The menu displays only the actions associated with the current section.
-  **Export to File** export symbol settings to a file.
-  **Import from File** import symbol settings to a file.
-  **Import from Server** start importing symbols from a remote MetaTrader 5 or MetaTrader 4 server.
-  **Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration automatically specified in the query field. You will only need to press the request button.
-  **Charts** request the minute history. This opens the relevant section with the name of the selected symbol automatically specified in the request field.
-  **Ticks** request the tick history. This opens the relevant section with the name of the selected symbol automatically specified in the request field.
-  **Find** open the search window.
- **Auto Arrange** if this option is enabled the size of columns is selected automatically.
- **Grid** this option shows/hides field separators in the table with symbols.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol
Settings

Symbol Settings

The symbol setting window appears when a symbol is added or modified. The window contains the following tabs:

- **Common**
- **Currency**
- **Quotes**
- **Trade**
- **Futures**
- **Options**
- **Bonds**
- **Execution**
- **Margin**
- **Margin Rates**
- **Swaps**
- **Sessions**

MetaTrader 5 trading platform

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Common

Common

Symbol: EURUSD

Common Currency Quotes Trade Execution Margin Margin Rates Swaps Sessions

 The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.

Symbol:	EURUSD	Description:	Euro vs US Dollar
Exchange:	Forex	International:	https://www.google.com/?finance?q=EURUSD
ISIN:	EUU0009652759	Sector:	Undefined
CFI:	JFTXCC	Industry:	Undefined
Basis:		Country:	
Source:		Category:	Currency
Background:	☐ None	Page:	https://www.google.com/?finance?q=EURUSD
Digits:	5	Market depth:	32
Spread:	off	Chart mode:	by bid price
Spread balance: 		0 bid / 0 ask	

OK Cancel Help

The common symbol parameters are set up on this tab:

- **Symbol** symbol name. It should not contain any punctuation marks or special characters except for ".", "_", "&" and "#". It's not recommended to use the following characters in the symbol names: <, >, :, ", /, |, ?, *. Do not use , and \ symbols (the latter is interpreted as the subgroup separator). Whitespaces set at the beginning or at the end of the name are automatically removed when saving. Do not create symbols with names only differing in case, such as for example Apple and APPLE. Otherwise problems may arise when working with price data.
- **Description** a brief description of the symbol.
- **ISIN** International Securities Identifying Number. A unique 12-character code that unambiguously identifies it.
- **International name** the international name of the symbol (in English, as a rule).
- **Exchange** the name of the exchange in which the symbol is traded.
- **Category** the symbol category. Categories are intended for additional marking of financial instruments. For example, this can be the market sector to which the symbol belongs: Agriculture, Oil & Gas and others.
- **CFI** instrument classification in accordance with the ISO 10962 standard. The parameter is used for EMIR reports.
- **Sector** economic sector the instrument belongs to, such as energy, finance, healthcare and others.
- **Industry** industry branch the instrument belongs to, such as sportswear and accessories, car manufacturing, restaurant business and others.
- **Country** country of the company whose shares are traded on the stock exchange.
- **Basis** the underlying asset of a derivative. For example, for gold futures contracts gold is the underlying asset. The parameter is used in the following calculations:
 - ◆ When splicing futures charts.
 - ◆ When calculating the margin for intercontract spread.
 - ◆ When building options boards in the client terminal for symbols with the Exchange Options and Exchange Margin Options calculation types.
- **Page** address of a web page with the information on this symbol. This address will be shown as a link when viewing symbol properties in the client terminal.
- **Source** a symbol, whose quotes should be used for the current symbol. Specify here the name of the symbol that exists in the system. Note that the quotes will be taken directly from the corresponding symbols in the data feed (without any transformation according to the settings of the corresponding symbol in the trading platform). After this field has been changed, the server needs to be restarted.
- **Digits** number of decimal places in the symbol price. After changing this parameter the server should be restarted.
- **Background** background color of the symbol. This will also be the background color of the symbol in the Market Watch window in the client and manager terminals.
- **Market depth** displaying of the market depth of this symbol and its depth. The depth is specified for one direction. For example, if you specify 16, then up to 16 buy orders and the same number of sell orders will be displayed in the Depth of Market. The maximum depth is 32 orders in one direction. If you set

"off" here, then a scalper depth of market instead of the exchange one will be used for the symbol. The scalper depth of market shows best Bid/Ask prices as well as a range of prices above and below them generated with the same step (minimal price step), which can be used to quickly place pending orders. If the depth of market is enabled for a symbol, its charts will be drawn using the prices of last executed deals (Last).

After changing this parameter the server should be restarted.

- **Spread** spread size in points. If the value of this field is non-zero, the spread will be considered fixed, and will be calculated for the symbol using the "Spread balance" parameter. If "0" is set here, the spread is considered floating, i.e. is formed based on quotes received from data feeds. The value is ignored if Market depth is enabled for the symbol (the value of the "Market depth" parameter is non-zero).
- **Spread balance** in this field you can specify how the Bid and Ask prices should be transformed or shifted. The first value indicates the Bid price shift in points (balance_bid) and the second value is for the Ask price shift in points (balance_ask). Three scenarios of transformation of quotes are possible:
 - ◆ If the spread of incoming quotes is equal to the spread specified for the symbol, then the prices are transformed according to the following formula: $\text{New Bid} = \text{Bid} + \text{points} * (\text{balance_bid} + \text{balance_ask})/2$, where points is $1/10^{\text{symbol digits}}$. $\text{New Ask} = \text{New Bid} + \text{spread}$.
 - ◆ If the spread of incoming quotes differs from the spread specified for the symbol, then the prices are transformed according to the following formula: $\text{New Bid} = (\text{Ask} + \text{Bid})/2 - \text{points} * (\text{spread}/2 - (\text{balance_bid} + \text{balance_ask})/2)$, where points is $1/10^{\text{symbol digits}}$. $\text{New Ask} = \text{New Bid} + \text{spread}$.
 - ◆ If the floating spread (value "0" or "off") is set for the symbol, then prices are shifted on the same number of points specified for the spread balance.
- **Chart mode** the mode of creation of the symbol chart: using the Bid or Last price. This parameter also sets the price used for building the bars of a financial instrument. Accordingly, this affects what charts will be shown to traders in client terminals. For symbol charts, appropriate symbol prices must be provided by datafeeds or gateways. When the chart drawing mode is changed, the accumulated price history will not be changed. Settings only apply to new received data.

- If there are no history data for the symbol, but the "Source" field is defined, then after the server restart these data will be created by automatically copying the history data of a symbol specified in the "Source" field.
- For the symbols, the charts of which are based on Bid prices, the history server does not accept Last prices and volumes from gateways and datafeeds. Such ticks are not saved and are not provided to other components of the platform.
- If the market depth is available for a symbol, the "Spread" and "Spread Balance" settings are not applied. In this case, the History server automatically generates the Bid/Ask ticks based on the best prices in the

market depth, so price markup settings are not applicable to them.

Automated enrichment of symbol data

MetaTrader 5 stores a huge data base from open sources, which enables the automated enrichment of financial symbol data: descriptions, ISIN, related trading venues, sectors and industries. With this data, your clients receive a detailed categorization of symbols, based on which they can analyze the entire sectors of the economy. No extra actions are required from brokers to enable the data.

The following symbol fields are checked during server restart: Description, ISIN, Exchange, Category, Sector, Industry, Country, Page. If any field is empty, the platform will try to fill it with the information from its database, based on the symbol name.

The platform journal will display the following log entries:

```
2020.09.11 16:29:46.541 Symbols symbol 'NAV' config enriched: Symbols - NAV - Description:
2020.09.11 16:29:46.541 Symbols symbol 'NAV' config enriched: Symbols - NAV - Sector: " -
2020.09.11 16:29:46.541 Symbols symbol 'NAV' config enriched: Symbols - NAV - Industry: "
```

If the provided description does not fit, you can replace it with your own information. The fields which the broker fills in manually, are not changed.

No enrichment is applied to symbols used in funds.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Currency

Currency

Symbol: EURUSD ? X

Common Currency Quotes Trade Execution Margin Margin Rates Swaps Sessions

 The setting up of base, profit, and margin currencies of the symbol.

Base currency: EUR ▼

Base currency digits: 2 ▼

Profit currency: USD ▼

Profit currency digits: 2 ▼

Margin currency: EUR ▼

Margin currency digits: 2 ▼

OK Cancel Help

The tab allows configuring symbol currency settings:

- **Base currency** base currency of the symbol.
- **Profit currency** currency that will be used to calculate the profit from deals on this symbol.
- **Margin currency** margin requirements for the symbols will be calculated in this currency.

For standard world currencies, such as USD, EUR, GBP, JPY, CHF, RUR, etc., the number of digits is set by the platform and thus it is not recommended to change this parameter. This may affect profit and margin calculation for the relevant instrument. The accuracy of cryptocurrencies (and other non-standard currencies) can be changed manually. A higher accuracy enables a proper calculation of profit and margin through cross-rates, in which such currencies are used. If the accuracy is small, the resulting profit or margin value can be equal to zero due to rounding.

MetaTrader 5 trading platform

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Quotes

Quotes

Symbol: Forex\EURUSD ?

Common Currency **Quotes** Trade Execution Margin Margin Rates Swaps Sessions

 The setting up of different filtration levels is intended for cutting off the incorrect data. Please specify the deviation of quotes and the repetition for filters to trigger.

☒ Allow realtime quotes from datafeeds ☐ Allow negative quotes

☐ Receive market statistics from datafeeds ☐ Save raw prices

Soft filtration level: points (must not be less than 4 or 5 times spread by default)

Filter: wrong quotes coming one after another

Hard filtration level: points (must not be less than soft filtration level)

Filter: wrong quotes coming one after another

Discard filtration level: points (must not be less than hard filtration level)

Gap mode level: points Disable gap after: ticks

Minimum spread: Maximum spread: points

Delay for subscriptions: min

OK Cancel

Parameters of processing of the symbol quotes received from data feeds are set up on this tab:

- **Allow realtime quotes from data feeds** allow/prohibit receiving of quotes for this symbol from data feeds in the real time mode. If this option is disabled, a dealer can manually through in quotes for this symbol.
- **Allow negative prices** the trading platform supports operations with negative security prices. If such a situation occurs in the market, all the appropriate functions will work correctly. This includes the display of quotes, calculation of profit margin, placing of orders, etc. However, such market situations are extremely rare, and negative values in the price stream are usually caused by an error on the data source side. Therefore, the history server rejects negative prices by default. To allow negative prices in the platform, you should enable the appropriate option.

Negative prices are only allowed for Futures contracts (Futures, Exchange Futures, Exchange FORTS Futures). Do not enable this option for other financial symbol types.

- **Save raw prices** save quotes in the way they come from data feeds. Ticks accepted by the platform in accordance with all filters and symbol settings are unconditionally saved. This option enables the additional saving of prices that the platform has not accepted (has discarded). You can use this option for the addition control and verification of filter operations. When enabling this options, please note that the stream of raw quotes will take up additional disk space. You can view the raw and accepted streams separately under the "Ticks Bid/Ask/Last" section, by specifying the appropriate type of quotes in the request.
- **Receive market statistics from data feeds** if this option is enabled, the market statistics (Ask High, Bid Low, etc.) is received directly from data feeds without its calculation by the history server. Before enabling the option, please make sure your data feeds support the delivery of the relevant data. If disabled, the following statistical parameters are calculated by the history server: High and Low for Bid, Ask and Last; current session Open and previous session Close. Besides, the price change percentage relative to the previous session is calculated on client terminals. After disabling this option, a restart of the history server is required. Statistics recalculation will resume only after the restart.

Filters

Filters allow controlling the correctness of quotes received from a data feed. The detailed description of filters is given in a separate section.

- **Soft filtration level** the soft filtration level is the first border of the channel of allowed symbol prices. If a new price (Bid or Ask) differs from the previous one by more than the specified value (in points), it is deleted from the price stream translated to clients. However, if such a price difference appears again the number of times specified in the "Filter" field, the new price level is accepted, and the filtration level is shifted by the specified value. Such quotes

will be translated again.

- ◆ **Filter** the number of quotes (of Bid or Ask prices), that come in succession and exceed the soft filtration level, after which the new channel of allowed prices limited by the soft filtration level is set;
- **Hard filtration level** the hard filtration level is the second border of the price channel. If a received quote exceeds the level both of the soft and of the hard filtration, it is cut out from the price stream translated to clients. For a new level of accepted price to be set, the quotes must be repeat the number of times specified in both levels.
 - ◆ **Filter** the number of quotes, received after the soft filtration level is broken and that exceed the hard filtration level, after which the new channel of allowed prices limited by the specified levels is set;
- **Discard filtration level** if the difference between prices of the previous and new quote exceed the specified value, such new prices are definitely removed from the stream.
- **Minimum spread** here you can set the minimum difference between Bid and Ask prices. If the spread is less than the value specified, the quote will be cut out.
- **Maximum spread** here you can set the maximal difference between Bid and Ask prices. This value being exceeded, the quote will be cut out.

- Filters cannot be applied to instruments with the enabled Depth of Market **and** with the one of the following properties: Exchange calculation type (begins with "Exchange") **or** Last price based charting mode. Also, filtering is not applied to splice symbols.
- Filtering is not applied to the depth of market data.
- To completely disable filtering, set 0 for all levels: Soft, Hard and Discard.
- The platform automatically filters out negative and zero Bid and Ask prices of OTC instruments, regardless of whether their Market Depth is enabled or not. OTC (over-the-counter or off-exchange) symbols include financial instruments whose calculation type does not start with "Exchange".
- Analysis and filtration are performed separately for Bid and Ask.
- Filters are not applied to the first tick after a break in the quotes stream: after platform restart, after a break in quoting sessions, after off-hours and after holidays. Filtration cannot be applied because it is not known in advance what was the price preceding the break. This rule does not apply to other price checks, including allowable spread.
- Filters do not apply to ticks added via the Manager terminal and Manager API.
- The platform allows using a symbol as the source of quotes for another symbol. The source symbol can be specified in the Source field. Source symbol filtering settings do not apply to quotes provided to the target symbol.

For example, EURUSD quotes are used for the EURUSDm symbol. A quote which is not added to the EURUSD stream due to filtering parameters can be added to the EURUSDm quotes stream.

Gap mode level

Gap is a considerable difference between the previous and the next quote in the price flow. The symbol settings allow you to define what price differences are to be considered as gaps.

Unlike the filters, quotes are not removed from the flow when enabling the gap mode. They are passed to clients. However, this check can be used in the routing rules in order to process trade requests in a special way under the market conditions that differ from normal ones. For example, after a gap, client requests can be rejected or requoted during a certain number of subsequent ticks.

Specify a value in points in the "Gap mode level" field. If the difference between the previous and the next quote exceeds the specified value, a gap is considered to be formed. The check is performed for Bid and Ask prices separately.

Next, specify the number of ticks after the gap, after which the gap mode is disabled. For example, suppose that we have the values of 100 points and 3 ticks. When a gap is detected, the tick counter is increased by 1. If no new price spike for the specified number of points occurs at the two subsequent ticks, the gap mode will be disabled at the next (fourth) tick:

- The first tick, on which a gap occurred. The tick counter is set to 1.
- The second tick without a gap. The tick counter is increased to 2. It is the first tick, which confirms the new price.
- The third tick without a gap. The tick counter is increased to 3. It is the second tick, which confirms the new price.
- The fourth tick without a gap. It is the third tick, which confirms the new price. The gap mode is switched off.

The "Disable gap after" parameter must be filled in, otherwise the gap mode will not be activated.

After launching the server, the first tick is always considered a gap, since the previous price is not known in advance. The gap counter is immediately set to maximum in that case. Thus, the gap mode is disabled if no gap occurs on the second tick after launching the server.

Gaps detection and tick calculation are displayed in the Journal of the trade server:

```
2017.03.16 12:21:20.323 TickSymbol 'EURUSD' gap by ask from 1.07149 to 1.07250 [diff:100]
2017.03.16 12:21:20.323 TickSymbol 'EURUSD' gap by ask from 1.07139 to 1.07240 [diff:100]
```

The 'diff' value shows the actual difference between the previous and the next quote. The 'gap level' value sets the number of ticks specified in the symbol settings for disabling the gap mode.

MetaTrader 5 trading platform

If the difference between the current and the next price (here, it is Ask) does not exceed the specified value (100), the Journal entry states that an ordinary tick has arrived and the counter has been increased:

```
2017.03.16 12:21:20.428 TickSymbol 'EURUSD' normal tick 1.07247 after gap by ask [tick
```

Please note that the counter is already equal to 2. The tick, on which the gap started, has already been taken into account. If the next tick does not differ by the specified value, the gap mode for the Ask price is disabled:

```
2017.03.16 12:21:20.730 TickSymbol 'EURUSD' normal tick 1.07248 after gap by ask [tick
2017.03.16 12:21:20.730 TickSymbol 'EURUSD' gap by ask mode disabled
```

Delay for Subscriptions

Financial symbol price data can be provided by subscription. One of the data providing options is the delayed delivery of quotes. This option is usually used for providing data to traders for free, for strategy testing and other purposes.

The default delay is 15 minutes, which can be overridden for each separate trading symbol. For further details please visit the Subscription Settings section.

- A restart of access servers is required for new delay settings to take effect.
- The delay does not affect data delivery for non-subscription instruments.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Trade

Trade

Symbol: EURUSD

Common Quotes Trade Execution Margin Swaps Sessions

 The setting up of symbol trading. Please specify the type of profit calculation, type of placed orders, allowed trade volumes, etc.

Contract size: 100000

Calculation: Forex

Trade: Full access

GTC: Good till cancelled

Filling: All

Expiration: All

Orders: All

Limit & stop level: 30 pt

Freeze level: 0 pt

Max quote delay: off sec

Convert profit: by deal

Enable Trading Signals ☒

Volumes

Minimum: 0.01 Step: 0.01 Maximum: 100.00 Limit:

OK Cancel Help

Symbol trading parameters are configured on this tab:

- **Contract size** amount of the underlying asset in one lot/contract (for currency pairs it is actually the value of one lot in the base currency of the instrument).

It is strongly recommended not to change the contract size if there are open client positions. This may lead to incorrect operation of some functions (for example, wrong correction of positions based on the history of deals).

- **Tick size** the minimal step of the price change. This value is used for calculating profit, margin and swaps for CFD Index and Futures. Also, this value is used for rounding the quotes passed from data feeds: for all financial instruments with the disabled Market Depth and non-exchange instruments (other than Exchange*) with the enabled Market Depth. After changing this parameter the server should be restarted.
- **Tick value** the cost of one point of the price change. This value is used for calculating profit, margin and swaps for CFD Index and Futures. After changing this parameter the server should be restarted;
- **Calculation** type of calculation of margin requirements and profit on a symbol. This parameter also influences the calculation of swaps. The following calculation types are available:
 - **Forex** calculation for Forex symbols.
 - **Forex No Leverage** totally equivalent to the Forex type except that the client leverage is not taken into account when calculating the margin. Forex No Leverage can be used if it is necessary to calculate the margin with a fixed leverage for all clients regardless of their actual leverage. If this type is selected, the client leverage is not taken into account when calculating the margin. A fixed leverage can be set through margin rates (for example, the rate of 0.01 is similar to the leverage of 1:100).
 - **CFD** calculation for CFD.
 - **Futures** calculation for Futures.
 - **CFD Index** calculation for CFD indexes.
 - **CFD Leverage** calculation for CFD with the account of the leverage.
 - **Exchange Stocks** calculation for exchange stocks.
 - **Exchange MOEX Stocks** calculation for stocks traded at Moscow Exchange.
 - **Exchange Bonds** calculation for exchange bonds.
 - **Exchange MOEXBonds** calculation for bonds traded at Moscow Exchange.
 - **Exchange Futures** calculation for exchange futures.
 - **Exchange FORTS Futures** calculation for the FORTS exchange. This mode must only be used together with the MOEX Derivatives Gateway. Otherwise the margin requirements calculation results may be unpredictable.

- **Exchange Option** calculation for exchange options.
- **Collateral** non-tradable instruments of this type are used as client's assets to provide the required margin for open positions of other instruments. For these instruments the margin and profit are not calculated.

In all types of calculation except the Exchange types, all types of pending orders as well as stop loss and take profit orders are triggered (activated) according to Bid or Ask price. In Exchange Stocks, Exchange Bonds, Exchange Futures and FORTS Exchange mode, orders are triggered according to the last deal price (Last). A similar division by prices is used when displaying floating profits in client terminals. In exchange modes, the floating profit is calculated using Last prices.

- **Trade** setup of how the symbol is traded:
 - ♦ **Disabled** trade is disabled;
 - ♦ **Long only** only long positions can be opened;
 - ♦ **Short only** only short positions can be opened;
 - ♦ **Close only** only position closing is allowed. On hedging accounts, only market operations that close positions are allowed. On netting accounts, in addition to the market Close operation, it is possible to place a pending order in an opposite direction. As a result of this pending order triggering, an existing position will be closed instead of opening a new position (unlike hedging).
 - ♦ **Full access** any trade is allowed;
- **GTC** mode of keeping the orders at a change of a trade day:
 - ♦ **Good till today including SL/TP** orders effective only within one trading day. As soon as it is over, all Stop Loss and Take Profit levels of open positions, as well as all pending orders are deleted.
 - ♦ **Good till canceled** as a trade day changes, pending orders and stop levels are preserved.
 - ♦ **Good till today excluding SL/TP** as a trade day changes, only pending orders are deleted, Stop Loss and Take Profit levels of open positions are preserved.
- **Limit & stop level** price channel (in points) from the current market price, within which it is prohibited to place Stop Loss, Take Profit and pending orders. At the attempt to place an order within this channel, the server will return message "Invalid S/L or T/P" and will not accept the order;
- **Freeze level** level for freezing orders that are close to the current price. When an order price is as close to the market price as the value specified here or less than that, its modification or deletion is prohibited. This limitation is also applied to positions. When current market price is close to a Stop Loss of Take Profit level of a position, modification of these levels is prohibited as well as modification of the position volume and closing of the position;
- **Max quote delay** time (in seconds) of delay in the receipt of quotes, after which trading will be automatically disabled for this symbol. As quotes start coming again, trade will be enabled automatically.
- **Filling** additional rules of order filling that can be set to traders. The necessary filling type should be ticked off:
 - ♦ **Fill or Kill** this fill policy means that a deal can be executed only with the specified volume.

- ◆ **Immediate or Cancel** in this case a trader agrees to execute a deal with the volume maximally available in the market within that indicated in the order. In case the order cannot be filled completely, the available volume of the order will be filled, and the remaining volume will be canceled.
- ◆ **Book or Cancel** the order can only be placed in the Depth of Market (order book). If the order can be filled immediately when placed, this order is canceled.
- ◆ **Return** this mode is not available in the list. It is always enabled for the market (Buy and Sell) orders in the Exchange Execution mode, as well as for limit and stop limit orders in the Market Execution and Exchange Execution modes.

In the client and manager terminal, the fill policy can be selected only when placing market orders. The exceptions are Limit and Stop Limit orders by symbols that have the "Exchange" type of execution.

- **Convert profit** this option is available only for the Forex type symbols. It determines the conversion of profit/loss from the symbol profit currency to a client deposit currency:
- **by deal** this mode is used on default. For the conversion of profit/loss from the profit currency to the deposit currency, the price at which a deal (exit from position) is performed is used, the profitability/unprofitability of the deal is not taken into consideration;
- **by market** in this mode, the conversion is performed using the current Bid/Ask price depending on the profitability/unprofitability of a deal. The Bid price is taken for calculations for profitable deals, because as a result of a profitable deal, a trader obtains a certain amount of the profit currency and needs to sell it for the deposit currency. The Ask price is taken for losing deals, because as a result of a losing deal, a trader needs to buy a certain amount of currency for the deposit currency.

More detailed information can be found in the "Profit calculation" section.

- **Expiration** condition for the expiration of orders. Necessary expiration types should be ticked off:
 - ◆ **Good till canceled** the order will stay in the queue until it is manually canceled;
 - ◆ **Day** the order will be effective only during the current trading day. When orders are processed on the MetaTrader 5 platform side, intraday orders are canceled at the end of the calendar day considering the trading server time and the symbol trading sessions. For example, if trading ends at 21:00, the untriggered intraday orders will be canceled at the same time. When orders are processed in external systems via gateways, the expiration rules can be different and depend on the specific system.
 - ◆ **Specified time** the order will be effective till the date specified by the trader;

- ◆ **Specified day** the order is active till 00:00 of the specified day. If that time appears to be out of a trade session, the expiration will be processed at a nearest trading time.
- **Orders** types of allowed orders: Market, Limit, Stop, Stop Limit, Stop Loss, Take Profit, Close By. Close By permission of a symbol does not mean that this type of orders will be available on netting accounts. Close By orders can only be used on hedging accounts.
- **Enable Trading Signals** if this option is disabled, clients will not be able to copy trade operations by this symbol using the Signals service. Trading signals can also be disabled on a client group level.
- **Volumes** setup of volumes of placed orders for this symbol:
 - ◆ **Minimum** minimal volume of the symbol order. Not applied when closing positions.
 - ◆ **Maximum** maximal volume of the symbol order. It is considered not only when traders place orders but also when a position is closed by stop out.
 - ◆ **Step** volume change step.
 - ◆ **Limit** maximum allowed cumulative volume of an open position and pending orders by a symbol in one direction (buy or sell). For example, if the limit is set to 5 lots, a user can have an open buy position of 5 lots and place a Sell Limit order of 5 lots. But in this case the trader cannot place a pending Buy Limit order, since the total volume in one direction would exceed the limit. Also, a client having a 5-lot Buy position cannot place a Sell Limit order with a volume exceeding 5 lots: if the pending order triggered after the closing of the initial position, the client would have a short position exceeding the specified limit.
- **Prices** price setting for futures contracts (available for the Exchange Futures and FORTS Futures trade modes):
 - ◆ **Settlement** calculated price of the futures contract (the clearing price of the previous trade session).
 - ◆ **Minimum** minimum price of the futures contract for the current trade session. A price no lower than specified can be specified when placing an order.
 - ◆ **Maximum** maximum price of the futures contract for the current trade session. A price no greater than specified can be specified when placing an order.

- The price parameters of the Exchange Futures and FORTS Futures contracts determine the margin requirements for them. The minimum and maximum prices for the current session, which are calculated at the exchange as a percentage of the settlement price, determine the maximum possible profit and loss of a client for the financial instrument. Thus, the maximum possible loss is used as the margin requirements for the symbol instead of the entire price of the contract.
- Usually, the price parameters and the margin requirements (initial and maintenance) are passed from and exchange through a gateway. In this case, you shouldn't modify those parameters manually.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Trade / Margin Calculation

Margin Calculation

Margin is charged to provide collateral for traders' open positions and orders. It serves as a guarantee that the conditions of the concluded deal will be fulfilled. The trading platform provides different risk management models, which define the type of pre-trade control. The model can be defined separately for each client group on the "Margin" tab in the "Risk Management" field.

Group: real\real

Common Company News & Mail Permissions **Margin** Symbols Commissions Reports

Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management: for Retail Forex, CFD, Futures

Margin call level: 50.00 Stop out level: 30.00 in %, percent

☐ Stop out fully hedged accounts

☐ Compensate negative balance after stop out

☐ Withdraw credit after negative balance compensation

Floating leverage profile: None

Profit/loss in free margin

Unrealized profit: use unrealized profit/loss

Daily fixed profit: use daily fixed profit/loss

☐ Release fixed profit at the end of day

Virtual credit: (applies only to opening new positions)

OK Cancel Help

The following models are currently available:

- **for Retail Forex, CFD, Futures** used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Netting position accounting system is used.
- **for Retail Forex, CFD, Futures with hedging** used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Hedging position accounting system is used.
- **for Stock Exchange, based on margin discount rates** used for the exchange market. Margin calculation is based on the discounts specified in symbol settings. Discounts are set by the broker, however they cannot be lower than the exchange set values.

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Basic Margin Calculation

The trading platform provides several margin requirement calculation types depending on the financial instrument. You can select the calculation type in [Calculation](#) field of Trade tab. Margin calculation formula for each instrument depends on the [risk management formula](#) used for the client group.

The basic calculation is only the first stage in the calculation of the final margin value. The value calculated by formulas is always converted to the deposit currency, after which the margin rate and other parameters are additionally applied. A detailed calculation scheme is provided in separate sections for each risk management system:

- [Retail Forex, CFD, Futures Netting](#)
- [Retail Forex, CFD, Futures Hedging](#)
- [Exchange Model](#)

If the [Initial Margin](#) parameter value is specified in symbol settings, this value will be used. The formulas described in this section will not be applied.

Forex

The margin for Forex market symbols is calculated using the following equation:

All modes: Volume in lots * Contract size / Leverage

For example, let's calculate the margin requirements for buying one lot of EURUSD, while [the size of one contract](#) is 100,000 and the leverage is 1:100. After placing the appropriate values to the equation, we will obtain the following result:

$$1 * 100000 / 100 = \text{EUR } 1000$$

So, now we have the margin requirements value in [base currency](#) (or [margin currency](#)) of the symbol.

- Generally, margin requirements currency and symbol's base currency are the same. If the margin currency is different, calculation results are displayed in that currency instead of the symbol's base one.
- In this mode, a client leverage is taken into account even if a [fixed margin](#) is set.

Forex No Leverage

This type of calculation is also used for Forex symbols. But unlike the previous one, it does not take into account the client's leverage:

All modes: Volume in lots * Contract size

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For example, let's calculate the margin requirements for buying one lot of EURUSD, while the size of one contract is 100,000 and the leverage is 1:100. After placing the appropriate values to the equation, we will obtain the following result:

$$1 * 100,000 = \text{EUR } 100,000$$

So, now we have the margin requirements value in base currency (or margin currency) of the symbol.

Generally, margin requirements currency and symbol's base currency are the same. If the margin currency is different, calculation results are displayed in that currency instead of the symbol's base one.

CFDs

The margin requirements for CFDs are calculated using the following equation:

All modes: Volume in lots * Contract size * Open market price

The current market Ask price is used for buy deals, while the current Bid price is used for sell ones.

For example, let's calculate the margin requirements for buying one lot of oil, the size of the contract is 100 barrels, the current Ask price is USD 80. After placing the appropriate values to the equation, we will obtain the following result:

$$1 * 100 * 80 = \text{USD } 8,000$$

So, now we have the margin value in base currency (or margin currency) of the symbol.

CFD Leverage

The leverage is also considered in this type of margin requirement calculation for CFDs:

All modes: Volume in lots * Contract size * Open market price / Leverage

CFD Index

For index CFDs, the margin requirements are calculated according to the following equation:

All modes: Volume in lots * Contract size * Open market price * Tick value / Tick size

In addition to the usual calculation for CFDs, the formula takes into account the tick value to tick size ratio.

Futures, Exchange Futures

There are two types of the margin requirements for futures contracts:

- **Initial margin** is the amount that must be available on the account at the moment of attempting to enter the market. Further maintenance of the same sum may not be obligatory.
- **Maintenance margin** is the minimum amount that must be available on the account for holding a position open.

Both values are specified in Margin tab of the symbol settings. The final size of the margin depends on the volume:

Volume in lots * Initial margin

Volume in lots * Maintenance margin

The calculation is the same in all risk management modes.

If the amount of the maintenance margin is not specified, the initial margin value will be used instead.

Exchange Options

There are two types of margin requirements for futures contracts:

- **Initial margin** is the amount that must be available on the account at the moment of attempting to enter the market. Further maintenance of the same amount may not be obligatory.
- **Maintenance margin** is the minimum amount that must be available on the account for holding a position open.

Both values are specified in Margin tab of the symbol settings. The final size of the margin depends on the volume:

Volume in lots * Initial margin

Volume in lots * Maintenance margin

If the amount of the maintenance margin is not specified, the initial margin value will be used instead. If neither the initial nor the maintenance margin is specified, the appropriate value will be calculated according to the following formula:

Volume in lots * Contract size * Open market price

The current market Ask price is used for buy deals, while the current Bid price is used for sell deals.

The same calculation method is applied for all risk management modes.

Exchange Stocks, Exchange MOEX Stocks

The margin requirements for stocks are calculated using the following equation:

Retail Forex, CFD, Futures: Volume in lots * Contract size * Open market price

Retail Forex, CFD, Futures with hedged position: Volume in lots * Contract size * Open market price

The current market Ask price is used for buy deals, while the current Bid price is used for sell ones.

For example, let's calculate the margin requirements for buying one lot of oil, the size of the contract is 100 barrels, the current Ask price is USD 80. After placing the appropriate values to the equation, we will obtain the following result:

$$1 * 100 * 80 = \text{USD } 8,000$$

So, now we have the margin value in base currency (or margin currency) of the symbol.

In the "Stock Exchange, based on margin discount rates" mode, margin for stocks is calculated as an indicative value, for assessing the trading account state relative to available positions. In this case, the margin amount is not blocked on the account and therefore does not reduce available funds. For details, please see the [appropriate section](#).

The only difference for the Exchange Stocks and Exchange MOEX Stocks modes is the [calculation of the adjusted initial margin](#) when using the exchange model for risk management.

Exchange Bonds, Exchange MOEX Bonds

The bond margin is calculated as part of the position value. Bond prices are provided as a face value percentage, so the position value is calculated as follows:

Retail Forex, CFD, Futures: Volume in lots * Contract size * Face value * Open price / 100

Retail Forex, CFD, Futures with hedged position: Volume in lots * Contract size * Face value * Open price / 100

The part of the position value to be reserved for maintenance is determined by [margin rates](#).

In the "Stock Exchange, based on margin discount rates" mode, bond margin is calculated as an indicative value, for assessing the trading account state relative to available positions. In this case, the margin amount is not blocked on the account and therefore does not reduce available funds. For details, please see the [appropriate section](#).

- The position value is calculated in the [symbol's base currency](#) (not in the profit currency).

- The only difference for the Exchange Bonds and Exchange MOEX Bonds modes is the calculation of the adjusted initial margin when using the exchange model for risk management.

Exchange FORTS Futures

The margin for the futures contracts of the Moscow Exchange derivative section is calculated separately for each symbol: First, the margin is calculated for the open position and all Buy orders. Then the margin for the same position and all Sell orders is calculated. The calculation is the same in all risk management modes.

$$\text{MarginBuy} = \text{MarginPos} + \text{Sum}(\text{MarginBuyOrder})$$

$$\text{MarginSell} = \text{MarginPos} + \text{Sum}(\text{MarginSellOrder})$$

The largest one of the calculated values is used as the final margin value for the symbol.

Thus, the same position is used in the calculation of both values. In the first formula (which includes Buy orders), the position margin is calculated as follows:

$$\text{MarginPos} = \text{Volume} * (\text{InitialMarginBuy} + (\text{Open Price} - \text{SettlementPrice}) * \text{Tick Value} / \text{Tick Size} * (1 + 0.01 * \text{Margin Currency Rate}))$$

The volume is used with a positive sign for long positions and with a negative sign for short positions.

In the second formula (which includes Sell orders), the position margin is calculated as follows:

$$\text{MarginPos} = \text{Volume} * (\text{InitialMarginSell} + (\text{SettlementPrice} - \text{Open Price}) * \text{Tick Value} / \text{Tick Size} * (1 + 0.01 * \text{Margin Currency Rate}))$$

The volume is used with a positive sign for short positions and with a negative sign for long positions.

This approach provides the trader a discount on margin, when there is an open position in the opposite direction with respect to the orders placed (the position acts as collateral for orders).

Margin on orders is calculated by the following formulas:

$$\text{MarginBuyOrder} = \text{Volume} * (\text{InitialMarginBuy} + (\text{Price} - \text{SettlementPrice}) * \text{Tick value} / \text{Tick size} * (1 + 0.01 * \text{Margin currency rate}))$$

$$\text{MarginSellOrder} = \text{Volume} * (\text{InitialMarginSell} + (\text{SettlementPrice} - \text{Price}) * \text{Tick value} / \text{Tick size} * (1 + 0.01 * \text{Margin currency rate}))$$

'Price' here depends on the order time and can be equal to:

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- The Highest and the Lowest price of the contract for the current session is used for not yet executed market or stop Buy and Sell orders, respectively. Since the price is not specified in market orders, the trader is charged the maximum possible margin. Once triggered, stop orders behave similar to market orders.
- The order price is used for limit orders.
- The Stop Limit price is used for stop limit orders.

Other parameters in the formulas:

- InitialMarginBuy the initial margin for the Buy operation.
- InitialMarginSell the initial margin for the Sell operation.
- Currency margin rate is the rate change radius of the currency, a futures contract is denominated in, relative to the Russian ruble
- SettlementPrice settlement price of an instrument for the current session.

All these parameters for calculation are provided by the Moscow Exchange.

InitialMarginBuy is written to the "Initial margin" field, InitialMarginSell is written to the "Maintenance Margin" field in symbol properties.

Calculation example

The below example shows the calculation of margin requirements for the following trading account state:

- Position Buy 3.00 Si-6.18 at 73640
- Order Buy Limit 2.00 Si-6.18 at 73000
- Order Sell Limit 10.00 Si-6.18 at 74500

Current session parameters

- Clearing price = 73638
- InitialMarginBuy = 7665.41
- InitialMarginSell = 7739.59
- Tick value = 1
- Tick size = 1
- Margin currency rate = 0

We substitute the values in the formulas

$$\text{MarginBuy} = 3 * (7665.41 + (73640 - 73638) * 1/1) + 2 * (7665.41 + (73000 - 73638) * 1/1) = 37057.05$$

$$\text{MarginSell} = -3 * (7739.59 + (73638 - 73640) * 1/1) + 10.0 * (7739.59 + (73638 - 74500) * 1/1) = 45563.13$$

$$\text{Margin} = \text{Max}(37057.05, 45563.13) = 45563.13$$

The resulting margin for the Si-6.18 symbol is 45563.13.

The Exchange FORTS Futures mode must only be used together with the MOEX Derivatives Gateway. Otherwise the margin requirements calculation results may be unpredictable.

Collateral

Non-tradable instruments of this type are used as trader's assets to provide the required margin for open positions of other instruments. For these instruments the margin is not calculated.

Fixed Margin

If a non-zero value is specified in the "Initial margin" field, then no calculations by formulas specified in the "Calculation" field are performed (except for the calculation of futures, which is not affected). In this case, for all types of calculations (except for Forex and CFD Leverage), the margin is obtained as if "Futures" option is selected:

Volume in lots * Initial margin

Volume in lots * Maintenance margin

For Forex and CFD Leverage calculation types, the leverage is additionally considered:

Volume in lots * Initial margin / Leverage

Volume in lots * Maintenance margin / Leverage

The calculation is the same in all risk management modes.

If the amount of the maintenance margin is not specified, the initial margin value will be used instead.

End-of-day margin conversion rate recalculation

This option assists brokers in meeting regulatory requirements (in particular, NFA requirements), which limit the maximum allowed leverage for different symbol types.

In currency pair trading, the margin determined in the symbol's base currency is converted into the account deposit currency. For example, when trading EURUSD with a contract size of 100,000 and a leverage of 1:100, the margin will be EUR 1,000 per lot. If the trader's deposit currency is USD, the margin will be converted from EUR to USD at the current rate. The conversion rate is registered in the position's Margin rate field.

The EURUSD exchange rate will change over time, and the USD equivalent of EUR 1,000 will change accordingly. Thus, the trader's actual leverage may become greater or less than the original one.

To prevent this from happening, the trading platform provides a mechanism for updating margin conversions. It is activated by the "Recalculate margin exchange rate

at the end of day" option in common symbol settings or in group-specific symbol settings.

Symbol: Forex\EURUSD

Common Currency Quotes Trade Execution **Margin** Margin Rates Swaps Sessions

The setting up of margin requirements for the symbol. Please specify the initial and maintenance margin and multipliers for different types of trade operations.

Initial margin:

Hedged margin:

Maintenance margin:

☐ Calculate hedged margin using larger leg

☐ Exclude long position PnL from free margin and margin level

☐ Recalculate margin exchange rate at the End of Day

Additional margin checks:

OK Cancel Help

At the end of the trading day, the server will update the "Margin rate" field in all positions for the instruments with the recalculation enabled. The new value will be calculated using the current market prices for the relevant client groups (similar procedures performed during trade execution).

Please note the following features:

- The margin rate is not recalculated on the days marked as non-working in the platform (according to the Time section settings)
- The margin rate is recalculated after swaps calculations (if no swaps are charged, the rate is recalculated too)
- If the required price is not available (zero) at the time of recalculation, no recalculation is performed

The recalculation process can be tracked in the trade server log. It will show the following records:

```
margin rates update started
...
margin rates update finished
```

The recalculation is carried out only when a symbol margin currency is different from the account deposit currency. If the margin currency and the deposit currency are the same, then no recalculation is actually carried out. However, a record on updating the margin ratio value is still sent to the server journal. In this case, it will always be equal

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to 1.00000. For example:

margin rates for 'EURUSD' positions updated - buy: 1.000000000, sell: 1.000000000

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Trade / Margin Calculation / Retail Forex, CFD, Futures Netting

Retail Forex, CFD, Futures Netting

This margin calculation model is used for Retail Forex, CFD and Futures with the netting position accounting system. Unlike to the same model for hedging accounts, spreads are additionally taken into account in this calculation method.

The first stage in margin calculation is defining if an account has positions or pending orders for the symbol the deal is performed for.

- If that account has no positions and orders for the symbol, the margin is calculated using the formulas.
- If the account has an open position, and a new order of any type with the volume being less or equal to the current position is placed in the opposite direction, the total margin is equal to the current position's one. Example: we have a 1 lot EURUSD Buy position and place an order to Sell 1 lot EURUSD (similarly for Sell Limit, Sell Stop and Sell Stop Limit).
- If the account has an open position and an order of any type is placed in the same direction, the total margin is equal to the sum of the current position's and placed order's margins.
- If the account has an open position, and an order of any type with the volume exceeding the current position is placed in the opposite direction, two margin values are calculated - for the current position and for the placed order. The final margin is taken according to the highest of the two calculated values.
- If the account has two or more oppositely directed market and limit orders, the margin is calculated for each direction (Buy and Sell). The final margin is taken according to the highest of the two calculated values. For all other order types (Stop and Stop Limit), the margin is summed up (charged for each order).

The margin amount is calculated in several stages:

- Basic calculation for a certain symbol
- Conversion of margin currency into deposit currency
- Multiplication by rate
- Calculations for trading symbols in spread

Basic Calculation for a Symbol

The basic margin value in accordance with the symbol type is calculated first. The type is defined by symbol settings in the Calculation field.

Converting into Deposit Currency

This stage is common for all calculation types. Conversion of margin requirements calculated using one of the methods mentioned above is performed in case their currency is different from the account deposit currency.

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The current exchange rate of margin currency to deposit currency is used for conversion. The Ask price is used for buy deals, and the Bid price is used for sell deals.

Suppose that the basic size of the margin previously calculated for buying one lot of EURUSD is 1000 EUR. If the account deposit currency is USD, the current Ask price of EURUSD pair is used for conversion. For example, if the current rate is 1.2790, the total margin size is 1279 USD.

The margin to deposit currency conversion rate is shown in the "Margin rate" field in orders and positions.

For details, please visit the Conversion section.

Margin rate

You can specify in symbol settings additional multipliers (rates) for the margin requirements depending on the position/order type. This can be done in the Margin rates tab. The final size of the margin requirements previously calculated regarding conversion into the deposit currency is additionally multiplied by the appropriate rate. In addition, you can completely disable margin charging for any desired types of trading operations. To do this, set the zero margin ratio.

For example, the previously calculated margin for buying one lot of EURUSD is 1279 USD. This sum is additionally multiplied by long margin rate. For example, if it is equal to 1.15, the final margin is $1279 * 1.15 = 1470.85$ USD.

Calculations for Spread Trading

The margin can be charged on preferential basis in case trading positions are in spread relative to each other. The spread is defined as the presence of the oppositely directed positions at related symbols. Reduced margin requirements provide traders with more trading opportunities. Configuration of spreads is described in a separate section.

Application of floating margin rules

Additional rates may be applied to the calculated margin value in accordance with the rules configured in the Leverages section.

Checking Margin

When placing new orders, a check is performed of whether there is enough free margin on the trader's account for the appropriate order. An additional free margin check can be performed after the execution of pending orders. This depends on symbol settings. Up to 3 checks can be performed in total:

1. Check during an order placement. The margin is checked according to the margin rates specified for the symbol. If the margin rate is 0, the margin is not checked.
2. Check during the pending order activation. If an order is activated by a trade server or a dealer (via the Manager terminal), sufficiency of funds for the deal

to be performed according to that order is checked. Such check is not performed if the order is activated by the gateway when activation is actually tracked by the external trading system.

3. Check after an order is confirmed by a dealer before that order is actually executed. The check is performed if Additional margin check: check before executing orders option is enabled in the trade symbol settings.

Check rules:

1. After the execution of an appropriate operation (order), the trader's free margin considering the virtual credit should not be negative (it should exceed or be equal to 0).
2. After executing an order, which is opposite to the present position (closing or reversal order), the margin should not increase. In fact, it means that a trader can close their position or reverse it considering available funds under any circumstances (even when there is not enough margin to place a new order).

The system will allow placing an order if any of the rules is met. If the trader has enough free margin (1), it does not matter if the margin will increase or not (2) the system will allow placing an order.

The orders that have been placed but have not yet been accepted for processing or have not yet been executed (in the "started" state) can only increase the margin but not decrease it. Example:

- The client account has the following position: Sell 0.02 lot USDJPY. The current margin is 6.67 USD.
- Case 1. The client places an order in the opposite direction: Buy 0.03 lot USDJPY. As long as the order is in the "started" state, the client's margin is still equal to 6.67 USD. As soon as the order is executed, the margin will be halved, i.e. 3.33 USD, since the position volume will decrease from 0.02 to 0.01 lots.
- Case 2. The client places an order in the same direction: Sell 0.03 lot USDJPY. The client's margin will increase immediately after placing the order, even while the order is in the "started" state. The margin value will become 16.67 USD, since the potential total volume may become 0.05 lots.

This prevents clients from placing orders for a total volume larger than the funds allow.

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Retail Forex, CFD, Futures Hedging

This margin calculation model is used for Retail Forex, CFD and Futures with the hedging position accounting system. Unlike to the same model for netting accounts, the presence of multiple positions/orders of one symbols is additionally taken into account in this calculation method.

The margin amount is calculated in several stages:

- Basic calculation for a certain symbol
- Conversion of margin currency into deposit currency
- Multiplication by rate
- Accounting multiple positions/orders of the same symbol

Basic Calculation for a Symbol

The basic margin value in accordance with the symbol type is calculated first. The type is defined by symbol settings in the Calculation field.

Converting into Deposit Currency

The main calculations by formulas is performed in the symbol margin currency. Conversion is applied if it differs from the client's deposit currency. For example, the basic calculation for the EURUSD pair with the EUR margin currency will produce the margin value in EUR. If the user's deposit currency is USD, the calculated value will be converted from EUR to USD.

The current exchange rate of margin currency to deposit currency is used for conversion. The Ask price is used for buy deals, and the Bid price is used for sell deals.

Suppose that the basic size of the margin previously calculated for buying one lot of EURUSD is 1000 EUR. If the account deposit currency is USD, the current Ask price of EURUSD pair is used for conversion. For example, if the current rate is 1.2790, the total margin size is 1279 USD.

The margin to deposit currency conversion rate is shown in the "Margin rate" field in orders and positions.

For details, please visit the Conversion section.

Margin rate

You can specify in symbol settings additional multipliers (rates) for the margin requirements depending on the position/order type. This can be done in the Margin rates tab. The final size of the margin requirements previously calculated regarding conversion into the deposit currency is additionally multiplied by the appropriate rate. In addition, you can completely disable margin charging for any desired types of

trading operations. To do this, set the zero margin ratio.

For example, the previously calculated margin for buying one lot of EURUSD is 1279 USD. This sum is additionally multiplied by long margin rate. For example, if it is equal to 1.15, the final margin is $1279 * 1.15 = 1470.85$ USD.

Accounting multiple positions/orders of the same symbol

The hedging system allows having simultaneously multiple open positions of the same financial instrument. This is reflected in margin calculation.

Positions/orders opened in the same direction

Their volumes are summed up and the weighted average open price is calculated for them. The resulting values are used for calculating margin by the formula corresponding to the symbol type. Margin is calculated separately for pending orders (if the margin rate is non-zero).

For example, the following positions and orders exist on the account:

- Buy 1 lot at 15.436
- Buy 2 lot at 15.432
- Buy Limit 1 lot at 15.412

The CFD Leverage calculation type is used for the instrument, its contract size is 5,000, and the account leverage is 1:100. The margin rate for all types of operations is 1.

- Calculating the average weighted Open price of Buys positions: $(1 * 15.436 + 2 * 15.432) / (1 + 2) = 15.433333333$
- Calculating margin using the CFD Leverage formula for the aggregate volume: $3 * 5\,000 * 15.433333333 / 100 = 2\,315.00$
- Taking into account the margin rate: $2\,315.00 * 1 = 2\,315.00$
- Calculating margin using the CFD Leverage formula separately for the pending order: $1 * 5\,000 * 15.412 / 100 = 770.60$
- Taking into account the margin rate: $770.60 * 1 = 770.60$
- Calculating the final value as the sum of position and order margin values: $2\,315.00 + 770.60 = 3\,085.60$

Further, this value can be converted to the account deposit currency, if necessary.

Opposite Positions/Orders

Oppositely directed open positions of the same symbol are considered hedged or covered. Two margin calculation methods are possible for such positions. The calculation method is determined by the symbol settings.

Basic calculation	Using the larger leg
Used if the "Calculate hedged margin using larger leg"	Used if the "Calculate

Basic calculation	Using the larger leg
option is disabled in the <u>symbol settings</u>. The calculation consists of several steps: • For uncovered volume • For covered volume (if hedged margin size is specified) • For pending orders The resulting margin value is calculated as the sum of margins calculated at each step. Calculation for uncovered volume • Calculation of the total volume of all positions and market orders for each of the legs buy and sell. • Calculation of the weighted average position and market order open price for each leg: (open price of position or order 1 * volume of position or order 1 + ... + open price of position or order N * volume of position or order N) / (volume of position or order 1 + ... + volume of position or order N). • Calculation of uncovered volume (smaller leg volume is subtracted from the larger one). • The calculated volume and weighted average price are used then to calculate margin by the appropriate formula corresponding to the <u>symbol type</u>. • When considering a <u>margin rate</u>, the larger leg rate (buy or sell) is used. • The weighted average rate value is used when <u>converting from a margin currency to a deposit one</u>. Calculation for covered volume Used if the "Hedged margin" value is specified in the <u>symbol settings</u>. In this case margin is charged for hedged, as well as uncovered volume.	hedged margin using larger leg" option is enabled in the <u>symbol settings</u>. • Calculation of margin for shorter and longer legs for all open positions and market orders. • Calculation of margin for each pending order type separately (Buy Limit, Sell Limit, etc.). • Summing up a longer leg margin: long positions and market orders + long pending orders. • Summing up a shorter leg margin: short positions and market orders + short pending orders. • The largest one of all calculated values is used as the final margin value. Long side calculation • The total volume of long positions and market orders is calculated • Calculation of the weighted average Open price of long positions and market orders:

Basic calculation	Using the larger leg
If the initial margin is specified for a symbol, the hedged margin is specified as an absolute value (in monetary terms). If the initial margin is not specified (equal to 0), the contract size is specified in the "Hedged" field. The margin is calculated by the appropriate formula in accordance with the type of the financial instrument, using the specified contract size. For example, we have two positions Buy EURUSD 1 lot and Sell EURUSD 1 lot, the contract size is 100,000. If the value of 100,000 is specified in the "Hedged field", the margin for the two positions will be calculated as per 1 lot. If you specify 0, no margin is charged for the hedged (covered) volume. If you migrate margin calculation settings from MetaTrader 4, you should multiply the value of the "Hedged" field by 2. Thus you will receive the same margin value during calculations. For example, if the hedged margin in MetaTrader 4 was 50,000, it should be set to 100,000 in MetaTrader 5. Per each hedged lot of a position, the margin is charged in accordance with the value specified in the "Hedged Margin" field in the <u>symbol settings</u>: • Calculation of hedged volume for all open positions and market orders (uncovered volume is subtracted from the larger leg). • Calculation of the weighted average Open price of all positions and market orders: (open price of position or order 1 * volume of position or order 1 + ... + open price of position or order N * volume of position or order N) / (volume of position or order 1 + ... + volume of position or order N). • The calculated volume, weighted average price and the hedged margin value are used then to calculate margin by the appropriate formula corresponding to the <u>symbol type</u>. • When considering a <u>margin rate</u>, the average value	(open price of position or order 1 * volume of position or order 1 + ... + open price of position or order N * volume of position or order N) / (volume of position or order 1 + ... + volume of position or order N). • The calculated volume and weighted average price are used then to calculate margin by the appropriate formula corresponding to the <u>symbol type</u>. • The <u>margin rate</u> for long positions is taken into account. • The weighted average rate value is used when <u>converting from a margin currency to a deposit one</u>. A similar calculation is performed for each type of pending Buy orders: separately for Buy Limit, Buy Stop and Buy Stop Limit orders. All calculated values are summed.

Basic calculation	Using the larger leg
of the buy and sell order rate is used: (Buy rate + Sell rate)/2 - The weighted average rate value is used when <u>converting from a margin currency to a deposit one</u>. Calculation for pending orders - Calculation of margin for each pending order type separately (Buy Limit, Sell Limit, etc.). - The weighted average rate value and the conversion rate for each pending order type is used when taking into account the <u>margin rate</u> and the <u>margin to deposit currency conversion rate</u>. Calculation specifics for hedging orders when using fixed margin When an order opposite to an existing position is placed, the margin on the hedged volume is always calculated using the "Hedge margin" value. For the non-hedged volume, the "Initial margin" value is used when placing an order, and "Maintenance margin" is applied after the appropriate position is opened. These specifics are only valid for symbols with the specified initial and maintenance margin (<u>Fixed Margin</u> or <u>Futures</u> calculation type). For example, the following parameters are used for EURUSD: - Initial margin = 1000 - Maintenance margin = 500 - Hedge margin = 500 A trader has a position Buy 1.00 BR-12.18 on a USD account. A margin of 500 USD (as per the "Maintenance margin") is reserved on the trader's account for this position. - To open Sell 2.00 BR-12.18, the trader needs the margin of 2000 USD: 500 USD for the existing	Short side calculation - The total volume of short positions and market orders is calculated - Calculation of the weighted average Open price of short positions and market orders: (open price of position or order 1 * volume of position or order 1 + ... + open price of position or order N * volume of position or order N) / (volume of position or order 1 + ... + volume of position or order N). - The calculated volume and weighted average price are used then to calculate margin by the appropriate formula corresponding to the <u>symbol type</u>. - The <u>margin rate</u> for short positions is taken into account. - The weighted average rate value is used when <u>converting from a margin currency to a deposit one</u>. A similar calculation is performed for each type

Basic calculation	Using the larger leg
position, 500 for 1 hedged lot of the new position (in accordance with the "Hedged margin" parameter) and 1000 for 1 non-hedged lot of the new position (as set in the "Initial margin" parameter). • Once the position is opened, a margin of 1000 USD will remain reserved on the trader's account: 500 USD for 1 hedged lot (in accordance with "Hedged margin") and 500 USD for 1 non-hedged lot (as specified in the "Maintenance margin").	of pending Sell orders: separately for Sell Limit, Sell Stop and Sell Stop Limit orders. All calculated values are summed. Total value The largest one of the calculated values is used as the final margin value.
Example The following positions are present: • Sell 1 lot at 1.11943 • Buy 1 lot at 1.11953 • Sell 1 lot at 1.11943 • Buy 1 lot at 1.11953 • Sell 1 lot at 1.11943 Hedged margin size = 100000. Buy margin rate = 2, for Sell = 4. Leverage 1:500. Calculate uncovered volume: Sell volume (3) - Buy volume (2) = 1 Calculate the weighted average Open price for the hedged volume by all positions: $(1.11943 * 1 + 1.11953 * 1 + 1.11943 * 1 + 1.11953 * 1 + 1.11943 * 1) / 5 = 5.59735 / 5 = 1.11947$ Calculate the weighted average Open price for the non-hedged volume by all positions of the larger leg: $(1.11943 * 1 + 1.11943 * 1 + 1.11943 * 1) / 3 = 1.11943$ Calculate the margin rate for the hedged volume: $(\text{buy rate} + \text{sell rate}) / 2 = (2 + 4) / 2 = 3$	Example The following positions and orders exist on the account: • Buy EURUSD 1.00 lot at 1.16214 • Buy EURUSD 2.00 lot at 1.16207 • Sell EURUSD 1.00 lot at 1.16184 • Sell EURUSD 1.00 lot at 1.16189 • Buy Limit EURUSD 1.00 lot at 1.16053 • Sell Stop EURUSD 1.00 lot at 1.15808 • Sell Stop EURUSD 1.00 lot at 1.16067 The Forex calculation type is used for the instrument, its contract size is 100000, the account leverage is 1:100 and the deposit currency is USD. The margin rate for all types of operations

Basic calculation	Using the larger leg
The larger leg (sell) margin rate is used for the non-hedged volume (sell): 4. Calculate the hedged volume margin using the equation: $(2.00 \text{ lots} * 100000 \text{ EUR} * 1.11947 * 3) / 500 = 1343.36$ Calculate the non-hedged volume margin using the equation: $(1.00 \text{ lot} * 100000 \text{ EUR} * 1.11943 * 4) / 500 = 895.54$ The final margin size: $1343.36 + 895.54 = 2238.90$	is 1. Calculating the margin for the long side: • Calculating the total volume of Buy positions and market orders: $1 + 2 = 3$ • Calculating the average weighted Open price of Buys positions: $(1 * 1.16214 + 2 * 1.16207) / (1 + 2) = 1.16209333$ • Calculating margin using the <u>Forex formula</u> for the total volume: $3 * 100000 / 100 = 3000 \text{ EUR}$ • Converting the resulting value to USD using the weighted average conversion rate: $3000 * 1.16209333 = 3486.28$ • Calculating margin for the Buy Limit order using the <u>Forex formula</u>: $1 * 100000 / 100 = 1000 \text{ EUR}$ • Converting margin to USD using the rate from the order: $1000 * 1.16053 = 1160.53 \text{ USD}$ • Calculating the total margin value for the long side: $3486.28 + 1160.53$

Basic calculation	Using the larger leg
	= 4646.81 Calculating the margin for the short side: • Calculating the total volume of Sell positions and market orders: $1 + 1 = 2$ • Calculating the average weighted Open price of Sell positions: $(1 * 1.16184 + 1 * 1.16189) / (1 + 1) = 1.161865$ • Calculating margin using the <u>Forex formula</u> for the total volume: $2 * 100000 / 100 = 2000 \text{ EUR}$ • Converting the resulting value to USD using the weighted average conversion rate: $2000 * 1.161865 = 2323.73$ • Calculating the total volume of Sell Stop orders: $1 + 1 = 2$ • Calculating the average weighted Open price of the orders: $(1 * 1.15808 + 1 * 1.16067) / (1 + 1) = 1.159375$ • Calculating margin using the <u>Forex formula</u> for the total volume of Sell Stop orders: $2 * 1161.865 = 2323.73$

Basic calculation	Using the larger leg
	$100000 / 100 = 2000 \text{ EUR}$ • Converting margin to USD using the weighted average order price: $2000 * 1.159375 = 2318.75 \text{ USD}$ • Calculating the total margin value for the short side: $2323.73 + 2318.75 = 4642.48$ The largest one of the calculated values will be used as the final margin: 4646.81.

Application of floating margin rules

Additional rates may be applied to the calculated margin value in accordance with the rules configured in the the Leverages section.

Checking Margin

When placing new orders, a check is performed of whether there is enough free margin on the trader's account for the appropriate order. An additional free margin check can be performed after the execution of pending orders. This depends on symbol settings. Up to 3 checks can be performed in total:

1. Check during an order placement. The margin is checked according to the margin rates specified for the symbol. If the margin rate is 0, the margin is not checked.
2. Check during the pending order activation. If an order is activated by a trade server or a dealer (via the Manager terminal), sufficiency of funds for the deal to be performed according to that order is checked. Such check is not performed if the order is activated by the gateway when activation is actually tracked by the external trading system.
3. Check after an order is confirmed by a dealer before that order is actually executed. The check is performed if Additional margin check: check before executing orders option is enabled in the trade symbol settings.

Check rules:

1. After the execution of an appropriate operation (order), the trader's free

margin considering the virtual credit should not be negative (it should exceed or be equal to 0).

2. After executing an order, the margin should not increase. In fact, it means that a trader can close their position or reverse it considering available funds under any circumstances (even when there is not enough margin to place a new order).

The system will allow placing an order if any of the rules is met. If the trader has enough free margin (1), it does not matter if the margin will increase or not (2) the system will allow placing an order.

The orders that have been placed but have not yet been accepted for processing or have not yet been executed (in the "started" state) can only increase the margin but not decrease it. Example:

- The client account has the following position: Sell 0.02 lot USDJPY. The current margin is 6.67 USD. Margin is calculated using the larger leg.
- Case 1. The client places an order in the opposite direction: Buy 0.03 lot USDJPY. As long as the order is in the "started" state, the client's margin is still equal to 6.67 USD. As soon as the order is executed, the margin will be halved, i.e. 3.33 USD, since the uncovered volume will decrease from 0.02 to 0.01 lots.
- Case 2. The client places an order in the same direction: Sell 0.03 lot USDJPY. The client's margin will increase immediately after placing the order, even while the order is in the "started" state. The margin value will become 16.67 USD, since the potential uncovered volume may become 0.05 lots.

This prevents clients from placing orders for a total volume larger than the funds allow.

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Exchange Risk Management Model

Exchange risk management model is the calculation of collateral on positions on the basis of symbol discounts which the broker defines based on data provided by the exchange. This model can be enabled in the "Margin" tab of group settings. In the "Risk management" field select "for Stock Exchange, based on margin discount rates".

Basic Terminology

Assets

Assets – the current value of purchased financial instruments (of long positions) defined in client's deposit currency. The value is determined dynamically based on the price of the latest symbol deal taking into account the liquidity margin rate. In fact, the amount of assets is equivalent to the amount of money that the trader would receive in case of immediate closure of long positions.

$$\text{Assets} = \text{Value1} * L1 + \text{Value2} * L2 + \dots + \text{ValueN} * LN$$

Here:

- Value is the market value of the position. It is determined depending on the financial instrument type.
- L – the liquidity rate of the instrument.

Only liquid instruments, i.e. those with the liquidity rate >0, can be used as collateral.

Liabilities

Liabilities – obligations on current short positions calculated as the value of these positions at the current market price. In fact, the amount of liabilities is equivalent to the amount of money that the client would pay in case of immediate closure of short positions.

$$\text{Liabilities} = \text{Value1} + \text{Value2} + \dots + \text{ValueN}$$

Here Value is the market value of the position. It is determined depending on the financial instrument type.

Balance (own funds)

Balance – the client's own funds on the account.

Equity (portfolio value)

Equity is calculated by the following formula:

$$\text{Equity} = \text{Own Funds} + \text{Assets} - \text{Liabilities} - \text{Commission}$$

Margin

- **Initial margin** is the minimum value of trader's own funds with which the trader is allowed to enter the market.
- **Adjusted initial margin** is the minimum value of own funds with which a trader is allowed to enter the market, including client's current market positions and limit orders .
- **Maintenance margin** is the minimum amount of funds that must be available on the account for maintaining an open position. If the equity level falls below the maintenance margin, the broker starts closing trader's positions. The position closing procedure is determined by the broker's regulations.

Determining the market value of a position

For instruments with the Exchange Stocks calculation type, the market value is determined as follows:

Volume in lots * Contract size * Market price

For Exchange Bonds instruments, the position value is calculated using the following formula:

Volume in lots * Contract size * Face value * Market price / 100 + AI

Here AI is the accrued interest.

The market value is calculated in symbol's base currency. If necessary, the resulting value can be converted to the account deposit currency.

Calculation Features

On the spot market, as opposed to the futures and forward markets (where there is only the movement of collateral), payment and receipt of assets (or liabilities in the event of repurchase) occur immediately at the moment of deal conclusion. Accordingly, the transaction value is immediately reflected on the trader's balance.

Since the payment for the instrument purchase or sale is always made in full, the margin is only used as an indication of the trading account state, which determines the possibility of opening new positions or necessity to close out existing positions.

Margin calculation for Exchange Stocks, Exchange Bonds, Exchange MOEX Stocks and Exchange MOEX Bonds

The margin is the discounted assessment of client's positions:

$$\text{Margin} = \text{Value1} * \text{MarginRate1} + \text{Value2} * \text{MarginRate2} + \dots + \text{ValueN} * \text{MarginRateN}$$

Here:

MetaTrader 5 trading platform

- Value is the market value of the position. It is determined depending on the financial instrument type.
- MarginRate is the margin rate of discount of the instrument, for which the position is open. Individual margin rates can be used for the initial and maintenance margin, as well as for short and long positions.

If margin rates of an instrument are equal to 0, such an instrument is considered to be non-marginable. For long positions, the charged margin is equal the value of such positions; short positions on non-margin instruments are not allowed.

Example of Opening a Long Position

Assume initially the client's balance is 1,000,000 RUR. The initial and maintenance margin rates are equal to 0.1 and 0.05. For simplicity, we do not take into account the commission size.

Trade operations and price fluctuations	Trader's account state
Buying 1000 shares of LKOH 150 RUR each	• Balance: $1,000,000 \text{ RUR} - 1000 * 150 \text{ RUR} = 850,000 \text{ RUR}$ • Assets: $1000 * 150 = 150,000 \text{ RUR}$ • Liabilities: 0 RUR • Equity: $850,000 \text{ RUR} + 150,000 \text{ RUR} = 1,000,000 \text{ RUR}$ • Initial margin: 15,000 RUR • Maintenance margin: 7,500 RUR
Price drop to 50 RUR per share	• Balance: 850,000 RUR • Assets: $1000 * 50 = 50,000 \text{ RUR}$ • Liabilities: 0 RUR • Equity: $850,000 \text{ RUR} + 50,000 \text{ RUR} = 900,000 \text{ RUR}$ • Initial margin: 5,000 RUR • Maintenance margin: 2,500 RUR
Buying 20,000 shares 50 RUR each	• Balance: $850 \text{ 000 RUR} - 20 \text{ 000} * 50 \text{ RUR} = -150 \text{ 000 RUR}$ (uses borrowed money) • Assets: $(1,000 + 20,000) * 50 \text{ RUR} = 1,050,000 \text{ RUR}$ • Liabilities: 0 RUR • Equity: $1,050,000 \text{ RUR} - 150,000 \text{ RUR} = 900,000 \text{ RUR}$ • Initial margin: 105,000 RUR

Trade operations and price fluctuations	Trader's account state
	• Maintenance margin: 52,500 RUR
Price drop to 10 RUR per share	• Balance: -150 000 RUR • Assets: 21,000 * 10 RUR = 210,000 RUR • Liabilities: 0 RUR • Equity: 210,000 RUR - 150,000 RUR = 60,000 RUR • Initial margin: 21,000 RUR • Maintenance margin: 10,500 RUR
Price drop to 7.8 RUR per share	• Balance: -150 000 RUR • Assets: 21,000 * 7.8 RUR = 163,800 RUR • Liabilities: 0 RUR • Equity: 163,800 RUR - 150,000 RUR = 13,800 RUR • Initial margin: 16,380 RUR • Maintenance margin: 8,190 RUR Note: equity below the initial margin. The client cannot open new positions, only close existing ones.
Price drop to 5 RUR per share	• Balance: -150 000 RUR • Assets: 21,000 * 5 RUR = 110,000 RUR • Liabilities: 0 RUR • Equity: 110,000 RUR - 150,000 RUR = -40,000 RUR • Initial margin: 11,000 RUR • Maintenance margin: 5,500 RUR Note: equity below the maintenance margin. Broker forcibly closes the client's position.

Example of Opening a Short Position

Assume initially the client's balance is 1,000,000 RUR. The initial and maintenance margin rates are equal to 0.1 and 0.05. For simplicity, we do not take into account the commission size.

Trade operations and price fluctuations	Trader's account state
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Selling 1000 shares of LKOH 150 RUR each	• Balance: $1,000,000 \text{ RUR} + 1,000 * 150 \text{ RUR} = 1,150,000 \text{ RUR}$ • Assets: 0 RUR • Liabilities: $-1,000 * 150 \text{ RUR} = -150,000 \text{ RUR}$ • Equity: $1,150,000 \text{ RUR} - 150,000 \text{ RUR} = 1,000,000 \text{ RUR}$ • Initial margin: 15,000 RUR • Maintenance margin: 7,500 RUR
Price grows to 300 RUR per share	• Balance: 1,150,000 RUR • Assets: 0 RUR • Liabilities: $-1000 * 300 \text{ RUR} = -300,000 \text{ RUR}$ • Equity: $1,150,000 \text{ RUR} - 300,000 \text{ RUR} = 850,000 \text{ RUR}$ • Initial margin: 30,000 RUR • Maintenance margin: 15,000 RUR
Price grows to 1000 RUR per share	• Balance: 1,150,000 RUR • Assets: 0 RUR • Liabilities: $-1000 * 1000 \text{ RUR} = -1,000,000 \text{ RUR}$ • Equity: $1,150,000 \text{ RUR} - 1,000,000 \text{ RUR} = 150,000 \text{ RUR}$ • Initial margin: 100,000 RUR • Maintenance margin: 50,000 RUR
Price grows to 1100 RUR per share	• Balance: 1,150,000 RUR • Assets: 0 RUR • Liabilities: $-1000 * 1100 \text{ RUR} = -1,100,000 \text{ RUR}$ • Equity: $1,150,000 \text{ RUR} - 1,100,000 \text{ RUR} = 50,000 \text{ RUR}$ • Initial margin: 110,000 RUR • Maintenance margin: 55,000 RUR Note: equity below the initial margin. The client

Short side calculation:

$$\text{Margin Sell} = -\text{PositionSize} * (\text{PriceMax} - \text{PriceMarket}) - (\text{PositionSize} - \text{OrdersSellSize}) * \text{PriceMarket}$$

Here:

- **PositionSize** position size calculated as the product of the volume in lots and the contract size.

Trade operations and price fluctuations	Prader's account state
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- **PriceMax** maximum price among all current sell limit orders of the client.
- **OrdersSellSize** the size of the client's sell limit orders calculated as the product of the total volume of orders in lots and the contract size.
- **OrdersSellValue** the value of the sell limit orders if they were executed at the prices specified in them. It is calculated as the sum of the products of order sizes and their limit price.
- **MarginRate** the amount of the symbol discount.

If the client's current position is long, and its size is greater than or equal to **OrdersSellSize**, **Margin Buy** is not calculated and is considered to be 0. In fact, this is a situation where even after the execution of all limit **Sell** orders, the client will have a long position or the position will be completely eliminated.

Consider the following example. The client has:

- **Position Buy** 1 lot LKOH, contract size is 1000 shares, the current price is 100 RUR, initial margin rate is 0.1
- **Order Buy Limit** 0.5 lot LKOH (500 shares), order price is 80 RUR
- **Order Buy Limit** 0.3 lot LKOH (300 shares), order price is 60 RUR
- **Order Buy Limit** 0.1 lot LKOH (100 shares), order price is 40 RUR

Calculations:

PriceMin = 40
Price Market = 100
OrdersBuySize = 500 + 300 + 100 = 900
OrdersBuyValue = 500 * 80 + 300 * 60 + 100 * 40 = 62 000
Margin Buy = 1000 * (100 - 40) + (1000 + 900) * 40 * 0.1 + (62 000 - 900 * 40) = 87 900

The total amount of the adjusted initial margin is equal to 87,900.

Adjusted initial margin calculation for Exchange MOEX Stocks and Exchange MOEX Bonds

The Exchange MOEX Stocks and Exchange MOEX Bonds calculation modes are only used for the financial instruments which are traded on Moscow Exchange. The adjusted margin calculation rules applied in these modes are valid on Moscow Exchange since 01.07.2019.

If a client has limit orders, then the below formulas are used for calculating the initial margin when opening a position.

The adjusted margin is always calculated on the larger side, i.e. the aggregate amount of Buy or Sell positions and orders.

$$\text{Corrected Margin} = \text{Max}(\text{Margin Buy}; \text{Margin Sell})$$

Long side calculation:

$$\text{Margin Buy} = (\text{PositionSize} + \text{OrdersBuySize}) * \text{PriceMarket} * \text{MarginRate}$$

Here:

- **PositionSize** position size calculated as the product of the volume in lots and the contract size.
- **PriceMarket** the current market price of the financial instrument (last deal price).
- **OrdersBuySize** the size of the client's buy limit orders calculated as the product of the total volume of orders in lots and the contract size.
- **MarginRate** the amount of the symbol discount.

If the current position of a client is short and its size is greater than or equal to **OrdersBuySize**, then Margin Buy is not calculated and is considered 0. This is because even if all the client's limit buy orders were executed, a short position would still remain after that or the position would be completely eliminated.

Short side calculation:

$$\text{Margin Sell} = (-\text{PositionSize} + \text{OrdersSellSize}) * \text{PriceMarket} * \text{MarginRate}$$

Here:

- **PositionSize** position size calculated as the product of the volume in lots and the contract size.
- **PriceMarket** the current market price of the financial instrument (last deal price).
- **OrdersSellSize** the size of the client's sell limit orders calculated as the product of the total volume of orders in lots and the contract size.
- **MarginRate** the amount of the symbol discount.

If the current position of a client is long and its size is greater than or equal to **OrdersSellSize**, then Margin Sell is not calculated and is considered 0. This is because even if all the client's limit sell orders were executed, a long position would still remain after that or the position would be completely eliminated.

Consider the following example. The client has:

- **Position Buy** 1 lot LKOH, contract size is 1000 shares, the current price is 100 RUR, initial margin rate is 0.1
- **Order Buy Limit** 0.5 lot LKOH (500 shares), order price is 80 RUR
- **Order Buy Limit** 0.3 lot LKOH (300 shares), order price is 60 RUR
- **Order Buy Limit** 0.1 lot LKOH (100 shares), order price is 40 RUR

Calculations:

$$\begin{aligned}\text{Price Market} &= 100 \\ \text{OrdersBuySize} &= 500 + 300 + 100 = 900 \\ \text{Margin Buy} &= (1000 + 900) * 100 * 0.1 = 19,000\end{aligned}$$

The total amount of the adjusted initial margin is equal to 19,000.

Adjusted margin calculation for non-marginal instruments

A financial instrument is considered non-marginal if zero margins are specified for this instrument. Thus, no margin is charged for the positions of such instruments, while the trader pays full collateral when opening such positions. Accordingly, the calculation of initial margin when a position for a non-marginal instrument is opened is actually involves the check of whether the trader has enough money to buy the asset (short positions for non-marginal instruments are usually prohibited).

Margin Buy = OrderBuySize * PriceAsk
Margin Sell = (- PositionSize + OrderSellSize) * PriceMarket

Here:

- PositionSize position size calculated as the product of the volume in lots and the contract size.
- OrdersBuySize the size of the client's buy limit orders calculated as the product of the total volume of orders in lots and the contract size.
- OrdersSellSize the size of the client's sell limit orders calculated as the product of the total volume of orders in lots and the contract size.
- PriceMarket the current market price of the financial instrument (last deal price).
- PriceAsk the current Ask price of the instrument.

Note that the current Ask (not Last) price is used in the long position margin calculation formula. Non-marginal instruments are usually low-liquid. Accordingly, deals with such instruments are rare and thus the last deal price may differ significantly from the current market price. Therefore, the current Ask price is used for the correct estimation of the asset value.

This calculation is used for non-marginal instruments with type Exchange Stocks, Exchange Bonds, Exchange MOEX Stocks and Exchange MOEX Bonds.

Margin calculation for other types of instruments

For all financial instruments except Exchange Stocks and Exchange Bonds, margin is calculated similar to Retail Forex, CFD, Futures for netting. However, this calculated margin value is not recorded in a separate field of the account state, but increases the client's liabilities. The margin amount is actually blocked in the trader's account as collateral. Floating profit from positions is included in the account equity (and thus it affects the free margin amount).

Unlike the Retail mode, the exchange system does not take into account whether positions are in spread.

For example, there is an open position: Buy EURUSD 1.00 lot at 1.16135. Forex calculation type is used for the instrument, and its contract size is 100,000. The account deposit is USD, the leverage is 1:200.

- Calculate the margin using the Forex formula: $1 * 100\,000 / 200 = 500$ EUR

MetaTrader 5 trading platform

- Convert the margin from EUR to USD at the price as of the deal time: $500 * 1.16135 = 580.68 \text{ USD}$

The resulting value is not recorded to the Margin field, but is written to the Trading Account Liabilities field instead.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Trade / Profit Calculation

Profit Calculation

Profit calculations differ for different types of trade instruments. The final amount of profit is calculated in two stages:

- The main calculation for a symbol;
- Conversion of a profit currency to the deposit currency.

The main calculation for a symbol

The trading platform provides five types of profit calculation, which depend on the financial instrument. The calculation type is selected in the "Calculation" of the "Trade" tab. When using calculation formulas, please consider the following features:

- Mathematical rounding is used when calculating the profit. The calculated value is rounded to the number of decimal places in the price of the financial instrument. In the formulas given below the rounding is denoted as Normalize() function.
- On netting accounts, the position opening price is equal to the weighted average price.

Forex

Profit of Buy trades (execution of operations opposite to them) for Forex symbols is calculated by the following formula:

$\text{Normalize}(\text{Close Price} * \text{Contract Size} * \text{Volume in Lots}) - \text{Normalize}(\text{Open Price} * \text{Contract Size} * \text{Volume in Lots})$

For Sell trades, open and close price change over in the formula:

$\text{Normalize}(\text{Open Price} * \text{Contract Size} * \text{Volume in Lots}) - \text{Normalize}(\text{Close Price} * \text{Contract Size} * \text{Volume in Lots})$

Let's consider closing of 1 EURUSD lot Buy position. The position was opened at the price of 1.2000; by the moment of closing the price is equal to 1.2050; the contract size is 100,000. Substitute the values to the formula:

$$(1.2050 * 100\,000 * 1) - (1.2000 * 100\,000 * 1) = 500 \text{ USD}$$

As a result we obtain the profit size in the symbol profit currency.

CFD, CFD Index, CFD Leverage, Exchange Stocks, Exchange MOEX Stocks

Profit of Buy trades (execution of operations opposite to them) for CFD and Stocks symbols is calculated by the following formula:

MetaTrader 5 trading platform

$\text{Normalize}((\text{Close Price} - \text{Open Price}) * \text{Contract Size} * \text{Volume in Lots})$

For Sell trades, open and close price change over in the formula:

$\text{Normalize}((\text{Open Price} - \text{Close Price}) * \text{Contract Size} * \text{Volume in Lots})$

Let's consider closing of 1 EURUSD lot Buy position. The position was opened at the price of 1.2000; by the moment of closing the price is equal to 1.2050; the contract size is 100,000. Substitute the values to the formula:

$$(1.2050 - 1.2000) * 100,000 * 1 = 500 \text{ USD}$$

As a result we obtain the profit size in the symbol profit currency.

Futures, Exchange Futures, FORTS Futures, Exchange Options, Exchange Margin Options

Profit on Futures Buy trades is calculated by the following formula:

$\text{Normalize}((\text{Close Price} - \text{Open Price}) * \text{Volume in Lots} * \text{Tick Value} / \text{Tick Size})$

For Sell trades, open and close price change over in the formula:

$\text{Normalize}((\text{Open Price} - \text{Close Price}) * \text{Volume in Lots} * \text{Tick Value} / \text{Tick Size})$

Calculation of the profit on Futures trades also takes into account the ratio of the value and size of one tick.

Exchange Bonds, Exchange MOEX Bonds

The profit is calculated as the difference between the prices of a position at the time of opening and closing. Position price is defined as follows:

$\text{Price}/100 * \text{Face value} * \text{Volume in lots} * \text{Contract size} + \text{Accrued interest} * \text{Volume in lots} * \text{Contract size}$

The price is divided by 100 since bond prices are passed as a face value percentage.

The current floating profit at bonds is calculated without an accrued interest:

$(\text{Current Price} - \text{Open Price})/100 * \text{Face value} * \text{Volume in lots} * \text{Contract size}$

- Accrued interest history is not stored in the platform, it is accounted by the exchange when processing deals.
- The position value is calculated in the symbol's base currency (not in the profit currency).

Collateral

Non-tradable instruments of this type are used as client's assets to provide the required margin for open positions of other instruments. For these instruments the profit is not calculated.

Conversion to the Deposit Currency

The stage of the calculated profit conversion appears if the profit currency differs from the deposit currency of the account.

For details, please visit the Conversion section.

All Types of Symbols Except Forex

The conversion process slightly differs for profitable and losing trades.

- **Profitable deals** the conversion is performed at the current rate of the profit currency to the deposit currency. The Bid price is taken for calculations, because as a result of a profitable deal, a trader obtains a certain amount of the profit currency and needs to sell it for the deposit currency.
- **Losing deals** The conversion is also performed at the current rate of the profit currency to the deposit currency. However, in this case the Ask price is taken, because as a result of a losing deal, a trader needs to buy a certain amount of currency for the deposit currency.

Forex

For the Forex symbol, the conversion type can be selected using the "Convert profit" option on the "Trade" tab.

By deal

This is a default mode for Forex symbols. To convert a profit when closing a position, the system uses the prices of the XXXYYY currency pair, where XXX is a trade profit currency, while YYY is a client deposit currency. The Ask price is used when closing buy positions and the Bid price is used when closing sell positions. It does not matter whether the position that is being closed is profitable or not. If possible, the price, at which the market exit has been performed, is used during the conversion (provided that the appropriate currency pair takes part in the conversion).

For example, a customer with a USD deposit currency closed the buy USDCHF position. In this case, the platform should apply the Ask price of the CHFUSD pair. Since there is only a reverse rate (USDCHF), the platform calculates the desired price the following way: $\text{Ask CHFUSD} = 1/\text{Bid USDCHF}$. Here, the price, at which the position was actually closed, is used as Bid USDCHF.

If there is no direct exchange rate between profit and deposit currencies, the conversion is performed via USD. For example, a customer with a TRY deposit currency performs a trade on USDRUR. Since there is no TRYRUR rate, the profit is converted via USDTRY.

By market

In this mode, the conversion is performed using the current Bid/Ask price depending of the profitability/unprofitability of a deal. The Bid price is taken for calculations for profitable deals, because as a result of a profitable deal, a trader obtains a certain amount of the profit currency and needs to sell it for the deposit currency. The Ask price is taken for losing deals, because as a result of a losing deal, a trader needs to buy a certain amount of currency for the deposit currency.

Conversion features for Exchange-based settlement groups

In the exchange model, the payment and receipt of assets (or the incurrence of liabilities in case of repurchase operations) occur immediately upon conclusion of the deal. The deal cost is immediately reflected in the client's balance. Accordingly, the profit currency is immediately converted into the deposit currency, both when entering and exiting a position.

Floating profits displayed in client terminals is calculated based on this feature. Let's consider the an example:

1. Suppose you have a euro account (EUR). You buy shares that have US dollars (USD) as the profit currency.
2. Upon completing a trade, the cost of the shares is immediately debited from your balance, and the shares become your asset.
3. Since you do not have USD, you need to buy them with EUR to complete the trade. Therefore, the Ask price of the EURUSD pair at the time of opening the position is used for conversion.
4. When closing a position, you will sell the shares for USD. To credit the funds to your balance, the cost of the shares must be converted into euros. This conversion will use the Bid price of the EURUSD pair at the time of closing the position.
5. Thus, when calculating floating profit in the exchange mode, the profit currency is converted into the deposit currency at different rates at existing different points in time.

Normalization

For all instruments except Forex, the total profit is mathematically rounded by the number of decimal places in a symbol profit currency. For Forex type symbols, a position price during closing and opening is normalized separately:

Buy trade profit = Normalize(Close price * Volume * Contract size, Digits)
Normalize(Open price * Volume * Contract size, Digits).

Sell trade profit = Normalize(Open price * Volume * Contract size, Digits)
Normalize(Close price * Volume * Contract size, Digits).

Here, Normalize is a mathematical rounding function. Its second argument (Digits) is the number of rounded decimal places, and it is equal to the number of digits after the decimal point in a symbol profit currency.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Symbols](#) / [Symbol Settings](#) / [Trade](#) / Conversion

Conversion

The calculated value of the margin or profit (loss) is always converted to the account deposit currency. Conversion is based on the following principles:

Forex and Forex No Leverage instruments	Other instruments
• Profit/loss conversion rate depends on whether the position is profitable or not. • During the profit conversion, a trader sells the profit currency for the deposit currency (at Bid price). During the loss conversion, a trader should buy the loss currency for the deposit currency (at Ask price). The worst price is always used during the conversion. • To perform conversion, the system searches for a Forex symbol (symbol with the <u>Forex/Forex No leverage calculation type</u>) with the base and quoted currency coinciding with the profit (margin) and deposit currency. For example, USDCHF symbol is used to convert USD to CHF. • If there is no such a symbol, the attempt is made to convert via USD. Suppose that the profit currency is XYZ, while the deposit one is ABC. If the server has no XYZABC or ABCXYZ symbol, two symbols (XYZUSD and ABCUSD) are used for conversion. Thus, the conversion is performed in two stages. The worst rate for a client is used at each stage (Bid or Ask price). • If a postfix is used in the name of the trading symbol participating in the deal (for example, EURUSDx), the server will try to use currency pairs with the same postfix during	• Profit/loss conversion rate depends on whether the position is profitable or not. • During the profit conversion, a trader sells the profit currency for the deposit currency (at Bid price). During the loss conversion, a trader should buy the loss currency for the deposit currency (at Ask price). The worst price is always used during the conversion. • To perform conversion, the system searches for a Forex symbol (symbol with the <u>Forex/Forex No leverage calculation type</u>) with the base and quoted currency coinciding with the profit (margin) and deposit currency. For example, USDCHF symbol is used to convert USD to CHF. The following search orders is applied when selecting an appropriate symbol: 1. A symbol is searched among all the symbols available to a client group in accordance with the <u>group settings</u>. The presence of a postfix in the instrument name is ignored. For example, to convert USD to CHF the system can use either USDCHF or USDCHFx. 2. If several symbols suitable for conversion are available to a client group, the one, which is earlier alphabetically, will be used. For example, among "GBPUSD.xyz, GBPUSD.abc, GBPUSD" GBPUSD will be selected; GBPUSD.abc will

Forex and Forex No Leverage instruments	Other instruments
direct conversion and conversion via USD. 1. The server tries to find the conversion rate with the postfix matching the postfix of the original trading instrument. Symbol visibility for the client group is taken into account. If the instrument is found and its prices are available, this instrument will be used. For example, to convert GBPCHF.xyz profit to USD, the server will try to use GBPUSD.xyz and USDCHF.xyz. 2. Otherwise, the server will search for currency pairs among those available to the client group in accordance with its <u>settings</u>. The presence of a postfix in the instrument name is ignored. For example, to convert USD to CHF the system can use either USDCHF or USDCHFx. 3. If several symbols suitable for conversion are available to a client group, the one, which is earlier alphabetically, will be used. For example, among "GBPUSD.xyz, GBPUSD.abc, GBPUSD" GBPUSD will be selected; GBPUSD.abc will be selected among "GBPUSD.xyz, GBPUSD.abc". 4. If no suitable instrument is found among the symbols available to the client group, the system will continue search among all the symbols in the platform. If several suitable symbols are found, the alphabetically earlier symbol will be used. Please note that in this case conversion will only be performed on the platform side, while in terminal the floating profit and margin will have zero values. The appropriate values are calculated by terminals using	be selected among "GBPUSD.xyz, GBPUSD.abc". 3. If no suitable instrument is found among the symbols available to the client group, the system will continue search among all the symbols in the platform. If several suitable symbols are found, the alphabetically earlier symbol will be used. Please note that in this case conversion will only be performed on the platform side, while in terminal the floating profit and margin will have zero values. The appropriate values are calculated by terminals using incoming quotes. Accordingly, if the required instruments are not available to the terminals, they will not be able to calculate profit and margin. • If there is no such a symbol, the attempt is made to convert via USD. Suppose that the profit currency is XYZ, while the deposit one is ABC. If the server has no XYZABC or ABCXYZ symbol, two symbols (XYZUSD and ABCUSD) are used for conversion. Thus, the conversion is performed in two stages. The worst rate for a client is used at each stage (Bid or Ask price).

Forex and Forex No Leverage instruments	Other instruments
incoming quotes. Accordingly, if the required instruments are not available to the terminals, they will not be able to calculate profit and margin.	

The above principles are applied both when converting the profit and when converting the margin. The features of each case are described below:

- Profit conversion: select the profit calculation mode for Forex type symbols: by deal or by market.
- Margin conversion: if the deal's currency pair is used to convert the margin currency to the deposit one, the deal execution price is always used to calculate the margin. For example, in case of EURUSD deal and USD deposit currency, conversion is conducted at EURUSD price, at which the deal is performed.

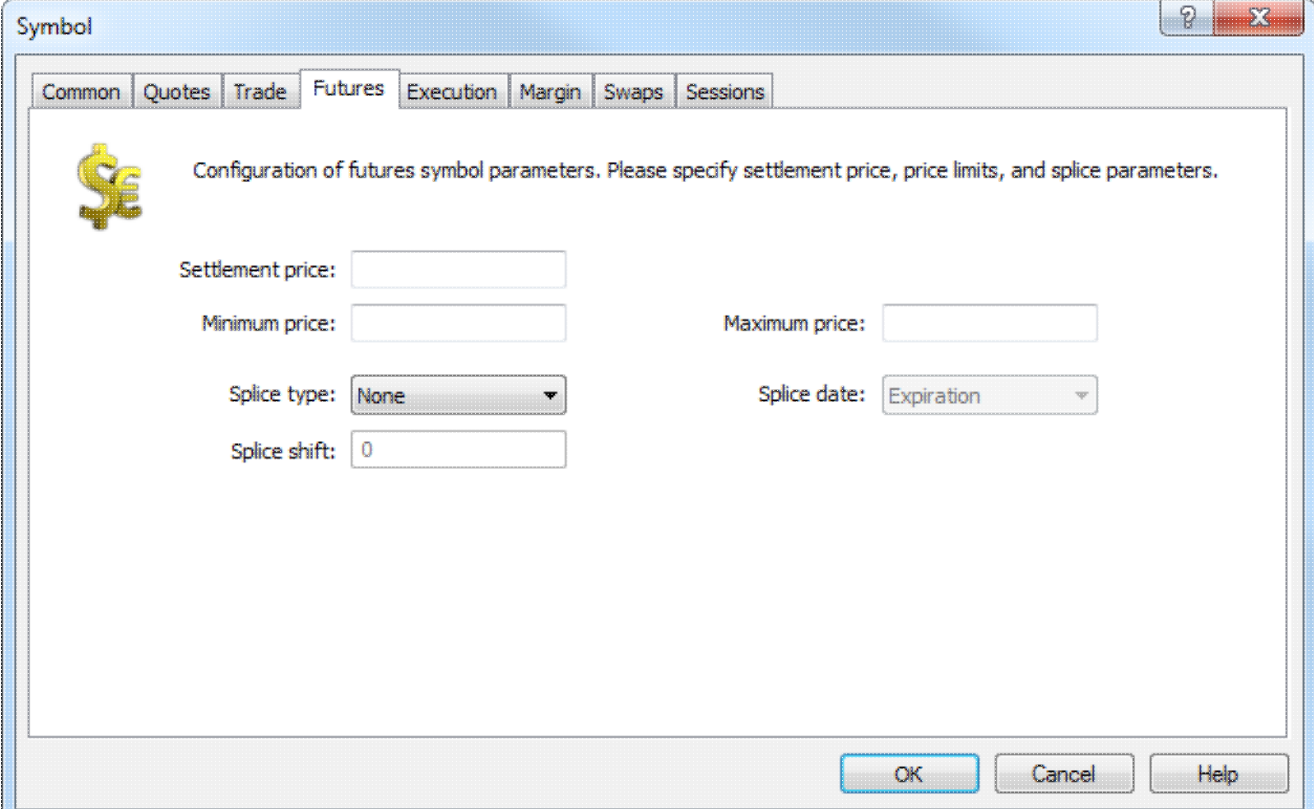
- The floating profit and margin values on the terminal side are calculated using incoming quotes. To ensure the correct display of these variable, the currency pairs that are used for the conversion should be available to the group of the account, for which the calculation is being performed.
- Additionally, to be able to view profit on a client account via a Manager or Administrator terminal, the manager account must also have access to the currency pairs used for conversion (they must be available for the manager group).

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Futures

Futures

This tab contains the settings specific to futures contracts.

It appears only if Exchange Futures or FORTS Futures symbol calculation mode is enabled.



The screenshot shows the 'Symbol' dialog box with the 'Futures' tab selected. The dialog has a title bar with a question mark and a close button. Inside, there are tabs for 'Common', 'Quotes', 'Trade', 'Futures', 'Execution', 'Margin', 'Swaps', and 'Sessions'. The 'Futures' tab is active, displaying a configuration area for futures symbol parameters. A yellow dollar sign icon is on the left. The text reads: 'Configuration of futures symbol parameters. Please specify settlement price, price limits, and splice parameters.' Below this, there are several input fields: 'Settlement price:', 'Minimum price:', 'Maximum price:', 'Splice type:' (a dropdown menu currently showing 'None'), 'Splice date:' (a dropdown menu currently showing 'Expiration'), and 'Splice shift:' (a text box containing '0'). At the bottom right, there are three buttons: 'OK', 'Cancel', and 'Help'.

Symbol

Common Quotes Trade **Futures** Execution Margin Swaps Sessions

 Configuration of futures symbol parameters. Please specify settlement price, price limits, and splice parameters.

Settlement price:

Minimum price: Maximum price:

Splice type: Splice date:

Splice shift:

OK Cancel Help

Symbol price settings can be specified in the upper part:

- **Settlement price** estimated (clearing) price of a symbol's previous trading session;
- **Minimum price** minimum symbol price in the current trading session;
- **Maximum price** maximum symbol price in the current trading session.

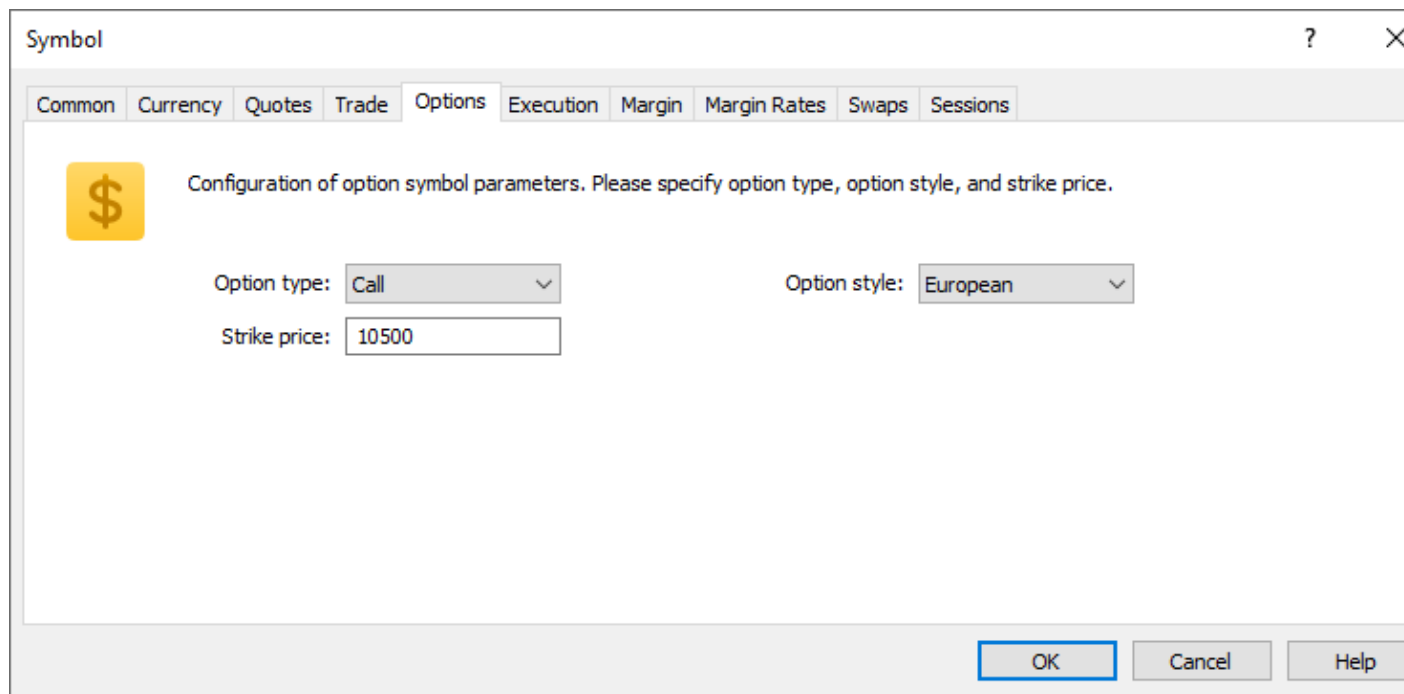
The settings described above are followed by futures contracts splicing ones. This process is described in details in the [separate section](#).

- **Splice type** Unadjusted mode allows splicing futures quotes "as is" without converting a previous contract's price level to the current (front) contract's prices. In Adjusted mode, the difference between the previous contract's last quote and the front contract's first one is calculated. Then all previous futures' quotes are changed by that value. In that case, the obtained spliced chart is smooth and has no sharp variations on the borders. For ordinary symbols, set None.
- **Splice date** only Extension option is currently provided. The splicing is performed at the expiration of the symbol specified in To field of [Sessions](#) tab.
- **Splice shift** splice date shift can be set here. The shift is set in the number of days to the past from the symbol's expiration date.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Symbols](#) / [Symbol Settings](#) / Options

Options

This tab contains the settings specific to options. It appears only if Exchange Options calculation mode is enabled for the symbol.



Symbol

Common Currency Quotes Trade Options Execution Margin Margin Rates Swaps Sessions

Configuration of option symbol parameters. Please specify option type, option style, and strike price.

Option type: Call

Option style: European

Strike price: 10500

OK Cancel Help

The tab contains the following settings:

- **Option type** there are two main option types: Call option gives its holder the right to buy the underlying asset by a certain date for a certain price, while Put one gives the right to sell it.
- **Option style** options can be American and European. American option can be exercised at any time up to the expiration date. European option can be exercised only on the expiration date itself.
- **Strike price** price, at which an option gives the right to buy or sell an asset.

Options Board

The client terminal features options trading functionality, which includes the Options Board. This tool becomes available under the following conditions:

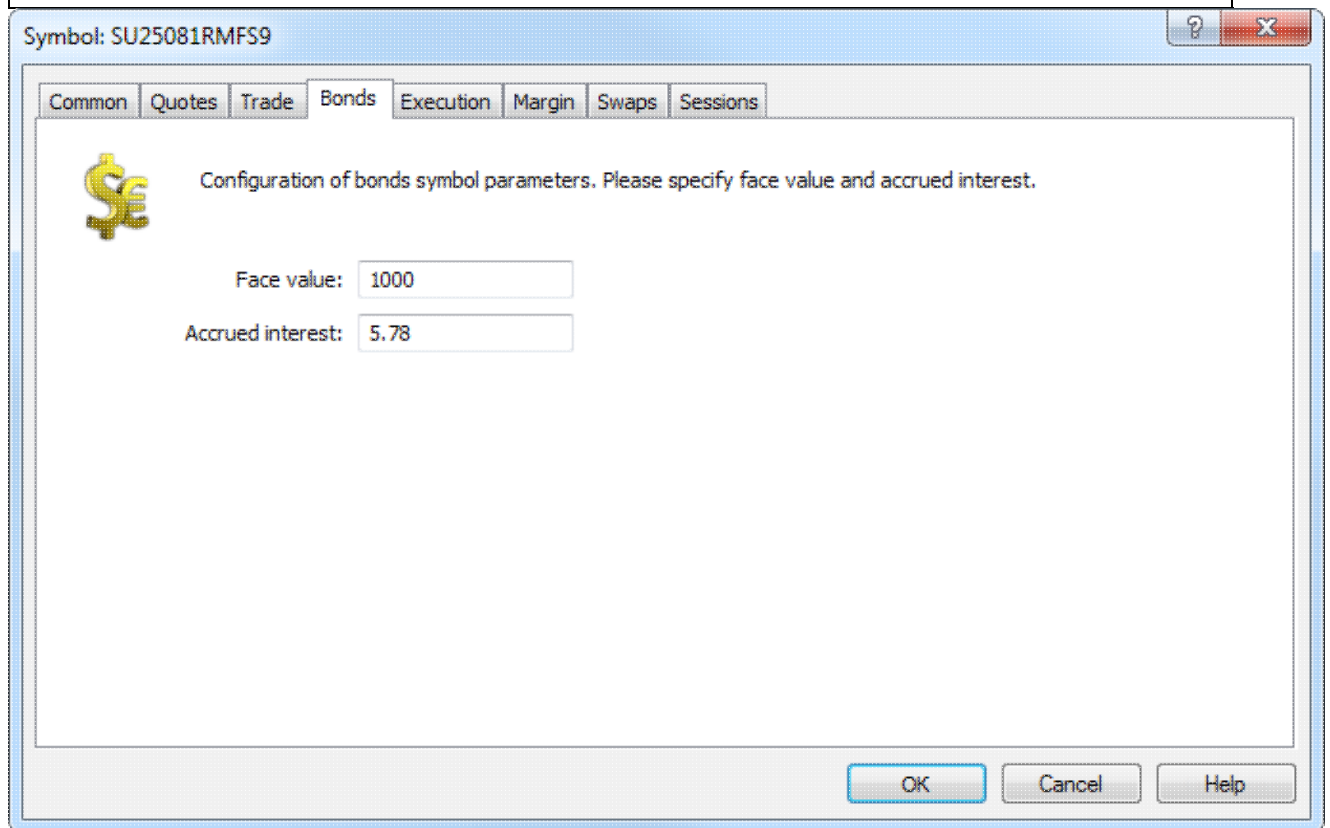
- Exchange Options calculation mode is selected in symbol settings.
- The underlying asset symbol (basis) existing on the server is specified in symbol settings.
- The option has not expired, the date in the "To" field is later than the current date.
- The symbol of the options and of its underlying asset are available to the group, in which the client account is located.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Bonds

Bonds

This tab contains the settings specific to bonds.

This tab appears only if Exchange Bonds symbol calculation mode is enabled.



The screenshot shows the 'Symbol Settings' dialog box for the symbol 'SU25081RMFS9'. The 'Bonds' tab is selected, and the window displays the configuration for bond parameters. The 'Face value' is set to 1000 and the 'Accrued interest' is set to 5.78. The dialog includes a title bar with the symbol name, a tabbed interface with options like Common, Quotes, Trade, Bonds, Execution, Margin, Swaps, and Sessions, and a main area with a dollar sign icon and instructional text. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons.

Symbol: SU25081RMFS9

Common Quotes Trade **Bonds** Execution Margin Swaps Sessions

 Configuration of bonds symbol parameters. Please specify face value and accrued interest.

Face value:

Accrued interest:

OK Cancel Help

The tab contains the following settings:

- **Face value** initial value set by an issuer.
- **Accrued interest** accumulated coupon interest (part of the coupon interest calculated in proportion to the number of days since the coupon bond issuance or the last coupon interest payment).

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MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Execution

Execution

Symbol: EURUSD ? X

Common Quotes Trade Execution Margin Margin Rates Swaps Sessions

 The setting up of execution of orders by the symbol. Please specify the execution type and its parameters.

Execution: Instant ▾

Max time deviation: 7 seconds

Max profit deviation: 2 points

Max losing deviation: 2 points

☐ Fast confirmation of requotes within client deviation

Maximum volume: 0.00 before switching to Request execution

OK Cancel Help

The symbol order execution mode is set up on this tab:

Instant

In this mode a client places an order at prices from the Market Watch window without requesting prices before that. The following settings are available for this mode:

- **Max time deviation** the maximal difference (in seconds) between the time when the price specified in the client's order was received, and the time of the latest price. If this limit is exceeded, requote is performed and new prices are sent to the client;
- **Max profit deviation** maximal deviation of the order price from the current symbol price in the direction profitable for a client. This value being exceeded, new prices will be sent to the client;
- **Max losing deviation** maximal deviation of the order price from the current symbol price in the direction losing for a client. This value being exceeded, new prices will be sent to the client;
- **Maximum volume** the maximum volume of the order that can be accepted in the instant execution mode. All clients' orders larger than that will be automatically switched to the request execution mode.
- **Fast confirmation of requotes within client deviation** the option is disabled by default. When creating a trade request in the instant execution mode, the client can specify tolerance, i.e. by how many points the request execution price can differ from the price specified in the order. If a dealer returns a requote in response to such a request, and the requote price is within the tolerable deviation specified by the client, the order will be returned to the dealer for confirmation after the client confirms this new price. If the option is enabled, the order will be executed immediately after the client confirms the price. No additional confirmation from a dealer is required.

The following options are used only if the deal volume exceeds the maximum volume that can be accepted in the instant execution mode. Such deals are processed in the "Request" execution mode.

- **Timeout** time in seconds, during which the price provided by the dealer is valid;
- **Confirm orders** if this option is enabled, then after a trader receives quotes from a dealer and agrees to execute a deal at them, the dealer will have to additionally confirm the execution of this order.

Checking prices and requotes

If the price in the trade request exceeds the maximum deviation in the profit or loss direction, the server returns a requote to the client and prints the following message to the Journal:

```
'7000009': requote 1.15309 / 1.15329 (instant buy 0.01 EURUSD at 1.15328)[slippage]
```

If the requote price provided by the dealer exceeds the maximum deviation specified by the trader in the trade request (order), another requote is sent to the trader. the following message is printed to the journal:

```
'7000009': requote 1.15313 / 1.15333 (instant buy 0.01 EURUSD at 1.15331)[dealer deviation]
```

Apart from checking the price value and time deviation according to symbol settings, the server performs additional checks.

Checking for presence in the flow. The price used by a client to set an order, should be present in the quote flow. In other words, a deal cannot be performed at a non-existing price. The price search depth is defined by the "Max time deviation" parameter. If there is no price, the server sends a requote to a client and the appropriate message (deviation is blocked with 'count' on 'request') appears in the journal. Here 'count' means the counter of invalid client requests, while 'request' is a request description. In each case of that type, the invalid client request counter is increased.

Checking for obsolescence. The platform checks if the prices used by a client to place an order are obsolete. Depending on the market rate, the server calculates the range of the last ticks, within which the requested price is located. On the fast market, the acceptable range is expanded, while on the slow market, it is narrowed. If the price does not fall within the acceptable range, the server sends the client a requote and leaves the following message in the journal: previous price 'x' on 'request.' Here, 'x' is a location of the requested price in the quote flow (0 means the last price), while 'request' is a request description. In each case of that type, the invalid client request counter is increased.

Invalid request counter. The server features the invalid request counter for each client. As soon as the counter exceeds the threshold value, the server starts rejecting subsequent invalid client requests leaving the 'off quotes' message. The appropriate message (deviation is blocked with 'count' on 'request') appears in the journal. Here, 'count' means the counter of invalid client requests, while 'request' is the last request description. When a valid client request is received, the counter is decreased.

Request

In this mode, a client requests the filling price before placing an order. The following options are available here:

- **Timeout** time in seconds, during which the price provided by the dealer is valid;
- **Confirm orders** if this option is enabled, then after a trader receives quotes from a dealer and agrees to execute a deal at them, the dealer will have to additionally confirm the execution of this order.

Market

In this mode, a client places an order at the broker's price, "agreeing with it in advance".

Exchange

In this execution mode a trader's request is passed to an external gateway connected to an exchange or another liquidity provider. Operation specifics:

MetaTrader 5 trading platform

- Different order execution policies can be used
- The Limit & Stop Level parameter does not work

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Margin

Margin

Symbol: Forex\EURUSD

Common Currency Quotes Trade Execution **Margin** Margin Rates Swaps Sessions

 The setting up of margin requirements for the symbol. Please specify the initial and maintenance margin and multipliers for different types of trade operations.

Initial margin:

Hedged margin:

Maintenance margin:

☐ Calculate hedged margin using larger leg

☐ Exclude long position PnL from free margin and margin level

☐ Recalculate margin exchange rate at the End of Day

Additional margin checks:

OK Cancel Help

On this tab you can set up parameters of margin requirements of a symbol:

- **Initial margin** amount of money (margin) deposited for futures contracts to ensure performance of the contract to execute a deal with the volume of 1 lot. Value is specified in the margin currency. If the initial margin value is specified for a symbol, then this exact value is used. Formulas of margin calculation for the corresponding trade type are not used.
- **Maintenance margin** the minimal amount of money (margin) that must be available on a client's account to support a position with the volume of 1 lot. Value is specified in the margin currency.
- **Hedged margin** the amount of margin charged per each hedged lot of a position. If initial margin is set for an instrument, the hedged margin is specified as an absolute value (in margin currency) terms. If the initial margin is not set (equal to 0), the contract size is specified. The margin is calculated by the appropriate formula corresponding to the type of the instrument and using the specified contract size. For more details please see the special section. To enable correct risk management, the hedged margin amount must be greater than 0. Otherwise, traders will have almost no restrictions on increasing oppositely directed positions (if the same volume is used for the two directions). It is recommended to use a value equal to the contract size or the initial or maintenance margin (depending on how the hedging margin is set).
- **Calculate hedged margin using larger leg** determines the hedged margin calculation mode: basic or using larger position leg.
- **Exclude long position PnL from free margin and margin level** the option affects the display of positions purchased with own funds, on accounts with margin trading enabled. If the option is enabled, profit/loss of positions for the symbol will be ignored when calculating free margin, margin level and account equity. Also, Stop Out will not be applies to such positions.
- **Recalculate margin exchange rate at the End of Day** if this option is enabled, at the end of the trading day, the server will update the position margin conversion rate at current market prices. For further details, please see Basic Margin Calculation.
- **Additional margin check** on default, the margin is checked when any order is placed and when a pending order triggers. This parameter allows enabling additional margin checks:
- **check before executing orders** in this mode, another check of margin is added to those described above: the margin is checked before executing an order after it is confirmed (checked) by the server (at automated execution), by a dealer or by a gateway.
- **check on SL-TP trigger** enables an additional check of margin before a position is closed by stop loss or take profit. If the position close results in reducing the margin level, and it becomes insufficient to maintain open positions and orders, the stop loss/take profit will not trigger, the position will stay open. This check must be enabled in case the trade operations are transmitted to an external system (exchange).

- If the initial margin is specified, then no calculations by formulas specified in the "Calculation" field are performed.

- If the maintenance margin value is not specified, the value of the initial margin is used for it.

The margin calculation is described in details in a separate section.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Margin Rates

Margin Rates

Symbol: EURUSD
?
X

Common
Quotes
Trade
Execution
Margin
Margin Rates
Swaps
Sessions


Please specify margin rates for different types of trade operations.

Liquidity margin rate: 1.000

Currency margin rate: 1.00000000

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
Buy	1.00000000	0.00000000	0.00000000	0.00000000
Sell	1.00000000	0.00000000	0.00000000	0.00000000
Maintenance margin				
Buy	0.00000000	0.00000000	0.00000000	0.00000000
Sell	0.00000000	0.00000000	0.00000000	0.00000000

OK
Cancel
Help

This section contains margin rates for various order types. The rates are set for the initial and maintenance margin individually. If there is no rate for the maintenance margin (equal to zero), the initial margin value is used instead.

- **Market buy order** multiplier for calculating margin requirements for long positions relative to the main amount of margin;
- **Market sell order** multiplier for calculating margin requirements for short positions relative to the main amount of margin;
- **Buy limit order** multiplier for calculating margin requirements for Buy Limit orders relative to the main amount of margin;
- **Sell limit order** multiplier for calculating margin requirements for Sell Limit orders relative to the main amount of margin;
- **Buy stop order** multiplier for calculating margin requirements for Buy Stop orders relative to the main amount of margin;
- **Sell stop order** multiplier for calculating margin requirements for Sell Stop orders relative to the main amount of margin;
- **Buy stop limit order** multiplier for calculating margin requirements for Buy Stop Limit orders relative to the main amount of margin;
- **Sell stop limit order** multiplier for calculating margin requirements for Sell Stop Limit orders relative to the main amount of margin.

Liquidity margin rate in the exchange risk management model, clients can use their own assets as collateral for open positions. The liquidity margin rate determines the amount of the current value of an asset for the specified financial instrument, which will be taken into account as collateral (accounted for in client's equity). If the value is set to 0, the instrument cannot be used as collateral.

Currency margin rate rate change radius of the currency, a futures contract is denominated in, relative to the Russian ruble. The parameter is used when calculating security deposit for futures contracts (Exchange FORTS Futures) traded on Moscow Exchange (calculating variation margin and security deposit using the current USD exchange rate - in Russian). The values are sent by the Moscow Exchange when using the gateway MetaTrader 5 to MOEX Derivatives.

If the exchange risk management model is used for the client group, and a symbol has 0 margin rates, then the margin is not charged for operations by that symbol.

The margin calculation is described in details in a separate section.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Swaps

Swaps

Swap (Rollover) is an operation of transferring a trader's open position on the next trading day. This operation is performed at the end of the trading day if the position remains open at that moment. A fee is charged for the transfer of the positions. The swap amount is accumulated in a separate field of the trading position. Once the position is closed, this amount is accounted in the balance.

Symbol: Forex\EURUSD

Common Currency Quotes Trade Execution Margin Margin Rates **Swaps** Sessions

 The setting up of charging swaps by the symbol. Please specify swaps for long and short positions and swap multipliers for each day of the week.

☒ Enable swaps

Type: in points

Long positions: 0 Short positions: 0

Days in year: 360 ☐ Automatically consider holidays

Swap multipliers:

Day of week	Multiplier
Sunday	
Monday	1
Tuesday	1
Wednesday	3
Thursday	1
Friday	1
Saturday	

Forex

All week

From symbol AUDCAD, Australian Dollar vs Canadian Dollar

OK Cancel Help

Position rollovers for a symbol are set up on this tab:

- **Enable swaps** enable/disable charging of swaps.
- **Type** one of swap charging types, described below.
- **Long positions** swap for Buy positions.
- **Short positions** swap for Sell positions.
- **Days in year** the number of days in a year to be used for swap percent calculation. Depending on the country and market in which the broker operates, as well as on the financial instrument type, different number of days in a year can be used when calculating annual percent. This parameter operates with such calculation specifics. The most common option of 360 days is used by default. You can change the value to 365 or 366, as well as specify a different value manually.
- **Swap multipliers** swap multiplier for each day of the week. This multiplier will be applied to the calculated swap value before charging. Specify 1 to charge the regular amount, 3 for triple swap or 0 to cancel swap. Conveniently manage the settings using the following commands:
 - **Forex** sets standard settings for Forex instruments: standard single swaps on weekdays and triple swap on Wednesday.
 - **All week** standard single swap seven days a week.
 - **From symbol** copies swap multipliers from the selected symbol.

Automatically consider holidays

If the corresponding option is enabled, the platform checks all holiday configurations. The day before the holiday, the swap is doubled. No swap is charged on the day of the holiday. The calculations still use the swap multipliers specified for the corresponding days. For example:

- Swap multiplier is set to 1 for Tuesday
- Wednesday swap multiplier is 3
- Wednesday is a holiday

On Tuesday, the swap will be charged with a coefficient of 4: a sum of the the multipliers of the current day and of the holiday. No swap will be charged on Wednesday as it is a holiday.

The coefficients are applied similarly if you have multiple holidays in a row. If the holidays are set to Wednesday and Thursday, the platform will add the multipliers for Tuesday, Wednesday and Thursday and will charge the corresponding swap on Tuesday.

- Charging of swaps can be disabled for groups of accounts.
- Swap calculations are globally controlled at the trading server level. If swaps for certain days are disabled on the server, it cannot be enabled for individual symbols. If swaps are enabled on the server, they can be disabled for a specific symbol by specifying a zero multiplier for the certain day.

- By default, swap calculations are not performed on Saturday and Sunday. When working days are shifted to weekends, specify swap multipliers for the relevant days and enable weekend swap in trading server settings.

The following types of swap charging are available:

Calculation in points

Swap size is set in points and calculated by the following formula:

Swap size * Point price

Depending on the position direction, a swap size is taken from the "Long positions" or "Short positions" field. The calculation is performed in the following order:

- Calculating the point price in a symbol profit currency considering a position volume
- Converting the obtained price into a client's deposit currency
- Calculating a swap sum using the formula shown above

A single point price is a profit obtained when the price moves one point for a position of a specified volume. The price calculation (including converting to the deposit currency) is performed according to the type of the profit calculation by symbol depending on the position direction.

For example, in case of Buy 5.00 USDTRY with the contract volume of 100 000 and Forex profit calculation type, the point price is calculated as follows:

$$5 * 100,000 * 0.00001 = 5 \text{ TRY}$$

If the deposit currency is USD, the price is converted at the current rate:

$$5 * 0.2274587 = 1.14 \text{ USD (with rounding)}$$

If the swap for long positions is -11.35, the total sum is as follows:

$$1.14 * (-11.35) = 12.94 \text{ USD}$$

The swap may be equal to zero, if the sum is below the minimum estimated currency unit at one of the stages. This may happen on small volume trades at micro-accounts (with a reduced contract size). For example, in case of Buy 0.01 EURUSD and the contract size of 1 000, the point price is calculated as follows: $0.01 * 1\,000 * 0.00001 = 0.0001 \text{ USD}$. When rounding, the result is 0, thus the final swap value is also zero.

Calculation in money

If you select swap charging in money, the amount of money charged for the transfer of each lot of a position is specified in fields "Long positions" and "Short positions". Three types of currencies can be used for specifying swap amount:

- **Using base currency** charging in the base currency of the symbol a position is opened for;
- **Using margin currency** charging in the margin currency of the symbol a position is opened for;
- **Using profit currency** charging in the profit currency of the symbol a position is opened for;
- **Using group currency** charging in the deposit currency of the group the account belongs to.

In all cases except for charging the swaps in deposit currency, the swap size is converted into deposit currency. The conversion is performed using the unfavorable price for trader: if the swap is positive, a trader has to sell the swap currency for the deposit currency; if the swap is negative, a trade has to buy the swap currency for the deposit currency.

Calculation in percentage

In this case the annual interest rate is specified for long and short positions. Since swaps are calculated and charged every day at the end of day time, the calculated amount of the annual interest rate is divided by the number of days in a year.

When charging swaps, first the cost of one symbol lot is calculated (the symbol of the opened position), and then the specified percent is calculated, the obtained amount is multiplied by the position volume (in lots) and the result is divided by the number of days in a year. There are two calculation methods:

- **In percentage terms, using current price** in this case the swap is calculated as the percent from the current position cost: $(\text{cost of 1 lot of position} * \text{volume in lots} * \text{specified swap size} / 100) / [\text{days in year}]$;
- **In percentage terms, using open price** in this case the swap is calculated as the percent from the position cost at the moment of its opening: $(\text{cost of 1 lot of position} * \text{volume in lots} * \text{specified swap size} / 100) / [\text{days in year}]$.

The calculation of the cost of 1 lot of a position depends on the type of profit/margin calculation of the symbol:

- For the symbols with the Forex type of calculation the cost of 1 lot of a position is calculated in the base currency and is equal to the contract size. For example, for the EURUSD symbol that has contract size 100 000 the cost of 1 lot of a position is equal to 100 000 EUR. The feature of Forex symbols is the cost of 1 lot of a position in the base currency for them does not change in time. Thus, both modes of charging swaps in percentage terms work identically for the Forex symbols.
- For the symbols with CFD, CFD Leverage, CFD Index and Futures type of calculation the cost of 1 lot of a position is also calculated in the base currency. Since the contract size of such symbols is not represented in money (but in the number of securities for example), then to represent the position cost in money the contract size is additionally multiplied by the symbol price. For symbols of the Futures type, the resulting value is multiplied by the tick value to tick size ratio. For example, if a Futures symbol has the base currency USD, contract size 100, its price is equal to 33 and the tick value/tick size ratio is 1/0.1, then

MetaTrader 5 trading platform

the cost of 1 lot of the position is equal to $100 \times 33 \times 10 = 33\,000$ USD. For a CFD symbol with the same parameters, one lot size would be $100 \times 33 = 3300$ USD.

If the currency of swap calculation (the base currency of the symbol) is different from the client group currency, it is converted using the current rate when charging swaps. For Buy positions, the Bid price of the currency pair [swap currency|deposit currency] is used; Ask price is used for Sell positions.

Charging swaps by reopening positions

In this case all positions are forcedly closed at the end of a trading day, and on the next day they will be reopened. Reopen conditions depend on the swap settings:

- **In points, reopen position by close price** all positions will be opened at their close price +/- specified swap (in points);
- **In points reopen position by bid price** all positions will be opened at the current Bid price +/- specified swap (in points).

Thus swap is charged by way of adjusting the reopen price.

- Positions are closed at a time set in the "End of day" parameter. The open time of new (reopened) positions depends on the "Daily reports" setting of the trade server. If "End of day" option is selected, the open time of the reopened positions is as the end of day time of current trade day ("End of day"). If "Start of day" option is selected, the open time of the reopened positions is set as the start of the next trade day. At that, if the open time of a position does not match a trading session of the symbol, then the open time is set to the start of a closest trading session.
- When re-opening a position, the new position has the same ticket as the old one. This position ticket is also written to the deals, which close the old position and open a new one (in the "Position" field of the deals).

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Symbol Settings / Sessions

Sessions

As a rule, trading is disabled on weekends. Most banks and stock exchanges do not work, which means there is no liquidity in the markets. The platform supports a trading schedule, which can prevent traders from sending trade requests in non-trading hours. You can also limit trading at a certain time for each individual instrument.

Symbol: EURUSD

Common Quotes Trade Execution Margin Swaps Sessions

 The setting up of trade and quotation sessions of the symbol by days. During the quotation session it is possible to view the price dynamics but trading is prohibited.

Day	Quotes	Trade
01 Sunday		
01 Monday	00:00-24:00	00:00-24:00
01 Tuesday	00:00-24:00	00:00-24:00
01 Wednesday	00:00-24:00	00:00-24:00
01 Thursday	00:00-24:00	00:00-24:00
01 Friday	00:00-24:00	00:00-24:00
01 Saturday		

Edit

☐ Use time limits

From: 1970.01.01 00:00 To: 1970.01.01 00:00

OK Отмена Справка

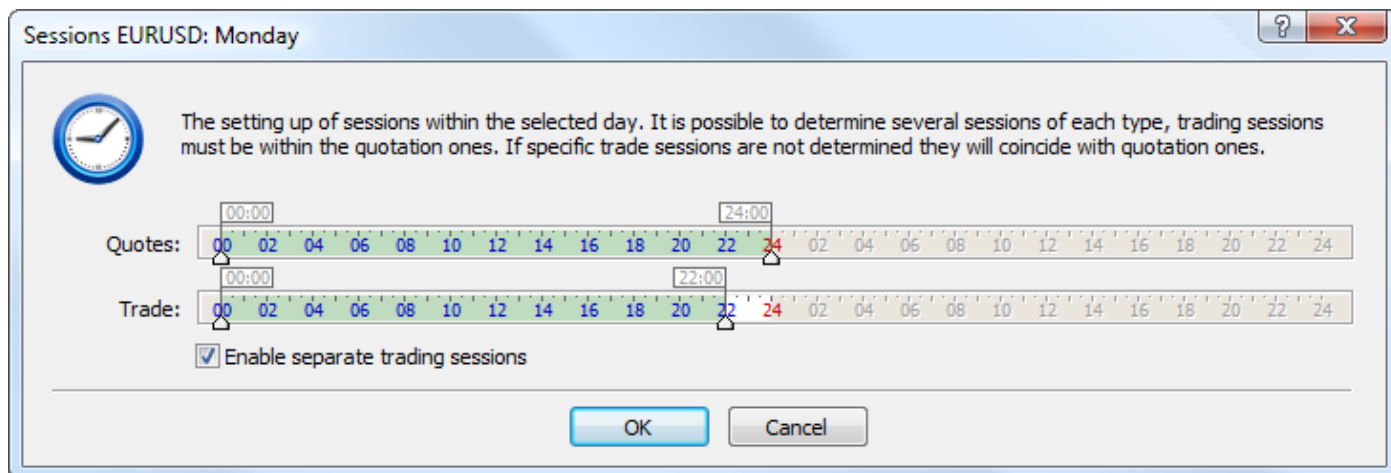
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The following session types can be configured here:

- Quoting sessions. During this time, the platform will receive (and broadcast to traders) quotes from data sources.
- Trading sessions. During this time, users can perform trading operations for the symbol. When trying to place an order outside the session, the trader will receive the "Market closed" warning.

- Trade and quotation sessions do not affect the possibility of trading using the manager terminal, Manager API and Web API.
- Session settings do not apply to spliced symbols, because the sessions of original symbols may differ.

To start setting up a day, click twice on a corresponding line in the table:



The following settings are available here:

- **Quotes** setup of a quoting session;
- **Trade** setup of a trading session;
- **Enable separate trading sessions** if you need to set up quoting and trading sessions separately, select this option.

- To set the same sessions for several week days, select them using the mouse while holding Ctrl or Ctrl+Shift. Then click "Edit".
- Several quoting and trading sessions can be set up within one weekday. Click with your left mouse button on the time scale and holding the button pressed drag it till the necessary time.

The time adjusting levers can be moved both with a mouse or using a keyboard. If your hold Shift pressed, the speed of lever moving will be slowed down. Thus you can set up the time of sessions maximum precisely, including minutes.

Beside that, the following parameters are available on the "Sessions" tab:

- **Use time limits** if a symbol will exist only during a certain timeperiod, select this option and then set the time limits when it will exist. As soon as this period is over, trading of this symbol will be disabled. At the attempt to open an order with this symbol, "Trading disabled" will be returned to clients.
- **From** start date and time of the symbol existence;
- **To** end date and time of the symbol existence;

We strongly recommend to perform any operations on modification of financial symbols only on holidays when markets are closed.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Splicing
Futures

Splicing Futures

MetaTrader 5 trading platform allows users to splice the quotes of the symbols having a single underlying but different validity periods. Futures can serve as examples of such symbols. Spliced symbol charts allow performing technical analysis of prices for an underlying asset.

It is possible to splice not only bars but tick data as well. Please note that tick data take much more disk space compared to minute history. After splicing, same ticks are stored separately for source symbols and a spliced one.

To splice the futures, a new symbol should be created. The history of quotes of this symbol will represent splicing of several futures with one underlying asset. Also, quotes of the front futures will be broadcast in real time at this symbol.

The front futures is a futures with the closest expiration date.
--

MetaTrader 5 trading platform

Symbol

?

×

Common Quotes Trade Futures Execution Margin Margin Rates Swaps Sessions



Configuration of futures symbol parameters. Please specify settlement price, price limits, and splice parameters.

Settlement price:

Minimum price: Maximum price:

Splice type: Splice date:

Splice shift:

Symbol

?

×

Common Quotes Trade Futures Execution Margin Margin Rates Swaps Sessions



The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.

Symbol: Description:

ISIN: International:

Basis: Page:

Source: Digits: Base currency:

Background: ☐ None Market depth: Profit currency:

Spread: pt Chart mode: Margin currency:

Spread balance: 0 bid / 0 ask

- Since the splicing symbol is created solely for analytical purposes, the ability to trade it should be disabled at "Trade" tab.
- Splicing symbols are not influenced by session settings, because the sessions of original symbols may differ.

"Exchange Futures" calculation type should be set on the symbol's "Trade" tab. Also, the following parameters should be set for the symbol:

- name should be specified in "Symbol" field of "Common" tab ;
- additional information can be specified in "Description" field (optional);
- a link to the web page can be specified in "Page" field (optional);
- market depth should be enabled;
- spread, spread balance, prices, etc. settings on the "Common" tab should be disabled;
- quote sessions from 00:00 to 24:00 on all days should be set on "Sessions" tab (session borders are often defined by an exchange shortly before the sessions themselves; these settings are set to ensure getting into the exchange's quote sessions);
- set underlying asset and splicing settings according to the instruction below.

All other symbol parameters are of no importance.

The following settings of the splicing symbol must match the settings of the symbols used for splicing:

- Tick size
- Digits
- Market depth (size)

Setting an Underlying Asset

The underlying asset, according to which the splicing will be performed, should be specified in "Basis" field of "Common" tab. The quotes of all symbols with the specified underlying asset will participate in splicing.

Configuring Splicing

As a rule, several futures having different expiration dates are simultaneously traded for the same underlying asset. A point, at which the quotes of one futures are replaced with the quotes of another, is called a splicing point.

The splicing type is set first. Despite a single underlying asset, the level of price for futures having different expiration dates may differ. Thus, considerable drops may form in their splicing points. The trading platform allows splicing the prices both with and without reducing the price to a single level.

- **Unadjusted** splicing of quotes is performed without reducing the price level of the previous (expiring) contract to the prices of the next one (or a frontal one futures with the closest expiration date). In this case, real quotes will be available in history.

- **Adjusted** the difference between the last quote of the previous (expiring) contract and the first quote of the new front contract is calculated during splicing. The calculated difference is added/deducted from all previous quotes leading to the adjustment of the price level. This method allows forming a smooth final chart convenient for full-scaled technical analysis. However, the history will contain reduced quotes instead of actual ones.

The main step in setting up splicing is defining a splicing point. The trading platform has the ability to define a splicing point relative to a symbol's expiration date. Each symbol has its expiration date shown in "To" field of "Sessions" tab.

Futures splicing can be performed either at the moment of a current (front) contract's expiration or with a shift from that moment by the specified number of days. The point of switching to the quotes of the next contract depends on the "Splice shift" parameter. If it is set to 0, the quotes switch at the moment of the current contract's expiration, otherwise the specified number of days earlier than its expiration.

After saving the symbol, the platform will start splicing and its chart will become available for viewing. Also, quotes of the front futures will be broadcast in real time at this symbol. And depending on the "Splice shift" parameter the symbol will switch to the quotes of the next contract at the moment of the current contract's expiration or the specified number of days earlier than its expiration.

Detailed information about splicing points and transformation prices can be found in the journal of the history server.

Example of splicing

Let's consider the example of splicing of three symbols:

- RTS-9.12 futures for RTS underlying with the expiration date of 2012.09.15 15:45
- RTS-12.12 futures for RTS underlying asset with the expiration date of 2012.12.15 15:45
- RTS-3.13 futures for RTS underlying asset with the expiration date of 2013.03.15 15:45.

Let's specify the following settings for the splicing symbol:

- Symbol @RTS;
- Basis RTS;
- Splice Adjusted;
- Date Expiration;
- Shift 1.

After saving the symbol, check the splicing by the history server's journal:

2013.02.15 09:24:35	HistoryBase	@RTS continuous build from RTS-3.13 [2012.12.14-2013.03.15 15:45]
2013.02.15 09:24:37	HistoryBase	@RTS continuous build from RTS-12.12 [2012.09.14-2012.12.15 15:45]
2013.02.15 09:24:39	HistoryBase	@RTS continuous build from RTS-9.12 [2011.06.06-2012.09.15 15:45]

The example above shows the splicing of three symbols. The splicing is performed the day before the symbol's expiration.

The quotes' period that has got into the final spliced symbol, as well as the value, by which the previous quotes have been shifted, while splicing with that symbol, should be specified for each symbol.

Splicing may take some time. A large volume of the accumulated tick data on spliced symbols may greatly affect a splicing time. However, splicing is performed asynchronously charts are completed step by step without interfering with the platform operation.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Import of Symbols

Import of Symbols

Using the administrator terminal you can import symbols with all their settings from any remote servers of MetaTrader 5 or MetaTrader 4.

In order to import symbols from another server, there should be any active account (demo or real, but not a manager one).

To start importing symbols use the "👉 Import from Server" command of the context menu of the Symbols section. After that the following window will appear:

Import

Connection details
Choose server type, fill server address and authorization information.

This wizard will help you to import settings from a MetaTrader 4 or MetaTrader 5 server. Please specify server type, address and authorization details below.

Server Type: MetaTrader 5

Server: 192.168.1.230:443

Login: 1000

Password: ●●●●●●●●

☐ Use certificate from file

NOTE: When importing settings from a MetaTrader 4 server, part of settings will be set to defaults. For importing settings from a MetaTrader 5 server with the advanced authorization mode, you must have a confirmed certificate for the account.

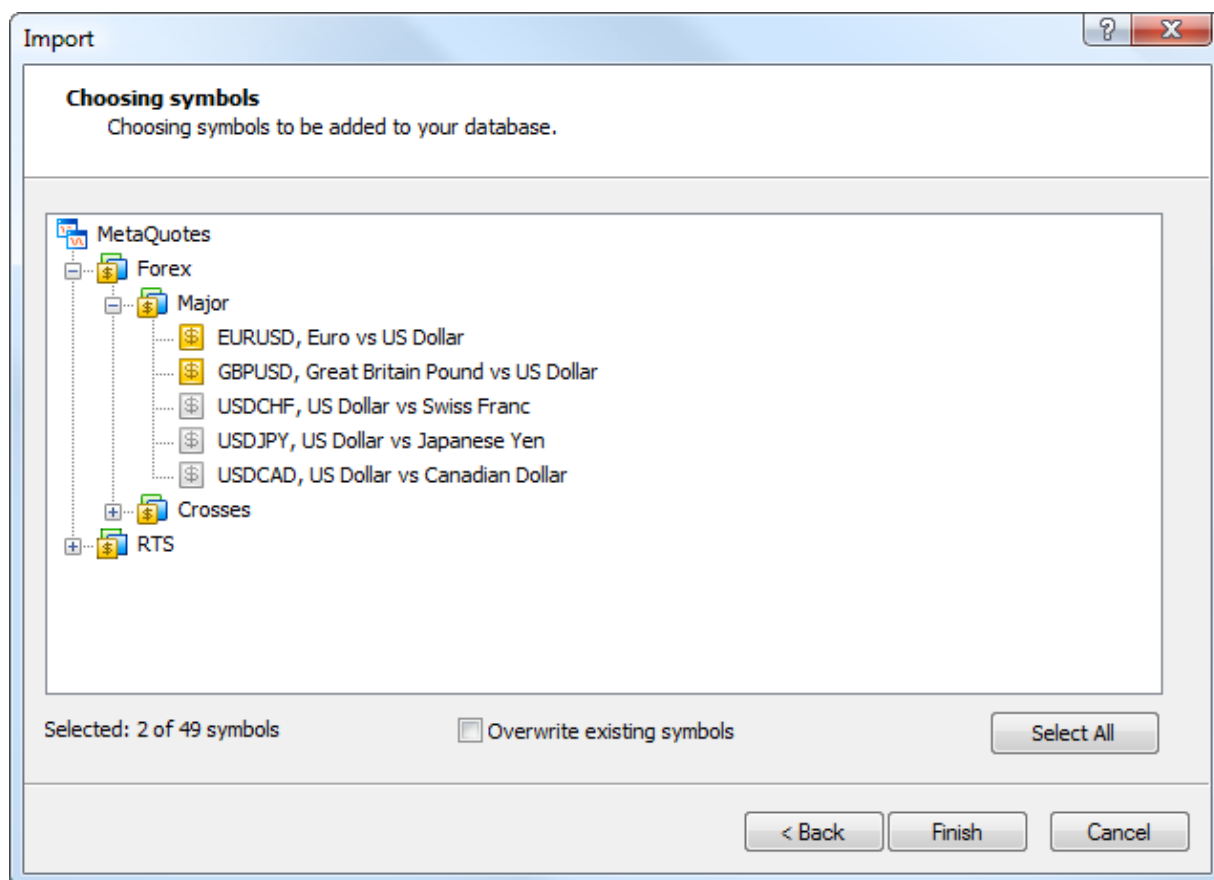
< Back Next > Cancel

On the first step specify the details of the server from which symbols and details for connecting it will be imported:

- **Server Type** in this field you should select the type of the server from which symbols will be imported: MetaTrader 5 or MetaTrader 4;
- **Server** IP address and port of the server separated by a colon;
- **Login** login for authorizing on the server;
- **Password** password for authorizing on the server;
- **Use certificate from file** if symbols are imported from a MetaTrader 5 server with the enabled mode of extended authorization, connection using the details of the account specified above will require a confirmed certificate. If the certificate has been installed in the operating system storage, it will be automatically recognized by the importing system. If the certificate hasn't been installed yet, this option should be enabled and path to the file should be specified manually;
- **Certificate file** press "Browse" and specify the pfx file of the certificate. The file certificate obtained in the terminal is saved to */terminal data folder/profiles/server name/certificates/*.
- **Certificate Password** password of the certificate.

- Import in the extended authorization mode is available only from MetaTrader 5 servers.
- Symbols from a MetaTrader 4 server are imported to the current selected group. When importing from a MetaTrader 5 server, location of groups is preserved the same as that on the source server.

After you have specified all the data, press "Next". In case of a successful connection to the server, the list of available symbols will appear in the next window:



There are several ways to select symbols:

- By double clicking on a symbol icon - the icon will become yellow at that;
- By double clicking on a group icon - all symbols on the group will be selected at that;
- Using the "Select all" button.

Here you can also view settings of imported symbols. To do it click twice on their names. To start the import press "Done".

- If you enable the "Overwrite existing symbols" option, then in case the name of an imported symbol matches one of the existing symbols, all its settings will be overwritten (including path in the hierarchy). If this option is disabled, such symbols will not be imported.
- All symbols are imported with the disabled trading possibility;
- After importing symbols, please check their settings.

Specifics of Import

When importing symbols from a MetaTrader 4 server, parameters are converted, while missing parameters are assigned default values:

- **Filters**

The filtering level of a symbol from MetaTrader 4 is set as soft. The hard

MetaTrader 5 trading platform

filtering is 5 times larger, while the discard level is 100 times larger.

- **Volumes**

Default volume levels are set.

Symbols from a MetaTrader 4 server are imported to the current group. When importing symbols from a MetaTrader 5 server location of symbols by groups is preserved the same as that on the source server.

MetaTrader 5 Trading Platform / Platform Setup / Symbols / Collateral Symbols

Collateral Symbols

The trading platform supports a special type of non-tradable assets, which can be used as client's assets to provide the required margin for open positions of other instruments. For example, a certain amount of gold in physical form can be available on a trader's account, which can be used as a margin (collateral) for open positions.

Such instruments have calculation type "Collateral". Features of such symbols:

- Clients can't perform any operations with these symbols, except for closing. A position can be opened only by a manager.
- No profit can be accrued for the positions of these symbols, they have no stop loss or take profit.
- The value of such a position, and thus the amount of money that a client can use as collateral, is calculated by the below formula:

Such assets are displayed as open positions. Their value is calculated by the formula:
 $\text{Contract size} * \text{Lots} * \text{Market Price} * \text{Liquidity Rate}$.

Liquidity Rate is the share of the asset that a broker allows to use for the margin, it is set in the symbol properties.

For groups with the risk management mode "for Stock Exchange, based on margin discount rates", the current Last price is used for Market Price, instead of Bid and Ask.

The Assets are added to the client's Equity and increase Free Margin, thus increasing the volumes of allowable trade operations on the account.

Toolbox									
Symbol ▲	Ticket	Time	Type	Volume	Price	S / L	T / P	Price	Pro
[-] eurUSD		2015.05.20 13:31:37	buy	1.00	1.11441	0.00000	0.00000	1.11514	73.00
⊕ Balance: 4 952.25 USD Equity: 6 234.33 Margin: 1 114.41 Free Margin: 5 119.92 Margin Level: 559.43 % 71.52									
[-] gold 1oz		2015.05.14 13:32:23	buy	1.00				1210.56	
⊕ Assets: 1 210.56 USD									
Trade Exposure History News 23 Mailbox 7 Calendar Company Alerts Articles 3 Code Base Experts Jour									

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In the example above, a trader has 1 ounce of gold having the current market value of 1 210.56 USD. This value is added to the equity and the free margin.

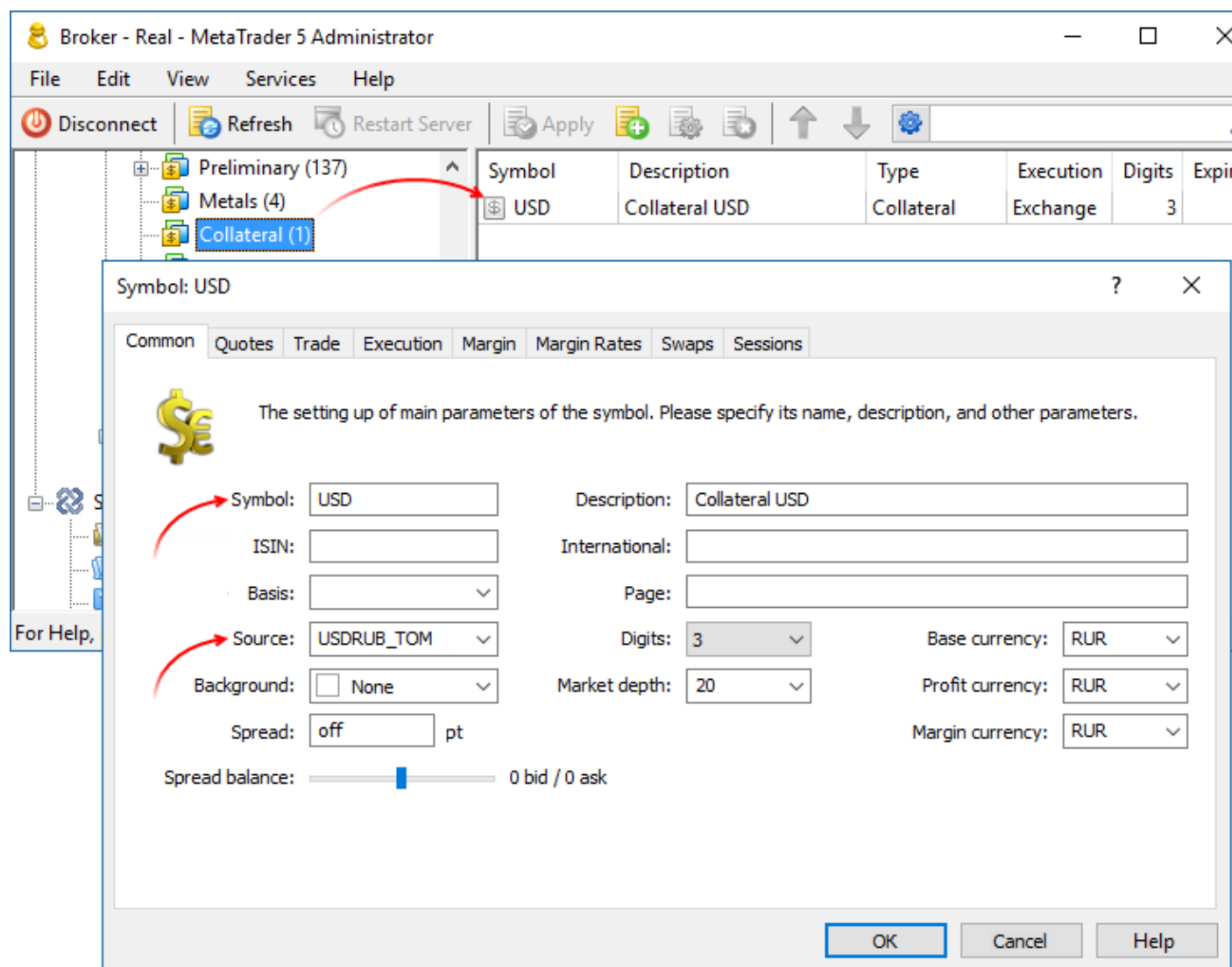
Symbol settings may allow clients and managers close such positions (Trade = Close only). In this case a trader will be able to convert the asset in the deposit currency at the current market rate and use that money for trading. A position can be closed only if the conversion rate of the assets currency into the deposit currency is available.

Charging a client account

An asset can be added to a client account via the manager terminal or gateways. For example, MetaTrader 5 Gateway to MOEX Securities supports importing the clients' money limits in currencies other than the ruble as Collateral positions.

Configuring collateral symbols

Create a separate symbol group via the Administrator terminal and then create a currency symbol in it. Select "Collateral symbol currency" "Client deposit currency" pair as a quote source for it.



Next:

- Set Collateral for the symbol type
- Disable trading for the symbol
- Set contract size 1 to let the price be displayed down to the last cent
- Specify zero size and tick value
- Specify the liquidity margin rate - amount of the current value of an asset, which will be taken into account as collateral
- Remove all margin ratios (set zero values)

Symbol: USD

Common Quotes Trade Execution Margin Margin Rates Swaps Sessions

The setting up of symbol trading. Please specify the type of profit calculation, type of placed orders, allowed trade volumes, etc.

Contract size: 1 Tick size: 0.000 Tick value: 0.0000000

Calculation: Collateral Limit & stop level: 0 pt

Trade: Disabled Freeze level: 0 pt

Symbol: USD

Common Quotes Trade Execution Margin Margin Rates Swaps Sessions

Please specify margin rates for different types of trade operations.

Liquidity margin rate: 1.000

Currency margin rate: 0.00000000

Margin rates:

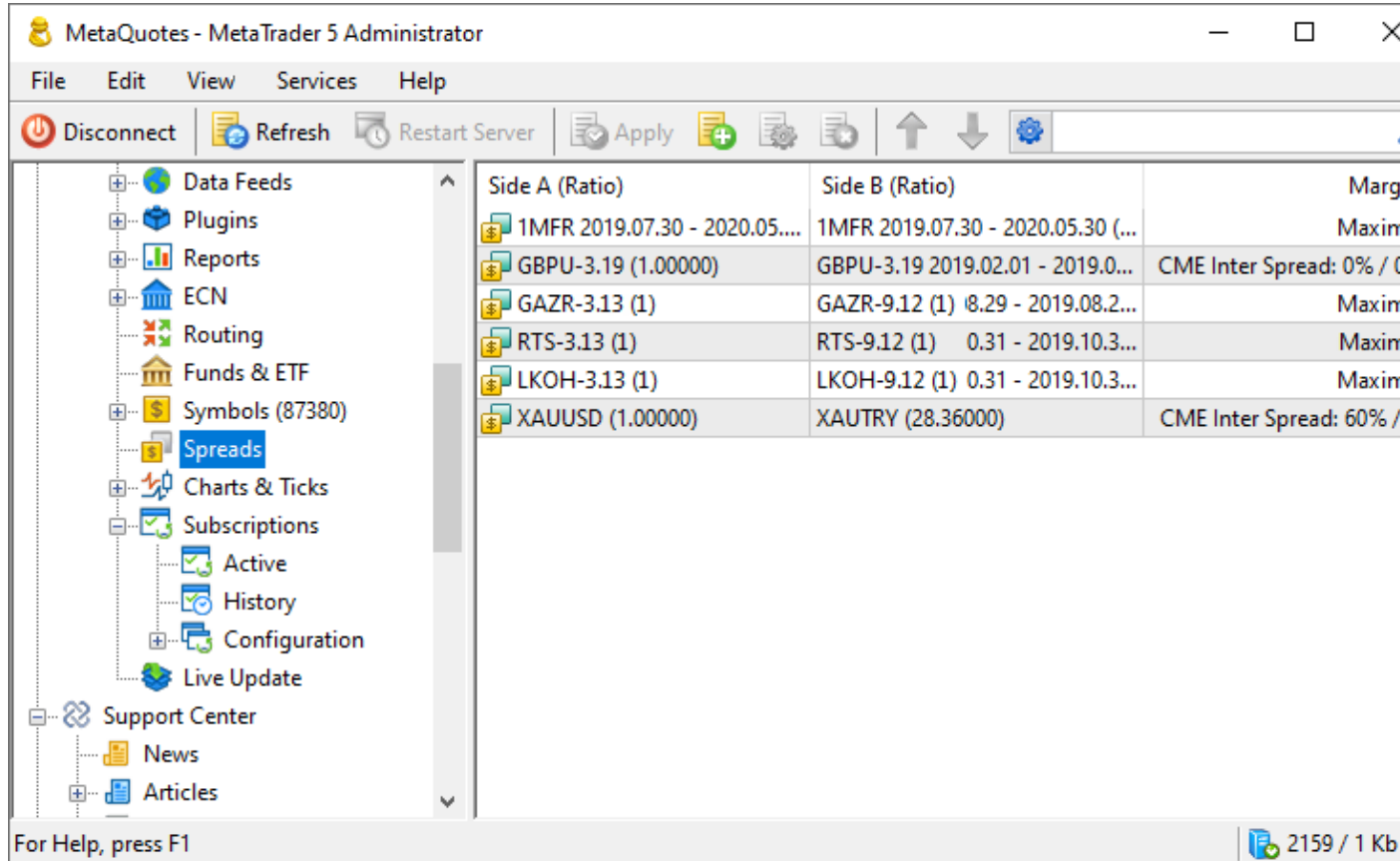
	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
Buy	0.0000000	0.0000000	0.0000000	0.0000000
Sell	0.0000000	0.0000000	0.0000000	0.0000000
Maintenance margin				
Buy	0.0000000	0.0000000	0.0000000	0.0000000
Sell	0.0000000	0.0000000	0.0000000	0.0000000

OK Cancel Help

MetaTrader 5 Trading Platform / Platform Setup / Spreads

Spreads

This section allows users to configure the charging of the margin in case client's trading positions are in a spread to one another. Being in a spread means the presence of the oppositely directed positions on related symbols. Reduced margin requirements for positions in a spread present traders with wider trading opportunities.



The screenshot shows the 'MetaQuotes - MetaTrader 5 Administrator' window. The left sidebar contains a tree view with the following items: Data Feeds, Plugins, Reports, ECN, Routing, Funds & ETF, Symbols (87380), **Spreads** (selected), Charts & Ticks, Subscriptions (with sub-items: Active, History, Configuration), Live Update, Support Center (with sub-items: News, Articles). The main panel displays a table for configuring spreads.

Side A (Ratio)	Side B (Ratio)	Margin
1MFR 2019.07.30 - 2020.05....	1MFR 2019.07.30 - 2020.05.30 (...)	Maxim
GBPU-3.19 (1.00000)	GBPU-3.19 2019.02.01 - 2019.0...	CME Inter Spread: 0% / 0
GAZR-3.13 (1)	GAZR-9.12 (1) 8.29 - 2019.08.2...	Maxim
RTS-3.13 (1)	RTS-9.12 (1) 0.31 - 2019.10.3...	Maxim
LKOH-3.13 (1)	LKOH-9.12 (1) 0.31 - 2019.10.3...	Maxim
XAUUSD (1.00000)	XAUTRY (28.36000)	CME Inter Spread: 60% /

At the bottom of the window, there is a status bar with the text 'For Help, press F1' on the left and '2159 / 1 Kb' on the right.

Spread settings are only used on netting accounts.

Creating and Editing a Spread

To create a Spread, click "+ Add" or "⚙ Edit" in "Edit" menu of "Standard" panel or in the context menu of this section.

Spread

The setting up of spread trades. Please specify spread legs and margin requirements of spread trades.

Margin type: Maximal

Initial margin: Maintenance margin:

Legs:

Symbol	Ratio	Side	Period
RTS-3.13	1	A	
RTS-9.12	1	B	

Add Edit Delete

OK Cancel

Margin Calculation Type

First of all, a type of margin charging should be specified for a spread.

- For all calculations types except Maximal, the specified margin is charged per unit of spread specified combination of positions. If any part of the position does not fit in the spread, an additional margin is charged according to symbol settings. If the client's current positions have the volume the mentioned combination fits in several times, the charged margin is increased appropriately. For example, two instruments A and B are in a spread with the ratio of 1 and 2. If a client has positions by those instruments of 3 and 4 lots respectively, the total margin will be equal to the doubled value of the spread configuration (two spreads: 1 lot of A and 2 lots of B, 1 lot of A and 2 lots of B) plus the margin for one left lot of A instrument.
- The values are displayed in the margin currency (excluding CME Inter Spread where ratios are specified). It is assumed that the margin currency of all spread symbols is the same.

Value

When this type is selected, the appropriate margin values that will be charged at the specified combination of positions are set in "Initial margin" and "Maintenance margin" fields.

Example:

- Initial and maintenance margins are set at 2 000;
- two symbols (GAZR-9.12 having the ratio of 2 and GAZR-3.13 having the ratio of 1) form the spread;

When a client has an opposite directed positions at GAZR-9.12 and GAZR-3.13 with the volume 2 and 1 lot respectively, the margin of 2 000 units will be charged. With the volume of 4 and 2 lots, the margin of 4 000 units will be charged.

Maximal

When this type is selected, the values of initial and maintenance margins will be calculated for each spread leg. The calculation is performed by summing up the margin requirements for all symbols of the leg. The margin requirements of the leg having a greater value will be used for the spread.

In this mode the total volume of positions on each side is considered, not only covered volume. Ratio for symbols in the spread legs cannot be specified in this mode.

Example:

- two symbols GAZR-9.12 and GAZR-3.13 form the spread;
- for GAZR-9.12 symbol, the values of initial and maintenance margins are equal to 2 000;
- for GAZR-3.13 symbol, the values of initial and maintenance margins are equal to 2 100.

When a client has an opposite directed positions at GAZR-9.12 and GAZR-3.13 with the volume of 2 and 1 lot respectively, the margin of 4 000 units will be charged.

CME Inter Spread

When this option is selected, ratios (multipliers) for the appropriate margin types should be selected in "Initial" and "Maintenance" fields. The final value of the margin is defined by summing up the margin requirements for all spread symbols and multiplying the obtained value by the specified ratio.

Example:

- two symbols (GAZR-9.12 having leg A and ratio of 2 and GAZR-3.13 having leg B and the ratio of 1) form the spread;
- for GAZR-9.12 symbol, the values of initial and maintenance margins are equal to 2 000;
- for GAZR-3.13 symbol, the values of initial and maintenance margins are equal to 2 100;

- The ratios of 0.5 are specified in spread settings in Initial and Maintenance fields.

The final value of the initial and maintenance margins will be calculated the following way: $(2\ 000 * 2 + 2\ 100) * 0.5 = 3\ 050$.

CME Intra Spread

When this option is selected, the margin is calculated the following way: the difference between the total margin of A leg symbols and the total margin of B leg symbols is calculated (the difference in absolute magnitude is used, so that it does not matter what leg is a deductible one). According to the type of the calculated margin, the value specified in "Initial" or "Maintenance" field is added to the obtained difference.

Example:

- two symbols (GAZR-9.12 having leg A and ratio of 2 and GAZR-3.13 having leg B and the ratio of 1) form the spread;
- for GAZR-9.12 symbol, the values of initial and maintenance margins are equal to 2 000;
- for GAZR-3.13 symbol, the values of initial and maintenance margins are equal to 2 100;
- The values of 500 are specified in spread settings in Initial and Maintenance fields.

Maintenance and initial margin will be calculated the following way: $(2\ 000 * 2 - 2\ 100) + 500 = 2\ 400$.

Spread Legs

The next step is configuring spread legs. The legs are oppositely directed positions in a spread – buy or sell. To create a leg, click "Add".

Spread Leg

The leg is one side of spread. Please specify side, symbol or basis security with periods, and ratio of the leg.

Leg: Side A Ratio: 1.00000000

Symbol: Symbol GAZR-3.13

Start period: ☒ 20.10.15 End period: ☒ 20.10.15

OK Cancel

Select A or B leg to be configured. The leg type is not connected to some definite position direction (buy or sell). Note that client's positions at all leg's symbols should be either long or short.

Several symbols can be configured for each leg. A volume ratio at this spread leg can be specified for each of them.

Example:

- leg A consists of GAZR-9.12 and GAZR-3.13 symbols having the ratios of 1 and 2 respectively
- leg B consists of GAZR-6.13 symbol having the ratio of 1.

To keep the client's positions in the spread, the client should open positions of 1 and 2 lots for GAZR-9.12 and GAZR-3.13 respectively to one direction and position of 1 lot for GAZR-6.13 to another direction.

Symbols for a leg can be specified as some definite symbol ("Symbol" option) or as an underlying asset ("Basis") in case there are quite a lot of symbols in the spread.

If an underlying asset is specified for the leg, all symbols with this underlying asset are considered ("Basis" field in symbol settings). In this case, symbols can be additionally filtered by the time of their operation (specified in "From" and "To" fields of "Sessions" tab). To do this, specify the time interval in "Start period" and "End period" fields. To be able to use the symbol, its expiration date ("To" field) should be in the specified interval. To make the date open, uncheck the box in the Start or End period field.

If the start and end dates are not specified, all symbols with the specified underlying asset will be used without considering their expiration date.

Context Menu

The context menu of "Spreads" section contains the following commands:

-  **Add** add a new spread configuration;
-  **Edit** edit the selected configuration;
-  **Delete** delete the selected configuration;
-  **Move up** move the selected configuration up relative to others;
-  **Move down** move the selected configuration down relative to others;
-  **Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
-  **Export to File** export the settings of spreads to a file.
-  **Import from File** import the settings of spreads to a file.
-  **Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration automatically specified in the query field. You will only need to press the request button.
-  **Find** open the search window;
- **Auto Arrange** if this option is enabled, the size of columns will be selected automatically;

MetaTrader 5 trading platform

- **Grid** show/hide grid to separate the table fields.

MetaTrader 5 Trading Platform / Platform Setup / 1 Minute History Charts

Charts

This section allows managing price data of financial instruments. The history of prices is stored in the platform in the form of one-minute bars and ticks. One-minute bars are used for displaying history charts in client terminals. Such charts allow traders to analyze price dynamics, perform technical analysis and test trading robots.

Only one-minute bars are stored in the platform; all other timeframes are built in client terminals (including mobile and web platforms) using these bars. The use of 1-minute bars allows saving the disk space and network traffic, as well as preserve data consistency on all timeframes.

The trading platform (the history server) builds bars using ticks received from datafeeds and gateways. Depending on the "Charts" parameter of a trading symbol, bars are based on Bid or Last prices (the price of the last executed trade). As a rule, charts of exchange instruments with the enabled Market Depth feature are based on the Last price.

Bars are only formed and accumulated using newly received ticks. To meet traders' needs, it is highly recommended to find a high-quality deep history of all your financial instruments and import it to the platform. The history can also be synchronized with another MetaTrader 4/5 server.

The screenshot shows the 'MetaQuotes - MetaTrader 5 Administrator' window. The left sidebar is expanded to 'Charts & Ticks', with '1 Minute History Charts' selected. The main panel displays a table of price data for EURUSD.

Date	Open	High	Low	Close
2020.11.26 00:00	1.19144	1.19163	1.19142	1.19144
2020.11.26 00:01	1.19155	1.19159	1.19155	1.19155
2020.11.26 00:02	1.19157	1.19161	1.19157	1.19157
2020.11.26 00:03	1.19160	1.19162	1.19157	1.19160
2020.11.26 00:04	1.19158	1.19167	1.19158	1.19158
2020.11.26 00:05	1.19165	1.19165	1.19161	1.19165
2020.11.26 00:06	1.19163	1.19163	1.19157	1.19163
2020.11.26 00:07	1.19157	1.19161	1.19157	1.19157
2020.11.26 00:08	1.19158	1.19163	1.19158	1.19158
2020.11.26 00:09	1.19162	1.19163	1.19162	1.19162
2020.11.26 00:10	1.19162	1.19165	1.19162	1.19162
2020.11.26 00:11	1.19164	1.19164	1.19162	1.19164
2020.11.26 00:12	1.19163	1.19163	1.19162	1.19163
2020.11.26 00:13	1.19163	1.19163	1.19162	1.19163

At the bottom of the table, there is a dropdown menu for the symbol (EURUSD), a date range selector (2020.11.26 00:00 to 2020.11.26 23:59), and a 'Request' button.

Each bar can store the following data:

- **Date** the bar formation date (minute).
- **Open** the price at the beginning of bar formation (the beginning of the minute).
- **High** the highest price inside the bar.
- **Low** the lowest price inside the bar.
- **Close** the price at the end of bar formation (the end of the minute).
- **Tick Volume** the number of ticks received during bar formation. The variable only counts the ticks that change the price based on which the bar is constructed (Bid or Last, depending on symbol settings). Several ticks in a row with the same price will be counted as one.
- **Volume** the real volume of trades executed during bar formation.
- **Spread** the lowest symbol spread recorded during the bar formation time.

- When the chart constructing mode is changed, the accumulated price history will not be re-built. Settings only apply to new received data.
- For the symbols, the charts of which are based on Bid prices, the history server does not accept Last prices and volumes from gateways and datafeeds. These ticks are not saved and are not delivered to other components of the platform.
- The price data received from gateways and data feeds is stored under [history server installation directory\history]. If the free disk space on the history server drops below 2 gigabytes, no saving of new arriving price data is performed. This prevents data from occupying of all available disk space. Thus, the overall server performance is preserved. If data cannot be written to disk, the following message is added into the history server's log:
"EURUSD' save skipped due not enough free space on disk", "bars and ticks are not saved due not enough free space on disk", "synchronization was stopped due not enough free space on disk ". In this case, you should immediately increase free disk space on the server.

Requesting Data

In order to view or edit bars of a symbol, you should request them.

- **Choosing a symbol**

In the first field, specify one of financial symbols from the platform. The symbol can be specified manually or chosen from the list which opens by clicking on the down arrow.

- **Choosing a period**

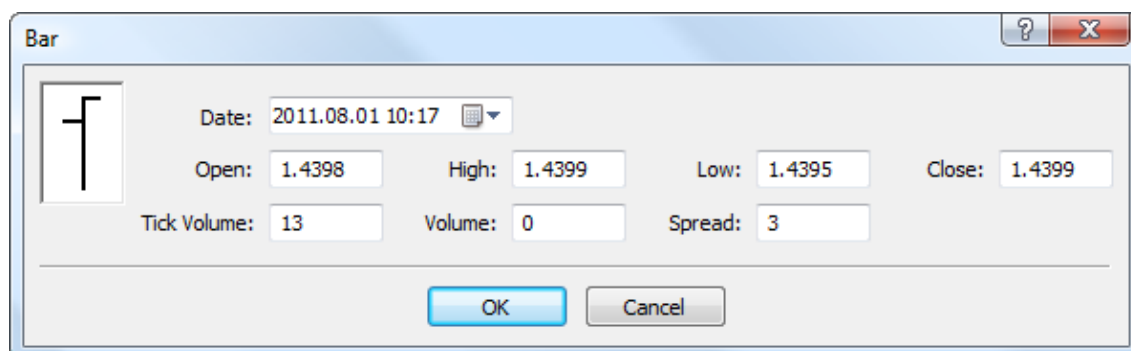
Then specify the period, for which you want to request bars. You can choose one of the predefined periods by clicking on  (today, last 3 days, last week, last month, last 3 months, last 6 months or the entire history). You can also specify a custom time interval.

- **Request execution**

To receive the history data, click "Request" or choose the same command from the context menu .

Adding and Editing Data

To add a bar, click "+ Add Bar" in the Edit menu, on the Standard toolbar or in the context menu. To edit a bar, click "⚙️ Edit Bar" or double-click on it.



Each bar can store the following data:

- **Date** the bar formation date (minute).
- **Open** the price at the beginning of bar formation (the beginning of the minute).
- **High** the highest price inside the bar.
- **Low** the lowest price inside the bar.
- **Close** the price at the end of bar formation (the end of the minute).
- **Tick Volume** the number of ticks received during bar formation.
- **Volume** the real volume of trades executed during bar formation.
- **Spread** the lowest symbol spread recorded during the bar formation time.

To delete a bar, select it and click "✖ Delete Bars". To delete multiple bars, select them with the mouse while holding down "Shift" or "Ctrl", and then click "✖ Delete Bars".

Context Menu

The following commands are available from the context menu of this section:

- **+ Add** add a new bar.
- **⚙️ Edit** edit a selected bar.
- **✖ Delete** delete selected bars.
- **🔄 Request** request history data.
- **📁 Export** export current requested history data to a file in CSV or HTML format.
- **📁 Import from File** import history data to the current requested symbol from a file.
- **📁 Import from Folder** import history data to the current requested symbol from multiple files.
- **📊 Split Stock** transform minute bars after splitting stocks.
- **🔍 Find** open the Search window.
- **Auto Arrange** if this option is enabled the size of columns is selected automatically.
- **Grid** this option shows/hides field separators in the table.

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- **Columns** select columns to display. You can additionally enable "Volume", "Spread" and "Tick volume".

MetaTrader 5 Trading Platform / Platform Setup / 1 Minute History Charts / Import of History Data

Import of History Data

The import of history data implements the following tasks:

- creating history when a new symbol is added;
- filling up missing parts of history of an existing symbol;
- correcting separate parts of history.

History data can be imported from a single file or from multiple files at once (from a folder).

Only one-minute bars are stored in the platform; all other timeframes are built in client terminals based on these bars. The use of 1-minute bars allows saving the disk space and network traffic, as well as preserve data consistency on all timeframes.

You can import data from higher timeframes (for example when switching from MetaTrader 4). However please note that lower timeframe charts will be incomplete. For example, if you import D1 data, these data will be written as 1-minute bars as of the beginning of corresponding days. The D1 and higher timeframe charts will be complete, but charts on lower timeframes will have gaps.

Import from File

In order to start importing data, click "📁 Import from File" in the context menu of the "Charts" section. After that the following window will be opened:

Import: EURUSD

File: C:\Users\Desktop\EURUSD.csv Browse...

Separator: , Skip: 0 columns 0 rows

Shift: 0 hours ☐ Use selected only ☐ Tick volumes

Date	Open	High	Low	Close	Tick Volume	Volume	Spread
📅 2009.03.23 10:25	1.3710	1.3710	1.3707	1.3707	1	0	0
↑ 2009.03.23 10:24	1.3713	1.3713	1.3708	1.3709	1	0	0
📅 2009.03.23 10:23	1.3707	1.3713	1.3707	1.3712	1	0	0
📅 2009.03.23 10:22	1.3708	1.3708	1.3705	1.3706	1	0	0
📅 2009.03.23 10:21	1.3707	1.3707	1.3705	1.3707	1	0	0
📅 2009.03.23 10:20	1.3702	1.3707	1.3700	1.3706	1	0	0
↑ 2009.03.23 10:19	1.3699	1.3701	1.3698	1.3701	1	0	0

OK Cancel

Specify the following information here:

- **File** specify a file to import using the "Browse" button. You can import files of different formats: *.hst and *.hsc history data of the MetaTrader 4 servers and terminals; *.csv, *.prn and *.txt text files with separators;
- **Separator** separator of elements in a text file;
- **Skip columns and rows** number of columns (left to right) and rows (top to bottom) that should be skipped when importing;
- **Shift** time shift in hours. This option is used when importing data from other time zones;
- **Use selected only** this option allows importing only lines selected in the preview window. Strings can be selected using the mouse, holding "Ctrl" or "Shift".
- **Tick volume** if this option is not checked, the information on tick volumes will not be imported from the file.

Data are imported for the symbol that is currently selected by request in the "Charts" section.

Import from Folder

You can import history data in bulk from multiple files. To do it, click "📁 Import from Folder" in the context menu of the "Charts" section. After that the following window will be opened:

Import Folder

Folder: C:\Import Browse...

Separator: ; Shift: 0 hours

Skip: 0 columns 0 rows

Select symbols to import: ☐ Tick volumes

Symbol	Period	Bars
☒ EURUSD	2013.03.08 00:00 - 2013.03.08 11:16	668
☒ GBPUSD	2013.03.08 00:00 - 2013.03.08 11:17	663

OK Cancel

Specify the following information here:

- **Folder** using the "Browse" button choose a folder containing history data files that should be imported. Only CSV format of data files is supported. File names must correspond to the names of the symbols created in the trading platform. For example, EURUSD.csv, GBPUSD.csv, etc.;
- **Separator** separator of elements in a text file;
- **Skip columns and rows** number of columns (left to right) and rows (top to bottom) that should be skipped when importing;
- **Shift** time shift in hours. This option is used when importing data from other time zones;
- **Tick volume** if this option is not checked, the information on tick volumes will not be imported from the files.

Once a folder is selected, the recognized price data will be displayed in the bottom part of the window. Selected symbols that should be imported by putting a check mark against them and click OK.

Record Format

The following types of entries in text formats are supported by import:

- YYYY.MM.DD HH:MM O H L C V
- YYYY-MM-DD HH:MM O H L C V
- YYYY/MM/DD HH:MM O H L C V
- DD.MM.YYYY HH:MM O H L C V
- DD-MM-YYYY HH:MM O H L C V
- DD/MM/YYYY HH:MM O H L C V
- MM.DD.YYYY HH:MM O H L C V
- MM-DD-YYYY HH:MM O H L C V
- MM/DD/YYYY HH:MM O H L C V
- YYYY.MM.DD,HH:MM O H L C V
- YYYY-MM-DD,HH:MM O H L C V
- YYYY/MM/DD,HH:MM O H L C V
- DD.MM.YYYY,HH:MM O H L C V
- DD-MM-YYYY,HH:MM O H L C V
- DD/MM/YYYY,HH:MM O H L C V
- MM.DD.YYYY,HH:MM O H L C V
- MM-DD-YYYY,HH:MM O H L C V
- MM/DD/YYYY,HH:MM O H L C V
- YYYYMMDD,HHMMSS,O,H,L,C,V

Time can be also specified in the format HH:MM:SS.

Copying History Data from Another Symbol

For newly created symbols without any price data, one can copy the data from another symbol in the platform. To do it, specify the name of the symbol from which you need to copy the data in the Source field of the new symbol. The server will automatically copy the entire price history.

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- Once copying is finished, delete the value of the Source field from the new symbol settings.
- Copying works only if the new symbol doesn't have any price data. To make sure there are no historical prices available, check the history\[symbol_name] directory on the history server. This directory must be empty.

MetaTrader 5 Trading Platform / Platform Setup / 1 Minute History Charts / Split Stocks

Split Stocks

Split is an increase of the number of outstanding stocks by splitting them with a proportional decrease in their value. The opposite operation – consolidating (or reverse split) by decreasing a number of stocks with a proportional increase in their value – is possible as well.

The appropriate operations (stock splitting/consolidation) can be executed on the trading platform's side as well:

- Transforming the minute and tick symbol history means proportional increase or decrease in prices, spreads and volumes.
- Transforming the current client's positions – assigning a new volume and price with the ability to remove stop loss and take profit levels. Transforming is performed using MetaTrader 5 Manager.

Click "Split Stock" in the context menu of the "1 Minute History Charts" section.

Stock Split

Symbol: #AA, Alcoa Inc

From: 2019.05.10 00:00 To: 2019.05.14 23:59

Request

Date	Open	High	Low	Close	Spread	New Open	New High	New Low	New Close	New Spread
2019.05.10 16:30	24.48	24.49	24.44	24.44	3	12.24	12.24	12.22	12.22	1
2019.05.10 16:31	24.43	24.47	24.41	24.46	3	12.21	12.23	12.20	12.23	1
2019.05.10 16:32	24.46	24.49	24.46	24.49	3	12.23	12.24	12.23	12.24	1
2019.05.10 16:33	24.47	24.53	24.47	24.53	3	12.23	12.26	12.23	12.26	1
2019.05.10 16:34	24.53	24.54	24.51	24.54	3	12.26	12.27	12.25	12.27	1
2019.05.10 16:35	24.54	24.55	24.50	24.50	3	12.27	12.27	12.25	12.25	1
2019.05.10 16:36	24.50	24.51	24.50	24.50	3	12.25	12.25	12.25	12.25	1
2019.05.10 16:37	24.49	24.54	24.49	24.53	3	12.24	12.27	12.24	12.26	1
2019.05.10 16:38	24.52	24.52	24.46	24.46	3	12.26	12.26	12.23	12.23	1
2019.05.10 16:39	24.47	24.50	24.47	24.50	3	12.23	12.25	12.23	12.25	1

Stock Split Ratio

New Shares: 2 Old Shares: 1

Rounding Rule

Price: Down

Split From: 2019.05.12 00:00 To: 2019.05.14 23:59

Calculate Process Cancel

MetaTrader 5 trading platform

Select a trading instrument and a time interval, and click Request. Minute bars appear in the list.

Set the split settings:

- Use "New Shares" and "Old Shares" to set stock split/consolidation ratio so that prices can be transformed similarly.
- Set the rounding rule in case the number of digital places of a new price exceeds the value set in the symbol's Digits parameter. For example, when transforming the price of 35.15 with the ratio of 2:1, we obtain 17.575. When rounded down, the final price is 17.57, when rounded up, it is 17.58. Also, the "Standard" rounding option (standard rounding to the nearest integer) is available as well. For example, if the Digits is 2, the rounding is performed as follows: 17.234 -> 17.23, 17.235 -> 17.24.
- If necessary, you can specify the time interval in which the split will be performed, using the "Split from" and "To" fields. In this case the operation will only be performed for one-minute data in the specified range, while the remaining data will not be affected. Such an operation may be needed for performing a backdated split. If you want to split the entire available symbol history, leave these parameters unchanged.

After implementing all the settings, click Calculate to see the preliminary split results. New parameters for each bar are displayed in the table: new OHLC prices and spread. Click Process to execute a split.

- Splitting can take quite a long time. Do not launch the process during trading hours.
- Split is performed both for 1-minute and tick data, regardless of the section from which the process is launched. Only preliminary split results can be viewed separately.

The process can be tracked using the history server journal.

2017.04.18 17:30:27.339	HistoryBase	#AA split 2 for 1 started, up rounding
2017.04.18 17:30:27.366	HistoryBase	#AA split for 2014.04.26 completed [1239 bars up
2017.04.18 17:30:27.406	HistoryBase	#AA split for 2014.04.27 completed [1052 bars up
....		
2017.04.18 17:34:28.429	HistoryBase	#AA split for 2017.04.17 completed [1438 bars up
2017.04.18 17:34:28.888	HistoryBase	#AA split for 2017.04.18 completed [874 bars up
2017.04.18 17:34:28.888	HistoryBase	#AA split 2 for 1 finished [3497656 bars updated,

Split Scanner

Split scanner automatically analyzes the price history of trading instruments to identify possible stock splits. The scanner then adjusts the price history to generate a smooth chart corresponding to the instrument's current prices. During analysis, the system additionally checks an internal database of publicly known stock splits.

Go to the "1-Minute History Charts" section and select  "Split Scanner" in the context menu. Next, select a group of symbols and click "Scan". Check the received split

MetaTrader 5 trading platform

points. The list will display:

- Found split date
- Price before and after split, their ratio
- Number of stocks before and after split, their ratio (split ratio)
- Overall conversion rate for prices and stocks, taking into account all splits found for the instrument

To customize the information displayed, use the context menu.

Records found in the internal list of known splits are highlighted in green.

Stock Split Scanner

Symbols: Stock Markets\USA\NYSE\NASDAQ(Other)*

Scan

Symbol	Date	Old Shares	New Shares	Old Price	New Price	Recalculate Charts
SIBN: SI-BONE, Inc.	2015.02.05	10	1	333.5	3309	☐
TRMK: Trustmark Corporation	1990.01.01	3	2	34.6	57.66	☐
YNDX: Yandex N.V.	2011.05.22	65	1	38.2	2398.2	☐
BSL: Blackstone / GSO Senior Floating R...	2019.04.26	1	3	151	49.75	☐
KIE: SPDR S&P Insurance ETF	2019.04.17	1	1.5	266.2	166.2	☐
CME: CME Group Inc.	2007.04.26	1	5	256.71	51.17	☒
EBAY: eBay Inc.	2007.04.27	1	2	66.32	27.25	☐
FAST: Fastenal Company	2019.04.17	1	2	64.28	31.47	☒
FFIN: First Financial Bankshares, Inc.	2019.04.26	1	2	57.54	29.3	☒
FRAN: Francesca's Holdings Corporation	2019.04.17	12	1	0.41	4.65	☐
GOGL: Golden Ocean Group Limited	2007.05.04	5	1	0.66	3.22	☒
HSO: Hudson Global, Inc.	2019.04.17	10	1	1.18	12.37	☒
K: Kellogg Company	2019.04.17	14	1	7.85	108.4	☐
PBYI: Puma Biotechnology, Inc.	2019.04.17	1	1.5	30.28	19	☐
STMP: Stamps.com Inc.	2019.04.17	1	2	83.3	37.31	☐
APRN: Blue Apron Holdings, Inc.	2019.04.26	15	1	0.55	8.4	☒

Process split Cancel

MetaTrader 5 trading platform

Check the "Recalculate Charts" box for the desired splits. Please note that there can be multiple splits for each instrument. Click "Process split", and the platform will automatically convert minute and tick data.

- Splitting can take quite a long time. Do not launch the process during trading hours.
- Split is performed both for 1-minute and tick data, regardless of the section from which the process is launched. Only preliminary split results can be viewed separately.

MetaTrader 5 Trading Platform / Platform Setup / Bid/Ask/Last Ticks

Bid/Ask/Last Ticks

This section allows managing tick data of financial instruments. Quotes are received by the history server from data feeds and gateway as a stream of ticks, trade statistics and Market Depth changes. The server passes this data to clients, but only stores the history of tick data. Trade statistics and Market Depth changes are not saved.

It is recommended to find a high-quality deep history of all your financial instruments and import it to the platform. The ability to test Expert Advisors using ticks in the Strategy Tester is an important function for traders. This testing mode is the most precise one while it is close to real market conditions.

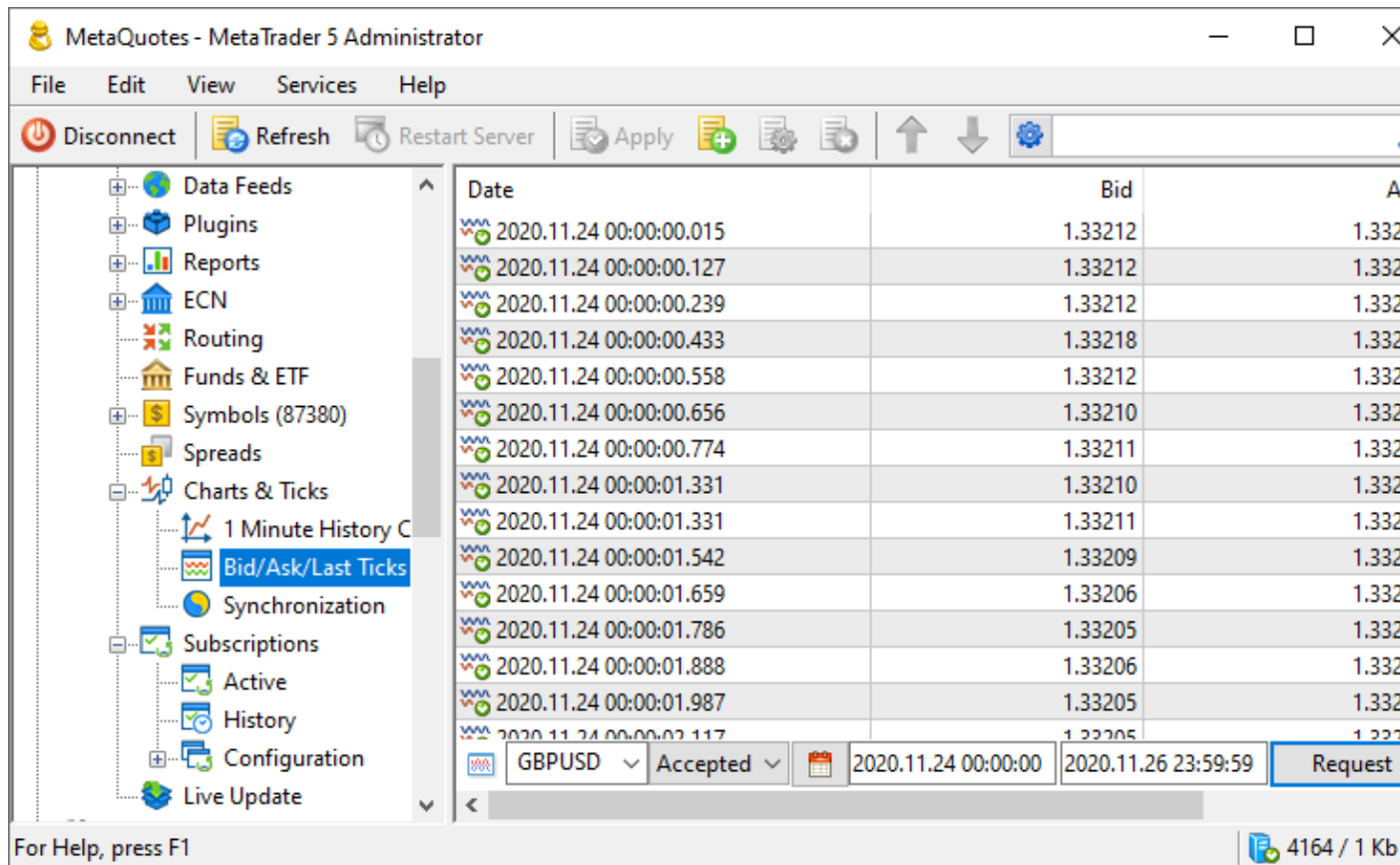
The Features of Tick History Generation

If a data feed or gateway sends symbol Market Depth changes to a platform, the history server automatically monitors changes of the best Bid and Ask price in it. If the best Bid or Ask price has changed, the history server generates a tick with the values of the best Bid and Ask prices. In this tick, the value of the last trade price and the volume will be zero.

The gateway/datafeed must only send ticks with the filled Last price and volume value. Otherwise, there will be duplicate Bid/Ask ticks (formed by history server and sent by the gateway/datafeed).

- At the beginning of each quoting session, a Bid/Ask quote is automatically added for symbols with the enabled Market Depth feature. The quote is generated based on the current Market Depth state. The purpose of this quote addition is to ensure that prices in the Market Watch window in client terminals correspond to the Market Depth state. An appropriate message is printed to the History server log in such case, for example: "AUDUSD tick added 0.75761 / 0.75779 due book session start". If the Market Depth is enabled, but one of the sides is absent (it has a zero Bid or Ask), a tick is added at the session beginning only if the symbol has the Exchange calculation type.
- For the symbols, the charts of which are based on Bid prices, the history server does not accept Last prices and volumes from gateways and datafeeds. These ticks are not saved and are not delivered to other components of the platform.
- The history server generates ticks using the best Market Depth prices only if the "Market depth" parameter in the appropriate symbol settings is not equal to "disabled".

MetaTrader 5 trading platform



Each tick can store the following data:

- **Date** the date and time of tick arrival.
- **Bid** the Bid price.
- **Ask** the Ask price.
- **Last** the price of the last executed trade (for exchange instruments).
- **Volume** the volume of the last executed trade (for exchange instruments).
- **Direction** direction of a deal, as a result of which the tick was generated: Buy or Sell. Data on direction is generally filled by the source of ticks, i.e. a gateway or a data feed. If the data source does not provide such information, the history server fills the direction automatically using the following algorithm:
 - ◆ If the Last deal price is greater than or equal to the last Ask price, the price is considered to be the result of a buy deal.
 - ◆ If the Last deal price is less than or equal to the last Bid price, the price is considered to be the result of a sell deal.
 - ◆ In other cases, it is considered that the direction can't be determined, and both directions, Buy and Sell, are assigned to the tick.
- **Data source** the name of the datafeed/gateway.

The price data received from gateways and data feeds is stored under [history server installation directory\history]. If the free disk space on the history server drops below 2 gigabytes, no saving of new arriving price data is performed. This prevents data from occupying of all available disk space. Thus, the overall server performance is preserved. If data cannot be written to disk, the following message

is added into the history server's log: "'EURUSD' save skipped due not enough free space on disk", "bars and ticks are not saved due not enough free space on disk", "synchronization was stopped due not enough free space on disk ". In this case, you should immediately increase free disk space on the server.

Raw and Accepted Ticks

The trading platform can store two types of ticks: raw and accepted. Raw (📶) are ticks received directly from datafeeds/gateways. Accepted (📶) are the ticks that were filtered and converted in accordance with symbol settings.

If you need to store raw ticks of a selected symbol on the server, the appropriate option must be enabled in symbol settings. Accepted ticks are always saved on the history server.

Trade Statistics

In addition to the basic price data of the trading instrument, the platform allows broadcasting to traders many useful statistics, such as session opening and closing prices, the highest and lowest Bid/Ask/Last prices for the day, and much more.

Calculated statistics can be transmitted through data sources and gateways. If ready statistical variables are not available, the appropriate metrics can be calculated on the history server and client terminal side. The statistical variables are calculated as follows:

- **Open Price** the open price of the last (recent) session. If the symbol chart is built by Bid prices, the Bid price value from the first tick of the session is remembered. If the tick has no Bid, the next tick is used, and so on. Similarly, the Last price is used for the instruments whose charts are based on last deal prices. If the value is provided by a gateway or data feed, exactly this will be used. No extra check or calculation is performed until history server restart.
- **Close Price** the close price of the last (recent) session. If the symbol chart is built by Bid prices, the Bid price value from the last tick of the session is remembered. If the tick has no Bid, the previous tick is used, and so on. Similarly, the Last price is used for the instruments whose charts are based on last deal prices. If the value is provided by a gateway or data feed, exactly this will be used. No extra check or calculation is performed until history server restart.
- **Price Change** indicates the difference between the last price of the instrument and the close price of the last session in percentage terms. The value is always calculated on the client terminal side. The calculation formula depends on the symbol charting mode:

By Last prices: $((\text{Last} - \text{Last price at session close}) / \text{Last price at session close}) * 100$

By Bid prices: $((\text{Bid} - \text{Bid price at session close}) / \text{Bid price at session close}) * 100$.

For futures symbols, the clearing price is used instead of the the session close price, if the clearing price is provided by the broker (non-zero):

MetaTrader 5 trading platform

By Last prices: $((\text{Last} - \text{Clearing price}) / \text{Clearing price}) * 100$

By Bid prices: $((\text{Bid} - \text{Clearing price}) / \text{Clearing price}) * 100$.

Requesting Data

To view or edit ticks of a symbol, request them:

- **Choosing a symbol**

In the first field, specify one of financial symbols from the system. The symbol can be specified manually or chosen from the list which opens by clicking on the down arrow.

- **Choosing a type**

Choose the type of ticks to be requested: All, Accepted or Raw.

- **Choosing a period**

Specify the period, for which you want to request ticks. You can choose one of the predefined periods by clicking on  (today, last 3 days, last week, last month, last 3 months, last 6 months or the entire history). You can also specify a custom time interval.

- **Request execution**

To receive ticks, click "Request" or choose the same command from the context menu .

Analyzing Data Feed

The information about each tick contains the index of a data feed it came from. Index is the position of a data feed in the list at the time of the tick coming. The "Data Feed" column of ticks displays a data feed that currently corresponds to the index written for a tick.

Thus, in most cases the "Data Feed" column will display a data feed that is currently the first one in the list. However, that is not always so.

An error in the operation of a data feed may occur, in that case the translation of quotes will be automatically switched to the next data feed in the list. That exact moment can be traced by analyzing the information displayed in the "Data Feed" column.

Switching of data feeds (or gateways that can also be used as sources of quotes) can be tracked in the journal of the history server operation. For example:

"18:32:44 Ticks datafeed 2: EURUSD activation"

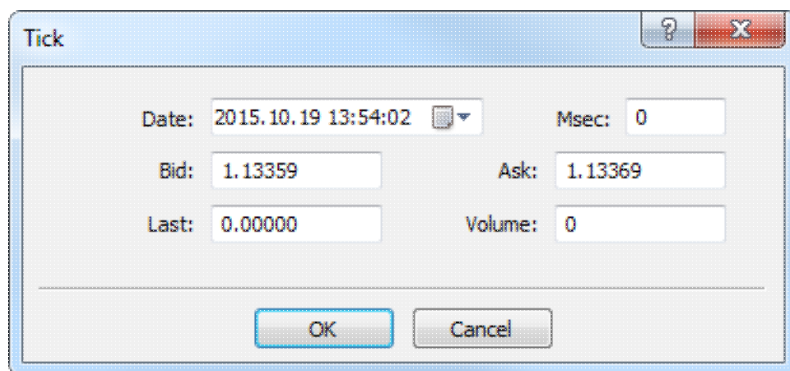
This entry tells that at 18:32:44 the flow of quotes from the data feed 2 in the list of configurations has been activated for EURUSD.

The numeration of gateways and data feeds in the list of configurations starts from 0.

Adding and Editing Tick Data

MetaTrader 5 trading platform

To add a tick, click "+ Add" button in the Edit menu, on the Standard toolbar or in the context menu. To edit a tick, click "⚙️Edit" or double-click on it.



The following data can be edited:

- **Date** the date and time of tick arrival.
- **Bid** the Bid price.
- **Ask** the Ask price.
- **Last** the price of the last executed trade (for exchange instruments).
- **Volume** the volume of the last executed trade (for exchange instruments).

Mass Deletion of Tick Data

To delete the tick data for a symbol:

- stop the history server using the console command `mt5history.exe /stop` or by execution the command `MetaTrader 5 Platform\History Server\Stop History Server` in the Start menu.
- go to the folder `\history\[symbol name]` on the history server and delete files with *.tkc extension. Tick data for each month are stored in a separate file. For example, 201203.tkc contains tick for March 2012.

Context Menu

The context menu of this section allows executing the following commands:

- **+ Add** add a new tick.
- **⚙️ Edit** edit a selected tick.
- **✖ Delete** delete selected ticks.
- **🔄 Request** request tick data.
- **📄 Export** export the current requested tick data in a file of CSV, HTM or HTML format.
- **📄 Import from File** import tick data to the current requested symbol from a file.
- **📁 Import from Folder** import tick data to the current requested symbol from multiple files.
- **📊 Split Stock** transform tick data after splitting stocks.
- **🔍 Find** open the search window.

MetaTrader 5 trading platform

- **Auto Arrange** if this option is enabled the size of columns is selected automatically.
- **Grid** this option shows/hides field separators in the table.
- **Columns** select columns to display.

MetaTrader 5 Trading Platform / Platform Setup / Bid/Ask/Last Ticks / Import Tick Data

Tick Data Import

Tick data import has the following objectives:

- generating a tick history when a new symbol is added;
- filling in missing parts of an existing symbol's tick history;
- correcting individual fragments of a tick history if needed.

Tick data can be imported both from a single file and from multiple files simultaneously (from a folder).

Import from File

Click "📁 Import from File" in the context menu of the "Ticks" section.

Import: EURUSD

File: D:\EURUSD.csv Browse...

Separator: ; ▼ Skip: 0 ▼ columns 0 ▼ rows

Shift: 0 ▼ hours ☐ Use selected only

Date	Bid	Ask	Last	Volume
2015.10.20 08:51:20.000	1.13424	1.13432	0.00000	0
2015.10.20 08:51:20.000	1.13425	1.13432	0.00000	0
2015.10.20 08:51:19.000	1.13424	1.13434	0.00000	0
2015.10.20 08:51:19.000	1.13424	1.13433	0.00000	0
2015.10.20 08:51:18.000	1.13424	1.13433	0.00000	0
2015.10.20 08:51:18.000	1.13426	1.13434	0.00000	0
2015.10.20 08:51:18.000	1.13426	1.13435	0.00000	0

OK Cancel

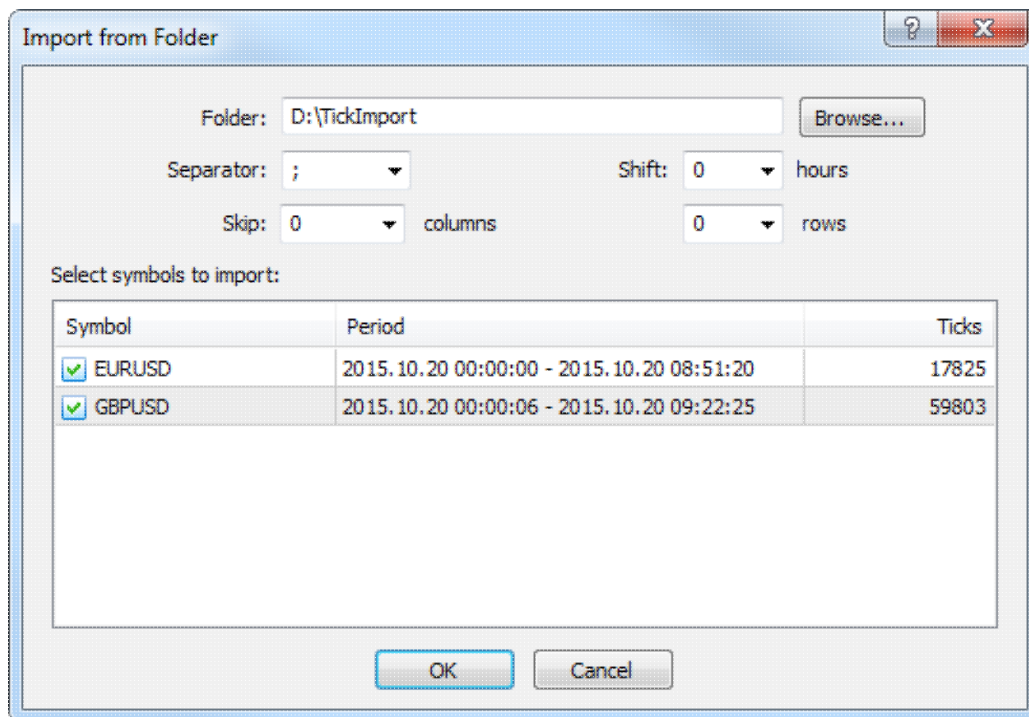
The following data should be specified here:

- **File** use the "Browse" button to specify a file to import. You can use history files of different formats: *.tkc MetaTrader 4 terminal and server tick data; *.csv, *.prn, and *.txt text files with separators;
- **Separator** element separator in a text file;
- **Skip columns and rows** amount of columns (from left to right) and rows (top to bottom) to be skipped during an import;
- **Shift** time shift by hours. The option is used when importing data from other time zones;
- **Use selected only** import only rows highlighted in the row view area. You can highlight rows with your mouse while holding Ctrl or Shift.

Data import is performed for a symbol currently selected in the "Ticks" section.

Import from Folder

You can import tick data from multiple files. Click "📁 Import from Folder" in the context menu of the "Ticks" section.



Import from Folder

Folder: D:\TickImport Browse...

Separator: ; Shift: 0 hours

Skip: 0 columns 0 rows

Select symbols to import:

Symbol	Period	Ticks
☒ EURUSD	2015.10.20 00:00:00 - 2015.10.20 08:51:20	17825
☒ GBPUSD	2015.10.20 00:00:06 - 2015.10.20 09:22:25	59803

OK Cancel

The following data should be specified here:

- **Folder** use the "Browse" button to select a folder containing CSV price history files to be imported. File names should match symbol names created in the trade platform. For example, EURUSD.csv, GBPUSD.csv, etc.;
- **Separator** element separator in a text file;
- **Shift** time shift by hours. The option is used when importing data from other time zones;
- **Skip columns and rows** amount of columns (from left to right) and rows (top to bottom) to be skipped during an import.

After selecting a folder, information on the found price data is displayed at the bottom of the window. Tick ☒ the symbols to be imported and click OK.

MetaTrader 5 Trading Platform / Platform Setup / Bid/Ask/Last Ticks / Split Stocks

Split Stocks

Split is an increase of the number of outstanding stocks by splitting them with a proportional decrease in their value. The opposite operation – consolidating (or reverse split) by decreasing a number of stocks with a proportional increase in their value – is possible as well.

The appropriate operations (stock splitting/consolidation) can be executed on the trading platform's side as well:

- Transforming the minute and tick symbol history means proportional increase or decrease in prices, spreads and volumes.
- Transforming the current client's positions – assigning a new volume and price with the ability to remove stop loss and take profit levels. Transforming is performed using MetaTrader 5 Manager.

Click "Split Stock" in the context menu of the "Bid/Ask/Last Ticks" section.

MetaTrader 5 trading platform

Broker-Demo - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Reports
Symbols (1290)
Spreads
Charts & Ticks
1 Minute History Chart
Bid/Ask/Last Ticks
Synchronization
Live Update

Date

2017.03.20 16:57:43.340		
2017.03.20 16:58:14.833		
2017.03.20 16:58:19.715		
2017.03.20 16:58:20.255		
2017.03.20 16:58:20.255		
2017.03.20 16:58:26.287		
2017.03.20 16:58:44.141		

Add... Ctrl+
Edit... Ctrl+
Delete Ctrl+
Request
Copy Ctrl+
Export...
Import from File...
Import from Folder...
Split Stock...

Stock Split

Symbol: #AA, Alcoa Inc

From: 2017.03.20 16:57:19 To: 2017.04.04 16:57:19 Request

Date	Bid	Ask	Last	New Bid	New Ask	New Last
2017.03.20 16:57:43.340	35.21	35.24	0.00	17.60	17.62	0.00
2017.03.20 16:58:14.833	35.20	35.23	0.00	17.60	17.61	0.00
2017.03.20 16:58:19.715	35.19	35.22	0.00	17.59	17.61	0.00
2017.03.20 16:58:20.255	35.18	35.21	0.00	17.59	17.60	0.00
2017.03.20 16:58:20.255	35.17	35.20	0.00	17.58	17.60	0.00
2017.03.20 16:58:26.287	35.16	35.19	0.00	17.58	17.59	0.00
2017.03.20 16:58:44.141	35.14	35.17	0.00	17.57	17.58	0.00
2017.03.20 16:58:45.202	35.13	35.16	0.00	17.56	17.58	0.00
2017.03.20 16:59:14.933	35.15	35.18	0.00	17.57	17.59	0.00
2017.03.20 16:59:15.833	35.14	35.17	0.00	17.57	17.58	0.00

Stock Split Ratio

New Shares: 2 Old Shares: 1 Price: Down

Split From: 2019.05.10 18:12 To: 2019.05.20 18:12

Calculate Process Cancel

For Help, press F1

Select a trading instrument and a time interval, and click Request. Ticks appear in the list.

Set the split settings:

- Use "New Shares" and "Old Shares" to set stock split/consolidation ratio so that tick data can be transformed similarly.
- Set the rounding rule in case the number of digital places of a new price exceeds the value set in the symbol's Digits parameter. For example, when transforming the price of 35.15 with the ratio of 2:1, we obtain 17.575. When rounded down, the final price is 17.57, when rounded up, it is 17.58. Also, the "Standard" rounding option (standard rounding to the nearest integer) is available as well. For example, if the Digits is 2, the rounding is performed as follows: 17.234 -> 17.23, 17.235 -> 17.24.
- If necessary, you can specify the time interval in which the split will be performed, using the "Split from" and "To" fields. In this case the operation will only be performed for tick data in the specified range, while the remaining data will not be affected. Such an operation may be needed for performing a backdated split. If you want to split the entire available symbol history, leave these parameters unchanged.

After implementing all the settings, click Calculate to see the preliminary split results. New Bid, Ask and Last prices for each tick are displayed in the table. Click Process to execute a split.

- Splitting can take quite a long time. Do not launch the process during trading hours.
- Split is performed both for 1-minute and tick data, regardless of the section from which the process is launched. Only preliminary split results can be viewed separately.

The process can be tracked using the [history server journal](#).

2017.04.18 17:30:27.339	HistoryBase	#AA split 2 for 1 started, up rounding
2017.04.18 17:30:27.366	HistoryBase	#AA split for 2014.04.26 completed [1239 bars up
2017.04.18 17:30:27.406	HistoryBase	#AA split for 2014.04.27 completed [1052 bars up
....		
2017.04.18 17:34:28.429	HistoryBase	#AA split for 2017.04.17 completed [1438 bars up
2017.04.18 17:34:28.888	HistoryBase	#AA split for 2017.04.18 completed [874 bars upd
2017.04.18 17:34:28.888	HistoryBase	#AA split 2 for 1 finished [3497656 bars updated,

Split Scanner

Split scanner automatically analyzes the price history of trading instruments to identify possible stock splits. The scanner then adjusts the price history to generate a smooth chart corresponding to the instrument's current prices. During analysis, the system additionally checks an internal database of publicly known stock splits.

Go to the "Bid/Ask/Last Ticks" section and select  "Split Scanner" in the context menu. Next, select a group of symbols and click "Scan". Check the received split

MetaTrader 5 trading platform

points. The list will display:

- Found split date
- Price before and after split, their ratio
- Number of stocks before and after split, their ratio (split ratio)
- Overall conversion rate for prices and stocks, taking into account all splits found for the instrument

To customize the information displayed, use the context menu.

Records found in the internal list of known splits are highlighted in green.

Stock Split Scanner

Symbols: Stock Markets\USA\NYSE\NASDAQ(Other)*

Scan

Symbol	Date	Old Shares	New Shares	Old Price	New Price	Recalculate Charts
SIBN: SI-BONE, Inc.	2015.02.05	10	1	333.5	3309	☐
TRMK: Trustmark Corporation	1990.01.01	3	2	34.6	57.66	☐
YNDX: Yandex N.V.	2011.05.22	65	1	38.2	2398.2	☐
BSL: Blackstone / GSO Senior Floating R...	2019.04.26	1	3	151	49.75	☐
KIE: SPDR S&P Insurance ETF	2019.04.17	1	1.5	266.2	166.2	☐
CME: CME Group Inc.	2007.04.26	1	5	256.71	51.17	☒
EBAY: eBay Inc.	2007.04.27	1	2	66.32	27.25	☐
FAST: Fastenal Company	2019.04.17	1	2	64.28	31.47	☒
FFIN: First Financial Bankshares, Inc.	2019.04.26	1	2	57.54	29.3	☒
FRAN: Francesca's Holdings Corporation	2019.04.17	12	1	0.41	4.65	☐
GOGL: Golden Ocean Group Limited	2007.05.04	5	1	0.66	3.22	☒
HSO: Hudson Global, Inc.	2019.04.17	10	1	1.18	12.37	☒
K: Kellogg Company	2019.04.17	14	1	7.85	108.4	☐
PBYI: Puma Biotechnology, Inc.	2019.04.17	1	1.5	30.28	19	☐
STMP: Stamps.com Inc.	2019.04.17	1	2	83.3	37.31	☐
APRN: Blue Apron Holdings, Inc.	2019.04.26	15	1	0.55	8.4	☒

Process split Cancel

MetaTrader 5 trading platform

Check the "Recalculate Charts" box for the desired splits. Please note that there can be multiple splits for each instrument. Click "Process split", and the platform will automatically convert minute and tick data.

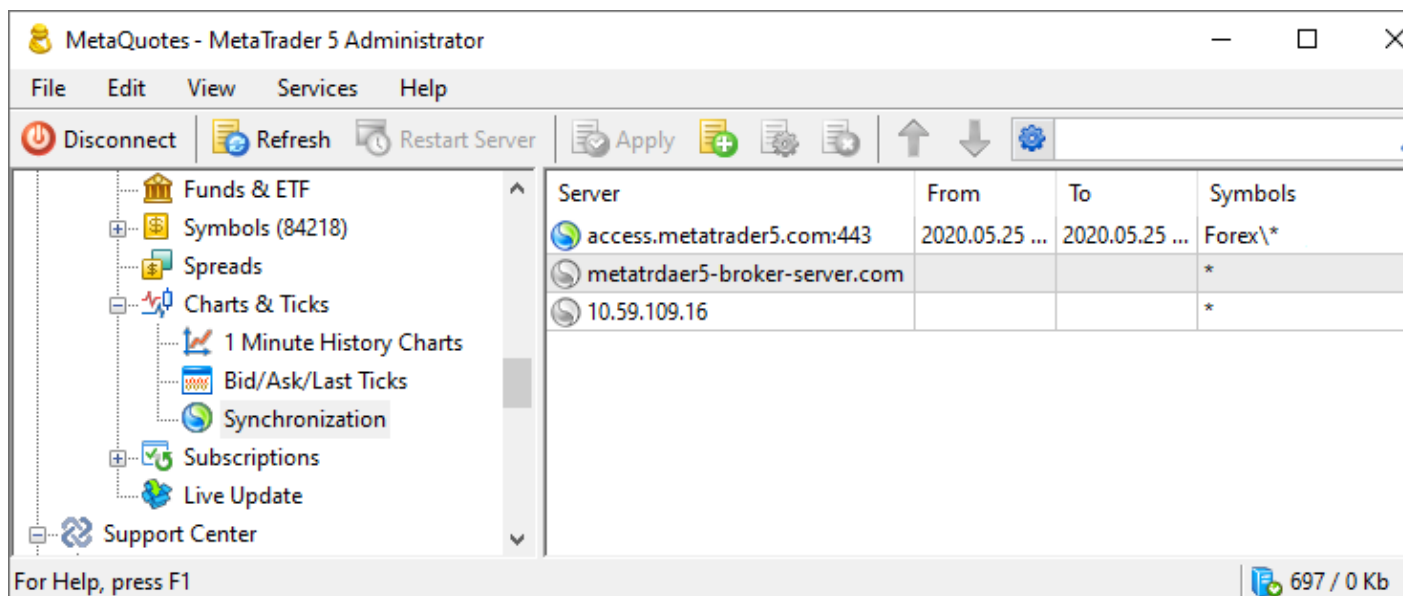
- Splitting can take quite a long time. Do not launch the process during trading hours.
- Split is performed both for 1-minute and tick data, regardless of the section from which the process is launched. Only preliminary split results can be viewed separately.

MetaTrader 5 Trading Platform / Platform Setup / Synchronization

Synchronization

Deep high-quality history is essential for traders. Historical data is used for technical analysis, as well as for creating and testing trading strategies. If you do not provide historical data, potential clients are likely to choose your competitors who do provide history.

With the MetaTrader 5 platform, you can easily solve the missing data issue. Minute bar and tick history on your server can be quickly synchronized with any other MetaTrader 5 and MetaTrader 4 server. For more details, please read the article [Deep Price History in MetaTrader 5 in a Few Clicks](#).



Setup

Data synchronization is performed through a regular client connection to an external MetaTrader 4/5 server. To access the data, you should open a demo account on the source server. The only special requirement for this account is ability to access trading symbols which your are going to synchronize. Connect to the external server via a client terminal using the created account to check all symbols available on this server. Next, execute the Symbols command in the context menu of the Market Watch window.

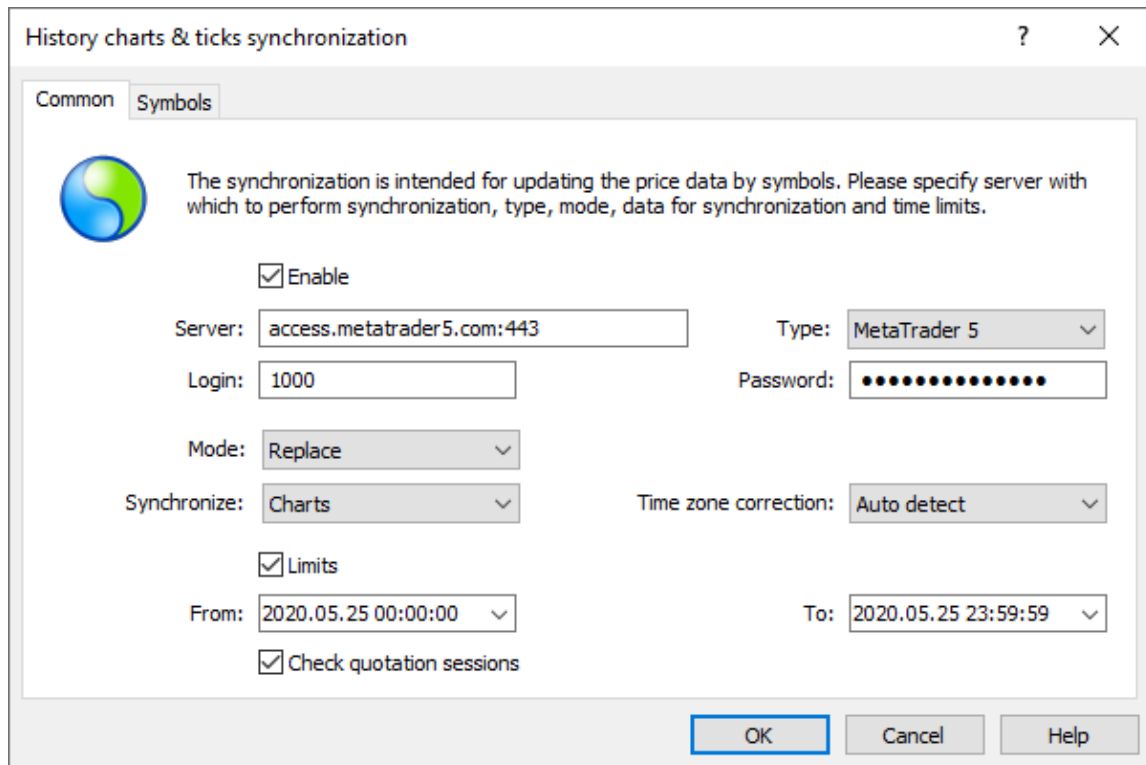
- Please note that price data is protected as intellectual property. Therefore you should make sure that you have all the necessary agreements before you proceed with the synchronization.
- By default, your platform is already ready to synchronize data with the MetaQuotes-Demo server, and the corresponding configuration is available in

the settings. This server stores deep history for 35 Forex pairs, 35 CFDs, as well as XAUUSD and XAGUSD metal price data. Historical data include Bid prices, spreads (which actually means the availability of Ask prices), and tick volumes. We recommend running synchronization by executing the "Synchronize History" command from the Services menu. This will provide the minimum required data to your traders. For more details, please read the article [Deep Price History in MetaTrader 5 in a Few Clicks](#).

Prior to starting synchronization, please make sure that the system time of your MetaTrader 5 server is accurate. The use of correct time will prevent discrepancies in quotes. Check platform settings under the [Time](#) section.

After completing the preparation, create a new configuration. Click "+Add" in the toolbar and proceed with the setup.

Common



The screenshot shows the 'History charts & ticks synchronization' dialog box with the 'Common' tab selected. The dialog contains the following fields and options:

- Enable:** A checked checkbox.
- Server:** A text field containing 'access.metatrader5.com:443'.
- Type:** A dropdown menu set to 'MetaTrader 5'.
- Login:** A text field containing '1000'.
- Password:** A text field filled with dots.
- Mode:** A dropdown menu set to 'Replace'.
- Synchronize:** A dropdown menu set to 'Charts'.
- Time zone correction:** A dropdown menu set to 'Auto detect'.
- Limits:** A checked checkbox.
- From:** A date/time field set to '2020.05.25 00:00:00'.
- To:** A date/time field set to '2020.05.25 23:59:59'.
- Check quotation sessions:** A checked checkbox.

At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Help'.

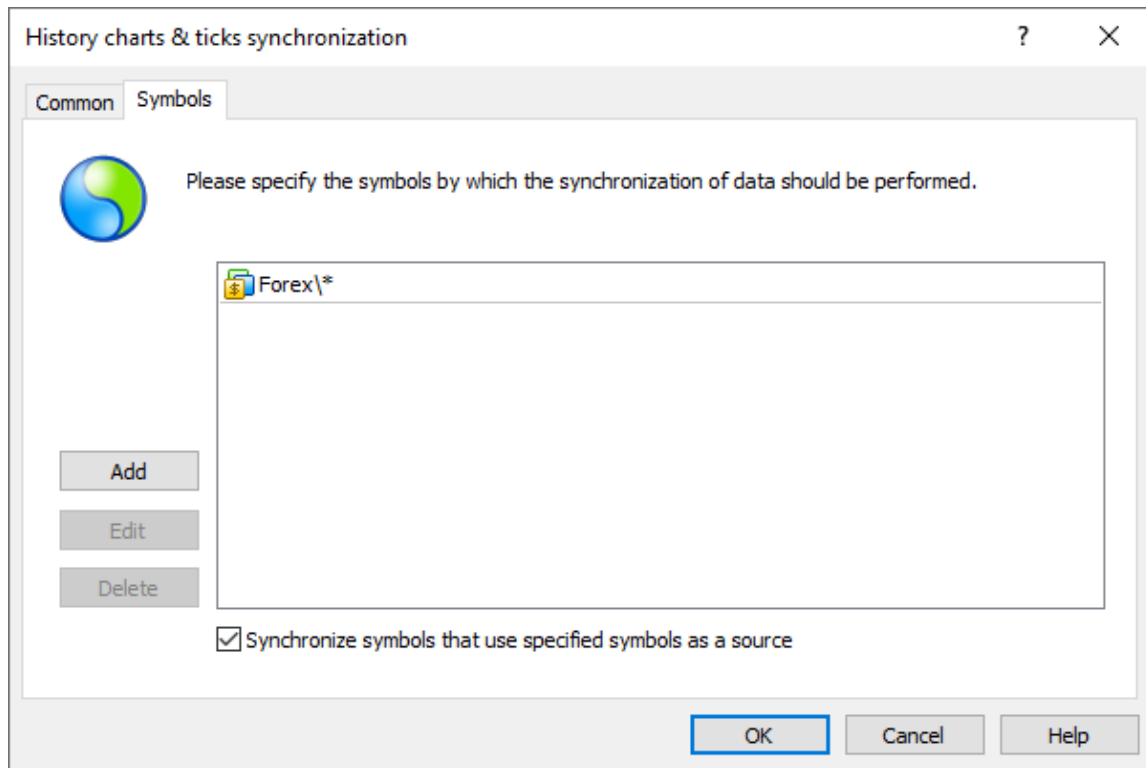
MetaTrader 5 trading platform

Specify here data for connection to the external server and synchronization parameters:

- **Enable** enable or disable synchronization with this server.
- **Server** server address and port separated by a colon.
- **Type** the type of the server, with which synchronization will be conducted - MetaTrader 4 or MetaTrader 5.
- **Login** the account number on the server with which you synchronize. Use a regular client account having access to required symbols, for connection.
- **Password** password to the account on the server with which you synchronize.
- **Mode** synchronization mode: Replace (replace all data and add what is missing), Merge (add only missing data).
- **Synchronize** data type to synchronize: charts, ticks or both.
- **Time zone correction** if MetaTrader servers are located in different time zones, the time in the received data can be corrected. The system can determine the time correction automatically. Optionally, the relevant value can be specified manually. Values from -24:00 to 24:00 are valid. Please see Time zone correction for further details.
- **Limits** time interval for history data synchronization. If this option is enabled, fields for entering starting and ending dates will be active. You can specify the date using your keyboard or a calendar that is opened at a click on button .
- **Check quotation sessions** if this option is enabled, then the quotation sessions set for the symbols in your trading platform will be considered during the synchronization of history data. If a source server has price data that falls outside of quotation sessions of your symbols, such data will not be synchronized (will be ignored). Thus only the price data within the set quotation sessions is imported in the platform.

When conducting synchronization, take into account specific features of this process for MetaTrader 5 and MetaTrader 4 servers.

Symbols



Specify here symbols, history data of which will be synchronized:

- **Add** add a symbol or a group of symbols. After you press this button, a new field will appear. A click on this field will open a list where you should specify a symbol or group of symbols from those available on the server. For more details please read "[Symbol specification](#)".
- **Delete** delete a selected symbol.
- **Edit** modify a selected symbol. The same action can be performed by a double click on an entry.

If you enable the option "Synchronize symbols that use specified symbols as a source", the system will additionally synchronize historical data for the symbols for which one of the selected symbols is specified in the [Source](#) field.

Synchronization launch

To start synchronization, click " Synchronize History" in the [Services](#) menu. The process will run for all [enabled configurations](#). If there is currently no need to synchronize with any server, disable its configuration before starting.

To control the process and to view the result, request server [logs](#) for the "Synchronization" keyword. The log appears as follows:

```
'1000': start history synchronization
history synchronization with 'access.metatrader5.com' started
'EURUSD' synchronization started
...
```

history synchronization with 'access.metatrader5.com' finished
--

Context Menu

The following commands are available in the context menu of Synchronization:

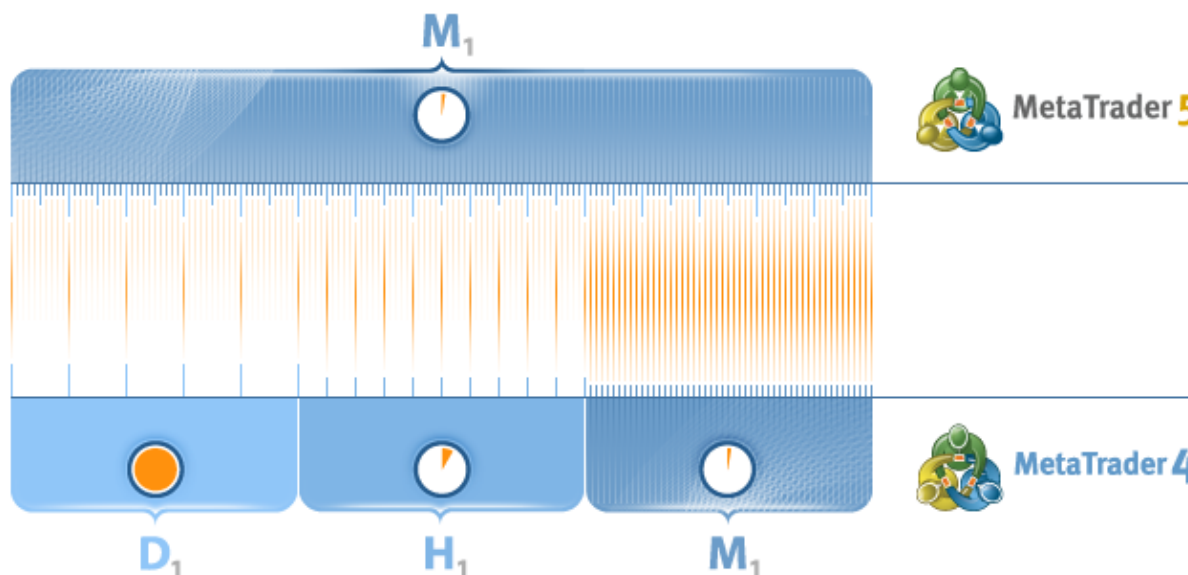
-  **Add** add a new synchronization server;
-  **Edit** change a selected synchronization server;
-  **Delete** delete a selected synchronization server;
-  **Move Up** move a selected server up relative to others;
-  **Move Down** move a selected server down relative to others;
-  **Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
-  **Enable** enable the selected configuration.
-  **Disable** disable the selected configuration.
-  **Export to File** export synchronization settings to a file.
-  **Import from File** import synchronization settings to a file.
-  **Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration automatically specified in the query field. You will only need to press the request button.
-  **Find** open the Search window;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators in the table with synchronization servers.

MetaTrader 5 Trading Platform / Platform Setup / Synchronization / Synchronization Features

Synchronization Features

There are some specific features of synchronization that should be taken into account before you start this process.

Specifics of synchronization of MetaTrader 5 and MetaTrader 4 servers



There is a considerable difference between synchronization with MetaTrader 4 and MetaTrader 5 servers.

- History data in MetaTrader 5 are stored only as 1 minute data and are converted to larger timeframes by request in the client terminal. In MetaTrader 4 different timeframes are stored.
- During synchronization with MetaTrader 4 servers, the successive selection of the most complete data starting from 1 minute timeframe is performed. First all M1 data are taken. Then it tries to obtain the missing part from H1 data. The hour value of the MetaTrader 4 server is written for the first minute on the MetaTrader 5 server. Then an attempt is made to obtain missing data from D1 timeframe. The daily data of MetaTrader 4 (e.g. 01.03.2009) are written for the first minute of the day of MetaTrader 5 (e.g. 01.03.2009 00:00). Therefore, clients can get a complete chart for the history period only on the timeframe, from which the data were imported.

To synchronize data with the MetaTrader 4 server, at least one demo group should be available on it. The group can be disabled.

Time zone correction

Synchronization settings provide the special function of correcting time zones. It allows to easily synchronize history data by MetaTrader servers located in different time zones. There are two variants of operation of this function:

- **Auto detect**

If this variant is selected, the time difference will be detected automatically. Besides, it will also be detected, whether the option of Daylight saving time is enabled on both servers. This parameter is taken into account only when small parts of history data are synchronized, because switch to DST may differ in different countries. At the synchronization of larger parts of data, the current time shift (+1 hour if it has been performed already) and is added to the time zone difference. History data are synchronized with the difference by the obtained time shift.

- **Setting a precise correction**

When this variant is chosen, history data will be shifted strictly by this value, not taking into account the DST.

To synchronize larger periods of history data we recommend using servers with disabled DST switch.

MetaTrader 5 Trading Platform / Platform Setup / Subscriptions

Subscriptions for Additional Trader Services

With the "Subscriptions" service, you can offer additional paid services to traders directly in the client terminals. For example, you can sell subscriptions for high-quality market data from well-known providers, offer personal manager services to assist traders in understanding the basics of trading, deliver one-time services such as position transferring or currency conversion, and much more.

Service Advantages

- A ready-made showcase of products which can be viewed by all your traders. You set up products via the Administrator terminal, and the subscriptions appear straight in client terminals. It means that you will not have to additionally attract traders to your site.
- Payment is performed directly from a trading account. Traders can easily pay for the product, as they do not have to think which payment methods to use. The easy procedure can increase the probability of a successful purchase.
- Automated delivery of market data and news. The trader starts receiving the data immediately after purchasing a subscription. Once you set up the subscriptions, you do not need to perform any additional manual actions. For more details, please refer to the article "Market data subscriptions in MetaTrader 5".
- Unified Database. Information on subscriptions is linked to trading accounts in the platform. You can easily view all services purchased by the trader and generate a report.

Set up subscriptions and offer more services to your traders.

35878581 - MetaQuotes-Demo: Demo Account - Hedge - [Subscriptions]

File View Insert Charts Tools Window Help

Algo Trading New Order

Navigator

- MetaTrader 5
 - Accounts
 - Subscriptions
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 - Indexes
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 - Scripts
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Common Favorites

North America

Search

Subscriptions / Market Data



NASDAQ Global Index Data Service

3.5 USD, monthly

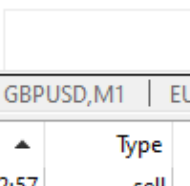
[Subscribe](#)



Canadian Exchange Group (TSX/TSXV)

20 USD, monthly

[Subscribe](#)



Canadian Securities Exchange (CSE)

Free

Symbol	Ticket	Time ▲	Type	Volume	Price	S / L	T / P	Price	Profit
eurusd	746740427	2020.10.23 11:52:57	sell	0.5	1.18395	1.19395 ×	1.17395 ×	1.18386	3.80
Balance: 99 991.13 EUR Equity: 99 994.93 Margin: 500.00 Free Margin: 99 494.93 Margin Level: 19 998.99 %									
gbpusd	746739728	2020.10.23 11:52:08	buy limit	0.5 / 0	1.30769	1.29769 ×	1.31769 ×	1.30881	placed

Trade Exposure History News Mailbox 6 Calendar Company Market 85 Alerts Signals Articles 2 Code

For Help, press F1 Default

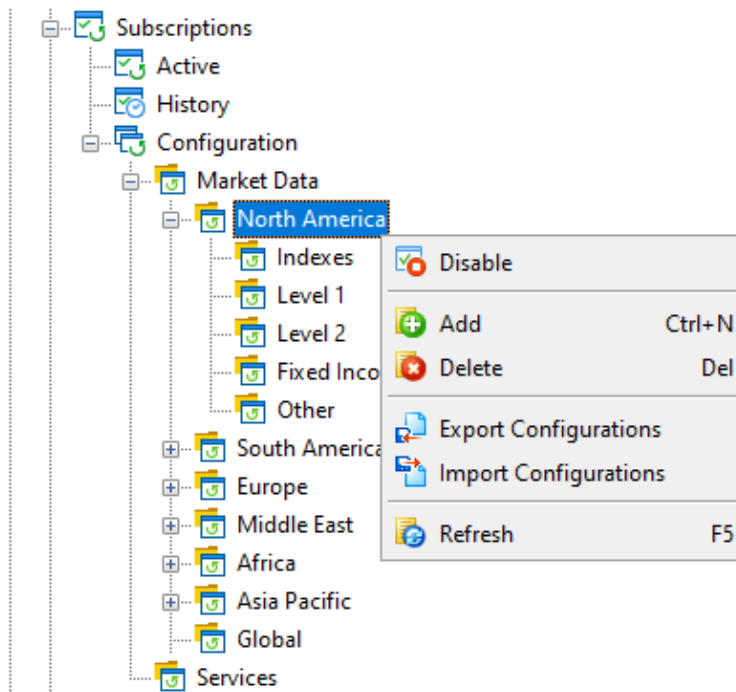
Setting up subscriptions

Go to the new Subscriptions service in the Administrator terminal, create a configuration and set the following:

- Subscription price and period
- Product description for a showcase in client terminals
- Subscription access permissions for certain groups and countries
- Trading instruments for market data subscriptions
- News categories for news subscriptions

The platform features a few ready-made examples, which can help you in understanding the service operation and setup principles.

Before setting up specific configurations, we recommend defining the service structure and creating appropriate categories. For example, "Market Data" and "Manager Services". Categories can be created using the context menu in the sections tree.



Subscription Payments

Subscription fees are paid directly from the trading account balance. Traders will not need to visit other websites, as the payment can be made straight in the platform.

A payment is deducted from the account as a balance operation of "charge" type. The deal comment will specify "Subscription 'Subscription name'".

To start receiving data, the trader purchases a subscription in the terminal:

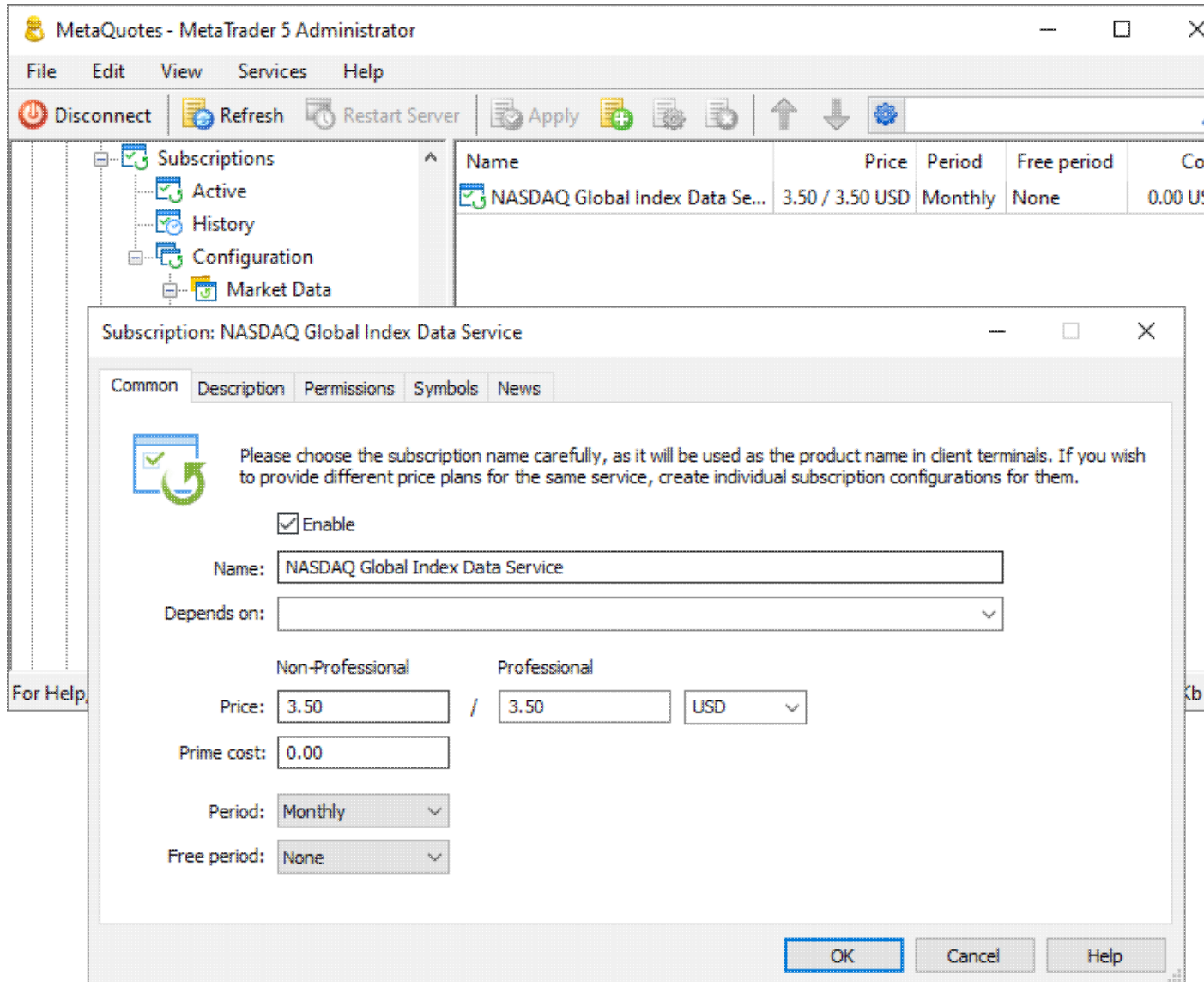
- At this moment, the appropriate amount is debited from the user's trading account balance (for paid subscriptions).
- An appropriate subscription record is created, indicating the subscription expiry date.
- If automated renewal is enabled for a subscription, an attempt to deduct the appropriate amount from the account will be performed upon expiration. If the attempt is successful, the subscription will remain active.
- If the required amount is not available on the account, the suspended status is set for the subscription. A repeated payment attempt will be made an hour later. Repeated attempts are made within a day. If a payment cannot be made during this time, the subscription is canceled.

Since the payment can only be made from an account balance, subscriptions are only available for real and preliminary accounts.

MetaTrader 5 Trading Platform / Platform Setup / Subscriptions / Common

Common

Set the service name, price and payment frequency in the main section. Choose a clear and understandable name for the service.



The following settings are available:

- **Name** the name of the service. It should be short and straightforward. This name will be displayed as the product name in client terminals.
- **Depends on** sets if another product subscription is required in order to subscribe to the current product. Using this option you can create product packages. For example, you can create a paid subscription for US market price data, and a dependent free subscription for Asian market data. In this case, traders having the first subscription will be able to receive the second one for free.
- **Price** subscription cost. Another function enabling separate pricing for non-professional (ordinary traders) and professional (legal entities, brokers, banks) clients is currently under development. The price from the Non-Professional fields is currently used for all subscriptions.
- **Prime cost** specify your expenses relating to the service provision. For example, if you are reselling market data, you can specify here how much you are paying for the subscription. This will allow you to build reports on the service profitability.
- **Period** the period of the subscription. If the subscription is canceled before the end of the paid period, data delivery will stop immediately, while the cost of unused time will not be refunded.
- **Free period** subscription trial period. The trading account owner can use the free period once. If the subscription period is extended after the free period (including automatic renewals), the appropriate amount will be paid from the user's account.

To quickly create similar services, use the "📄Add copy" command of the context menu. Instead of creating each service from scratch, create a copy of the existing one and adjust the required parameters.

MetaTrader 5 Trading Platform / Platform Setup / Subscriptions / Description

Description

In this section, you should provide a detailed description of a subscription, including its benefits for traders. This information will be displayed in the subscriptions showcase in client terminals.

In the "URL" field, add a link to more detailed information about the service. Select a logo you can access images for the most popular services, as well as for well-known data providers. In the future, you will be able to upload your own logos.

Subscription: NASDAQ Global Index Data Service

Common
Description
Permissions
Symbols
News


Please specify image URL or choose one of the predefined images for the product logo in client terminals. Describe all the benefits of the product in details.

URL:

Image:  NASDAQ

Tahoma 2 (10 pt)









B
I
U






Actual and historical index data for the diverse set of NASDAQ global indexes.

NASDAQ is an American stock exchange for high-tech company stocks. It was founded on February 8, 1971. More than 3,200 companies are currently listed on the NASDAQ. It is considered one of the major stock exchanges in the U.S., the other one being NYSE. The NASDAQ operation is regulated by SEC. The exchange is owned by NASDAQ OMX Group, which also owns 8 European exchanges.

Global Index Data Service (GIDS) is a premier global index data feed service, offering real-time updates, daily summary messages, and access to the most widely followed indexes and ETFs in the world. Data available for indexes:

- Intraday Net Asset Values (NAVs)
- Settlement values
- Currency Spot values for select instruments
- Daily security weightings and components data
- End of day summary data

OK
Cancel
Help

MetaTrader 5 Trading Platform / Platform Setup / Subscriptions /
Permissions

Permissions

Specify here subscription access permissions: to which client groups and countries the subscription will be available.

Subscription: NASDAQ Global Index Data Service

Common Description Permissions Symbols News

 Manage access to the product here: set permissions to limit subscriptions, use countries and groups as a product filter on the client side.

☒ Auto prolong subscription

Agreement URL:

Permissions:

Countries:

Groups:

-  real\\forex-usd
-  real\\stocks-usd

Add

Edit

Delete

OK Cancel

Specify the following settings:

- Auto renew subscription this option enables automated subscription renewal upon expiration. During renewal, the subscription amount will be automatically debited from the client's trading account.
- Agreement URL you can specify here a link to additional Agreement/Rules to be accepted by the trader. An agreement link will be displayed in the subscription confirmation dialog in client terminals.
- Permissions select the subscription actions available to traders:
 - Full the client can subscribe and unsubscribe from the service.
 - Unsubscribe the client can only unsubscribe from the service. The subscription can only be created through the Manager terminal or the API. If the user is not subscribed to a service with this permission, the appropriate subscription will not be shown to the trader in the client terminal.
 - View the client can view the service in the terminal, but subscribe and unsubscribe options are only available through the Manager terminal or the API. If the user is not subscribed to a service with this permission, the appropriate subscription will not be shown to the trader in the client terminal.
 - Hidden clients cannot view the service in the terminal, even if they are subscribed to this services (the subscription fee is charged from the trading account, as for regular subscriptions). The subscription can only be created through the Manager terminal or the API.

Access to subscriptions by country and group

You can create subscriptions for traders from certain countries. To limit access, specify the list of countries in the appropriate field. To select/deselect all countries in the list, right-click on the list.

Note:

- When opening a demo account from the client terminal, the client's country is determined automatically.
- When opening a preliminary account, the country is also determined automatically, but the client can specify another country in the registration form.
- When creating a real account, you specify the country manually.

Similarly, you can limit access, allowing subscriptions only for clients from individual groups.

Restrictions by countries and groups also affect the ability to subscribe through Manager terminals or the API. If the service is not available for the trader, there is no way to subscribe to it.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Subscriptions](#) / Symbols

Symbols

For market data services, set the list of trading instruments in the Symbols section. After purchasing a subscription, the trader will have access to these symbols in Market Watch, including their real-time quotes and available price history.

Preparatory Steps

Before you start offering market data to your traders, you should configure receiving of appropriate symbol data from liquidity providers. The process is similar to adding trading symbols available to traders in the platform:

- Create the required [symbols](#)
- Set up [gateways](#) and/or [data feeds](#) to receive appropriate symbol data.

- To start offering market data, you will need to sign an agreement with appropriate exchanges and trading systems, as the data is proprietary.
- The support website [App Store](#) features a variety of solutions enabling receiving of data from popular liquidity providers.
- If you are further transmitting subscription data between your servers using gateways or data feeds, enable the "[Allow access to subscription data via data feeds](#)" permission in the settings of the trading account you are using for connecting to the source server.

Setup

MetaTrader 5 trading platform

Subscription: NASDAQ Global Index Data Service

Common

Description

Permissions

Symbols

News



If you are setting up a subscription for price data, specify a list of financial instruments for it. Make sure these symbols are available for the corresponding client groups.

Symbol	Level	Available ticks history
 NASDAQ*	Level 2 realtime	All
	15 minutes delayed	
	Level 1 realtime, best Bid/Ask	
	Level 2 realtime	

Add

Edit

Delete

OK

Cancel

Help

MetaTrader 5 trading platform

Specify the symbols and/or symbol groups which data will be available by subscription. Specify, which data will be available for each of the symbols or groups:

- Level data type that will be available by subscription:
- Delayed data in the Market Watch in client terminals will be displayed with a delay. Depth of Market and tick history will not be available.
- Level 1 realtime, best Bid/Ask data in the Market Watch will be displayed in real time. Depths of Market will not be available.
- Level 2 realtime the subscriber will receive in real time the best Bid/Ask prices in the Market Watch, as well as the Depth of Market data.
- Available tick history the tick history depth that will be available by subscription.

To provide Depth of Market or tick history by subscription, the relevant data must be available in your platform (must be provided by the gateways/data feeds that you use).

Access to symbols on the client terminal side

Subscription limit general access to trading symbols available in the platform. If a symbol is not used in subscriptions, this symbol can be accessed in a regular way in accordance with the settings of the group to which the account belongs.

If a symbol subscription exists in the platform, then access to the symbol data (quotes, history, Depth of Market, tick history) depends on whether the subscription is available to the trader in accordance with the subscription permission settings.

- If permission settings include the trader's group, then the trader will be able to access the real-time symbol data by subscription. Without a subscription, the trader will only receive delayed data. It means that you do not disable access to symbol data by setting up a subscription. Without a subscription, the trader can still receive the symbol data, but only with a delay.
- If there are no subscriptions available to this client group, the trader will be able to receive the relevant market data without a subscription. The access will be provided in accordance with the group settings.

Example: you have a EURUSD symbol subscription with access for the 'real-usd' group.

- A trader from the 'real-usd' group will need a subscription to receive the EURUSD data.
- Traders from any other groups will receive the EURUSD data in accordance with the appropriate group settings, without a subscription.

If the trader has multiple intersecting subscriptions, the maximum subscribed data will be available. For example, if a trader has a subscription for the delayed EURUSD data and a subscription for EURUSD Level 2, then Level 2 data will be available to the trader.

Subscriptions restrict access to data that is originally available to the user group. For example, if only Forex* symbols are available to a trader's group, then no subscription can provide access to data from the Stocks* group.

License Control

Each subscription allows up to three connections: one from a desktop terminal, one from a mobile terminal, and one from a web terminal. Additional connections will receive subscription data with a 15-minute delay. Accounts connected in investor mode will always receive delayed data and will not have access to tick history.

Delayed data

You can provide market data with a minute delay. For example, this can serve as a free trial version of your service. A certain delay in quote delivery can also be demanded by some exchanges providing data.

The default delay is 15 minutes, however you can override this value for each trading instrument:

Symbol: Forex\EURUSD ?

Common Currency **Quotes** Trade Execution Margin Margin Rates Swaps Sessions

 The setting up of different filtration levels is intended for cutting off the incorrect data. Please specify the deviation of quotes and the repetition for filters to trigger.

☒ Allow realtime quotes from datafeeds ☐ Allow negative quotes

☐ Receive market statistics from datafeeds ☐ Save raw prices

Soft filtration level: points (must not be less than 4 or 5 times spread by default)

Filter: wrong quotes coming one after another

Hard filtration level: points (must not be less than soft filtration level)

Filter: wrong quotes coming one after another

Discard filtration level: points (must not be less than hard filtration level)

Gap mode level: points Disable gap after: ticks

Minimum spread: Maximum spread: points

Delay for subscriptions min

OK Cancel

MetaTrader 5 trading platform

The delay is specified at this level, because exchanges can have different requirements for each trading instrument. Therefore, you cannot create multiple subscriptions for the same trading instrument with different delay values. The value specified in symbol settings will be applied for all subscriptions connected to this instrument.

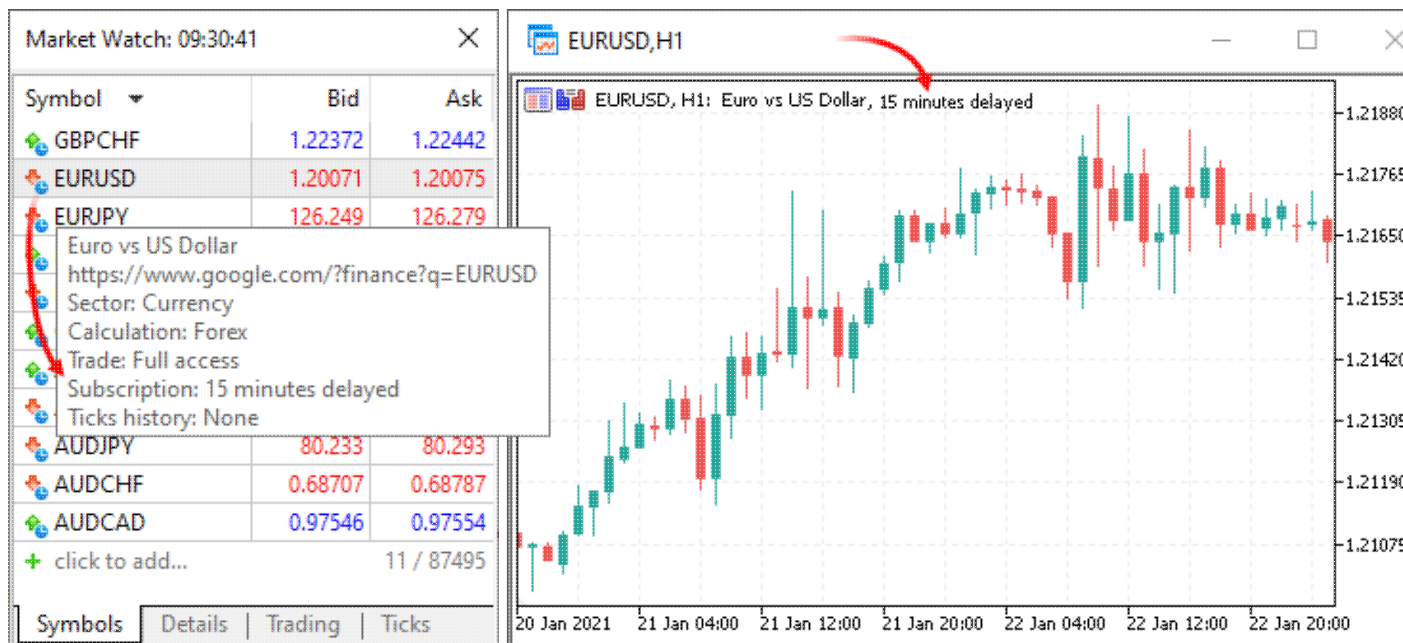
A restart of access servers is required for new delay settings to take effect.

This works as follows on the client terminal side:

- Quotes in the Market Watch will be received with a delay.
- Charts will be updated with a delay.
- The current prices for positions and orders in the Trade section will be updated with a delay.
- Profit by position and account balance will be updated at current prices, without delay.
- The tick history is only available for the current terminal operation session (deep history is not available, regardless of the subscription settings).
- Depths of Market is not be available.

A special indication is used in client terminals to inform traders about which data is provided with a delay:

- Symbols in Market Watch are marked with a clock icon, and the delay value is displayed in the tooltip
- Data delay notification is displayed on charts



MetaTrader 5 Trading Platform / Platform Setup / Subscriptions / News

News

For news data services, set the news categories in this section.

Preparatory Steps

Before you start offering news subscriptions, you should configure news delivery via data feeds. The news categories specified in subscription settings must be available in the platform.

- To start offering news, you will need to sign an agreement with appropriate agencies, as news constitute proprietary data.
- The support website App Store features a variety of solutions enabling receiving of news data from popular providers.

Setup

Subscription

Common

Description

Permissions

Symbols

News



If you are setting up a subscription for financial news, specify a list of news categories. Make sure the categories are available for the corresponding client groups.

Category	Language
 premium\finance	Any
 premium\technical-analysis	Any
 premium\fundamental	Any

Add

Edit

Delete

OK

Cancel

Help

In the settings, specify those news categories, which will be available by subscription.

- Usually, news categories are provided by data feeds. News providers supplement news with the information about categories to which the news belong. The same applies to the news language.
- A news category can be additionally specified in the News Category parameter of the data feed. This is a standard parameter supported by all data feeds. The parameter is applied as follows:
 - If a data source does not provide news categories, news items will be added to the category specified in this parameter (the category will be specified in the news).
 - If the data source provides a category, then the data feed will add the "News Category" parameter value before the original category name. The resulting category will look like "[value from News Category] \ [category from data source]". The "\" characters in a category name are interpreted as subcategory separators. Therefore, it means that all news items are added to the category specified in the parameter, within which original categories from the data source will be used.

Access to news on the client terminal side

Subscriptions limit access to news in the platform. A subscription is needed to access news if the following conditions are met:

- There is a subscription with a list of groups containing the trader group.
- There is a subscription, having the news category and language in the "News" section.

If there are no appropriate subscriptions in the platform, it is considered that the subscription is not required and the news will be delivered in a normal way, in accordance with the parameters of the group to which the account belongs.

MetaTrader 5 Trading Platform / Platform Setup / Subscriptions / Controlling Subscriptions

Subscriptions Control

The service usage can be monitored using the Active and History sections. The sections feature active subscriptions and events concerning all subscriptions, respectively.

MetaQuotes - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Security Time Holidays Groups (731) Clients & Accounts Orders & Deals Gateways Data Feeds Plugins Reports ECN Routing Funds & ETF Symbols (87377) Spreads Charts & Ticks Subscriptions Active History Configuration Market Data Services Live Update

ID	Login	Subscription name	Status	Expiration time
46	3133346	Forex - Delayed	Active	2020.11.05 17:06:
9	3133347	Forex - Level 1	Active	2020.10.24 10:20:
30	3133348	Forex - Level 2	Active	2020.10.23 15:28:
48	3133348	Forex - Level 1	Active	
69	3133348	Forex - Level 2	Active	2020.10.09 11:14:
70	3133348	NASDAQ Global Index Data Service	Active	2020.10.24 11:32:
43	3133416	Forex - Delayed	Active	2020.11.05 17:02:
112	3133416	NASDAQ Global Index Data Service	Active	2020.10.23 16:05:
54	3133492	Personal manager	Active	2020.11.06 10:23:
50	3133492	Canadian Securities Exchange (CSE)	Active	2020.11.06 10:16:
53	3133492	Johannesburg Stock Exchange	Active	2020.11.06 10:22:
55	3133492	Forex - Delayed	Active	2020.11.06 11:15:
76	3133492	Personal manager	Active	2020.11.07 17:38:
56	3133493	Forex - Level 1	Active	2020.10.24 11:16:
57	3133494	Forex - Level 2	Active	2020.10.24 11:16:
63	3133496	NASDAQ Global Index Data Service	Active	2020.11.06 15:32:
73	3133496	Personal manager	Active	2020.11.14 14:21:
108	3133498	NASDAQ Global Index Data Service	Active	2020.10.23 15:55:
109	3133498	Personal manager	Active	2020.10.23 15:57:
111	3133498	Forex - Level 1	Active	2020.10.23 16:03:

Group or Accounts (1,2,...,1000) Request

For Help, press F1 5749 / 4 Kb

Request subscriptions from the server to view them:

- By trader group select a group from the list or specify the group name manually.
- By login specify one or several accounts separated by commas.

Active Subscriptions

The following information is available for each active subscription:

- ID the internal identifier of the subscription configuration.
- Login subscriber's account number.
- Subscription name the name of the subscription.
- Status current subscription status:
 - Active the subscription is active and the trader is receiving the service.
 - Suspended the subscription has been suspended. This status is activated if the subscription renewal fee cannot be payed from a trading account.
 - Canceled the subscription has been canceled.
- Subscription time the date and time when the subscription was activated.
- Renewal time the latest renewal date and time.
- Expiration time subscription expiration date and time.

Using the context menu, you can quickly jump to viewing the subscribed account and the subscription configuration.

Here you can edit the parameters of an active subscription:

- Service and subscription status. If you change the subscription state to "Canceled", this subscription will not be available in the list during the next request of active subscriptions.
- Free period. This is an indication that the user has received the subscription as a free trial. If you disable this option, the user will be able to use a free period once again at the next subscription.
- Subscription, next renewal and expiration dates. When the subscription period is changed, no additional fees are debited from the account.

Active Subscription #54

Login: 3133492

Subscription: Services\Personal manager

Price: 15.00 USD

Status: Active

☐ Free period

Subscribed: 2020.10.07 10:23:08

Renewal: 2021.02.04 10:23:20

Expiration: 2021.03.06 10:23:08

Update Cancel

Delete and unsubscribe

When a subscription is canceled or deleted, the user will no longer be able to use the service. However, these actions have fundamental differences:

- When a subscription is canceled, the relevant information remains in the database, and the history will contain a separate record about subscription cancellation.
- When deleted, the whole subscription record is removed from the database. If you delete a trial subscription record, the trader will be able to use the free period again, since there will be no record in the database that it has already been used.

Both actions can be performed using the context menu of the active subscriptions section.

The Manager account needs appropriate permissions in order to access subscription editing options.

Subscriptions History

All actions related to subscriptions can be viewed under the "History" tab:

- ID the internal identifier of the subscription configuration.
- Login subscriber's account number.
- Subscription name the name of the subscription.
- Record Id the unique identifier of the subscription action.
- Action action type:
 - Subscription subscribing to a service.
 - Renewal renewal of an existing subscription.

MetaTrader 5 trading platform

- Suspension subscription suspension if the relevant renewal fee cannot be paid from the user's account.
- Cancellation subscription cancellation by the user.
- Deletion subscription deletion by the system, for example when the appropriate configuration is deleted.
- Amount subscription cost.
- Deal Id the identifier of the deal by which the appropriate subscription fee amount was charged from the account.

To adjust the amount of information displayed, use the context menu. Using the menu, you can also quickly jump to viewing the subscribed account and the subscription configuration.

If you have appropriate permissions, you can edit actions in history. To do this, double-click on the line.

The following parameters can be edited:

- Service and type of action. Changes do not affect active trader subscriptions. If you change the action type from "Cancel" to "Subscription", the trader's subscription will not be restored.
- The date of the action.
- Subscription fee. The change does not affect the deal by which the subscription payment was conducted.

Subscription History #2055

Login: 3133348

Subscription: Market Data\Forex - Level 2

Price: 0.10 USD

Action: Renewal

Created: 2021.03.01 15:28:52

Amount: -0.10 USD

Deal: 14126874

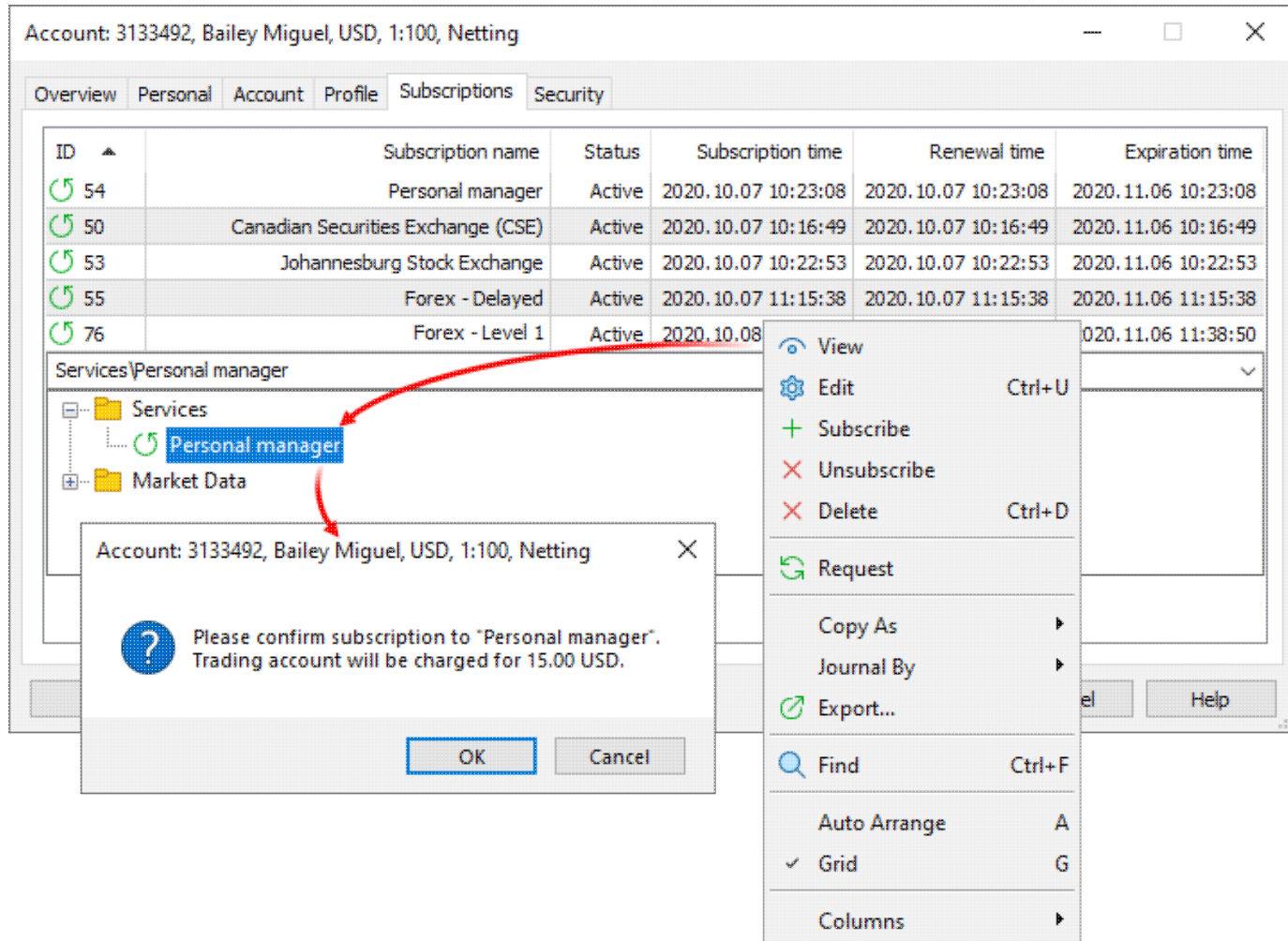
Update Cancel

Trader Subscriptions

To view subscription data of a specific trader, open his or her account and navigate to the "Subscriptions" tab.

To navigate to general subscription settings, double-click on the line.

From this section, you can add a new subscription or delete an existing one:



MetaTrader 5 trading platform

When adding a paid subscription, the system will notify the user that the payment for the subscription will be debited from the trading account.

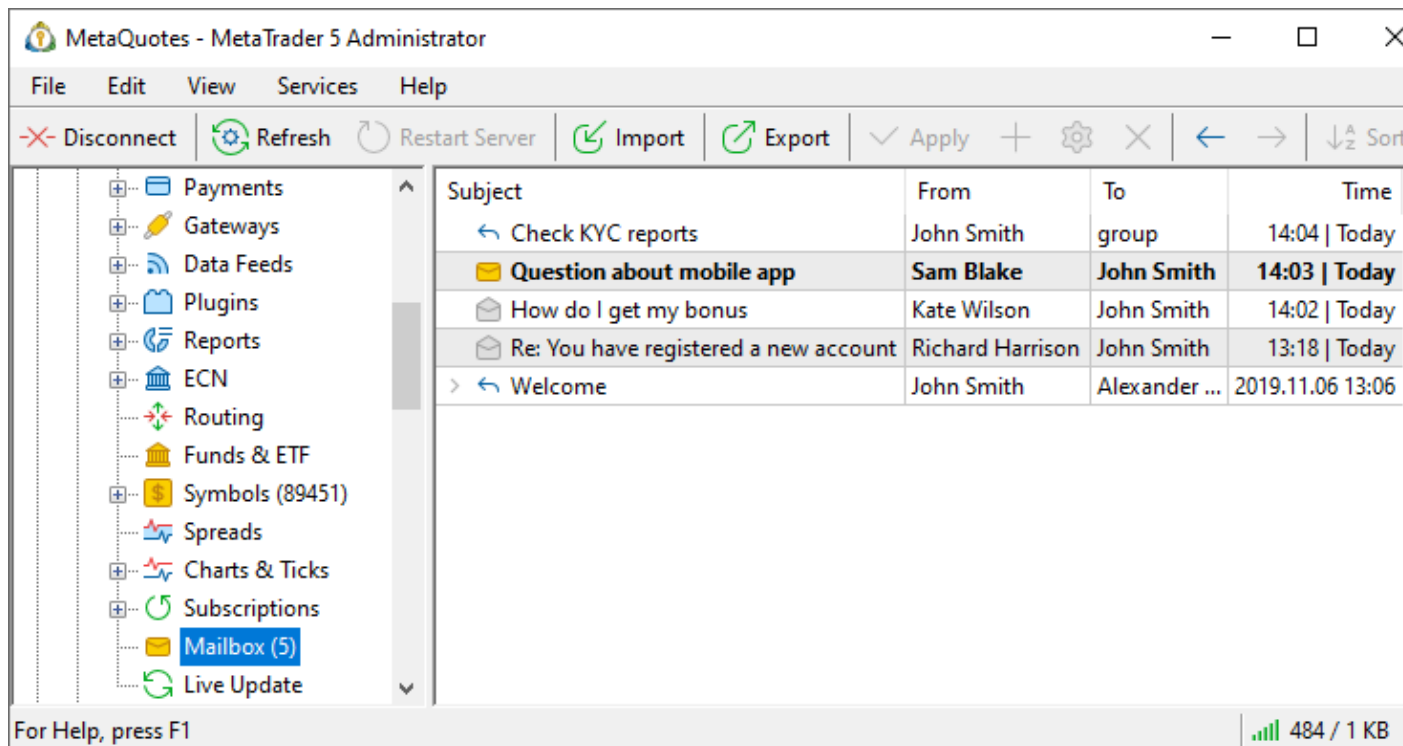
MetaTrader 5 Trading Platform / Platform Setup / Mailbox

Mailbox

The MetaTrader 5 trading platform features an internal email system which enables communications with clients and between employees. Using emails, you can effectively send important information about your services and company events to traders. The email system is integrated directly into trading terminals. When a message is received in the terminal, a sound notification is played, and thus the trader will not miss important information. In addition, the built-in email system allows you to reach potential clients for whom you do not have reliable contact information, such as phone number or email addresses. If a trader has opened a demo account and connected to your server, your managers can contact the trader offering a real accounts.

Using the Automations service, you can automate any mailings. For example, send promotional materials immediately after opening an account. This will save employee time, eliminate human errors, and increase customer conversion.

All emails received and sent from your account are displayed under the "Mail" section:



The screenshot shows the MetaQuotes - MetaTrader 5 Administrator window. The left sidebar contains a tree view with various categories: Payments, Gateways, Data Feeds, Plugins, Reports, ECN, Routing, Funds & ETF, Symbols (89451), Spreads, Charts & Ticks, Subscriptions, Mailbox (5), and Live Update. The 'Mailbox (5)' item is selected. The main area displays a table of emails.

Subject	From	To	Time
← Check KYC reports	John Smith	group	14:04 Today
✉ Question about mobile app	Sam Blake	John Smith	14:03 Today
✉ How do I get my bonus	Kate Wilson	John Smith	14:02 Today
✉ Re: You have registered a new account	Richard Harrison	John Smith	13:18 Today
> ← Welcome	John Smith	Alexander ...	2019.11.06 13:06

At the bottom of the window, there is a status bar that says "For Help, press F1" and a network status indicator showing "484 / 1 KB".

Each email contains the subject, sender, recipient, and time of sending/receiving.

Unread mails are indicated by icon , while read ones by . In order to start viewing the message, click in the field of its subject with the left mouse button. Outgoing mails are marked with the  icon. When using the function of replying to a message, mails are grouped in a chain which allows to navigate through mailing with different clients easily. Email threads have an additional icon . Click on it to view the entire conversation.

Context Menu

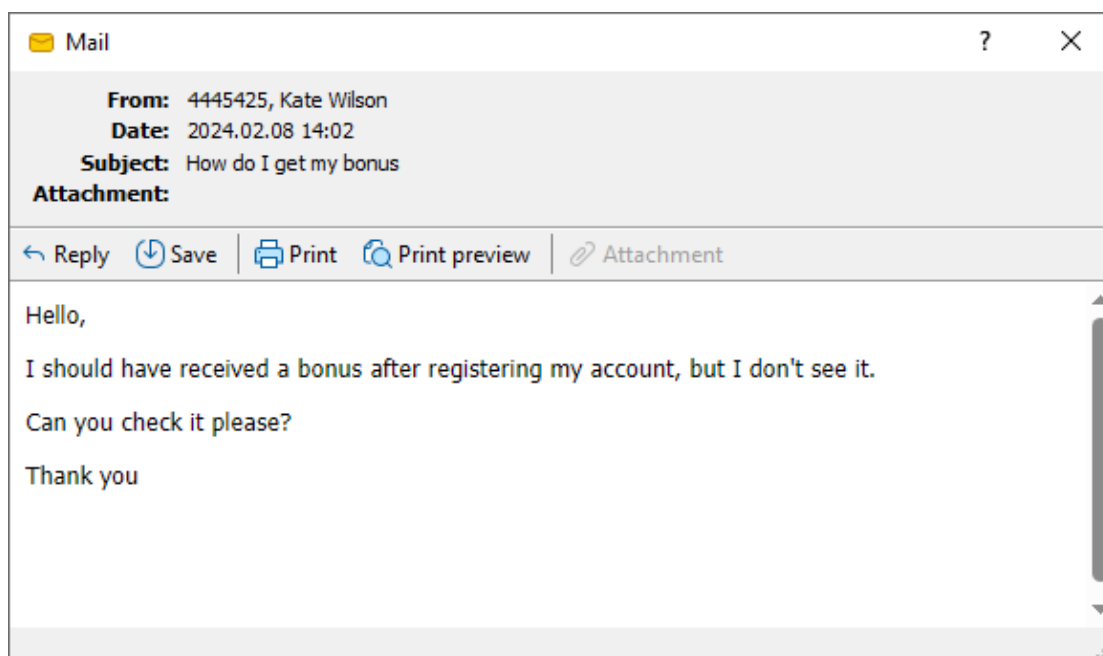
The context menu of the "Mailbox" tab contains the following commands:

- **+ Create** create a message;
- ** View** open a selected mail;
- ** Delete** delete a selected message;
- ** Expand** expand a selected mail chain;
- **Auto Arrange** if this option is enabled, the size of columns is selected automatically;
- **Grid** this option shows/hides grid to separate fields of the mail table.

- The command of creating a mail is active only if the mailbox name is specified in the settings of the manager's account.
- Mails are stored at the trade server. If a mail is deleted in the terminal interface, it won't be re-downloaded. However, if you delete the mail base of the terminal (the file `"/profiles/server_name/mail/mail-account_number.dat"`) or connect using another terminal, all the mails for the last 30 days will be downloaded again.

Viewing Messages

To start viewing a message, click with the left mouse button in the field of its subject. After that the following window will be opened:



The upper part of the message contains the client's account and name, the date of mail coming, its subject and attached files (if there are).

The toolbar of this window contains the following commands:

- **Reply** open the window of writing a message with filled address of recipient and the quotation of received message;
- **Save** save the message on a computer as a HTML file or a text file of the Unicode standard;
- **Print** print the message;
- **Print Preview** view the message before printing it;
- **Attachment** save files attached to the message. Also, an attached file can be opened and saved by simply clicking on its name.

Configure the directory, to which email attachments will be saved. When you open an attached file, the terminal checks its extension and the corresponds of file contents to this extension. If the file type is allowed and its contents are checked, the terminal will open the file and save it to the specified directory. Otherwise, a warning will be displayed, notifying that the file may harm the computer, and the file will not be saved. The following file types are allowed: PNG, JPG/JPEG, BMP, ZIP, 7Z, GIF, DOC, XLS, DOCX, XSLX, ODT, RTF, CSV, TXT, LOG.

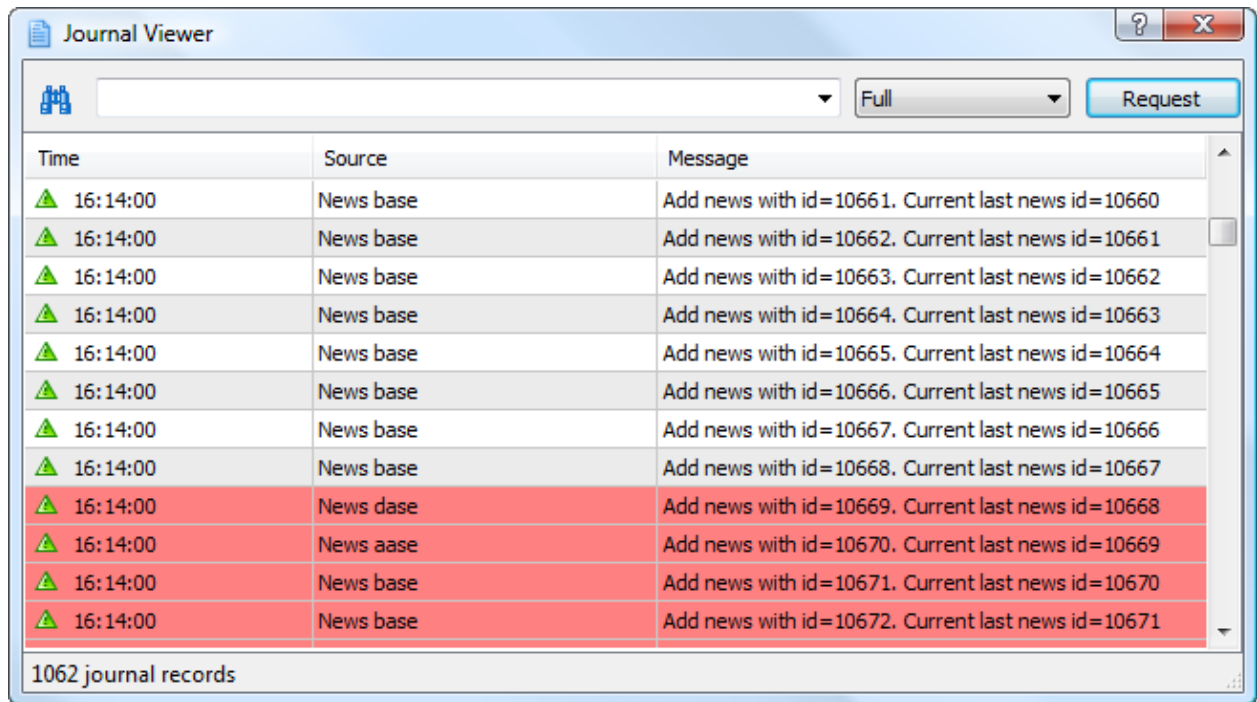
If a file with the same name and a different content exists in the directory, the manager's login and email arrival date/time up to a second will be added to the saved file name: [login]-[file name]-[date and time].[extension]. For example, the log file 20170501.log will be saved as 1001-20170501-20170502-170038.log.

By default, files are saved to C:\Users\[Windows username]\Downloads\MetaTrader.

Viewing Attached Log Files

MetaTrader 5 trading platform

If a client's message has a log file (a file that has the *.log extension) attached then one can view it using a special function. To do it, it is necessary to click on its name in the "Attachment" field. The following window will be opened:



The upper part of the window of viewing contains the line of searching through the log (the search is performed only by the exact words and it is case sensitive) and the filter of entries (Standard, Errors, Full). Once having the searched word and filter specified, one should press the "Request" button.

If a client has changed even one entry of the journal then all the entries starting from it are shown with the red color. At the attempt of saving such a file, the warning that says which line in it was changed, is shown.

The context menu of the window of viewing the journal allows to execute the following commands:

- **Copy** copy selected entries to the clipboard;
- **Save** save the received logs to the disk as a text document;
- **Auto Arrange** if this option is enabled, the size of columns is selected automatically;
- **Grid** this option shows/hides grid to separate table fields.

Sending Messages

In order to be able to send messages it is necessary to:

- Enable corresponding option in the manager's group settings;
- Grant manager the permission of sending mails;

- Specify the name of manager's mailbox.

To create a message, one should press the "+ Create" button of the context menu or use the "Insert" hot key having the "Toolbox" window active. The window of creating the mail will appear as soon as it is pressed:

In this window one should fill out the following fields:

- **To** the account number of the message recipient.
- **Subject** title of the message.
- **Template** in this field one can select a mail template.
- **Internal mail** select this option to send a message using the internal mail system.
- **Email** use this option to send an email. The address specified in account personal data will be used. Please note:
 - A mail server must be configured in the platform to enable email sending.
 - Messages sent by email are not displayed under the "Toolbox \ Mail" section. To check the sending, request the trade server journal using the 'email' keyword. If an email has been sent successfully, the recipient address and the configuration name of the mail server through which the email has been sent, are shown in the log: email to someaddress@mail.net sent [MailServer].

The attachment of files has the following limitations:

- The size of one attached file cannot exceed 2MB;

- The total size of attached files cannot exceed 4MB;
- Up to 5 files can be attached.

The window for working with the mail text is located lower. The editor features commands for creating lists, as well as inserting images, links and tables. To view or edit the source HTML code of the message, click  on the toolbar.

The context menu of the window of text editing contains the standard commands of working with text: "Copy", "Cut", "Paste", "Insert Link", "Insert Table" and "Insert Image". The context menu also allows to work with mail templates and macros.

To send the message, one should press the "Send" button.

Bulk Mailing

To send emails to multiple recipients, enter a range of accounts through a hyphen in the "To" field. For example, 1000-9000. You can also select multiple accounts in the list and click 'Email' in the context menu. The numbers of the selected accounts will be inserted into the "To" field.

When using Email instead of the internal mail, it could be more efficient to send emails to a list of clients rather than accounts. Clients often have multiple accounts that use the same email address. Because of this, when sending emails to a list of accounts, the client may receive several identical messages. To avoid this, send emails from the Clients section. Select the desired client records and click "Internal mail/Email" in the menu. The mailing list will only include unique email addresses from all the trading accounts linked to the selected clients.

Templates

To manage templates, use the context menu of the mail editing window (text editing area). Open the "Templates" submenu:

- **Save Template** save the current text of the message as a template in the *.htm format. All templates are stored in the /templates/mail folder of the administrator terminal;
- **Load Template** open the window of choosing a previously saved template for loading it;
- **Remove Template** delete the currently selected template.

When executing the "Remove Template" command a template is irrecoverably deleted from PC.

Macros

Using the "Macros" submenu of the context menu of the mail creating window, one can insert macros into the text. They allow to substitute various information depending on the message recipient. The following macros are available:

- Login (#LOGIN#) the email/messenger recipient's Account number.

MetaTrader 5 trading platform

- Name (#USERNAME#) the first name and the last name of the recipient.
- Currency (#USER_CURRENCY#) the recipient's deposit currency. The currency is determined by the group, in which the user account is created.
- Balance (#USER_BALANCE#) the recipient's balance.
- Credit (#USER_CREDIT#) the credit amount on the recipient's account.
- Equity (#USER_EQUITY#) the equity amount on the recipient's account.
- Leverage (#USER_LEVERAGE#) account leverage.
- Margin (#USER_MARGIN#) the amount of funds reserved on the recipient's account.
- Free margin (#USER_MARGIN_FREE#) the free margin amount on the recipient's account.
- Margin level (#USER_MARGIN_LEVEL#) the margin level on the recipient's account.

These macros substitute email sending time in the message text:

- Trading time time used in the platform, taking into account its settings.
- Local time time used in the computer on which the platform is installed.
- UTC time UTC time.

MetaTrader 5 Trading Platform / Platform Setup / Live Update

Live Update

The system of the live update of the trading platform is a reliable and efficient way to deliver the updated versions of all its components. There are three modes of the system operation. You can choose one of them on the starting page of the server:

- **Disabled** no updates to the platform components are performed, either automatically or manually.
- **Enabled** in this mode the platform components will be updated automatically when the final versions of updates are released.
- **Enabled with beta versions** besides the release versions, there are beta versions that are released more often. If this option is enabled, all types of updates will be installed. However, it should be noted that intermediate versions can be unstable, so they should be installed only in extreme cases.

It is strongly recommended to use beta updates only in extreme cases and only on demo servers.

The "Live Update" section contains the list of all updated components of the platform:

Component	Version	Date	Size (Kb)
MetaTrader 5 Administrator	2699	2020.11.25	16 9
MetaTrader 5 Access Server	2699	2020.11.25	7 0
MetaTrader 5 Backup Server	2699	2020.11.25	33 1
MetaTrader 5 Client	2699	2020.11.25	28 2
MetaTrader 5 Client IDE	2699	2020.11.25	23 1
MetaTrader 5 Client Terminal Tester	2699	2020.11.25	10 0
MetaTrader 5 History Server	2699	2020.11.25	134 1
MetaTrader 5 Server Updater	2699	2020.11.25	5 8
MetaTrader 5 Manager	2699	2020.11.25	13 8
MetaTrader 5 Trade Server	2699	2020.11.25	46 3

For Help, press F1

4316 / 1 Kb

Information is represented as a table with the following fields:

- **Component** name of the platform component;
- **Version** current version and build number of the component;
- **Size** size of the component distribution in Kb.

The Live Update procedure

- On each night of Saturday/Sunday, at optimization time, the history server starts the "mt5srvupdater64.exe" file that connects to the update server and checks the availability of updates of all the platform components;
- If there are any updates, they are downloaded;
- Then the history server informs other platform components about the availability of updates;
- Server components are immediately updated automatically;
- In terminals, a window appears that offers to update to the latest version.

The "🔄 Start Live Update" command in the Services menu and in the context menu, allows the manual start of this process.

- Terminal updates are cached on access servers to reduce the load on the history server;
- Update files are hidden and protected by digital signatures. They are transferred through a protected channel;
- During unpacking, updates are also checked by the platform components;
- To check the result of updating, request the entries of the history server journal by the LiveUpdate type.

Update Rollback

It is not recommended to rollback to previous versions unless there is a strong necessity.

In case there is a need to rollback the trade platform to a previous version after an update, you need to keep several safety measures:

- Before you start a rollback, make a full backup of the current state of the platform.
- Considering that some updates are connected with conversion (change of format) of data bases and configuration files, it is prohibited to rollback only executable files of the platform components. A rollback must be made for the entire platform.
- You should take into consideration, that some updates may introduce some changes into MetaTrader 5 API. Therefore, third-party applications developed using API should be recompiled with their proper versions.

Context Menu

The following commands are available in the context menu of this section:

MetaTrader 5 trading platform

-  **Start Live Update** start the updating process;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators in the table with components.

MetaTrader 5 Trading Platform / Migration from MetaTrader 4

Migration from MetaTrader 4 to MetaTrader 5

MetaTrader 5 is the next generation of the most popular MetaTrader 4 trading platform. The platform allows connection to a variety of markets, trading Forex and exchanges, transactions with currencies, futures, stocks, bonds and other instruments. MetaTrader 5 supports two position accounting systems – hedging (similar to MetaTrader 4 one) and netting for exchanges. Distributed architecture, services for traders, advanced trading and analytical functions, rich integration capabilities - all this and more is available in the fifth generation platform.

We have done our best to ensure the simplicity and convenience of migration to the modern multi-asset MetaTrader 5 platform. Transferring trading and client databases, as well as group and symbol settings is performed automatically via the network. The general platform settings are similar, and it is quite easy to compare them.

- Financial instruments
- Trade groups
- Accounts and trades
- General settings
- Manager accounts
- Data feeds

MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / General Settings

Migrating the Platform General Settings

Here you can find out how to relocate the general settings of the platform.

Common

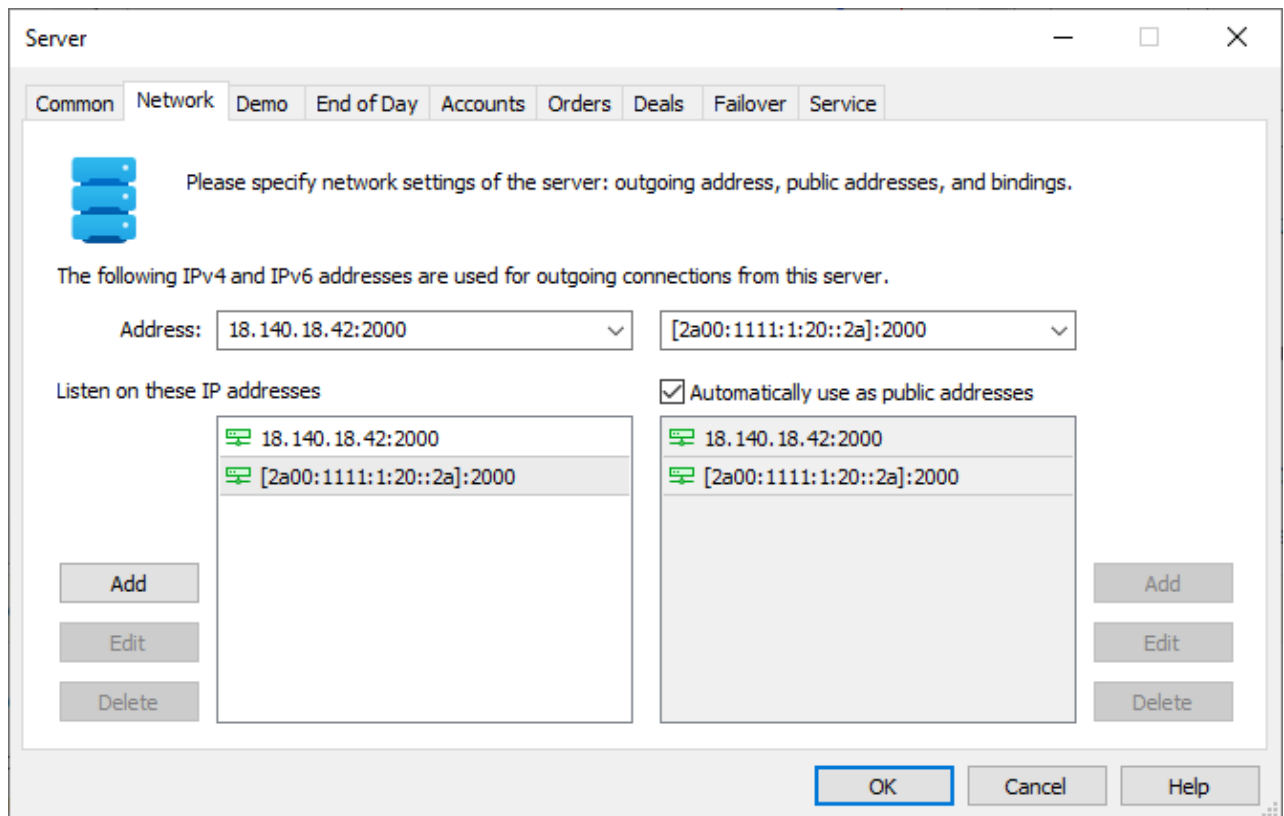
The Common section of MetaTrader 4 contains quite a large number of various platform settings. In MetaTrader 5 these settings are allocated among different sections depending on their destination.

License data and server name

You can view license data and set the server name to be displayed to clients at the start page of the platform.

Server IP address and communication port

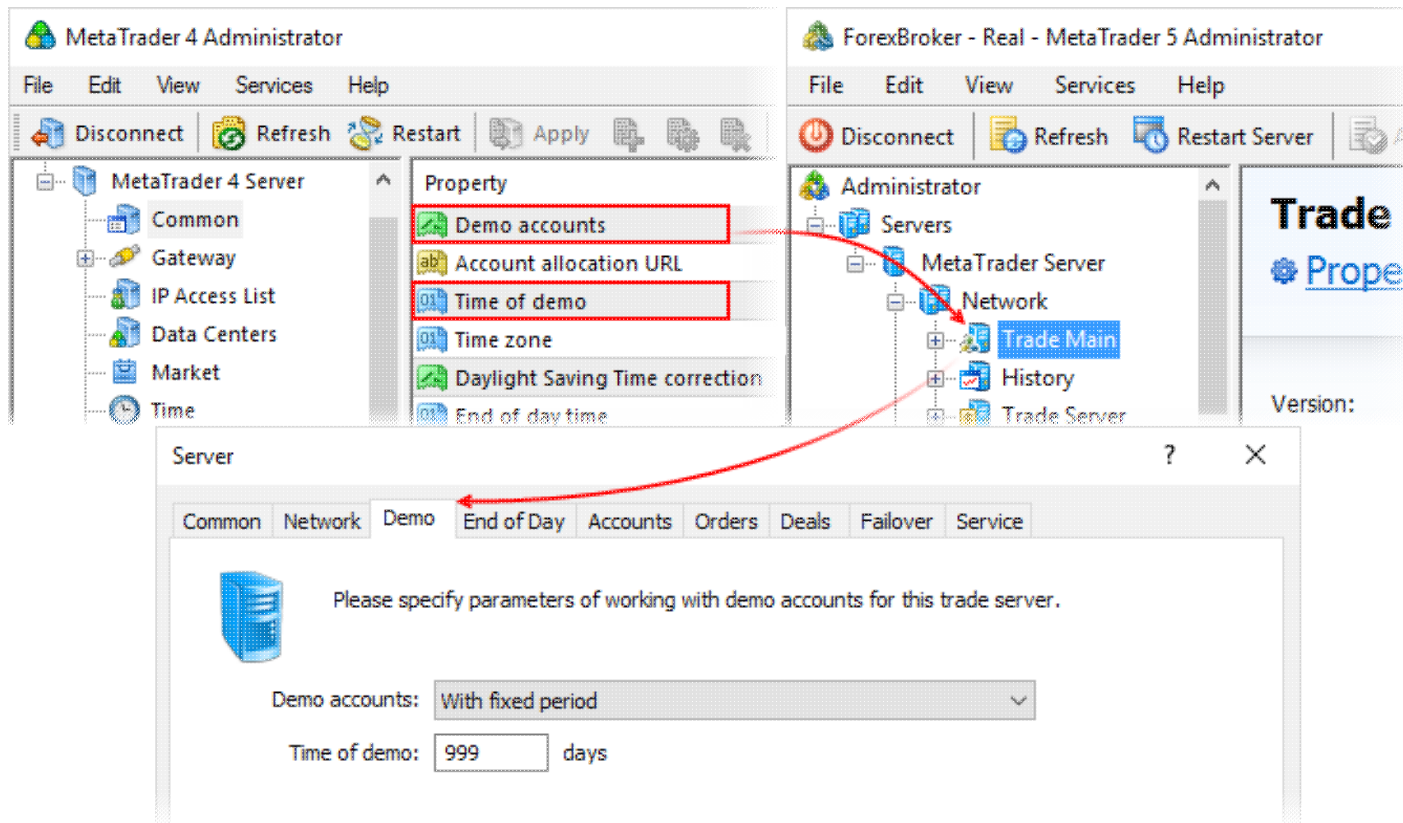
It is specified separately for each of the platform servers on the Network tab:



MetaTrader 5 trading platform

Demo accounts

Demo account settings are specified separately for each trade server of the platform:



Account allocation URL is set at the platform start page. From there you can also specify certain groups, in which clients are to open demo accounts via the terminals.

Time zone and daylight saving time

General time settings are specified together with the operation schedule in the Time section:

MetaTrader 5 trading platform

The image shows two windows side-by-side. The top window is 'MetaTrader 4 Administrator' and the bottom is 'ForexBroker - Real - MetaTrader 5 Administrator'. Both have a tree view on the left and a properties table on the right. Red boxes and arrows highlight the 'Time synchronization with' settings in both.

MetaTrader 4 Administrator

Property	Value
Time zone	GMT+02:00
Daylight Saving Time correction	Yes
End of day time	08:00
Rollover mode	Normal
Statements mode	End of day
Monthly statements mode	Last day of month
Generate statements at weekends	No
Keep internal emails	30 days
Keep ticks	60 days
Optimization time	10:40
Antiflood control	No
Antiflood max. connections	n/a
Feeder switch timeout	30 seconds
LiveUpdate mode	Enabled with beta versions
Time synchronization with	clock.psu.edu
IP access list of web services	

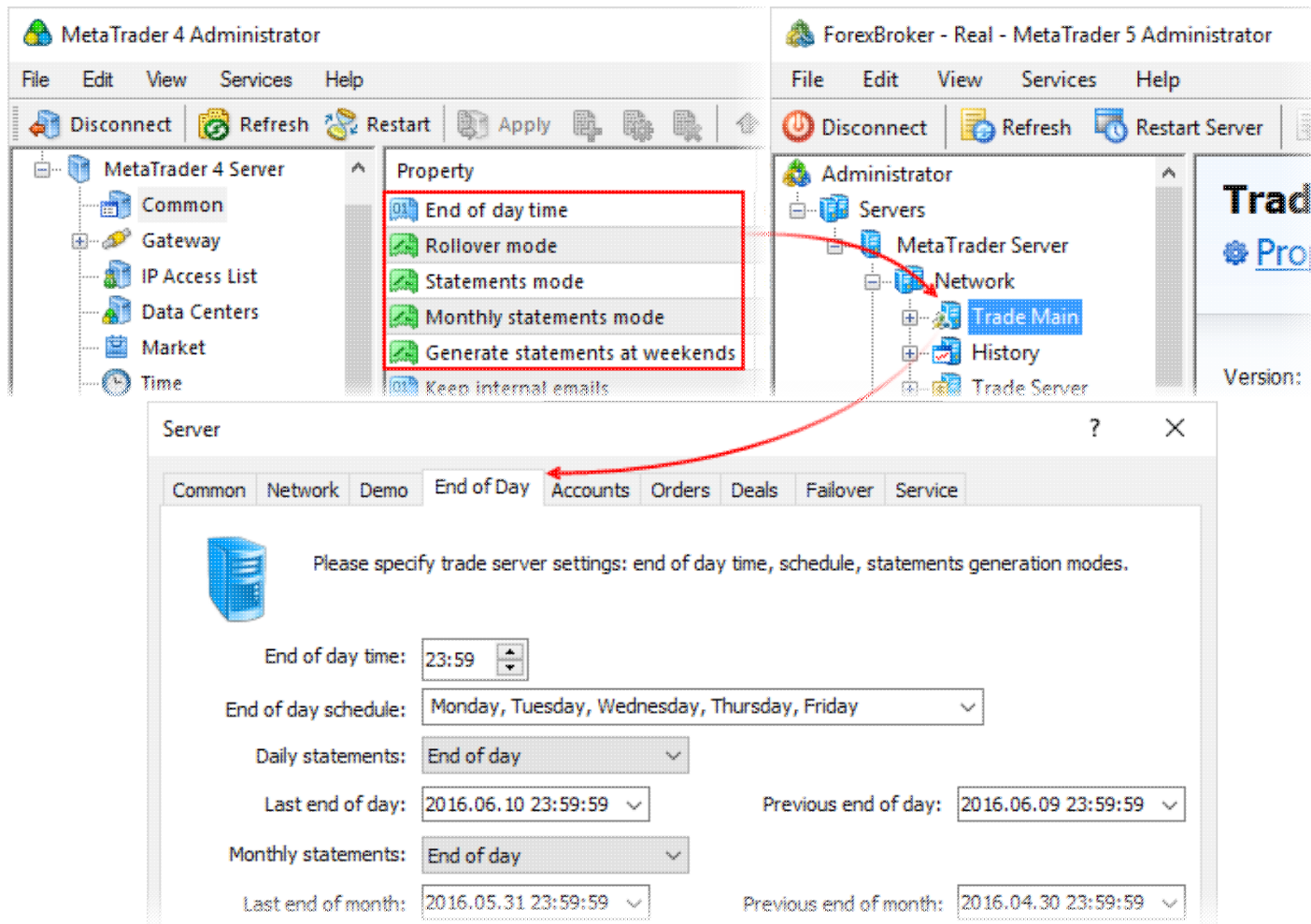
ForexBroker - Real - MetaTrader 5 Administrator

Property	Value
Common settings	
Time zone	GMT+01:00
Daylight Saving Time correction	Yes
Time synchronization with	pool.ntp.org
Day settings	
Sunday	
Monday	00:00-24:00

MetaTrader 5 trading platform

End of day, swap operation and report generation

Trading days and months closing settings are located in the trade server. The main distinguishing feature of the settings block is that swap operation method in MetaTrader 5 is defined separately for each trading instrument, rather than for the entire server.



Storing emails and ticks

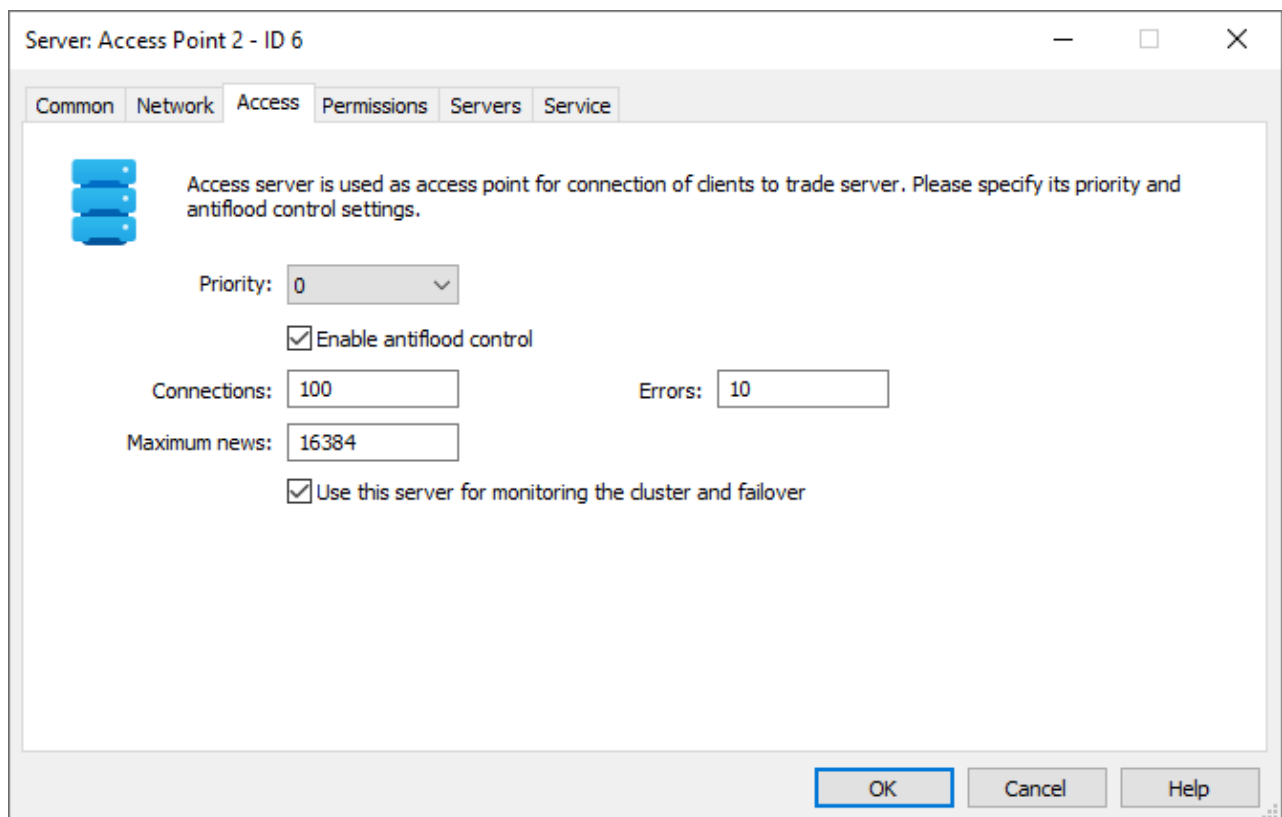
MetaTrader 5 does not allow specifying email and tick data storage parameters. This data is always stored without depth limitations.

Time optimization

The time of conducting all operations necessary to increase performance and reliability is set separately for each platform server on the Service tab.

Protection against DDoS attacks

These settings are located in the access server. Apart from the number of allowed connections, MetaTrader 5 enables you to configure the number of connection errors.



The screenshot shows a window titled "Server: Access Point 2 - ID 6" with standard Windows window controls (minimize, maximize, close). The window has a tabbed interface with the following tabs: "Common", "Network", "Access" (selected), "Permissions", "Servers", and "Service".

On the "Access" tab, there is a blue server icon on the left. To its right, a text box states: "Access server is used as access point for connection of clients to trade server. Please specify its priority and antiflood control settings."

The configuration options include:

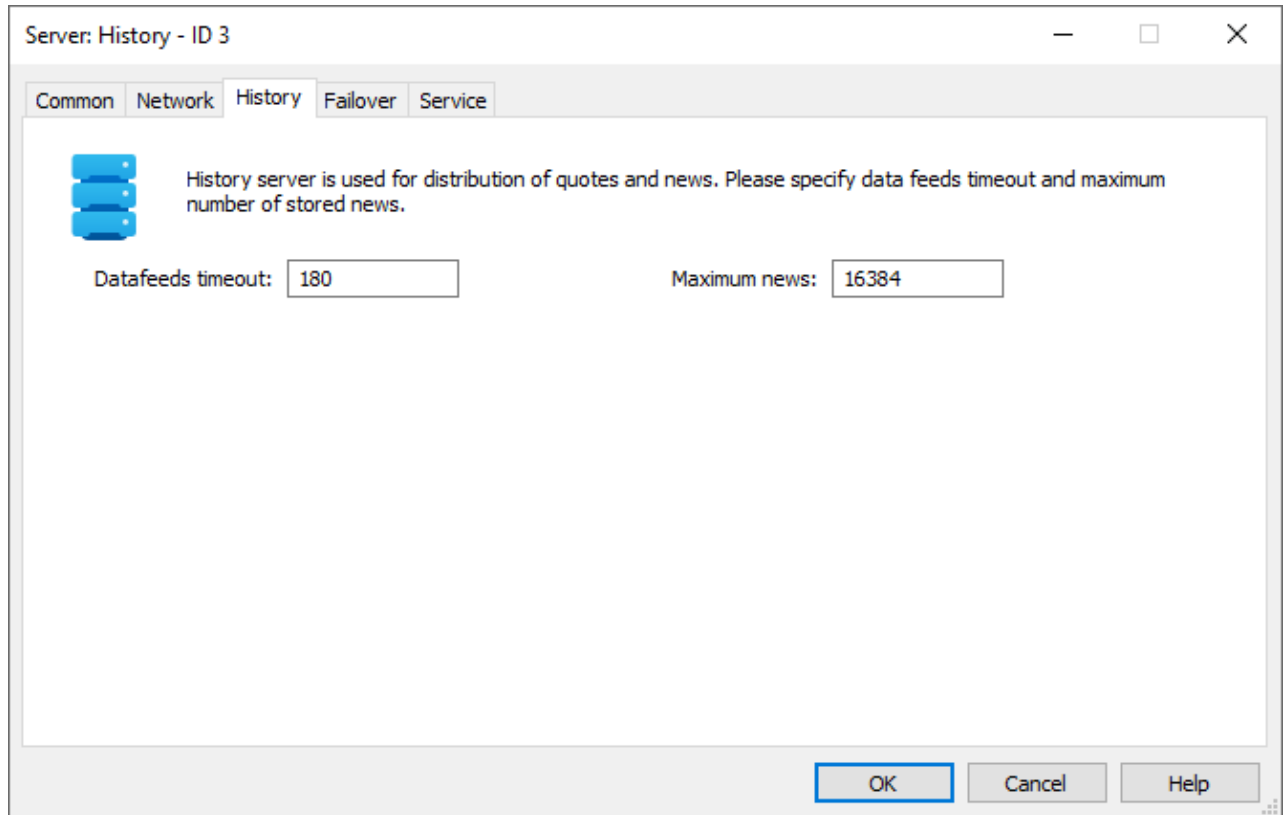
- "Priority:" with a dropdown menu showing "0".
- A checked checkbox labeled "Enable antiflood control".
- "Connections:" with a text input field containing "100".
- "Errors:" with a text input field containing "10".
- "Maximum news:" with a text input field containing "16384".
- A checked checkbox labeled "Use this server for monitoring the cluster and failover".

At the bottom right of the window are three buttons: "OK", "Cancel", and "Help".

MetaTrader 5 trading platform

Data feed switch timeout

This parameter is located in the history server:



Version update mode

The mode is set at the platform start page. In MetaTrader 5, the update settings are similar: you can disable them completely, allow updates to release versions or allow updates both to release and beta versions.

List of IPs for access of Web services

There is no such parameter in MetaTrader 5. In order to connect to the trade platform via Web API, create a special manager account having the right to "Enable API/FIX connection" and an API password.

Paths to databases

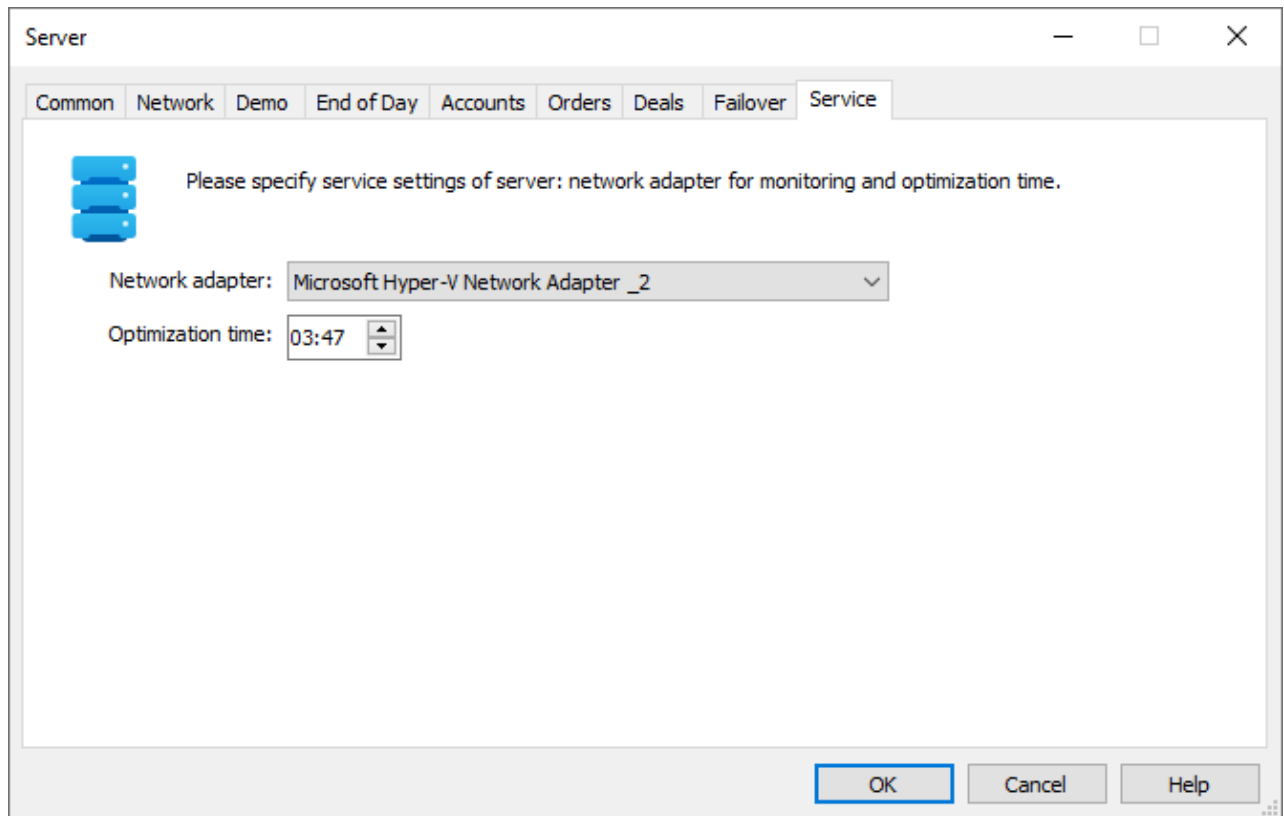
In MetaTrader 5, these parameters are not available since data storage has been allocated among different platform components. If you need to allocate the databases, simply install the platform components on different servers.

- Trade and client databases [trade server directory]\bases\
- History data [history server directory]\history\
- Logs [server directory]\logs\

Network adapter for monitoring

MetaTrader 5 trading platform

The network controller for monitoring is set separately for each platform server on the Service tab:

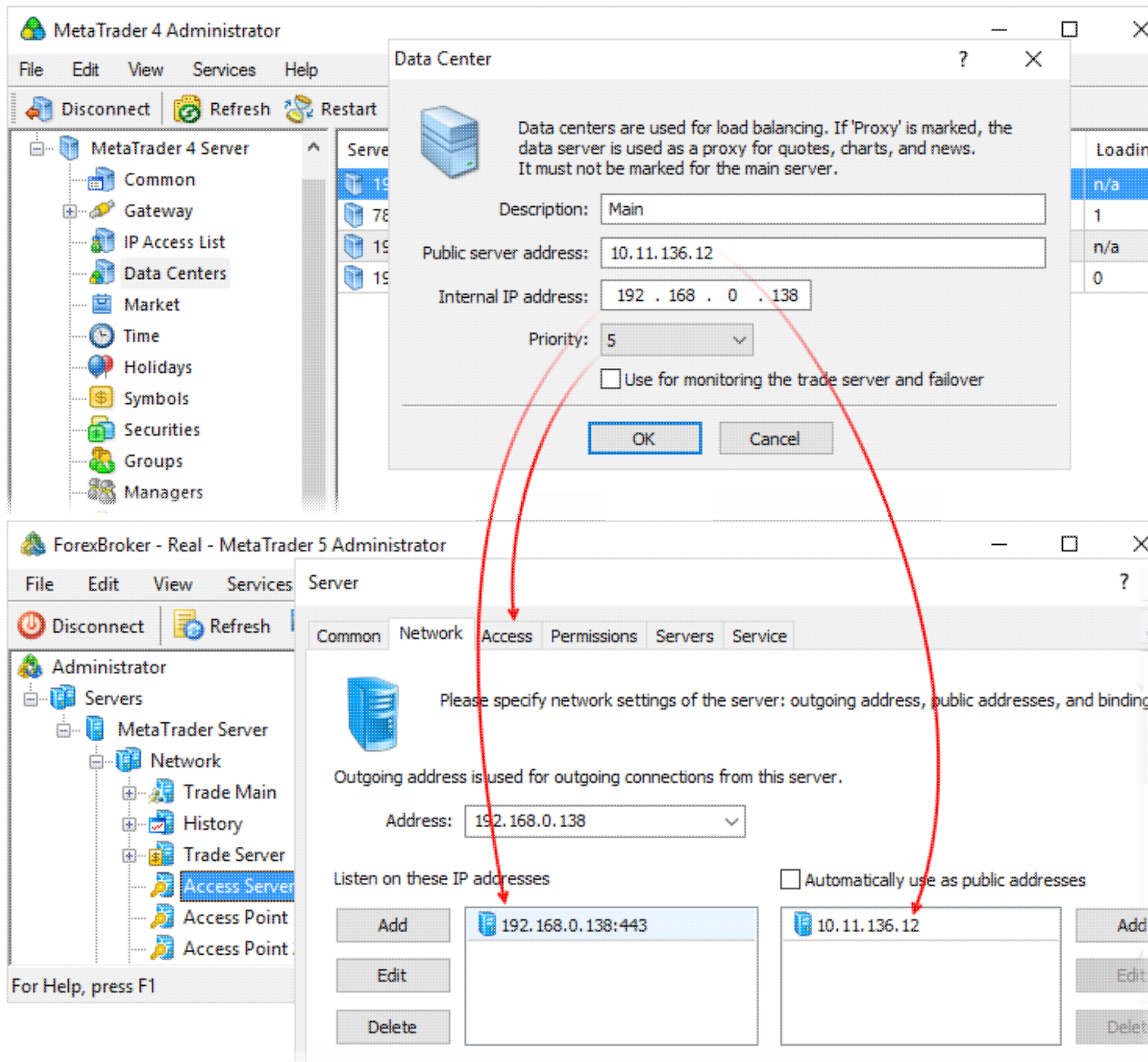


Data Centers

MetaTrader 5 Access Servers are similar to data centers. They also act as intermediate servers reducing the load on the main one and protecting it against DDoS attacks. Access servers can also be installed on dedicated PCs and enabled as a new access point via MetaTrader 5 Administrator.

Unlike MetaTrader 4, the fifth version of the platform does not allow direct connection to the trade server. You need at least one access server to do that.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

Each access server can run at multiple IP addresses of a dedicated server (if the server provides such an ability) and listen to multiple ports. If the system has multiple access servers, you can set MetaTrader 5 access server priorities on the Access tab.

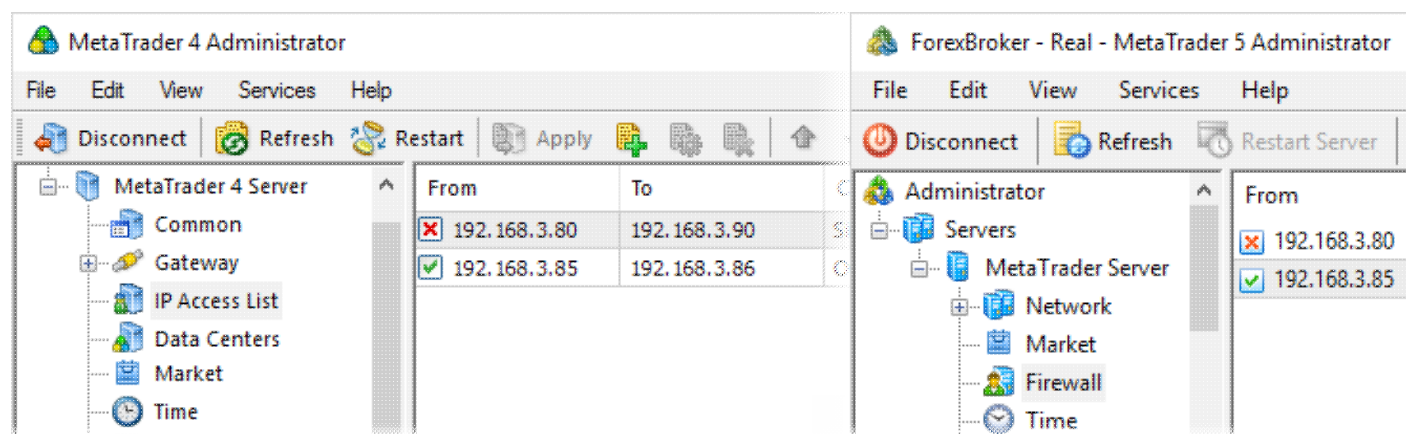
In MetaTrader 5, all users (administrators, managers and traders) can connect to and work in the system only via MetaTrader 5 Access Server. Permissions tab allows you to configure access to the system to various connection types, thus allocating manager and client connections among different access servers.

Servers tab allows you to specify trade servers the current access server is used with.

In MetaTrader 5, the number of access servers is not limited.

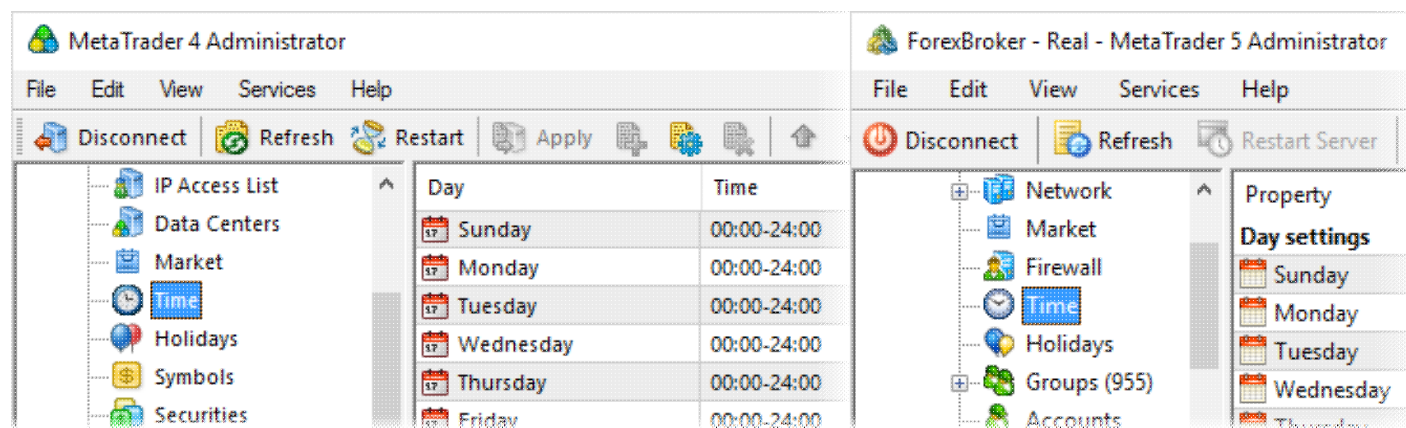
Access by IP

Relocation of settings for access by an IP address from MetaTrader 4 Server to MetaTrader 5 is simple since the settings are similar in both systems.



Working Time

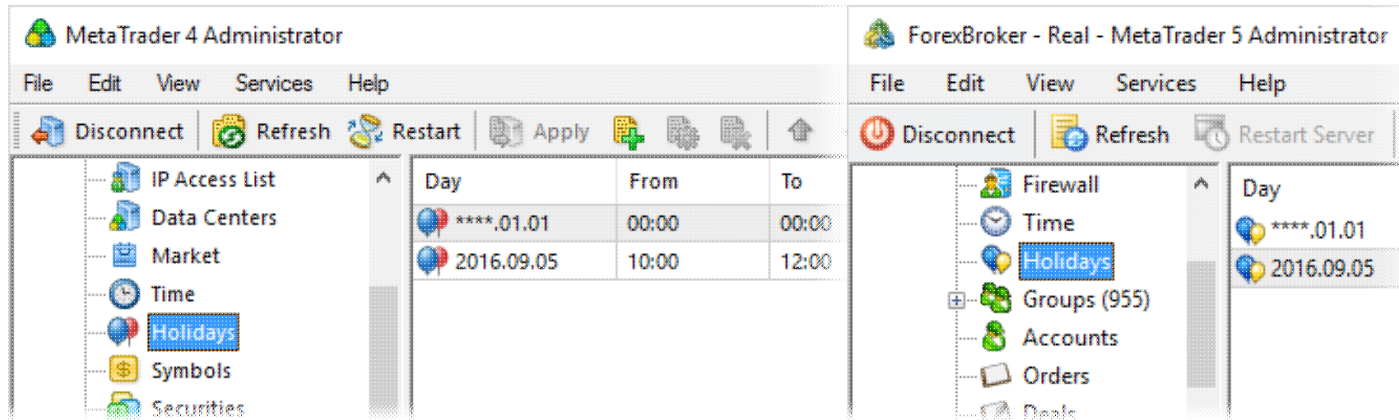
The working time settings in MetaTrader 4 and MetaTrader 5 are similar:



MetaTrader 5 trading platform

Holidays

Holiday settings in MetaTrader 4 and MetaTrader 5 are similar.



MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / Financial Instruments

Importing and Setting Financial Instruments

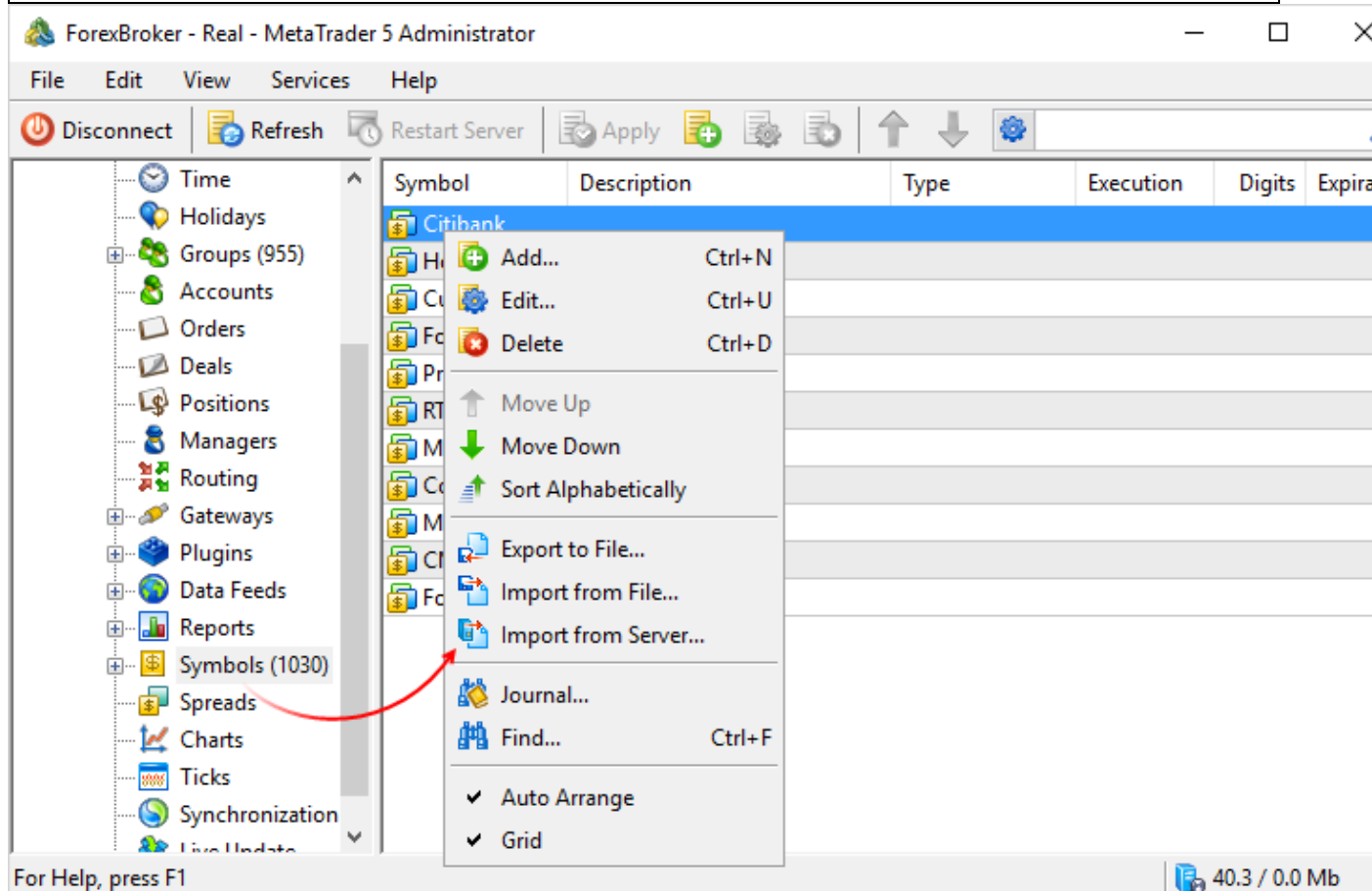
Please keep in mind the following when working with MetaTrader 5 symbols:

- Unlike MetaTrader 4, the fifth version allows you to create symbol groups (similar to Securities in MetaTrader 4) directly in Symbols section. Moreover, you can create a tree-like symbol structure.
- In MetaTrader 5, it is not necessary to collect individual symbol into groups. However, it is still recommended to divide symbols into some logical groups to make system administration more convenient.
- MetaTrader 5 has no limitations on the number of financial instruments and their groups.

Importing Symbols

Use the import of symbols to copy all symbols and their settings from MetaTrader 4 server to the MetaTrader 5 platform.

When you import symbols, existing settings are converted and default values are assigned to missing settings.



MetaTrader 5 trading platform

Select "MetaTrader 4" server type and specify connection data: IP address and server port, as well as account login and password. The account used for importing symbols may be opened in any group except for a manager one ("manager"). The group, in which the account is opened, should have access to all symbols to be imported from MetaTrader 4 server.

The image shows two overlapping windows from the MetaTrader 5 'Import' wizard.

The background window is titled 'Import' and shows the 'Connection details' step. It includes a dropdown for 'Server Type' set to 'MetaTrader 4', and text boxes for 'Server' (192.168.0.138:443), 'Login' (810918), and 'Password' (masked with dots). A note at the bottom left states: 'NOTE: When importing settings from a MetaTrader 4 account, the account must be opened in any group except for a manager one ("manager"). The group, in which the account is opened, should have access to all symbols to be imported from MetaTrader 4 server.'

The foreground window is also titled 'Import' and shows the 'Choosing symbols' step. It displays a tree view of symbol groups under 'Forex'. The following symbols are listed:

- AUDCAD, Australian Dollar vs Canadian Dollar
- AUDCHF, Australian Dollar vs Swiss Franc
- AUDJPY, Australian Dollar vs Japanese Yen
- AUDNZD, Australian Dollar vs New Zealand Dollar
- AUDUSD, Australian Dollar vs US Dollar
- CADJPY, Canadian Dollar vs Japanese Yen
- CHFJPY, Swiss Franc vs Japanese Yen
- CHFJPY.1, Swiss Franc vs Japanese Yen
- EURAUD, Euro vs Australian Dollar
- EURCAD, Euro vs Canadian Dollar** (highlighted)
- EURCHF, Euro vs Swiss Franc
- EURGBP, Euro vs Great Britain Pound
- EURJPY, Euro vs Japanese Yen

At the bottom of the foreground window, it shows 'Selected: 10 of 373 symbols', an unchecked 'Overwrite existing symbols' checkbox, a 'Select All' button, and navigation buttons: '< Previous', 'Import' (highlighted), and 'Cancel'.

MetaTrader 5 trading platform

The next window shows the list of all symbols that can be imported from MetaTrader 4 server. Selected symbols are imported to MetaTrader 5 platform after pressing Import button.

In case of successful import, the symbols appear in the Symbols section of the group, from which the import command has been called.

- By default, all imported symbols are not available for trading.
- Check the settings carefully before making any symbol available for trading.

In order to distribute imported symbols among symbol groups, select the necessary symbols and drag them by mouse to a necessary group (or subgroup).

Configuring the Symbols

All symbol settings from MetaTrader 4 are automatically converted to the appropriate symbol settings for MetaTrader 5. Since the MetaTrader 5 platform features the new parameters that are not present in MetaTrader 4, such parameter values are set to default ones for each symbols imported from MetaTrader 4.

"Percentage" value used to calculate the margin in the MetaTrader 4 symbol settings (Calculation tab) has been redefined and expanded in MetaTrader 5. Therefore, it is not imported. In MetaTrader 5, you can assign the necessary ratio for each trading operation type and direction separately:

MetaTrader 5 trading platform

Symbol: EURUSD

Symbol

Filtration

Calculation

Swaps

Sessions



Adjustment of parameters for calculations of margins and profit.
Initial data and final formulas must be specified.

Contract size:

100000

Initial margin:

0.00

Maintenance:

0.00

Hedged:

50000.00

Tick size:

0.00000

Tick price:

0.00000

Percentage:

100.0

Symbol: EURUSD

Common

Quotes

Trade

Execution

Margin

Margin Rates

Swaps

Sessions



Please specify margin rates for different types of trade operations.

Liquidity margin rate:

0.000

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	0.0000000	0.0000000	0.0000000
 Sell	1.0000000	0.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	1.0000000	0.0000000	0.0000000	0.0000000
 Sell	1.0000000	0.0000000	0.0000000	0.0000000

OK

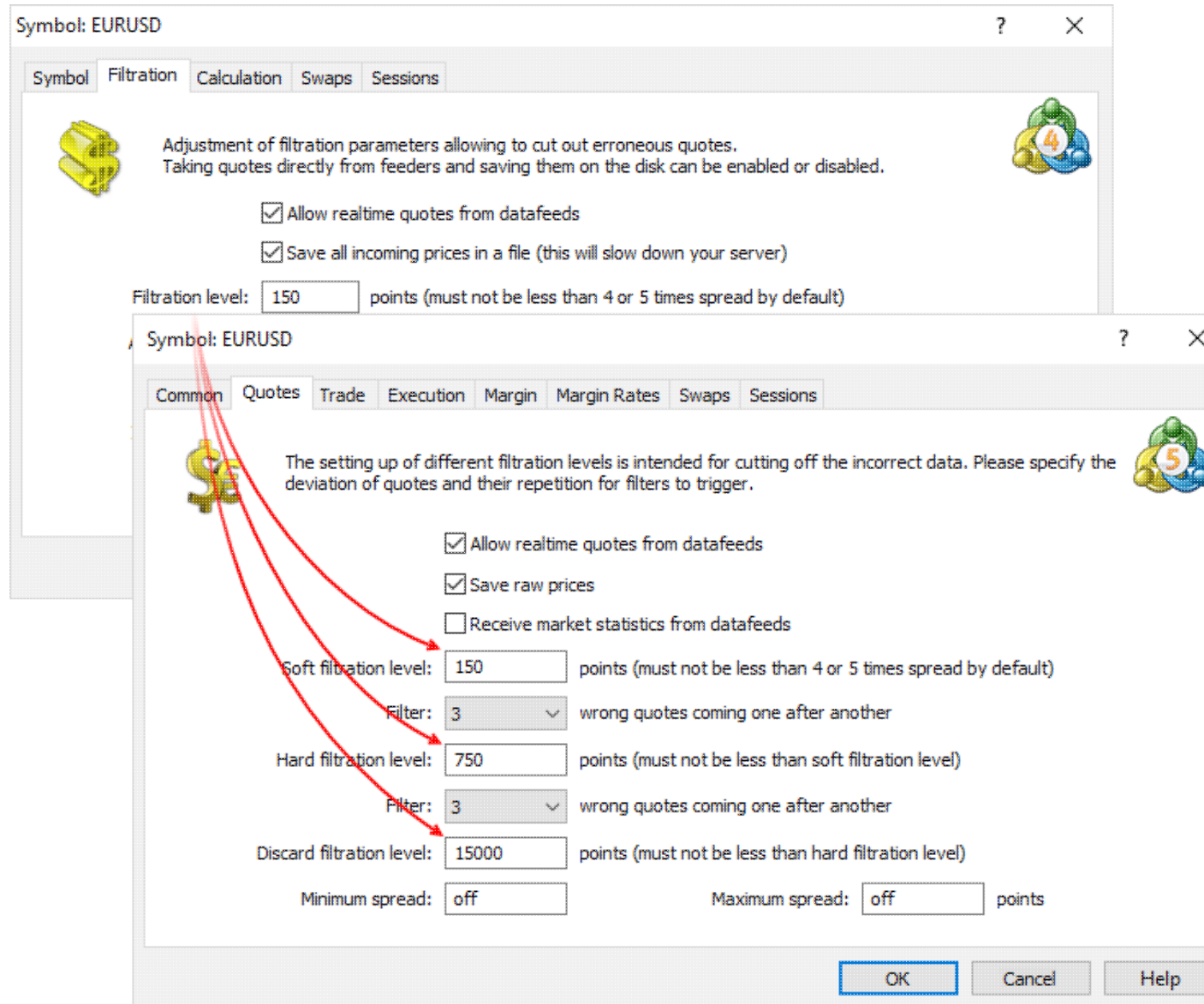
Cancel

Help

2085

Due to differences in hedged margin calculation between the platforms, the hedged margin value is doubled during the import making margin calculation in MetaTrader 5 similar to the one in MetaTrader 4.

"Filtration level" in MetaTrader 4 symbol filtration settings is inserted into "Soft filtration level" in the MetaTrader 5 symbol settings. Its five-fold value is used as "hard filtration level" parameter, while the hundred-fold one as "Discard filtration level".

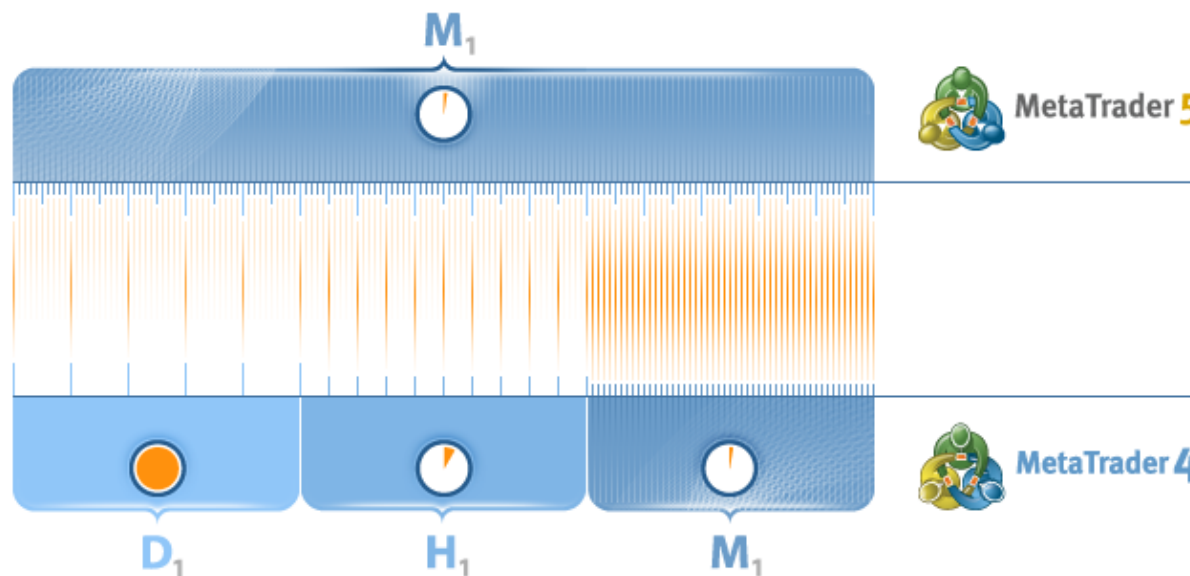


The minimum and maximum volume, as well as step in MetaTrader 5 are set by default.

History Data Synchronization between MetaTrader 5 and MetaTrader 4

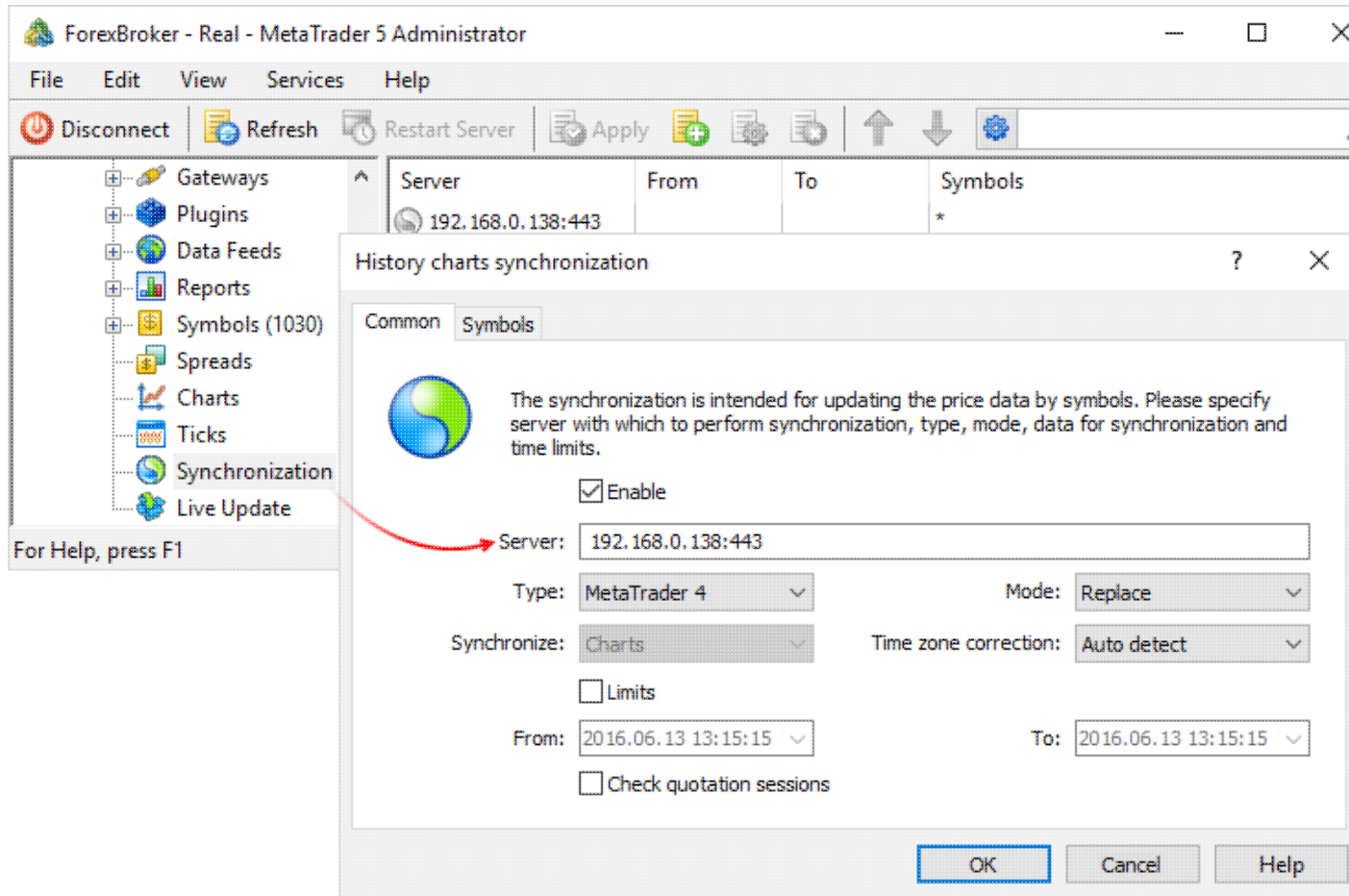
After importing the symbols, you can download history data for them from your MetaTrader 4 Server. There are some synchronization features to be considered before the start of the process:

- In order to synchronize history data with MetaTrader 4 server, the latter should have at least one demo group. The group can be disabled.
- MetaTrader 5 history data is stored only as 1-minute data and converted programmatically into larger timeframes on request from the client terminal, while MetaTrader 4 stores different timeframes.
- When synchronizing with MetaTrader 4 servers, the system consistently selects the most complete data beginning from one-minute timeframe. M1 data is fully taken first. Next, the system attempts to receive the missing part from H1 data. MetaTrader 4 server's hour reading is recorded to the first minute of an hour at MetaTrader 5 server. After that, the attempt is made to receive missing data from D1 timeframe. Daily data of MetaTrader 4 server (for example, 01.03.2009) is recorded to the first minute of a day at MetaTrader 5 server (for example, 01.03.2009 00:00). Therefore, clients will be able to receive a complete chart for a historical period only on the timeframe, from which the data has been imported.



MetaTrader 5 trading platform

Before launching synchronization, specify the source server (your MetaTrader 4 Server) in Synchronization section of MetaTrader 5 Administrator. The Symbols tab allows you to specify the symbols, the data for which is to be synchronized.



MetaTrader 5 trading platform

To start synchronization, select Services -> Synchronize History in the terminal's main menu.

Data synchronization process can be tracked in the history server journal. To do this, enter "Synchronization" (without the quotes) in the search bar of the filtration settings, set Full for a request type and All for an event type, specify the current day and click Request. The following journal entry indicates the completion of a synchronization process:

history synchronization with xxx.xxx.xxx.xxx:xxx finished

Synchronized data can be seen in Charts section of MetaTrader 5 Administrator.

MetaTrader 5 has no limitations on the depth of history data and the number of symbols.

MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / Trade Groups

Importing and Setting Trade Groups

Group parameters in MetaTrader 5 provide more flexibility as compared to MetaTrader 4. The total number of groups in MetaTrader 5 is not limited.

MetaTrader 5 supports the same functions as MetaTrader 4 except for a few functions related to hedging. MetaTrader 5 does not have the "Multiple Close by orders" and "Auto close-out" functions.

Import of Groups

Use the import of groups to copy all groups and their settings from MetaTrader 4 server to the MetaTrader 5 platform.

- When you import groups, existing settings are converted and default values are assigned to missing settings.
- Trade settings of symbols for the groups are not imported.
- Groups from the MetaTrader 4 server are imported into the currently selected groups section.

MetaTrader 5 trading platform

ForexBroker - Real - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Holidays
 Groups (955)
 contest (1)
 demo (912)
 managers (18)
 real (12)
 Accounts
 Orders
 Deals
 Positions
 Managers
 Routing
 Gateways
 Plugins
 Data Feeds
 Reports
 Symbols (1030)
 Spreads
 Charts
 Ticks

Group	Server	Company	Authen...	Margin	Currency
demo\dem...	Tr	EV	Normal	80 / 30 %	USD
demo\dem...	Tr	EV	Normal	50 / 30 %	USD
demo\forex...	Tr	softwa...	Normal	50 / 30 %	GBP
demo\atc20...	Tr	EV	Normal	50 / 30 %	USD
demo\forex...	Tr	EV	Normal	80 / 30 %	EUR
demo\dem...	W	EV	Normal	50 / 30 %	USD
demo\dem...	Tr	EV	1024-bit ...	80 / 30 %	USD
demo\forex...	Tr	softwa...	Normal	50 / 30 %	GBP
demo\dem...	17	EV	1024-bit ...	50 / 30 %	USD
demo\dem...	18	EV	Normal	50 / 30 %	USD
demo\dem...	Ex	EV	Normal	50 / 30 %	USD
demo\dem...	Ac	EV	Normal	50 / 30 %	USD
demo\forex...	Tr	EV	Normal	50 / 30 %	JPY
demo\TestS...	Tr	EV	Normal	50 / 30 %	USD
demo\dem...	Tr	EV	2048-bit ...	50 / 30 %	USD
demo\dem...	Tr	EV	Normal	50 / 30 %	USD
demo\dem...	Tr	EV	1024-bit ...	50 / 30 %	USD
demo\dem...	Tr	EV	Normal	80 / 30 %	USD

Servers
 Add... Ctrl+N
 Edit... Ctrl+U
 Delete Ctrl+D
 Move Up
 Move Down
 Export to File...
 Import from File...
 Import from Server...
 Request
 Find... Ctrl+F
 Auto Arrange
 Grid

Import config 42.1 / 0.0 Mb

MetaTrader 5 trading platform

Select "MetaTrader 4" server type and specify connection data: IP address and server port, as well as account login and password. The account used for importing symbols should be opened in the manager group and have Administrator right.

The image shows two overlapping windows from the MetaTrader 5 'Import' wizard. The background window is at the 'Connection details' step, and the foreground window is at the 'Choosing groups' step.

Import - Connection details
Choose server type, fill server address and authorization information.

This wizard will help you to import settings from a MetaTrader 4 or MetaTrader 5 server. Please specify server type, address and authorization details below.

Server Type: MetaTrader 4
Server: 192.168.0.138:443
Login: 810918
Password: ●●●●●●●●

Import - Choosing groups
Choosing groups to be added to your database.

NOTE: When importing settings from a MetaTrader 4 account.

manager
coverage
demo forex
demo forex-otp
demo forex1
demo forex2
demo forex3
demo forex4
demo forex5
demo forex6
demo forex7
demo forex8
preliminary
test

Selected: 0 of 20 groups

Select All

< Previous Import Cancel

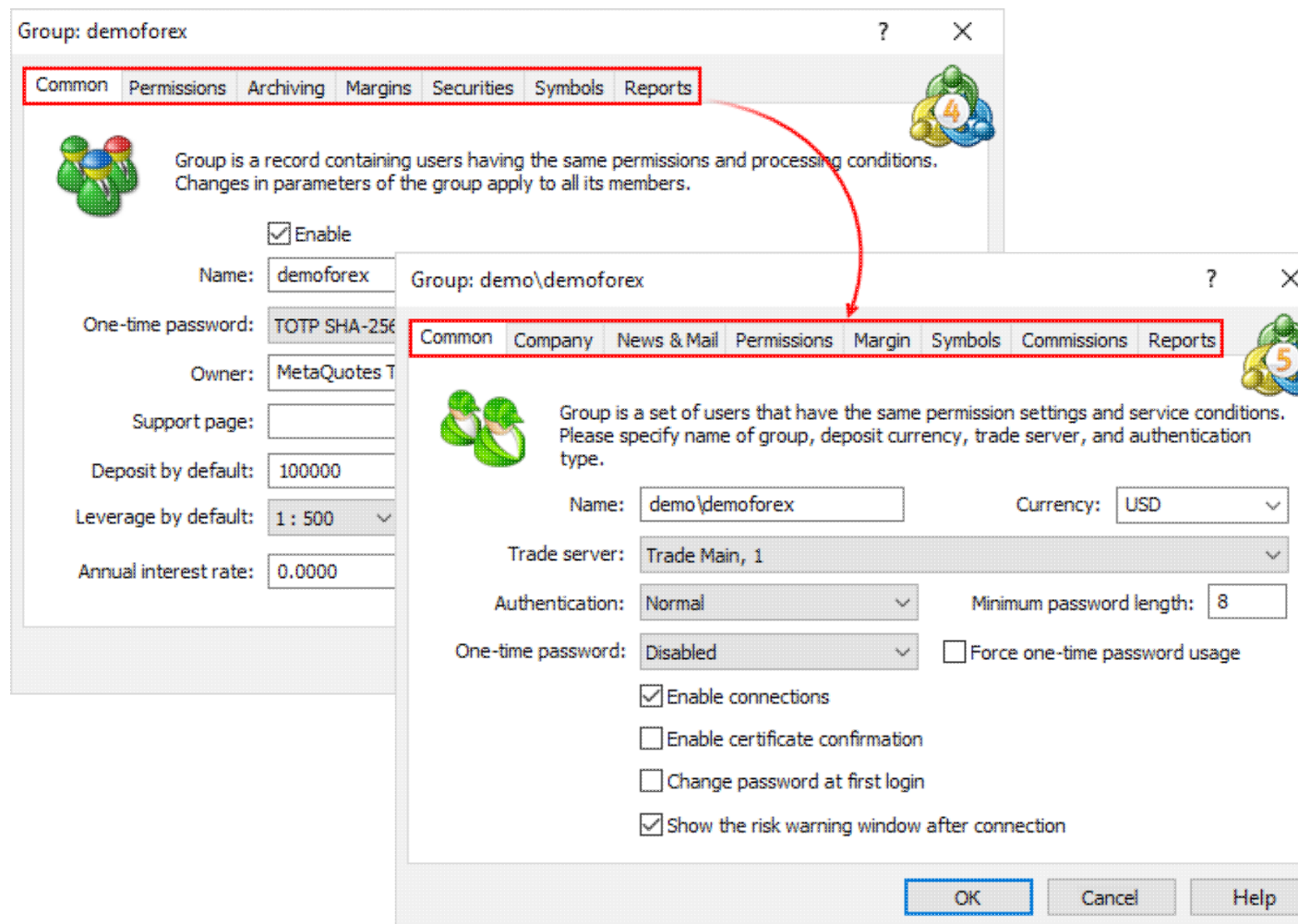
MetaTrader 5 trading platform

The next window shows the list of all groups that can be imported from MetaTrader 4 server. Selected groups are imported to MetaTrader 5 platform after pressing Import button.

- After importing, check the settings of all groups.
- All groups are bound to the main trade server. After importing the binding can be changed.
- Only new groups are imported. Settings of existing groups of the same name are not overwritten.

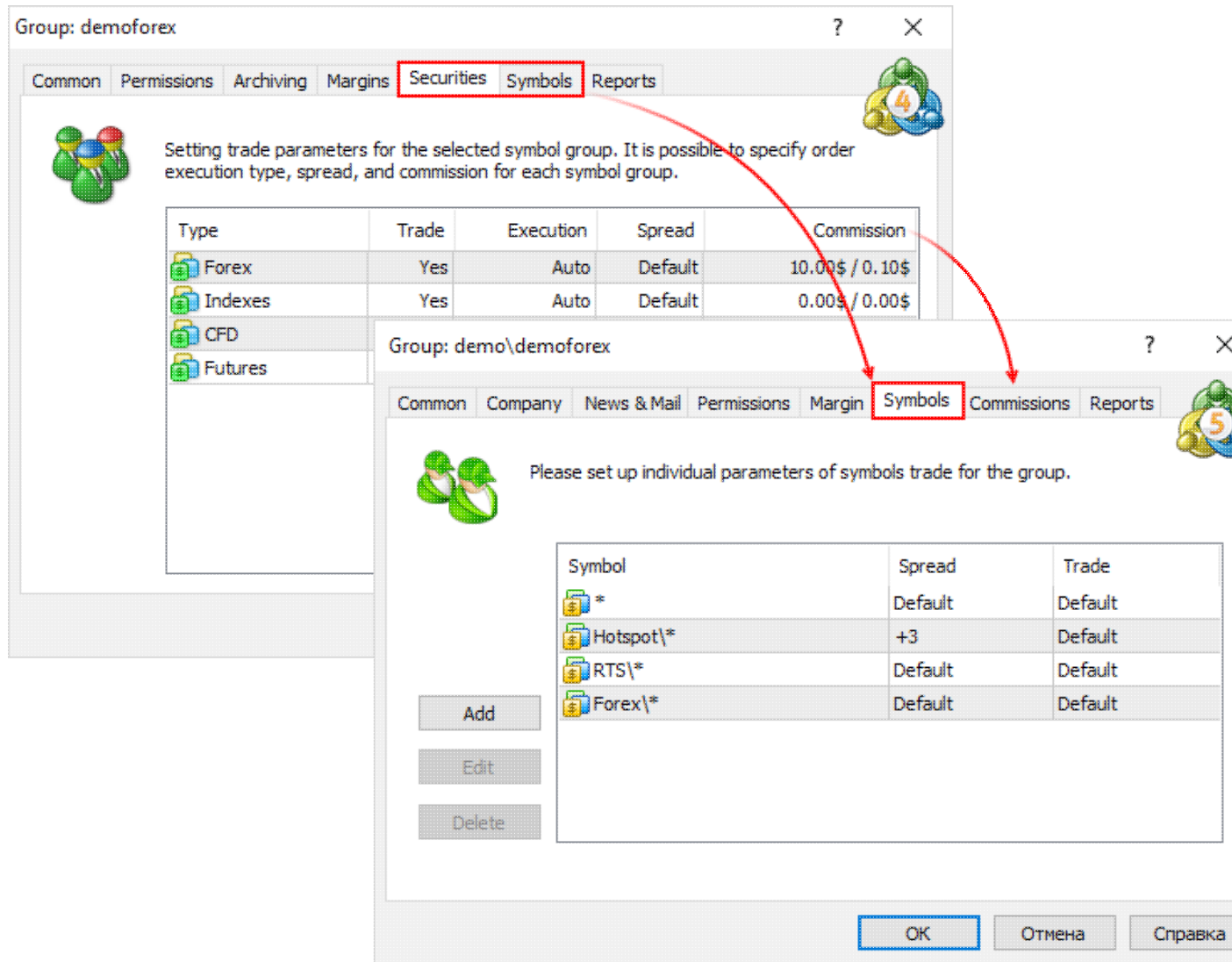
Account Group Settings

MetaTrader 5 platform group settings have more tabs. This is due to the fact that MetaTrader 5 features additional group settings not present in MetaTrader 4. Most MetaTrader 4 settings have remained in MetaTrader 5.



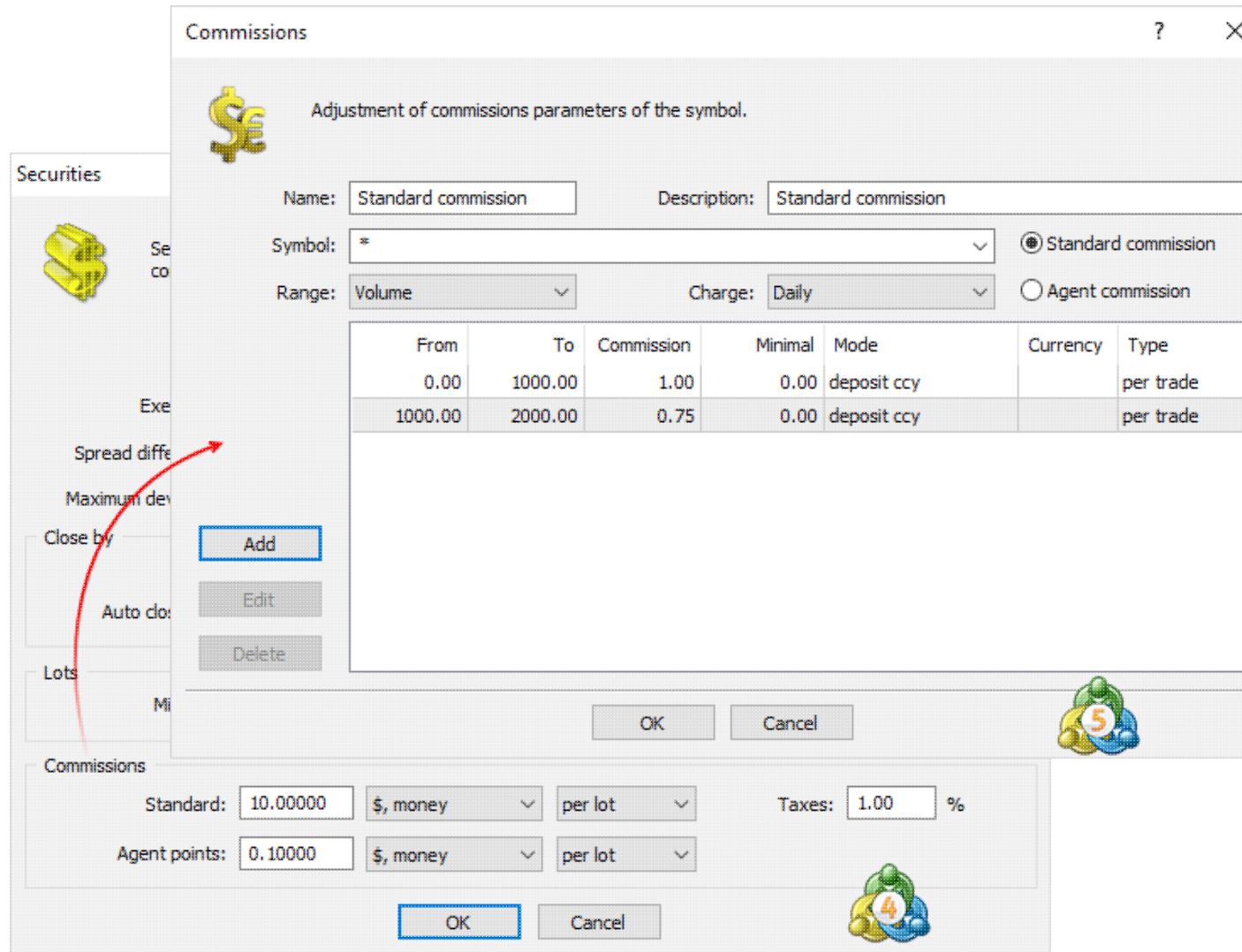
Individual Symbol Settings

MetaTrader 5 features considerably expanded trading symbol settings redefining options.



Commission Settings

In MetaTrader 5, commission settings can be found on a separate tab. The number of settings has significantly increased: charging time, dependence on trade volume and turnover, etc.



Commissions

Adjustment of commissions parameters of the symbol.

Name: Standard commission Description: Standard commission

Symbol: * ☒ Standard commission ☐ Agent commission

Range: Volume Charge: Daily

From	To	Commission	Minimal	Mode	Currency	Type
0.00	1000.00	1.00	0.00	deposit ccy		per trade
1000.00	2000.00	0.75	0.00	deposit ccy		per trade

Add Edit Delete

OK Cancel

Commissions

Standard: 10.00000 \$, money per lot Taxes: 1.00 %

Agent points: 0.10000 \$, money per lot

OK Cancel

Processing Trade Requests

Now, trade request execution settings are not limited to three types ("Manual only, no automation", "Automatic only" and "Manual, but automatic if no dealers online") like in MetaTrader 4. Routing section introduced in MetaTrader 5 allows you to create custom rules and conditions with virtually any combination of settings and execution types.

The screenshot displays the MetaTrader 5 interface. In the background, the 'Securities' window is open, showing settings for trading with symbols. A red arrow points from the 'Execution' dropdown menu in the 'Securities' window to the 'Routing: Auto Execution' dialog in the foreground.

Securities Window:

- Enable:** ☒
- Trade:** ☒
- Use confirmations in REQUEST mode:** ☒
- Execution:** automatic only (dropdown menu)
- Spread difference:** automatic only (dropdown menu)
- Maximum deviation:** 4 points (IE only) ☐ Fast confirmation on IE with deviation specified

Routing: Auto Execution Dialog:

- Common Tab:**
 - ☒ Enable this rule
 - Name:** Auto Execution
 - Perform action:** Confirm by request price (dropdown menu)
 - Where request is:** All

Dealers Tab:

Name	Action	Dealers
Auto Execution	Confirm by request price	

Left Panel:

- Deals
- Positions
- Managers
- Routing (highlighted)
- Gateways
- Plugins
- Data Feeds
- Reports
- Symbols (F4)
- Spreads
- Charts
- Ticks

For Help, press F1

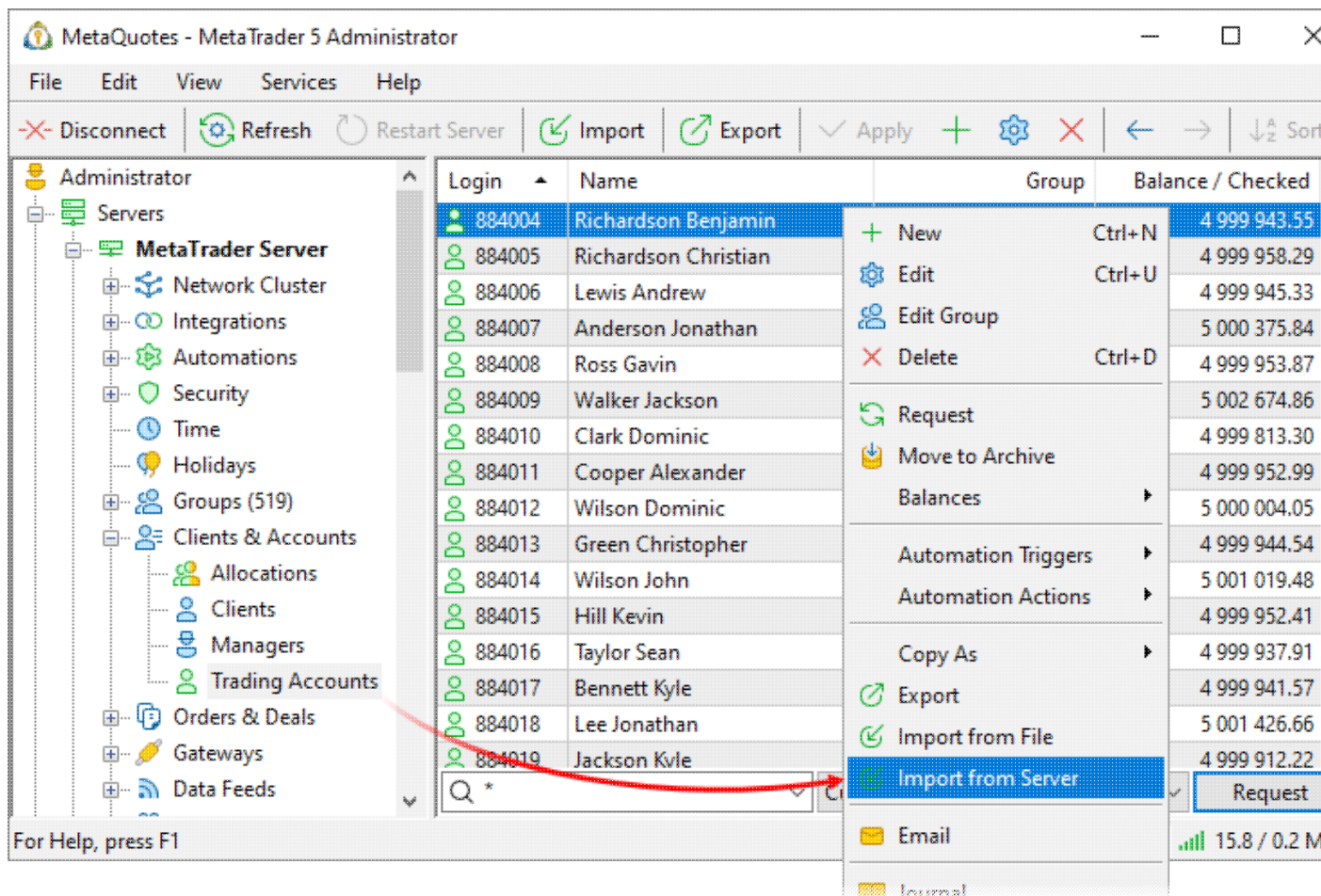
MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / Import of Accounts and Trades

Import of Accounts and Trades from MetaTrader 4

You can easily import your client database from MetaTrader 4 to MetaTrader 5, including clients' trading operations and history. You will need Administrator accounts in MetaTrader 4 and MetaTrader 5 with permissions to access clients, groups and trading operations.

- When importing accounts and orders from MetaTrader 4, all related data is transferred as well. Some new order and trade fields that were not present in MetaTrader 4 are filled with default values.
- Before importing clients and trading operations, be sure to configure/import all required trading instruments and groups.

Open the "Trading Accounts" section in MetaTrader 5 Administrator and click "Import from Server":



MetaTrader 5 trading platform

Specify the server address and the details of the administrator/management account on it. The account must have the following permissions:

- Manager (add/edit/delete accounts)
- Supervise trades
- Personal details

Import

Connection details
Choose server type, fill server address and authorization information.

This wizard will help you to import settings from a MetaTrader 4 or MetaTrader 5 server. Please specify server type, address and authorization details below.

Server Type:

Server:

Login:

Password:

NOTE: When importing settings from a MetaTrader 4 server, part of settings will be set to defaults. For importing settings from a MetaTrader 5 server with the advanced authorization mode, you must have a confirmed certificate for the account.

< Back Next > Cancel

MetaTrader 5 trading platform

On the receiving server, the account must have the following permissions:

- Connection type: MetaTrader 5 Administrator
- Accounts section: "Accountant", "Access accounts", "Access account personal details"
- Dealing section: Access to trading orders (if import includes trading history)
- Configuration settings section: group settings

Request Accounts

The first step is to request the accounts you want to import from the source server. Enter your request in the "Choose groups" field. Here you can specify a comma separated list of logins or a more complex query using the "*" masks and the "!" negation symbol.

The "Choose groups" field contains the default request of "!demo*,!manager*,!coverage*,!contest*,*", which allows to select all accounts except those from the groups of demo and manager accounts, as well as coverage and contest groups.

To execute the request click "Request".

Import

?

×

Choosing trading accounts

Select trading accounts to be added to your database.

Choose groups: demo*

Request

Login	New Login	Group	Name	Balance	!
☒ 871318	881318	demo\demo forex	Spencer Jayden	5 000 000.00	
☒ 871319	881319	demo\demo forex	Patterson Landon	5 000 000.00	
☒ 871320	881320	demo\demo forex	Martin Eric	5 000 000.00	
☒ 871321	881321	demo\demo forex	Wood Jordan	5 000 000.00	
☒ 871322	881322	demo\demo forex	Taylor Kevin	5 000 000.00	
☒ 871323	881323	demo\demo forex	Cook Nicholas	5 000 000.00	
☒ 871324	881324	demo\demo forex	Parker Luke	5 000 000.00	
☒ 871325	881325	demo\demo forex	Account login already exists on the current server	0	
☒ 871326	881326	demo\demo forex	Turner Mason	5 000 000.00	
☒ 871327	881327	demo\demo forex	Jones Nicholas	5 000 000.00	
☒ 871328	881328	demo\demo forex	Johnson Jayden	5 000 000.00	

Move to group:

Accounts: 63093 Valid: 47578 Selected: 47578

Login shift: 10000

☒ Import balance, trades and trade history

< Back

Import

Cancel

MetaTrader 5 trading platform

Some of the accounts may have a red background, which means they cannot be imported. You can check the reason for that from the tooltip. The following reasons are possible:

- The currency of the account deposit does not match the deposit currency of the group to which you want to import the account.
- An account with the same number already exists on the current server or on one of other trade servers of the platform.
- The account number is out of the range of accounts allowed for the current server.
- The symbol, for which the account to import has orders or positions, does not exist on the server.
- The symbol, for which the account to import has orders or positions, has different numbers of decimal places on the source and target servers.
- The manager account group on the MetaTrader 4 server side does not have access to the financial instrument settings or the group of symbols.

Accounts from MetaTrader 4 can only be imported to groups with the hedging system of position accounting. Select a group with the hedging system in the "Move to group" field.

Double click on an account to view how it would look like after import. This will open a standard account viewing window.

You can disable import of individual accounts from the list. To do this, click on the icon at the beginning of the account row, and then it will change to .

If a client's name in MetaTrader 4 contains non-Latin characters, it may be imported incorrectly. During import, the current code page of MetaTrader 5 Administrator is used, which corresponds to the selected interface language. For example, a name with Chinese characters will be imported incorrectly if the Administrator terminal is used with English interface.

Depending on import settings, the login of a migrated account can be preserved or converted with an offset. For example, login 101980 is converted to 1101980, if the parameter "Login shift" is set to 1 000 000. If login shift is applied, the numbers of agent accounts in client records are shifted by the same value.

Enable Import of Trading Operations

To import trading operations of selected clients, enable the "Import balance, trades and trade history" option.

In MetaTrader 4, all trading operations are represented as a single entity – an order, which can be open, closed and pending. Additional notions are used in the 5th generation platform for a more detailed and clear presentation: deal and position. During import, each operation can be divided into an order, deal and position depending on the type of a source operation.

MetaTrader 5 trading platform

Source operation	History in MetaTrader 4	History in MetaTrader 5
Open order	1 operation	2 operations: an order to open a position, an opening deal
Closed order	1 operation	4 operations: an order to open a position, an order to close a position, an opening deal, a closing deal
Pending order	1 operation	1 operation

Note when importing orders

- "Migration" is specified in the "Reason" field of orders
- Fill or Kill policy is set for all orders
- If the expiry type is not specified in the source order, Good Till Canceled is set, otherwise the specified date is used.
- Standard commission specified in the source open order in MetaTrader 4 is imported to the "Swap" field of the relevant position on the MetaTrader 5 side

Note when importing deals

- Balance operations are imported "as is"
- Entry and exit trades are created based on the orders in accordance with the above table

Importing Tickets

Up to four history records in MetaTrader 5 can correspond to one MetaTrader 4 history record. That is why MetaTrader 4 tickets of orders and positions (including history orders) are not written directly to the appropriate fields of MetaTrader 5 orders, deals and positions during import. New tickets are given to all imported trading records. Ticket numbers are assigned in the ascending order from the appropriate range of order, deal and position tickers set for the MetaTrader 5 trade server.

Tickets of orders from MetaTrader 4 are copied to the 'ID' field (ID in the external trading system) of orders, deals and positions created on their basis. For example, when you import an open order from MetaTrader 4, three trade records are created in MetaTrader 5: an opening order, an opening deal, and an open position. The ticket of the MetaTrader 4 order will be written to the ID field of each of these trading records. The # character is added before the ticket number.

MetaTrader 5 trading platform

Terminal

Order /	Time	Type	Size	Symbol	Price	S / L	T / P	Price	Swap	Pro
257558455	2017.11.27 14:51:36	buy	1.00	eurusd	1.19405	1.19037	1.19676	1.19518	0.00	113.00
Balance: 5 000.00 USD Equity: 5 113.00 Margin: 1 194.05 Free margin: 3 918.95 Margin level: 428.21%										113.00

Trade | Exposure | Account History | News | Alerts | Mailbox **8** | Company | Market **55** | Signals | Code Base | Experts

Toolbox

Symbol /	Ticket	ID	Time	Type	Volume	Price	S / L	T / P	Price	Pro
eurusd	00279084	#257558455	2017.11.27 14:21:42	buy	1.00	1.19405	1.19037 ×	1.19676 ×	1.19550	14.50
Balance: 99 999.00 USD Equity: 100 012.50 Commission: -1.00 Margin: 59.73 Free Margin: 99 952.77										14.50
Margin Level: 167 440.98 %										

Trade | Exposure | History | News | Mailbox **7** | Calendar | Company | Market | Alerts | Signals | Code Base | Experts

Toolbox

Time	Order	ID	Symbol	Type	Volume	Price	S / L	T / P	Time	Sta
2017.11.27 14:51:36	100279084	#257558455	eurusd	buy	1.00 / 1.00	market	1.19037	1.19676	2017.11.27 14:21:42	fill
Total: 1 Filled: 1 Canceled: 0										

Trade | Exposure | History | News | Mailbox **7** | Calendar | Company | Market | Alerts | Signals | Code Base | Experts

Toolbox

Time	Deal	ID	Order	Symbol	Type	Direction	Volume	Price	Pro
2017.11.27 14:21:18	100289032				balance				100 000.
2017.11.27 14:21:42	100289033	#257558455	100279084	eurusd	buy	in	1.00	1.19405	
Profit: -1.00 Credit: 0.00 Deposit: 100 000.00 Withdrawal: 0.00 Balance: 99 999.00									0.

Trade | Exposure | History | News | Mailbox **7** | Calendar | Company | Market | Alerts | Signals | Code Base | Experts

If the comment of the source order contains non-Latin characters, it can be imported incorrectly. During import, the current code page of MetaTrader 5 Administrator is used, which corresponds to the selected interface language. For example, a comment written in Chinese will be imported incorrectly if the Administrator terminal is used with English interface.

Result of Import

The total number and the number of imported accounts will be shown on the last step:

Import?×

Importing trading accounts
Results of importing trading accounts to your database.

Login	New Login	Group	Name	Status ▲
810911	810911	demo\demo forex-hed...	Lewis Matthew	Done
810912	810912	demo\demo forex-hed...	Smith Thomas	Done
810911	810913	demo\demo forex-hed...	James Brandon	Done
810912	810914	demo\demo forex-hed...	Patterson Lucas	Done
810911	810915	demo\demo forex-hed...	Patterson Michael	Done
810912	810916	demo\demo forex-hed...	Evans Jayden	Done
810911	810917	demo\demo forex-hed...	Green Ashton	Done
810912	810918	demo\demo forex-hed...	James Brandon	Done
810911	810919	demo\demo forex-hed...	Lewis Matthew	Done
810912	810920	demo\demo forex-hed...	Clark Alejandro	Done
810911	810931	demo\demo forex-hed...	Evans Jayden	Done

Accounts: 104 / 104

< Back

Finish

Cancel

MetaTrader 5 trading platform

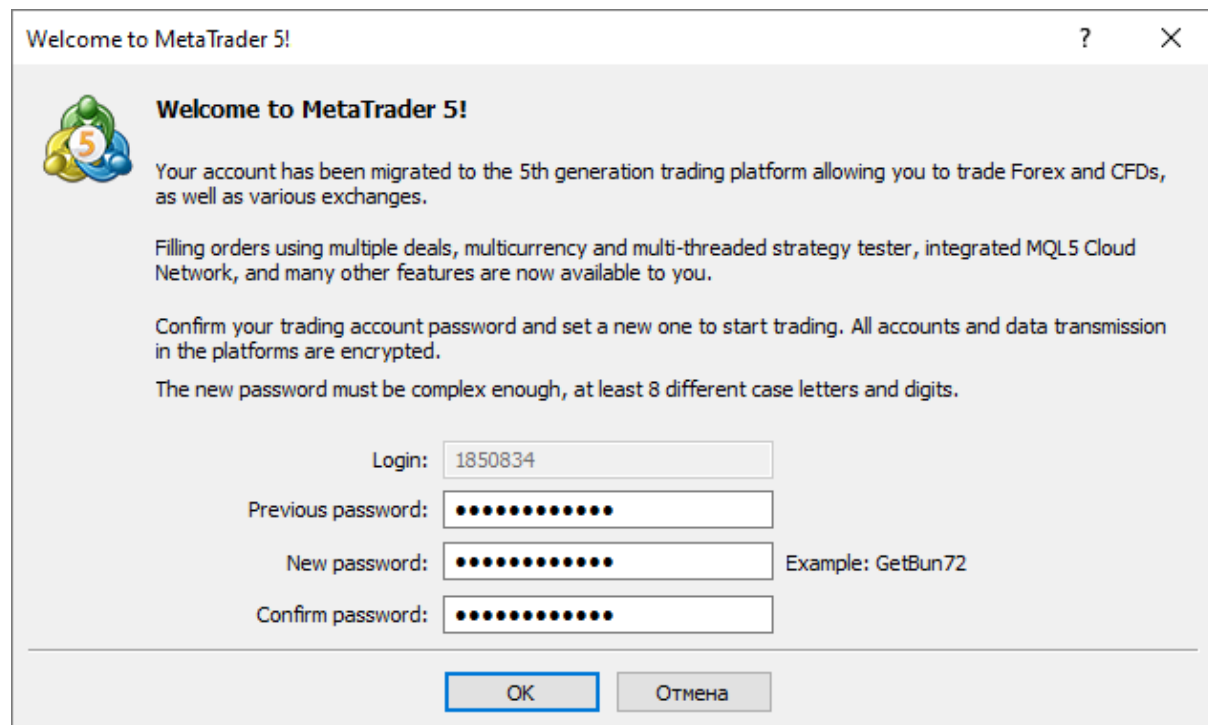
Import details are also available in the trade server journal.

- After import, the appropriate accounts on the MetaTrader 4 server must be disabled or set to the "read-only" mode to avoid trading state mismatch with MetaTrader 5.
- After completing the import for accounts having open positions, you can see the differences in equity and free margin in the Manager terminal. This may happen if quotes in your MetaTrader 5 platform are significantly different from the ones in MetaTrader 4 or a data feed is not configured at all. The Manager terminal calculates these values in real time according to the current prices in the platform. In the Administrator terminal, there are no differences (the values coincide with MetaTrader 4) since the data is taken directly from the accounts and are not recalculated.

Client connection and passwords

Random passwords are generated for imported accounts. Hashes of passwords that were used on the MetaTrader 4 accounts are saved in the client record. They are used for verification during the first account connection in MetaTrader 5.

During the first connection to the imported account, the client will attempt to connect using the old password. Upon entering the incorrect password (since a new random password has been generated), a welcome dialog will be displayed:



Welcome to MetaTrader 5!

Welcome to MetaTrader 5!

Your account has been migrated to the 5th generation trading platform allowing you to trade Forex and CFDs, as well as various exchanges.

Filling orders using multiple deals, multicurrency and multi-threaded strategy tester, integrated MQL5 Cloud Network, and many other features are now available to you.

Confirm your trading account password and set a new one to start trading. All accounts and data transmission in the platforms are encrypted.

The new password must be complex enough, at least 8 different case letters and digits.

Login: 1850834

Previous password:

New password: Example: GetBun72

Confirm password:

OK Отмена

MetaTrader 5 trading platform

The old password used in MetaTrader 4 should be specified here. The client will not be able to continue without this password. A new password should be set here, which will later be used to connect to the account.

If authenticated successfully, the dialog will not be displayed again. Once connected, the client will be able to continue using the account, just as if it has been opened in MetaTrader 5.

You can set a new password for the imported account via the administrator or manager terminal and provide it to the client. The welcome dialog will not be displayed in this case. After password change, the hash of the MetaTrader 4 password is deleted from the client record.

Import Features

Accounts are imported in packages each containing 100 account. If an account from the package could not be imported, you will need to repeat the import. If errors occur, the previous state will not be restored. The accounts and trade operations from the package successfully imported before the error will not be removed. Before re-import, remove them manually in the following order:

- Orders
- Deals
- Positions
- Accounts

Importing may take a long time: it depends on the database size and your Internet connection speed.

MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / Manager Accounts

Manager Accounts

The essential difference between the MetaTrader 5 system from the MetaTrader 4 server is an ability to create multiple manager* groups. This means that it is now possible to divide manager accounts so that different symbols (symbol groups) are available to them. Besides, they may receive different news, have different security settings and connect to different trade servers.

After installing the MetaTrader 4 server, the manager "1" with the highest server access level rights is created. In MetaTrader 5 platform, such an account is created by default as well but it has the login "1000".

Similarly to the fourth platform version, open an account in the manager group and add a new entry in the Managers section to create a manager account in MetaTrader 5. After that, configure the rights.

Setting the Manager Rights

Most MetaTrader 4 access rights have retained their functions in MetaTrader 5. When you start working in MetaTrader 5, you are able to open manager accounts identical to MetaTrader 4 server ones in terms of access right settings.

- Login, Groups and Email fields on the MetaTrader 4 server match the Common tab settings of MetaTrader 5 manager account.
- "Access Rights" block corresponds to the Permissions tab in MetaTrader 5.
- "IP filter" field matches the "IP Access List" tab.

MetaTrader 5 trading platform

Manager: John Smith

The Manager is a privileged user having additional rights, such as administration of other users and process of trading operations.

Login: 5 Available data for: unlimited days

Groups: * Mailbox name: John Smith

☐ IP filter From: 0 . 0 . 0 . 0 to: 0 . 0 . 0 . 0

Role: Manager: First Admin

Access Rights

- ☒ Manager (add/edit/delete a
- ☒ Administrator (full access to
- ☒ Reports
- ☒ Internal mail system
- ☒ Send news
- ☒ Connections (show online cli
- ☒ Configure server plugins
- ☒ Access to technical support
- ☒ Push notifications

Common Permissions IP Access List

Managers are privileged users that have additional permissions, such as administration of users and processing trade operations. Please specify account, mailbox, and groups to be served.

Login: 1000 Mailbox name: Administrator

Groups: *

Add Edit Delete

OK Cancel Help

MetaTrader 5 trading platform

The table below shows the correlation between MetaTrader 4 and MetaTrader 5 access rights:

MetaTrader 4	MetaTrader 5
Manager (add/edit/delete accounts)	Access accounts Access the account personal details Edit accounts
Administrator (full access to server configuration)	It corresponds to enabling all permissions.
Reports	Receive reports
Internal mail system	Send emails
Send news	Send news
Connections (show online clients)	View currently connected clients
Configure server plugins	Configure plugins
Access to technical support page	Access technical support page
Push notifications	Push notifications
Supervise trades	Access orders and positions
Accountant (deposit/credit/withdrawal money)	Accountant (deposit/withdraw)
Risk manager	Risk manager
Journals (direct access to server journals)	Access server logs
Edit prices, spreads, execution types	Throw in quotes
Personal details	Access the account personal details
Automatic server reports	Receive automatic server reports
Access to Applications Market	Access to Applications Market

Request Routing

The MetaTrader 5 manager settings have no parameters similar to symbol routing table of MetaTrader 4. However, the identical parameters can be configured in a separate Routing section of the Administrator terminal. The new section allows configuring routing in MetaTrader 5 in a more flexible and efficient manner.

Below you can see how to reproduce the settings of the MetaTrader 4 routing table in the MetaTrader 5 Routing section. In our example, the manager "1" is responsible for processing all trade requests with the volumes from 0 to 10 lots at "Forex" group

MetaTrader 5 trading platform

symbols on MetaTrader 4 server:

Trading Transactions

☒ Dealer ☒ Edit/Delete trades

Routing table for securities:

Type	Use	Minimum Lots	Maximum Lots
Forex	Да	0.00	10.00
Indexes	Нет	0.00	10.00
CFD	Нет	0.00	10.00
Futures	Нет	0.00	10.00
Forex23	Нет	0.00	10.00
Forex DD	Нет	0.00	10.00

OK Cancel

First, create a custom rule and set conditions for processing trade operations. Then use the Dealers tab to specify the manager who will process trade requests that meet the rule conditions.

MetaTrader 5 trading platform

Routing

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Perform action: ☒ skip this rule if no dealers online

Where request is:

Where order is:

Where conditions are:

Type	Condition	Value
 Client group	Equal (=)	real/real
 Request volume	Less or equal (<=)	10.00
 Symbols	Equal (=)	Forex*

Add

Edit

Delete

Routing

Common

Dealers



Please specify dealers who will process requests that meet the rule conditions.

Login	Name
 1000	First Admin

Specify the following parameters:

- Rule name.
- Perform action "Process to dealers". Enable "skip this rule if no dealers online" option to avoid missing trade requests if there are no managers online. In this case, if there is no appropriate manager online, all requests meeting the rule are processed according to the next rule in the list (with the lower priority).
- Select All for "Where request is:" and "Where order is:" options.
- Add the following rules for "Where conditions are:" section:
 - "Client group" set "real\real" in order to limit the manager operation by processing client requests from the real account group.
 - "Request volume" this condition corresponds to the level of maximum and minimum lots of the MetaTrader 4 routing table (from 0 to 10).
 - "Symbols" select Forex symbols similar to the MetaTrader 4 settings.

Next, add the manager "1000" on the Dealers tab of the rule. Please note that you can assign more than one dealer for a single rule.

MetaTrader 5 Trading Platform / Migration from MetaTrader 4 / Data Feeds

Data Feeds

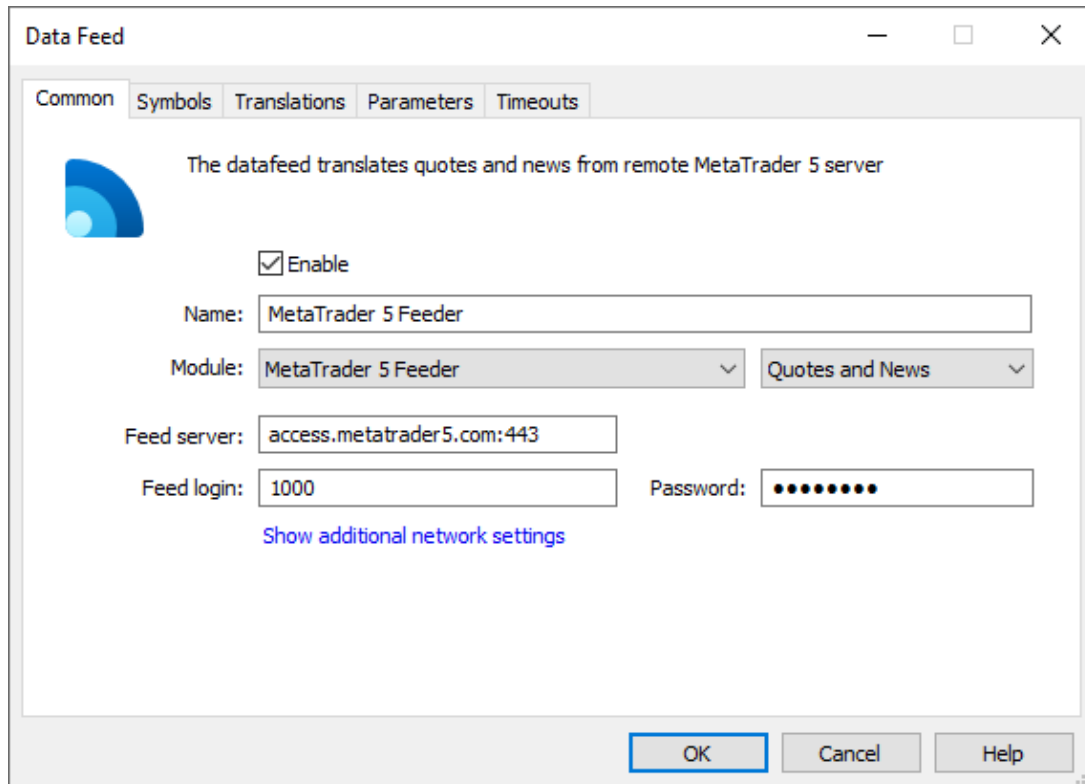
Any sources of quotes and news used in the MetaTrader 4 platform can be easily moved to MetaTrader 5. There are three options to receive a stream of quotes:

- From the MetaTrader 4 server having a trading account via MetaTrader 4 Feeder.
- From other data terminals via UniDDE Connector.

Apart from quote and news already working in the MetaTrader 4 platform, you can receive data via the new feeders designed specifically for MetaTrader 5. You can find out the details in the Data Feeds section.

Receiving Quotes from the MetaTrader 4 Server

If both platforms work simultaneously, use MetaTrader 4 Feeder to receive quotes and news from the MetaTrader 4 server. Add the new data source:



The screenshot shows the 'Data Feed' window with the 'Common' tab selected. The window title is 'Data Feed'. Inside, there is a description: 'The datafeed translates quotes and news from remote MetaTrader 5 server'. Below this, there is a checkbox labeled 'Enable' which is checked. The 'Name' field is set to 'MetaTrader 5 Feeder'. The 'Module' dropdown is set to 'MetaTrader 5 Feeder'. The 'Quotes and News' dropdown is set to 'Quotes and News'. The 'Feed server' field is set to 'access.metatrader5.com:443'. The 'Feed login' field is set to '1000'. The 'Password' field is masked with dots. There is a link 'Show additional network settings' below the password field. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons.

Data Feed

Common Symbols Translations Parameters Timeouts

The datafeed translates quotes and news from remote MetaTrader 5 server

☒ Enable

Name: MetaTrader 5 Feeder

Module: MetaTrader 5 Feeder Quotes and News

Feed server: access.metatrader5.com:443

Feed login: 1000 Password:

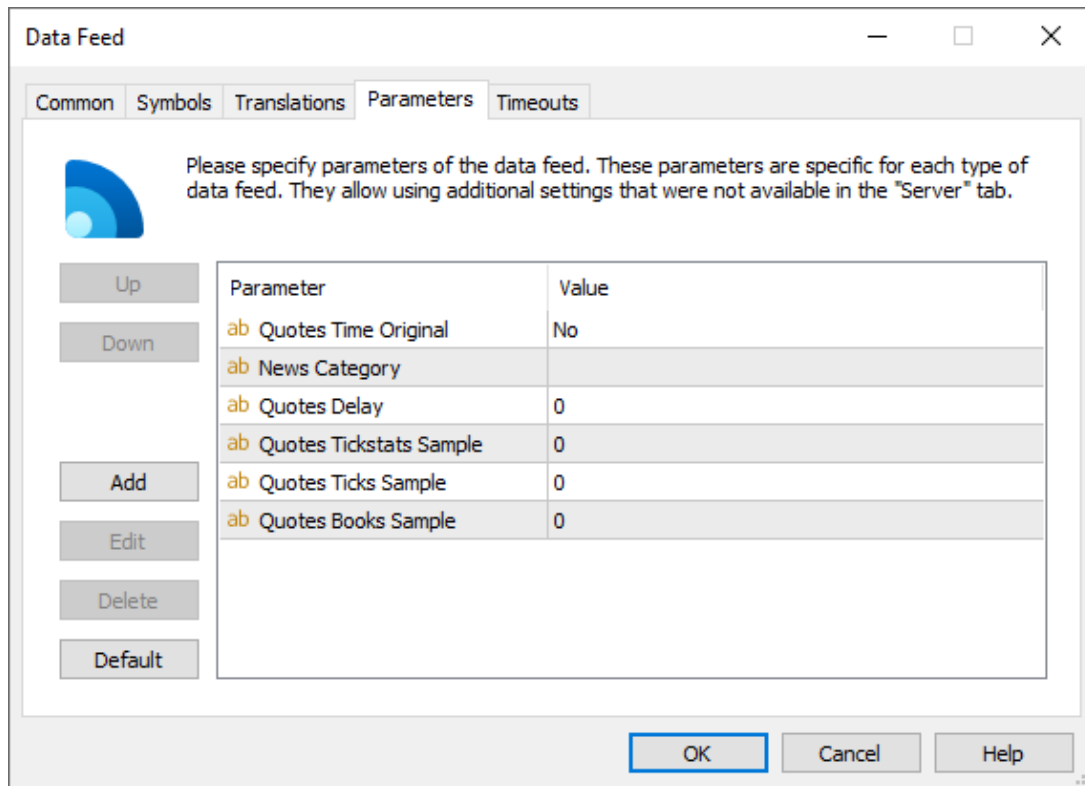
[Show additional network settings](#)

OK Cancel Help

MetaTrader 5 trading platform

Select MetaTrader4Feeder in the Module field and enter server's IP address and port, as well as account login and password in the Feed server, Feed login and Password fields to connect to the MetaTrader 4 server.

MetaTrader 5 allows sorting out news by language. To do this, set the Language parameter on the Parameters tab:



MetaTrader 5 trading platform

News coming from MetaTrader 4 servers can be in text or HTML formats. In case of a text format, setting the Language parameter is desirable for a proper language recognition and correct news display.

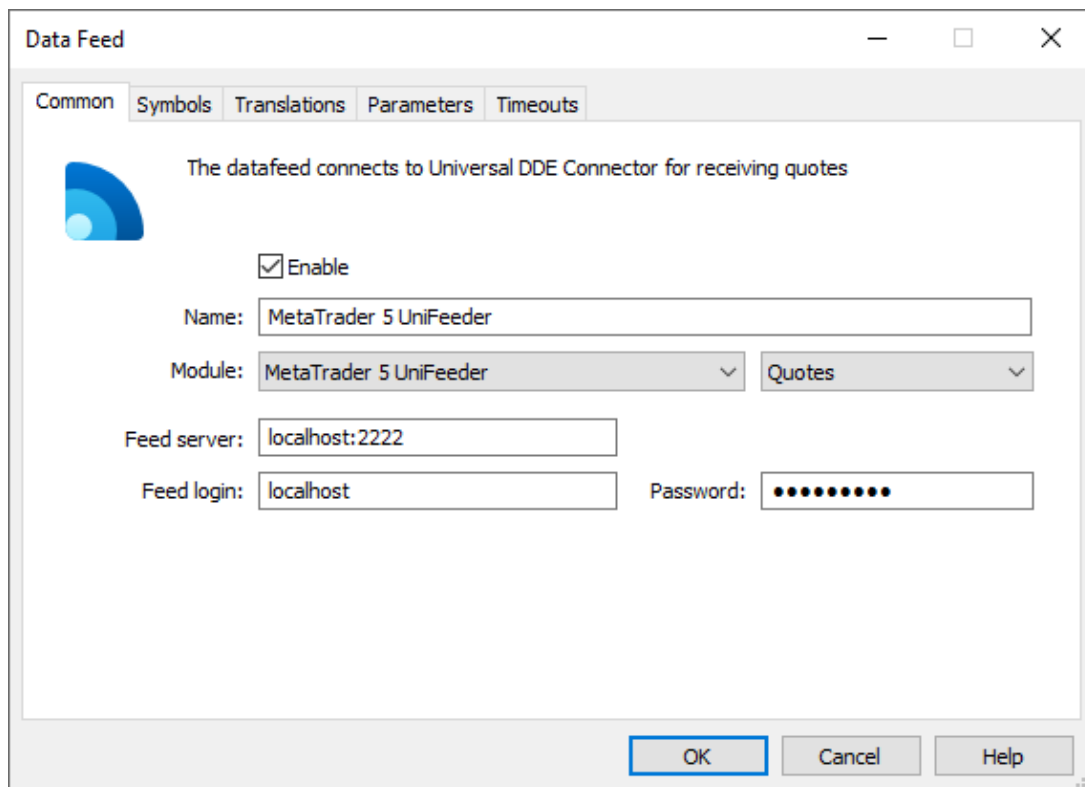
The news language in HTML format is automatically defined by the "charset" attribute built in the news. The language name is specified in the format that is standard for Windows operating systems without defining the dialectical features by geographic location, for example, English, Russian, etc.

Receiving Quotes via UniDDE Connector

If your MetaTrader 4 server receives quotes via UniDDE Connector, you can use MetaTrader5UniFeeder component to receive them in MetaTrader 5.

DDE Connector allows collecting quotes from various data sources that support the DDE (Dynamic Data Exchange) protocol. The feeder translates quotes received from it to the MetaTrader 5 History Server.

Add the necessary data source via the corresponding section of MetaTrader 5 Administrator:



Data Feed

Common Symbols Translations Parameters Timeouts

The datafeed connects to Universal DDE Connector for receiving quotes

☒ Enable

Name: MetaTrader 5 UniFeeder

Module: MetaTrader 5 UniFeeder Quotes

Feed server: localhost:2222

Feed login: localhost Password:

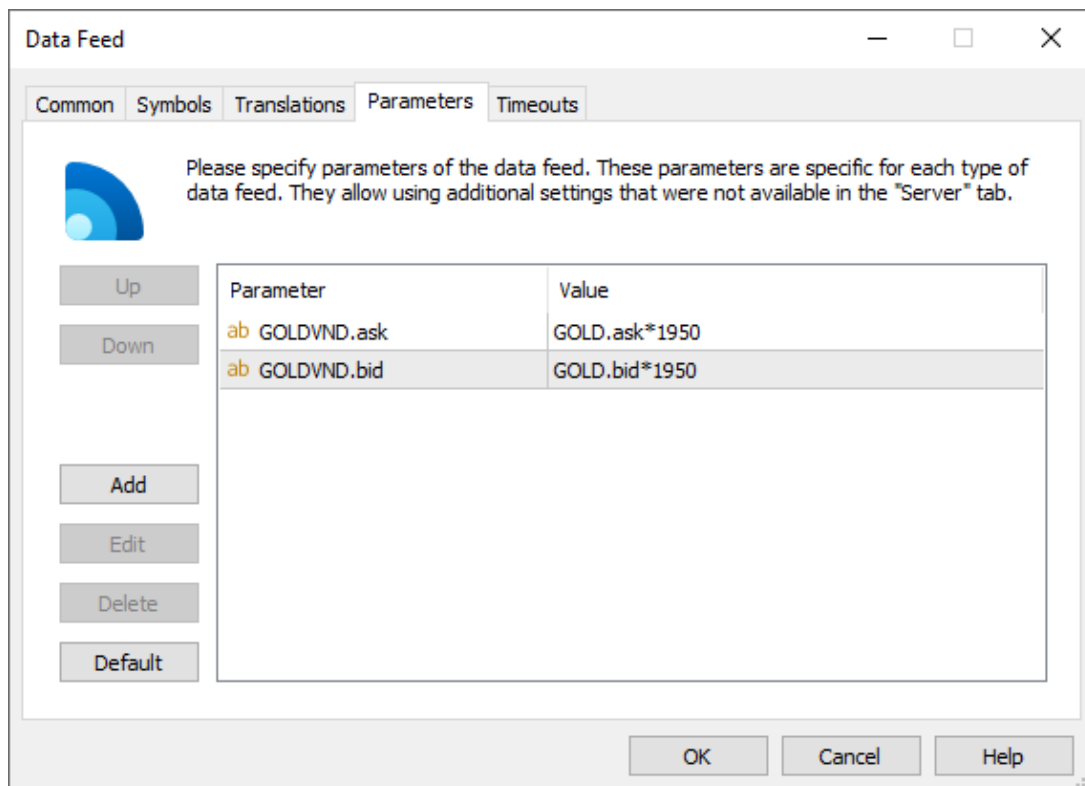
OK Cancel Help

MetaTrader 5 trading platform

Set the server address where UniDDE Connector is installed, as well as connection port and account preliminarily created in UniDDE Connector (login and password) in the source settings.

MetaTrader5UniFeeder allows receiving quotes for the symbol data not transmitted via UniDDE Connector. This is achieved by mathematical conversion of other symbols' quotes transmitted via UniDDE Connector. This method of calculation can be applied to non-freely convertible currencies with their exchange rates to the major world currencies defined by the central bank.

To receive quotes for such symbols, specify the quote calculation equations on the Parameters tab of a data source:



Data Feed

Common Symbols Translations **Parameters** Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

Parameter	Value
ab GOLDVND.ask	GOLD.ask*1950
ab GOLDVND.bid	GOLD.bid*1950

Up Down Add Edit Delete Default

OK Cancel Help

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator

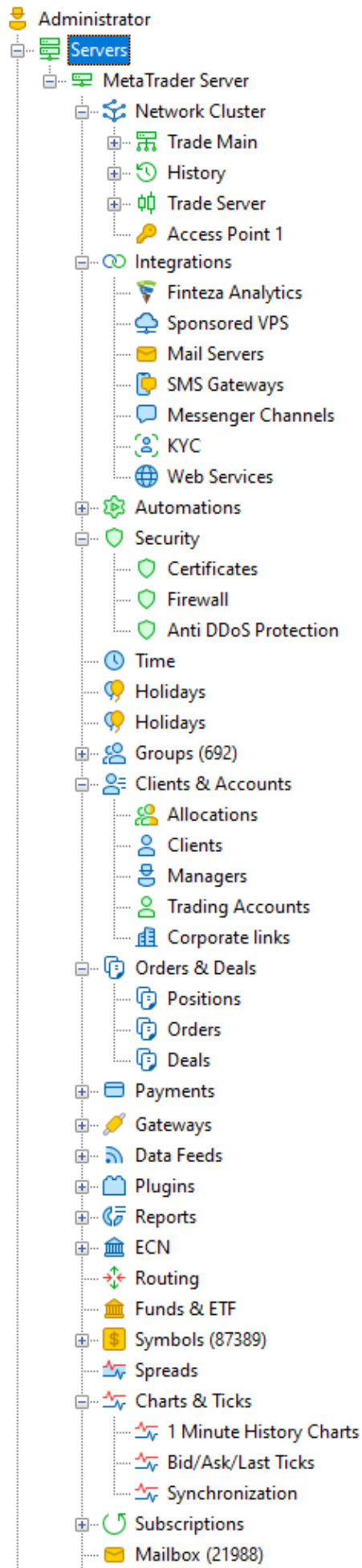
MetaTrader 5 Administrator User Guide

MetaTrader 5 Administrator is a software program for the remote administration of the MetaTrader 5 trading platform. It enables control over servers and other components of the platform, management of user group settings, manager configurations, financial symbol and data feed parameters. Also, the Administrator Terminal allows managing databases of quotes and tick data, configuring trading schedule and adjusting backup parameters.

The trading platform is provided with enhanced configuration options. All platform settings are grouped into categories, which are displayed as a tree in the left part of the Administrator Terminal:

- Start Page general information about the platform, live update settings
- Network Cluster settings of platform components
- Integrations configuring mail servers, messengers and access to Finteza analytics
- Automations configuring scenario-based automated actions
- Certificates configuring certificates for extended authentication
- Firewall configuring access by IP addresses
- Anti DDoS Protection integration with external protection providers
- Time general time settings of the server and setting of trade time
- Holidays holiday settings
- Leverage floating margin settings
- Groups account group settings
- Allocations configuring the opening of demo and preliminary accounts through client terminals
- Clients working with customer data, documents and accounts
- Managers administration of manager accounts
- Trade Accounts working with accounts
- Corporate links adding custom links to the client terminals
- Positions working with positions
- Orders working with orders

MetaTrader 5 trading platform



- Deals working with deals
- Payments integration with payment systems
- Gateways administration of gateways that provide interaction with exchanges
- Data Feeds configuring quotes and news sources
- Plugins modules of expanding the platform functionality
- Reports report settings
- ECN configuring the built-in electronic communication network
- Routing rules of processing of coming trade requests
- Funds & ETF controlling fund settings: commissions, managers, investors
- Symbols configuring financial instruments
- Spreads configuring margin requirements for positions in spread
- 1 Minute History Charts working with price data stored at the server
- Bid/Ask/Last Ticks viewing tick data
- Synchronization configuring synchronization with other MetaTrader servers
- Subscriptions configuring subscriptions for additional trader services
- Mailbox built-in email system for communicating with traders and colleagues
- Live Update configuring automatic updating of the system components
- Support Center this section offers help for the MetaTrader 5 platform users: articles, news, answers to frequently asked questions and a possibility to contact the technical support team directly
- App Store store of additional components for the trading platform: gateways, liquidity, CRM and others

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started

Getting Started

This section contains basic information you need to know when starting to work with the administrator terminal:

- **Install Terminal**
Step-by-step instructions on how to install the terminal.
- **Start Terminal**
How to start the administrator terminal depending on where it is installed and the parameters of the operating system.
- **Structure of Directories and Files**
Description of the structure of information storage in the administrator terminal.
- **Add or Remove Servers**
Working with administered servers.
- **Connect to Server**
Connecting to a server to administer it.
- **Change Password**
Changing password of the currently connected account.
- **Live Update**
How to receive new versions of the administrator terminal.
- **Uninstall Terminal**
Step-by-step instructions on how to uninstall the administrator terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Install Terminal

Install Terminal

In order to install the administrator terminal, it is necessary to download its distribution and launch it.

- The installation program is a web installer; this means that the major portion of the distributive will be downloaded from the Internet during installation.
- The administrator terminal installer determines type of an operating system and installs the corresponding version of the terminal.
- The terminal can operate under Microsoft Windows 10/11 or Windows Server 2008R2/2012/2012R2/2016. A processor with SSE2 support (Pentium 4/Athlon 64 or higher) is required for operating as well. Other hardware requirements are limited with software ones.
- The terminal can run on devices with ARM64 (for examples, on Microsoft Surface tablets), but in this case, the minimum OS version is Windows 11.

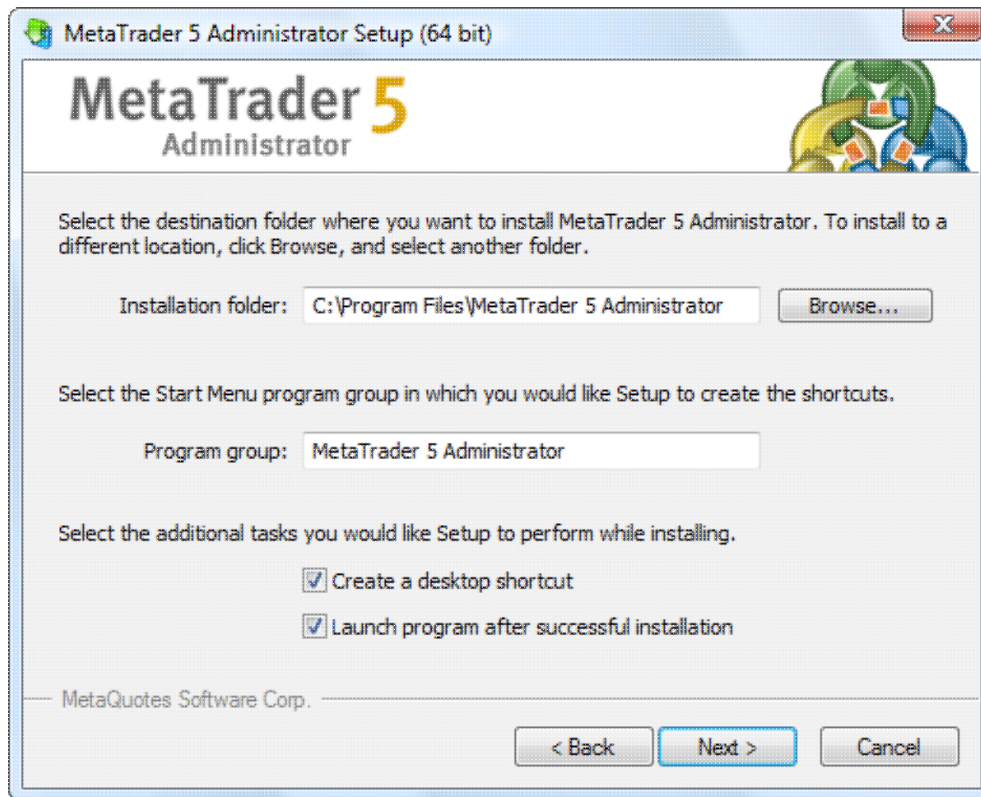


MetaTrader 5 trading platform

At the first step of installation, read the program description and the copyright notice. To get acquainted with the full text of the End User License Agreement, press the "Read License Agreement" button. If you agree with the license terms without seeing, press the "Next" button.



If you agree with all terms of the agreement, press the "Next" button. If you do not agree, exit the terminal installation program.



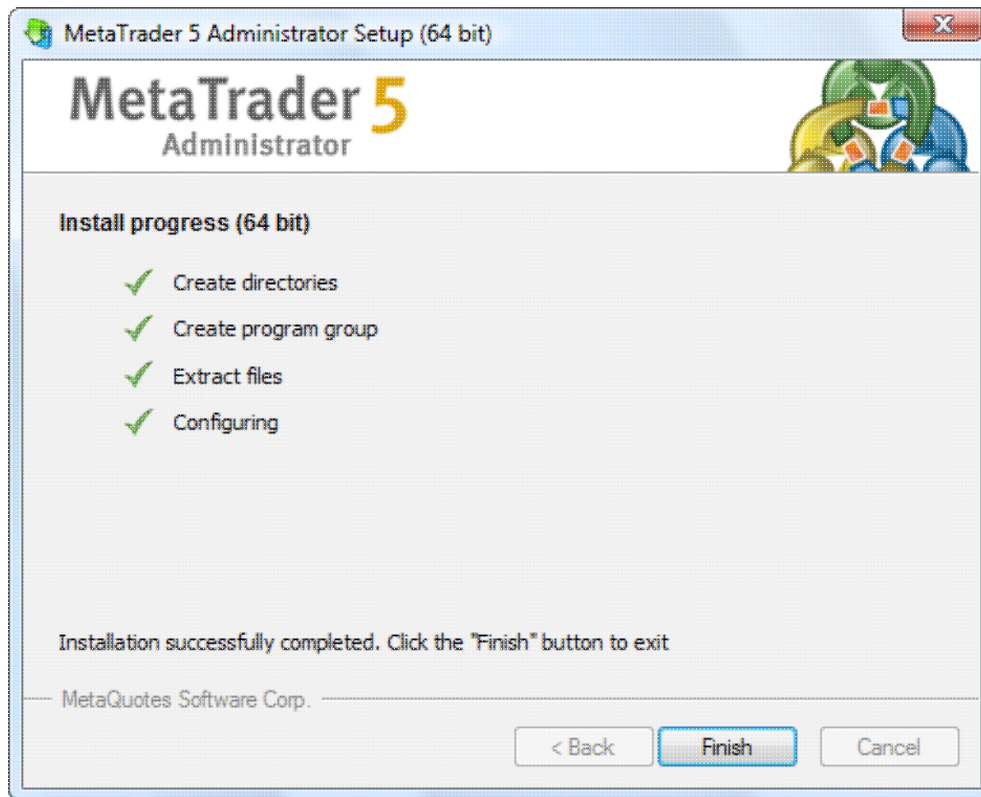
The following parameters should be selected at this installation stage:

- **Installation folder** specify a folder to install the terminal to. Another folder can be selected by writing a path to it manually or using the "Browse" button.
- **Program group** the name of the program group that will be created in the "Start" menu is indicated in this field.
- **Create a desktop shortcut** if this field is checked, a shortcut for launching the terminal will be additionally created on the desktop.
- **Launch program after successful installation** if this field is checked, the terminal will be automatically launched after the successful installation.

Terminal can be installed over an installed version of it. At that, all terminal settings remain the same.

After "Next" is pressed, the terminal downloading and installation process will start.

MetaTrader 5 trading platform



To complete the installation process, press "Done". After that you can launch the terminal.

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [Getting Started](#) / Start Terminal

Start Terminal

After the installation has been completed, a group of the administrator terminal programs is created in the "Start" menu, and the program shortcut is additionally located on the desktop. They will help to start the terminal.

You cannot run two copies of the administrator terminal from the same directory simultaneously, including running in the console mode. If you need to run multiple terminals at the same time, install the appropriate number of programs in different directories.

The administrator terminal can be launched in two modes.

Main launch mode

Starting from MS Windows Vista, applications installed to Program Files are not allowed by default to store their data in the installation folder. All data should be stored in a separate Windows user directory.

Thus, if the terminal is installed in the Program Files directory and user rights to write to that directory are limited, it is run in the main mode. The main mode is also used in the following situations:

- If the UAC (User Activity Control) system is enabled.
- If remote connection to a computer is used (RDP, Remote Desktop Protocol).

In this mode, the editable files of the terminal are stored in a specific Windows user directory, and the immutable files are stored in Program Files. The immutable files include the executable terminal file, built-in help files, etc. Editable files are:

- all settings of the terminal, configuration files
- all databases (news, mails)
- terminal operation logs
- email templates

All the editable files of the terminal are stored in the following directory:
C:\Users\username\AppData\Roaming\MetaQuotes\MetaTrader 5
Administrator\instance_id\.

Here 'C' is the logical drive letter on which Windows is installed, "username" is the account name in the operating system under which the terminal has been installed, "instance_id" means a unique identifier generated based on the path to the directory where you installed 0terminal.

For quick access to these folders, use the command " Open Data Folder" in the [File](#) menu. Each data folder contains a special text file origin.txt. This file contains the path to the terminal installation folder, which corresponds to this data directory.

- In the main mode, the catalog where editable files are stored is different for each Windows account.
- A detailed description of the file structure of the administrator terminal and of the files purpose are given in the appropriate section.

Portable Mode

When installed to Program Files, the terminal operates by default in the main mode described above. All data are stored in a special Windows user directory. However you can force the terminal to store its data in the installation folder. To do it, run the platform in the portable mode. To use this mode, start terminal from the command line with the additional /portable key. For example, "D:\Program Files\MetaTrader 5 Manager\mt5manager64.exe /portable".

To run the terminal in Portable mode, the following conditions must be met:

- If the terminal is installed in the Program Files folder, the user must have administrator rights on the computer. In addition, UAC (User Account Control) must be disabled in the operating system.
- If the terminal is installed in any other folder, the user must have permission to write data to that folder.

Console Mode

The administrator terminal can be used in the console mode, without the user interface. This mode enables the automation of some processes when operating with the trading platform.

Use the /console key to launch the terminal in the console mode:

```
/console /server:<trade server address:port> /login:<login> /password:<password> /action:
```

Specify platform connection details in the 'server', 'login' and 'password' parameters. The type of the action to perform is specified in 'action':

Server restart

Command /action:restart [/name:<server name>]. Optionally the server name can be specified (History, Access...). If the name is specified, the appropriate sever will be searched (case insensitive). If the name is not specified, the specified command will be performed for the currently connected server.

Platform configuration export to JSON

Command /action:export /file:<path> [/type:<type> /config:<mask>]. If optional arguments are not specified, the entire configuration of the connected server is exported.

Argument /file:<path> sets the destination file path. Argument /type:<type> is used for exporting a selected configuration branch and it may have the following values:

common network firewall time holidays groups managers routing gateways plugins feeders reports symbols spreads historysync
--

Argument /config:<mask> sets a search criterion to search for a particular configuration structure by a string mask (this may contain *,!). Search by mask can be used for the configurations of servers, groups, managers, trade requests, gateways, plugins, data feeds, reports and symbols. The following rule applies to other configurations: if a mask is set, the configuration should be skipped, if no mask is specified, the configuration should be processed.

Platform configuration import

Command /action:import /file:<path> [/type:<type> /config:<mask>]. All arguments are similar to those applied to exports, except for the /type argument. The common branch (common platform settings) cannot be imported.

In console mode, log files are saved in the directory where the Administrator terminal executable file is located: [directory with mt5admin64.exe]\Logs\.

MetaTrader 5 Trading Platform /
MetaTrader 5 Administrator /
Getting Started / Structure of
 Directories and Files

Structure of Directories and Files

This section contains the description of directory and file storing structure in the administrator terminal. In the guest terminal start mode storages are divided into those containing changeable and unchangeable files.

Unchangeable Files

These files are located in /Program Files/MetaTrader 5 Administrator/. They are:

- **MT5Admin.exe** executable file of the administrator terminal;
- **Uninstall.exe** program for deinstalling the administrator terminal;
- **/help/*.chm** built-in help files of the terminal.

Changeable Files

The main directory of the terminal contains several folders: *config*, *logs*, *profiles*, *roles*, *templates*. The special command "📁 Open Data Folder" located in the "File" of the terminal menu allows quick accessing the location of this data storage.

All text files are of Unicode format. The corresponding software should be used for their editing.

The **config** directory contains the files of terminal settings, including servers added for administration:

Files	Description
mt5admin.ini	Contains <u>settings</u> of the administrator terminal, all interface parameters of the terminal, last used values of window positions, etc.
servers.dat	Contains parameters of <u>connection</u> to the administered servers.

The **logs** directory contains log files of terminal operation, as well as crash logs:

Folders and Files	Description
/Crash/crash.log.*	Directory /crash contains files of terminal crashes. These files are automatically sent to the developer for finding out reasons for the crash and eliminating them.
yyyymmdd.log	<u>Log</u> files containing all the information about events occurring in the administrator terminal. Terminal logs are saved in separate files for each working day. Here: yyyy denotes a year, mm month, dd day.

The **profiles** directory contains databases of the terminal distributed by administered servers and some settings:

Folders	Description	Files	Description
certificates	Contains files of certificates.	*.pfx	Files of certificates that can be installed to the system storage or e-token for further <u>authorization</u> on the server.
config	Contains files of settings related to this server.	performance.dat	File that contains the history of server performance displayed at the " <u>Monitor</u> " tab.
		symbols-*.dat	List of <u>symbols</u> of the server and their settings.
mail	Contains mail database of the terminal.	mail-*.dat	Files of <u>mail database</u> separately for each login. The login number is specified after a hyphen in the file name.
news	Contains news database of the terminal.	news-*.dat	Files of <u>news database</u> separately for each login. The login is specified after a hyphen in the file name.

The **roles** folder contains sets of permissions for manager accounts:

Files	Description
*.rol	These files contain the sets of settings of permission for the <u>accounts of managers</u> .

The **templates** directory contains templates of news and messages for sending:

Folders	Description	Files	Description
mail	Contains mail templates.	*.htm	HTM-files of <u>mail templates</u> saved using the corresponding command in the window of message writing.
news	Contains news templates.	*.htm	HTM-files of <u>news templates</u> saved using the corresponding command in the window of news sending.

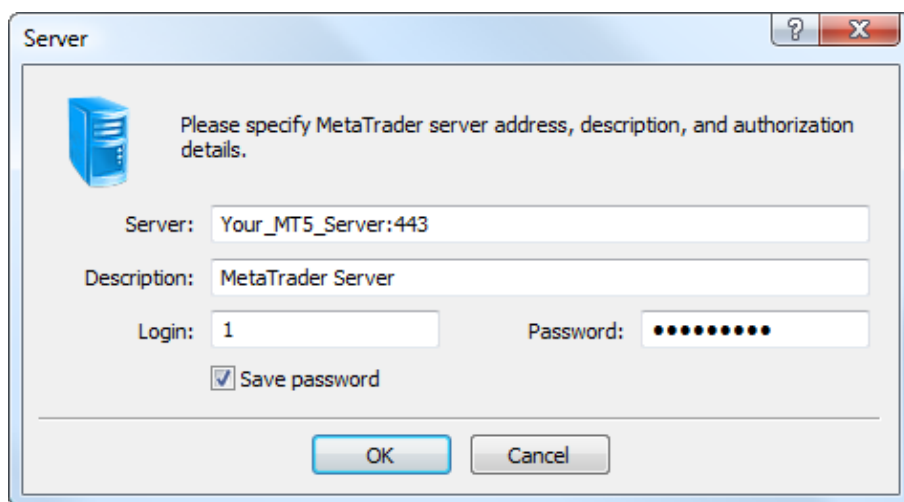
MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Add or Remove Servers

Add or Remove Servers

The administrator terminal allows managing operation of several servers. Servers and commands for working with them are located in the form of a tree in the left part of the terminal.

Adding a Server

To add a server, one should execute the "+ Add Server" command in the **"File"** menu or use the similar context menu command of the left part of the terminal. After that the following window will appear:



In this window the following fields should be filled out:

- **Server** IP-address or domain name of the necessary server and its port number. If port number is not specified then port 443 is used on default;
- **Description** description or name of the server (not more than 128 symbols). This information will be displayed in the tree of servers in the left part of the terminal window;
- **Login** administrator account (login), using which connection to the server will be performed. If such an account has not been created on the server, connection is impossible;
- **Password** administrator password;
- **Save password** save password between program restarts.

It is not necessary to specify "Login" and "Password" when adding a server. These details will be requested when connecting to it.

Password to the server must be no less than six symbols long and contain at least two of three types of symbols (uppercase letters, lowercase letters and digits).

MetaTrader 5 trading platform

If there are several servers added to the administrator terminal, they can be moved relatively to each other in the tree-like list. To do that, one should use the ↑ and ↓ buttons of the "Edit" menu.

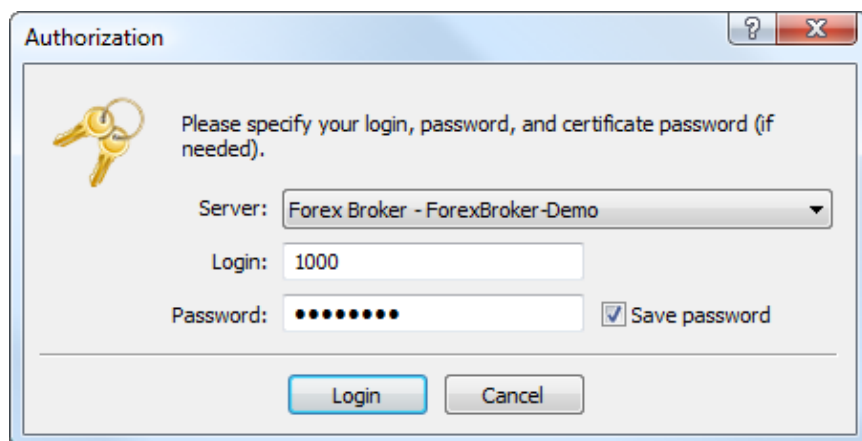
Deleting a Server

In order to delete a server, execute the "✗ Delete Server" command in the "File" menu or the similar context menu command in the left part of the administrator terminal. Further you can add this server again and start administering it.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Connect to Server

Connect to Server

To start administering a server, one should connect to it. The "→ Connect" command of the "File" menu, the similar command of the toolbar and the corresponding command of the context menu are used for it. Once one of them is pressed the following window will appear:



The window contains the following fields:

- **Server** this field contains the server the connection to which will be performed. From the list here one can choose any server from those added to the terminal.
- **Login** administrator account (login), using which the connection to the server will be performed. If such an account has not been created on the server, connection is impossible.
- **Password** administrator password.
- **Save Password** save password between program restarts.

In order to connect to a server, one should press the "Login" button.

- To be able to connect to a server, one should first add it to the administrator terminal.
- In case the mode of extended authorization is enabled for managers, the procedure of additional authorization is conducted.
- If the connection to the administered servers is performed through a proxy server, it is necessary to set up the terminal in the corresponding way.

If the authorization is successful, the connection icon  appears in the status bar, as well as two following entries appear in the journal:

1. 'login number' **authorized on** 'server name';
2. 'login number' **configuration synchronized with** 'server name'.

MetaTrader 5 trading platform

After the successful authorization you can start administering the server. If any problems occur while connecting to the server and you don't get one of the messages mentioned above, you should check the correctness of all specified connection settings, address and port of the server.

While performing full installation of the platform one administrator account with login "1001" and random password is automatically created.

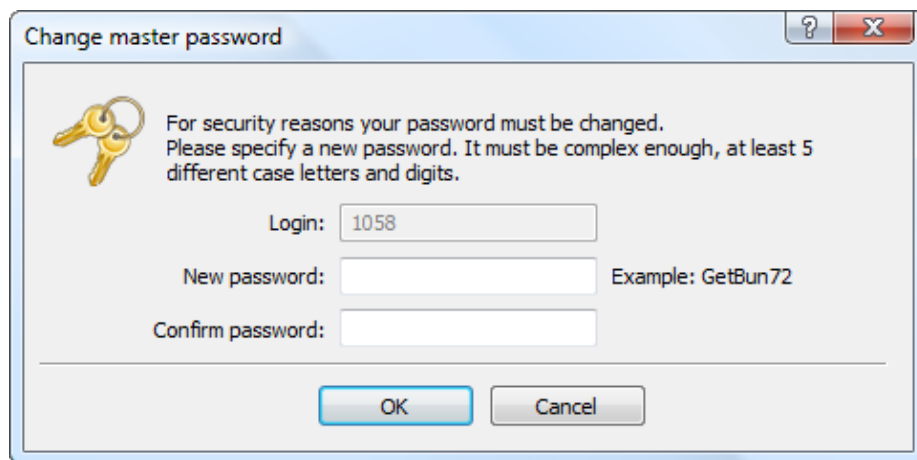
If there are several trade servers within the platform, note that different accounts are used to connect to each of them.

An administrator connected to the main trade server can set up the whole trading platform and manage trade databases and account databases of this server. The main server administrator configures groups, symbols, security settings, etc.

Administrator of additional trade servers do not have an access to the platform settings, but they can manage trade and account databases of their specific servers.

Forced Change of Password

After authorizing you may be asked to change master password to your account. The forced change of password can be enabled in the group settings or in the account settings. The procedure of master password changing, whether at first login or periodically, increases the security of working.



Type a new password in the "New password" field and then type it again in the "Confirm password" field. The new password must comply with the following requirements:

- It must be no shorter than the length requirement displayed in the password change dialog. The minimum number of characters is determined by group settings, while the lowest possible value is 8 characters. The maximum length is 16 characters.
- It must contain four character types: lowercase letters, uppercase letters, numbers, and special characters (#, @, ! etc.). For example, 1Ar#pqkj.

MetaTrader 5 trading platform

- It should not be the same as the previous password.

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [Getting Started](#) / [Connect to Server](#) / Extended Authorization

Extended Authorization

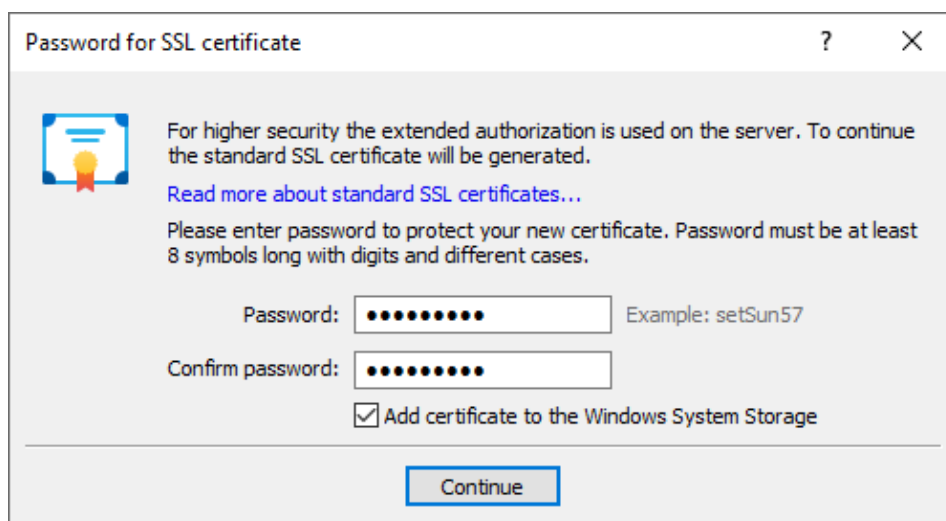
The trading platform offers the possibility of an extended authorization using SSL certificate which considerably increases the system security. The extended authorization is enabled in [group settings](#). If this connection mode is enabled, [standard authorization](#) is still enabled too. It means that a user will have to enter account details in any authorization mode.

- The authorization algorithm is generally accepted and secure. It is fully analogous to the SSL authorization.
- Connection between a terminal and a server is established by a custom protocol with the encryption of all the data transmitted.
- A public key can be freely distributed and used for verifying the message signed by the secret key. It is guaranteed that knowing a public key it is impossible to count the secret one within reasonable time. The calculation of a secret key based on a public one even with powerful computers can take tens or even hundreds of years.
- A detailed description of the entire process of the extended authorization is given in a [separate section](#).

Generating and Getting a Certificate

At the attempt to authorize through an account from a group with the enabled extended authorization, the [standard authorization](#) must be performed first. After that a trade server sends a request to an administrator terminal for the generation of two keys: private and public. A public key is sent to a trade server.

Based on the account data, a server generates a certificate and signs it by its private key (signing of a certificate by a server's private key guarantees that the certificate can't be falsified). After that a window appears in the administrator terminal, where the password must be indicated to protect the certificate:



Password for SSL certificate

For higher security the extended authorization is used on the server. To continue the standard SSL certificate will be generated.

[Read more about standard SSL certificates...](#)

Please enter password to protect your new certificate. Password must be at least 8 symbols long with digits and different cases.

Password: Example: setSun57

Confirm password:

☒ Add certificate to the Windows System Storage

Continue

The following fields and parameters are available in this window:

- **Password** password for the certificate installation;
- **Confirm Password** password confirmation to eliminate errors;
- **Add certificate to the Windows System Storage** if this option is enabled, the certificate will be automatically added to the operating system storage.

The password for the certificate must contain at least two types of symbols (lower case, upper case, digits), and be at least 5 symbols long.

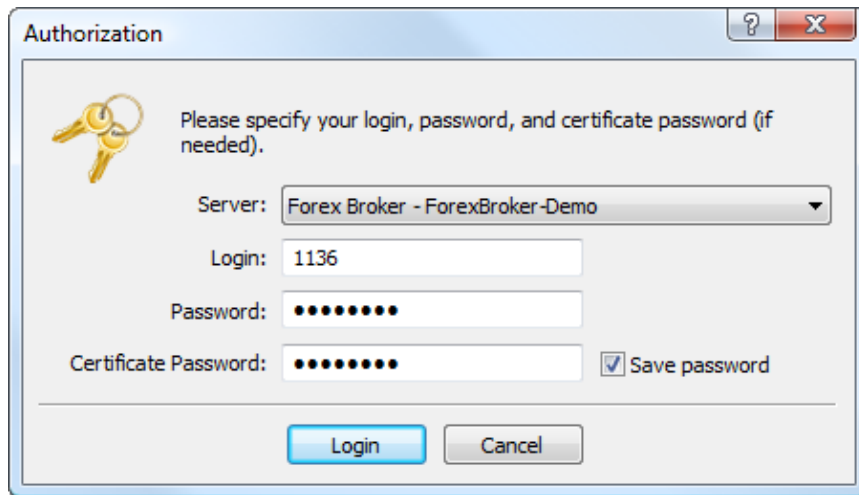
After all the required data are specified, press "Continue". After that the certificate is packed and protected by the specified password. The resulting *.pfx file of the certificate is saved in /profiles/server name/certificates of the administrator terminal to enable its further relocation. Certificate files are named according to the following rule: *Login_ID_Name.pfx*, where:

- **Login** account number;
- **ID** short company name where the account is opened;
- **Name** client's name specified during account creation.

- Even getting access to the *.pfx file one can't use the certificate without the password. The minimum length of a password is set in group settings.
- Generation of certificates is performed only during the first account connection or when a certificate was intentionally reset on the server.
- The certificate is not required when connecting using an investor password.

Authorization

In further attempts to connect with the extended authorization, the certificate password will be required together with the main account details:



Confirmation of Certificates

The additional mode of certificate confirmation can be enabled on a server – this considerably increases the platform operation security. Until the certificate is confirmed, connection using the account in a manager or administrator terminal is impossible, while connection in a client terminal can be established only in the investor mode without the possibility to trade.

In this mode, after a certificate is received, a special email is sent to the terminal describing actions that must be taken to confirm the certificate (for example, call at the specified number and identify a person). The email can be viewed on the "Mailbox" tab of the "Toolbox" window.

Once a client or manager performs the actions specified in the email, the administrator can confirm the certificate of the account on the "Security" tab.

- For demo accounts the certificate confirmation is performed automatically, as soon as the certificate is generated.
- After the certificate has been confirmed, the reconnection using account details is necessary in the client terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Connect to Server / 2FA/TOTP

2FA/TOTP Authentication Using One-Time Password

The use of one-time passwords (Time-based One-time Password Algorithm, TOTP) or two-factor authentication (2FA) provides an additional level of security for trading accounts. This technology protects a trading account from unauthorized access even if its login and password are leaked.

If 2FA/TOTP is enabled, users are required to enter a special one-time password every time they connect to their accounts, in addition to standard account login and password (as well as a certificate if the extended authentication is enabled). One-time passwords can be generated by:

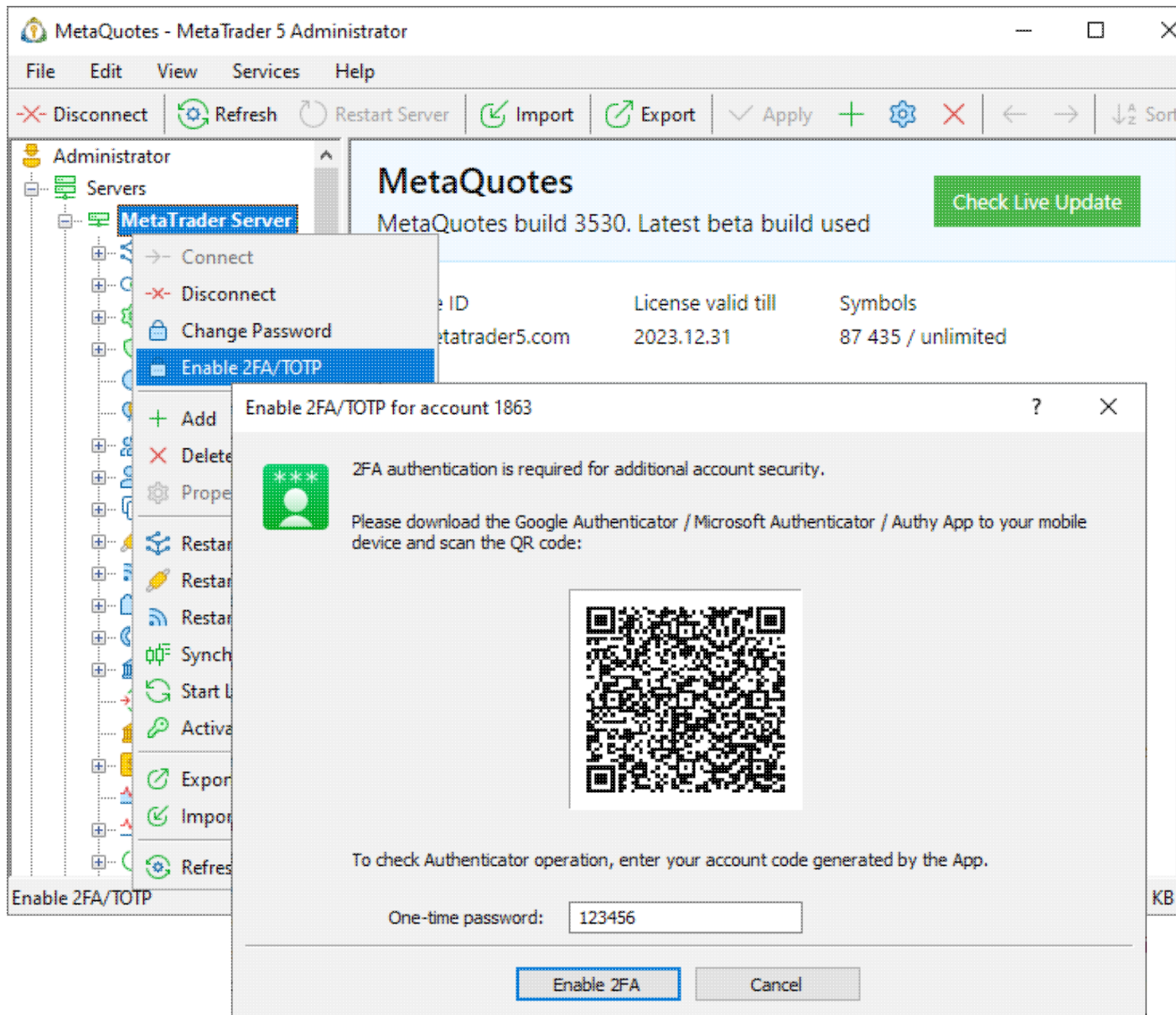
- Mobile terminals for iPhone and Android.
- The most popular 2FA applications, including Google Authenticator, Microsoft Authenticator, LastPass Authenticator and Authy. They can be downloaded to a mobile device from the App Store or Google Play.

The ability to use OTPs should be enabled on the trading server. This additional level of security can be optional or mandatory, depending on the specific regulatory or market requirements.

Enabling 2FA/TOTP in the desktop terminal

To enable the two-factor authentication option, connect to your account in the usual way. Then click "Enable 2FA/TOTP" in the server menu.

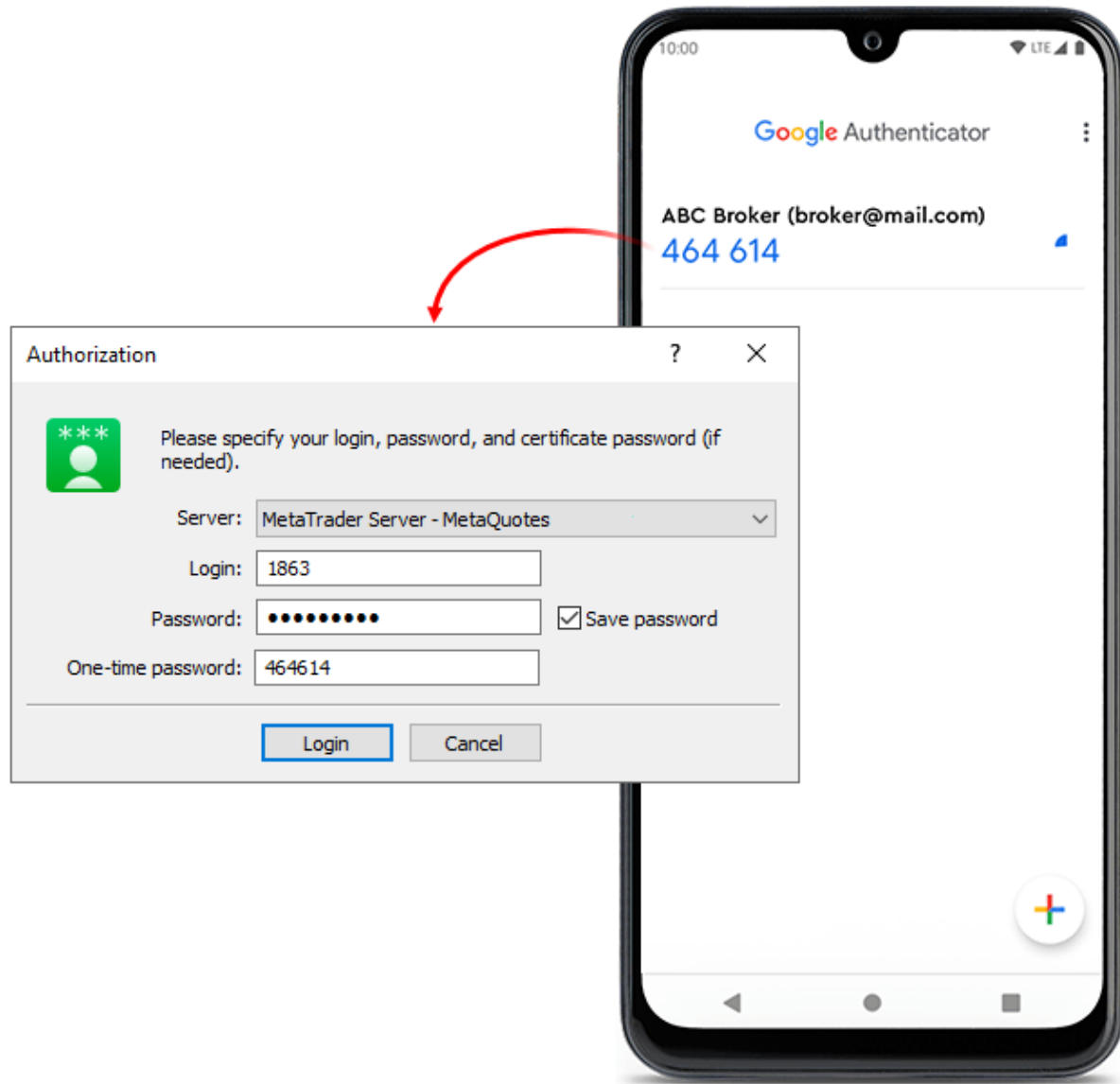
MetaTrader 5 trading platform



MetaTrader 5 trading platform

Run the Authenticator app on your mobile device, click "+" to add your account and scan the QR code from the terminal. Enter the received code in the "One-Time Password" field and click "Enable 2FA". This will link your account to the generator with a secret key.

A one-time password will be required for each connection to the account. Each OTP is valid for 30 seconds. After that a new one is generated.

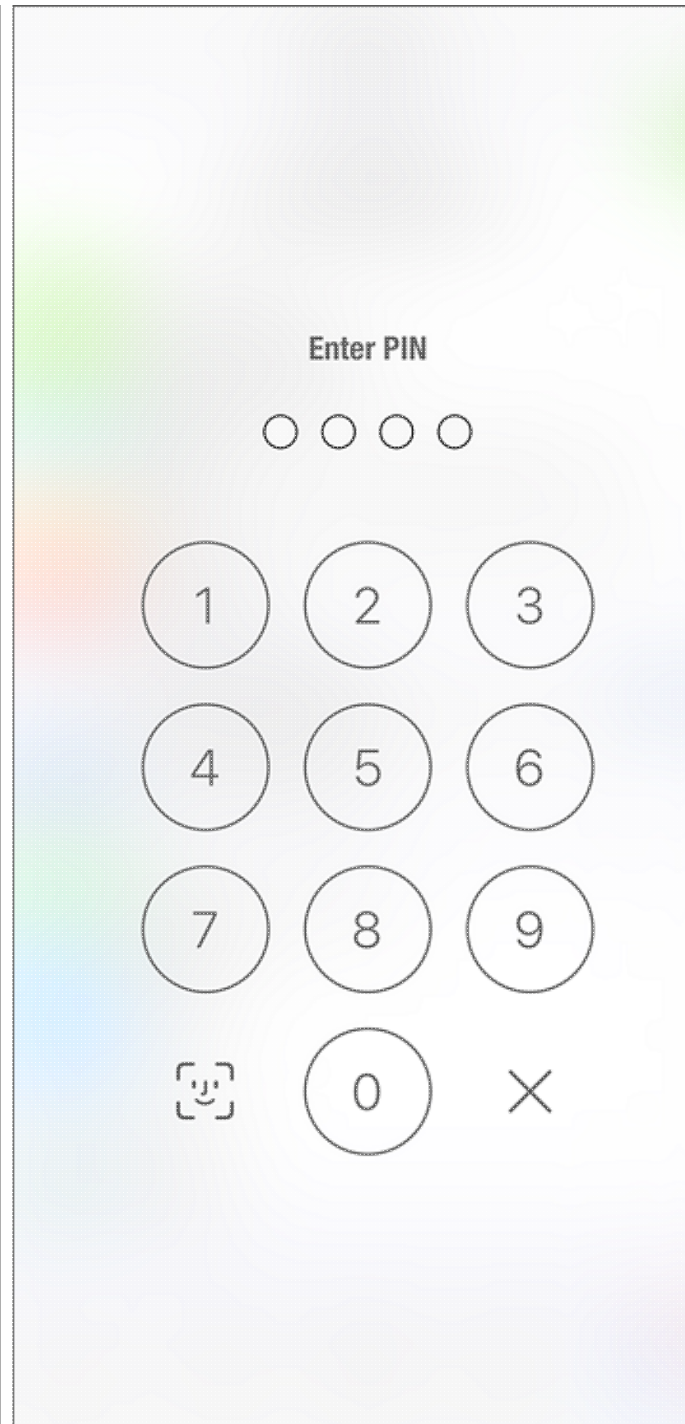
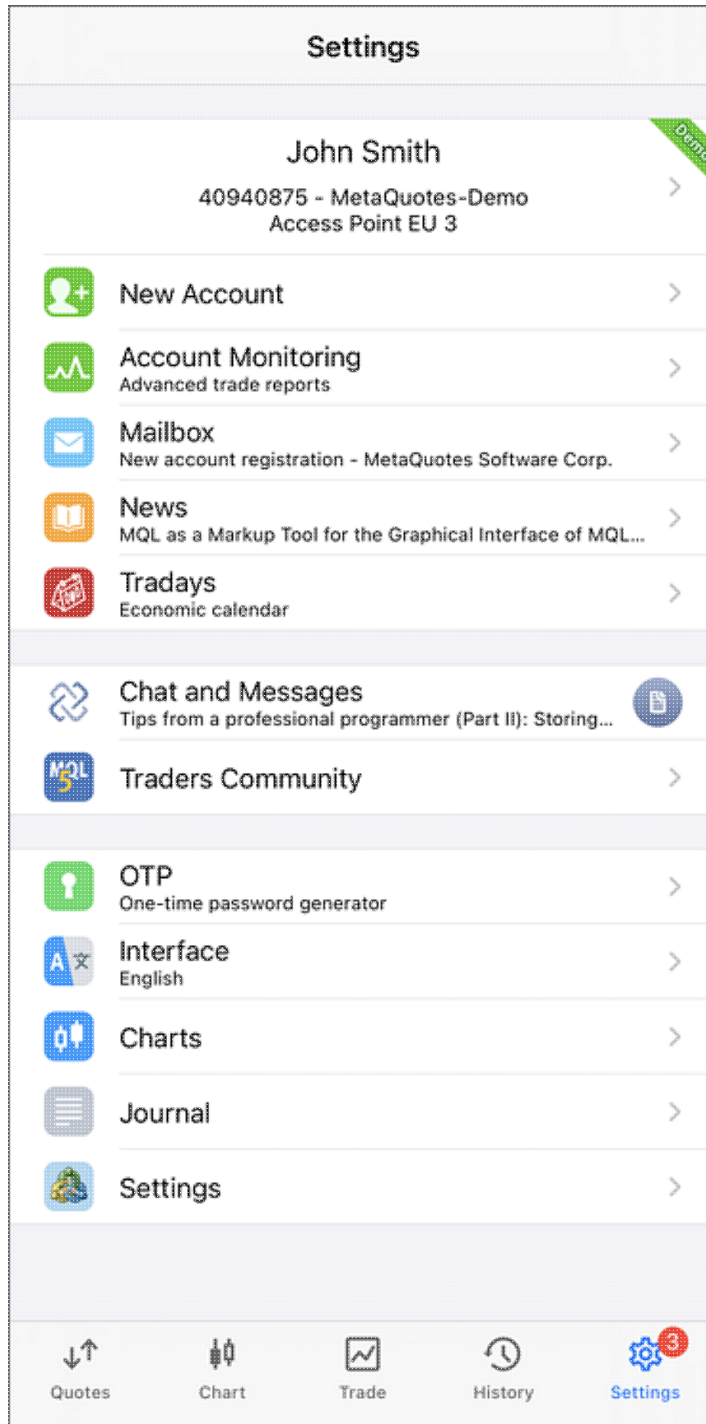


How to enable 2FA/TOTP on iPhone

To start using one-time passwords, you should bind your account with the password generator, which is the mobile terminal 5 for iPhone.

Go to the Settings of your mobile terminal and select OTP. For security reasons, when you first open this section, you will be required to set a four-digit password. The password must be entered every time you access the password generator.

If you forgot your password to the password generator but still use the same mobile device, reinstall the mobile platform and rebind your account to the generator. If you no longer have access to the mobile device, the binding to the password generator can be reset in the [account settings](#).



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In the window that opens, select "Bind to account".

OTP



321394

Enter this code when logging in to the bound trading accounts

Bind to account

Change Password

Synchronize Time

Last synchronization: 12:30 pm

OTP

Bind



Server

MetaQuotes-Demo

Login

40940875

Password

master password

Bind

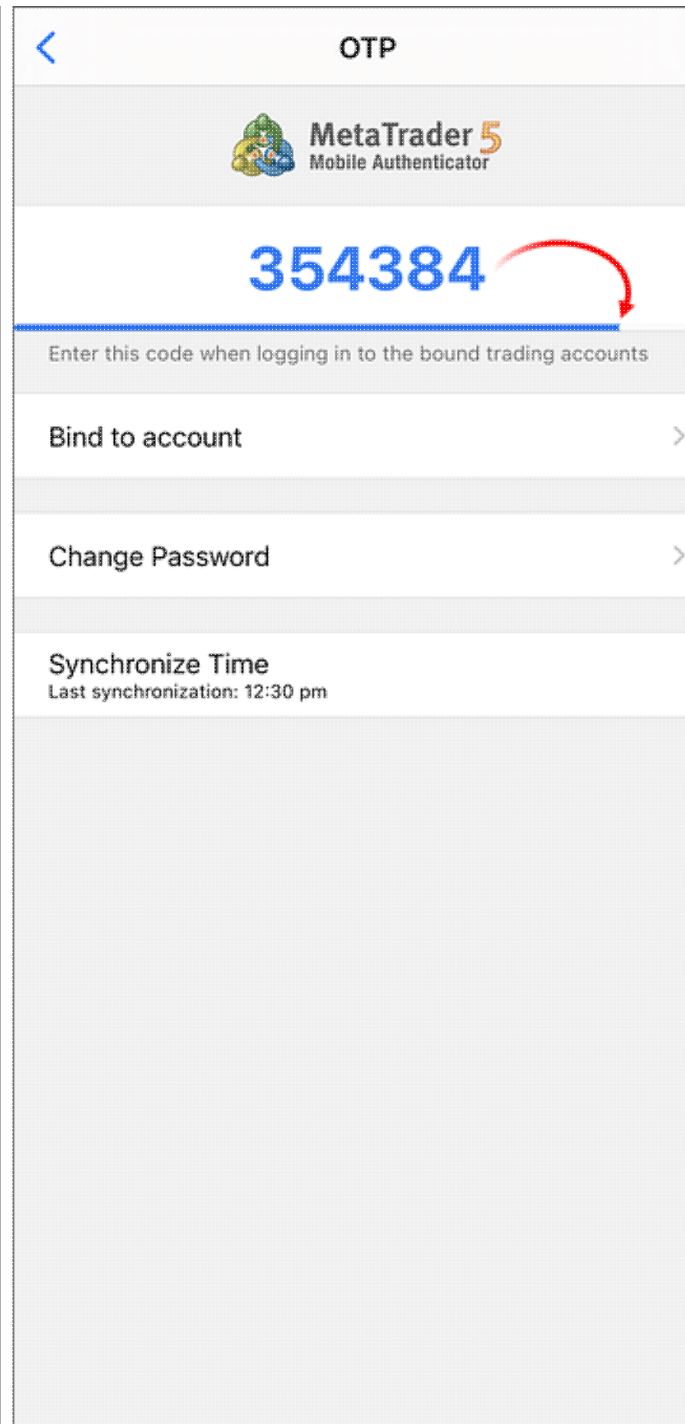
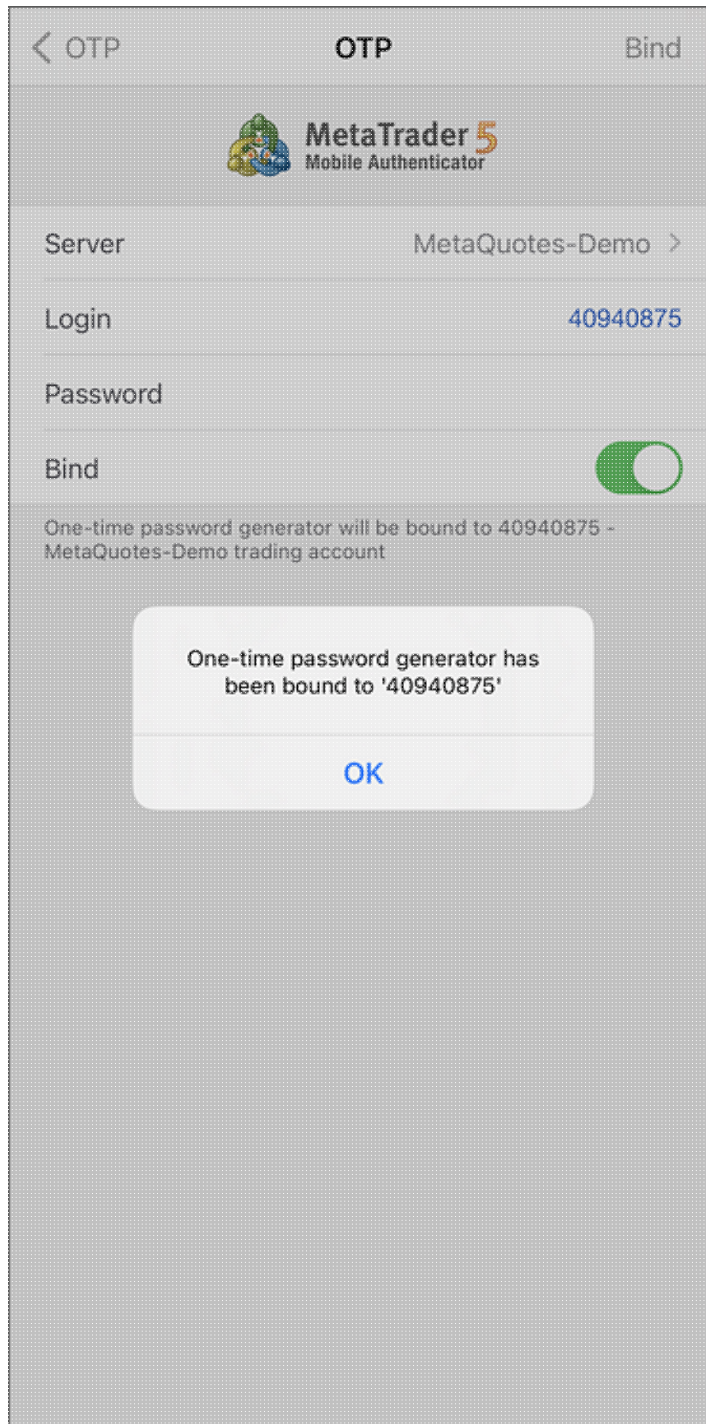
☒

One-time password generator will be bound to 40940875 - MetaQuotes-Demo trading account

MetaTrader 5 trading platform

Next, specify the name of the server on which the account was created, the account number and the master password to it. The "Bind" should be kept enabled. It must be disabled, if you are going to unbind the specified account from the generator and stop using one-time passwords.

After you tap the "Bind" button located in the upper part of the window, your trading account will be bound to the generator, and an appropriate message will appear.



Likewise, you can bind an unlimited number of accounts to the generator.

The one-time password is displayed at the top of the OTP section. Underneath, a blue bar visualizes the password lifetime. Once the password expires, it is no longer valid, and a new password will be generated.

Additional Commands:

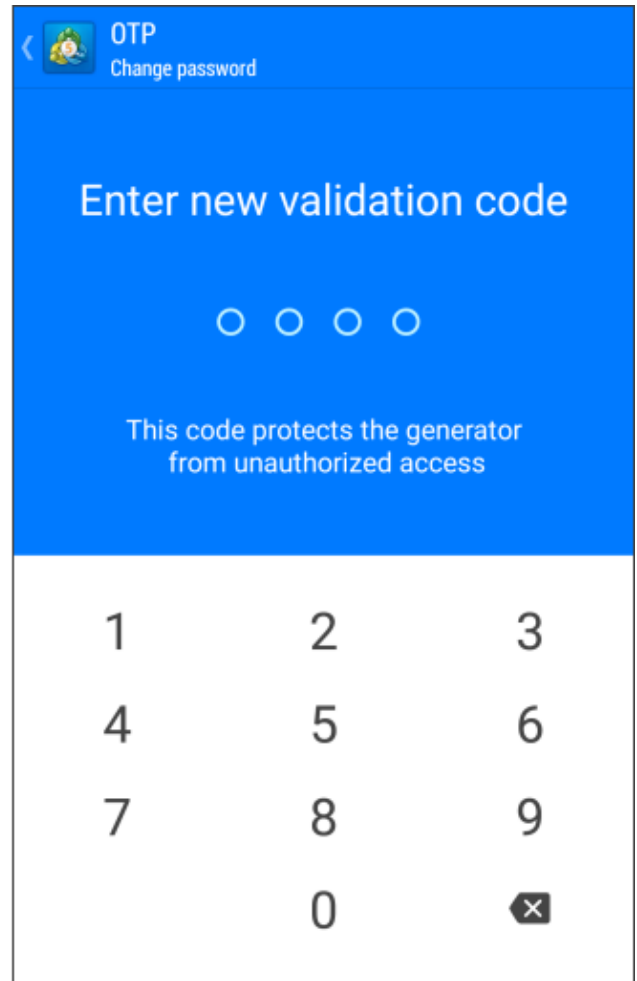
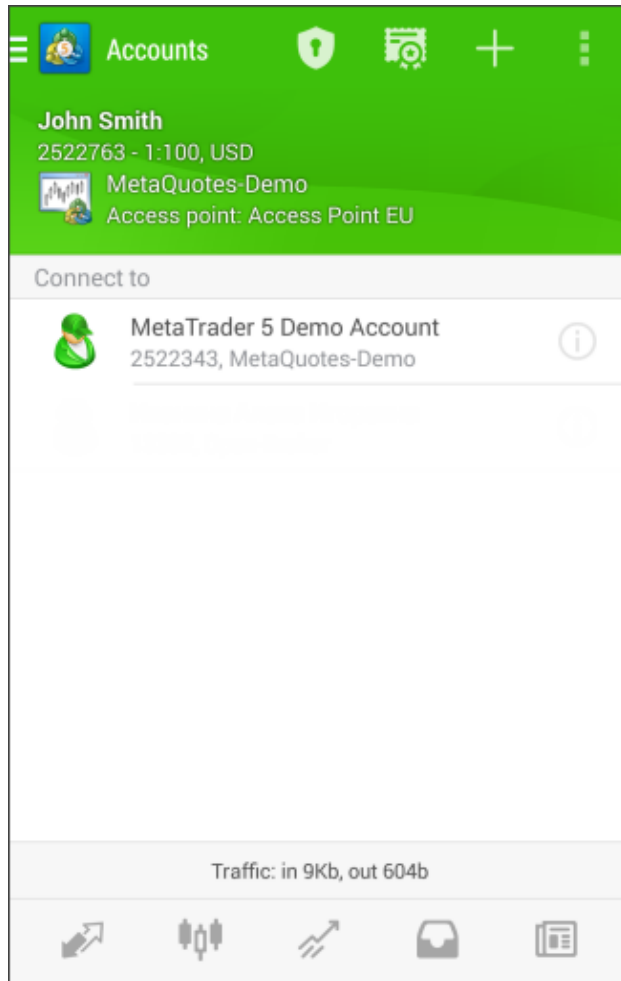
- **Change Password** change the generator password.
- **Synchronize Time** synchronize the time of the mobile device with the reference server. Accuracy requirement is connected with the fact that the one-time password is bound with the current time interval, and this time should be the same on the client terminal and the server side.

How to Enable 2FA/TOTP on Android Based Devices

Go to the Accounts of your mobile terminal and tap . For security reasons, when you first open this section, you will be required to set a four-digit password. The password must be entered every time you access the password generator.

If you forgot your password to the password generator but still use the same mobile device, reinstall the mobile platform and rebind your account to the generator. If you no longer have access to the mobile device, the binding to the password generator can be reset in the [account settings](#).

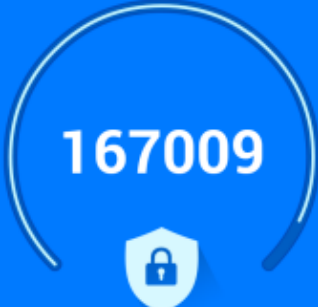
MetaTrader 5 trading platform



MetaTrader 5 trading platform

In the window that opens, select "Bind to account".

**OTP**



Enter this code when logging in to the bound trading accounts

 Bind to account

 Change password

 Synchronize time

Last in 16:30

**OTP**

Bind to account

BIND

MetaQuotes-Demo

MetaQuotes Software Corp.



Login

1822746

Master Password

Bind

☒

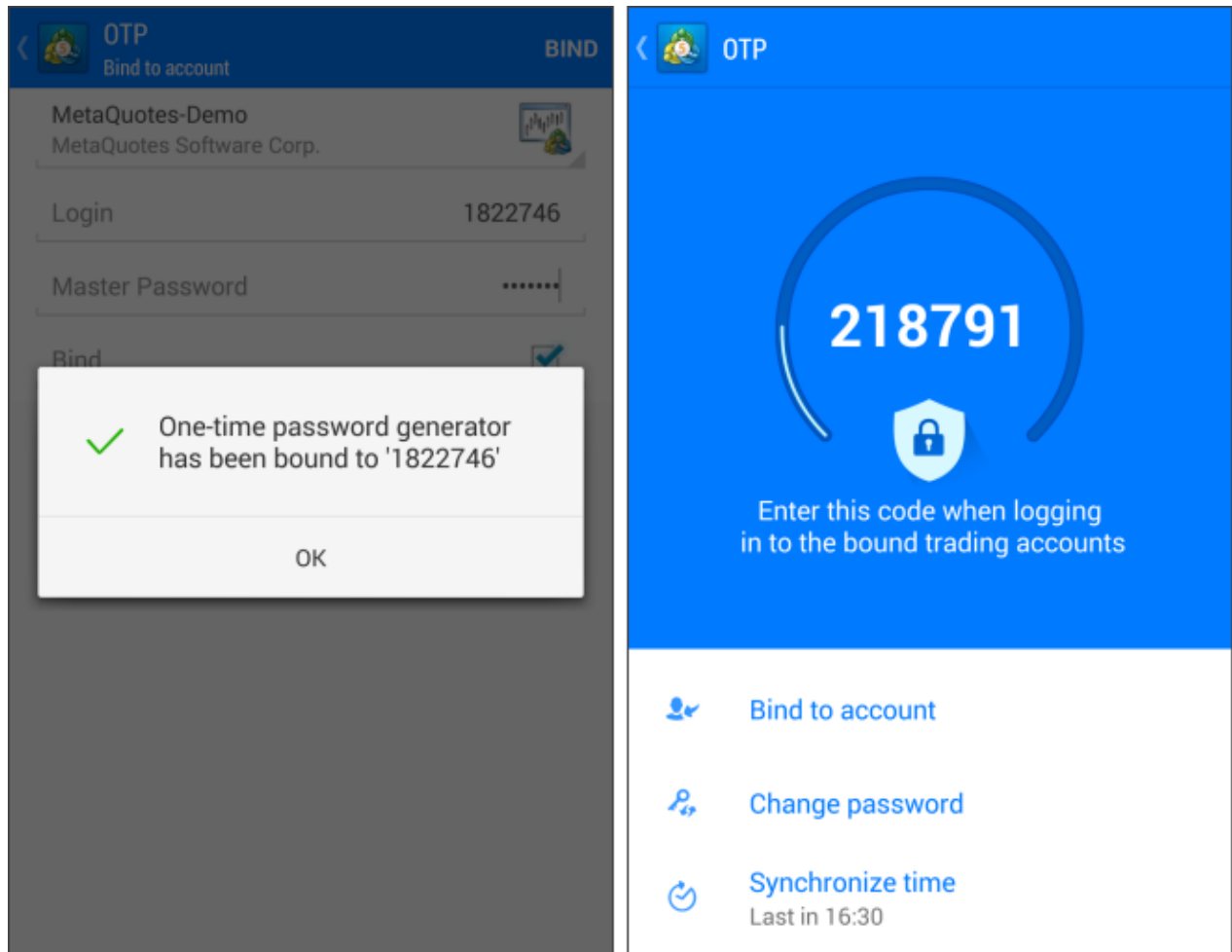


One-time password generator will be bound to 1822746 - 'MetaQuotes-Demo' trading account

MetaTrader 5 trading platform

Next, specify the name of the server on which the trading account was opened, the account number and the master password to it. The "Bind" should be kept enabled. It must be disabled, if you are going to unbind the specified account from the generator and stop using one-time passwords.

After you tap the "Bind" button located in the upper part of the window, your trading account will be bound to the generator, and an appropriate message will appear.



Likewise, you can bind an unlimited number of accounts to the generator.

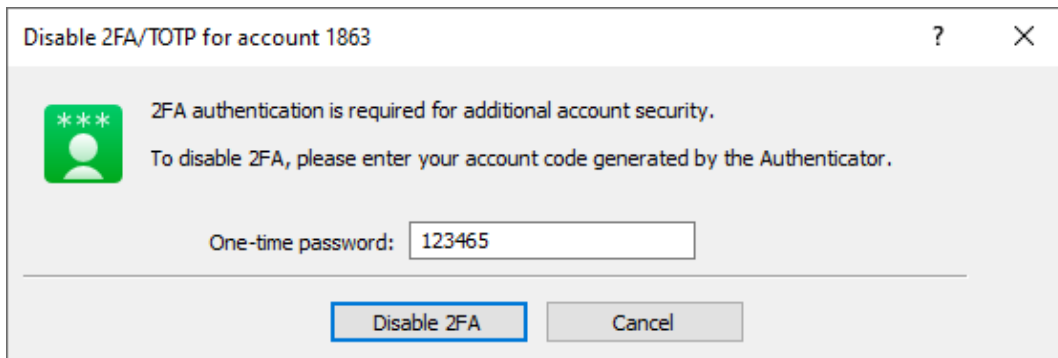
The one-time password is displayed at the top of the OTP section. Underneath, a blue bar visualizes the password lifetime. Once the password expires, it is no longer valid, and a new password will be generated.

Additional Commands:

- **Change Password** change the generator password.
- **Synchronize Time** synchronize the time of the mobile device with the reference server. Accuracy requirement is connected with the fact that the one-time password is bound with the current time interval, and this time should be the same on the trading platform and the server side.

Disabling TOTP/changing password generator

An account can only be linked to one OTP generator. To switch to another one, first disable OTP. If you use a mobile terminal as the password generator, go to the OTP section in it and select "Unbind". If you bound a third party Authenticator using the administrator terminal, select "Disable 2FA/TOTP" from the server context menu. You will need to enter a one-time password generated by the old generator.



If you do not have access to the device with the old password generator (for example, a mobile device is lost), you can reset the binding on the server side, via the account's "[Security](#)" section.

How to Use 2FA/TOTP in the Terminal

After binding an account to the generator, a one-time password will be additionally requested when connecting to it from the administrator terminal:

MetaTrader 5 trading platform

Authorization

Please specify your login, password, and certificate password (if needed).

Server: MetaTrader Server Dev - MetaQuotes-Dev

Login: 1863

Password: ●●●●●●●● ☒ Save password

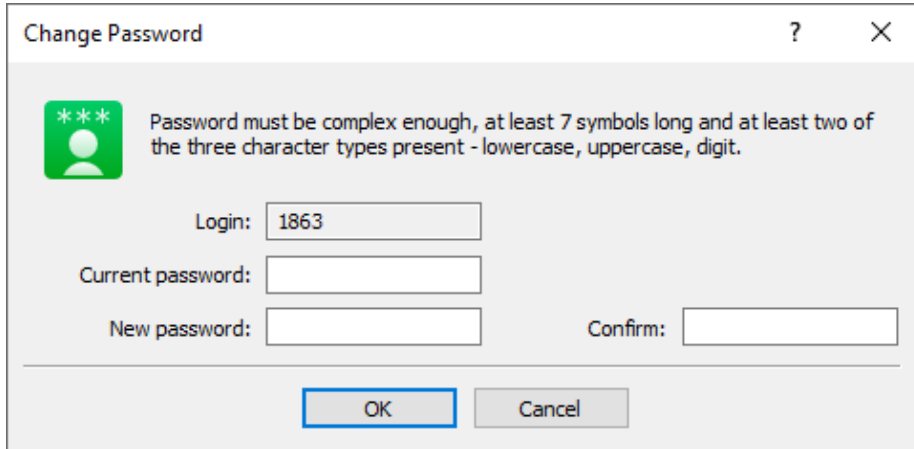
One-time password: 664649

Login Cancel

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Change Password

Change Password

To change the password for the Administrator account with which you are currently connected to the server, select "Change password" in the File menu.



The image shows a "Change Password" dialog box with a title bar containing a question mark and a close button. Inside the dialog, there is a green icon with three asterisks and a person silhouette. To the right of the icon, a message states: "Password must be complex enough, at least 7 symbols long and at least two of the three character types present - lowercase, uppercase, digit." Below this message, there are four input fields: "Login:" with the value "1863", "Current password:", "New password:", and "Confirm:". At the bottom of the dialog, there are two buttons: "OK" and "Cancel".

In the window that appears, enter the current account password, and then the new password twice.

The same action can be performed via the Accounts section. Find the account based on which your Administrator account was created, and change the password on the Security tab.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Live Update

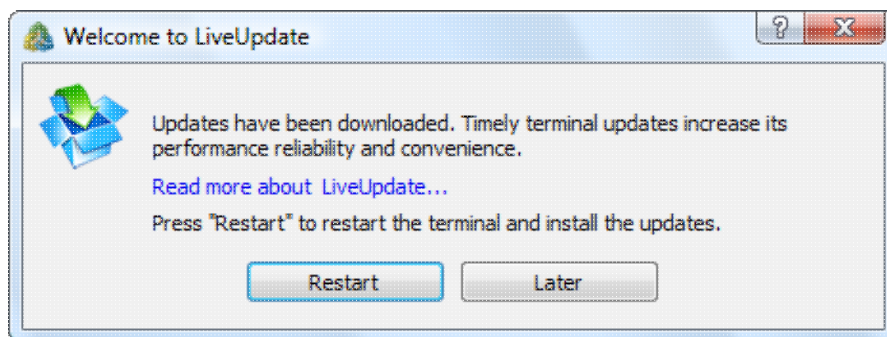
Live Update

A system of automatic updates is built into the terminal. It allows to get informed about and install new versions of the program promptly. This system is always enabled, it is impossible to disable it. Updating of all components of the trading platform can be managed at the "Live Update" section.

Updating Procedure

The terminal checks for new versions of the program when it connects to the server. If a new version of any of the terminal components has been discovered, it will be automatically downloaded in the background mode.

After all the updates have been downloaded, the below window offering to update the terminal will appear:



One of the buttons must be pressed in this window:

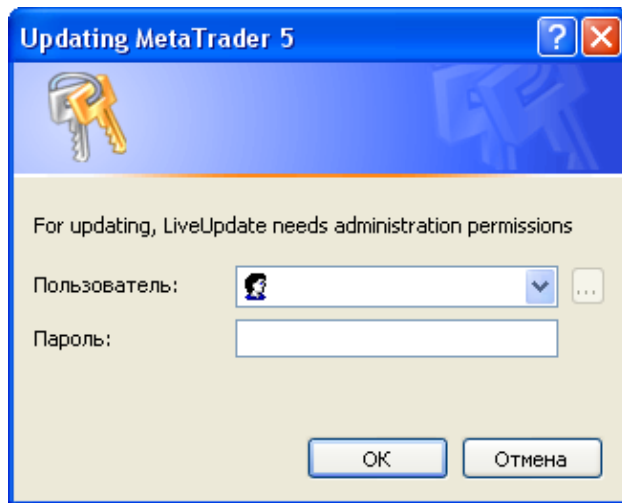
- **Restart** if this button is pressed, window of the terminal will be closed, the terminal will be updated and automatically re-opened.
- **Later** in this case the window will be hidden and the terminal will be automatically updated during the next launch.

- All the stages of the updating are reflected in the administrator terminal journal. "LiveUpdate" is specified in the Server column of such entries.
- In case of unsuccessful updating (server connection lost) the next attempt is performed one hour later. In this case only missing data will be additionally downloaded.

Update in the Guest Mode

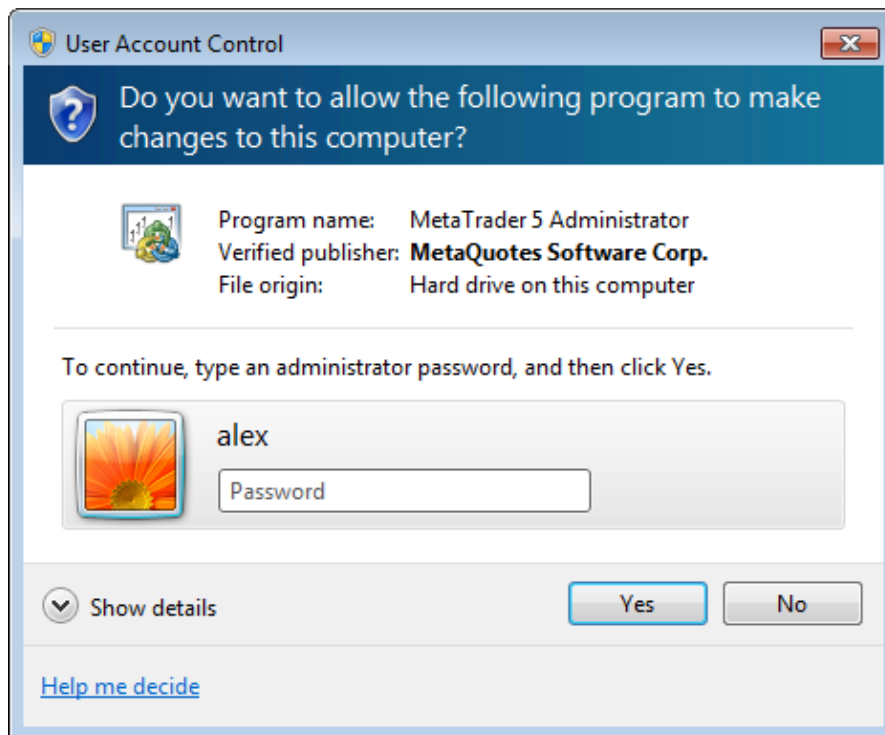
If an administrator terminal has been launched in the guest mode (if the OS user has insufficient rights), a window requesting the increase of the user's permissions will be shown at the attempt to update.

Microsoft Windows XP



In this case you are requested to specify details of the administrator account with enough permission to write files into the terminal installation directory.

Microsoft Windows Vista



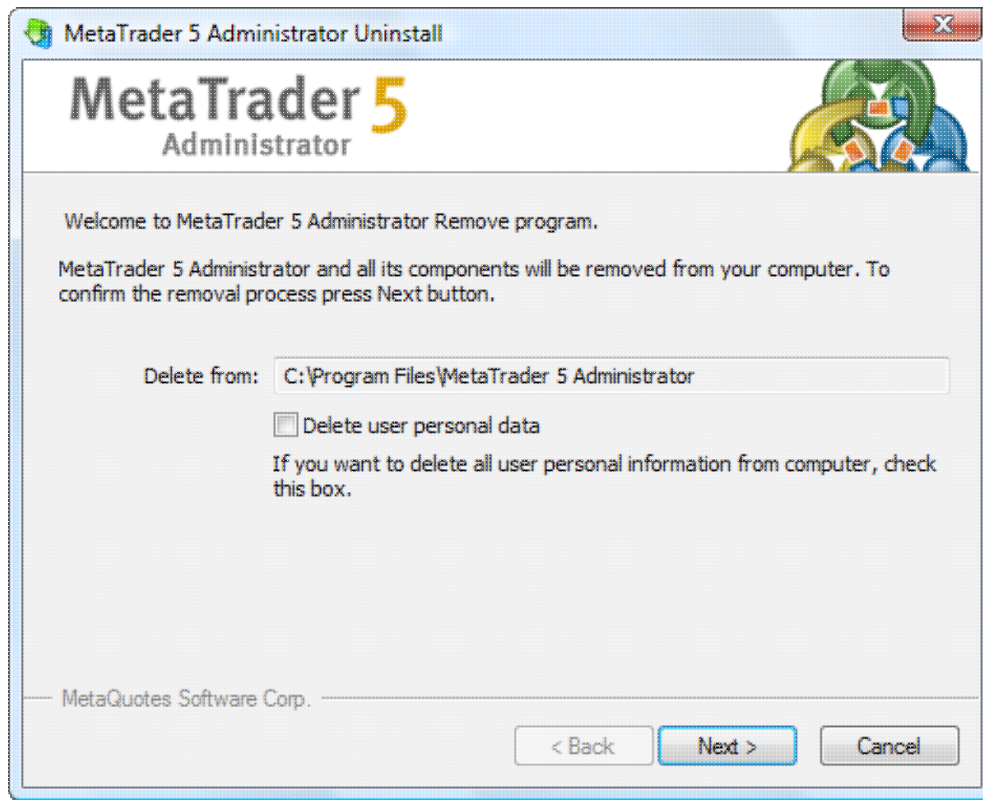
MetaTrader 5 trading platform

Depending on the user's permissions in MS Windows Vista, it is necessary either to allow the operation (if a user is an administrator) or specify administrator's account details.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Getting Started / Uninstall Terminal

Uninstall Terminal

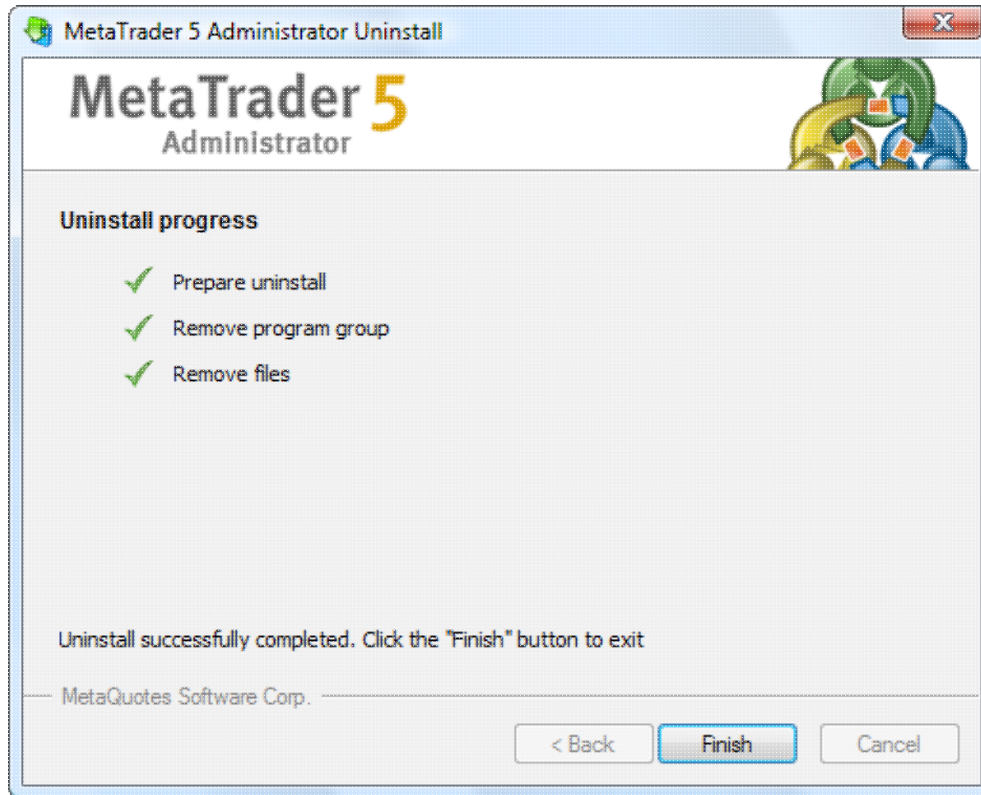
In order to start the process of terminal deletion from the PC one should launch the "Uninstall.exe" file in the folder, where the client terminal is installed, or execute the "Uninstall" commands in the "MetaTrader 5 Administrator" group of programs in the "Start" menu. The following window will be opened as soon as it is done:



MetaTrader 5 trading platform

The folder from which the administrator terminal will be removed is displayed at this window. Also there is the "Delete user personal data" option. If it is checked then all the user's information (terminal settings, mailing, news, etc.) will be deleted together with the permanent files of the terminal.

If one is sure to continue the deinstallation of the administrator terminal, it is necessary to press the "Next" button. The deletion process will be started as soon as it is done:



MetaTrader 5 trading platform

In order to finish the deinstallation one should press the "Finish" button.

- It is recommended to be careful with the deletion of the terminal. If the user data is deleted it will be impossible to recover it.
- If the "Delete user personal data" option is not enabled then further it is possible to recover the terminal operation with all its settings and information by installing it to the same folder.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface

User Interface

Window of the program is divided into several parts:

- **Main Menu**

The main menu contains different commands for administering servers and setting up the administrator terminal.

- **Toolbar**

The toolbar is implemented in the administrator terminal, it is divided into two parts: "Standard" and "Search". Commands of the main menu used for administering servers are duplicated in the standard part of the toolbar. It can be customized. The search part of the toolbar allows to perform a global search through the administrator terminal.

- **Status Bar**

The information line that displays the status of connection to the server and tips for different interface elements when the mouse cursor is put over them.

- **Toolbox**

"Toolbox" service window is intended for working with news, mail, search system and journal of the terminal.

The title of the terminal window contains the name of the server that is currently administered. The main part of the administrator terminal is intended for managing servers. The description of administering is given at the separate section.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Main Menu

Main Menu

Main menu is located below the title of the program. It contains different commands and functions that can be executed in the terminal. Main menu consists of:

- **File**
Adding and deleting servers to administer in the terminal, connecting and disconnecting from them.
- **Edit**
Commands for changing parameters in the different sections of administration and the search system.
- **View**
Managing elements of the terminal (toolbars, status bar, "Toolbox" window) and choosing the interface language.
- **Services**
Different service commands.
- **Help**
User guide, information about the program and useful links.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Main Menu / File

File

The "File" menu contains the following commands:

	Command	Description
+	Add Server	<u>Add a new server</u> to be administered through the terminal.
×	Remove Server	<u>Delete the selected server</u> from the terminal.
⚙️	Server Properties	Set up connection to the server. The command is inactive if the selected server is currently connected.
→	Connect	<u>Connect to a server</u> .
→×	Disconnect	Disconnect from the server.
🔒	Change Password	<u>Change password</u> of the current Administrator account.
↗️	Export	<u>Export</u> platform settings to a JSON file. To export settings of only the selected section, for example, managers, select this section in the tree and execute the command.
↖️	Import	<u>Import</u> platform settings from a JSON file.
📁	Open Data Folder	Open the folder where the administrator terminal data is stored.
🖨️	Print	Proceed to print the information from the selected tab, for example, accounts or orders.
🔍	Print Preview	View information before printing. The preview uses the settings of the selected printer. With this command, you can check whether all the required data will be printed.
⚙️	Print Setup	Configure print settings: printer, paper size, page orientation etc.
↩️	Exit	Close the Administrator terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Main Menu / Edit

Edit

The following commands are available in the "Edit" menu:

	Command	Description
✓	Apply Changes	Apply the new settings after changes. For example, when editing parameters in the Time section, you should click this button for the changes to take effect.
+	Add	Add an instruction. Depending on the selected section in the server tree, the command executes different commands. For example, it adds a holiday in the Holidays section or a new financial symbol in the Symbols section. The command applies to all sections except Time, Orders, Deals, Positions, Ticks and Update.
⚙	Edit	Edit an existing instruction. To edit the settings, select an element in the right part of the terminal window in the relevant section. The command applies to all sections except Ticks and Update.
✗	Delete	Delete an existing instruction. To delete an instruction, select an element in the right part of the terminal window in the relevant section. The command applies to all sections except Time, Ticks and Update.
↑	Move Up	Move an instruction up relative to others. Use this command to manage the priority in lists of access settings and data feeds. For example, if a data feed is higher than others, then the server will receive all information from this data feed. The server will switch to others only if the first one fails. In the Symbols tab, this command can be used to change the default arrangement of symbols in the Market Watch window in client terminals. In addition, using this button, you can move the added servers and platform components in the Network tab.
↓	Move Down	Move an instruction down relative to others. The command is similar to the previous one.
↓ A-Z	Sort Alphabetically	<u>Sort</u> configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
▶	Enable	Enable the selected configuration.
◻	Disable	Disable the selected configuration.
🔍	Find	<u>Find</u> an element in the current section.
🔍	Find Next	Move to the next found element.

MetaTrader 5 trading platform

	Command	Description
	Find Previous	Move to the previous found element.

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [User Interface](#) / [Main Menu](#) / View

View

The following commands are available in the "View" menu:

	Command	Description
	Languages	Select the Administrator terminal interface language. Language change takes effect only after restart.
	Color Themes	Select a color theme for the Administrator terminal interface: light or dark.
	Toolbar	Show/hide the <u>toolbar</u> .
	Status Bar	Show/hide the <u>status bar</u> at the bottom of the terminal.
	Split	Select the separator between the left part displaying the server tree with administration sections and the right parts showing the parameters. Move the cursor and fix the desired position of the separator with a single mouse click.
←	Navigate Backward	Go to the previous viewed page.
→	Navigate Forward	Undo moving to the previous page.
	Toolbox	Show/hide the <u>Instruments</u> window.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Main Menu / Services

Services

The following commands are available in the "Services" menu:

	Command	Description
	Refresh Configuration	Request configuration changes from the server. If another administrator edited the server configuration during your session, administration inconsistencies may occur. To avoid such situations, it is recommended to run this command from time to time. The command is executed automatically upon login.
	Restart Server	<u>Restart server</u> .
	Deploy Server	<u>Deploy</u> the selected server in the specified configuration. The command is only available for the servers which have not yet been deployed.
	Restart Gateways	Restart all <u>gateways</u> .
	Restart Data Feeds	Restart all <u>data feeds</u> .
	Synchronize History	Synchronize <u>historical data</u> .
	Start Live Update	Run platform <u>update</u> .
	Activate License	<u>Activate</u> the trading platform. A non-activated platform operates with significant functional limitations.
	Options	Open Administrator terminal <u>settings</u> .

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [User Interface](#) / [Main Menu](#) / [Help](#)

Help

The following commands are available in the "Help" menu:

	Command	Description
	Help Topics	Call this Help of the administrator terminal.
	What's New	Link to the What's New list of the MetaTrader 5 platform's client side.
	Check Updates	Check and install the latest update of the administrator terminal. The following update options are available for installation: • Release version. Such versions are considered to be more stable, as they are subject to longer testing. Such versions are released less frequently. • Beta version. It is considered to be less stable. Such versions are released more often, compared to Release versions. Such updates can provide early bug fixes and access to new functions. This option does not execute the update of the entire platform.
	MQL5.community	Link to the community of Expert Advisor developers MQL5.Community .
	MetaTrader 5 Platform	Link to the official MetaTrader 5 platform website.
	MetaQuotes Ltd	Link to the official MetaQuotes company website.
	MetaQuotes Support Center	Link to MetaQuotes' technical support website .
	About MetaTrader Administrator	Show information about the Administrator terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbar

Toolbar

Toolbar is a part of the administrator terminal interface. It allows to organize working on the administration of the trading platform in a convenient way. The toolbar is divided into two parts:

- **Standard**

In this part of the toolbar key commands of the main menu are duplicated.

- **Search**

This part of the toolbar allows to perform a global search of elements through the administrator terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbar / Standard

Standard

The standard part of the toolbar is a set of commands for managing servers and the terminal. The set of commands that is displayed at the toolbar is adjustable.

	Command	Description
	Connect	<u>Connect to a server</u> .
	Disconnect	Disconnect from the server.
	Update	Request configuration changes from the server. If another administrator edited the server configuration during your session, administration inconsistencies may occur. To avoid such situations, it is recommended to run this command from time to time. The command is executed automatically upon login.
	Restart Server	<u>Restart</u> the selected server.
	Apply	Apply the new settings after changes. For example, when editing parameters in the Time tab, you should click this button for the changes to take effect.
	Add	Add an instruction. Depending on the selected section in the server tree, the command executes different commands. For example, it adds a holiday in the Holidays section or a new financial symbol in the Symbols section. The command applies to all sections except Time, Orders, Deals, Positions, Ticks and Update.
	Edit	Edit an existing instruction. To edit the settings, select an element in the right part of the terminal window in the relevant section. The command applies to all sections except Ticks and Update.
	Delete	Delete an existing instruction. To delete an instruction, select an element in the right part of the terminal window in the relevant section. The command applies to all sections except Time, Ticks and Update.
	Sort Alphabetically	<u>Sort</u> configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
	Move Up	Move an instruction up relative to others. Use this command to manage the priority in lists of access settings and data feeds. For example, if a data feed is higher than others, then the server will receive all information from this data feed. The server will switch to others only if the first one fails. In the Symbols tab, this command can be used to change the default

	Command	Description
		arrangement of symbols in the Market Watch window in client terminals. In addition, using this button, you can move the added servers and their components in the Network tab.
	Move Down	Move an instruction down relative to others. The command is similar to the previous one.
	Enable	Enable the selected configuration.
	Disable	Disable the selected configuration.
	Export	<u>Export</u> platform settings to a JSON file. To export settings of only the selected section, for example, managers, select this section in the tree and execute the command.
	Import	<u>Import</u> platform settings from a JSON file.
	Toolbox	Show/hide the <u>Instruments</u> window.
	Help	Use the context help of the Administrator terminal. When you select this command, a question mark will appear at the end of the cursor. Click on any interface element to view the relevant reference help.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbar / Search

Search

This part of the toolbar allows to perform the global search of elements through the administrator terminal.



Before starting the search one can specify sections to search in. To do it press the  button, then the list of available sections will be opened. Specify the necessary sections by clicking on the corresponding items. Besides it, using the "Select All" and "Select None" buttons, one can select or unselect all the sections respectively.

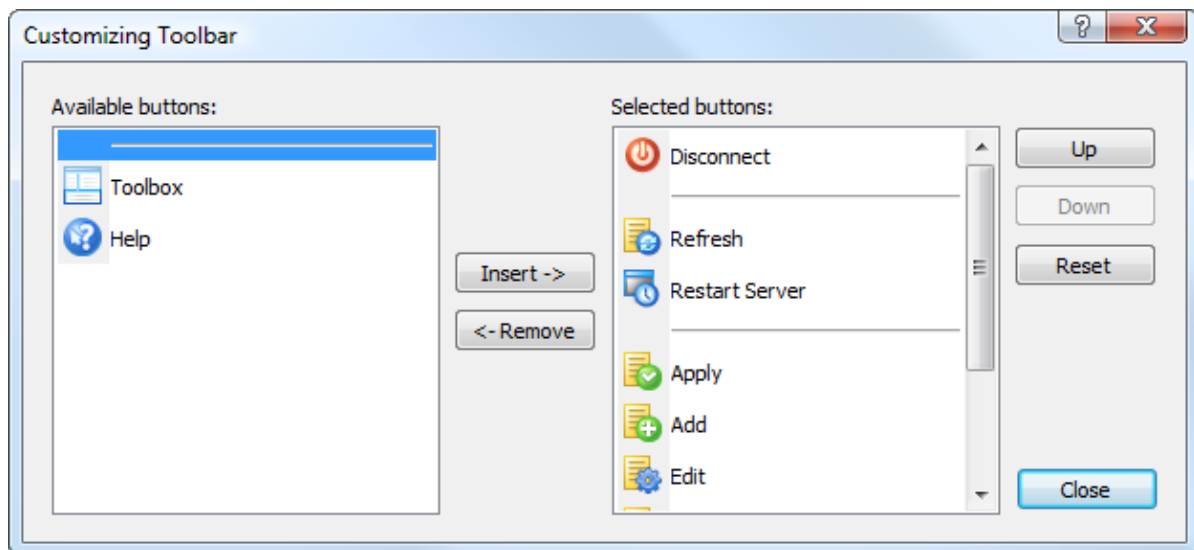
Once the necessary settings are made, it is necessary to type the desired word or phrase in the field to the left and press the  button. Also, using the  button, one can choose one of the previous search requests.

- In the "Accounts", "Orders" and "Deals" sections the search is performed only by the entries that are currently displayed in the list.
- You can search by serial numbers of SSL certificates issued to clients for extended authorization.
- The results of the global search are displayed at the "Search" tab of the "Toolbox" window".

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbar / Setup

Setup

In order to start customizing the set of commands in the standard part of the toolbar, one should click with the right mouse button on it and choose the "Customize" item in the context menu. After that the window of customizing the toolbar will be opened:



In order to add a button into a toolbar, it should be moved from the left part of the window into the right one. This can be done by a double click on the button, or by selecting it and pressing "Insert". A button can be deleted from a toolbar by a double click on it in the right part of the window, or by selecting it and clicking "Remove". Buttons "Up" and "Down" in the right part of the window are used for changing the order of buttons displaying in a toolbar. In order to return default settings, the "Reset" button should be pressed.

The list of available buttons also contains a separating line. Adding it one can divide toolbar buttons into groups.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Status Bar

Status Bar

Status bar is located in the lower part of the administrator terminal. It contains tips of program commands, the connection status and the amount of sent and received data:



The following information is shown here from left to right:

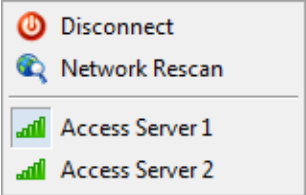
- Command hints;
- Status of connection to the trade server. Icon  means that the terminal is currently connected to the server. If connection to the terminal is not established, icon . Hover over the icon to view additional connection details: ping to the trade server and packet loss. These variables measure connection quality between the terminal and the server.
- Amount of incoming and outgoing traffic for the current session.

The status bar can be disabled by removing the check mark at the corresponding item of the "View" menu.

Access point changing menu

The administrator terminal is connected to the trade server through access servers. When connecting, the best access point is selected (the lowest load to server, the best quality of connection to it). The best access points are automatically checked every three hours of operation.

You can also switch between access servers using the menu that is opened by clicking on the connection status or traffic data in the status bar:

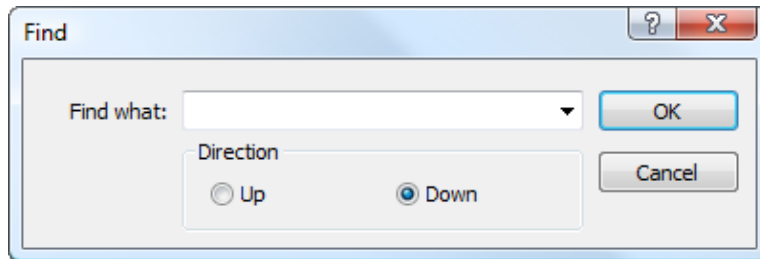
Menu	Commands
	•  Connect/ Disconnect <u>connect</u> or disconnect from the server. •  Network Rescan this command is intended to manually start scanning of access points. If a server with connection quality better than the current by 30% is found, it is automatically switched to. • List of access servers below all the available access servers are displayed. The indicator of the server quality is shown before the name of each access server. In order to switch to one of the access points, click on its name. It should be noted that in the next scanning of the network, the best server will be automatically choose.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Search

Search

Two types of search are implemented in the administrator terminal - one in the current section and a global one through the search part of the toolbar.

Search in the current section allows to find elements in the currently selected section of administration. To start searching, one should execute the "🔍 Find" command in the "Edit" menu or press the "Ctrl+F" key combination. The following window will be opened then:



It contains the following parameters:

- **Find what** the field for specifying a word or phrase to be searched;
- **Direction** the direction of searching from the current cursor position up or down. The searching of the currently chosen element is performed at the tab in the specified direction.

As soon as you press the "OK" button, search of the element will be performed. The first found element will be highlighted in the right part of the administrator terminal. In order to go to the next element it is necessary to execute "🔍 Find Next" command in the "Edit" menu or press the "F3" key.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Hot Keys

Hot Keys

Hot keys are keys and their combinations that allow executing various commands without using menus or toolbar. Hot keys execute various functions depending on what terminal interface window is currently active. Some hot keys execute certain functions independent of focus.

Common Actions in the Terminal

Hot keys	Description
Ctrl+N/Insert	Add an instruction. Depending on the section selected in the tree of servers it executes different commands. For example, in the "Holidays" section it allows to add a holiday; or if the "Symbols" section is chosen, it allows to add a new financial instrument.
Ctrl+U	Modify selected instruction.
Ctrl+D/Delete	Delete selected instruction in the current section.
Ctrl+F	Open the <u>search</u> window.
Ctrl+O	Open the window of <u>options</u> of the administrator terminal.
Ctrl+T	Show/hide the Toolbox service window.
F1	Open this user guide.
F3	Find the next element by the current search request.
F5	<u>Refresh configuration</u> of a server.

Hot Keys in the "Toolbox" Window

Hot keys	Description
Enter	View a selected <u>news</u> , <u>e-mail</u> or <u>search</u> result - depending on the tab selected.
Insert	Create a new <u>e-mail</u> or <u>news</u> - depending on the tab selected.
Delete	Delete a selected <u>e-mail</u> .
Space	Unwrap a selected <u>branch of mails</u> .
R	Show/hide the column of categories of <u>news</u> .
Ctrl+C	Copy the selected journal line of the <u>terminal</u> to the clipboard.
A	Enable/disable auto sizing of columns.
G	Show/hide grid.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbox

Toolbox

The "Toolbox" window is intended for working with news, mails and journal of the administrator terminal. This window can be hidden using the " Toolbox" command of the View menu or similar command of the toolbar. If you want to move the Toolbox window, click on its title and, holding it, drag the window to the necessary part of the terminal.

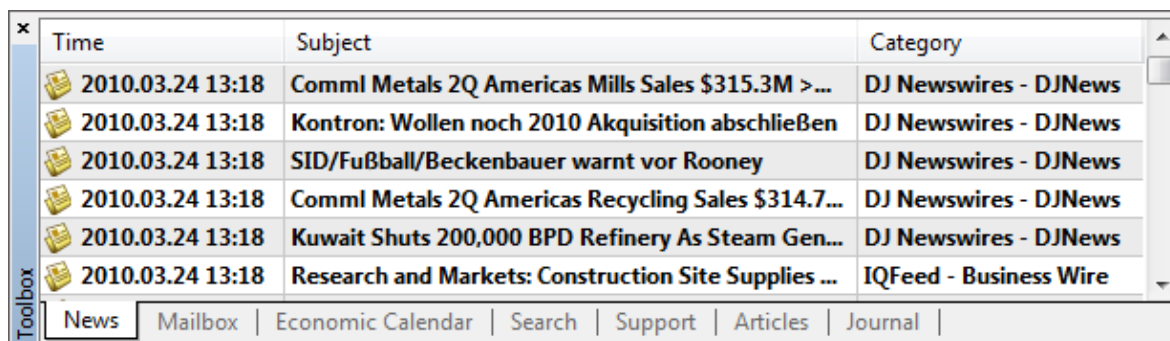
The Toolbox window contains the following tabs:

- **News**
This tab contains all the news messages that came to the terminal.
- **Economic Calendar**
The calendar features publications of over 600 macroeconomic indicators concerning 15 largest global economies, including USA, European Union, Japan, UK, Canada, Australia, China, etc.
- **Search**
This tab displays the results of the global search through the terminal performed via the toolbar.
- **Journal**
Information about terminal launch and events during its operation is stored in the journal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / User Interface / Toolbox / News

News

In this tab you can work with news messages that come to the terminal.



Time	Subject	Category
2010.03.24 13:18	Comml Metals 2Q Americas Mills Sales \$315.3M >...	DJ Newswires - DJNews
2010.03.24 13:18	Kontron: Wollen noch 2010 Akquisition abschließen	DJ Newswires - DJNews
2010.03.24 13:18	SID/Fußball/Beckenbauer warnt vor Rooney	DJ Newswires - DJNews
2010.03.24 13:18	Comml Metals 2Q Americas Recycling Sales \$314.7...	DJ Newswires - DJNews
2010.03.24 13:18	Kuwait Shuts 200,000 BPD Refinery As Steam Gen...	DJ Newswires - DJNews
2010.03.24 13:18	Research and Markets: Construction Site Supplies ...	IQFeed - Business Wire

Below the table is a toolbar with buttons: News, Mailbox, Economic Calendar, Search, Support, Articles, Journal.

News items are displayed in the form of a table with three columns:

- **Time** the time of news coming to the terminal;
- **Subject** title of news;
- **Category** category of news.

The news messages that has not been viewed yet are displayed with the bold font and have icon 📄. Read messages are marked by icon 📄. The 📄 icon means that the news has a high priority. In order to read a news, one should click with the left mouse button on it.

Context Menu

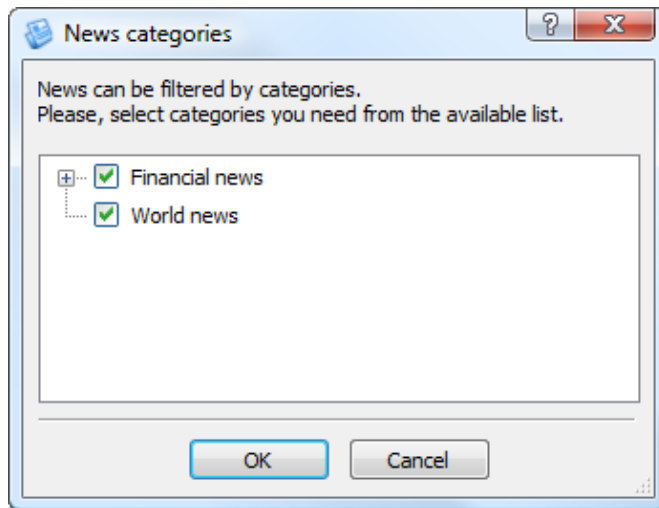
The context menu of news contains the following commands:

- **View** opens the window of viewing news;
- **Create** opens the window of sending news;
- **Categories** opens the submenu of choosing the news categories to be displayed in the list. To disable displaying of a category, remove the check against it. If the terminal receives only one category of news this menu is not displayed. If one of the categories contains two or more subcategories then the "Customize" command appears in this submenu. Using it one can adjust the news categories more precisely;
- **Category** show/hide the Category column in the list of news;
- **Auto Arrange** if this option is enabled, the size of columns is selected automatically;
- **Grid** this option shows/hides grid to separate table fields.

News Categories

Using the Customize command of the news categories submenu, you can open the window of their detailed setup:

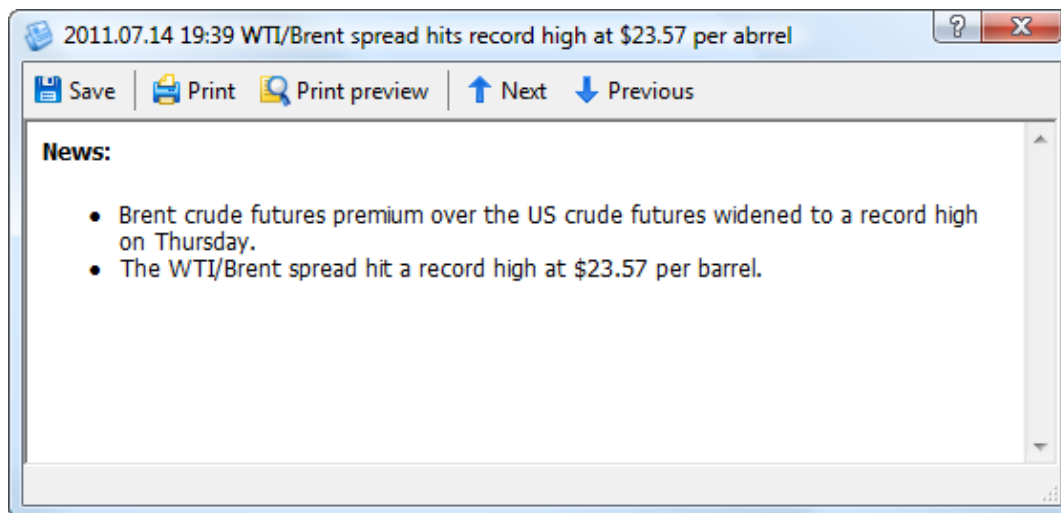
MetaTrader 5 trading platform



In the tree-like list, check the categories that shall be displayed in the administrator terminal.

Viewing News

To start viewing news, click on its subject in the list. After that the following window will be opened:



The window header contains the date and time when the news was received and its title. The main part of the window is occupied by the text of news. The toolbar that contains the following commands is located in the upper part of the window:

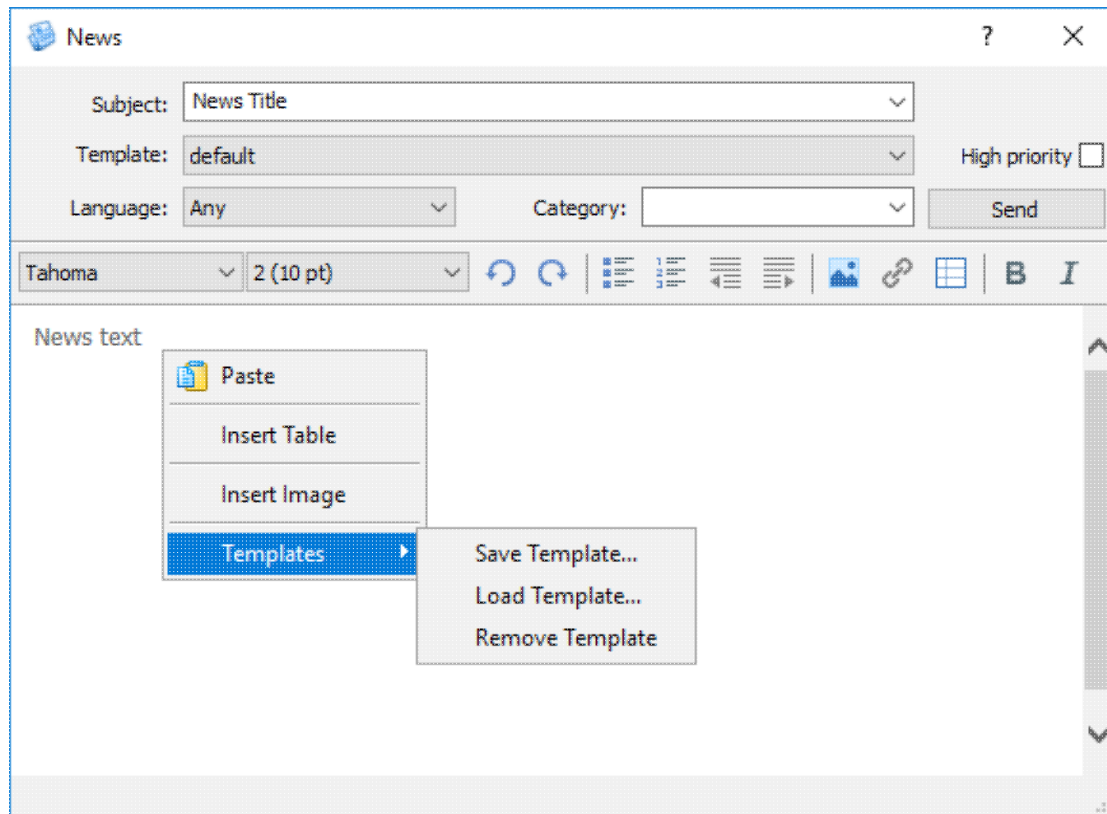
-  **Save** save the news on a computer as a HTML file or a text file of the Unicode standard;
-  **Print** print the news;
-  **Print Preview** open the preview window before printing the news;
-  **Next** view the next news. The same action can be performed using the "Page Up" key;
-  **Previous** view the previous news. The same action can be performed using the "Page Down" key.

Sending News

- To be able to send news, the group the manager belongs to must be allowed to work with news, and the manager must have the permission to send news.
- Also a manager should belong to a group that is created at the main trade server.

In order to write and send a news, press " Create" in the context menu of the "News" tab.

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```
font-family: tahoma, verdana, Arial, helvetica, sans-serif;  
font-size:.8em;  
height:100%;  
}  
div
```

This window contains the following fields:

- **Subject** subject of the news;
- **Template** in this field you can specify a previously created news template;
- **High priority** if this option is checked, a high priority will be assigned to the message. The priority news messages are marked with the special 🚨 icon;
- **Language** choice of the language. There is an option of receiving internal news in the terminals. A user can adjust the receiving of news only in one of the languages. The "Language" option allows to send the news by separate language groups of users. In this field you can also specify "Any", in that case the news will be sent to all users regardless of the language settings in their terminal;
- **Category** the field for entering or choosing the news category. In order to create a subcategory, specify its name with the "\" symbols after the name of the main category. For example, "Financial news\Free".

Below is a window for working with the news text. The editor features commands for creating lists, as well as inserting images, links and tables. To view or edit the source HTML code of the news, click  on the toolbar.

The context menu of the text editor window contains standard commands for working with a text: Copy, Cut, Paste, Insert Link, Insert Table and Insert Image. The context menu also allows working with the news templates.

In order to send the news, press the "Send" button.

Templates

The Templates submenu of the context menu of the news editing window allows working with templates:

- **Save Template** save the current text of the news as a template in the *.htm format. All templates are stored in the /templates/news folder of the administrator terminal;
- **Load Template** open the window of choosing a previously created template for loading it;
- **Remove Template** remove the currently selected template.

When executing the "Remove Template" command a template is irrecoverably deleted from PC.

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [User Interface](#) / [Toolbox](#) / Economic Calendar

Economic Calendar

The administrator terminal contains the [Economic Calendar](#). The calendar features publications of over 600 macroeconomic indicators concerning 15 largest global economies, including USA, European Union, Japan, UK, Canada, Australia, China, etc. Relevant data is collected from open sources in real time.

Macroeconomic indicators are parameters describing the state of the country they are calculated for. They characterize the level of economic development and may indicate either economic growth or a decline. By analyzing the macroeconomic indicators, it is possible to forecast future price movements.

The indicators and events can be viewed on the "Calendar" tab of the "Toolbox" window.

Toolbox							
Time	Currency	Event	Priority	Period	Actual	Forecast	Previous
Monday, 8 April							
01:50	JPY	Current Account n.s.a.	•	Feb	¥2676.8 B	¥1870.8 B	¥600.4 B
01:50	JPY	Goods Trade Balance	•	Feb	¥489.2 B	¥-891.5 B	¥-964.8 B
01:50	JPY	Adjusted Current Account	•	Feb	¥1957.6 B	¥1419.3 B	¥1833.0 B
02:30	JPY	BoJ Governor Kuroda Speech	•				
07:00	JPY	Consumer Confidence Index	•	Mar	40.5	42.3	41.5
08:00	EUR	Exports m/m	•	Feb	-1.3%	0.0%	0.1%
08:00	EUR	Imports m/m	•	Feb	-1.6%	0.1%	1.4%
08:00	EUR	Trade Balance	•	Feb	€18.6 B	€18.9 B	€18.6 B
08:00	JPY	Economy Watchers Index for Current Conditions	•	Mar	44.8	46.7	47.5
08:00	JPY	Economy Watchers Index for Future Conditions	•	Mar	48.6	49.3	48.9
09:30	CAD	BoC Senior Deputy Governor Wilkins Speech	•				
News 7 Mailbox Calendar Search Journal							

MetaTrader 5 trading platform

By default, the calendar displays current week events, including past and upcoming events. Use the context menu to switch to another period. You can access events for the previous, current and next week, as well as appropriate months. A deeper history is available in the [web version](#).

Every indicator is provided with the release time, priority, as well as its current, forecast and previous values. The current value appears as soon as the indicator is released. If this value is less than the predicted one, the indicator is highlighted in red. If the current value is larger, the indicator is shown in blue.

To view a detailed event description or the history of its values as a graph or table, double click on its name.

For easier search, filter events in the list using the context menu:

- by priority
- by the currency of the country for which the indicator is published
- by country for which the indicator is published

Economic calendar data can be saved as an HTM file. To do this, click  Export... in the context menu.

Install the
Calendar on
your site

Download the mobile calendar version

You can add the Economic Calendar in your site free of charge. This can be done by pasting [the ready widget code](#) in the desired web page.

The Economic Calendar is available as a separate [Tradays](#) application for mobile devices powered by iOS and Android.

The mobile version features a complete set of functions for the full-fledged operation and a number of additional features:

- More than 600 events related to the world's largest economies, including US, UK, European Union, Japan, Canada, Australia, China, among others.
- Detailed event descriptions.
- Access to historic indicator values in the form of tables and charts.
- Filters by priority and countries, to which the indicators are related.
- Ability to create event reminders in a couple of clicks.

You do not have to worry about licensing risks. The Calendar is based on data collected from public sources.



The advantages
of the Calendar:

- Useful content on your

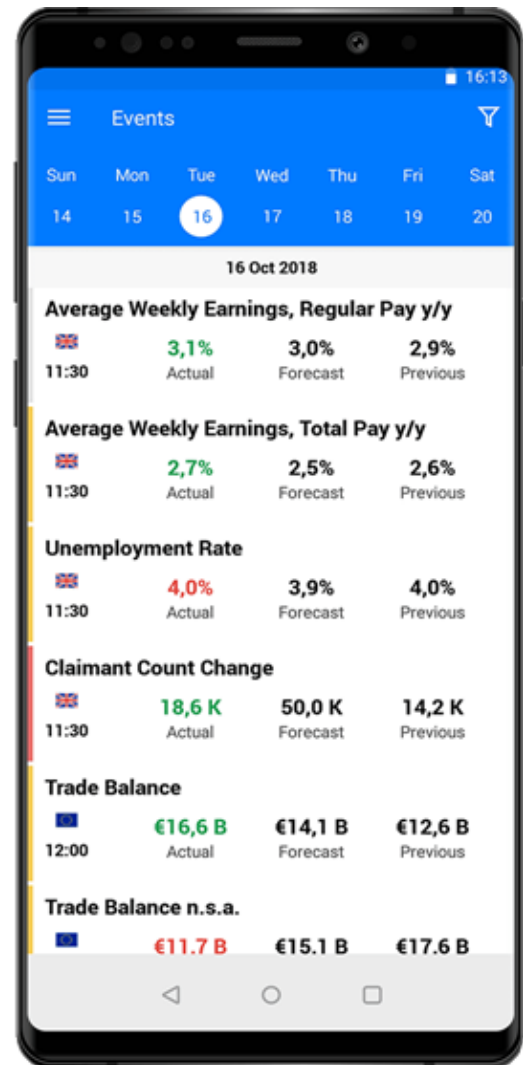
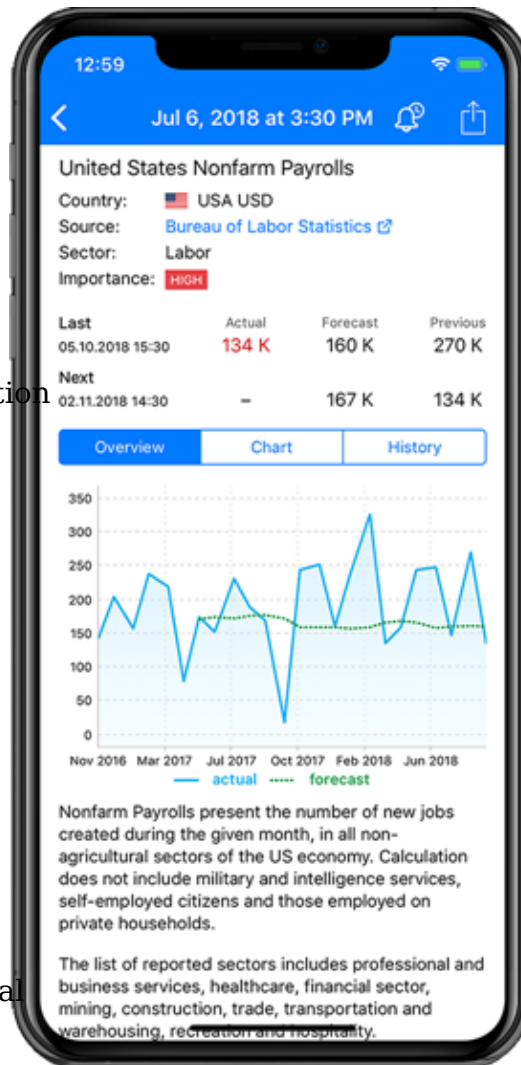
MetaTrader 5 trading platform

Install the Calendar on your site

site: offer your visitors a powerful service for tracking global economic events.

- Flexible customization for your website: set the desired widget width and height, the amount of data displayed, the language and date format.
- Multilingual interface: all events and descriptions are fully translated into 9 languages, including English, Chinese, Japanese, Russian, Spanish, Portuguese, German, Turkish and Arabic.

Download the mobile calendar version



Install the
Calendar on
your site

- Full representation: every event is provided with a detailed description and notes related to the release influence on individual currencies, as well as the source link, release date and charts.
- No ads: you will not have to pay for the service by showing third-party ads.
- Automatic time zone selection: event tracking is maintained via local time adaptation. Manual adaptation of time zones is also

Download the mobile calendar version

Install the
Calendar on
your site

possible.

- Automatic update: the calendar is automatically updated as soon as a new event appears in a source.
- Continuous development: when a country is added, all economic indices which have a significant impact on the national currency are included in the calendar.

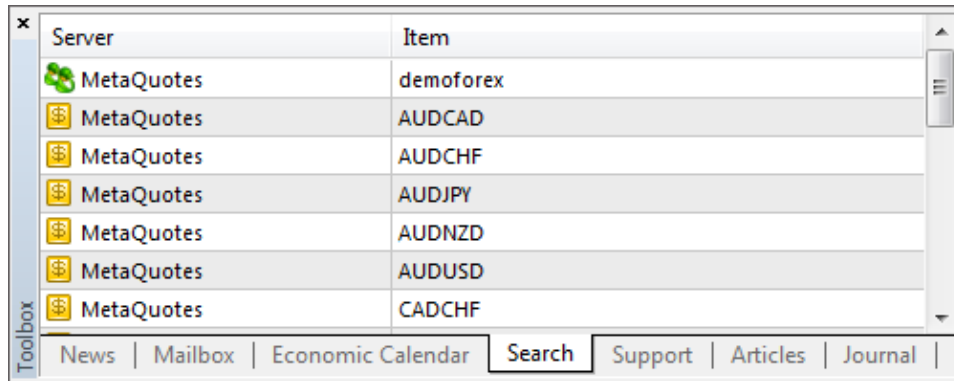
Download the mobile calendar version

Install the
Calendar in
your site >>

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [User Interface](#) / [Toolbox](#) / Search

Search

The result of global searching through the administrator terminal is displayed at this tab.



In this window elements found according to the search request are displayed. The information is shown in two columns:

- **Server** name of the platform component where the searched element was found.
- **Item** name of the found element.

If you click on the name of the found element, you will be moved to viewing it.

You can edit or delete records directly from search results, without opening the corresponding section of the Administrator terminal. The relevant operations can be performed using the context menu.

Also, the tab provides options for batch editing of same type elements. Select the required records by holding Shift or Ctrl+Shift, and then select "Edit" from the context menu.

Context Menu

The context menu of this tab allows to execute the following commands:

- **Open** open a selected element. When this command is executed the window of viewing and editing the selected element will be opened, also you will go to the corresponding section of the terminal. The same action can be performed by a double click with left mouse button on the line or by pressing the "Enter" key;
- **Auto Arrange** enable/disable the automatic setting of column size;
- **Grid** show/hide grid to separate fields.

[MetaTrader 5 Trading Platform](#) / [MetaTrader 5 Administrator](#) / [User Interface](#) / [Toolbox](#) / Journal

Journal

The "Journal" tab contains information about the registered actions of the administrator terminal for the current session of connection to the server. The information about the terminal start and events that occur during its working is registered in the journal. Only the last messages are displayed in the window. In order to view the folder that contains all the log files it is necessary to execute the "📁 Open" command in the context menu.

Time	Server	Message
2009.03.16 10:23:07	MetaQuotes-MQ Testers Server	'10': configuration synchronized with MetaQuotes-MQ Testers Server
2009.03.16 10:23:07	MetaQuotes-MQ Testers Server	'10': authorized on Access Server 5.00 build 11, Trade Server 5.00 build 11
2009.03.16 10:23:07	MetaQuotes-MQ Testers Server	'10': connected to MetaQuotes-MQ Testers Server
2009.03.16 10:23:04		MetaTrader Administrator 5.00 build 101 started

News | Mailbox | Economic Calendar | Search | Support | Articles | **Journal**

The journal entries are marked with the corresponding icons depending on their types:

- 📄 information message;
- ⚠ warning;
- ❌ error message.

Different types of errors returned by the server are described in the [separate section](#).

Context Menu

Context menu of the "Journal" tab contains the following commands:

- 📁 **Open** open the folder that contains the text files of the journal entries;
- 📋 **Copy** copy a selected entry of the journal to clipboard. You can copy the selected record using the "Ctrl+C" key combination;
- **Auto Arrange** if this option is enabled, the size of columns is selected automatically;
- **Grid** this option shows/hides grid to separate the fields of the journal entries table.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Terminal Settings

Terminal Settings

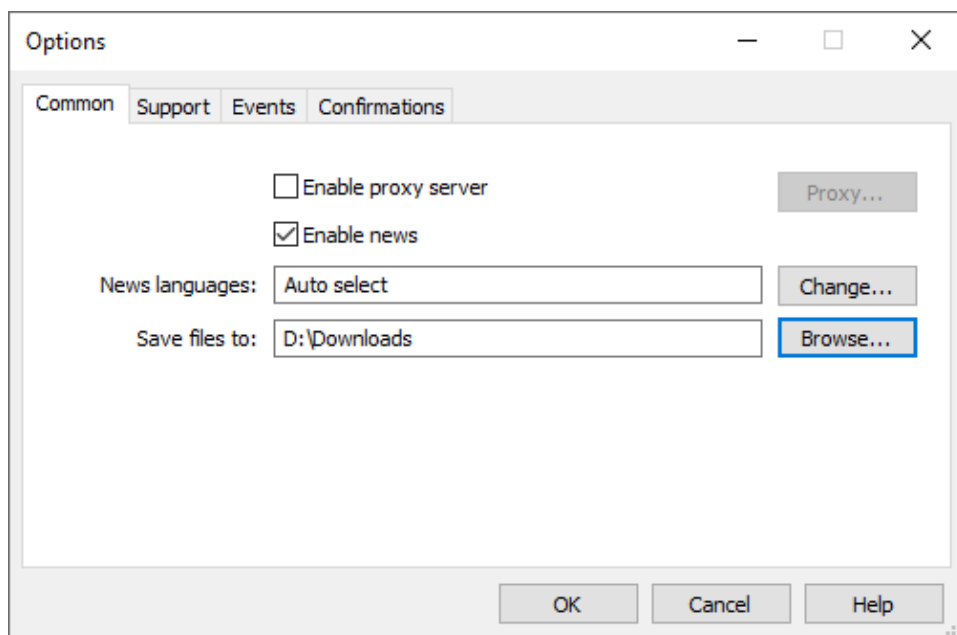
The window of the administrator terminal settings can be opened by pressing the "⚙️ Options" button in the "Services" menu or using the "Ctrl+O" key combination. All the settings are grouped in three tabs:

- **Common**
General settings of the administrator terminal.
- **Support**
The authorization details for contacting the technical support team.
- **Events**
Setup of notifications about different events in the terminal.
- **Confirmations**
Configuring confirmation requests for dangerous administration actions.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Terminal Settings / Common

Common

The "Common" tab contains the connection settings of the administrator terminal and the settings of receiving news.



The following options are present here:

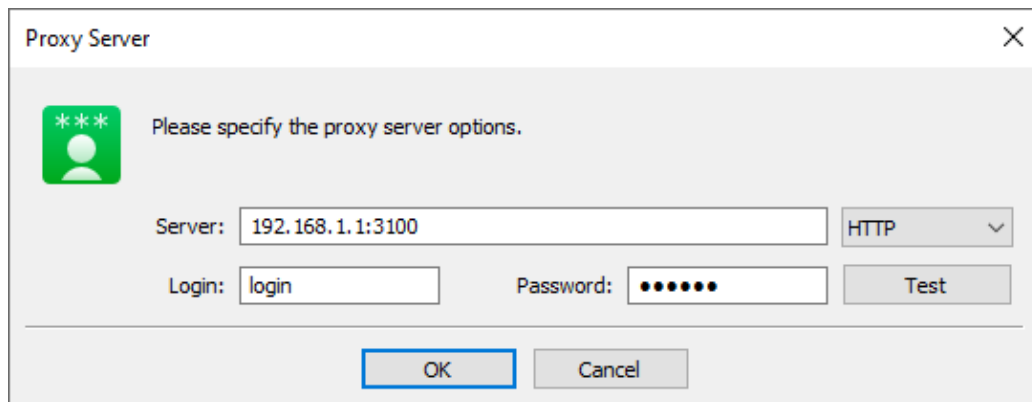
- **Enable proxy server** enable connection to administrated servers through a proxy server. If this option is checked, the "Proxy..." button becomes active. Using it one can set up connection to administered servers through a proxy server.
- **Enable news** this option allows to enable or disable news. If it is disabled, news won't come to the terminal;
- **Languages** this option allows to filter news by their language. The window of choosing the news language will appear as soon as the "Change" button is pressed. If the "Any language" parameter is indicated in this field then all news items will come to the terminal regardless of their language.
- **Save files to** the directory, to which files received via internal email system will be saved. When you open an attached file, the terminal checks its extension and the corresponds of file contents to this extension. If the file type is allowed and its contents are checked, the terminal will open the file and save it to the specified directory. Otherwise, a warning will be displayed, notifying that the file may harm the computer, and the file will not be saved. The following file types are allowed: PNG, JPG/JPEG, BMP, ZIP, 7Z, GIF, DOC, XLS, DOCX, XSLX, ODT, RTF, CSV, TXT, LOG.

If a file with the same name and a different content exists in the directory, the manager's login and email arrival date/time up to a second will be added to the saved file name: [login]-[file name]-[date and time].[extension]. For example,

the log file 20170501.log will be saved as
1001-20170501-20170502-170038.log.

By default, files are saved to C:\Users\[Windows
username]\Downloads\MetaTrader. You can change the directory by clicking on
"Browse".

Setting Up Connection through Proxy Server



In the "Proxy Server" window it is necessary to specify the following parameters:

- **Server** IP address and server port number separated by a colon are specified here. To the right of this field the type of proxy server is selected: HTTP, SOCKS4 or SOCKS5. In HTTP mode NTLM authorization is also supported;
- **Login** account to access the proxy server. If login is not needed, the field should be empty
- **Password** password to access the proxy server. If password is not needed, the field should be empty.

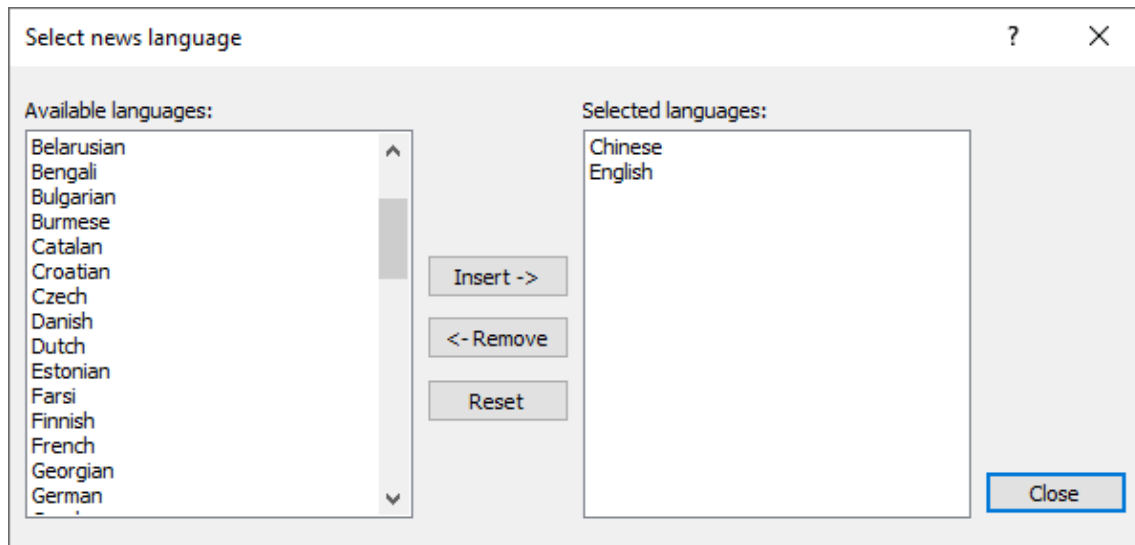
In order to verify the correctness of the settings of connection to the proxy server, press the "Test" button. In case of receiving the message that the settings are correct it is necessary to press the "OK" button to save the settings. The error message indicates that the proxy server is set up incorrectly. To find out the reason, contact the system administrator or internet provider.

The proxy server settings affect the connection to all the administered servers in the terminal.

Selecting News Language

In order to select only necessary languages of incoming news, the "Change" button should be pressed against the corresponding field. After that the following window will be opened:

MetaTrader 5 trading platform

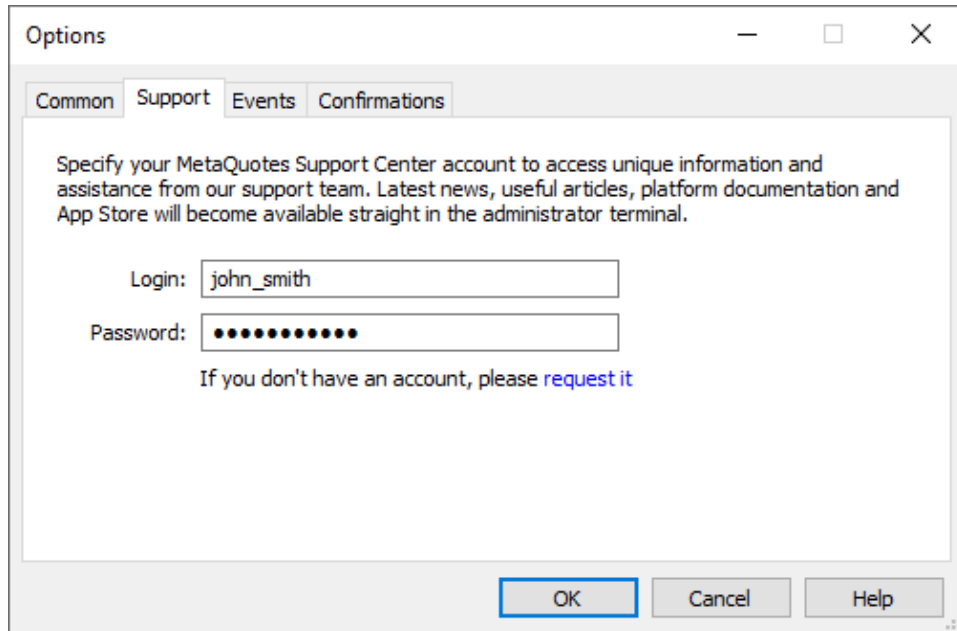


The left part of the window contains available languages, the right part - selected ones. In order to add a language, double-click on it or select it and press "Insert". Use "Remove" for removing languages from the list of selected ones. The "Reset" button returns default values. If "Any language" is indicated in the "News language" field, all news independent of the language will be received in the terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Terminal Settings / Support

Support

At this tab you can specify the authorization details (login and password) to access the special technical support website of MetaQuotes Software Corp.



Options

Common Support Events Confirmations

Specify your MetaQuotes Support Center account to access unique information and assistance from our support team. Latest news, useful articles, platform documentation and App Store will become available straight in the administrator terminal.

Login: john_smith

Password: ●●●●●●●●●●

If you don't have an account, please [request it](#)

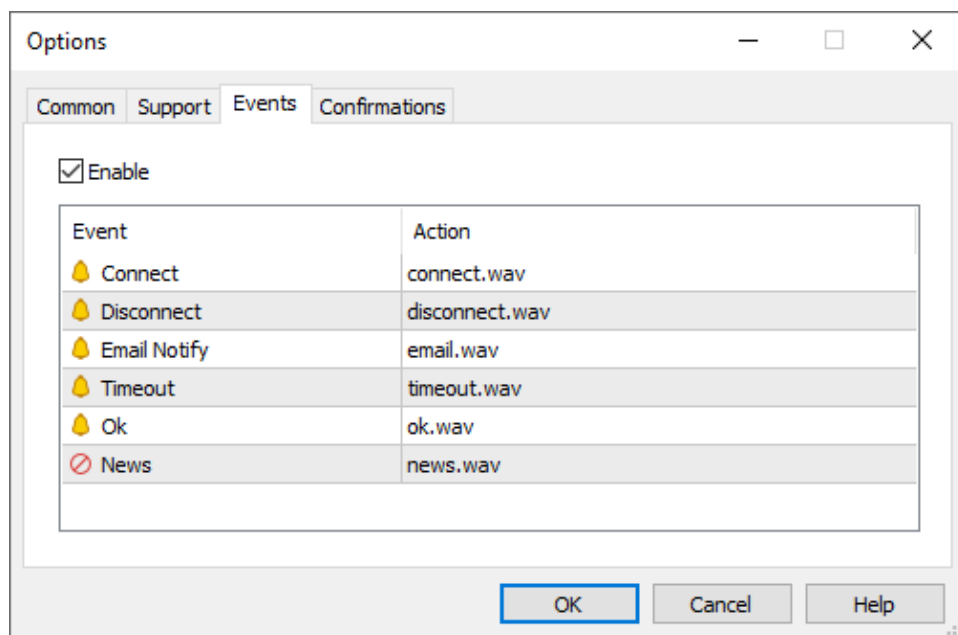
OK Cancel Help

This information will be used for contacting the technical support service right from the administrator terminal.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Terminal Settings / Events

Events

At the "Events" tab you can set up the sound notifications about different events occurring in the administrator terminal.



All the events are displayed in the form of a table that contains their names and the sound files (on default) that are played when they occur. The following types of events are represented here:

- **Connect** signal of successful connection to a server;
- **Disconnect** signal of losing connections with a server;
- **Email Notify** signal of receiving a message via the mail system;
- **Timeout** a certain time range is predefined for performing different operations (for example, requesting data from a server or refreshing the settings). If this range has been exceeded for some reason, the operation will not be performed, and this signal will trigger;
- **Ok** signal of a successfully performed operation;
- **News** signal of received news.

If there is a need to disable any of the signals, it is necessary to double-click on its icon 🔔 or double-click on its name. After that the icon will change to 🚫. To enable a signal the same operation must be performed.

In order to change a file played at the signal activation, double-click on its name or select it and press the "Enter" key. After that click "Choose other..." in the pop-up list and indicate the necessary file in the appeared window.

By default a file with *.wav extension is offered as a sound. However, another file can be indicated as a signal. If a *.wav file is selected, at the event triggering the file will be played. If another file is selected, it will be opened using application it is associated with in the operating system.

For changes to take effect the "OK" button should be pressed.

MetaTrader 5 Trading Platform / MetaTrader 5 Administrator / Terminal Settings / Confirmations

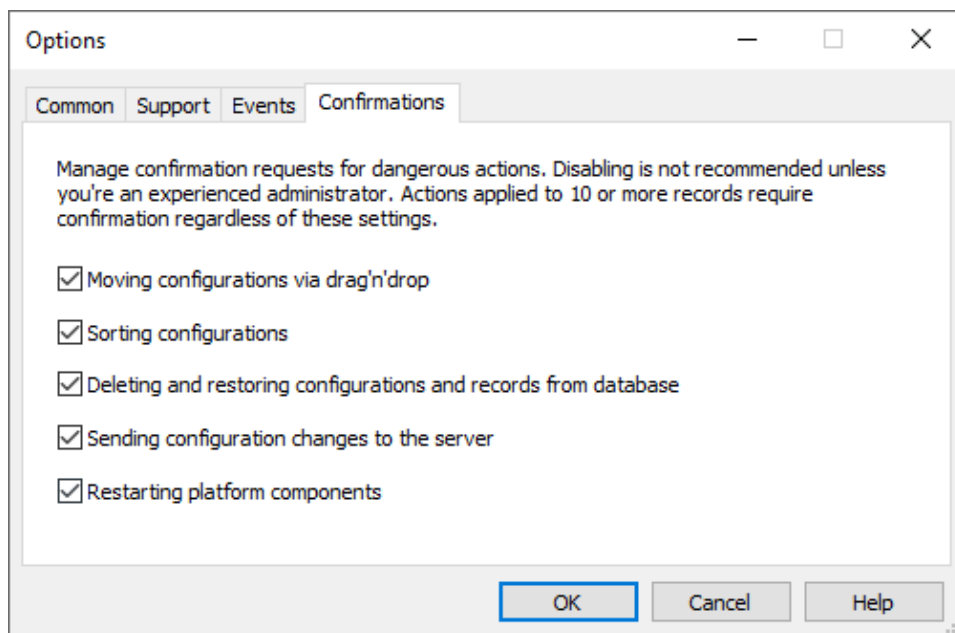
Confirmations

From this section, you can enable or disable additional confirmation requests for dangerous actions executed via the Administrator terminal.

The dangerous actions include:

- Moving configurations via drag'n'drop (to protect against accidental actions)
- Sorting and deleting configurations; sending configuration changes to the server (the Apply command)
- Deleting and restoring records from backup
- Restarting cluster components

When performing any of these actions, the terminal requests additional confirmation. Confirmations can be disabled for experienced administrators, or when performing a large platform reconfiguration.



MetaTrader 5 trading platform

The confirmation is always requested for actions applied to 10 or more entries.

MetaTrader 5 Trading Platform / App Store

App Store

App Store is the market of MetaTrader 5 solutions from third-party software providers. It features dozens of products from various categories, including gateways, liquidity solutions, CRM systems and traders' rooms, as well as trade server hosting services. Every product is provided with a detailed description of functionality and scope of application.

All software providers in the App Store are subject to a special registration procedure. They all sign an agreement with MetaQuotes Software Corp. Detailed information on all third-party companies is provided under the [Vendors section of the technical support website](#).

For convenience, all products are divided into categories. Use the Navigator window to switch between the categories.

- Aggregation and Liquidity: price aggregation software, liquidity sources and gateways
- News and Data feeds: price feeds, news and analytics sources
- White Labels and Partners: White Label for Introducing Brokers and Affiliate Programs
- Money Management: money management software, implementation of PAMM and MAMM accounts
- Reporting: expanded reports
- Risk Management software
- IT and Hosting services: hosting and MetaTrader 5 server management services
- Websites, CRM and Integrations: creation of websites, traders' rooms, integration with the MetaTrader 5 platform
- Business Consulting: legal and business advising services
- Supplementary apps: additional applications

MetaTrader 5 trading platform

Fill out a simple form to request a desired product. After that, a vendor's representative will contact you to complete the order procedure.

The screenshot shows the MetaTrader 5 Administrator window. On the left is a tree view of the system components, including Servers, Support Center, and App Store. The App Store is expanded, showing various categories like Aggregation and Liquidity, News and Data feeds, etc. The main panel displays the 'MetaTrader 5 App Store' header and a form to 'Send request to company MetaQuotes Software Corp.' for the 'Website Integration' item, priced at 10 000 / 1 000 USD. The form includes fields for name, company, position, and email, along with a text area to describe the request and a 'Send request' button.

MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Administrator

- Servers
 - MetaTrader
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 - Integrations
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 - Time
 - Holidays
 - Groups (185)
 - Clients & Orders
 - Managers
 - Routing
 - Gateways
 - Plugins
 - Data Feeds
 - Reports
 - Symbols (70159)
 - Spreads
 - Charts & Ticks
 - Live Update
- Support Center
- App Store
 - Aggregation and Liquidity (35)
 - News and Data feeds (5)
 - White Labels and Partners (3)
 - Money Management (5)
 - Reporting (1)
 - Risk Management (10)
 - IT and Hosting services (6)
 - Websites, CRM and Integrations
 - Business Consulting (2)
 - Supplementary apps (8)

MetaTrader 5 App Store

[App Store](#) / [Websites, CRM and Integrations](#)

Send request to company MetaQuotes Software Corp.

Web API

Item: **Website Integration**

Price: **10 000 / 1 000 USD**

Your name:

Company:

Position:

Email:

Describe your request

For Help, press F1

527 / 0 Kb

MetaTrader 5 trading platform

- The price of some applications may include mandatory monthly technical support fees.
- MetaQuotes support site account is required for accessing the App Store section. Specify the account credentials in the Administrator terminal settings.

MetaTrader 5 Trading Platform / Technical Support

Technical Support

The technical support is embedded directly in the Administrator terminal. It provides the complete documentation on all trading platform components, useful articles and forum. You can also directly contact the platform developers: ask question to a support team member via the chat or create a Service Desk request.

MetaTrader 5 trading platform

MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

Administrator

- Servers
 - MetaTrader Server
 - Network Cluster
 - Integrations
 - Firewall
 - Time
 - Holidays
 - Groups (653)
 - Clients & Accounts
 - Orders & Deals
 - Gateways
 - Data Feeds
 - Plugins
 - Reports
 - ECN
 - Routing
 - Funds & ETF
 - Symbols (84185)
 - Charts & Live Update
- Support Center**
 - News
 - Articles
 - Platform Documents
 - API Documents
 - Service Desk
 - Forum
 - Online Assistant
 - Search
- App Store

For Help, press F1

Service Desk

To list

Title:

Category:

Affected software:

Problem description:

MetaQuotes for MetaTrader 5 Platform

The App Store for MetaTrader 5 platform full of useful tools

Articles

MetaTrader 5

Workshops

MetaQuotes Support Center

London Summit 2019

13 November
09:00 — 17:30 GMT+0

Old Billingsgate, London, UK
1 Old Billingsgate Walk (Riverside, 16 Lower Thames St, Billingsgate, London EC3R 6DX)

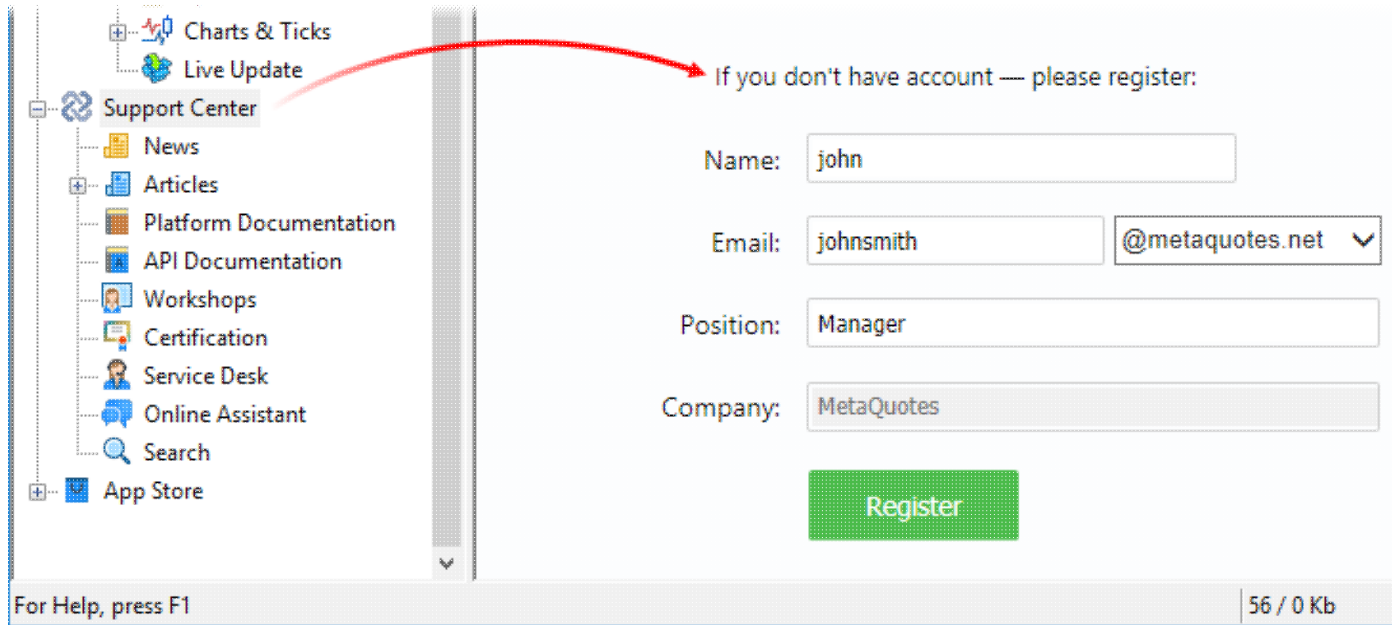
[Register](#)

Why is it important to update the 1st, 2020?

Keeping software up-to-date is critical for the failure-free operation of our platforms versions. This time we announce the end of support for versions below 1170 by January the 1st, 2020. The latest platform version build is 1220. Some brokers have been using old versions for years, skipping important releases. The longer you put off the platform updates, the more versions you miss. This may cause issues with the installation of the latest version after skipping previously released updates. Please keep your platforms up to date!

MetaTrader 5 trading platform

To access this section, you need an account on the [MetaQuotes technical support website](#). Specify it in the [Administrator terminal settings](#). If you do not have an account, register it straight from the terminal by filling out a simple form:



The screenshot shows the MetaTrader 5 interface. On the left is a vertical menu with the following items: Charts & Ticks, Live Update, Support Center (highlighted), News, Articles, Platform Documentation, API Documentation, Workshops, Certification, Service Desk, Online Assistant, Search, and App Store. A red arrow points from the 'Support Center' item to a registration form on the right. The form is titled 'If you don't have account — please register:'. It contains the following fields: 'Name:' with the value 'john', 'Email:' with the value 'johnsmith' and a dropdown menu showing '@metaquotes.net', 'Position:' with the value 'Manager', and 'Company:' with the value 'MetaQuotes'. A green 'Register' button is at the bottom of the form. At the bottom of the window, a status bar shows 'For Help, press F1' on the left and '56 / 0 Kb' on the right.

Charts & Ticks
Live Update
Support Center
News
Articles
Platform Documentation
API Documentation
Workshops
Certification
Service Desk
Online Assistant
Search
App Store

If you don't have account — please register:

Name: john

Email: johnsmith @metaquotes.net

Position: Manager

Company: MetaQuotes

Register

For Help, press F1 56 / 0 Kb

MetaTrader 5 Trading Platform / Additional Features

Additional Features

This section describes additional features of the components of the MetaTrader 5 trading platform:

- URL Schemes
- Marketing Campaigns

URL Schemes

Mobile terminals MetaTrader 5 for iPhone and MetaTrader 5 for Android support the Deep Linking technology. This technology allows using links that bring users to a specified location within a mobile app.

For example, quick account connection links can be added to registration emails and to the trader's room on the website. A click on this link opens a trader's mobile terminal with the selected account connection section, in which the account number and server are automatically specified. The trader will only need to enter a password.

Launch of the application and navigation to the selected section is implemented through links starting with `metatrader5://...`. The part of link before `://"` is called a URL scheme.

The `metatrader5` URL scheme is registered in the operating system of the mobile device during terminal installation. If the terminal is not installed, the links will not open.

The general scheme of links:

`metatrader5://path?par1=val1&...&parN=valN`

where:

- 'path' is the destination section of the application
- par is the list of parameters

Connection to a specified account on the server

The link should contain the application section, to which you want to link your traders, as well as the 'login' and 'server' parameters containing the account number and the name of the server to connect to. Example:

`metatrader5://account?login=123456&server=MetaQuotes-Demo`

Upon a click on such a link, the MetaTrader 5 app starts and tries to connect to the specified account. If the account already exists in the terminal and its password has been saved, connection is performed successfully. Otherwise, the account connection form is shown with the account and server already specified, and a user is prompted to enter the password. A server name should exactly match the name in the White

Label.

- If the server name is not specified, the server of the current account or of the first account from the list will be used.
- If the account number is not specified, and the server name is specified, any account from the specified server will be selected. If there are no such accounts, the scheme will not be executed.
- If both the account and the server are not specified, the current account or the first account from the list will be used.

Link to a specified section

In order to forward users to the specified section, specify the appropriate 'path' value in the link:

```
metatrader5://<path>?login=123456&server=MetaQuotes-Demo
```

After <path>, you can specify the account number and the server name, to which the terminal should connect before opening the section. If the account or server is not specified, the connection rules described above apply.

- account no specific section will be opened, it is used for connection to a specified account and server.
- marketwatch quotes. The example contains the additional 'symbols' parameter, in which financial instruments to be displayed in the window are specified.

```
metatrader5://marketwatch?server=MetaQuotes-Demo&symbols=USDRUB,EURRUB,GOLD,
```

- chart/symbol/period is a chart with the specified symbol and timeframe, e.g.:

```
metatrader5://chart/XAUUSD/H1?server=MetaQuotes-Demo
```

- trade Trade tab, the list of open positions and orders, e.g.:

```
metatrader5://trade?server=MetaQuotes-Demo
```

- trade/symbol a trade dialog, from which a trading operation on the specified symbol can be performed, e.g.:

```
metatrader5://trade/SILVER?server=MetaQuotes-Demo&login=1145990
```

Illegal characters in URI links must be encoded. For example, the #GOOG must be specified as %23GOOG. Names of financial instruments are case sensitive.

Some email services may erase links beginning with "metatrader4://" or "metatrader5://". In this case, instead of deep link, insert a link to a certain section of your website, which in turn re-directs users to the mobile application.

Extra parameters

It is possible to specify additional parameters for each deep linked section:

- 'login' and 'server' can be used for connection with the specified account number and server name.
- symbols one or more symbols separated by commas, which should be added to the Quotes section after connection to an account defined by the 'login' and 'server' parameters. Illegal characters in URI links must be encoded. For example, the #GOOG must be specified as %23GOOG. Names of financial instruments are case sensitive. Symbols passed do not replace those already displayed in the "Quotes" section, they are added in the current list.
- account_type run the demo or real account opening wizard on the first application start. 1 is used for demo accounts and 2 for real ones. To use this feature, you should correctly specify account allocation settings. If the required group for opening an account is not found, the wizard will not run.

Further information is provided in the article "[Deep Linking to MetaTrader mobile terminals and to their selected parts](#)"

Marketing Campaigns

The client record contains two special fields in the trading platform: Lead Source and Lead Campaign. They are used for marketing campaigns allowing you to track where a client came from. To receive the data, add the following labels to the client or mobile platform download link:

https://download.mql5.com/cdn/web/metaquotes.ltd/mt5/mt5setup.exe?utm_source=YourWebsite
https://download.mql5.com/cdn/mobile/mt5/ios?server=ABC-Demo,ABC-Real&utm_source=YourWebsite
https://download.mql5.com/cdn/mobile/mt5/android?server=ABC-Demo,ABC-Real&utm_source=YourWebsite

where YourCampaign is a campaign name, while YourWebsite is a website the link has been placed at. In the 'server' parameter of the mobile platform links, enter the list of your servers to be shown to traders when they open an account.

When opening a demo account and connecting to any trading account via the terminal downloaded using such a link, utm_source and utm_campaign values are set in a client record at the server side. If the fields are already filled, the label values are not overwritten when re-connecting to the account (even if the terminal used for connection was downloaded by a link containing other labels).

Lead Source and Lead Campaign data from terminal download links can be written to created accounts only if the broker has a Finteza license.